1	1001	MAIN ARITHMETIC	II
2	1002	REGISTERS	
3	1003	STATUS	II
4	1004	MEMORY DATA	
5	1005	MEMORY ADDRESS	IV
6	1006	OR LOGIC	III, IV
7	1007	DECODING	V
8	1008	INTERRUPT REGISTERS	I, II
9	1009	MICROADDRESSING	
10	1010	I/O ADDRESS	IV
11	1011	I/O DATA	II
12	1012	RACK DRIVER	II
13	1013	DEVICE-REGISTERS	IV
14	1014	DMA REGISTERS	IV
15	1015-17	DRUM SEQUENSER-CONTROL-TIMING	IV.E, III, III
16	1018	TAPE READER INTERFACE	VII
17	1019, 1021	RACK CONTROL 1-2	IV, IV
18	1020	TELETYPE BUFFER	IV
19	1022	BUS CONTROL	VI.G
20	1023	1 K PROM	II
21	1024	RTCL N-10	IV.G
22	1025	TIME COUNTER	V
23	1026	PUNCH INTERFACE	IV
24	1027	I/O CONTROL	VII
25	1028	INTERRUPT CONTROL	
26	1029-30	PANEL CONTROL 1-2	VI, II
27	1031-32	PANEL DATA 1-2	IV, IV
28	1033	PANEL DRIVER	I, IV
29	1034	PAGING SURROGAT	III
30	1035	MEMORY CONTROL	II
31	1036-39	DISC TXR, CONTROL, SEQUENCE, TIMING	IV, V, III, IV, V

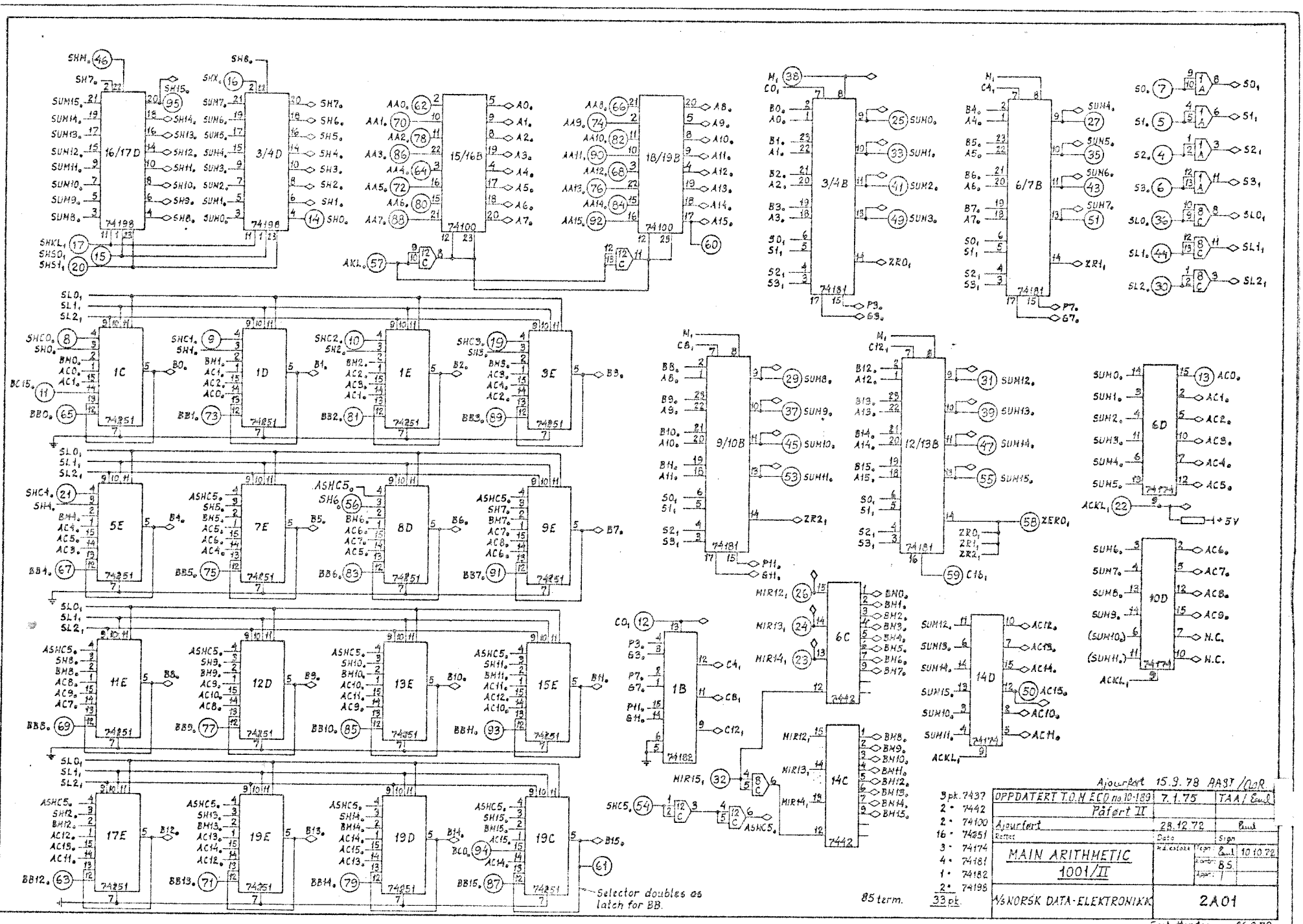
DOSAB Art.nr. 59207

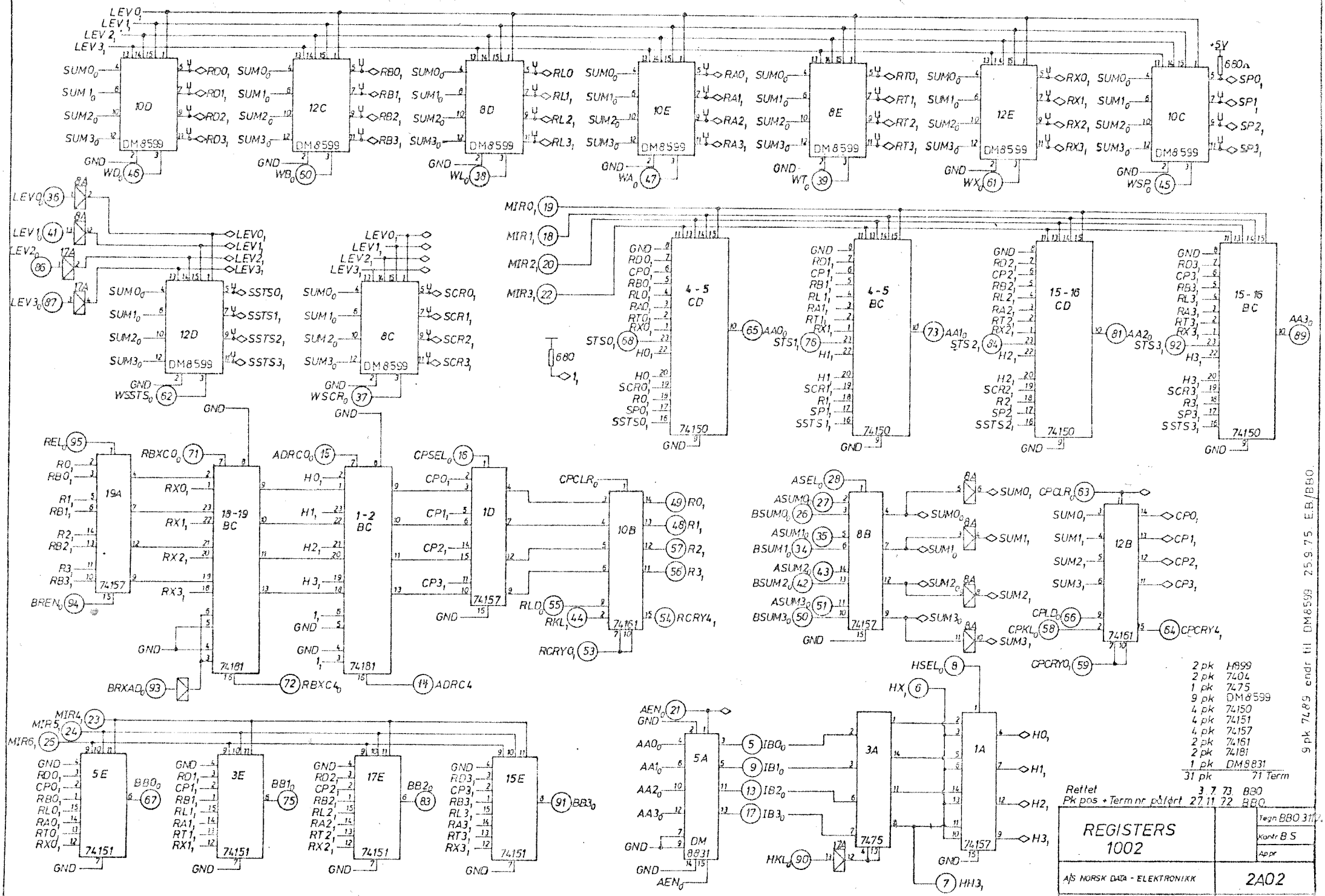
1	1040	PAGING	IV.V.F
2	1041-42	16-BIT DIGITAL INPUT-OUTPUT	III, III
3	1043	CARD READER BUFFER DOCUMATION	IV.F
4	1044		
5	1045	CDC 3000 CHANNEL XMIT AND REC.	I.II
6	1046	ASYNC. MODEM BUFFER	III.V.F
7	1047	CDC LINE PRINTER INTERFACE	III
8	1048-49	MAGTAPE CONTROL A-B	IV, II
9	1050	SYNCHRONOUS MODEM BUFFER	
10	1051-53	READ-WRITE-DRIVE CONTROL MODULE	
11	1054	PANEL CONTROL 2	VI
12	1055	MEMORY CONTROL	VI
13	1056-57	PROCESS INPUT-OUTPUT	II.C, III
14	1058	INTERRUPT CONTROL	III
15	1059		
16	1060	4 K RAM	IV
17	1061	HP PARITY	II.III
18	1062	STATUS	
19	1063-64	HP TIMING A-B	V, II.III.IV
20	1065		
21	1066	4 K MOS RAM MODULE "MOSTEK"	
22	1067	4 K PROM MEMORY	II
23	1068	ACM I/O	
24	1069		
25	1070	DATA AND ADDRESS BUS SELECTOR	I.B
26	1071	NORD 50 DATA	E.F
27	1072	HP CONTROL B	IV.V
28	1073	CONTROL-DRIVER	I.III.D.E
29	1074	CONTROL RECEIVER	III.D.E
30	1075	MICROADRESSING	
31	1076-80	SMD TIMING-CONTROL-RX-TX-UNIT CONTROL	

DORAB Art.nr. 59207

1	1081,83,89,90	MPM DATA-ADR. LINE DRIVER – REFRESH-CONTROL
2	1082	
3	1084	
4	1085	
5	1086	
6	1087	
7	1088	
8	1091	
9	1092	BUFFERED DMA REGISTER
10	1093	BRANCH TRANSCEIVER
11	1094	8 K RAM
12	1095	TERMINAL BUFFER
13	1096	DMA ADDRESS
14	1097	PROCESS CONSOLE CONTROL
15	1098	DEVICE SWITCH
16	1099	
17	1101	BRANCH CONTROL
18	1102	VERSATEC CONTROL MATRIX PRINTER/PLOTTER
19	1103	DIFF TRANSCEIVER
20	1104-05	CALCOMP PLOTTER/500-900
21	1106	
22	1107	DISC SEQUENCE
23	1108	REMOTE BYTE
24	1109	PARALLELL BYTE OUTPUT
25	1110	
26	1111	FLOPPY DISC CONTROL
27	1112	BUFFERED CALCOMP PLOTTER 500/900
28	1113	WATCHDOG
29	1114	PANEL TRANSCEIVER
30	1115	LAMP REGISTER
31	1116	

DORAB Art.nr. 59207


16/17D
3/4D
15/16B
18/19B
3/4B
6/7B
1C
1D
1E
9E
5E
7E
8D
9E
11E
12D
13E
15E
17E
19E
19D
19C
9/10B
12/13B
6D
6C
1B
14C
14D
9D
74198
74100
74181
74174
74442
74188
74851
Selector doubles as latch for BB.
85 term.
33 pk.
Ajourført 15.9.78 AAST/AoR
OPPDATERT T.O.H ECO no 10-189 7.1.75 TAA/Emil
Påført II
Ajourført 28.12.72 Emil
MAIN ARITHMETIC
1001/II
A/S NORSK DATA-ELEKTRONIKK
2A01
Erstatter tegn. av 26.9.72

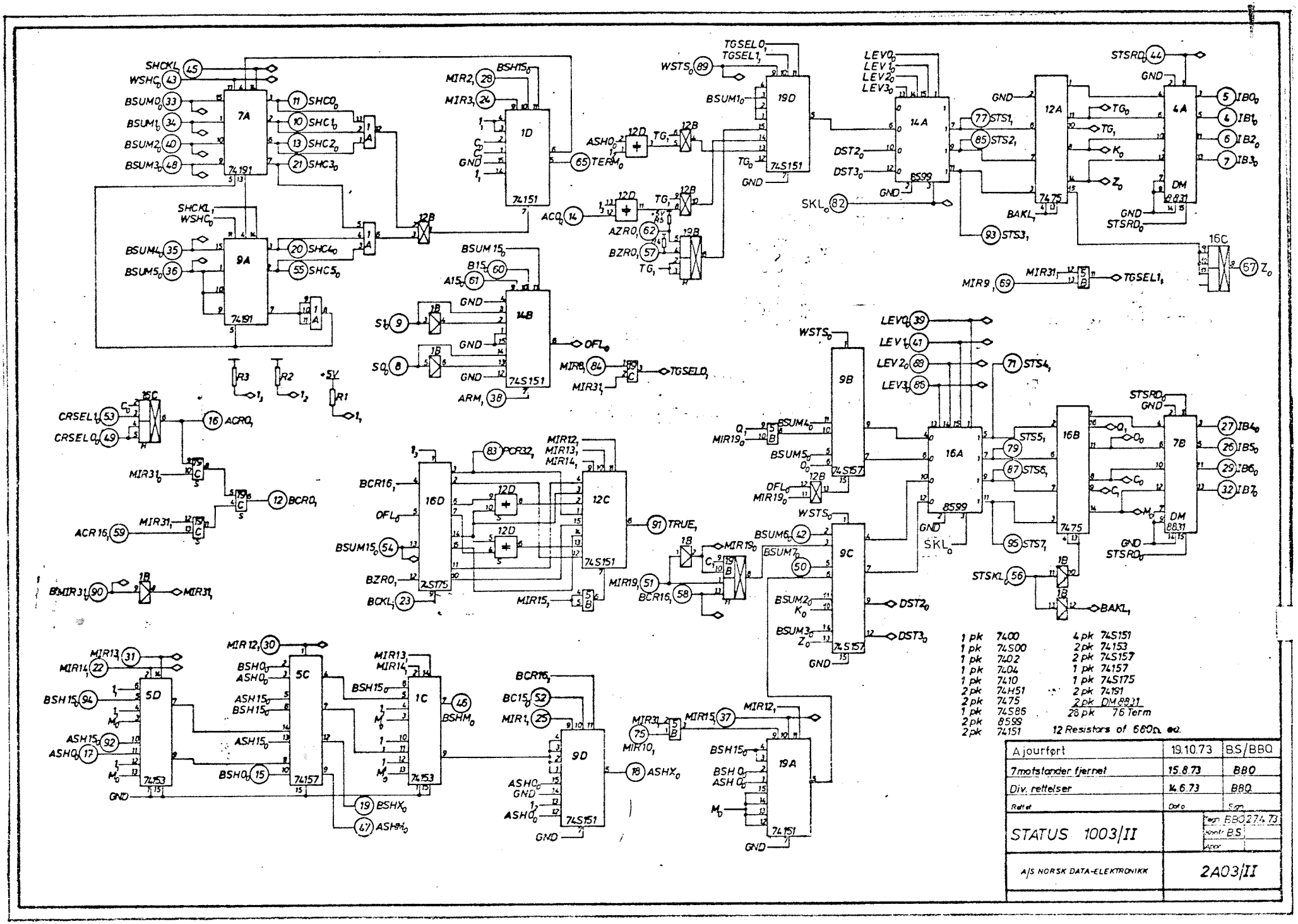


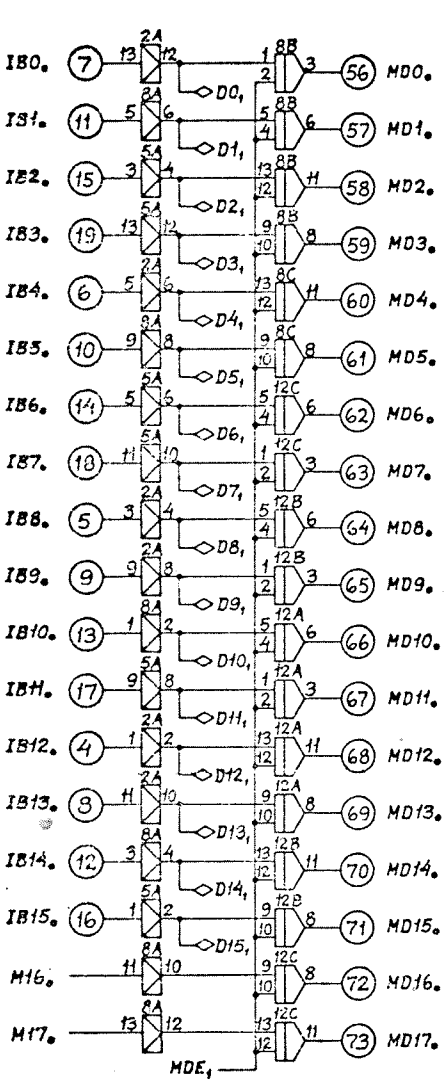
2 pk	H899
2 pk	7404
1 pk	7475
9 pk	DM8599
4 pk	74150
4 pk	74151
4 pk	74157
2 pk	74161
2 pk	74181
1 pk	DM8831
31 pk	71 Term

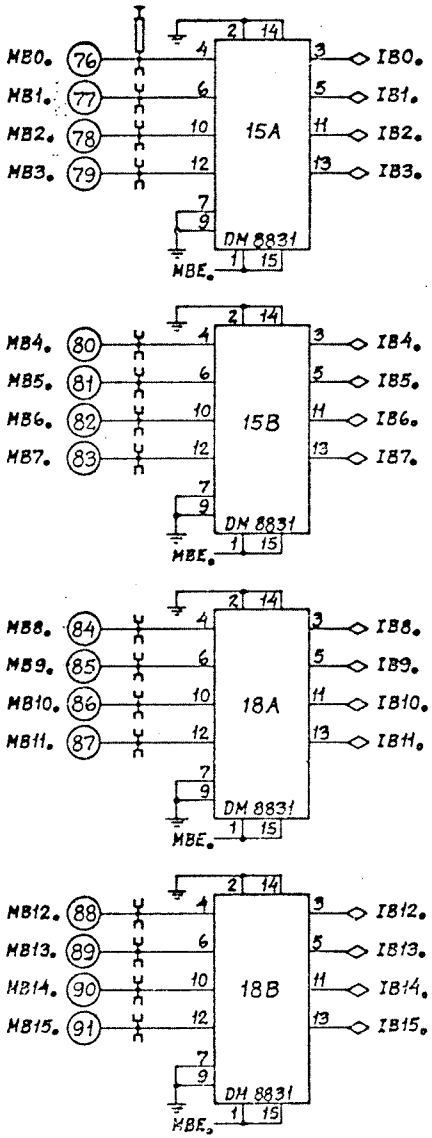
9 pk 7489 endr til DM8599 25.9.75 EB/BBO

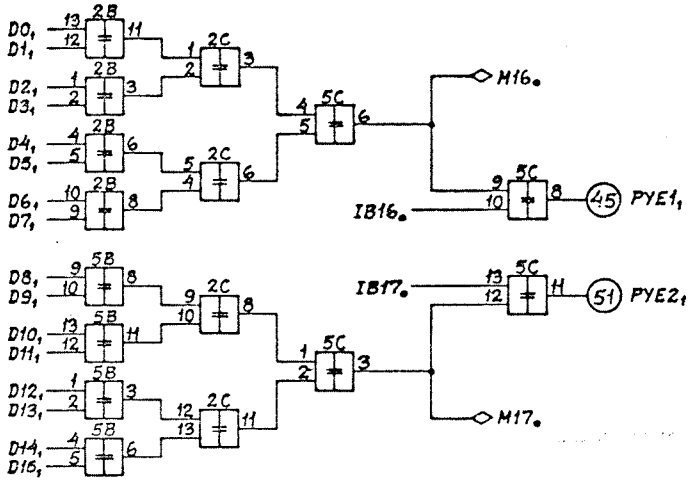
Rettet 3 7 73 BBO
Pk pos + Term nr påført 27 11 72 BBO

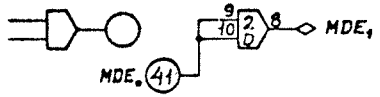
REGISTERS 1002	Tegn BBO 31.7.72
	Kontr B.S
	Appr
A/S NORSK DATA - ELEKTRONIKK	2A02

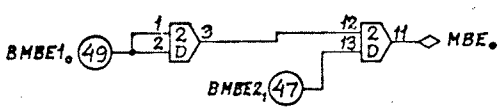

7A
74191
9A
74191
1D
74151
14B
74S151
19D
74S151
14A
8599
12A
74175
4A
DM 8831
16C
9B
74S157
16A
8599
16B
74175
7B
DM 8831
9C
74S157
16D
74S175
12C
74S151
5C
74157
5D
74153
1C
74153
9D
74S151
19A
74151
1 pk 7400
1 pk 74S00
1 pk 7402
1 pk 7404
1 pk 7410
2 pk 74H51
2 pk 7475
1 pk 74S86
2 pk 8599
2 pk 74151
4 pk 74S151
2 pk 74153
2 pk 74S157
1 pk 74157
1 pk 74S175
2 pk 74191
2 pk DM 8831
28 pk 76 Term
12 Resistors of 680Ω ea.
A jourført 19.10.73 BS/BBQ
7 motstander fjernet 15.8.73 BBQ
Div. rettelser 14.6.73 BBQ
Rettet Dato Sign
STATUS 1003/II
Tegn. BBQ 27.4.73
Kontr. B.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A03/II

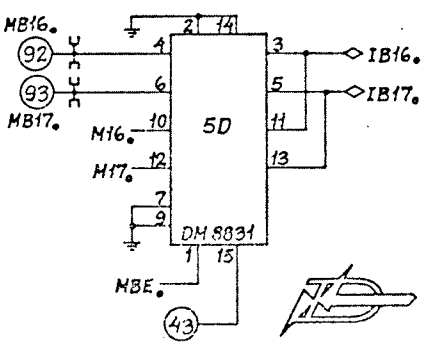


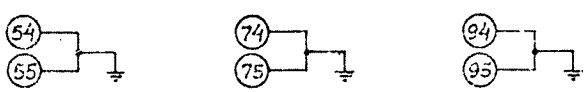












3pk. 7404
1 = 7437
5 = 7438
4 = 74S86
5 = DM 8831
2 = H-988-470
2 = H-988-390
22 pk.

64 term.

Term.nr. MD0. - MD17. forandr.	29.6.73	Emil
Ajourført, påført pakkepos. og term.nr.	17.1.73	Emil
Rettet	Dato	Sign.

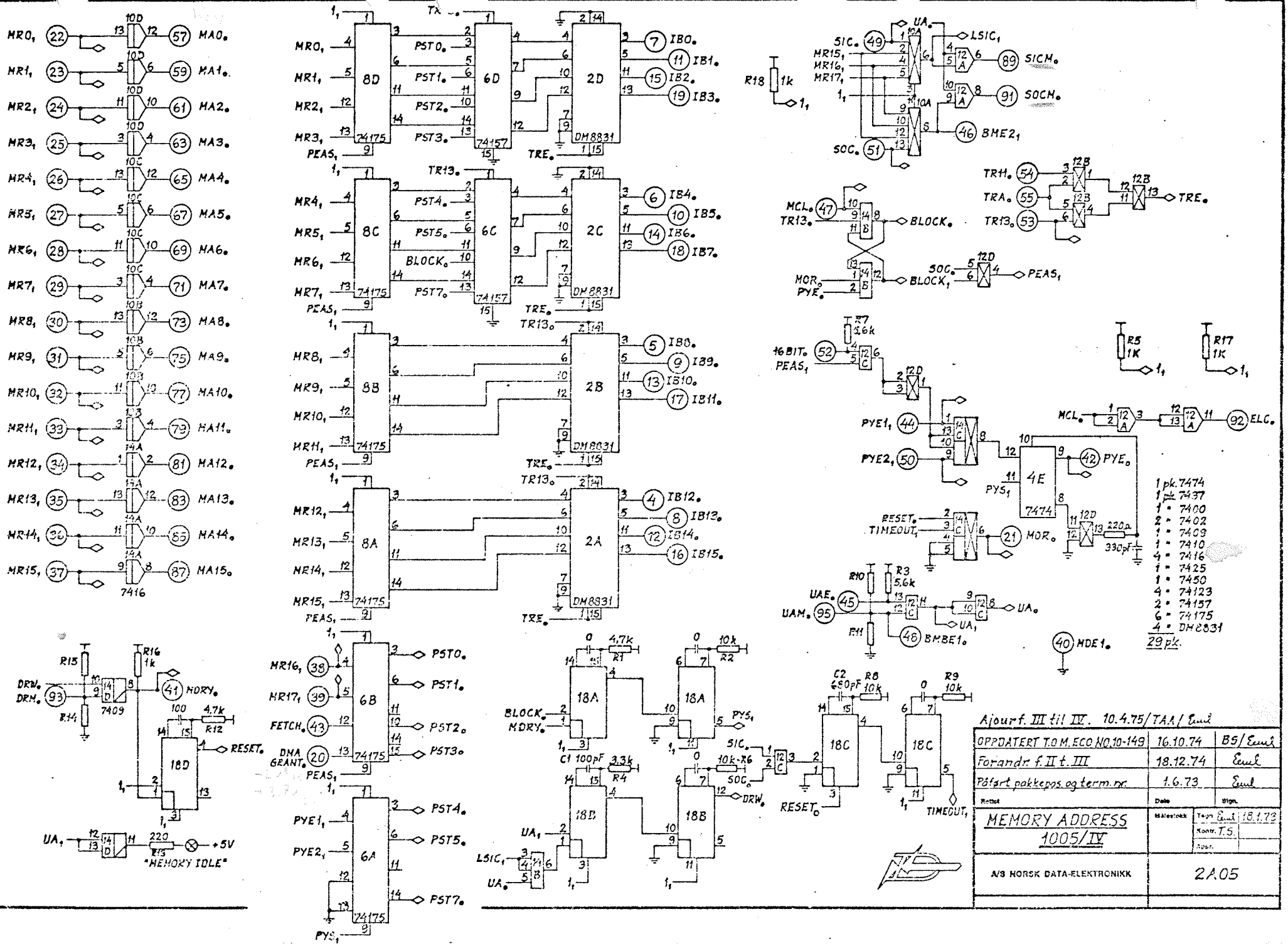
MEMORY DATA
1004

Målestokk

Tegn. Emil 10.1.73
Kontr. T.S.
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A04



Ajourf. III til IV. 10.4.75/TAA/ Eml

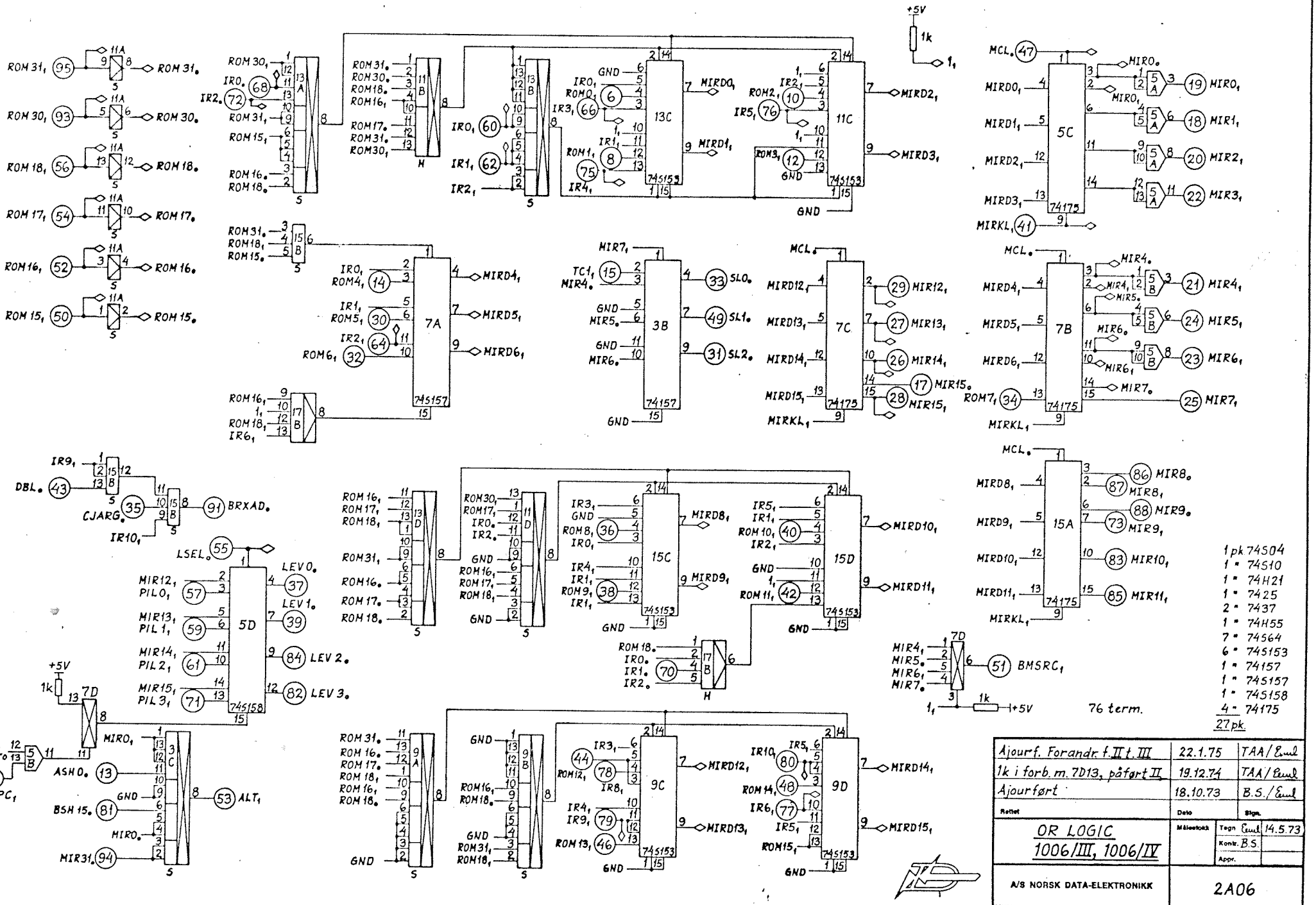
OPPDATERT T.O.M. ECO NO. 10-149	16.10.74	BS/ Eml
Forandr. f. II t. III	18.12.74	Eml
Påført pakkepos. og term. nr.	1.6.73	Eml
Retsat	Dato	Sign.

MEMORY ADDRESS 1005/IV

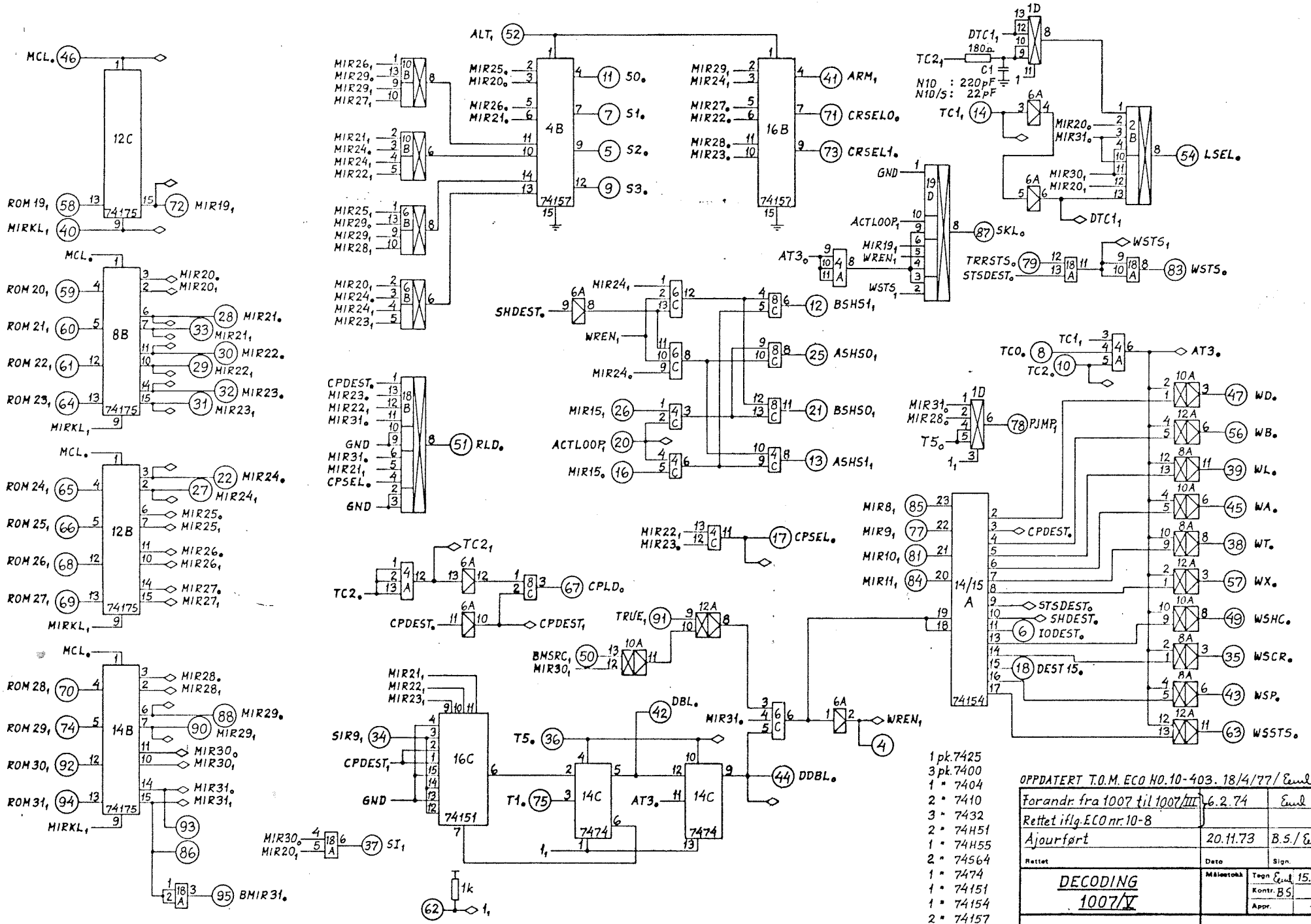
Tegn. Eml 18.1.73
Kontr. T.S.
Godk.

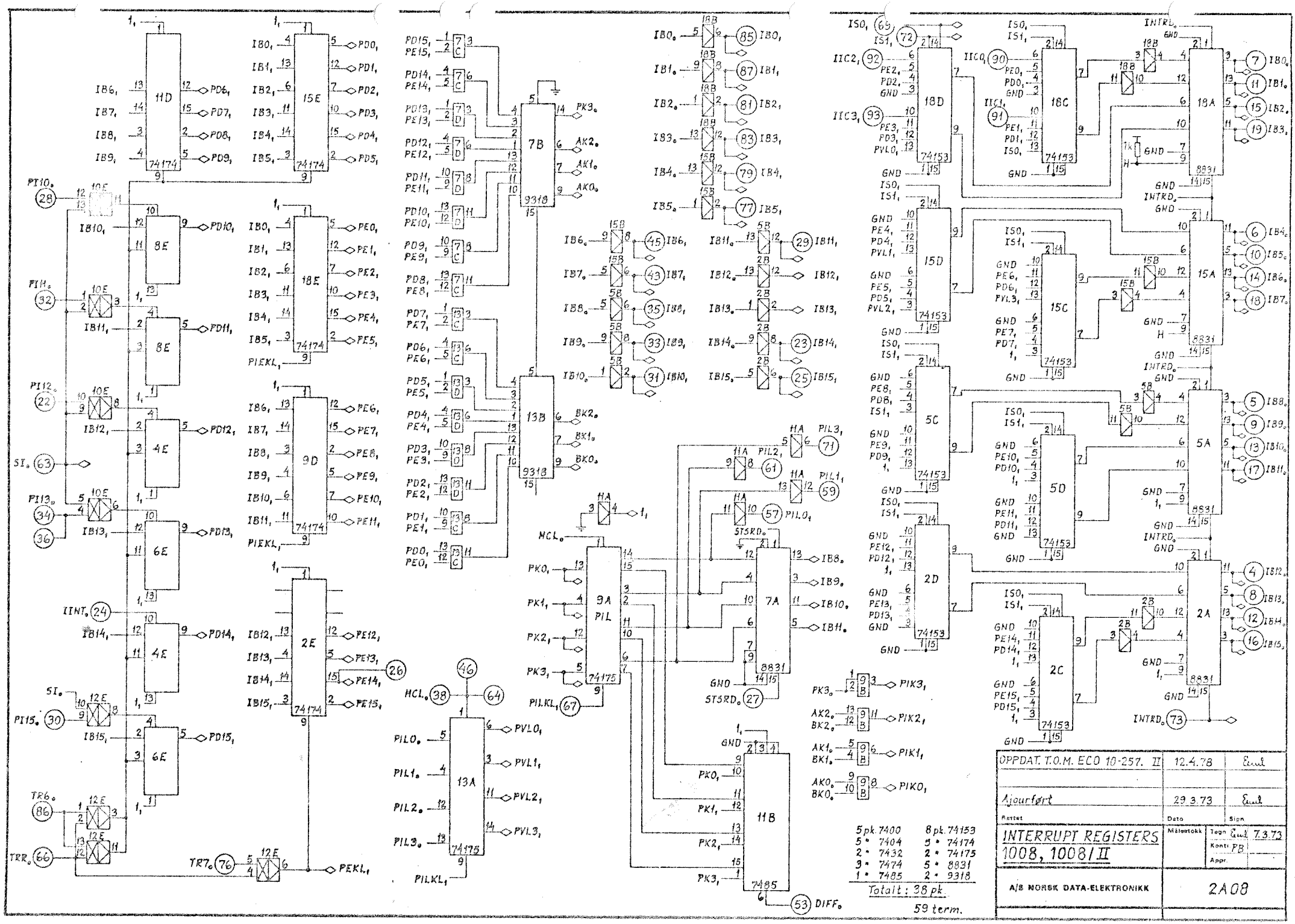
A/S NORSK DATA-ELEKTRONIKK

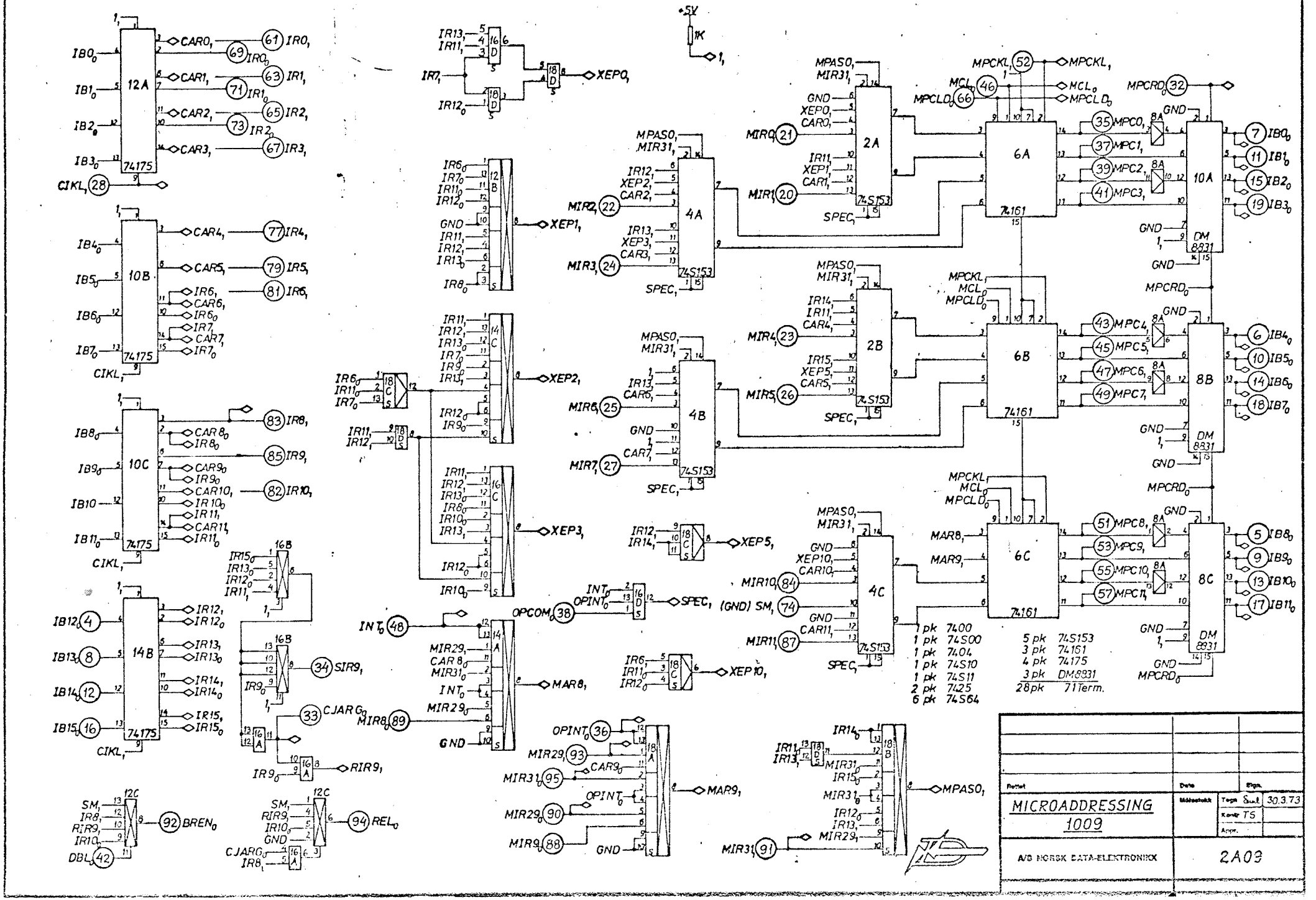
2A05

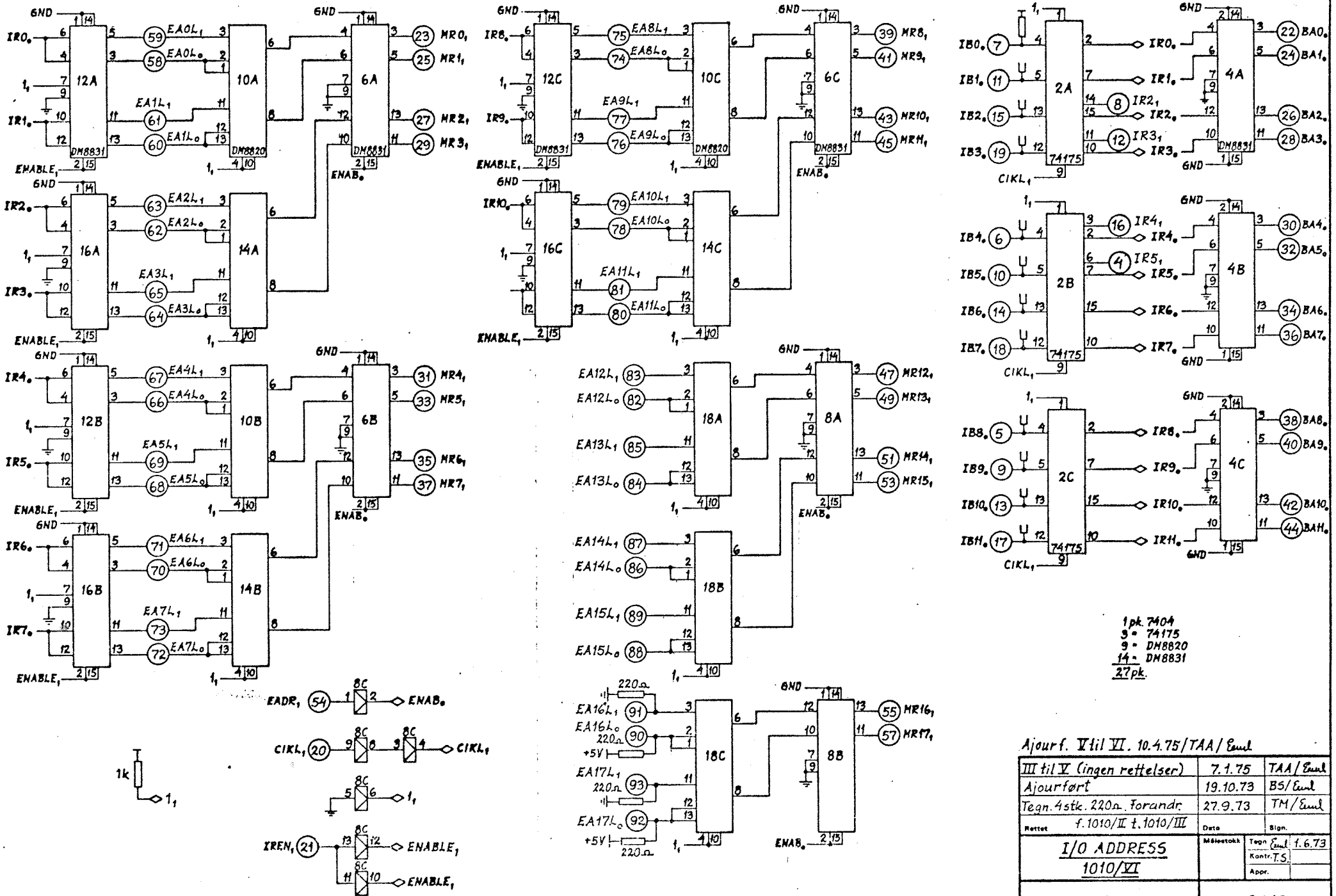


Fjernet 1stk. 1k (R3). 11/7/77/EB/Eml
Påført 1006/III (ingen rettelser) 9.12.75/TAA/Eml


MCL. 46
12C
ROM 19, 58
74175
72 MIR19,
MIRKL, 40
8B
ROM 20, 59
ROM 21, 60
ROM 22, 61
ROM 23, 64
MIR20.
MIR20,
28 MIR21.
33 MIR21,
30 MIR22.
29 MIR22,
32 MIR23.
31 MIR23,
12B
ROM 24, 65
ROM 25, 66
ROM 26, 68
ROM 27, 69
22 MIR24.
27 MIR24,
MIR25.
MIR25,
MIR26.
MIR26,
MIR27.
MIR27,
14B
ROM 28, 70
ROM 29, 74
ROM 30, 92
ROM 31, 94
MIR28.
MIR28,
88 MIR29.
90 MIR29,
MIR30.
MIR30,
MIR31.
MIR31,
93
86
95 BMIR31.
ALT, 52
4B
11 S0.
7 S1.
5 S2.
9 S3.
74157
16B
41 ARM,
71 CRSEL0.
73 CRSEL1.
MIR26,
MIR29.
MIR29,
MIR27,
MIR25.
MIR20.
MIR26.
MIR21.
MIR21,
MIR24.
MIR24,
MIR22,
MIR25,
MIR29.
MIR29,
MIR28,
MIR20,
MIR24.
MIR24,
MIR23,
DTC1,
180Ω
C1
TC2,
N10 : 220pF
N10/5 : 22pF
TC1, 14
MIR20.
MIR31.
54 LSEL.
MIR30,
MIR20,
DTC1,
GND
ACTLOOP, 10
87 SKL.
MIR19,
WREN,
WST5,
AT3. 9
WST5,
TRRSTS. 79
STSDEST.
83 WST5.
SHDEST. 9
MIR24,
WREN,
MIR24.
MIR15, 26
ACTLOOP, 20
MIR15. 16
12 BSH51,
25 ASH50,
21 BSH50,
13 ASH51,
CPDEST.
MIR23.
MIR22,
MIR31.
GND
MIR31.
MIR21,
CPSEL.
GND
51 RLD.
MIR22,
MIR23.
17 CPSEL.
TC2,
TC2.
67 CPLD.
CPDEST. 11
CPDEST,
TRUE, 91
BMSRC, 50
MIR30,
MIR21,
MIR22,
MIR23,
SIR9, 34
CPDEST,
GND
16C
74151
T5. 36
T1. 75
14C
7474
AT3.
42 DBL.
44 DDBL.
MIR31.
WREN,
4
MIR30.
MIR20,
37 SI,
1k
62
TC0. 8
TC2.
TC1, 14
AT3.
MIR31.
MIR28.
T5.
78 PJMP,
MIR8, 85
MIR9, 77
MIR10, 81
MIR11, 84
14/15 A
74154
CPDEST.
STSDEST.
SHDEST.
6 IODEST.
18 DEST 15.
47 WD.
56 WB.
39 WL.
45 WA.
38 WT.
57 WX.
49 WSHC.
35 WSCR.
43 WSP.
63 WSST5.
81 term.
1 pk.7425
3 pk.7400
1 * 7404
2 * 7410
3 * 7432
2 * 74H51
1 * 74H55
2 * 74564
1 * 7474
1 * 74151
1 * 74154
2 * 74157
4 * 74175
24 pk.
OPPDATERT T.O.M. ECO NO.10-403. 18/4/77/ Emil
Forandr. fra 1007 til 1007/III 6.2.74 Emil
Rettet iflg.ECO nr.10-8
Ajourført 20.11.73 B.S./ Emil
Rettet Dato Sign.
DECODING
1007/V
Målestokk
Tegn Emil 15.5.73
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A07


11D
15E
74174
IB6,
IB7,
IB8,
IB9,
PD6,
PD7,
PD8,
PD9,
IB0,
IB1,
IB2,
IB3,
IB4,
IB5,
PD0,
PD1,
PD2,
PD3,
PD4,
PD5,
PI10,
10E
8E
IB10,
PD10,
PIH,
IB11,
PD11,
PI12,
4E
IB12,
PD12,
SI,
PI13,
6E
IB13,
PD13,
IINT,
IB14,
PD14,
PI15,
IB15,
PD15,
TR6,
12E
TRR,
18E
PE0,
PE1,
PE2,
PE3,
PE4,
PE5,
PIEKL,
9D
IB6,
IB7,
IB8,
IB9,
IB10,
IB11,
PE6,
PE7,
PE8,
PE9,
PE10,
PE11,
2E
IB12,
IB13,
IB14,
IB15,
PE12,
PE13,
PE14,
PE15,
TR7,
PEKL,
PD15,
PE15,
PD14,
PE14,
PD13,
PE13,
PD12,
PE12,
PD11,
PE11,
PD10,
PE10,
PD9,
PE9,
PD8,
PE8,
PD7,
PE7,
PD6,
PE6,
PD5,
PE5,
PD4,
PE4,
PD3,
PE3,
PD2,
PE2,
PD1,
PE1,
PD0,
PE0,
7B
9318
PK3,
AK2,
AK1,
AK0,
13B
BK2,
BK1,
BK0,
IB0,
IB1,
IB2,
IB3,
IB4,
IB5,
IB6,
IB7,
IB8,
IB9,
IB10,
IB11,
IB12,
IB13,
IB14,
IB15,
9A
PIL
74175
MCL,
PK0,
PK1,
PK2,
PK3,
PIL0,
PIL1,
PIL2,
PIL3,
PILKL,
13A
PVL0,
PVL1,
PVL2,
PVL3,
HCL,
7A
ST5RD,
11B
7485
DIFF,
PIK3,
PIK2,
PIK1,
PIK0,
ISO,
IS1,
IIC2,
IIC3,
IIC0,
IIC1,
18D
74153
18C
18A
8831
15D
15C
15A
5C
5D
5A
2D
2C
2A
INTRL,
INTRD,
GND
H
1k
5pk 7400
5 " 7404
2 " 7432
3 " 7474
1 " 7485
8pk 74153
5 " 74174
2 " 74175
5 " 8831
2 " 9318
Totalt: 38 pk
59 term.
OPPDAT. T.O.M. ECO 10-257. II
12.4.73
Ajourført
29.3.73
Rettet
Dato
Sign
INTERRUPT REGISTERS
1008, 1008/II
Målestokk
Tegn
Kontr
Appr
7.3.73
PB
A/S NORSK DATA-ELEKTRONIKK
2A08


12A
10B
10C
14B
74175
CIKL
CAR0
CAR1
CAR2
CAR3
CAR4
CAR5
CAR6
CAR7
CAR8
CAR9
CAR10
CAR11
IR0
IR1
IR2
IR3
IR4
IR5
IR6
IR7
IR8
IR9
IR10
IR11
IR12
IR13
IR14
IR15
IB0
IB1
IB2
IB3
IB4
IB5
IB6
IB7
IB8
IB9
IB10
IB11
IB12
IB13
IB14
IB15
12B
14C
16A
16B
16C
18A
18B
18C
18D
XEP0
XEP1
XEP2
XEP3
XEP5
XEP10
SPEC
INT
OPCOM
MAR8
MAR9
SIR9
CJARG
RIR9
BREN
REL
MPASO
MIR31
MIR0
MIR1
MIR2
MIR3
MIR4
MIR5
MIR6
MIR7
MIR8
MIR9
MIR10
MIR11
2A
4A
2B
4B
4C
74S153
6A
6B
6C
74161
8A
8B
8C
10A
DM 8831
MPCKL
MCL
MPCLD
MPCRD
MPC0
MPC1
MPC2
MPC3
MPC4
MPC5
MPC6
MPC7
MPC8
MPC9
MPC10
MPC11
GND
+5V
1K
1 pk 7400
1 pk 74S00
1 pk 7404
1 pk 74S10
1 pk 74S11
2 pk 7425
6 pk 74S64
5 pk 74S153
3 pk 74161
4 pk 74175
3 pk DM8831
28 pk 71 Term.
MICROADDRESSING
1009
A/S NORSK DATA-ELEKTRONIKK
2A09
30.3.73



1 pk. 7404
3 • 74175
9 • DM8820
14 • DM8831
27 pk.

Ajourf. V til VI. 10.4.75/TAA/Eml

III til V (ingen rettelser)	7.1.75	TAA/Eml
Ajourført	19.10.73	BS/Eml
Tegn. 4 stk. 220Ω. Forandr.	27.9.73	TM/Eml
Rettet f. 1010/II t. 1010/III	Dato	Sign.

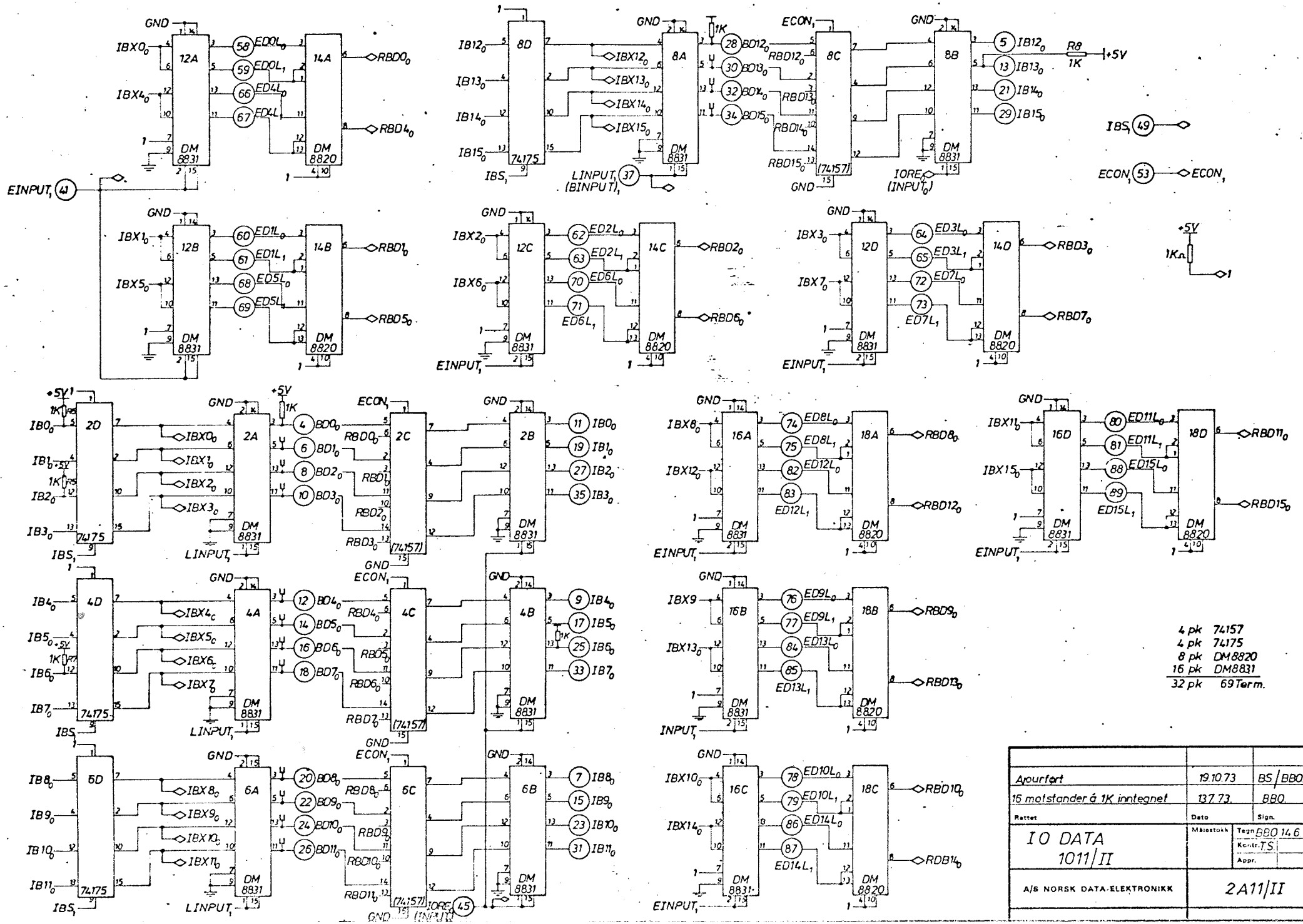
I/O ADDRESS
1010/VI

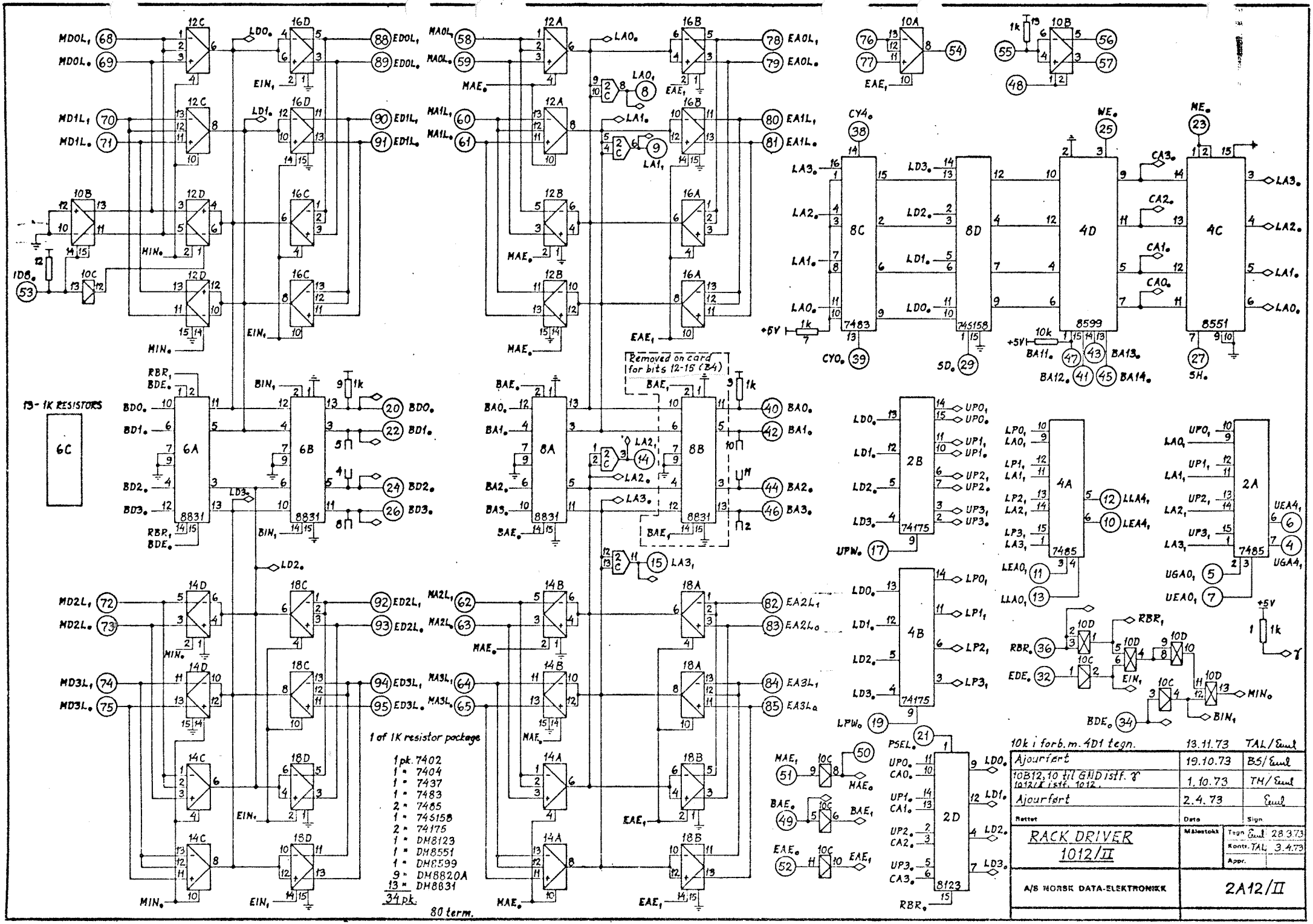
Målestokk | Tegn Eml 1.6.73 | Kontr. T.S. | Appr.

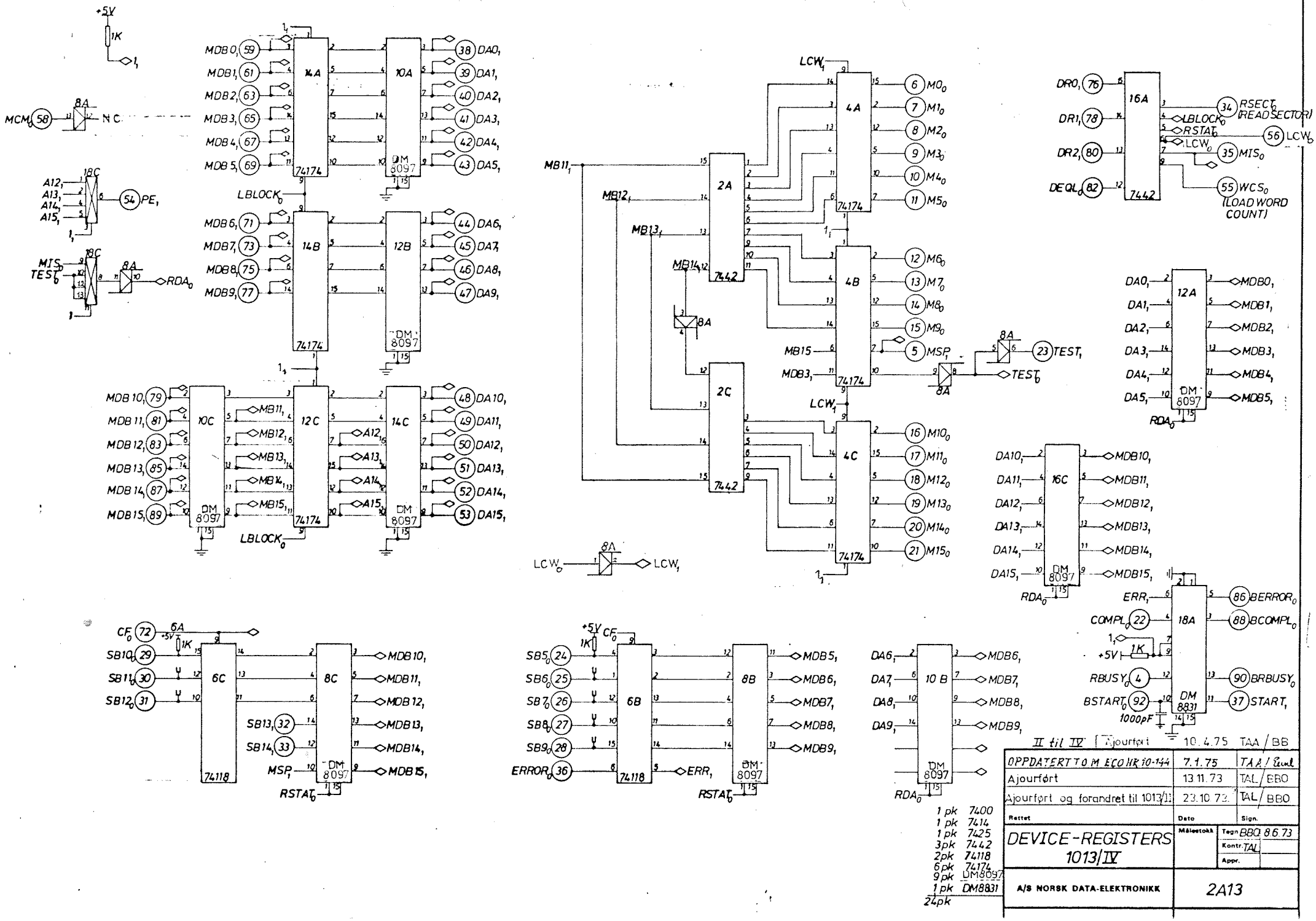
A/S NORSK DATA-ELEKTRONIKK

2A10

GJELDER OGSÅ 1010


GND
12A
14A
DM 8831
DM 8820
EINPUT1 41
IBX0 0
IBX4 0
EDOL 0
EDOL 1
ED4L 0
ED4L
RBD0 0
RBD4 0
8D
8A
8C
8B
74175
(74157)
IB12 0
IB13 0
IB14 0
IB15 0
IBS1
LINPUT1 (BINPUT)1
37
1K
BD12 0
BD13 0
BD14 0
BD15 0
ECON1
RBD12 0
RBD13 0
RBD14 0
RBD15 0
IORE (INPUT0)
R8
+5V
IB12 0
IB13 0
IB14 0
IB15 0
IBS 49
ECON 53
ECON1
12B
14B
IBX1 0
IBX5 0
ED1L 0
ED1L 1
ED5L 0
ED5L
RBD1 0
RBD5 0
12C
14C
IBX2 0
IBX6 0
ED2L 0
ED2L 1
ED6L 0
ED6L 1
RBD2 0
RBD6 0
12D
14D
IBX3 0
IBX7 0
ED3L 0
ED3L 1
ED7L 0
ED7L 1
RBD3 0
RBD7 0
1K Ω
2D
2A
2C
2B
IB0 0
IB1 0
IB2 0
IB3 0
IBX0 0
IBX1 0
IBX2 0
IBX3 0
BD0 0
BD1 0
BD2 0
BD3 0
RBD0 0
RBD1 0
RBD2 0
RBD3 0
IB0 0
IB1 0
IB2 0
IB3 0
4D
4A
4C
4B
IB4 0
IB5 0
IB6 0
IB7 0
IBX4 0
IBX5 0
IBX6 0
IBX7 0
BD4 0
BD5 0
BD6 0
BD7 0
RBD4 0
RBD5 0
RBD6 0
RBD7 0
IB4 0
IB5 0
IB6 0
IB7 0
6D
6A
6C
6B
IB8 0
IB9 0
IB10 0
IB11 0
IBX8 0
IBX9 0
IBX10 0
IBX11 0
BD8 0
BD9 0
BD10 0
BD11 0
RBD8 0
RBD9 0
RBD10 0
RBD11 0
IB8 0
IB9 0
IB10 0
IB11 0
45
16A
18A
16D
18D
IBX8 0
IBX12 0
IBX11 0
IBX15 0
ED8L 0
ED8L 1
ED12L 0
ED12L 1
ED11L 0
ED11L 1
ED15L 0
ED15L 1
RBD8 0
RBD12 0
RBD11 0
RBD15 0
16B
18B
IBX9
IBX13 0
ED9L 0
ED9L 1
ED13L 0
ED13L 1
RBD9 0
RBD13 0
INPUT1
16C
18C
IBX10 0
IBX14 0
ED10L 0
ED10L 1
ED14L 0
ED14L 1
RBD10 0
RDB14 0
4 pk 74157
4 pk 74175
8 pk DM8820
16 pk DM8831
32 pk 69 Term.
Ajourført 19.10.73 BS/BBO
15 motstander á 1K inntegnet 13.7.73 BBO
Rettet Dato Sign.
IO DATA 1011/II
Målestokk
Tegn BBO 14.6.73
Kontr. TS
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A11/II


13 - 1K RESISTORS
6C
1 of 1K resistor package
1 pk. 7402
1 = 7404
1 = 7437
1 = 7483
2 = 7485
1 = 74S158
2 = 74175
1 = DM8123
1 = DM8551
1 = DM8599
9 = DM8820A
13 = DM8831
34 pk.
80 term.
Removed on card for bits 12-15 (B4)
10k i forb.m. 4D1 tegn. 13.11.73 TAL/Eml
Ajourført 19.10.73 BS/Eml
10B12, 10 til GND istf. 8 10D12/K i stf. 10D2. 1.10.73 TH/Eml
Ajourført 2.4.73 Eml
Rettet Dato Sign
RACK DRIVER 1012/II
Målestokk
Tegn Eml 28.3.73
Kontr. TAL 3.4.73
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A12/II



II til IV Ajourført 10.4.75 TAA / BB

OPPDATERT TO M ECONR 10-144	7.1.75	TAA / Emil
Ajourført	13.11.73	TAL / BBO
Ajourført og forandret til 1013/II	23.10.73	TAL / BBO
Rettet	Dato	Sign.

DEVICE-REGISTERS 1013/IV

Målestokk

Tegn BBO 8.6.73

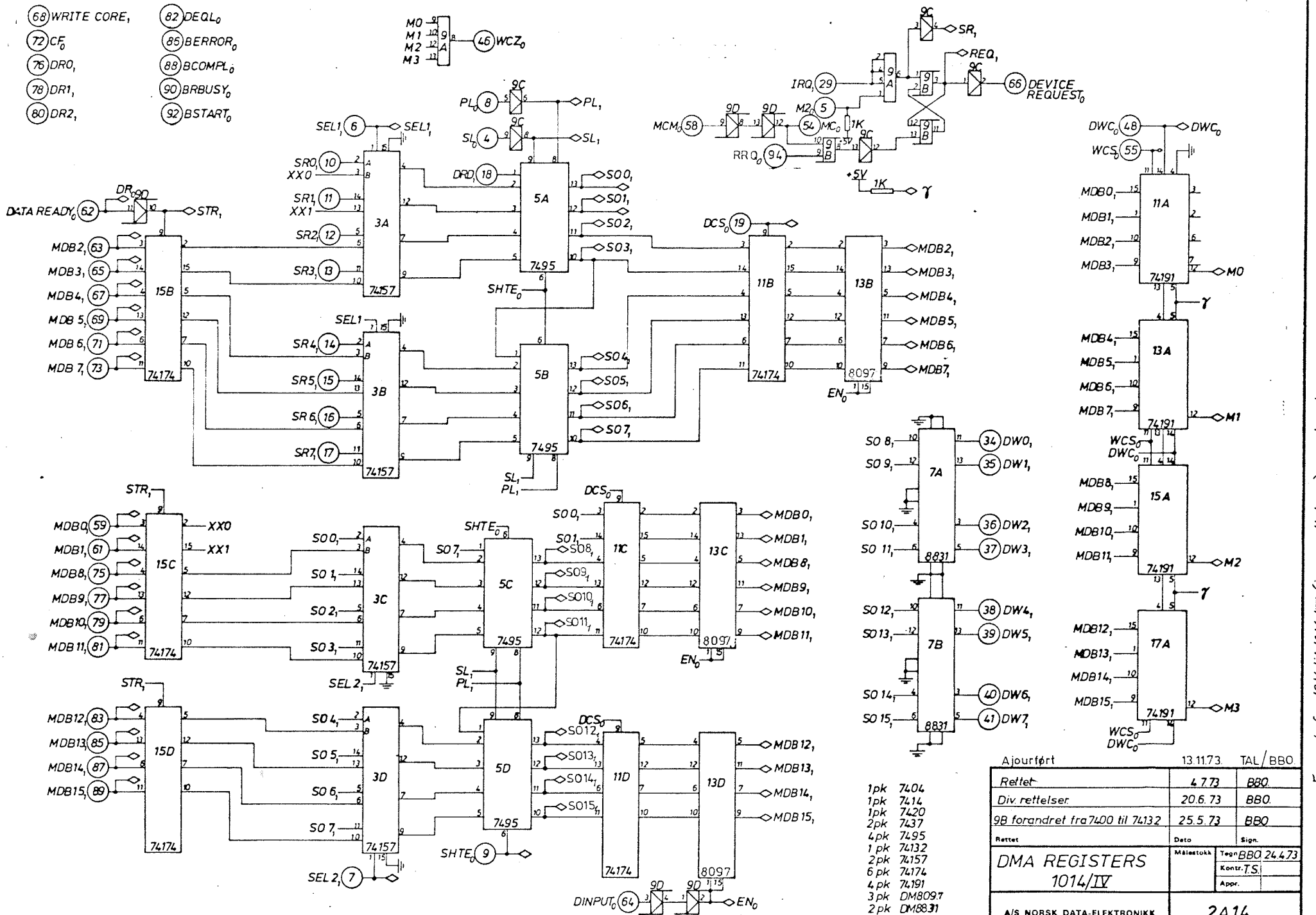
Kontr. TAL

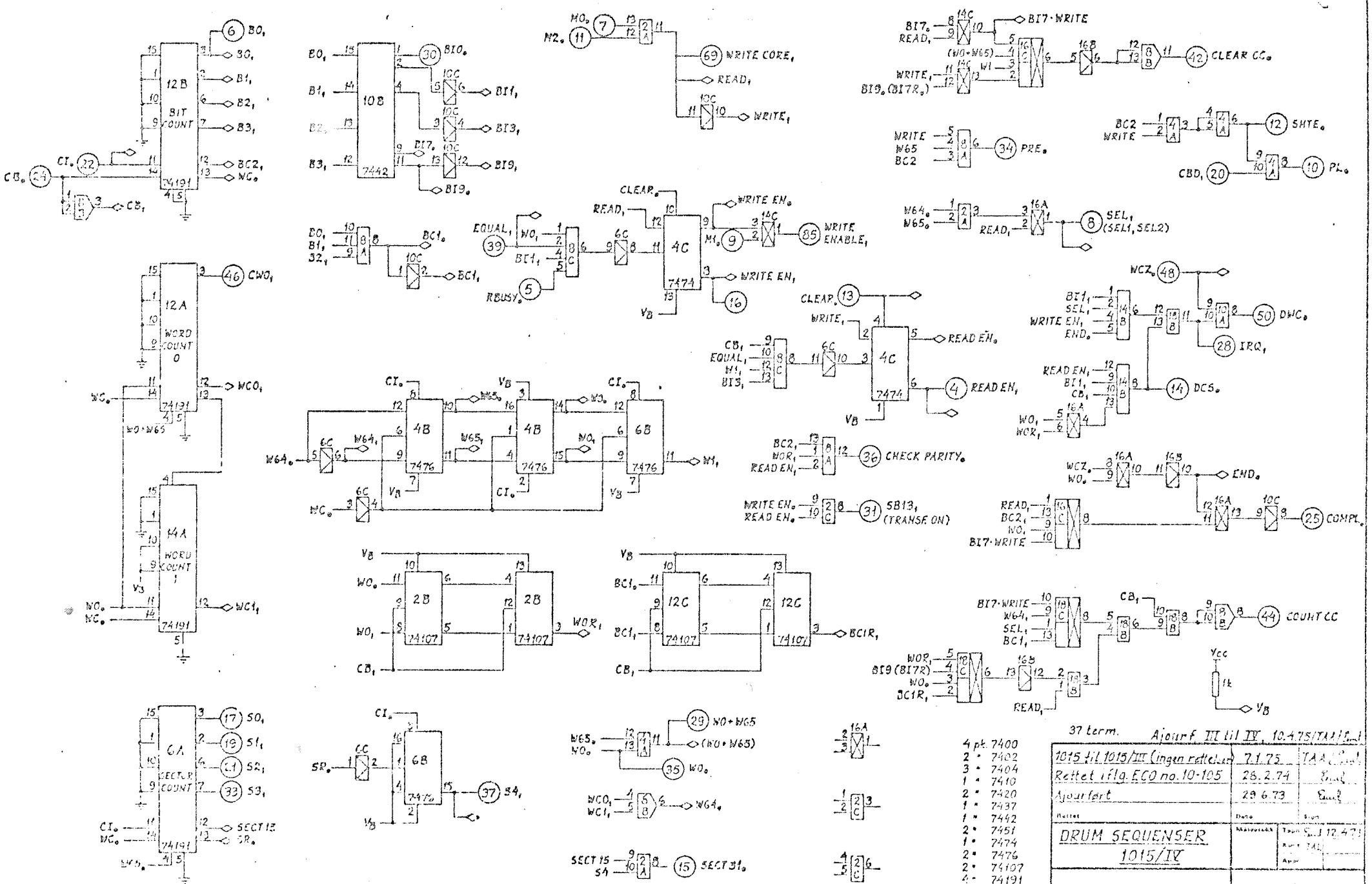
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A13

1 pk	7400
1 pk	7414
1 pk	7425
3pk	7442
2pk	74118
6 pk	74174
9 pk	DM8097
1 pk	DM8831
24pk	


68 WRITE CORE,
72 CE0
76 DR0,
78 DR1,
80 DR2,
82 DEQL0
85 BERROR0
88 BCOMPL0
90 BRBUSY0
92 BSTART0
M0
M1
M2
M3
9A
46 WCZ0
PL0 8
SL0 4
9C
PL1
SL1
SEL1 6
SEL1
SR0 10
XX0
SR1 11
XX1
SR2 12
SR3 13
3A
74157
DR0 18
5A
7495
SO0,
SO1,
SO2,
SO3,
SHTE0
DR0 9D
DATA READY0 62
STR1
MDB2 63
MDB3 65
MDB4 67
MDB5 69
MDB6 71
MDB7 73
15B
74174
SEL1
SR4 14
SR5 15
SR6 16
SR7 17
3B
74157
5B
7495
SO4,
SO5,
SO6,
SO7,
SL1
PL1
DCS0 19
11B
74174
13B
8097
EN0
MDB2,
MDB3,
MDB4,
MDB5,
MDB6,
MDB7,
STR1
MDB0 59
MDB1 61
MDB8 75
MDB9 77
MDB10 79
MDB11 81
15C
74174
XX0
XX1
SO0,
SO1,
SO2,
SO3,
3C
74157
SEL2,
SHTE0
SO7,
5C
7495
SO0,
SO1,
SO8,
SO9,
SO10,
SO11,
11C
74174
DCS0
13C
8097
EN0
MDB0,
MDB1,
MDB8,
MDB9,
MDB10,
MDB11,
SL1
PL1
STR1
MDB12 83
MDB13 85
MDB14 87
MDB15 89
15D
74174
SO4,
SO5,
SO6,
SO7,
3D
74157
SEL2, 7
5D
7495
SHTE0 9
SO12,
SO13,
SO14,
SO15,
11D
74174
DCS0
13D
8097
DINPUT0 64
EN0
MDB12,
MDB13,
MDB14,
MDB15,
IRQ1 29
M2 5
MCM0 58
MC0 54
RRQ0 94
1K
+5V
SR1,
REQ1,
66 DEVICE REQUEST0
γ
DWC0 48
WCS0 55
11A
74191
MDB0,
MDB1,
MDB2,
MDB3,
M0
13A
74191
MDB4,
MDB5,
MDB6,
MDB7,
M1
15A
74191
MDB8,
MDB9,
MDB10,
MDB11,
M2
17A
74191
MDB12,
MDB13,
MDB14,
MDB15,
M3
WCS0
DWC0
7A
8831
SO 8,
SO 9,
SO 10,
SO 11,
34 DW0,
35 DW1,
36 DW2,
37 DW3,
7B
8831
SO 12,
SO 13,
SO 14,
SO 15,
38 DW4,
39 DW5,
40 DW6,
41 DW7,
1pk 7404
1pk 7414
1pk 7420
2pk 7437
4pk 7495
1 pk 74132
2pk 74157
6 pk 74174
4 pk 74191
3 pk DM8097
2 pk DM8831
27 pk
2stk 1K
Ajourført 13.11.73 TAL/BBO
Rettet 4.7.73 BBO
Div. rettelser 20.6.73 BBO
98 forandret fra 7400 til 74132 25.5.73 BBO
Rettet Dato Sign.
DMA REGISTERS
1014/IV
Målestokk
Tegn BBO 24.4.73
Kontr. T.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A14
Forandr. fra 1014 til 1014/II (ingen rettelser) 7.1.75/TAA/Emil



4 pk. 7400	
2 " 7402	
3 " 7404	
1 " 7410	
2 " 7420	
1 " 7437	
1 " 7442	
2 " 7451	
1 " 7474	
2 " 7476	
2 " 74107	
4 " 74191	
25 pk	

37 term. Ajourf. III til IV. 10.4.75/TAA/IE

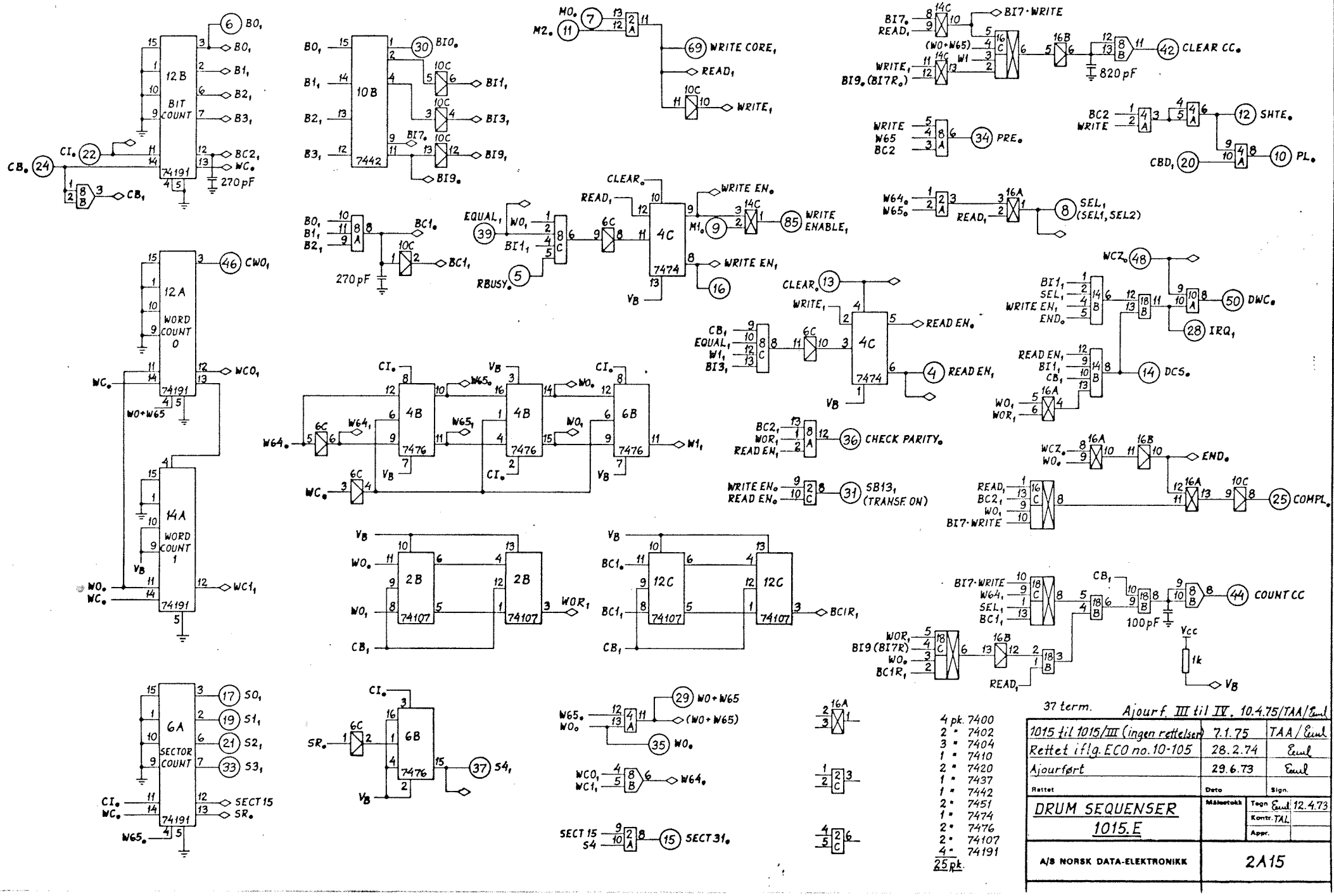
1015 I/I 1015/III (ingen rettelser	7.1.75	TAA
Rettet i/fg. ECO no. 10-105	26.2.74	
Ajourført	29.6.73	
Rettet	Dato	Sign

DRUM SEQUENSER 1015/IV

Tegn. nr. 12.473

A/S NORSK DATA ELEKTRONIKK

2A15



4 pk. 7400
2 " 7402
3 " 7404
1 " 7410
2 " 7420
1 " 7437
1 " 7442
2 " 7451
1 " 7474
2 " 7476
2 " 74107
4 " 74191
25 pk.

37 term.

Ajourf. III til IV. 10.4.75/TAA/Eml

1015 til 1015/III (ingen rettelser)	7.1.75	TAA/Eml
Rettet iflg. ECO no. 10-105	28.2.74	Eml
Ajourført	29.6.73	Eml
Rettet	Dato	Sign.

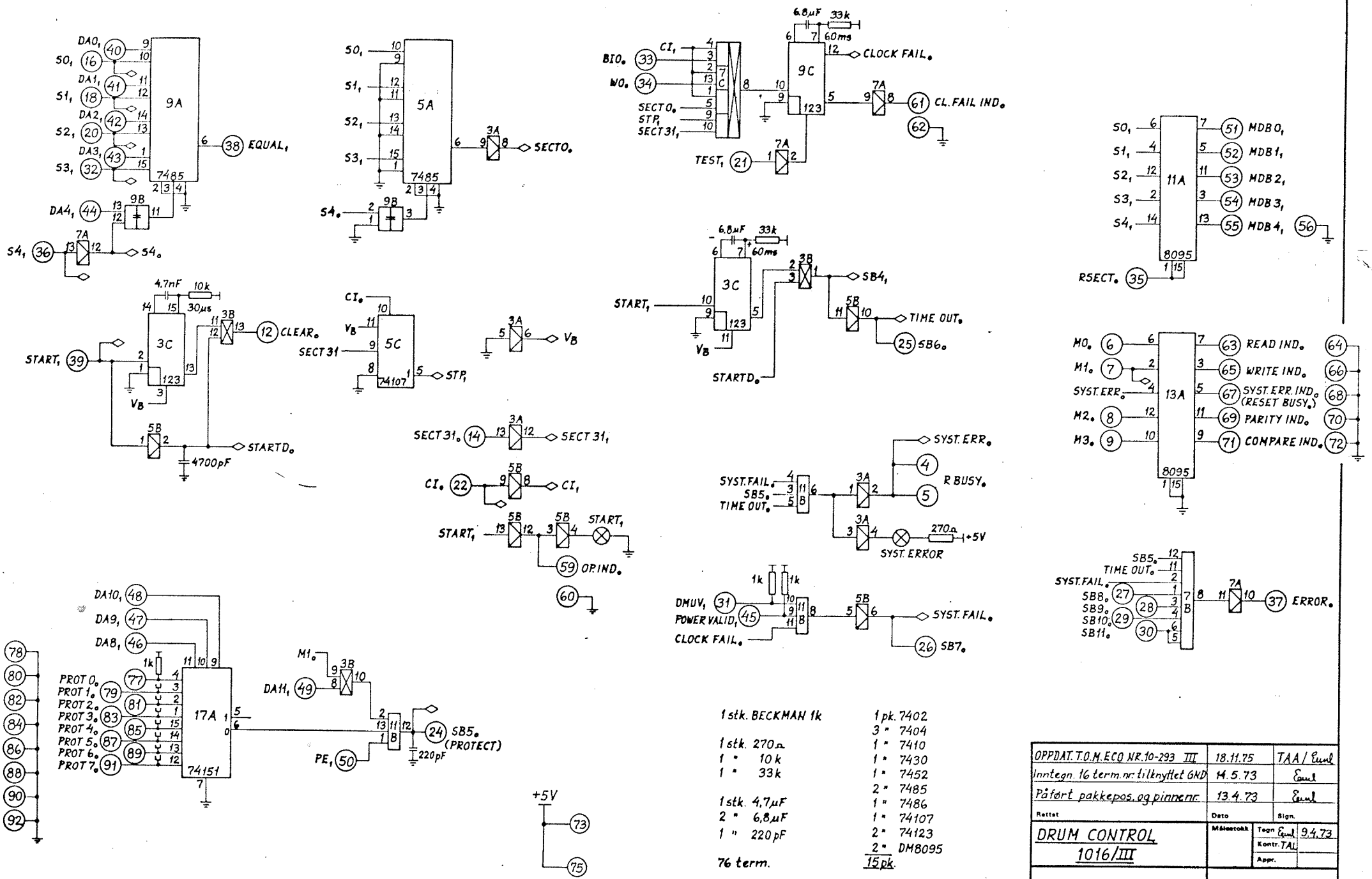
DRUM SEQUENSER 1015.E

Målestokk — Tegn Eml 12.4.73; Kontr. TAL; Appr.

A/S NORSK DATA-ELEKTRONIKK

2A15

OPPDAT. IFLG. ECO 10-411. IV TIL E. 20/1/78/Eml



1 stk. BECKMAN 1k

1 stk. 270Ω
1 " 10 k
1 " 33 k

1 stk. 4,7µF
2 " 6,8µF
1 " 220 pF

76 term.

1 pk. 7402
3 " 7404
1 " 7410
1 " 7430
1 " 7452
2 " 7485
1 " 7486
1 " 74107
2 " 74123
2 " DM8095
15 pk.

OPPDAT. T.O.M. ECO NR. 10-293 III	18.11.75	TAA / Eml
Inntegn. 16 term.nr. tilknyttet GND	14.5.73	Eml
Påført pakkepos. og pinnenr.	13.4.73	Eml
Rettet	Dato	Sign.

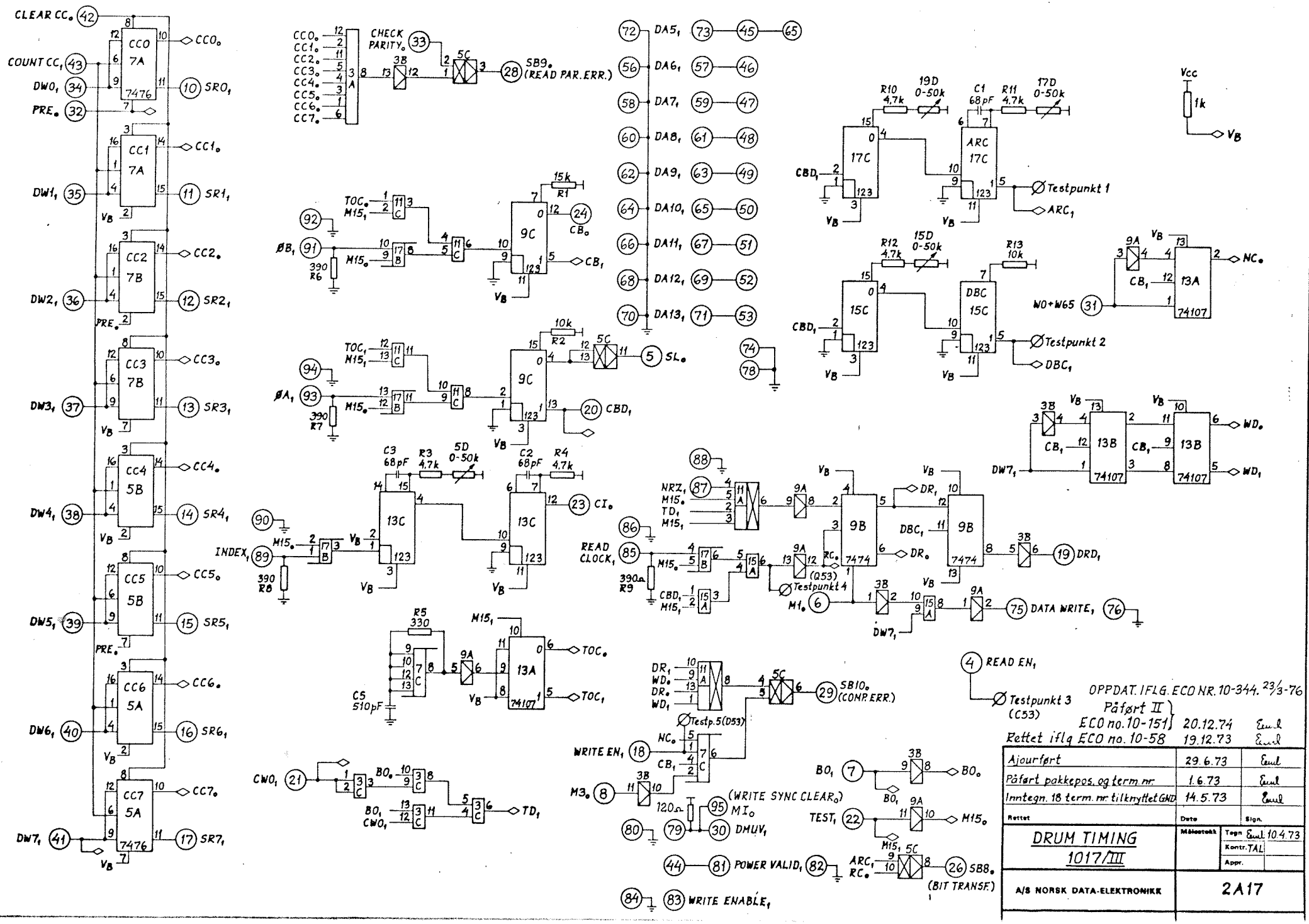
DRUM CONTROL
1016/III

Målestokk

Tegn Eml 9.4.73
Kontr. TAA
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A16



OPPDAT. IFLG. ECO NR. 10-344. 23/3-76

Påført II		
ECO no. 10-151	20.12.74	Eml
Rettet iflg ECO no. 10-58	19.12.73	Eml

Rettet	Dato	Sign.
Ajourført	29.6.73	Eml
Påført pakkepos. og term. nr.	1.6.73	Eml
Inntegn. 18 term. nr. tilknyttet GND	14.5.73	Eml

DRUM TIMING
1017/III

Målestokk

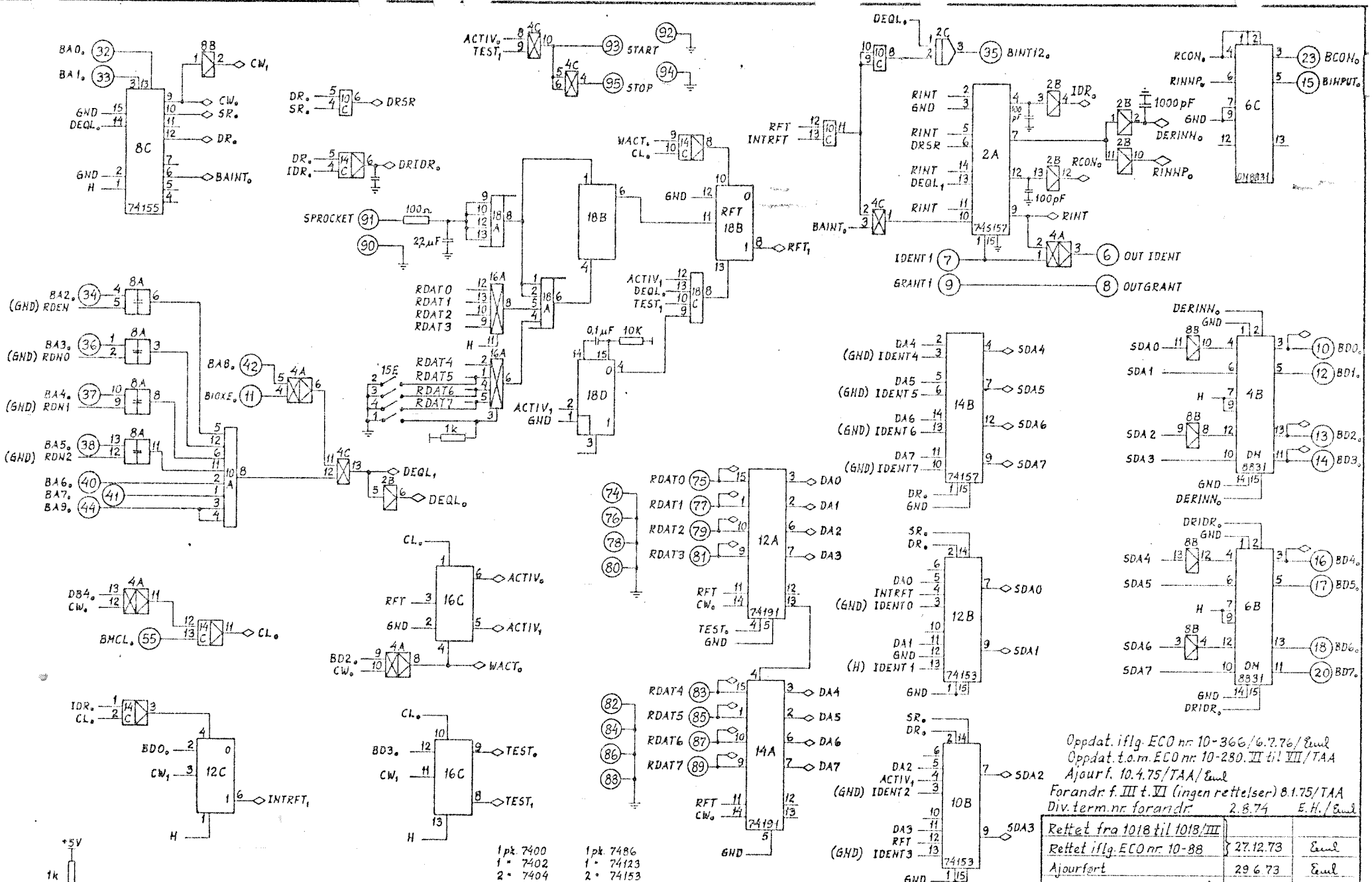
Tegn Eml 10.4.73
Kontr. TAL
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A17

Ajourf. og forandr. f. II t. III., 22.1.75/TAA/Eml

Plas. av term. nr. 56-72 forandr. 17.1.75/TAL/Eml



1 pk. 7400
1 " 7402
2 " 7404
1 " 7408
1 " 7413
1 " 7420
1 " 7425
1 " 7430
1 " 7432
1 " 7438
3 " 7474

1 pk. 7486
1 " 74123
2 " 74153
1 " 74155
1 " 74157
2 " 74191
3 " DM8831
1 " SWITCH

Totalt: 27 pk.

50 term.

Oppdat. iflg. ECO nr. 10-366/6.7.76/Eml
Oppdat. t.o.m. ECO nr. 10-280. III til VII/TAA
Ajourf. 10.4.75/TAA/Eml
Forandr. f. III t. VI (ingen rettelser) 8.1.75/TAA
Div. term. nr. forandr. 2.8.74 E.H./Eml

Rettet fra 1018 til 1018/III	27.12.73	Eml
Rettet iflg. ECO nr. 10-88		
Ajourført	29 6 73	Eml
Rettet	Dato	Sign

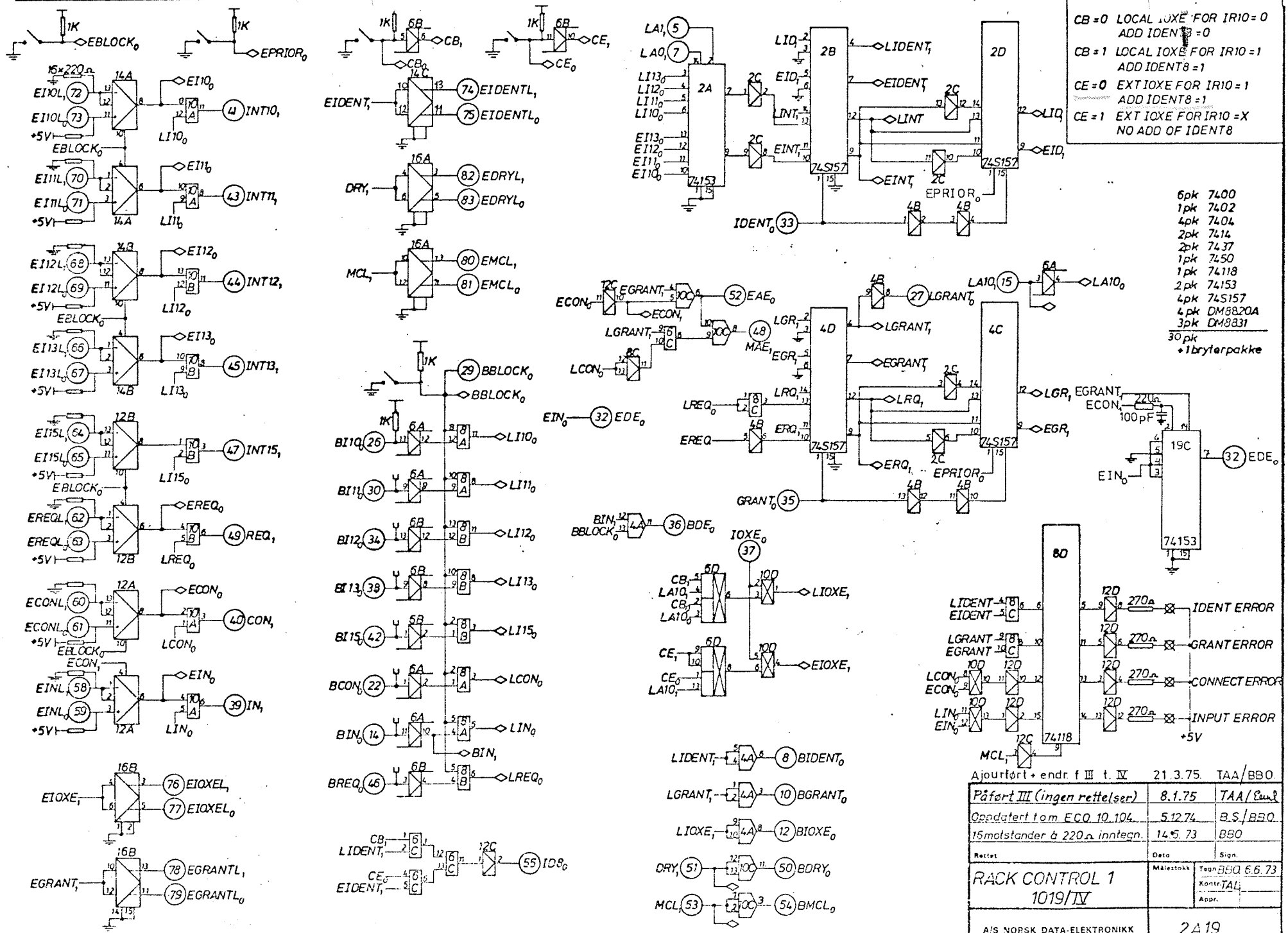
TAPE READER INTERFACE
1018/VII

Målestokk

Tegn Eml 8.5.73
Kontr TKF
Appr.

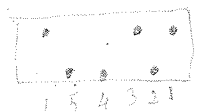
A/S NORSK DATA-ELEKTRONIKK

2A18

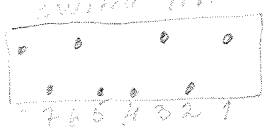

CB =0 LOCAL IOXE FOR IR10 = 0
ADD IDENT8 =0
CB =1 LOCAL IOXE FOR IR10 =1
ADD IDENT8 =1
CE =0 EXT IOXE FOR IR10 =1
ADD IDENT8 =1
CE =1 EXT IOXE FOR IR10 =X
NO ADD OF IDENT8
6pk 7400
1pk 7402
4pk 7404
2pk 7414
2pk 7437
1pk 7450
1pk 74118
2pk 74153
4pk 74S157
4pk DM8820A
3pk DM8831
30 pk
*1 bryterpakke
EBLOCK0
EPRIOR0
EI10L0
EI11L0
EI12L0
EI13L0
EI15L0
EREQL0
ECONL0
EINL0
INT10,
INT11,
INT12,
INT13,
INT15,
REQ1,
CON1,
IN1,
CB1,
CE1,
EIDENT1,
DRY1,
MCL1,
BBLOCK0
BI10 BI11 BI12 BI13 BI15 BCON BIN BREQ
BBLOCK0
EIOXE1
EGRANT1
EIOXEL0
EGRANTL0
ID80
LA1 LA0 LI13 LI12 LI11 LI10 EI13 EI12 EI11 EI10
LID1 EID1 LINT1 EINT1
LIDENT1 EIDENT1 LINT EINT EPRIOR0 IDENT0
LIQ EID
ECON0 EGRANT1 LGRANT1 LCON0 EIN0 LREQ0 EREQ
EAE0 MAE1 EDE0
LGR1 EGR1 LRQ1 ERQ1
LGRANT0 EGRANT1 LRQ1 ERQ1 EPRIOR0 GRANT0
LA10 BIN BBLOCK0 BDE0
IOXE0 CB1 LA10 CE1 LIOXE1 EIOXE1
LIDENT1 LGRANT1 LIOXE1 DRY1 MCL1
BIDENT0 BGRANT0 BIOXE0 BDRY0 BMCL0
IDENT ERROR
GRANT ERROR
CONNECT ERROR
INPUT ERROR
+5V
Ajourført + endr. f III t. IV 21.3.75 TAA/BBO
Påført III (ingen rettelser) 8.1.75 TAA/Emil
Oppdatert t.o.m. ECO 10.104 5.12.74 B.S./BBO
15 motstander à 220Ω inntegn. 14.6.73 BBO
Rettet Dato Sign.
RACK CONTROL 1
1019/IV
Målestokk
Tegn BBO 6.6.73
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A19

Kortet
Kort BR5

B26 E1



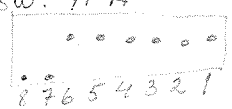
switch "11A"



Kort
A26 (kortet BR5)

switch
1E

sw. 11 A



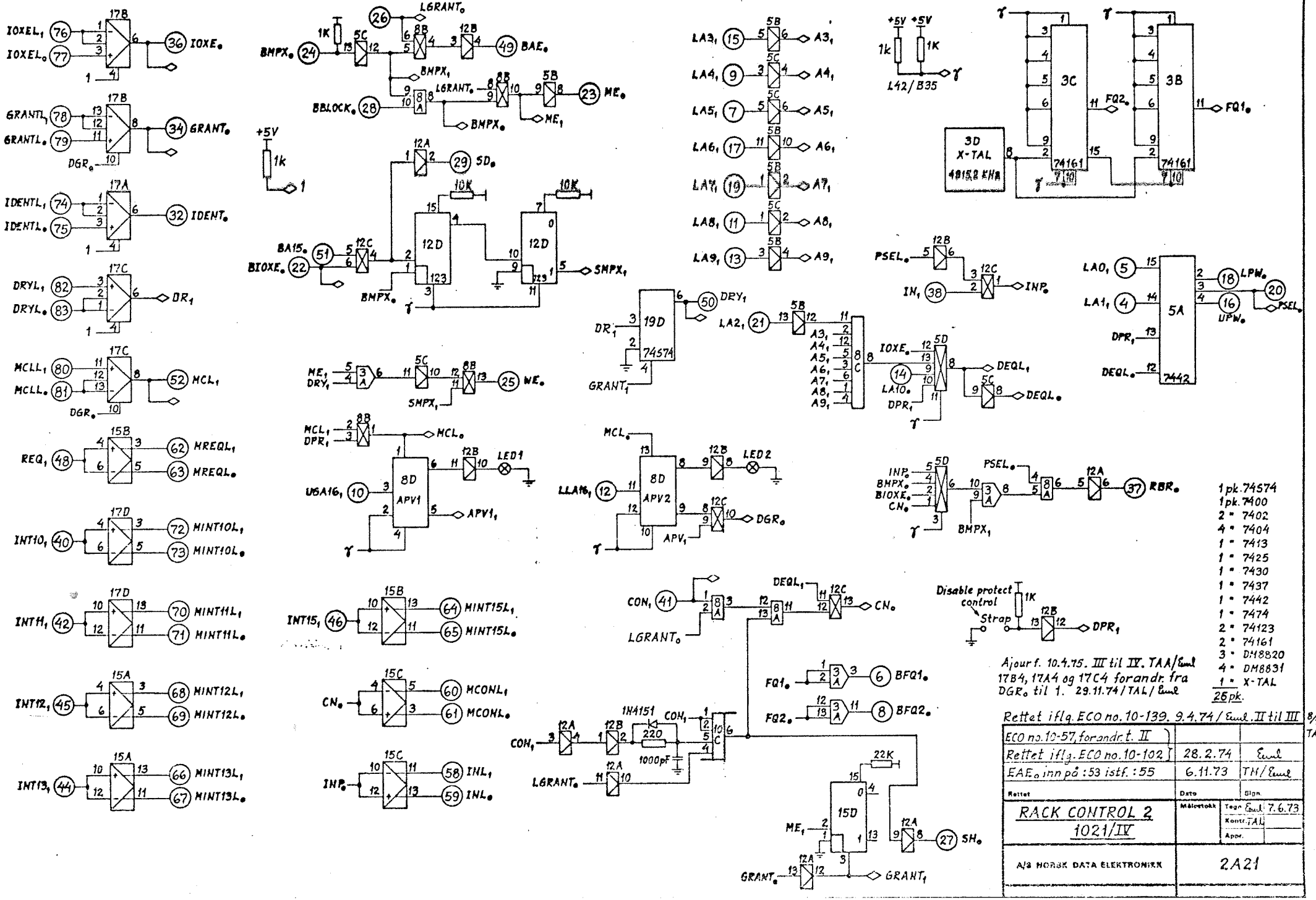
Kortnätet B15 — A26

OBS!

77-07-26 /Lan.

Om kort 1020/IV (Interface för Terminaler) demonteras, t ex p g a fel på kortet, kan ett DUMMY-KORT monteras

Om stiften ⑥-⑦ och ⑧-⑨ kortslutes.



Ajour f. 10.4.75. III til IV. TAA/Emil
17B4, 17A4 og 17C4 forandr. fra
DGR0 til 1. 29.11.74/TAL/Emil

1 pk.	74574
1 pk.	7400
2 *	7402
4 *	7404
1 *	7413
1 *	7425
1 *	7430
1 *	7437
1 *	7442
1 *	7474
2 *	74123
2 *	74161
3 *	DM8820
4 *	DM8831
1 *	X-TAL
25 pk.	

Rettet iflg. ECO no. 10-139. 9.4.74/Emil. II til III 8/1-75 TAA

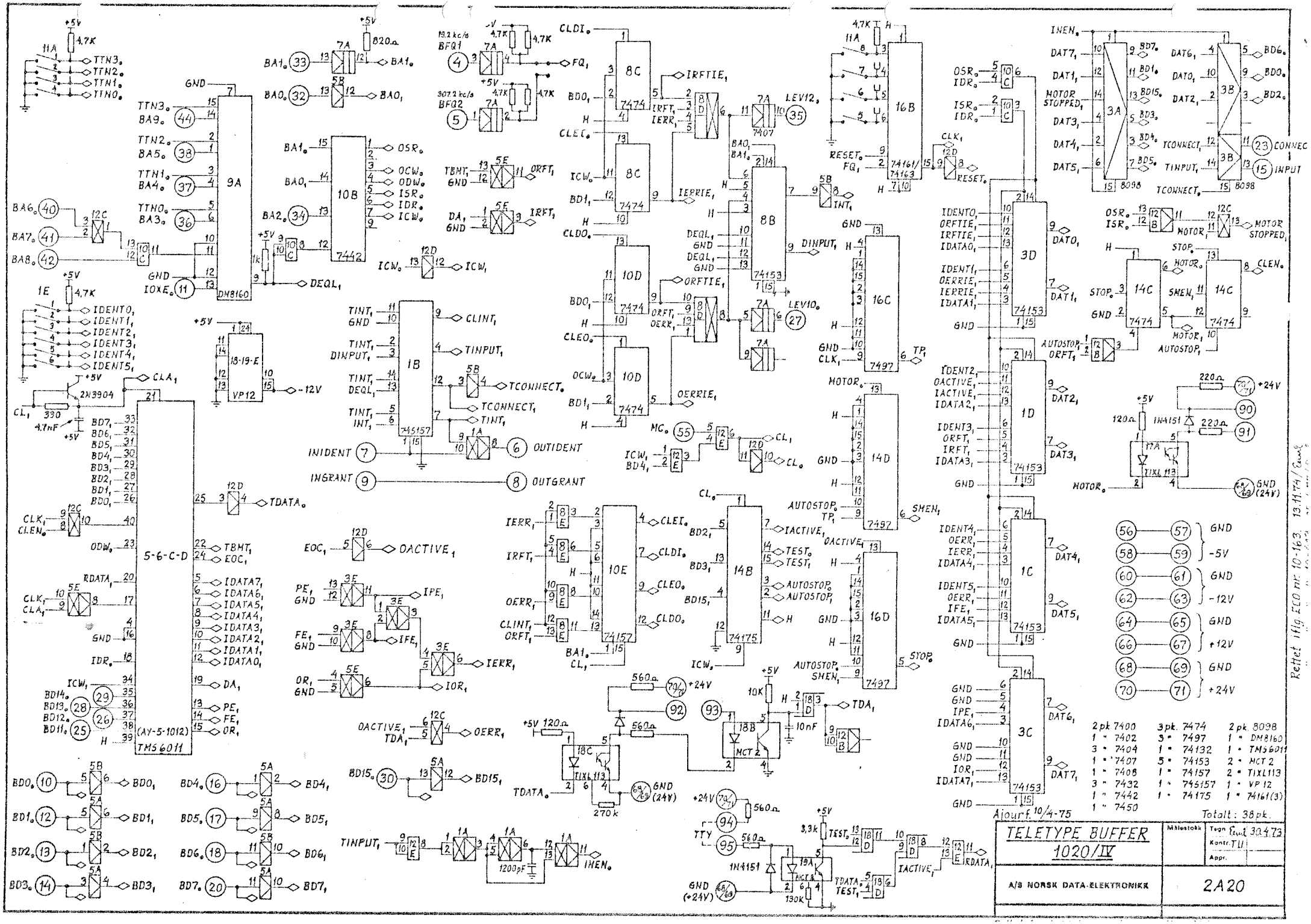
ECO no. 10-57, forandr. t. II		
Rettet iflg. ECO no. 10-102	28.2.74	Emil
EAE0 inn på : 53 istf. : 55	6.11.73	TH/Emil
Rettet	Dato	Sign

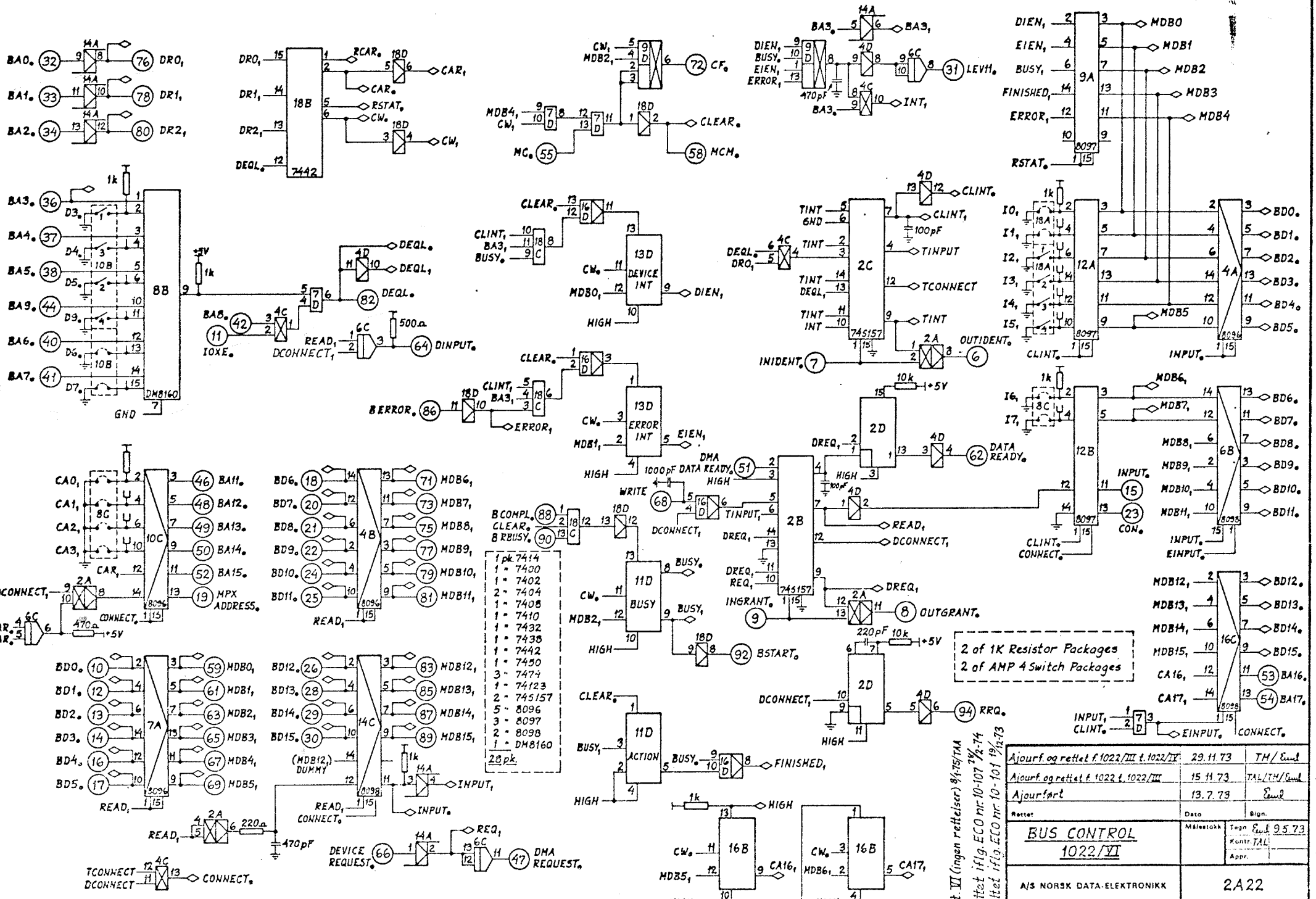
RACK CONTROL 2
1021/IV

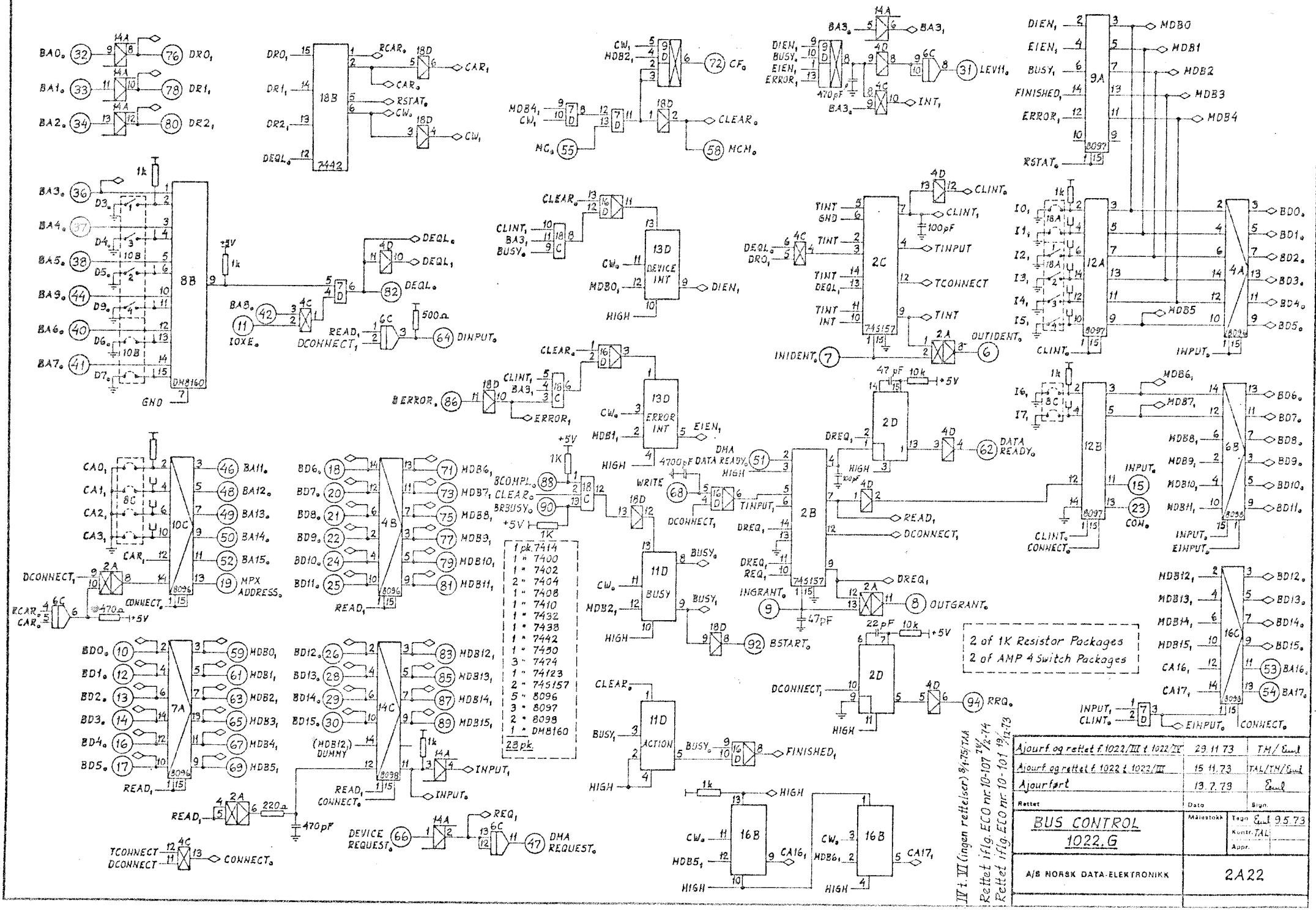
Målestokk
Tegn Emil 7.6.73
Kontr. TAL
Appr.

A/S NORSK DATA ELEKTRONIKK

2A21


TELETYPE BUFFER
1020/IV
A/S NORSK DATA-ELEKTRONIKK
2A20
Totalt: 38pk.
Ajourf. 10/4-75
Rettet iflg ECO nr. 10-163, 13.11.74 / Emil
2pk 7400
1 " 7402
3 " 7404
1 " 7407
1 " 7408
3 " 7432
1 " 7442
1 " 7450
3pk. 7474
3 " 7497
1 " 74132
5 " 74153
1 " 74157
1 " 74175
2pk 8098
1 " DM8160
1 " TMS6011
2 " MCT 2
1 " TIXL113
1 " VP 12
1 " 74161(3)
5-6-C-D
(AY-5-1012)
TMS6011
DM8160
2N3904
MCT 2
TIXL 113
1N4151
VP12
18-19-E


8B
DM8160
18B
7442
9A
8097
12A
4A
8096
2C
745157
13D
DEVICE INT
13D
ERROR INT
11D
BUSY
11D
ACTION
2B
745157
2D
12B
6B
16C
10C
4B
14C
7A
16B
16B
8C
10B
10B
2A
4C
4D
6C
14A
18D
18A
18C
1k
10k
470pF
100pF
220pF
500Ω
220Ω
+5V
GND
DMA
1000pF
DATA READY
HIGH
WRITE
DCONNECT
BCOMPL
CLEAR
BREUSY
BSTART
INGRANT
OUTGRANT
DREQ
REQ
DEVICE REQUEST
DMA REQUEST
FINISHED
INPUT
EINPUT
CONNECT
RRQ
LEV11
INT
CF
MCM
CLEAR
DIEN
IOXE
DINPUT
BERROR
INIDENT
OUTIDENT
TINPUT
TCONNECT
TINT
CLINT
DEQL
DRQ
RCAR
CAR
RSTAT
CW
MDB0
MDB1
MDB2
MDB3
MDB4
MDB5
MDB6
MDB7
MDB8
MDB9
MDB10
MDB11
MDB12
MDB13
MDB14
MDB15
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
BD8
BD9
BD10
BD11
BD12
BD13
BD14
BD15
BA0
BA1
BA2
BA3
BA4
BA5
BA6
BA7
BA8
BA9
BA11
BA12
BA13
BA14
BA15
BA16
BA17
MPX ADDRESS
DUMMY
READ
CONNECT
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
CA0
CA1
CA2
CA3
CA16
CA17
1 pk 7414
1 - 7400
1 - 7402
2 - 7404
1 - 7408
1 - 7410
1 - 7432
1 - 7438
1 - 7442
1 - 7450
3 - 7474
1 - 74123
2 - 745157
5 - 8096
3 - 8097
2 - 8098
1 - DM8160
28 pk
2 of 1K Resistor Packages
2 of AMP 4 Switch Packages
IV t. VI (ingen rettelser) 8/4-75/TAA
Rettet ifig. ECO nr 10-107 7/2-74
Rettet ifig. ECO nr 10-101 19/12-73
Ajourf. og rettet f. 1022/III t. 1022/IV 29.11.73 TH/Emil
Ajourf. og rettet f. 1022 t. 1022/III 15.11.73 TAL/TH/Emil
Ajourført 13.7.73 Emil
Rettet Dato Sign.
BUS CONTROL
1022/VI
Målestokk
Tegn Emil 9.5.73
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A22
Ajourf. 10.4.75/TAA/Emil



Ajourf. 10.4.75/TAA/Emil | Oppdat. t.o.m. ECO nr 10-376. VI til G. 14/9 77 / Emil Oppdat. iflg ECO nr 10 461 / 14 12 77 / BW

VI t. VI (ingen rettelser) 8/4-75/TAA
Rettet iflg ECO nr 10-107 24/2-74
Rettet iflg ECO nr 10-101 19/12-73

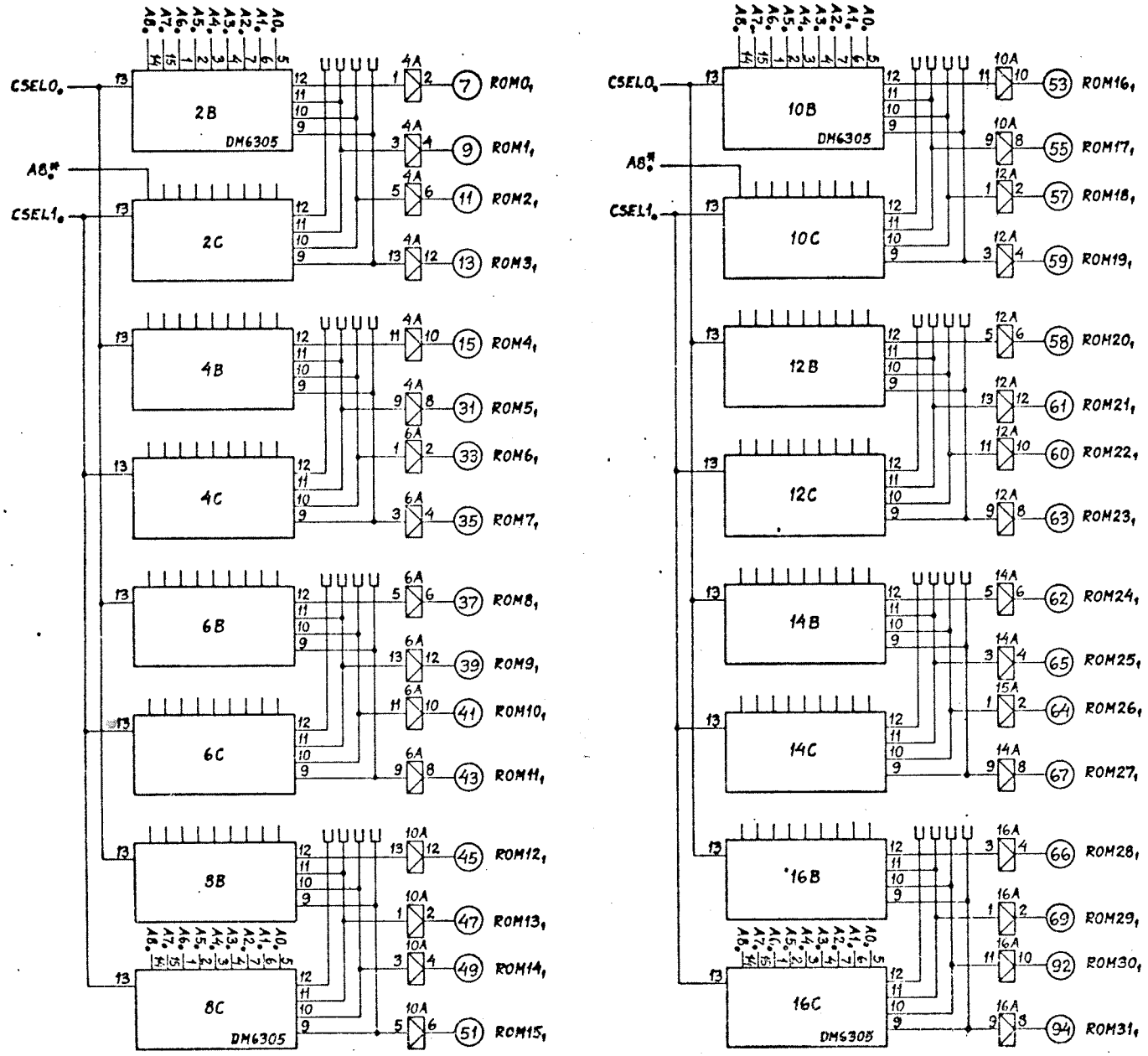
Ajourf og rettet f 1022/III t 1022/IV	29 11 73	TH / Emil
Ajourf og rettet f 1022 t 1022/III	15 11 73	TAL/TH/Emil
Ajourført	13. 7. 73	Emil
Rettet	Dato	Sign.

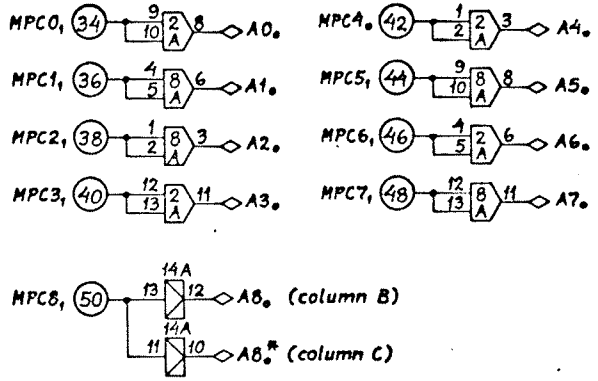
BUS CONTROL
1022.G

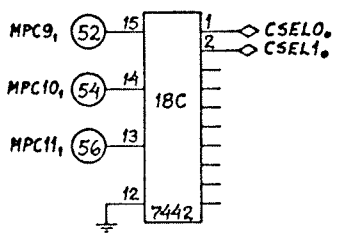
Målestokk | Tegn Emil 9.5.73 | Kontr. TAL | Appr.

A/S NORSK DATA-ELEKTRONIKK

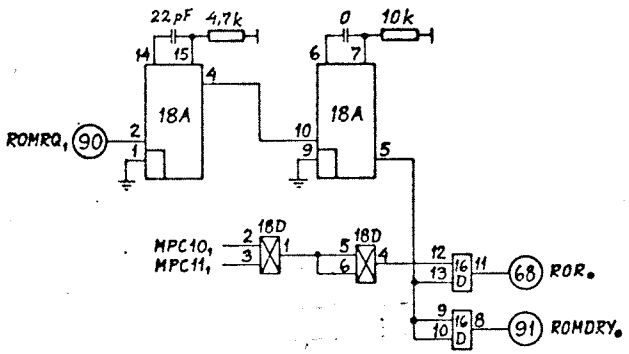
2A22







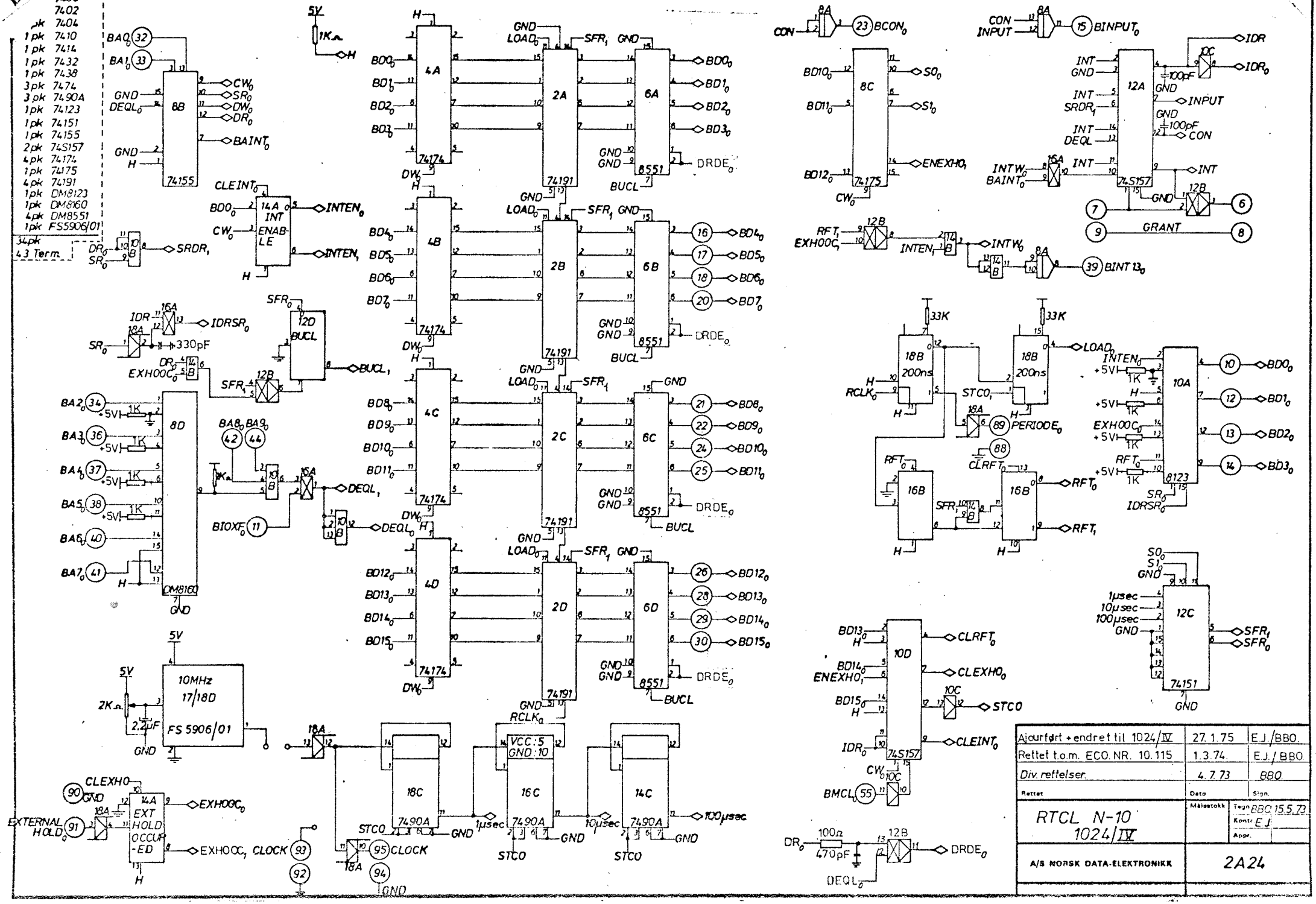
1 pk. 7400
1 * 7402
7 * 7404
2 * 7437
1 * 7442
1 * 74123
16 * DM6305
29 pk.



Ajourført	15.11.74	BS/Eml
Fjernet forb. m. 16D9 og 18D1,5,6.		
Forandr. f. 1023 t. 1023/II	28.9.73	TM/Eml
Rettet	Dato	Sign.

1K PROM 1023/II	Målestokk	Tegn Eml 8.6.73 Kontr. PB. Appr.
A/S NORSK DATA-ELEKTRONIKK		2A23/II

1pk	7400
1pk	7402
1pk	7404
1pk	7410
1pk	7414
1pk	7432
1pk	7438
3pk	7474
3pk	74S90A
1pk	74123
1pk	74151
1pk	74155
2pk	74S157
4pk	74174
1pk	74175
4pk	74191
1pk	DM8123
1pk	DM8160
4pk	DM8551
1pk	FS5906/01
34pk	
43 Term.	



	7400
	7402
pk	7404
1 pk	7410
1 pk	7414
1 pk	7432
1 pk	7438
3 pk	7474
3 pk	74S90A
1 pk	74123
1 pk	74151
1 pk	74155
2 pk	74S157
4 pk	74174
1 pk	74175
4 pk	74191
1 pk	DM8123
1 pk	DM8160
4 pk	DM8551
1 pk	FS5906/01
34 pk	
43 Term.	

Oppdat iflg. ECO 10-370 + endr. t. G 9.9.77 TEAA/BW

Ajourført + endret til 1024/IV	27.1.75	EJ/BBO
Rettet t.o.m. ECO.NR. 10.115	1.3.74	EJ/BBO
Div. rettelser	4.7.73	BBO
Rettet	Dato	Sign.

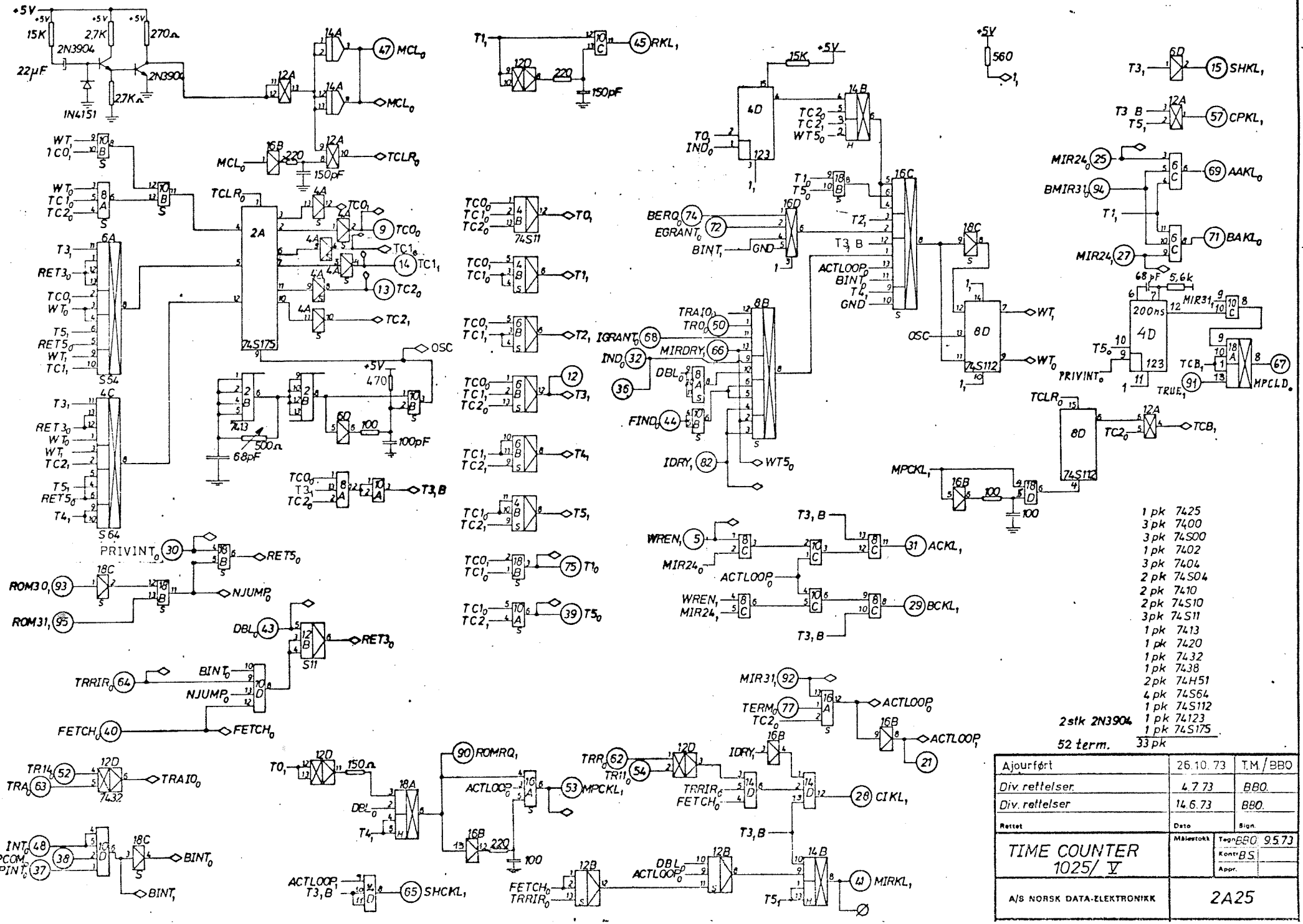
RTCL N-10
1024.G

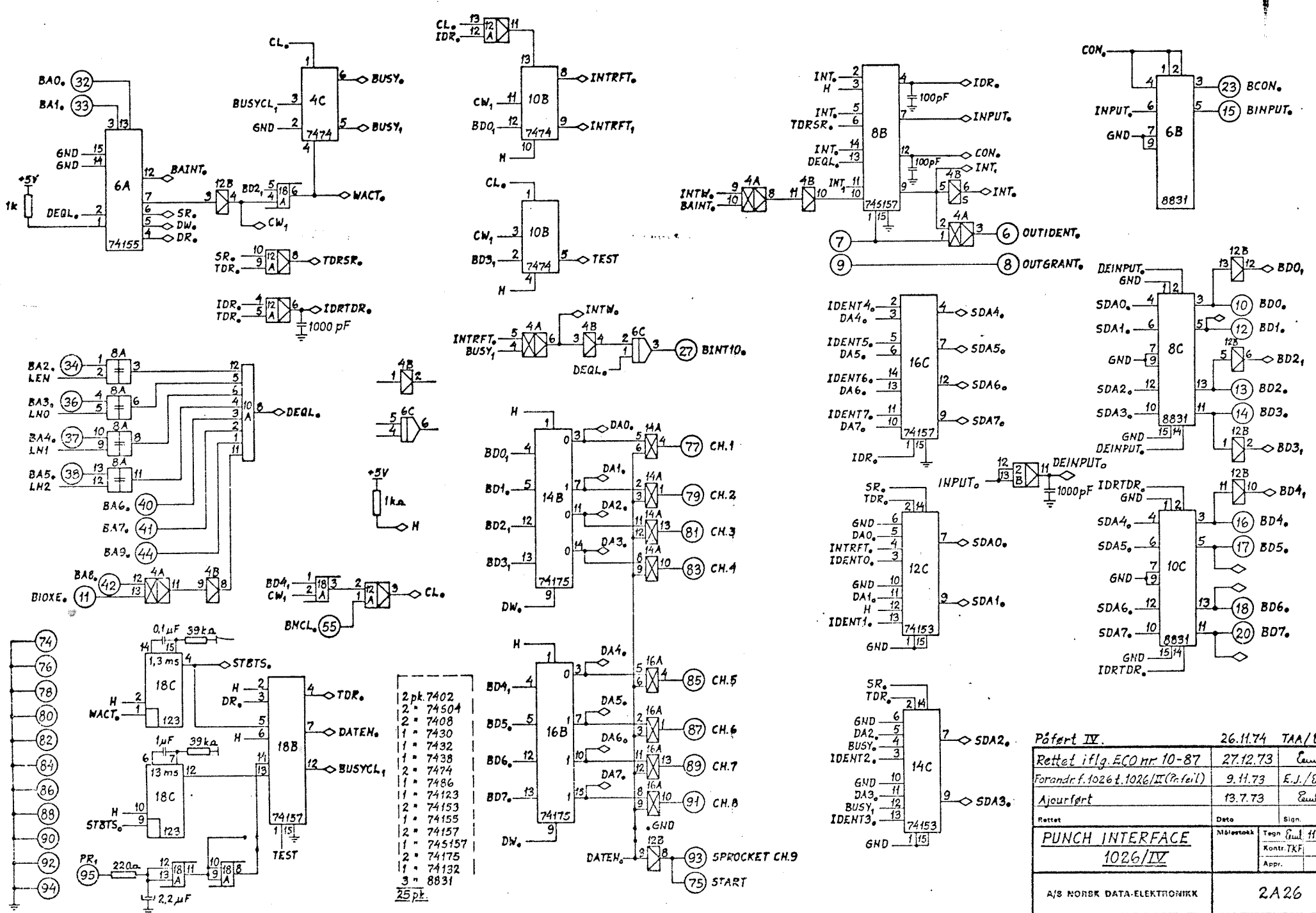
Målestokk

Tegn. BBO 15.5.73
Kontr. EJ
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A24


+5V
15K
2,7K
270Ω
2N3904
22µF
IN4151
2,7KΩ
MCL0
TCLR0
150pF
220
WT1
TCO1
WT0
TC10
TC20
T31
RET30
TCO0
TC11
TC21
T51
RET50
WT11
TC11
T41
RET31
WT01
TC21
S54
S64
74S175
74S11
74S112
74S13
500Ω
68pF
470
60
100
100pF
OSC
PRIVINT0
ROM30
ROM31
NJUMP0
DBL0
RET30
TRRIR0
BINT0
FETCH0
TR140
TRA0
TRAI00
INT0
OPCOM
OPINT0
BINT1
T01
ROMRQ1
ACTLOOP0
MPCKL1
TRR0
TRI10
IDRY1
TRRIR0
FETCH0
CIKL1
T3,B
DBL0
ACTLOOP0
MIRKL1
T51
ACTLOOP1
SHCKL1
T11
RKL1
BERQ0
EGRANT0
BINT1
GND
TC20
WT50
TRAI00
TRO0
IGRANT0
MIRDRY1
IND0
DBL0
FIND0
IDRY1
WT50
560
OSC
WT1
WT0
MIR240
MIR31
TERM0
TC20
ACTLOOP0
ACTLOOP1
WREN1
MIR240
ACTLOOP0
WREN1
MIR241
T3,B
ACKL1
BCKL1
T3,B
T3,B
T3 B
T51
SHKL1
CPKL1
MIR240
BMIR31
T11
MIR241
AAKL0
BAKL0
68pF
5,6k
200ns
MIR311
T50
PRIVINT0
TRUE1
TCB1
MPCLD1
TCLR0
TC20
TCB1
MPCKL1
100
1 pk 7425
3 pk 7400
3 pk 74S00
1 pk 7402
3 pk 7404
2 pk 74S04
2 pk 7410
2 pk 74S10
3pk 74S11
1 pk 7413
1 pk 7420
1 pk 7432
1 pk 7438
2 pk 74H51
4 pk 74S64
1 pk 74S112
1 pk 74123
1 pk 74S175
33 pk
2stk 2N3904
52 term.
Ajourført 26.10.73 T.M./BBO
Div. rettelser 4.7.73 BBO.
Div. rettelser 14.6.73 BBO.
Rettet Dato Sign.
TIME COUNTER 1025/ V
Målestokk
Tegn BBO 9.5.73
Kontr BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A25
OPPDATERT T.O.M. ECO NR. 10-218. III til IV. 17.1.75/TAA/Emil. Ajourført + endret fra 1025/III til 1025/IV 19.2.75 TAA/BBO



Påført IV.	26.11.74	TAA/Eml
Rettet iflg. ECO nr. 10-87	27.12.73	Eml
Forandr. f. 1026 t. 1026/II (Prefail)	9.11.73	E.J./Eml
Ajourført	13.7.73	Eml
Rettet	Dato	Sign.

PUNCH INTERFACE
1026/IV

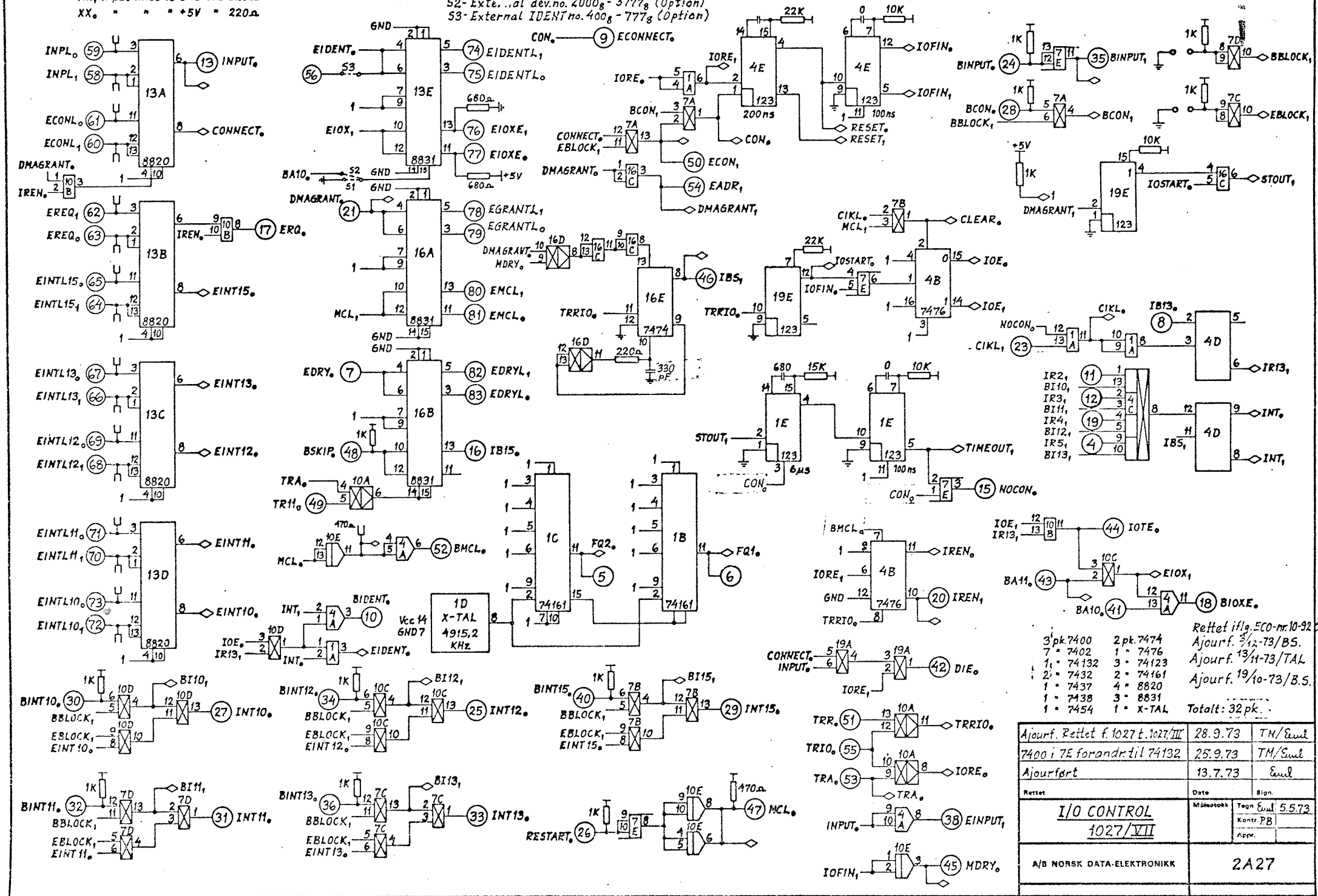
Målestokk | Tegn Eml 11.5.73 | Kontr. TKF | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A26

XX, input lines to GND via 220Ω
XX₀ " " " +5V " 220Ω

S P5: S1- External dev.no. 0-3777₈ (Standard)
S2- External dev.no. 2000₈ - 3777₈ (Option)
S3- External IDENT no. 400₈ - 777₈ (Option)



Ajourf. Rettet f. 1027 t. 1027/III	28.9.73	TN/Eml
7400 i 75 forandr. til 74132	25.9.73	TN/Eml
Ajourført	13.7.73	Eml
Rettet	Dato	Sign.

I/O CONTROL 1027/VII	Målestokk	Tegn Eml 5.5.73
		Kontr. PB
		Appr.
A/S NORSK DATA-ELEKTRONIKK	2A27	

3 pk. 7400	2 pk. 7474
7 " 7402	1 " 7476
1 " 74132	3 " 74123
2 " 7432	2 " 74161
1 " 7437	4 " 8820
1 " 7438	3 " 8831
1 " 7454	1 " X-TAL

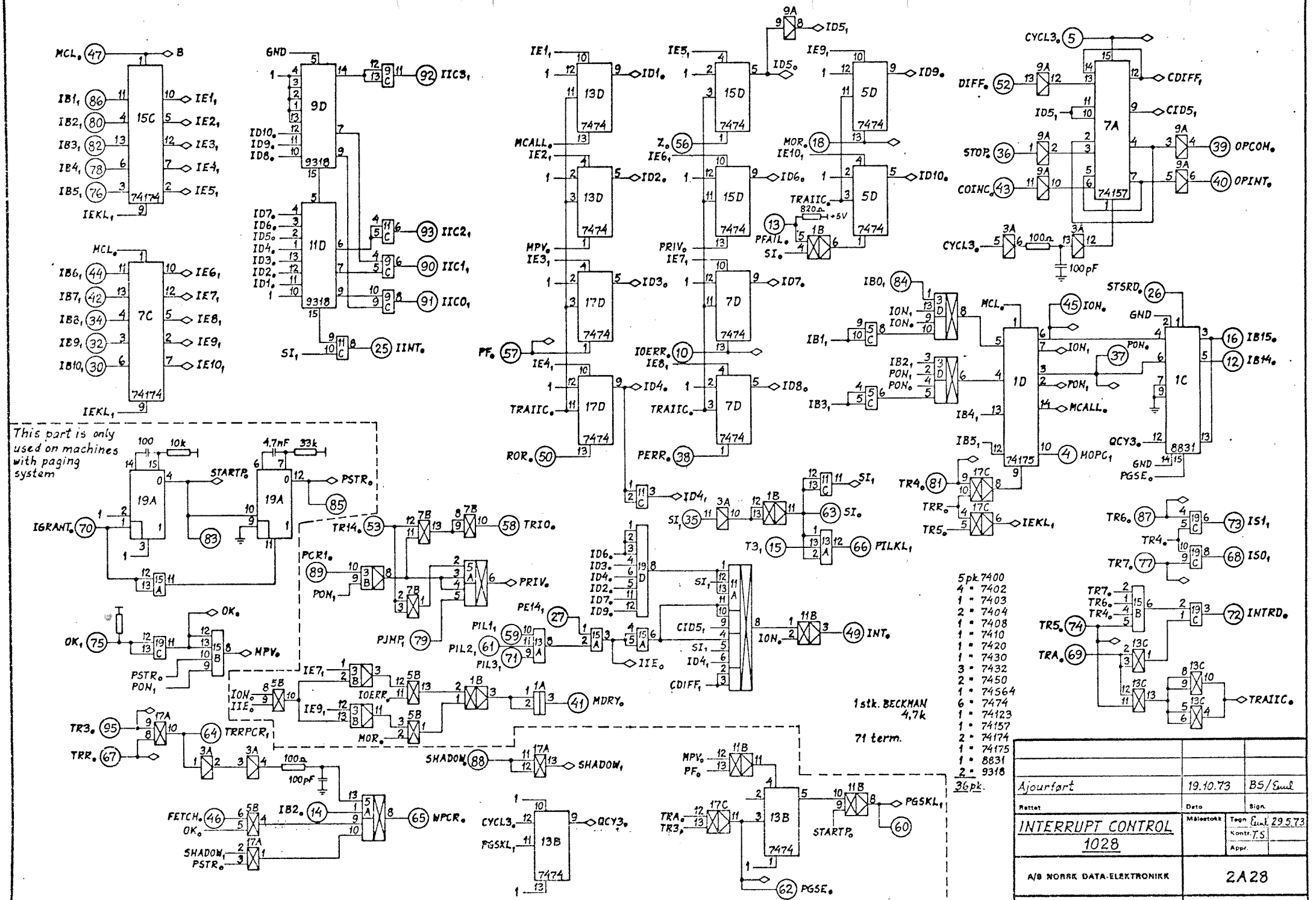
Totalt: 32 pk.

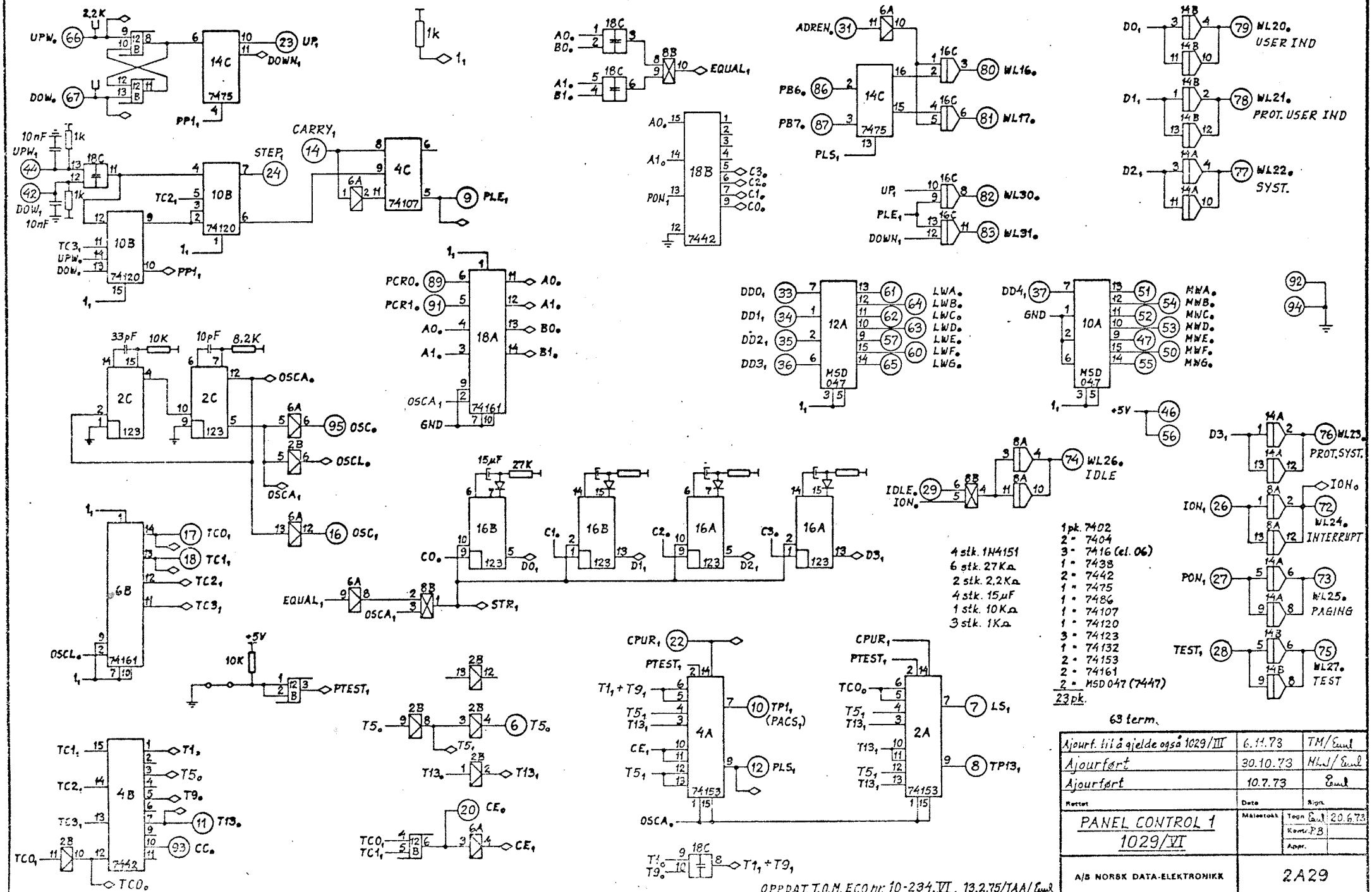
Rettet iflg. ECO-nr. 10-92 27/12-73
Ajourf. 3/12-73/BS.
Ajourf. 13/11-73/TAL
Ajourf. 19/10-73/B.S.

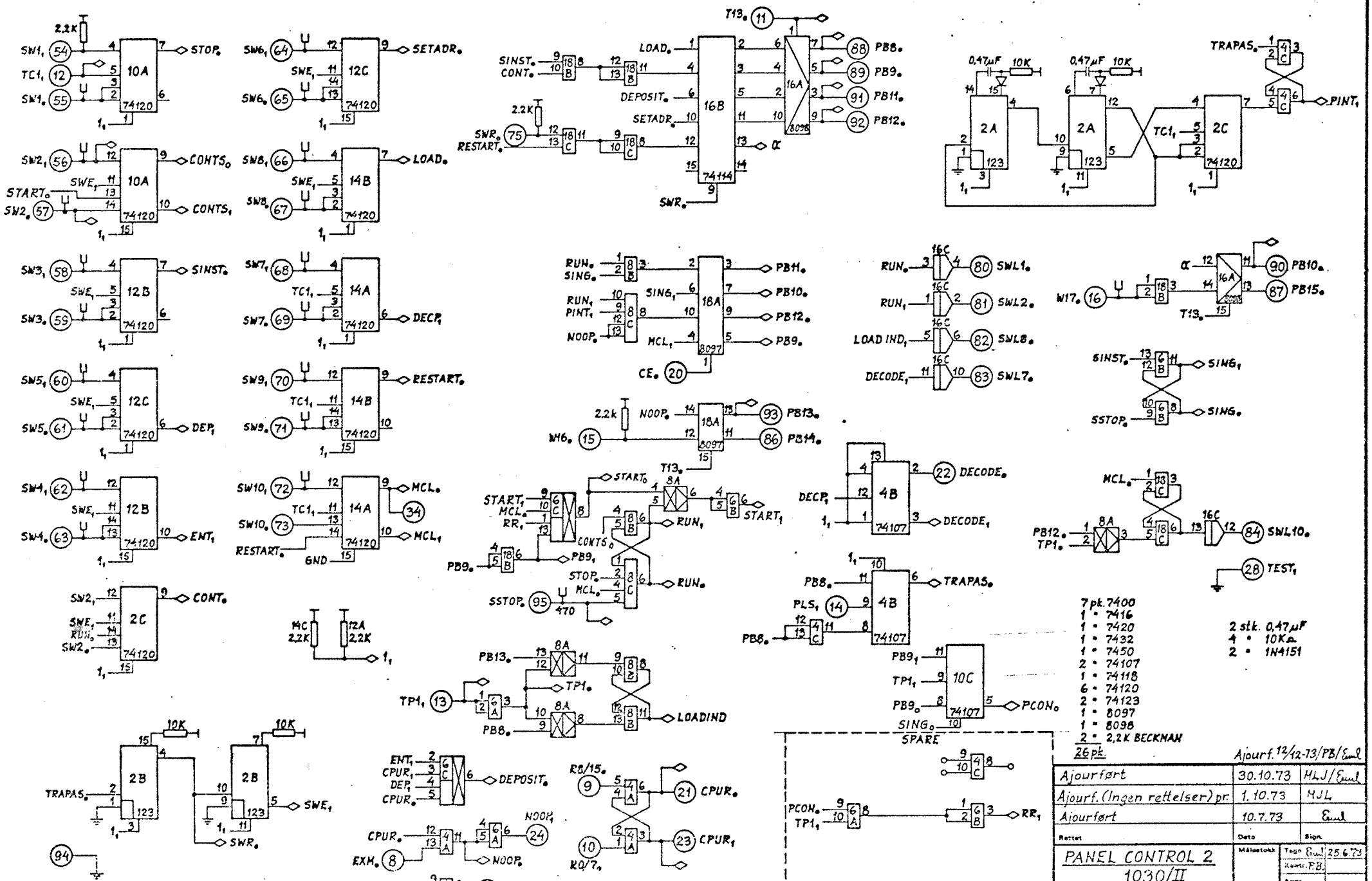
Ajourf. 10.4.75/TAA/Eml V til VII.

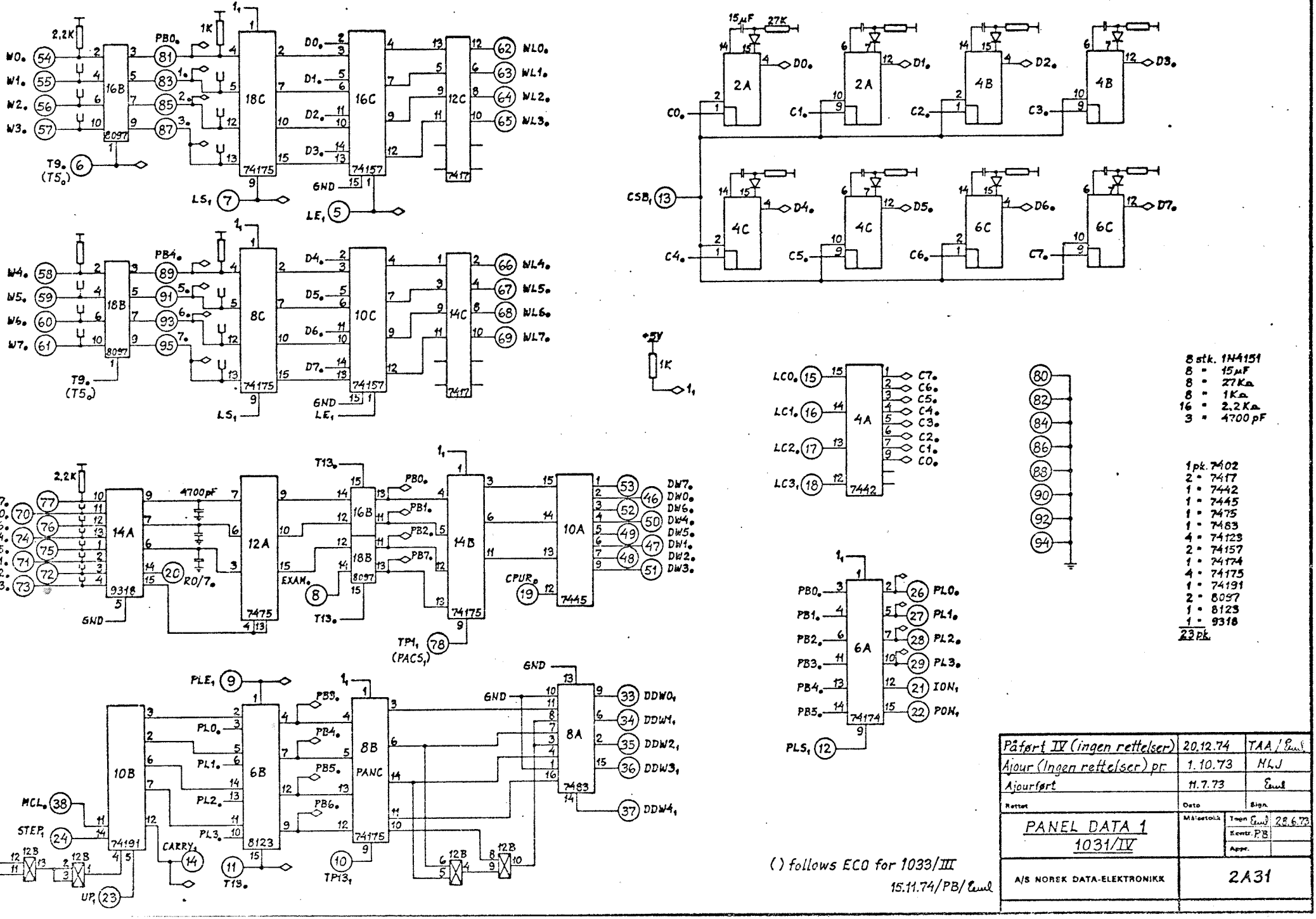
Rettet iflg ECO nr 10-244 III til V. 9/1-75/TAA/Eml

III til V. Rettet iflg ECO nr. 10-111 5/11-74/BS/Eml


This part is only used on machines with paging system
15C
9D
11D
7C
13D
15D
5D
17D
7D
7A
1D
1C
19A
74174
7474
74157
74175
8831
9318
5pk 7400
4 = 7402
1 = 7403
2 = 7404
1 = 7408
1 = 7410
1 = 7420
1 = 7430
3 = 7432
2 = 7450
1 = 74564
6 = 7474
1 = 74123
1 = 74157
2 = 74174
1 = 74175
1 = 8831
2 = 9318
36pk.
1 stk. BECKMAN 4,7k
71 term.
Ajourført
19.10.73
BS/Emil
Rettet
Dato
Sign.
INTERRUPT CONTROL
1028
Måleetikk
Tegn Emil 29.5.73
Kontr. TS
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A28


UPW
DOW
UP
DOWN
PP1
STEP
CARRY
PLE
TC2
TC3
EQUAL
PCR0
PCR1
OSCA
OSCL
GND
STR
TC0
TC1
TC2
TC3
PTEST
T1
T5
T9
T13
CC
CE
CPUR
PTEST
TP1 (PACS)
PLS
LS
TP13
T1 + T9
ADREN
PB6
PB7
PLS
UP
PLE
DOWN
WL16
WL17
WL30
WL31
D0
D1
D2
USER IND
PROT. USER IND
SYST.
WL20
WL21
WL22
DD0
DD1
DD2
DD3
LWA
LWB
LWC
LWD
LWE
LWF
LWG
DD4
GND
MWA
MWB
MWC
MWD
MWE
MWF
MWG
+5V
D3
PROT.SYST.
ION
PON
TEST
IDLE
WL23
WL24
WL25
WL26
WL27
INTERRUPT
PAGING
TEST
IDLE
4 stk. 1N4151
6 stk. 27Kω
2 stk. 2,2Kω
4 stk. 15μF
1 stk. 10Kω
3 stk. 1Kω
1 pk. 7402
2 - 7404
3 - 7416 (el. 06)
1 - 7438
2 - 7442
1 - 7475
1 - 7486
1 - 74107
1 - 74120
3 - 74123
1 - 74132
2 - 74153
2 - 74161
2 - MSD 047 (7447)
23 pk.
63 term.
Ajourf. til å gjelde også 1029/III 6.11.73 TM/Emil
Ajourført 30.10.73 MHJ/Emil
Ajourført 10.7.73 Emil
Rettet Dato Sign.
PANEL CONTROL 1
1029/VI
Målestokk
Tegn Emil 20.6.73
Kontr. PB
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A29
OPPDAT. T.O.M. ECO nr: 10-234.VI. 13.2.75/TAA/Emil
OPPDAT. T.O.M. ECO nr: 10-223.V 9.1.75/TAA/Emil


SW1, 54
TC1, 12
SW1. 55
2.2K
10A
74120
STOP.
SW6, 64
SW6. 65
12C
SETADR.
SW2, 56
START. SW2. 57
CONTS.
CONTS,
SW8, 66
SW8. 67
14B
LOAD.
SW3, 58
SW3. 59
12B
SINST.
SW7, 68
SW7. 69
14A
DECP,
SW5, 60
SW5. 61
DEP,
SW9, 70
SW9. 71
RESTART.
SW4, 62
SW4. 63
ENT,
SW10, 72
SW10. 73
RESTART. GND
MCL.
MCL,
SW2, 56
SW2. 57
2C
CONT.
MC 2.2K
12A 2.2K
SINST. CONT.
SWR. RESTART. 75
16B
74144
LOAD.
DEPOSIT.
SETADR.
SWR.
T13. 11
8098
PB8.
PB9.
PB11.
PB12.
RUN. SING.
RUN, PINT, NOOP.
18A
8097
SING,
MCL,
CE. 20
PB11.
PB10.
PB12.
PB9.
W6. 15
2.2k
NOOP.
PB13.
PB14.
START, MCL. RR,
CONTS.
RUN,
PB9,
STOP.
SSTOP. 95
470
START,
PB13.
TP1, 13
TP1.
PB8.
8A
LOADIND
ENT,
CPUR,
DEP,
CPUR.
DEPOSIT.
CPUR.
EXM. 8
NOOP,
24
NOOP.
CPUR. NOOP.
30 EXAM.
R5/15.
10
R0/7.
21 CPUR.
23 CPUR,
94
TRAPAS.
2B
SWE,
SWR.
10K
0,47µF 10K
2A
123
2C
TRAPAS.
PINT,
TC1,
RUN.
RUN,
LOAD IND,
DECODE,
16C
80 SWL1.
81 SWL2.
82 SWL8.
83 SWL7.
W17. 16
α
16A
90 PB10.
87 PB15.
T13.
SINST.
SSTOP.
SING,
SING.
DECP,
4B
74107
22 DECODE.
DECODE,
MCL.
PB12.
TP1.
84 SWL10.
28 TEST,
PB8.
PLS, 14
TRAPAS.
PB9,
TP1,
PB9.
10C
PCON.
SING.
SPARE
PCON.
TP1,
RR,
7 pk. 7400
1 • 7416
1 • 7420
1 • 7432
1 • 7450
2 • 74107
1 • 74118
6 • 74120
2 • 74123
1 • 8097
1 • 8098
2 • 2,2K BECKNAH
26 pk.
2 stk. 0,47µF
4 • 10KΩ
2 • 1N4151
Ajourf. 12/12-73/PB/Emil
Ajourført 30.10.73 HLJ/Emil
Ajourf. (Ingen rettelser) pr. 1.10.73 HJL
Ajourført 10.7.73 Emil
Rettet Dato Sign.
PANEL CONTROL 2
1030/II
Målestokk
Tegn Emil 25.6.73
Kontr. P.B.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A30
OPPDATERT T.O.M. ECO NR. 10-238. II. 14.3.75/Emil



() follows ECO for 1033/III

15.11.74/PB/Emil

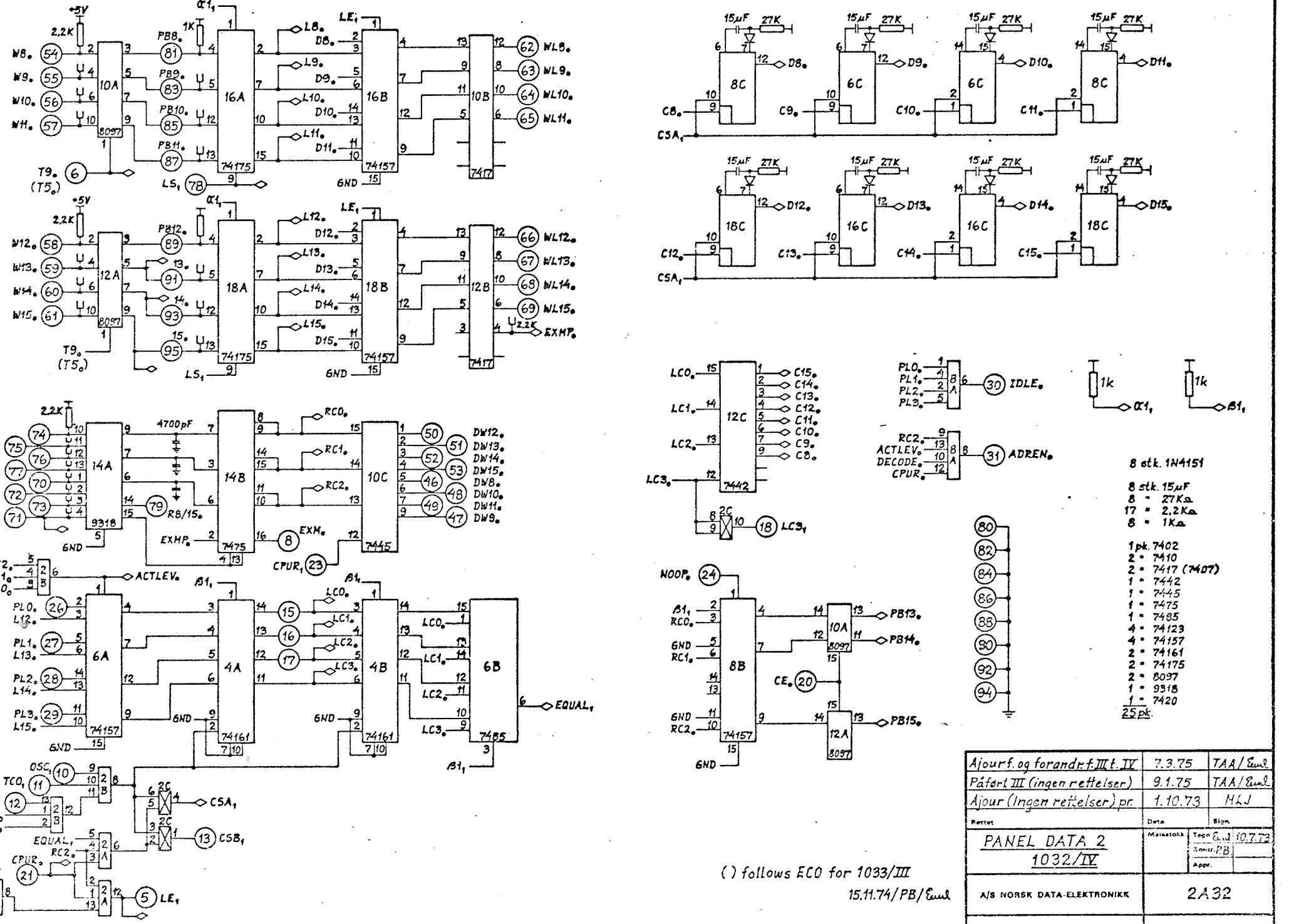
Påført IV (ingen rettelser)	20.12.74	TAA / Emil
Ajour (Ingen rettelser) pr	1.10.73	MLJ
Ajourført	11.7.73	Emil
Rettet	Dato	Sign.

PANEL DATA 1
1031/IV

Målestokk | Tegn. Emil 28.6.73 | Kontr. PB | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A31



8 stk. 1N4151

8 stk. 15µF
8 • 27KΩ
17 • 2,2KΩ
8 • 1KΩ

1 pk. 7402
2 • 7410
2 • 7417 (7407)
1 • 7442
1 • 7445
1 • 7475
1 • 7485
4 • 74123
4 • 74157
2 • 74161
2 • 74175
2 • 8097
1 • 9318
1 • 7420
25 pk.

() follows ECO for 1033/III
15.11.74/PB/Eml

Ajourf. og forandr.f.III t. IV	7.3.75	TAA/Eml
Påført III (ingen rettelser)	9.1.75	TAA/Eml
Ajour (Ingen rettelser) pr.	1.10.73	HLJ
Rettet	Dato	Sign.

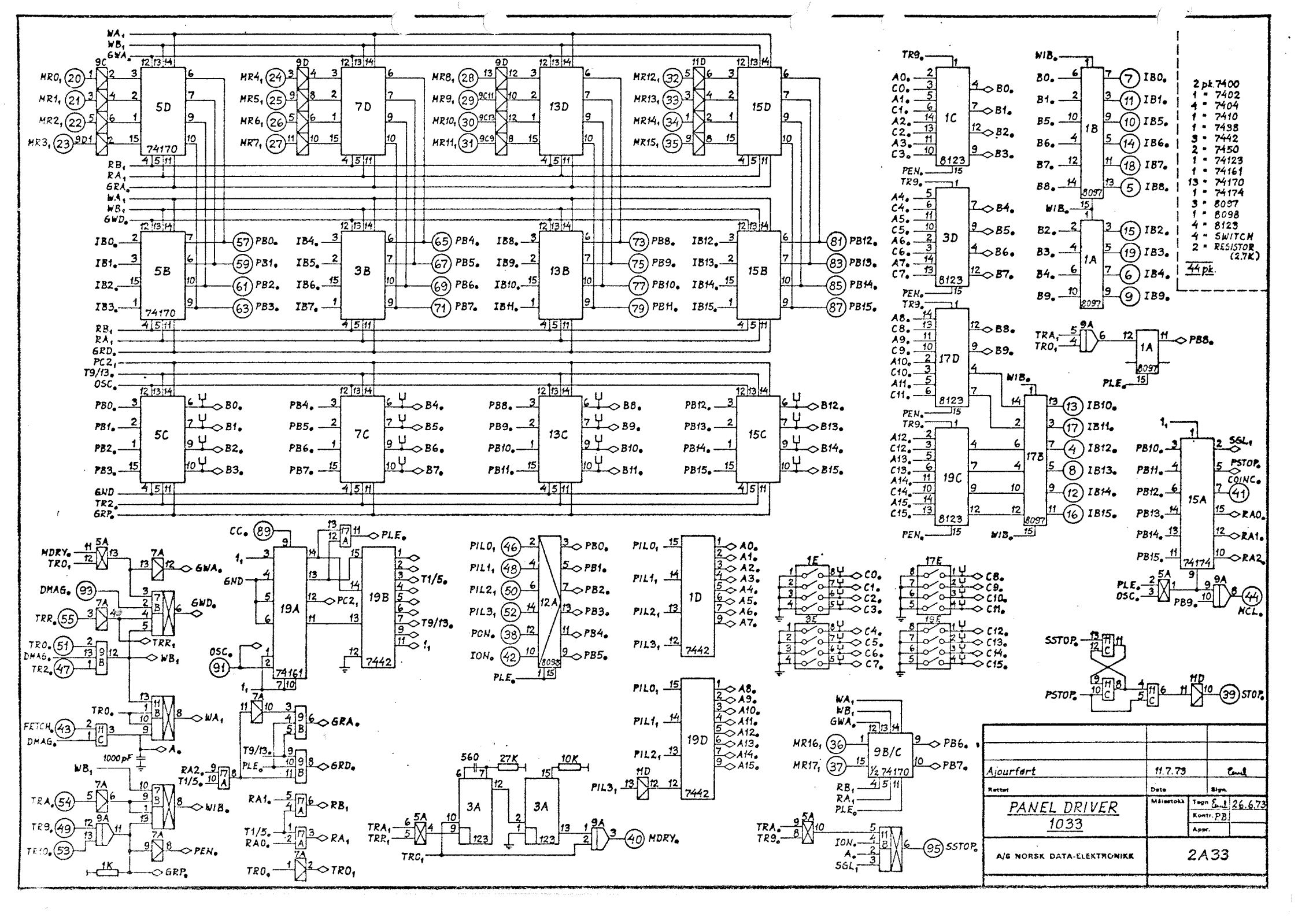
PANEL DATA 2
1032/IV

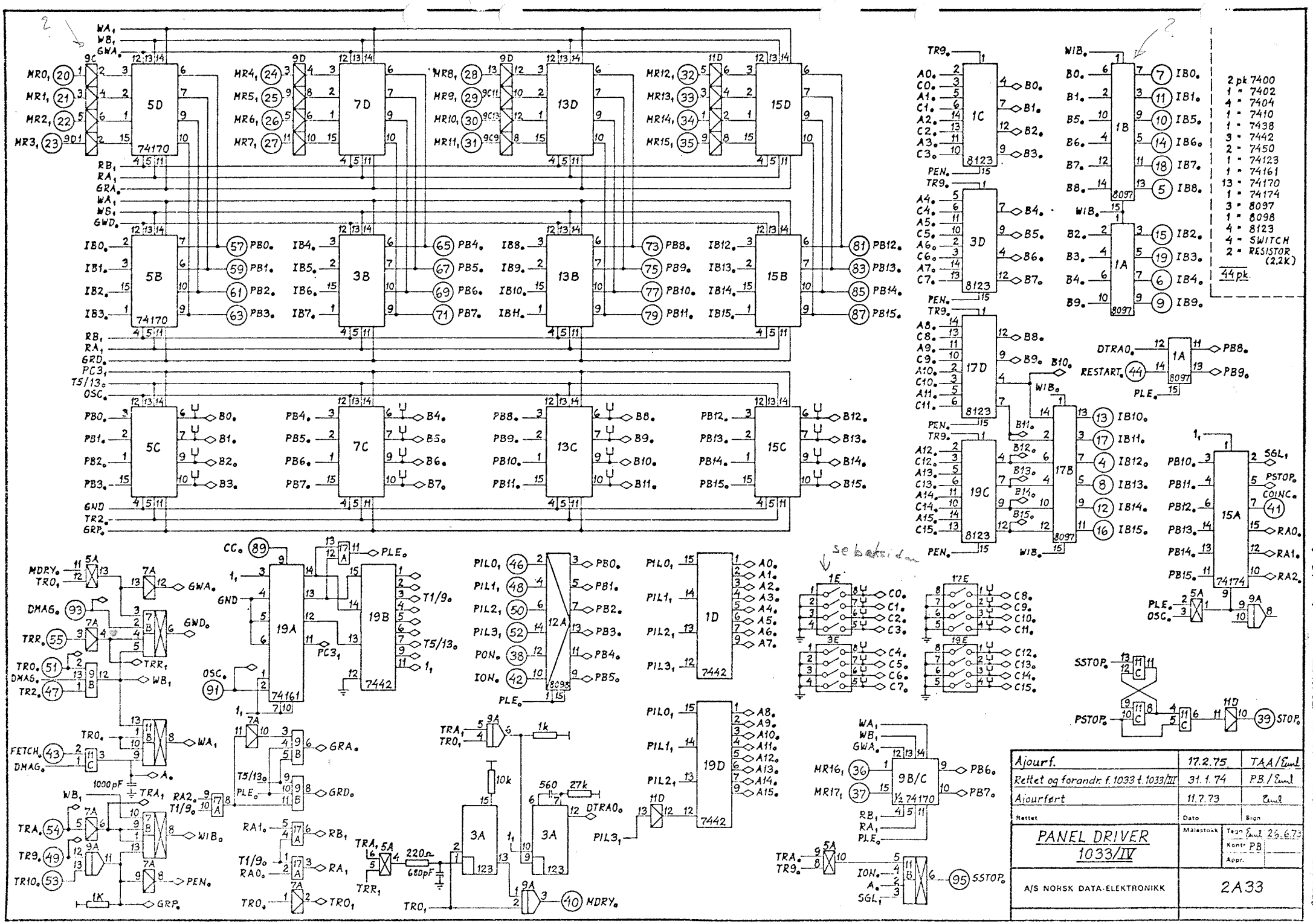
Målestokk

Tegn. Eml 10.7.73
Kontr. PB
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A32


5D
7D
13D
15D
5B
3B
13B
15B
5C
7C
13C
15C
74170
MR0
MR1
MR2
MR3
MR4
MR5
MR6
MR7
MR8
MR9
MR10
MR11
MR12
MR13
MR14
MR15
WA
WB
GWA
RB
RA
GRA
GWD
GRD
PC2
T9/13
OSC
GND
TR2
GRP
1C
1B
1A
3D
17D
19C
17B
1D
19D
9B/C
15A
19A
19B
74161
7442
12A
8098
3A
74174
PLE
SSTOP
PSTOP
COINC
NCL
MDRY
STOP
2 pk. 7400
1 = 7402
4 = 7404
1 = 7410
1 = 7438
3 = 7442
2 = 7450
1 = 74123
1 = 74161
13 = 74170
1 = 74174
3 = 8097
1 = 8098
4 = 8123
4 = SWITCH
2 = RESISTOR (2.7K)
44 pk.
Ajourført
11.7.73
Rettet
Dato
Sign.
PANEL DRIVER
1033
Målestokk
Tegn
26.6.73
Kontr. PB
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A33

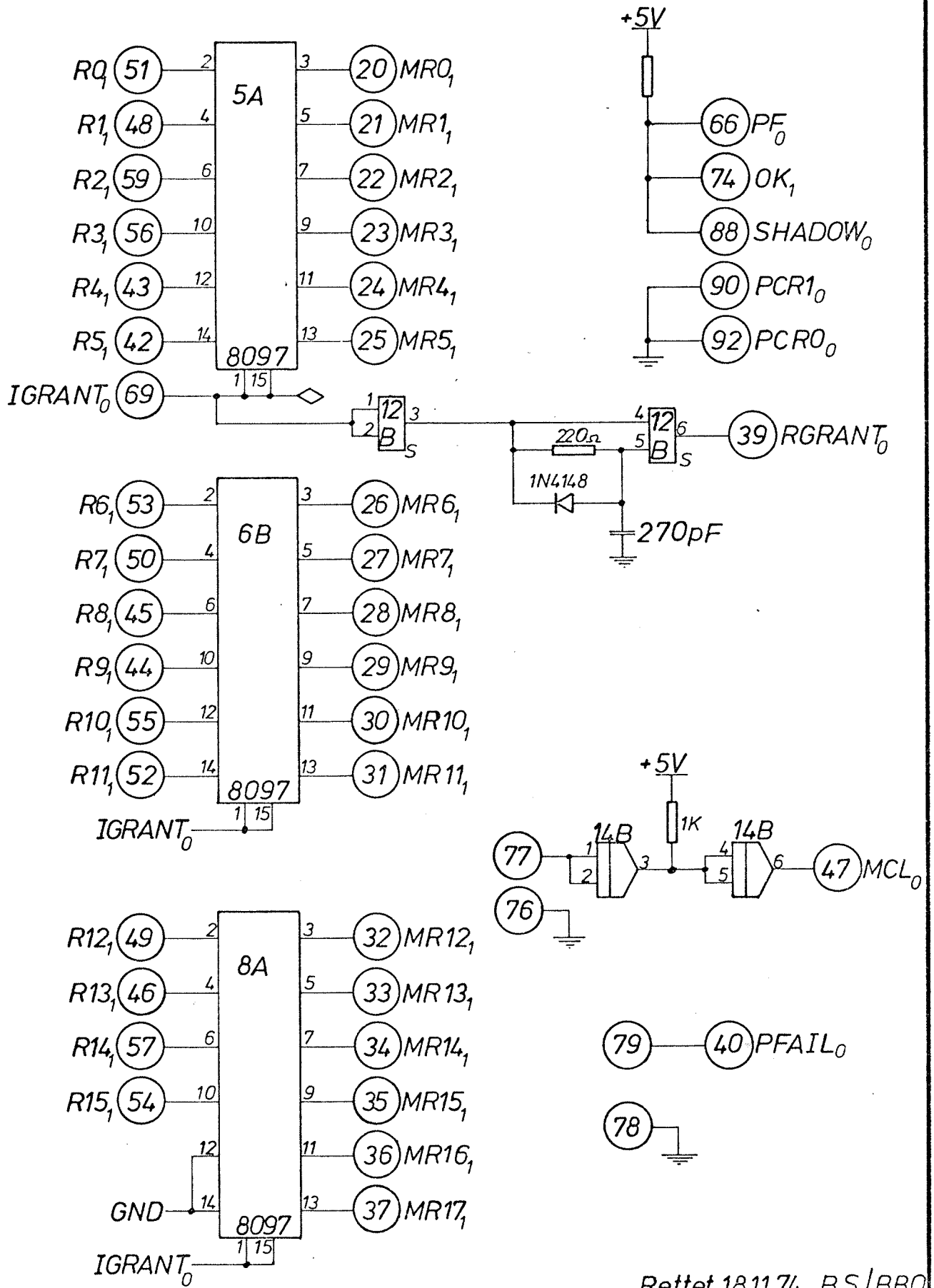

5D
7D
13D
15D
5B
3B
13B
15B
5C
7C
13C
15C
74170
1C
3D
17D
19C
1B
1A
15A
8123
8097
74174
19A
19B
74161
7442
12A
8098
1D
19D
9B/C
74170
3A
123
1E
17E
3E
19E
1K
1k
10k
27k
220n
680pF
1000pF
2 pk 7400
1 = 7402
4 = 7404
1 = 7410
1 = 7438
3 = 7442
2 = 7450
1 = 74123
1 = 74161
13 = 74170
1 = 74174
3 = 8097
1 = 8098
4 = 8123
4 = SWITCH
2 = RESISTOR (2,2K)
44 pk
se bakside
OPPDATERT T.O.H. ECO NR. 10-215. III til IV. 17.1.75/TAA/Eml
PANEL DRIVER
1033/IV
A/S NORSK DATA-ELEKTRONIKK
2A33
Ajourf. 17.2.75 TAA/Eml
Rettet og forandr. f. 1033 t. 1033/III 31.1.74 PB./Eml
Ajourført 11.7.73 Eml
Rettet Dato Sign
Målestokk
Tegn Eml 26.6.73
Kontr PB
Appr.

ALD 0020500

Dip switch	1	2	3	4
1E	0	0	0	0
3E	0	0	1	0
17E	1	0	0	0
19E	0	1	0	0

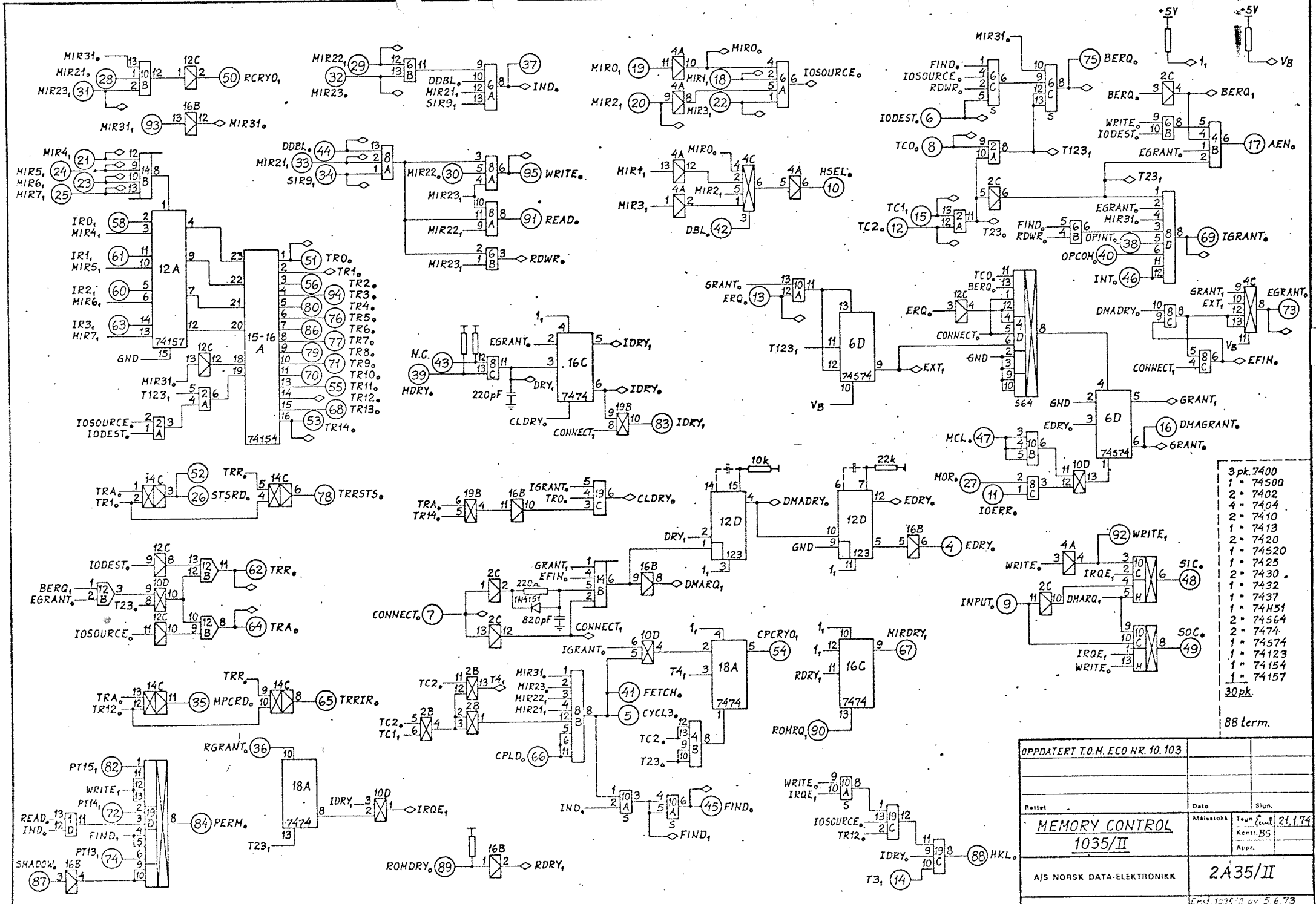
0 = OFF
1 = ON

A/S NORSK DATA-ELEKTRONIKK	Title: PAGING SURROGAT 1034/III	Drawing no.

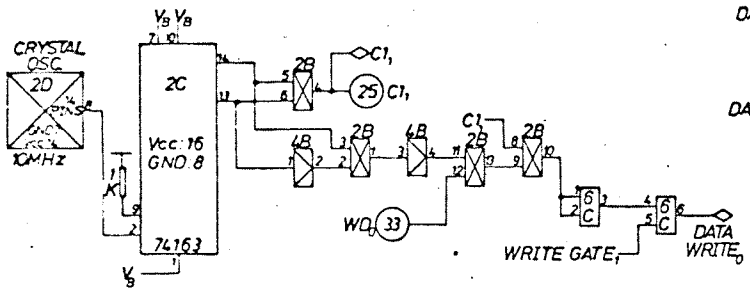


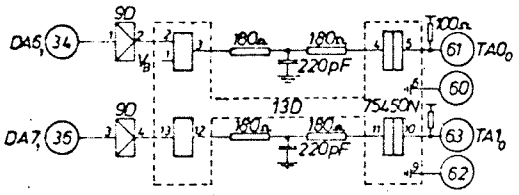
Rettet 18.11.74. B.S./BBO

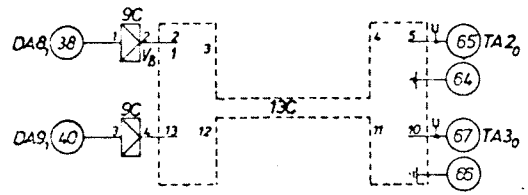
DRAWN BY BBO.	Remarks: Rettet fra II til III (ingen forandr.) 16.12.74 / TAA / Encl	Replacement for	Date
APPROVED BY B.S.			
DATE 13.11.74.		Replaced by	Date

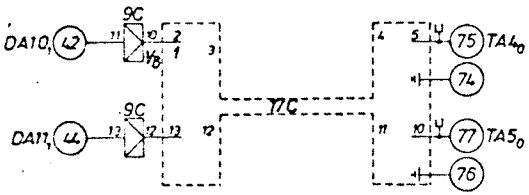

MIR31
MIR21
MIR23
RCRY0
MIR22
MIR23
DDBL
MIR21
SIR9
IND
MIR0
MIR1
MIR2
MIR3
IOSOURCE
FIND
IOSOURCE
RDWR
IODEST
BERQ
BERQ
WRITE
IODEST
EGRANT
AEN
TCO
T123
T23
TC1
TC2
T23
FIND
RDWR
OPINT
OPCOM
INT
IGRANT
MIR4
MIR5
MIR6
MIR7
DDBL
MIR21
SIR9
MIR22
MIR23
WRITE
READ
RDWR
MIR1
MIR3
MIR0
MIR2
DBL
HSEL
IR0
MIR4
IR1
MIR5
IR2
MIR6
IR3
MIR7
12A
74157
GND
MIR31
T123
IOSOURCE
IODEST
15-16 A
74154
TR0
TR1
TR2
TR3
TR4
TR5
TR6
TR7
TR8
TR9
TR10
TR11
TR12
TR13
TR14
N.C.
MDRY
EGRANT
220pF
DRY
16C
7474
IDRY
IDRY
CLDRY
CONNECT
GRANT
ERQ
T123
6D
74574
EXT
VB
TCO
BERQ
ERQ
CONNECT
GND
S64
GND
EDRY
6D
74574
GRANT
GRANT
DMAGRANT
MCL
MOR
IOERR
10D
DMADRY
EDRY
DMADRY
EXT
EGRANT
VB
CONNECT
EFIN
TRA
TR1
14C
STSRD
TRR
TRRSTS
TRA
TR14
19B
16B
IGRANT
TR0
CLDRY
10k
22k
12D
DRY
DMADRY
GND
EDRY
IODEST
BERQ
EGRANT
T23
IOSOURCE
12C
10D
TRR
TRA
GRANT
EFIN
220Ω
1N4151
820pF
CONNECT
DMARQ
CONNECT
IGRANT
10D
FETCH
CYCL3
CPLD
MIR31
MIR23
MIR22
MIR21
TC2
TC1
T4
18A
7474
CPCRY0
TC2
T23
TRA
TR12
14C
MPCRD
TRR
TRRIR
RGRANT
18A
7474
IDRY
10D
IRQE
T23
PT15
PT14
PT13
WRITE
READ
IND
FIND
SHADOW
16B
PERM
ROMDRY
16B
RDRY
IND
FIND
FIND
MIRDRY
16C
7474
RDRY
ROMRQ
WRITE
IRQE
IOSOURCE
TR12
IDRY
T3
HKL
WRITE
IRQE
SIC
INPUT
DMARQ
IRQE
WRITE
SOC
+5V
+5V
VB
1
3 pk. 7400
1 = 7450
2 = 7402
4 = 7404
2 = 7410
1 = 7413
2 = 7420
1 = 74520
1 = 7425
2 = 7430
1 = 7432
1 = 7437
1 = 74H51
2 = 74S64
2 = 7474
1 = 74S74
1 = 74123
1 = 74154
1 = 74157
30 pk.
88 term.
OPPDATERT T.O.M. ECO NR. 10.103
Rettet
Dato
Sign.
MEMORY CONTROL
1035/II
Målestokk
Tegnet
Kontr. BS
Appr.
21.1.74
A/S NORSK DATA-ELEKTRONIKK
2A35/II
Erst. 1035/II av 5.6.73

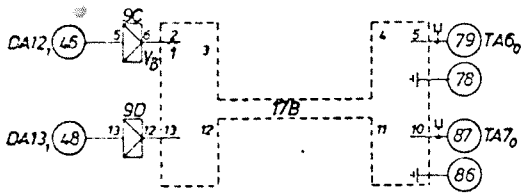
NOTE:
On all 75450N pin 6 and 9 should be connected to pin 7 (GND)

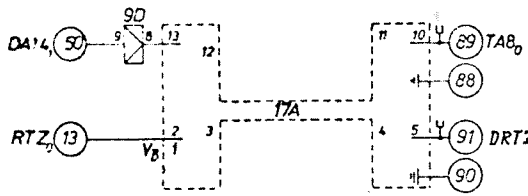


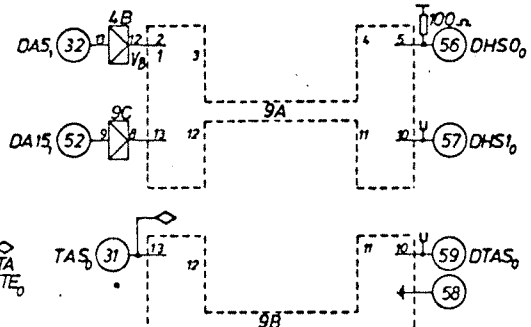


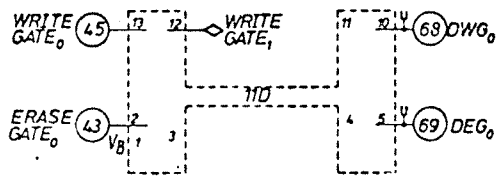


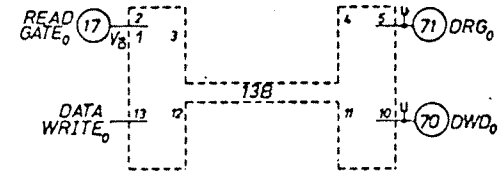


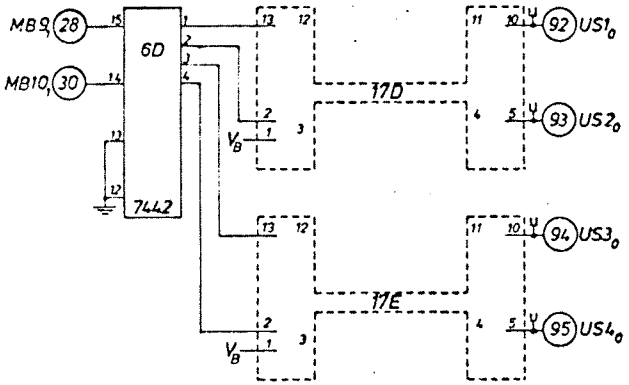


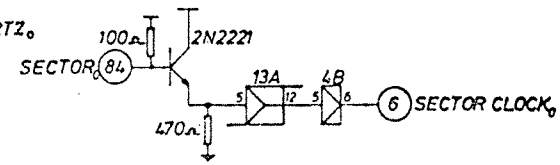


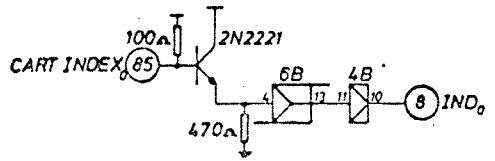


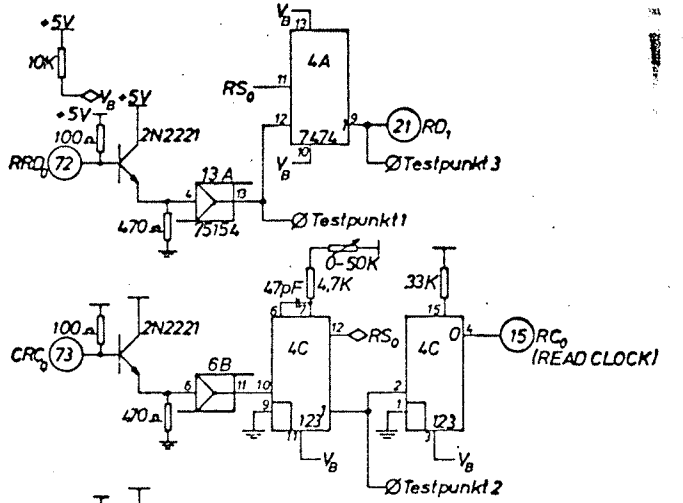


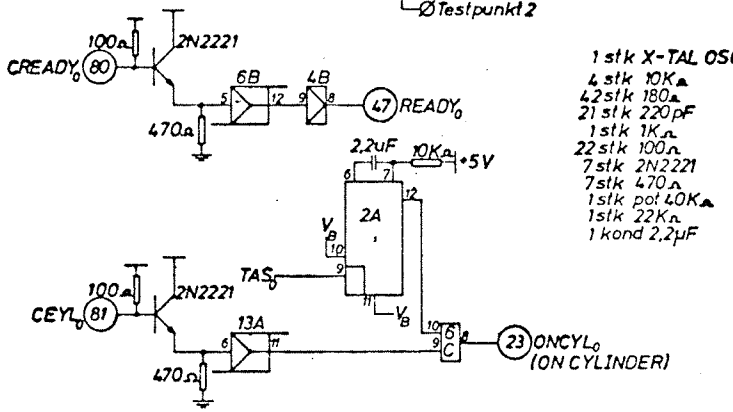


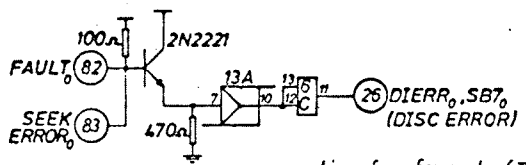












1 stk X-TAL OSC
4 stk 10KΩ
42 stk 180Ω
21 stk 220pF
1 stk 1KΩ
22 stk 100Ω
7 stk 2N2221
7 stk 470Ω
1 stk pot 40KΩ
1 stk 22KΩ
1 kond 2,2µF

1 pk 7400
1 pk 7402
3 pk 7404
1 pk 7442
1 pk 7474
2 pk 74123
2 pk 75154
1 pk 74163
11 pk 75450N
23 pk

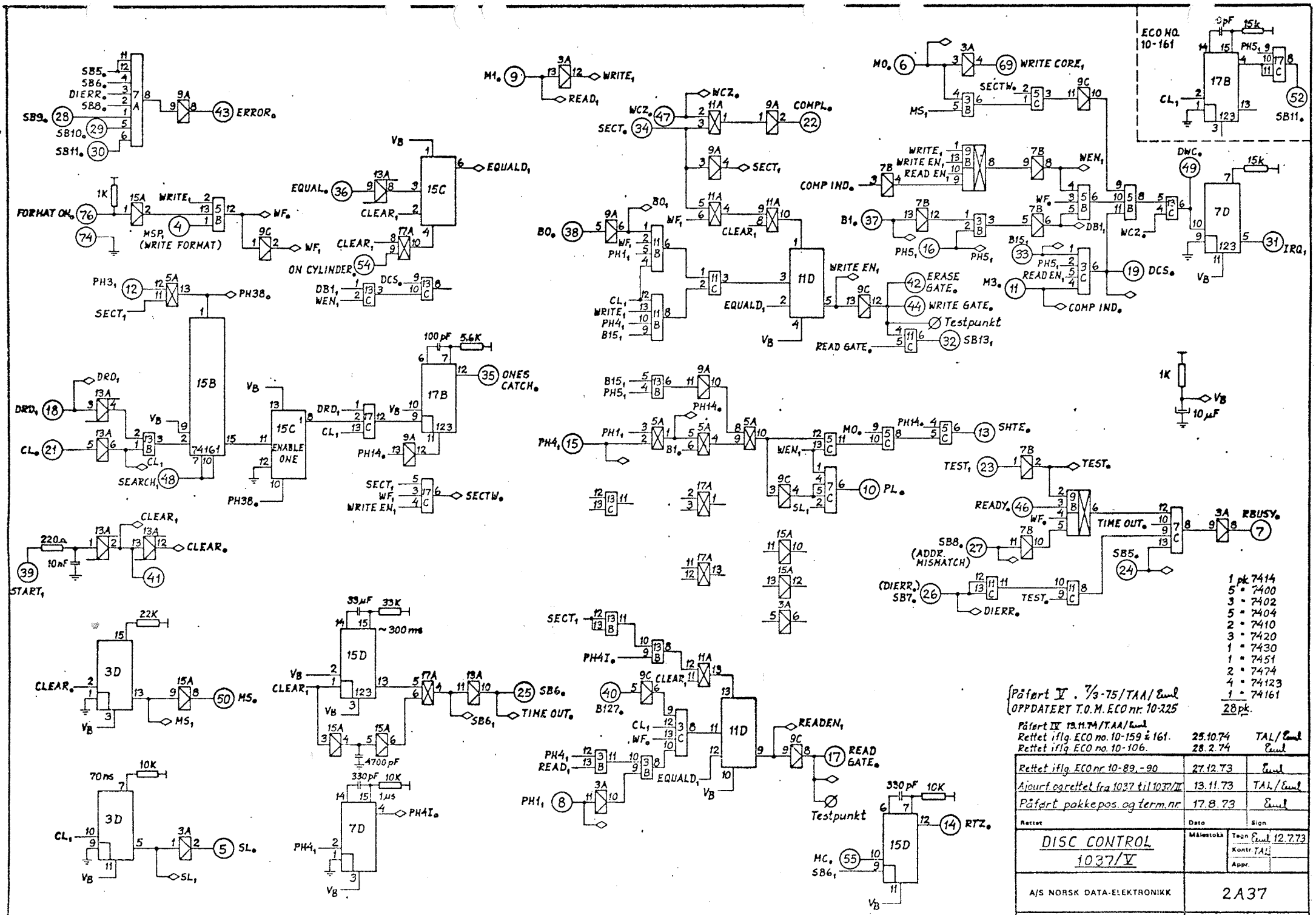
Ajourf. og forandr. f.III t.IV	7.3.75	TAA/Emil
Forandr. f.II t.III (ingen rettels.)	9.1.75	TAA/Emil
Ajourført og forandr. til 1036/II	13.11.73.	TAL/BBO
3 Term. flyttet fra 9B til 17A		
Rettet + påført pk. pos. og nr. på pk. bein.	Dato 5.10.73	Sign TAL/BBO

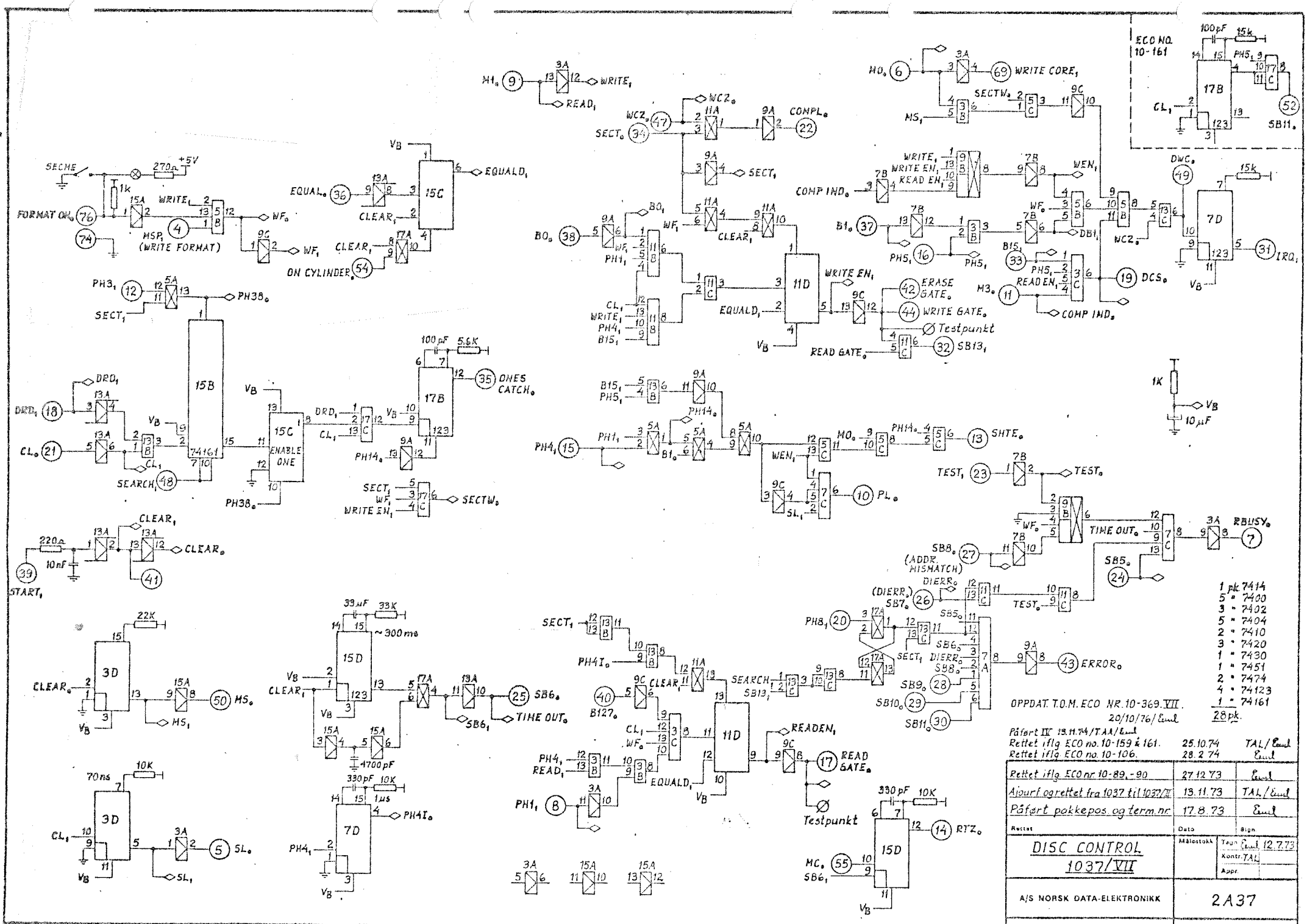
DISC TRANSCEIVER 1036/IV

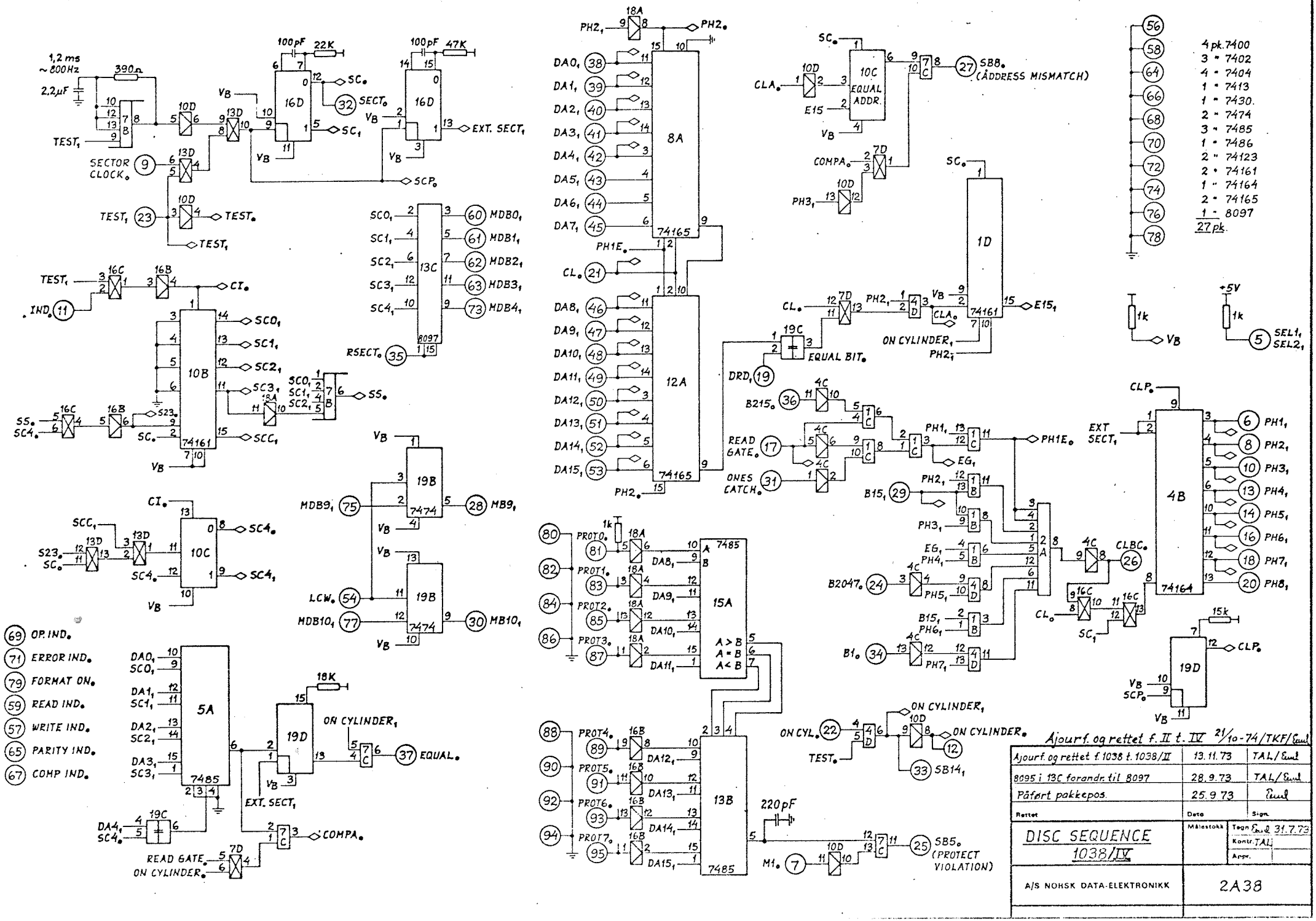
Målestokk | Tegn BBO 10.7.73 | Kontr TS | Appr

A/S NORSK DATA-ELEKTRONIKK

2A36


ECO NO. 10-161
SB5, SB6, DIERR, SB8, SB9, SB10, SB11, ERROR
FORMAT ON, MSP (WRITE FORMAT), WF, WF1
EQUAL, CLEAR1, ON CYLINDER, EQUALD1
DB1, WEN1, DCS
PH3, SECT1, PH38
DRD1, CL, SEARCH1, CL1
START1, CLEAR, 220Ω, 10nF
100pF, 5.6K, ONES CATCH
SECTW, WRITE EN1
M1, READ1, WRITE1
WCZ, SECT, COMPL
MO, MS1, WRITE CORE1, SECTW
WRITE1, WRITE EN1, READ EN1, WEN1
COMP IND, B1, PH5, B15, DB1, WC2
DWC, IRQ1
M3, READ GATE, ERASE GATE, WRITE GATE, Testpunkt, SB13
BO, PH1, WRITE EN1, EQUALD1
PH4, SHTE, PL, TEST, READY, TIME OUT, RBUSY
SB8 (ADDR. MISMATCH), SB5, (DIERR.) SB7, DIERR
1K, 10µF, VB
SECT1, PH4I, B127, CLEAR1, CL1, WF, EQUALD1, READEN1, READ GATE
PH4, READ1, PH1, Testpunkt
MC, SB61, RTZ
CLEAR, MS
CL1, SL
33µF, 33K, ~300 ms, 4700pF
330pF, 10K, 1µs, PH4I
70ms, 22K
SB6, TIME OUT
17B, 15A, 15B, 15C, 15D, 3A, 3D, 7B, 7D, 9A, 9C, 11A, 11B, 11C, 11D, 13A, 13B, 13C, 17A, 17B
1 pk 7414
5 = 7400
3 = 7402
5 = 7404
2 = 7410
3 = 7420
1 = 7430
1 = 7451
2 = 7474
4 = 74123
1 = 74161
28 pk.
Påført V. 7/3-75/TAA/Eml
OPPDATERT T.O.M. ECO nr. 10-225
Påført IV 13.11.74/TAA/Eml
Rettet iflg. ECO no. 10-159 & 161. 25.10.74 TAL/Eml
Rettet iflg. ECO no. 10-106. 28.2.74 Eml
Rettet iflg. ECO nr 10-89, -90 27.12.73 Eml
Ajourf og rettet fra 1037 til 1037/V 13.11.73 TAL/Eml
Påført pakkepos. og term.nr 17.8.73 Eml
Rettet Dato Sign
DISC CONTROL
1037/V
Målestokk
Tegn Eml 12.7.73
Kontr TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A37

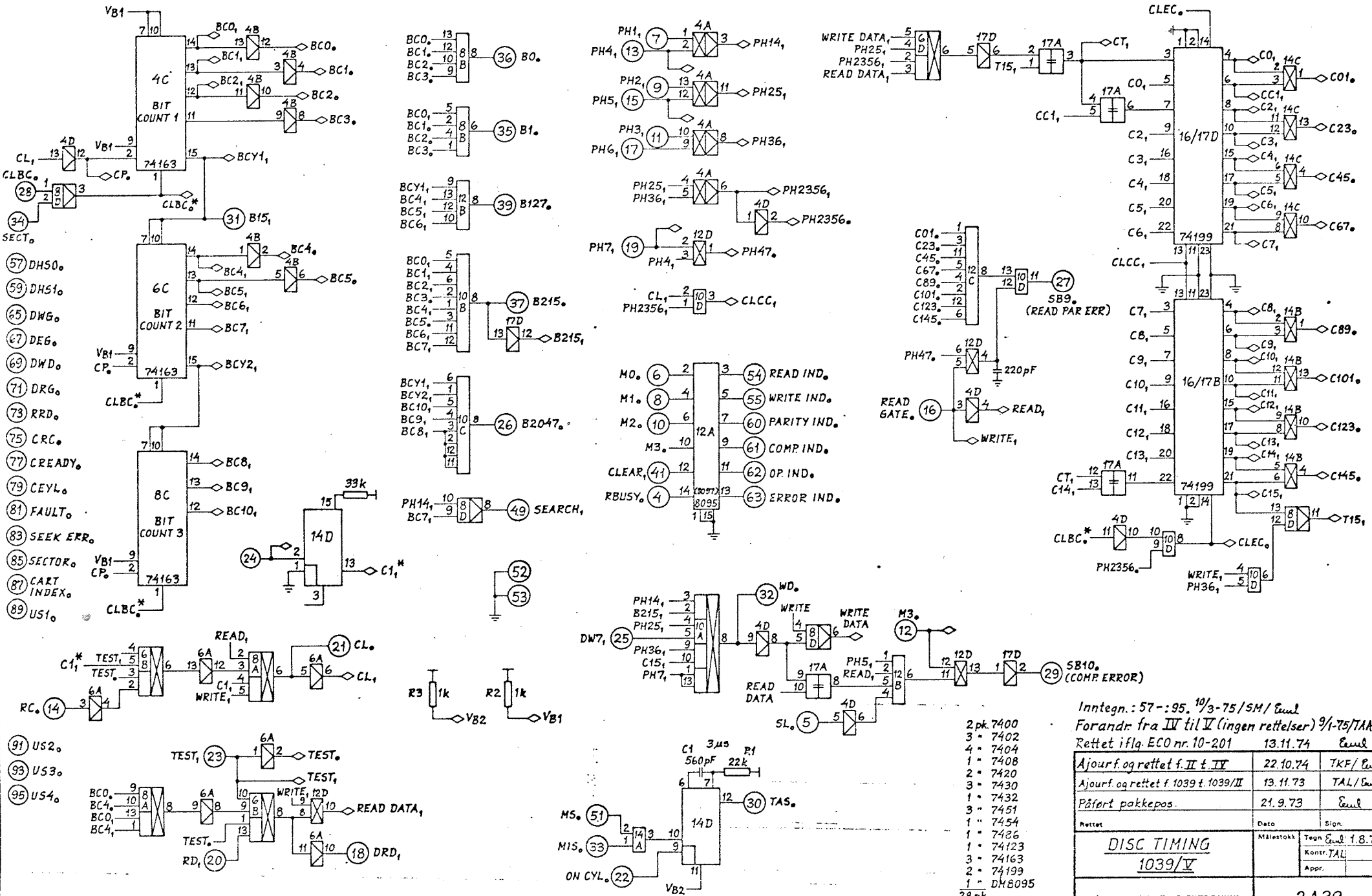

ECO NO. 10-161
100pF 15k
17B
PH5₁
CL₁
SB11₀
SECME
270Ω +5V
1k
FORMAT OK₀
WRITE₁
MSP₁ (WRITE FORMAT)
WF₀
WF₁
EQUAL₀
CLEAR₁
15C
EQUALD₁
ON CYLINDER₀
WCZ₀
SECT₀
SECT₁
CONPL₀
BO₀
BO₁
WF₁
CLEAR₁
11D
EQUALD₁
WRITE EN₁
ERASE GATE₀
WRITE GATE₀
Testpunkt
READ GATE₀
SB13₁
MO₀
WRITE CORE₁
SECTW₀
MS₁
COMP IND₀
WRITE EN₁
READ EN₁
WEN₁
B1₀
DBI₁
B15₁
M3₀
READ EN₁
COMP IND₀
DCS₀
DWC₀
7D
IRQ₁
PH3₁
SECT₁
PH38₀
15B
DRD₁
CL₀
SEARCH₁
ENABLE ONE
100pF 5.6K
17B
ONES CATCH₀
PH14₀
SECT₁
WF₁
WRITE EN₁
SECTW₀
B15₁
PH5₁
PH1₁
PH4₁
WEN₁
MO₀
SHTE₀
PL₀
1K
10µF
TEST₁
TEST₀
WF₀
TIME OUT₀
RBUSY₀
SB8₀ (ADDR. MISMATCH)
SB5₀
DIERR₀
(DIERR) SB7₀
TEST₀
PH8₁
SB6₀
SECT₁
DIERR₀
SB8₀
SB9₀
SB10₀
SB11₀
ERROR₀
CLEAR
220Ω
10nF
START₁
CLEAR₀
22K
3D
CLEAR₀
HS₀
HS₁
33µF 33K
~300ms
15D
CLEAR₁
SB6₀
SB6₁
TIME OUT₀
4700pF
330pF 10K
1µs
7D
PH4I₀
PH4₁
70ns
10K
3D
CL₀
SL₀
SL₁
SECT₁
PH4I₀
B12₇₀
CLEAR₁
SEARCH₁
SB13₁
READEN₁
READ GATE₀
Testpunkt
PH4₁
READ₁
EQUALD₁
PH1₁
11D
330pF 10K
15D
MC₀
SB6₁
RTZ₀
1 pk. 7414
5 " 7400
3 " 7402
5 " 7404
2 " 7410
3 " 7420
1 " 7430
1 " 7451
2 " 7474
4 " 74123
1 " 74161
28pk.
OPPDAT. T.O.M. ECO NR. 10-369. VII. 20/10/76/Emil
Påført IX 13.11.74/TAA/Emil
Rettet iflg ECO no. 10-159 & 161. 25.10.74 TAL/Emil
Rettet iflg ECO no. 10-106. 28.2.74 Emil
Rettet iflg ECO nr. 10-89, -90 27.12.73 Emil
Ajourf. ogrettet fra 1037 til 1037/II 13.11.73 TAL/Emil
Påført pakkepos. og term.nr. 17.8.73 Emil
Rettet Dato Sign
DISC CONTROL 1037/VII
Målestokk
Tegn. Emil 12.7.73
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A37



Ajourf. og rettet f. II t. IV 21/10-74/TKF/Emil

Ajourf. og rettet f. 1038 t. 1038/II	13.11.73	TAL/Emil
8095 i 13C forandr. til 8097	28.9.73	TAL/Emil
Påført pakkepos.	25.9.73	Emil
Rettet	Dato	Sign.

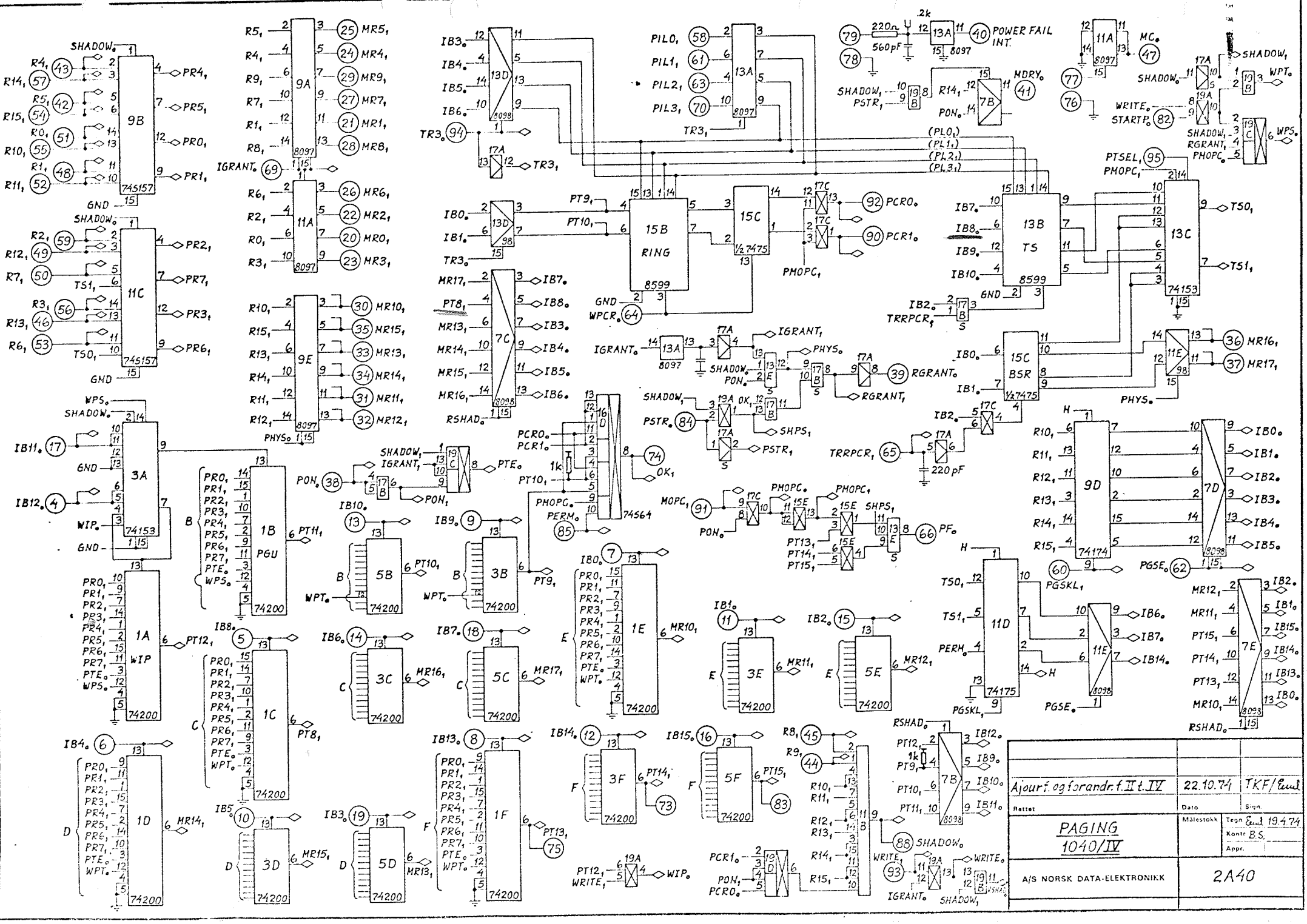
DISC SEQUENCE 1038/IV	Målestokk	Tegn Emil 31.7.73 Kontr. TAL Appr.
A/S NOHSK DATA-ELEKTRONIKK		2A38

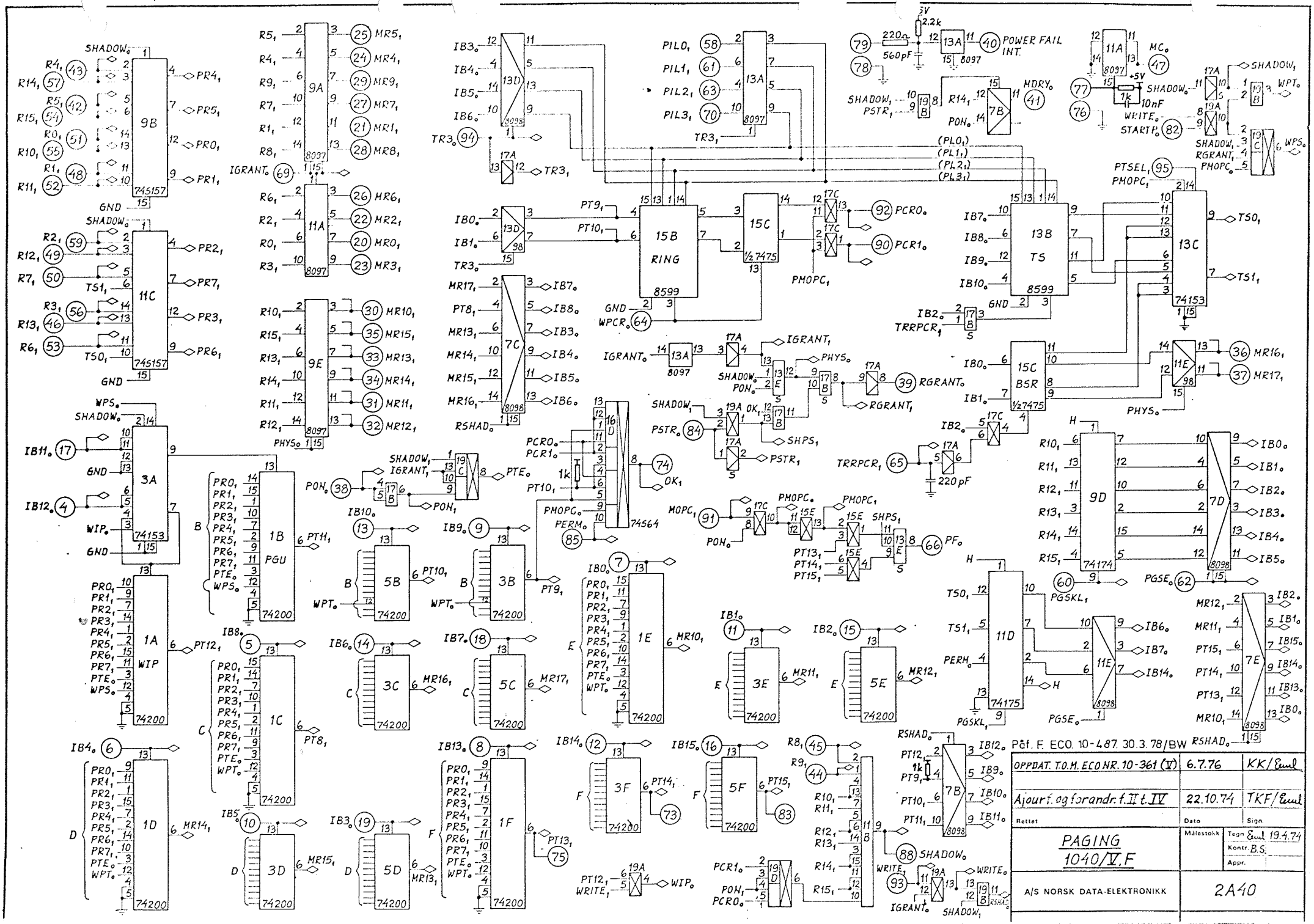


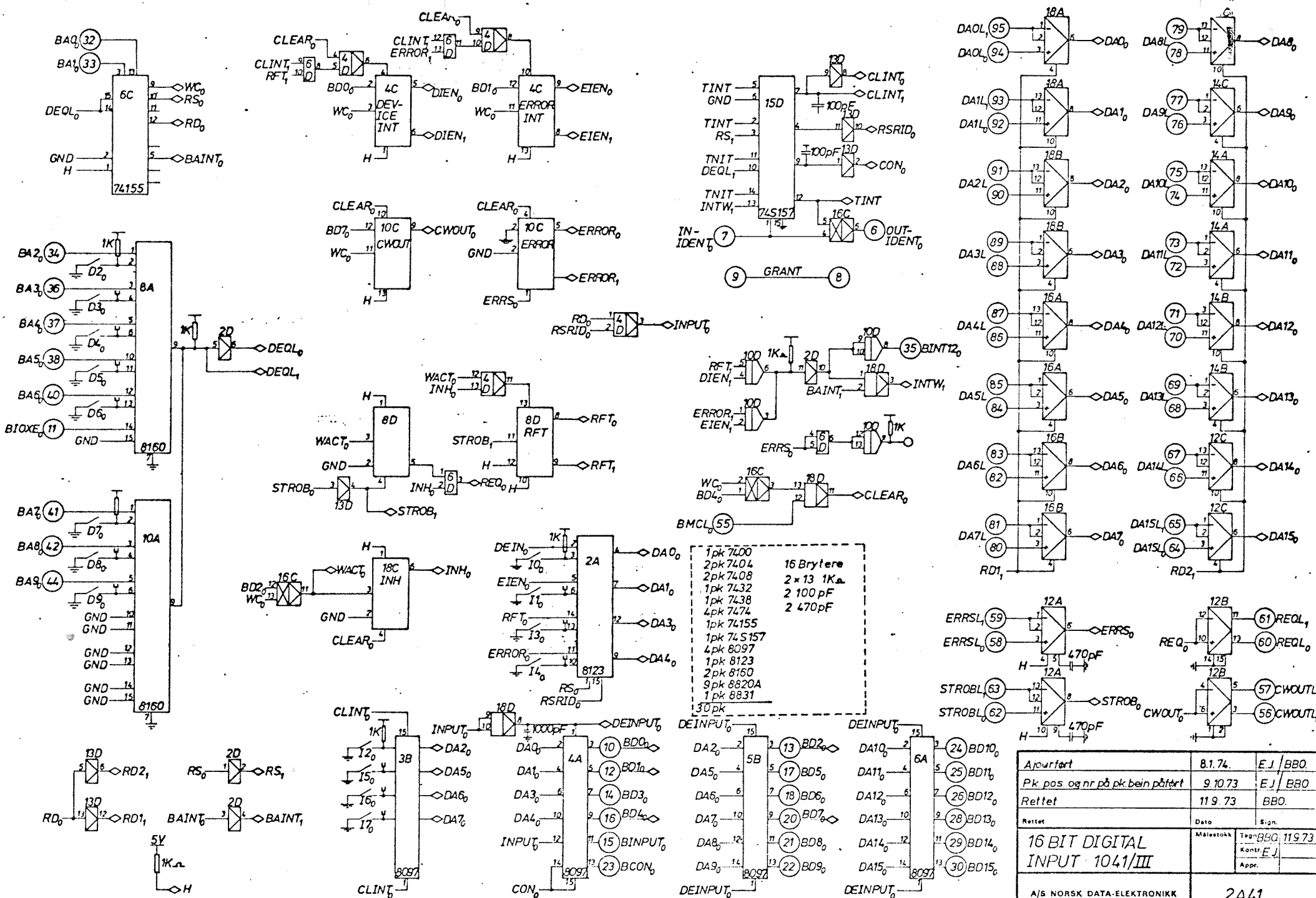
Inntegn.: 57-:95. 10/3-75/SM/Eml
Forandr. fra IV til V (ingen rettelser) 9/1-75/TAA
Rettet i flg. ECO nr. 10-201 13.11.74 Eml

Ajourf. og rettet f. II t. IV	22.10.74	TKF/Eml
Ajourf. og rettet f. 1039 t. 1039/II	13.11.73	TAL/Eml
Påført pakkepos.	21.9.73	Eml
Rettet	Dato	Sign.

DISC TIMING 1039/V	Målestokk	Tegn. Eml 1.8.73
		Kontr. TAL
		Appr.
A/S NORSK DATA-ELEKTRONIKK	2A39	


9B
9A
11A
11C
9E
745157
8097
8098
13D
13A
8097
7B
11A
8097
17A
19A
19C
15B
RING
8599
15C
7475
17C
13B
TS
8599
13C
74153
15C
BSR
7475
11E
98
3A
74153
1B
PGU
74200
1A
WIP
74200
7C
8098
74564
9D
74174
7D
8098
11D
74175
11E
8098
7E
8098
5B
74200
3B
74200
1E
74200
3E
74200
5E
74200
3C
74200
5C
74200
1C
74200
1D
74200
3D
74200
5D
74200
1F
74200
3F
74200
5F
74200
7B
8098
Ajourf. og forandr. f. II t. IV
22.10.74
TKF/Emil
Rettet
Dato
Sign
PAGING
1040/IV
Milestokk
Tegn Emil 19.4.74
Kontr B.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A40


9B
745157
11C
745157
9A
8097
11A
8097
9E
8097
7C
8098
15B
RING
8599
15C
½7475
13B
TS
8599
13C
74153
15C
BSR
½7475
9D
74174
7D
8098
11E
8098
7E
8098
11D
74175
3A
74153
1A
WIP
74200
1B
PGU
74200
1C
74200
1D
74200
1E
74200
1F
74200
3B
74200
3C
74200
3D
74200
3E
74200
3F
74200
5B
74200
5C
74200
5D
74200
5E
74200
5F
74200
7B
8098
74564
Påf. F. ECO. 10-487. 30.3.78/BW
OPPDAT. T.O.M. ECO NR. 10-361 (V)
6.7.76
KK/Eml
Ajourf. og forandr. f. II t. IV
22.10.74
TKF/Eml
Rettet
Dato
Sign.
PAGING
1040/V.F
Målestokk
Tegn Eml 19.4.74
Kontr. B.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A40



Ajourført	8.1.74.	E.J / BBO.
Pk. pos. og nr på pk. bein påført	9.10.73	E.J / BBO
Rettet	11 9. 73	BBO.
Rettet	Dato	Sign.

16 BIT DIGITAL INPUT 1041/III

Målestokk

Tegn BBO. 11.9.73

Kontr. E.J.

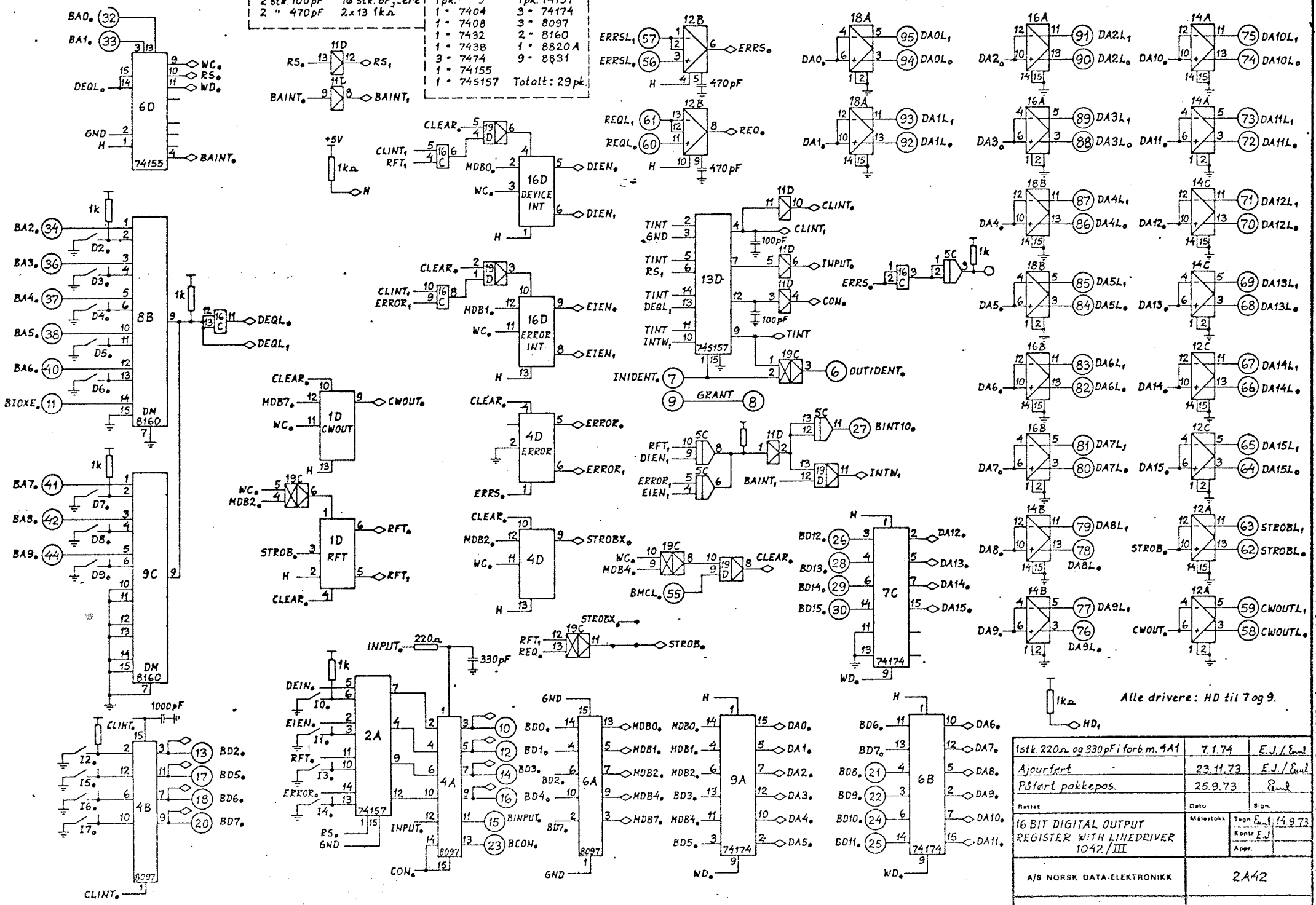
Appr.

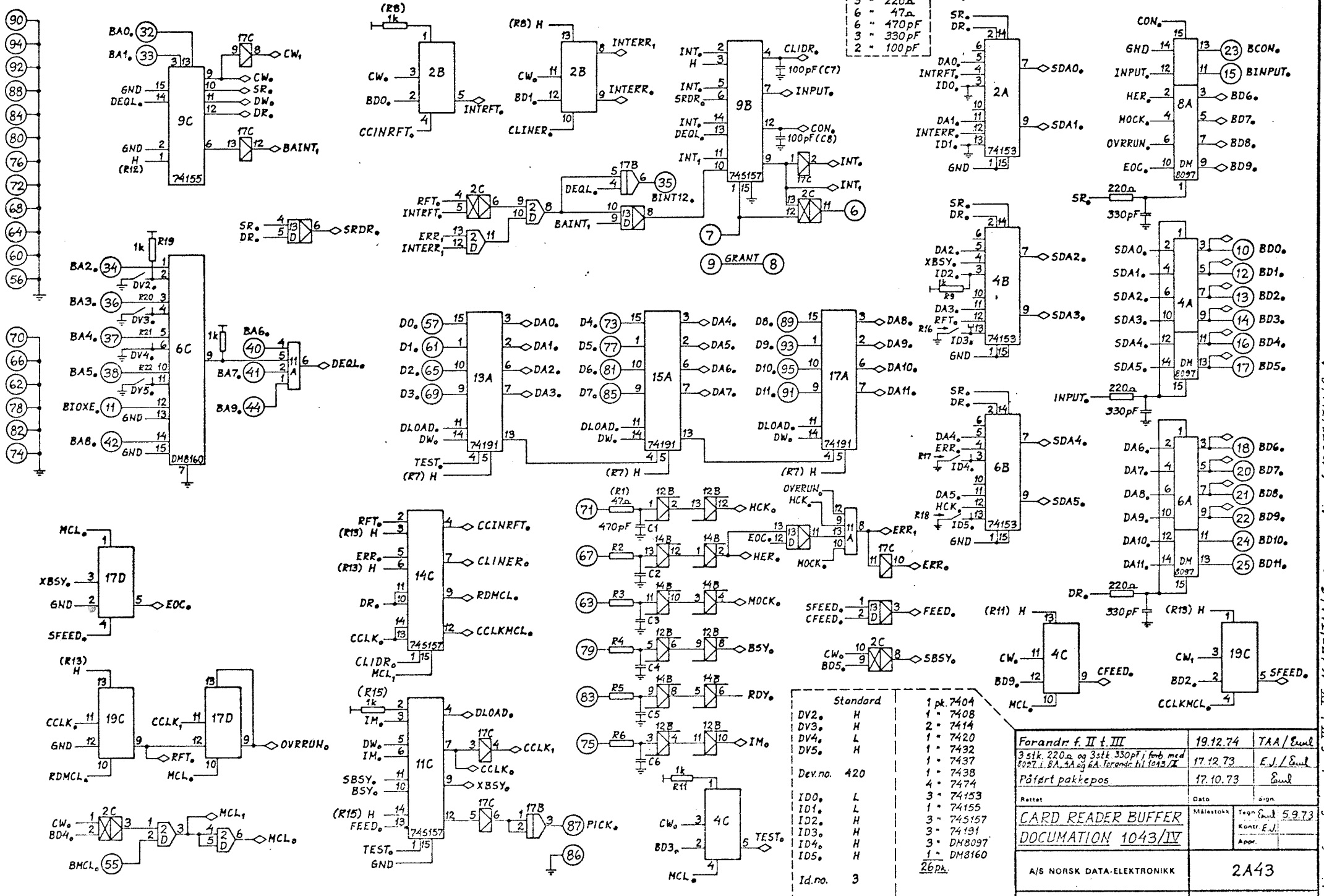
A/S NORSK DATA-ELEKTRONIKK

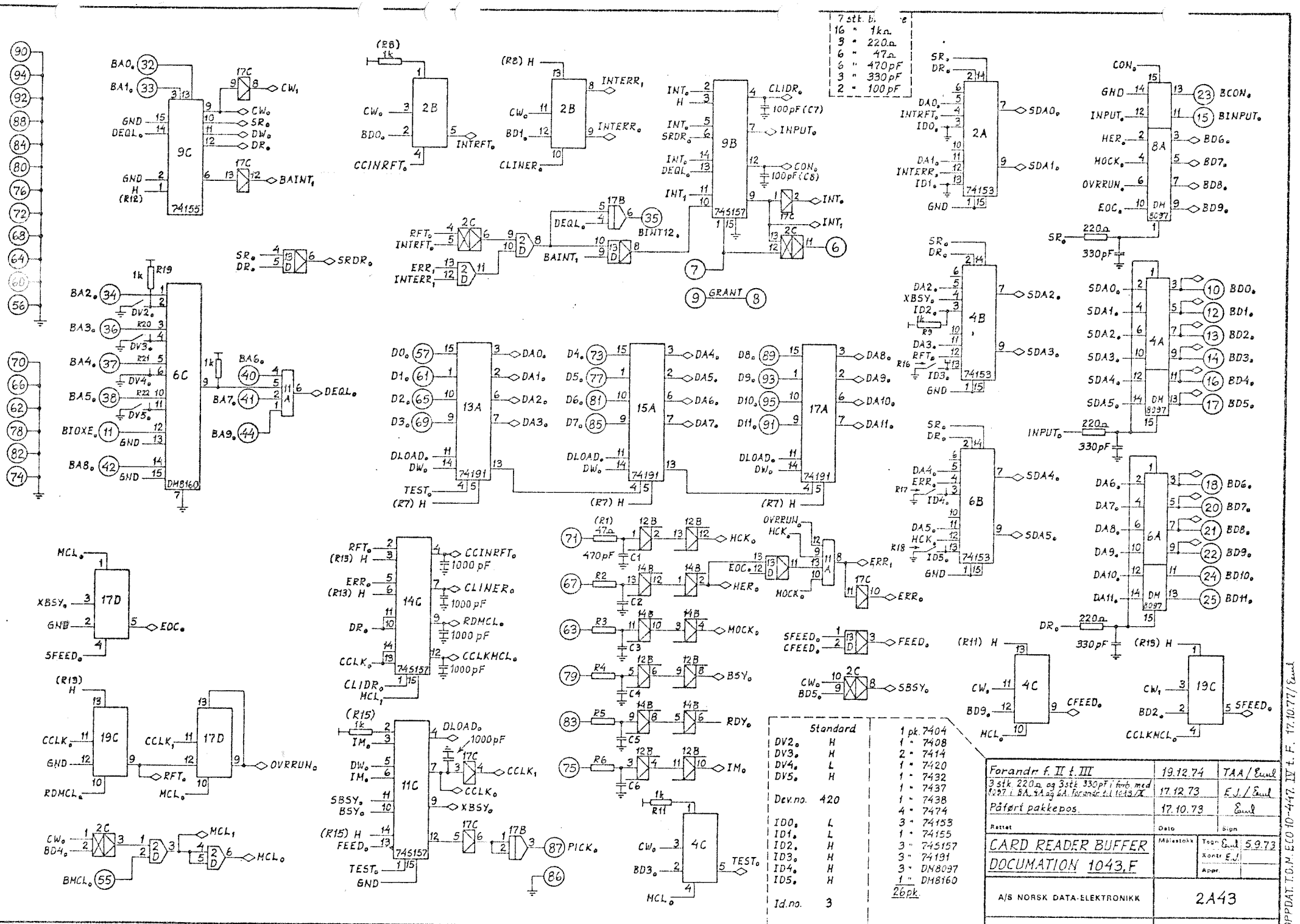
2A41

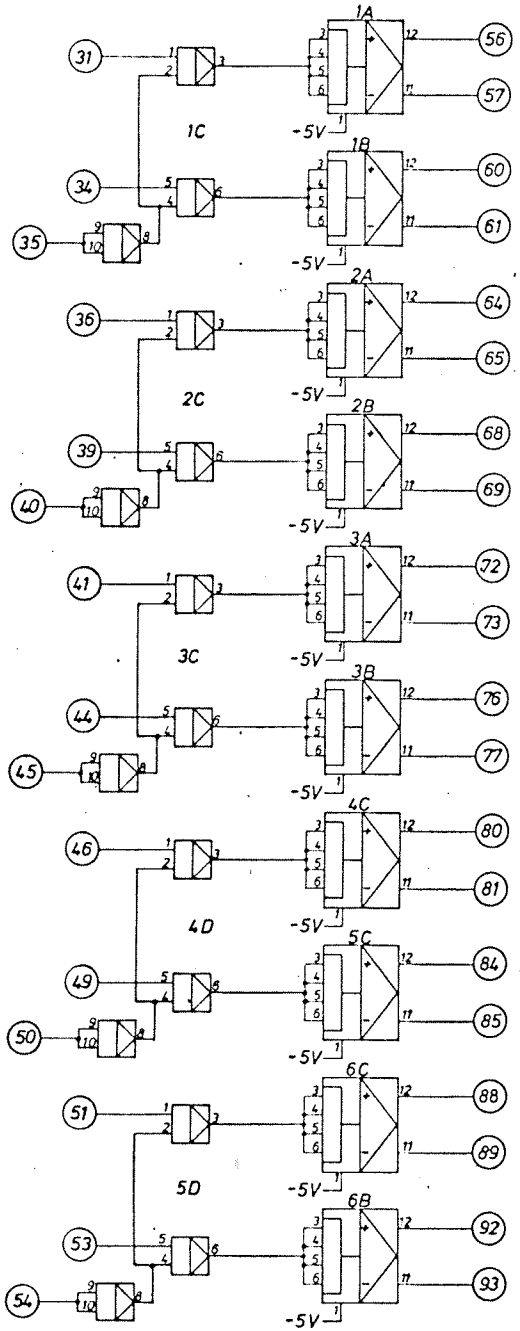
II til III (ingen rettelser) 10.3.75/TAA/Emil

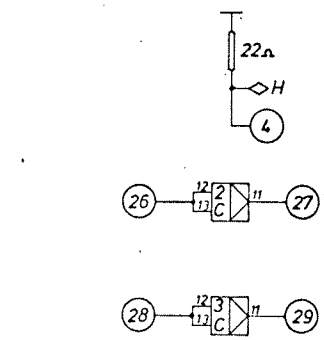
Påført II. 17.1.75/TAA/Emil

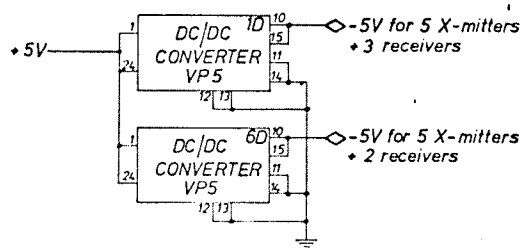

2 stk. 100pF 16 stk. brytere
2 " 470pF 2x13 1kΩ
1pk. 7404 1pk. 74157
1 " 7408 3 " 74174
1 " 7432 3 " 8097
1 " 7438 2 " 8160
3 " 7474 1 " 8820A
1 " 74155 9 " 8831
1 " 74S157 Totalt: 29pk.
6D 74155
8B DM 8160
9C DM 8160
4B 8097
2A 74157
4A 8097
6A 8097
9A 74174
6B 74174
7C 74174
16D DEVICE INT
16D ERROR INT
13D 74S157
1D CWOUT
1D RFT
4D ERROR
4D
11D
12B
18A
16A
14A
18B
16B
14B
14C
12C
12A
5C
19C
Alle drivere: HD til 7 og 9.
1stk. 220Ω og 330pF i forb.m. 4A1 7.1.74 E.J./Emil
Ajourført 23.11.73 E.J./Emil
Påført pakkepos. 25.9.73 Emil
Rettet Dato Sign.
16 BIT DIGITAL OUTPUT REGISTER WITH LINEDRIVER 1042/III
Målestokk
Tegn Emil 14.9.73
Konstr. E.J.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A42
Ajourf. og forandr. t. III t. III. 10.3.75/TAA/Emil

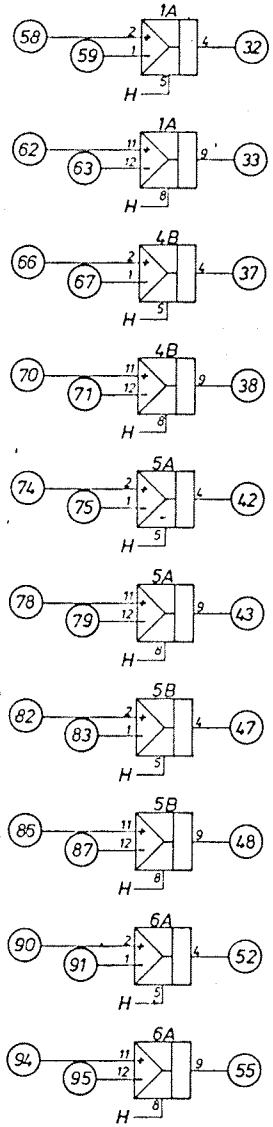

stk. bryte
5 = 1kΩ
3 = 220Ω
6 = 47Ω
6 = 470pF
3 = 330pF
2 = 100pF
9C
74155
6C
DM8160
2B
13A
15A
17A
74191
9B
745157
100pF (C7)
100pF (C8)
2A
8A
4B
4A
6B
6A
74153
DM 8097
14C
11C
17D
19C
4C
Standard
DV2. H
DV3. H
DV4. L
DV5. H
Dev.no. 420
ID0. L
ID1. L
ID2. H
ID3. H
ID4. H
ID5. H
Id.no. 3
1 pk. 7404
1 " 7408
2 " 7414
1 " 7420
1 " 7432
1 " 7437
1 " 7438
4 " 7474
3 " 74153
1 " 74155
3 " 745157
3 " 74191
3 " DM8097
1 " DM8160
26 pk.
Forandr. f. II t. III
19.12.74
TAA/Eml
3 stk. 220Ω og 3 stk. 330pF i forb. med 8097 i 8A, 4A og 6A. Forandr. til 1043/IV
17.12.73
E.J./Eml
Påført pakkepos.
17.10.73
Eml
Rettet
Dato
Sign.
CARD READER BUFFER
DOCUMATION 1043/IV
Målestokk
Tegn. Eml 5.9.73
Kontr. E.J.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A43
Ajourf. 10.3.75/TAA/Eml
Ajourf. og forandr. f. III t. IV. 16.1.75/TAA/Eml


7 stk. b. e
16 " 1kΩ
3 " 220Ω
6 " 47Ω
6 " 470pF
3 " 330pF
2 " 100pF
9C
74155
2B
9B
745157
2A
8A
4B
4A
6B
6A
6C
DM8160
13A
15A
17A
74191
14C
11C
19C
17D
4C
74153
DM8097
BA0
BA1
BA2
BA3
BA4
BA5
BA6
BA7
BA8
BA9
BIOXE
BINT12
CW0
CW1
SR0
DW0
DR0
BAINT1
INTERR1
CLIDR0
INPUT0
CON0
INT0
INT1
GRANT
SRDR0
DEQL0
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
DA8
DA9
DA10
DA11
SDA0
SDA1
SDA2
SDA3
SDA4
SDA5
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
BD8
BD9
BD10
BD11
BCON
BINPUT
HCK0
HER0
MOCK0
BSY0
RDY0
IM0
ERR1
EOC0
SBSY0
FEED0
CFEED0
CCINRFT0
CLINER0
RDMCL0
CCLKMCL0
OVRRUN0
PICK0
TEST0
MCL1
DV2
DV3
DV4
DV5
1k
220Ω
330pF
1000 pF
470pF
Standard
DV2 H
DV3 H
DV4 L
DV5 H
Dev.no. 420
ID0 L
ID1 L
ID2 H
ID3 H
ID4 H
ID5 H
Id.no. 3
1 pk. 7404
1 " 7408
2 " 7414
1 " 7420
1 " 7432
1 " 7437
1 " 7438
4 " 7474
3 " 74153
1 " 74155
3 " 745157
3 " 74191
3 " DM8097
1 " DM8160
26pk
Forandr f. II t. III
19.12.74
TAA/Eml
3 stk. 220Ω og 3 stk. 330pF i forb. med 8097 i 8A, 4A og 6A. Forandr f.t. 1043/F
17.12.73
EJ./Eml
Påført pakkepos.
17.10.73
Eml
Rettet
Dato
Sign
CARD READER BUFFER
DOCUMATION 1043,F
Målestokk
Tegn. Eml 5.9.73
Kontr. EJ.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A43
OPPDAT. T.O.M. ECO 10-447. IV t. F. 17.10.77/Eml



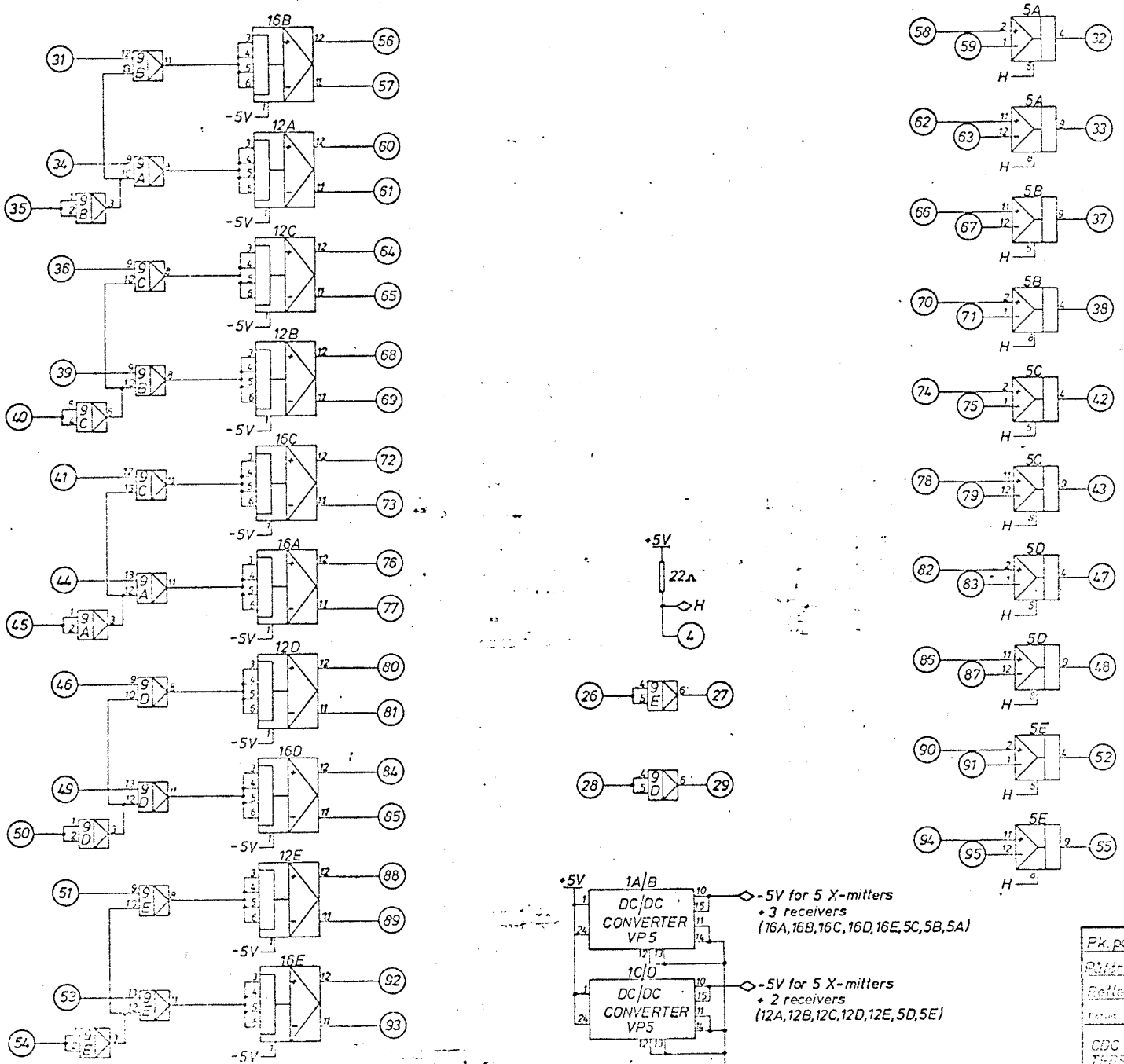






5 pk 7408
5 pk 75107
10 pk MC75113L
20 pk

Påført pkpos og term nr	15.3.74	JB/BBO
Rettet	5.11.73	JB/BBO
Rettet	Dato	Sign
CDC 3000 CHANNEL X-MITTERS AND RECEIVERS FOR NORD 10/45/I	Målestokk	Tegn BBO 21073 Kontr JB Appr
A/S NORSK DATA-ELEKTRONIKK	2A45/I	



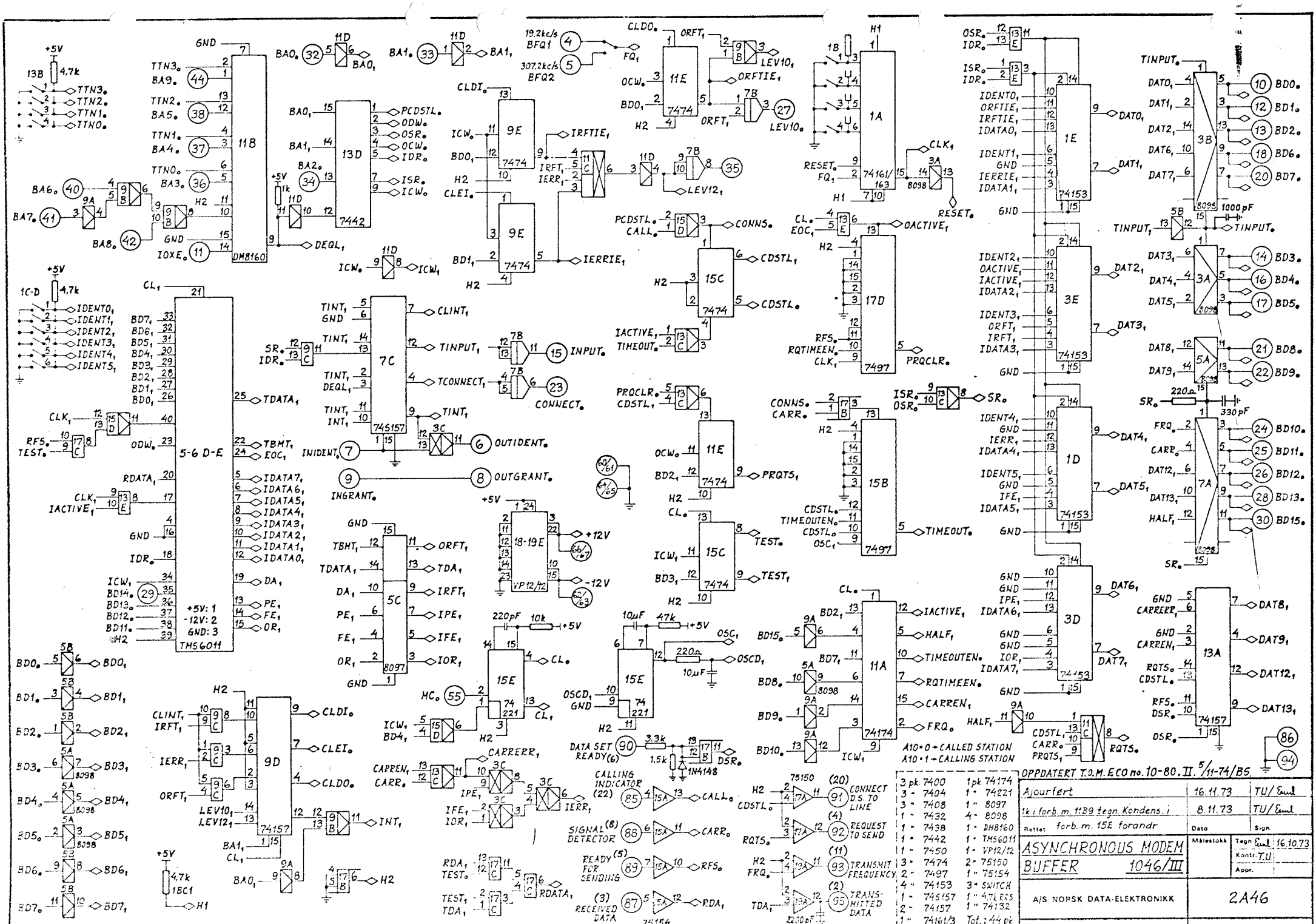
5 pk 7408
5 pk 75107
10 pk MC75113L
20 pk
1 stk. 22Ω
2 stk VP5

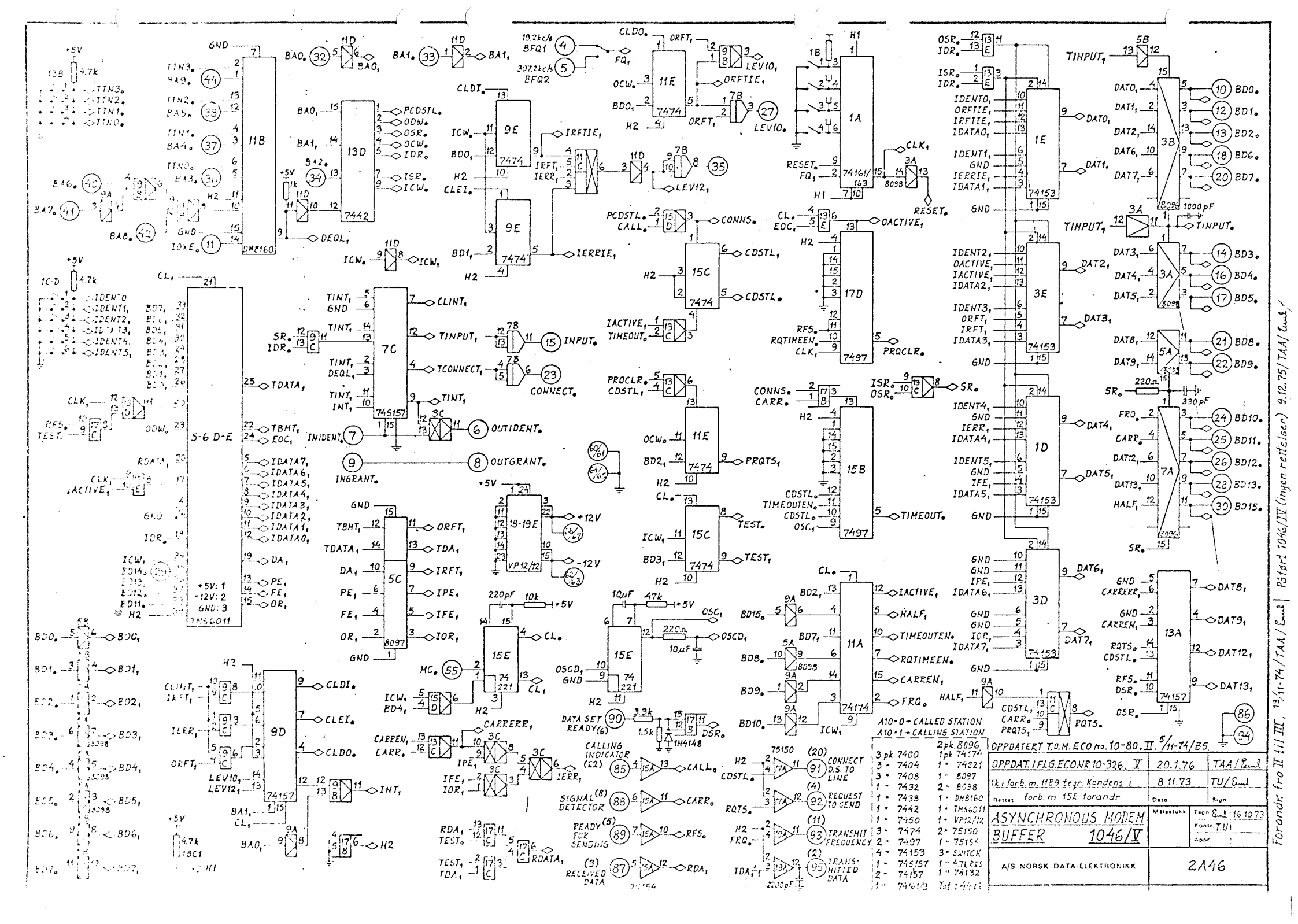
Pk.pos. og term.nr. endrett	15.11.74	TAA/BBO
Rettet pk.pos og term.nr	15.3.74	JB/BBO
Rettet	5.11.73	JB/BBO
Endret	Dato	

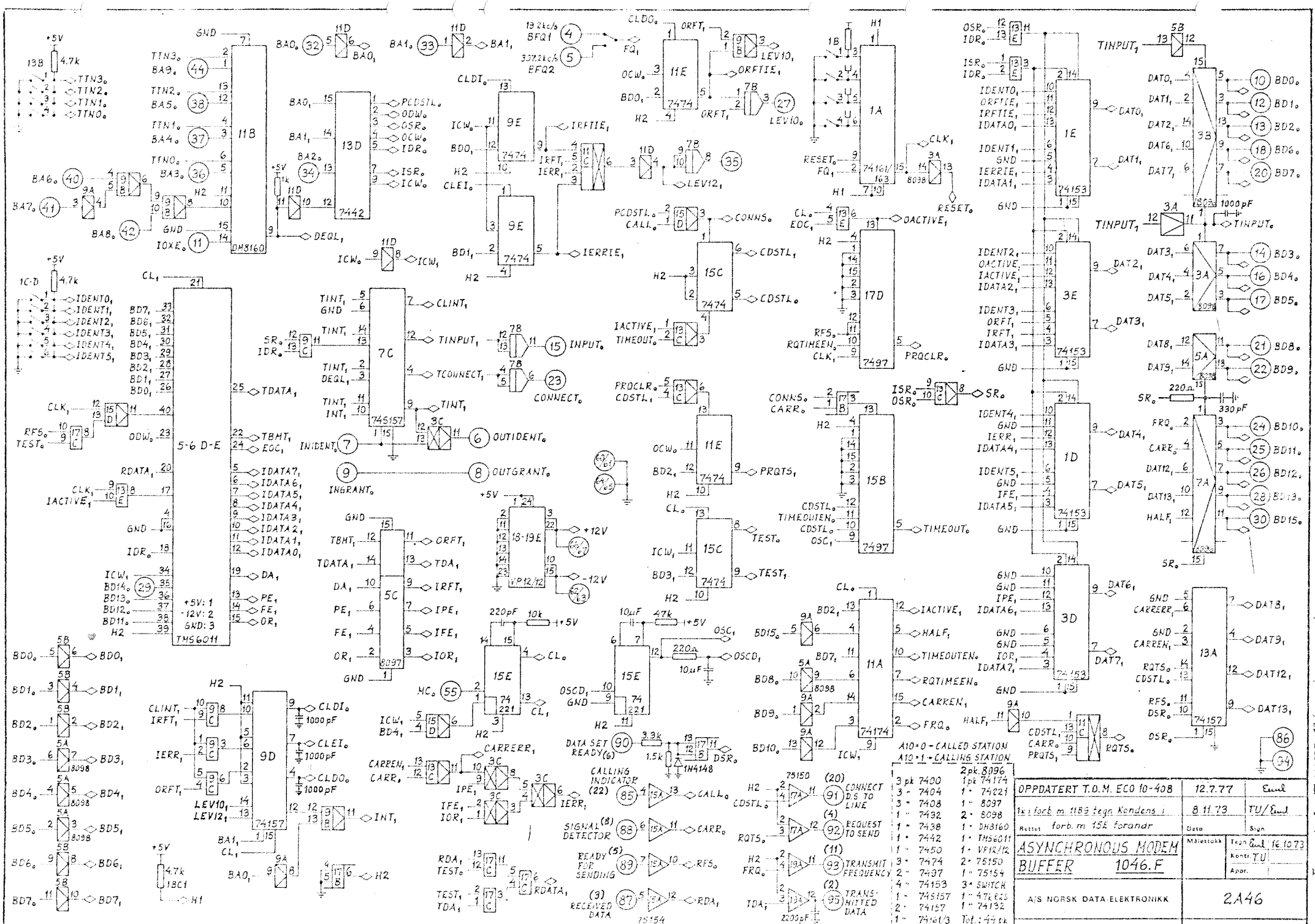
CDC 3300 CHANNEL X-MITTERS AND RECEIVERS FOR [illegible]

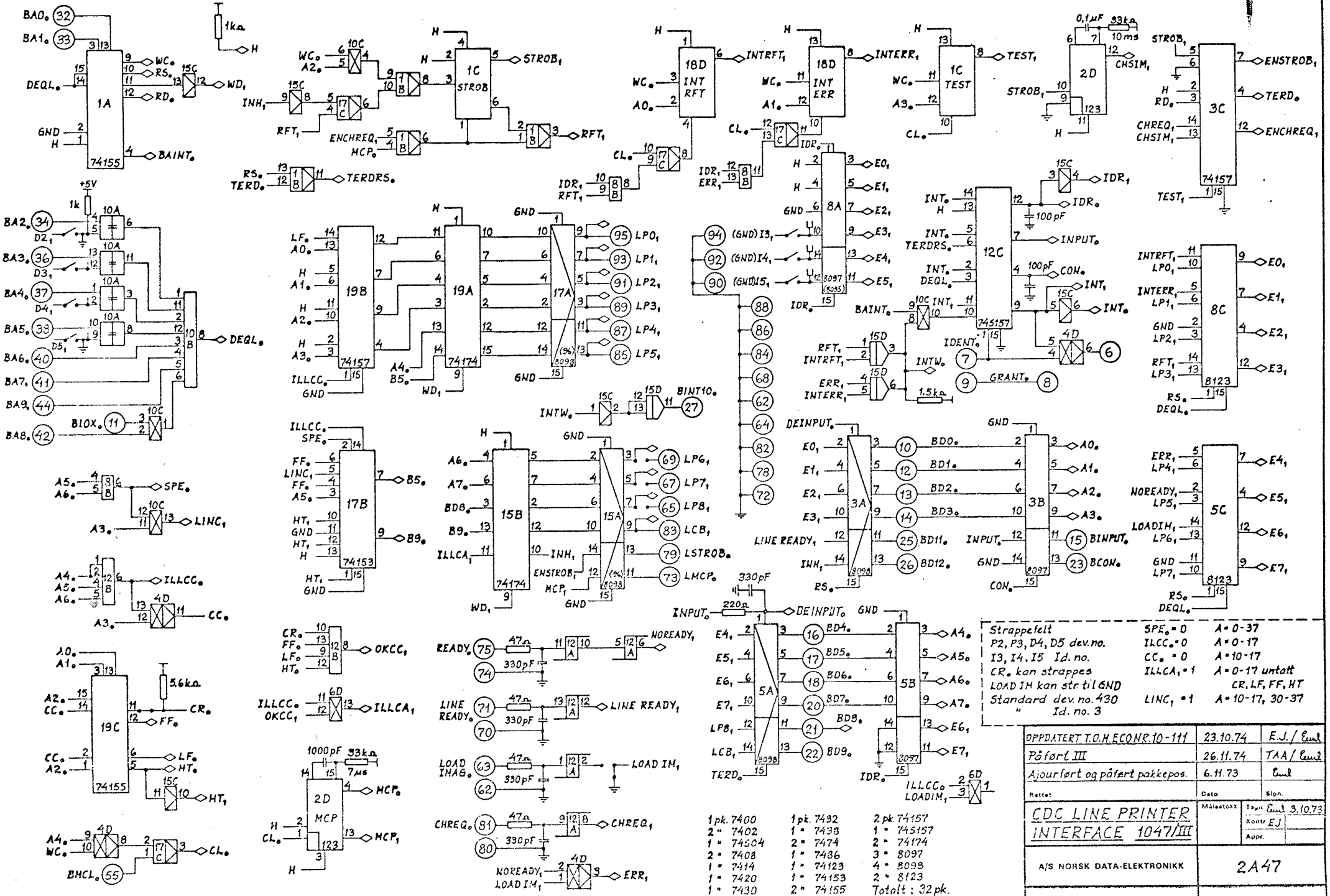
A/S NORSK DATA-ELEKTRONIKK

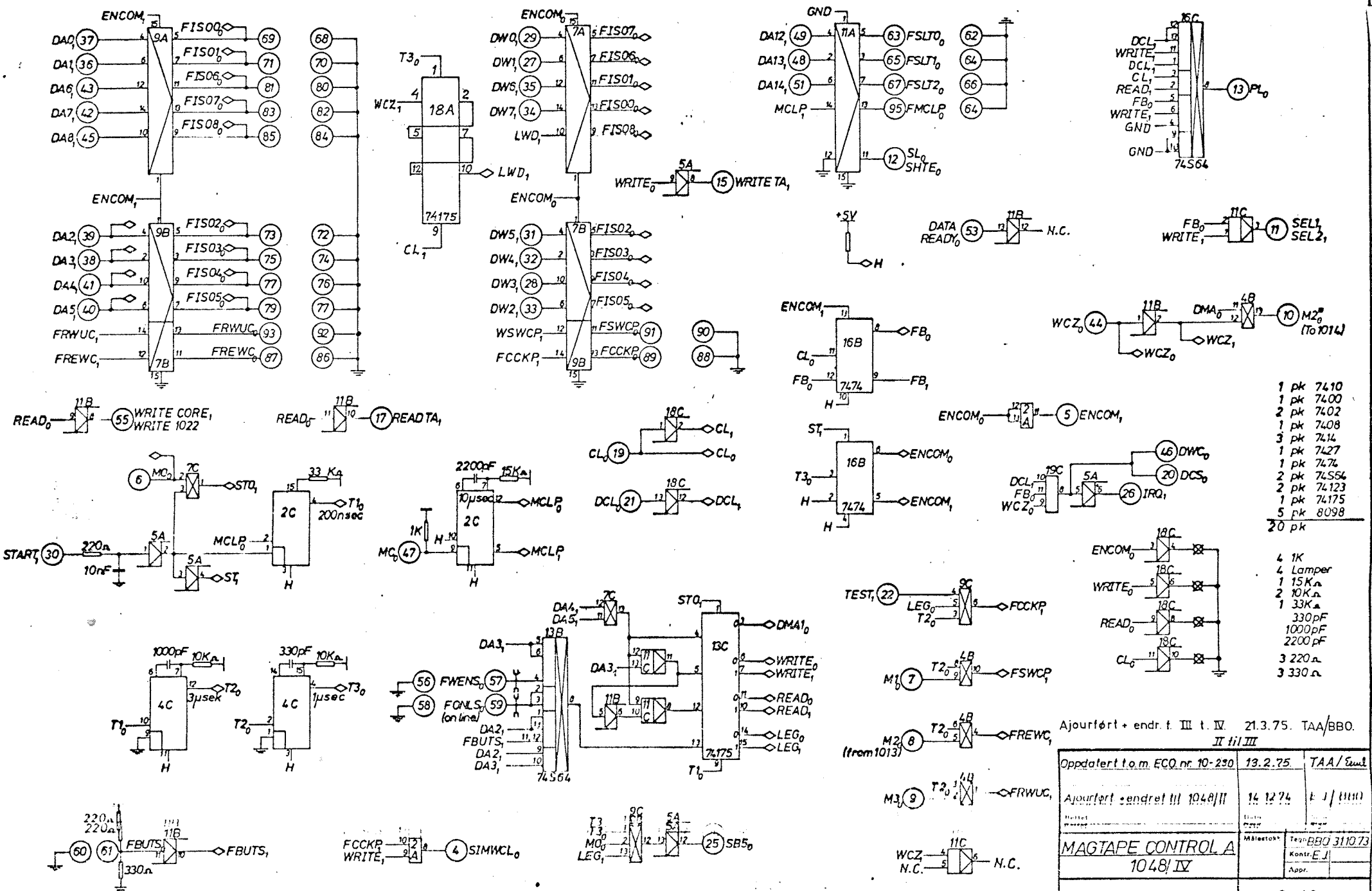
2A45


11B
13D
7442
9E
7474
11E
1A
74161/163
1E
74153
3B
8098
3A
3E
1D
5A
7A
17D
7497
15B
15C
7C
745157
5-6 D-E
TMS6011
5C
8097
18-19E
VP12/12
15E
74221
9D
74157
11A
74174
3D
13A
DM8160
1000 pF
220 n
330 pF
10 µF
47 k
3.3k
1.5k
220 pF
10 k
2200 pF
1N4148
+5V
+12V
-12V
GND
4.7k
TTN3
TTN2
TTN1
TTN0
BA0
BA1
BA2
BA3
BA4
BA5
BA6
BA7
BA8
BA9
IOXE
DEQL
ICW
CLDI
CLEI
CLDO
IRFTIE
IERRIE
IERR
IRFT
ORFT
ORFTIE
LEV10
LEV12
CONN5
CDSTL
CALL
OACTIVE
IACTIVE
TIMEOUT
RESET
CLK
PRQCLR
PRQTS
TEST
TIMEOUTEN
CARREN
CARRERR
RQTIMEEN
HALF
IACTIVE
TIMEOUT
FRQ
RQTS
DSR
SR
ISR
OSR
IDR
IDENT0
IDENT1
IDENT2
IDENT3
IDENT4
IDENT5
TDATA
TBMT
EOC
IDATA0
IDATA1
IDATA2
IDATA3
IDATA4
IDATA5
IDATA6
IDATA7
DA
PE
FE
OR
RDATA
ODW
RFS
TEST
IACTIVE
CLINT
TINPUT
TCONNECT
INPUT
CONNECT
TINT
INT
INIDENT
OUTIDENT
INGRANT
OUTGRANT
TBMT
TDATA
DA
IRFT
IPE
IFE
IOR
ORFT
MC
OSCD
OSC
CARREN
CARRERR
IERR
IRFT
ORFT
LEV10
LEV12
CLEI
CLDI
CLDO
INT
CALL
CARR
RFS
RDA
TDA
RQTS
DSR
FRQ
HALF
TIMEOUTEN
RQTIMEEN
CARREN
IACTIVE
TIMEOUT
PRQTS
CONNECT D.S. TO LINE
REQUEST TO SEND
TRANSMIT FREQUENCY
TRANSMITTED DATA
CALLING INDICATOR
SIGNAL DETECTOR
READY FOR SENDING
RECEIVED DATA
DATA SET READY
A10=0 → CALLED STATION
A10=1 → CALLING STATION
ASYNCHRONOUS MODEM
BUFFER 1046/III
A/S NORSK DATA-ELEKTRONIK
2A46
OPPDATERT T.D.M. ECO no. 10-80. II. 5/11-74/BS
Ajourført 16.11.73 TU/Emil
1k i forb. m. 11B9 tegn. Kondens. i 8.11.73 TU/Emil
Rettet forb. m. 15E forandr
Tegn Emil 16.10.73
Kontr. T.U
Appr.
3 pk. 7400
3 " 7404
3 " 7408
1 " 7432
1 " 7438
1 " 7442
1 " 7450
3 " 7474
2 " 7497
4 " 74153
1 " 74157
1 " 74161/3
1 pk 74174
1 " 74221
1 " 8097
4 " 8098
1 " DM8160
1 " TMS6011
1 " VP12/12
2 " 75150
1 " 75154
3 " SWITCH
1 " 4,7k RES
1 " 74132
Tot.: 44 pk
Forandr. fra II til III, 13/11-74/TAA/Emil


11B
DM8160
13D
7442
9E
7474
11E
7474
1A
74161/163
1E
74153
3B
1000pF
3A
3E
17D
7497
15C
15B
7497
1D
5A
7A
5-6 D-E
TMS6011
7C
74157
5C
8097
18-19E
VP12/12
15E
74 221
9D
74157
11A
74174
13A
74157
CLDO,
ORFT,
OCW,
BDO,
H2
CLDI,
ICW,
BD0,
IRFTIE,
IRFT,
IERR,
IERRIE,
CLEI,
BD1,
PCDSTL,
CALL,
CONN5,
CDSTL,
IACTIVE,
TIMEOUT,
RFS,
RQTIMEEN,
CLK,
PRQCLR,
OACTIVE,
RESET,
TINPUT,
INPUT,
TCONNECT,
CONNECT,
TINT,
CLINT,
INIDENT,
OUTIDENT,
OUTGRANT,
INGRANT,
TDATA,
TBMT,
EOC,
IDATA7,
IDATA6,
IDATA5,
IDATA4,
IDATA3,
IDATA2,
IDATA1,
IDATA0,
DA,
PE,
FE,
OR,
IDR,
ODW,
RDATA,
IACTIVE,
TEST,
+12V,
-12V,
PRQTS,
CONN5,
CARR,
ISR,
OSR,
SR,
DAT0,
DAT1,
DAT2,
DAT3,
DAT4,
DAT5,
DAT6,
DAT7,
DAT8,
DAT9,
DAT10,
DAT11,
DAT12,
DAT13,
FRQ,
HALF,
TIMEOUTEN,
RQTIMEEN,
CARREN,
CARREER,
IPE,
IFE,
IOR,
HC,
OSCD,
OSC,
DSR,
RQTS,
CALL,
RFS,
RDA,
CALLING INDICATOR (22)
SIGNAL DETECTOR (8)
READY FOR SENDING (5)
RECEIVED DATA (3)
DATA SET READY (6)
CONNECT D.S. TO LINE (20)
REQUEST TO SEND (4)
TRANSMIT FREQUENCY (11)
TRANSMITTED DATA (2)
A10=0 → CALLED STATION
A10=1 → CALLING STATION
2pk 8096
1pk 74174
3 pk 7400
3 - 7404
3 - 7408
1 - 7432
1 - 7438
1 - 7442
1 - 7450
1 - 7497
4 - 74153
1 - 745157
2 - 74157
1 - 74161/3
2pk 8098
1 - 74221
1 - 8097
2 - 8098
1 - DM8160
1 - TMS6011
1 - VP12/12
2 - 75150
1 - 7515*
3 - SWITCH
1 - 47 LEDS
1 - 74132
OPPDATERT T.O.M. ECO no. 10-80. II. 5/11-74/BS
OPPDAT. I FLG. ECO NR 10-326. V
20.1.76
TAA/Emil
1k i forb. m. 1189 fegn Kondens i
8.11.73
TU/Emil
rettet forb m 15E forandr
Dato
Sign
ASYNCHRONOUS MODEM BUFFER 1046/V
Målestokk
Tegn Emil 16.10.73
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A46
Forandr. fra II til III, 13/11-74/TAA/Emil | Rettet 1046/III (ingen rettelser) 9.12.75/TAA/Emil

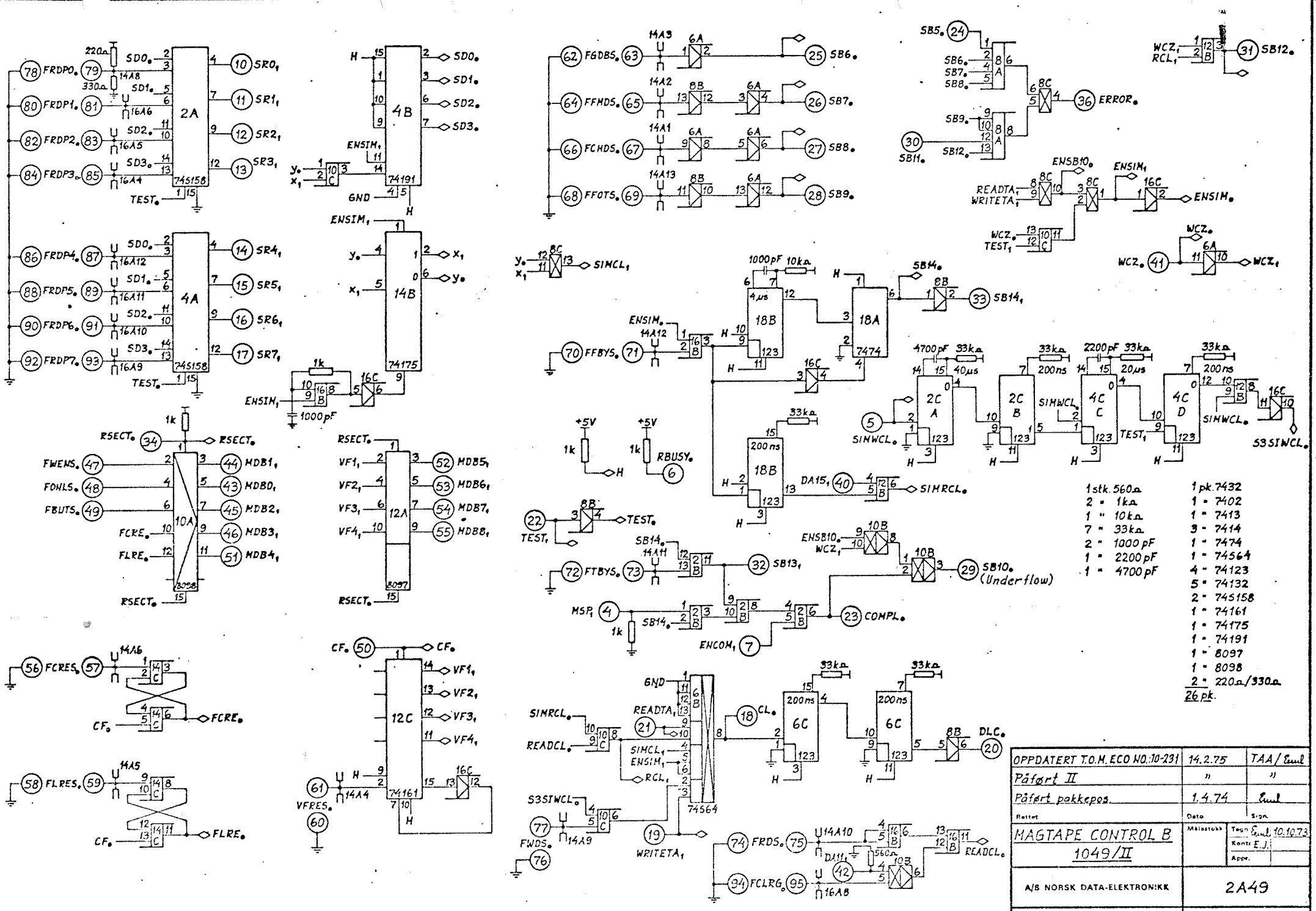

ASYNCHRONOUS MODEM
BUFFER 1046.F
A/S NORSK DATA ELEKTRONIKK
2A46
OPPDATERT T.D.M. ECO 10-408
12.7.77
8.11.73
TU/Emil
Rettet forb. m 15E foranar
Tegn Emil 16.10.73
Kontr. TU
Appr.
Forandr fra II til III, 13/11-74/TAA/Emil | Påført 1046/III (ingen rettelser) 9.12.75/TAA/Emil
A10-0 = CALLED STATION
A10-1 = CALLING STATION
3 pk 7400
3 " 7404
3 " 7408
1 " 7432
1 " 7438
1 " 7442
1 " 7450
3 " 7474
2 " 7437
4 " 74153
1 " 74157
2 " 74161/3
2 pk 8098
1 pk 74174
1 " 74221
2 " 8098
1 " DM8160
1 " TMS6011
1 " VFC12/12
2 " 75150
1 " 75154
3 " SWITCH
1 " 47k E23
1 " 74132
Tot.: 44 pk

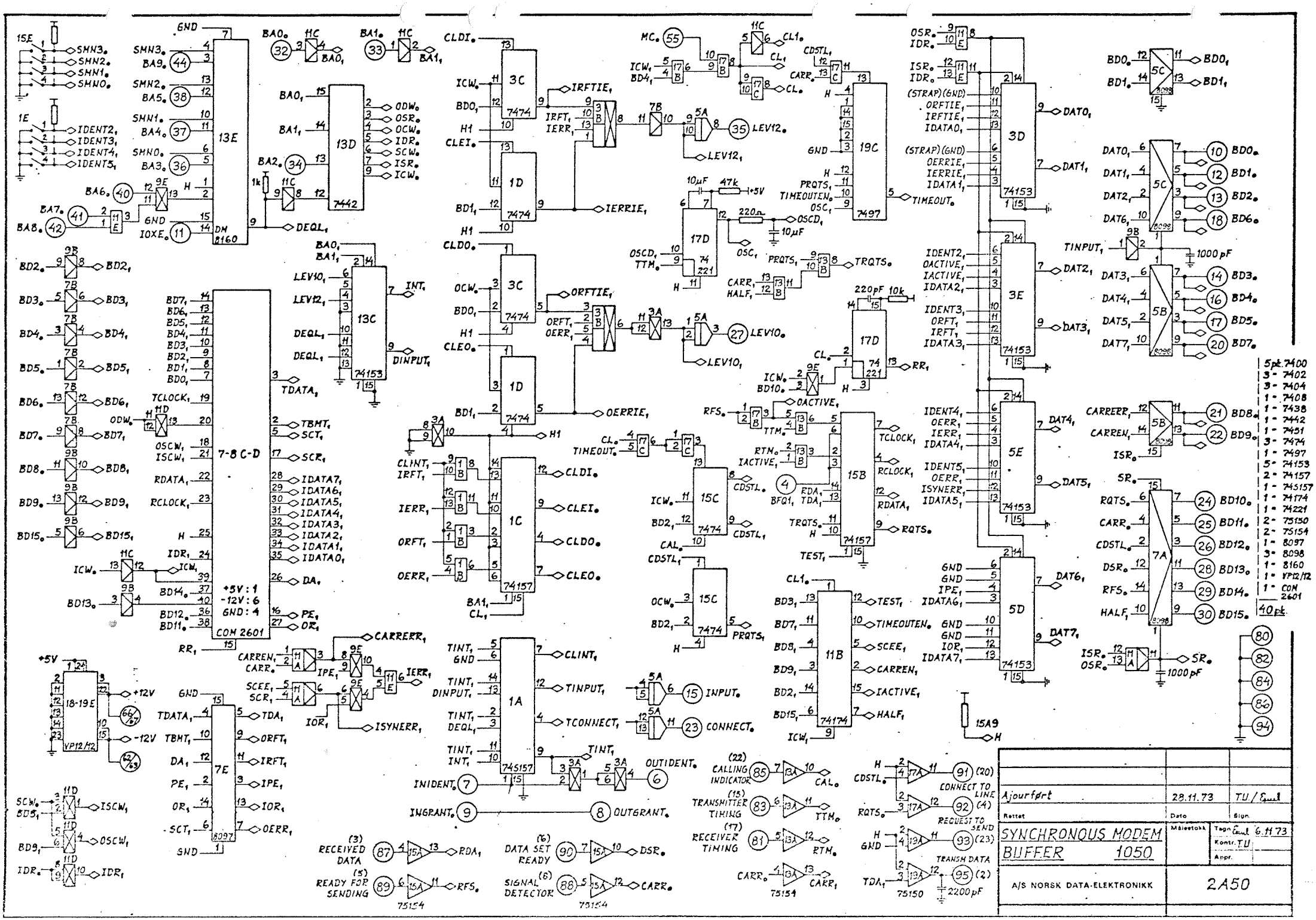

1A
74155
1C STROB
1C TEST
18D INT RFT
18D INT ERR
2D
3C
74157
19B
19A
74157
74174
17A
8098
8A
8097
12C
745157
8C
8123
17B
74153
15B
74174
15A
8098
3A
8098
3B
8097
5C
8123
19C
74155
2D MCP
5A
8098
5B
8097
1k
1kΩ
5.6kΩ
220Ω
1.5kΩ
47n
330pF
1000pF
33kΩ
7μs
0,1μF
10ms
100pF
+5V
H
GND
BA0
BA1
BA2
BA3
BA4
BA5
BA6
BA7
BA8
BA9
BIOX
DEQL
WC
RS
RD
WD
BAINT
INH
RFT
ENCHREQ
MCP
TERDRS
TERD
INTRFT
INTERR
CL
IDR
TEST
CHSIM
STROB
ENSTROB
TERD
ENCHREQ
CHREQ
CHSIM
TEST
LF
AO
A1
A2
A3
A4
A5
A6
A7
BD8
B9
ILLCC
SPE
FF
LINC
HT
B5
B9
LP0
LP1
LP2
LP3
LP4
LP5
LP6
LP7
INTW
BINT10
(GND)I3
(GND)I4
(GND)I5
E0
E1
E2
E3
E4
E5
INT
TERDRS
DEQL
CON
INPUT
IDENT
GRANT
RFT
INTRFT
ERR
INTERR
LINE READY
INH
RS
DEINPUT
BD0
BD1
BD2
BD3
BD11
BD12
BINPUT
BCON
ERR
NOREADY
LOADIM
LP4
LP5
LP6
LP7
E4
E5
E6
E7
CR
FF
LF
HT
OKCC
ILLCA
CC
READY
LINE READY
LOAD IMAG
LOAD IM
CHREQ
NOREADY
ERR
BD4
BD5
BD6
BD7
BD8
BD9
TERD
BNCL
CL
WC
RS
RD
WD
BAINT
INH
RFT
ENCHREQ
MCP
TERDRS
INTRFT
INTERR
CL
IDR
TEST
CHSIM
STROB
ENSTROB
TERD
ENCHREQ
CHREQ
CHSIM
TEST

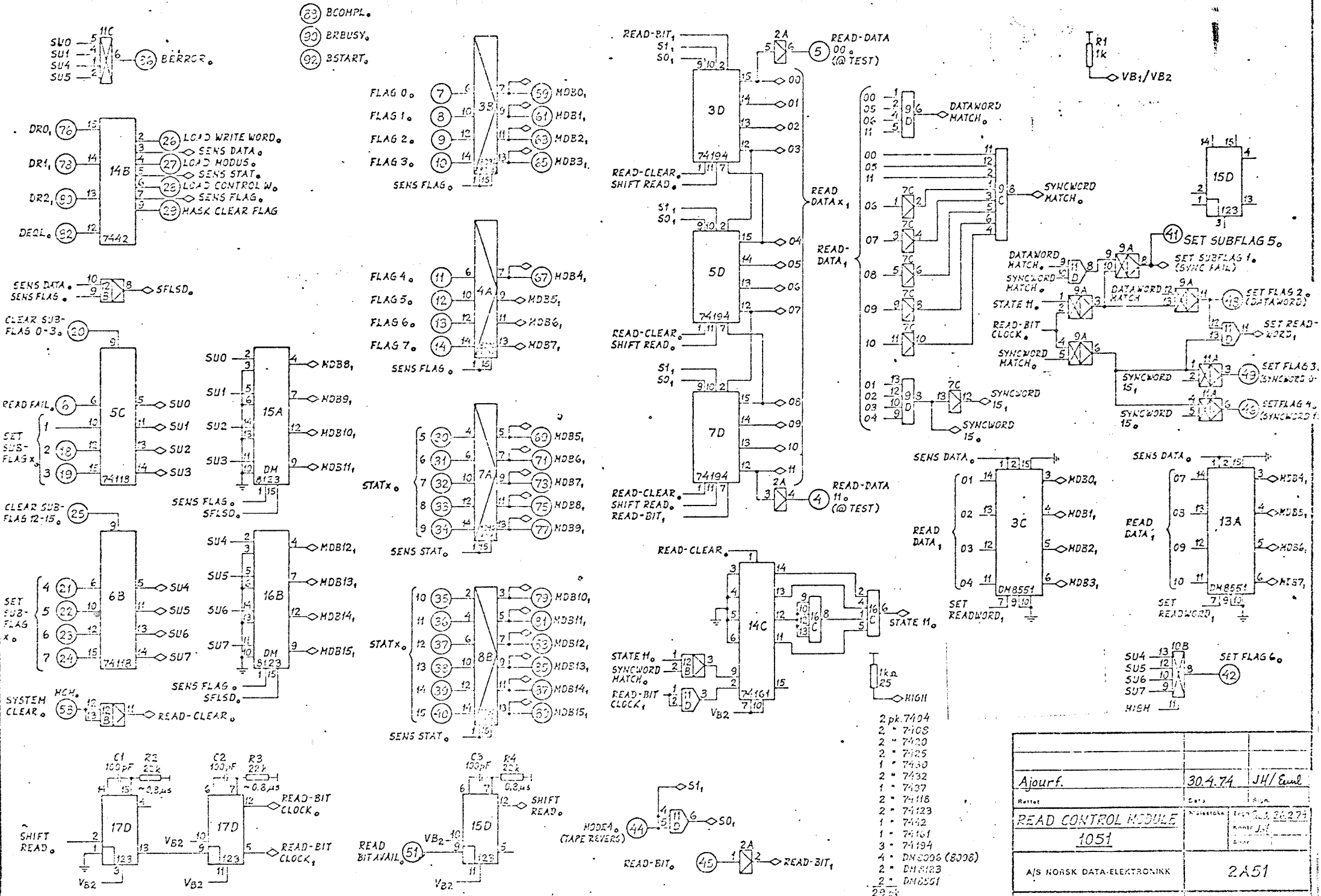


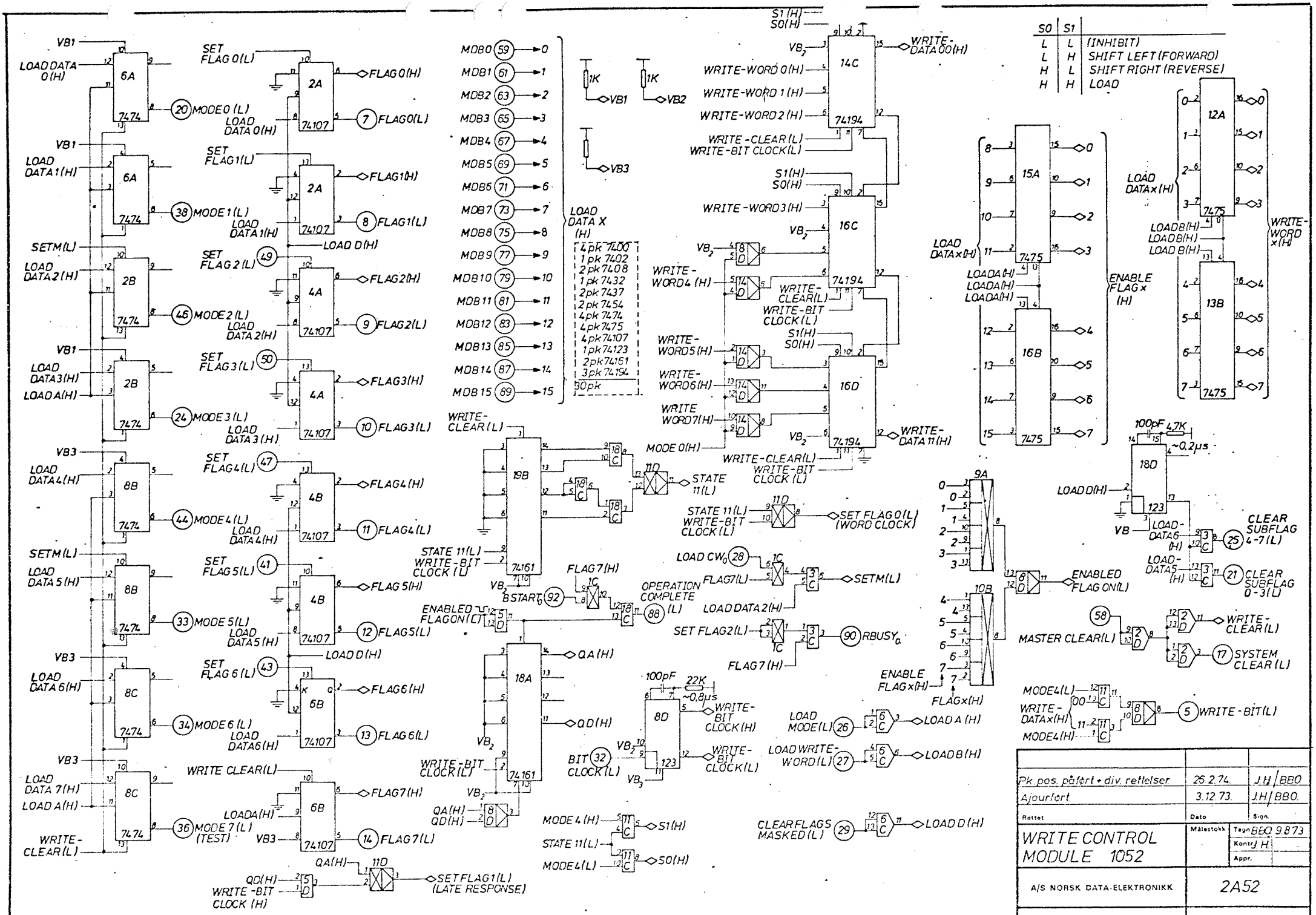
Ajourført + endr. f. III t. IV. 21.3.75. TAA/BBO.
II til III

Oppdatert t.o.m. ECO nr. 10-230	13.2.75	TAA/Sund
Ajourført + endret til 1048/II	14.12.74	E.J./BBO
MAGTAPE CONTROL A 1048/IV	Målestokk	Tegn. BBO 31.10.73 Kontr. E.J. Appr.
A/S NORSK DATA-ELEKTRONIKK	2A48	


2A
4A
4B
14B
10A
12A
12C
18A
18B
2C A
2C B
4C C
4C D
6C
74564
74161
74175
74191
74123
74132
74564
74161
74175
74191
8097
8098
SD0
SD1
SD2
SD3
SR0
SR1
SR2
SR3
SR4
SR5
SR6
SR7
TEST
RSECT
MDB1
MDB0
MDB2
MDB3
MDB4
MDB5
MDB6
MDB7
MDB8
VF1
VF2
VF3
VF4
CF
FCRE
FLRE
FRDP0
FRDP1
FRDP2
FRDP3
FRDP4
FRDP5
FRDP6
FRDP7
FWENS
FOWLS
FBUTS
FCRES
FLRES
FGDBS
FFMDS
FCHDS
FFOTS
FFBYS
FTBYS
FRDS
FCLRG
SB6
SB7
SB8
SB9
SB11
SB12
SB13
SB14
ERROR
SIMCL
SIMRCL
SIMWCL
S3SIWCL
RBUSY
COMPL
ENCOM
MSP
DA15
DA11
CL
DLC
READCL
READTA
WRITETA
FWDS
ENSIM
ENSB10
WCZ
RCL
GND
H
TEST
220Ω
330Ω
1k
1000pF
10kΩ
33kΩ
4700pF
2200pF
4μs
40μs
20μs
200ns
560Ω
100
14A1
14A2
14A3
14A4
14A5
14A6
14A7
14A8
14A9
14A10
14A11
14A12
14A13
14A14
16A1
16A2
16A3
16A4
16A5
16A6
16A7
16A8
16A9
16A10
16A11
16A12
1 stk. 560Ω
2 " 1kΩ
1 " 10kΩ
7 " 33kΩ
2 " 1000 pF
1 " 2200 pF
1 " 4700 pF
1 pk. 7432
1 " 7402
1 " 7413
3 " 7414
1 " 7474
1 " 74564
4 " 74123
5 " 74132
2 " 745158
1 " 74161
1 " 74175
1 " 74191
1 " 8097
1 " 8098
2 " 220Ω/330Ω
26 pk.
OPPDATERT T.O.M. ECO NO. 10-231
14.2.75
TAA/Eml
Påført II
"
"
Påført pakkepos.
1.4.74
Eml
Rettet
Dato
Sign.
MAGTAPE CONTROL B
1049/II
Målestokk
Tegn Eml 10.10.73
Kontr. E.J.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A49

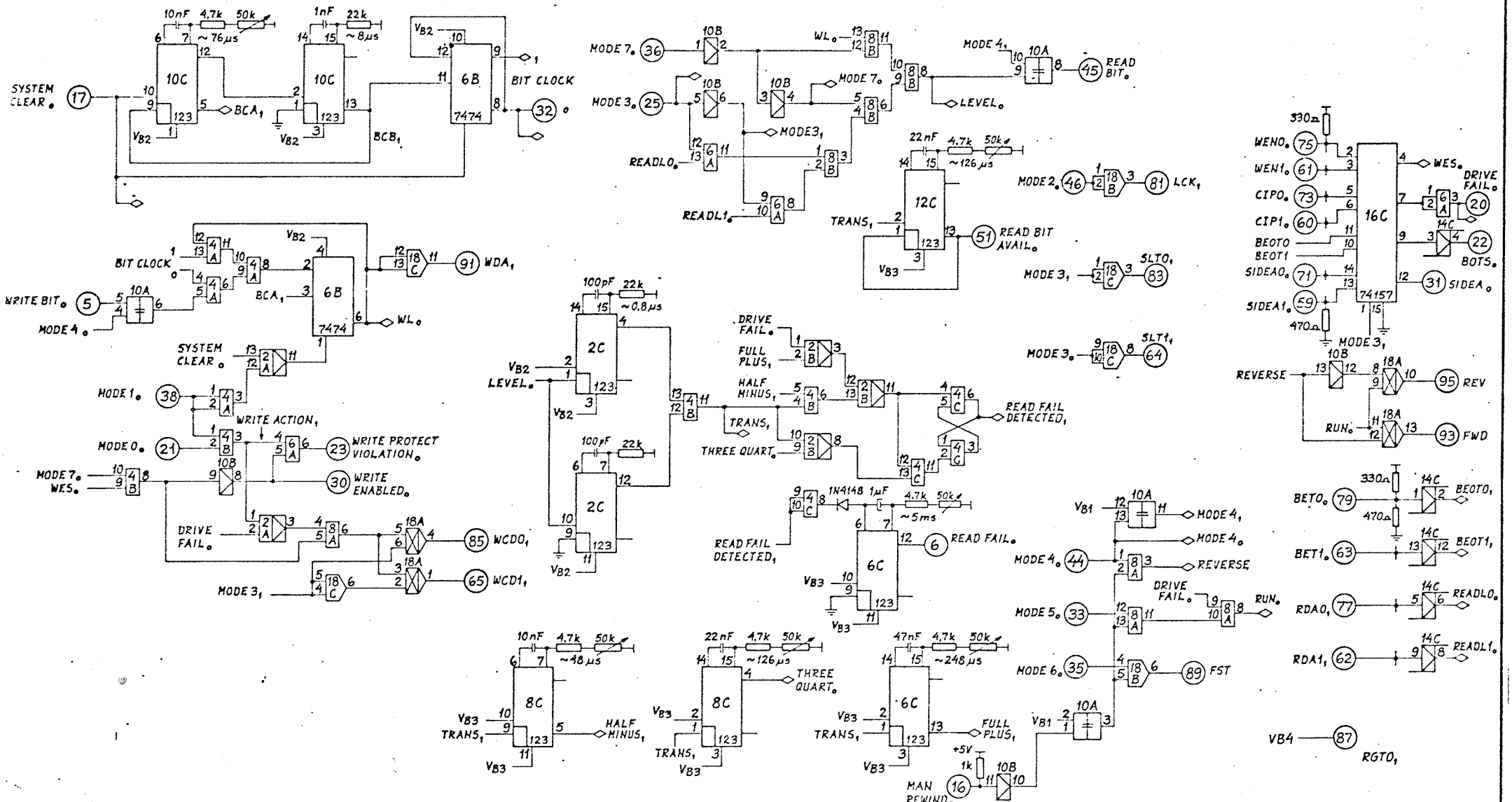

SYNCHRONOUS MODEM
BUFFER 1050
A/S NORSK DATA-ELEKTRONIKK
2A50
Ajourført
28.11.73
TU / Emil
Rettet
Dato
Sign.
Målestokk
Tegn. Emil 6.11.73
Kontr. TU
Appr.
5pk. 7400
3 - 7402
3 - 7404
1 - 7408
1 - 7438
1 - 7442
1 - 7451
3 - 7474
1 - 7497
5 - 74153
2 - 74157
1 - 745157
1 - 74174
1 - 74221
2 - 75150
2 - 75154
1 - 8097
3 - 8098
1 - 8160
1 - VP12/12
1 - CON 2601
40 pk.
13E
1E
15E
1D
1C
3C
7B
9B
11C
11D
9E
5A
3A
17D
19C
3D
3E
5E
5D
5B
5C
7A
15B
15C
11B
1A
7E
13C
13D
17B
18-19E
VP12/12
7-8 C-D
COM 2601
DM 8160
8097
8098
7442
7497
74157
74153
74221
74174
745157
75154
75150
SMN3
SMN2
SMN1
SMN0
IDENT2
IDENT3
IDENT4
IDENT5
BA0
BA1
BA2
BA6
BA7
BA8
BA9
BA5
BA4
BA3
GND
IOXE
DEQL
ODW
OSR
OCW
IDR
SCW
ISR
ICW
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
BD8
BD9
BD15
BD14
BD13
BD12
BD11
CLDI
CLEI
CLDO
CLEO
IRFTIE
IRFT
IERR
IERRIE
ORFTIE
ORFT
OERR
OERRIE
H1
LEV12
LEV10
INT
DINPUT
TDATA
TBMT
SCT
SCR
DA
PE
OR
RR
CARREER
CARREN
IPE
SCEE
SCR
IOR
ISYNERR
CLINT
IRFT
IERR
ORFT
OERR
TINT
DINPUT
DEQL
INT
TINPUT
TCONNECT
INIDENT
INGRANT
OUTIDENT
OUTGRANT
MC
CL
CARR
CDSTL
TIMEOUT
OSC
TRQTS
PRQTS
HALF
TTM
RFS
RTN
IACTIVE
EDA
TDA
BFQ1
TEST
CAL
INPUT
CONNECT
RDA
RTM
RQTS
TCLOCK
RCLOCK
RDATA
CDSTL
TIMEOUTEN
SCEE
CARREN
IACTIVE
HALF
ICW
BD4
BD10
+5V : 1
-12V : 6
GND : 4
+12V
-12V
10µF
47k
220Ω
1000 pF
220pF
10k
15A9
(22) CALLING INDICATOR
(15) TRANSMITTER TIMING
(17) RECEIVER TIMING
(3) RECEIVED DATA
(5) READY FOR SENDING
(6) DATA SET READY
(6) SIGNAL DETECTOR
CONNECT TO LINE (20)
REQUEST TO SEND (4)
TRANSM DATA (23)
(2)
2200pF
1000pF
DAT0
DAT1
DAT2
DAT3
DAT4
DAT5
DAT6
DAT7
TINPUT
(STRAP) (GND)
ORFTIE
IRFTIE
IDATA0
OERRIE
IERRIE
IDATA1
IDENT2
OACTIVE
IACTIVE
IDATA2
IDENT3
ORFT
IRFT
IDATA3
IDENT4
OERR
IERR
IDATA4
IDENT5
ISYNERR
IDATA5
IPE
IDATA6
IDATA7
CARREER
CARREN
ISR
SR
HALF
1000 pF

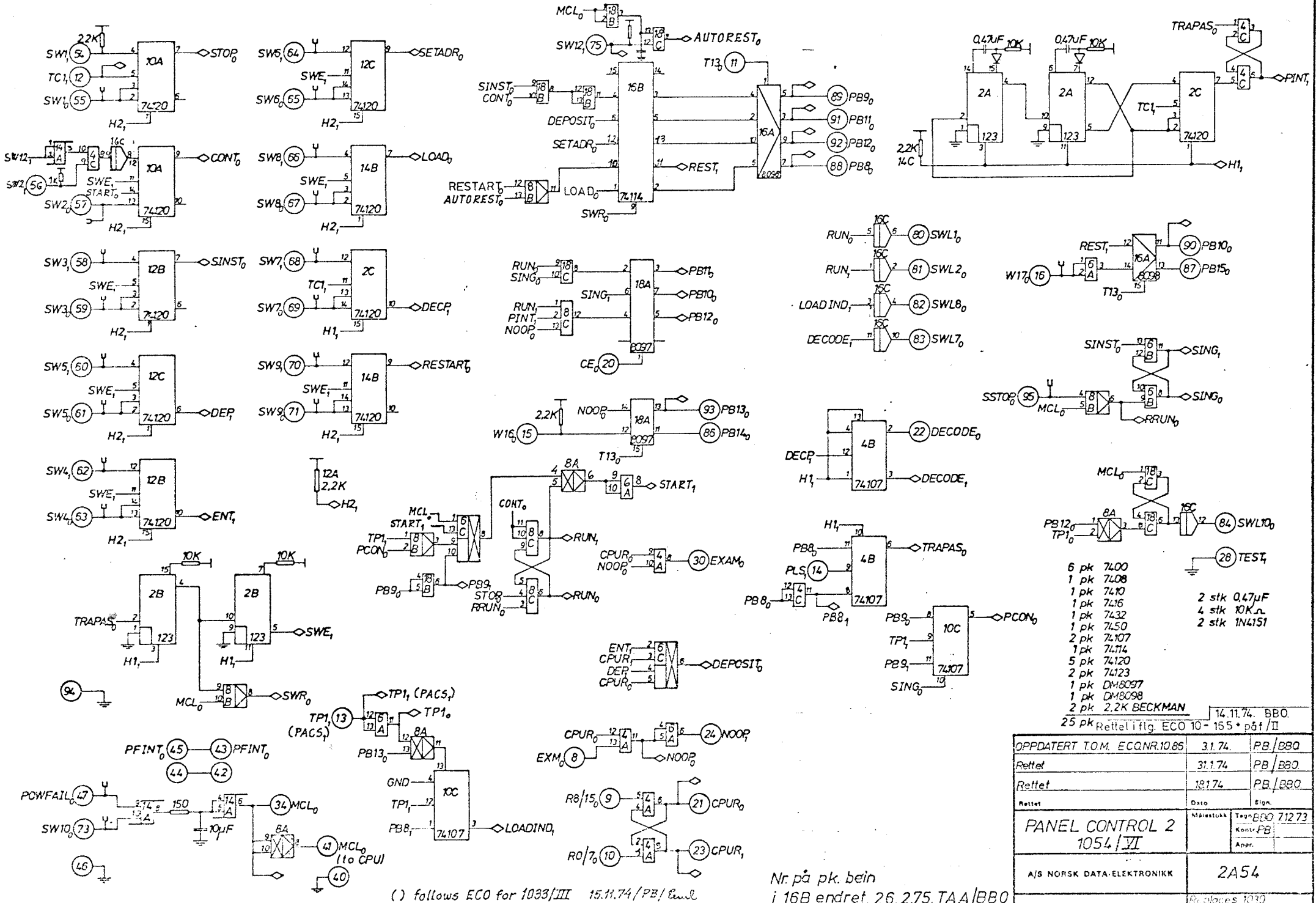

BCOMPL
BRBUSY
BSTART
SU0
SU1
SU4
SU5
11C
BERROR
DR0, DR1, DR2, DEQL
14B
7442
LOAD WRITE WORD
SENS DATA
LOAD MODUS
SENS STAT
LOAD CONTROL W
SENS FLAG
MASK CLEAR FLAG
SENS DATA
SENS FLAG
SFLSD
CLEAR SUB-FLAG 0-3
5C
74118
READ FAIL
SET SUB-FLAG x
SU0
SU1
SU2
SU3
CLEAR SUB-FLAG 12-15
6B
SU4
SU5
SU6
SU7
SYSTEM CLEAR
MCH
READ-CLEAR
15A
16B
DM 8123
MDB8
MDB9
MDB10
MDB11
MDB12
MDB13
MDB14
MDB15
SENS FLAG
SFLSD
FLAG 0
FLAG 1
FLAG 2
FLAG 3
3B
FLAG 4
FLAG 5
FLAG 6
FLAG 7
4A
SENS FLAG
MDB0
MDB1
MDB2
MDB3
MDB4
MDB5
MDB6
MDB7
STATx
7A
8B
SENS STAT
MDB5
MDB6
MDB7
MDB8
MDB9
MDB10
MDB11
MDB12
MDB13
MDB14
MDB15
READ-BIT
S1
S0
3D
5D
7D
74194
READ-CLEAR
SHIFT READ
READ-BIT
2A
READ-DATA 00 (@ TEST)
READ-DATA 11 (@ TEST)
READ DATA x
READ-DATA
R1 1k
VB1/VB2
DATAWORD MATCH
SYNCWORD MATCH
15D
SET SUBFLAG 5
SET SUBFLAG 1 (SYNC FAIL)
SET FLAG 2 (DATAWORD)
SET READ-WORD
SET FLAG 3 (SYNCWORD 0-
SET FLAG 4 (SYNCWORD 1-
STATE H
READ-BIT CLOCK
SYNCWORD 15
SYNCWORD 15
SENS DATA
3C
13A
DM8551
READ DATA
SET READWORD
SET FLAG 6
SU4
SU5
SU6
SU7
10B
HIGH
14C
74161
STATE 11
STATE H
SYNCWORD MATCH
READ-BIT CLOCK
1kΩ 25
HIGH
2 pk 7404
2 " 7408
2 " 7420
2 " 7425
1 " 7430
2 " 7432
1 " 7437
2 " 74118
2 " 74123
1 " 7442
1 " 74161
3 " 74194
4 " DM8096 (8098)
2 " DM8123
2 " DM8551
29 pk
C1 100pF R2 22k ~0.8µs
C2 100pF R3 22k ~0.8µs
C3 100pF R4 22k 0.8µs
17D
15D
SHIFT READ
READ-BIT CLOCK
READ BIT AVAIL
MODE4 (TAPE REVERS)
VB2
Ajourf.
30.4.74
JH/Emil
READ CONTROL MODULE
1051
A/S NORSK DATA-ELEKTRONIKK
2A51

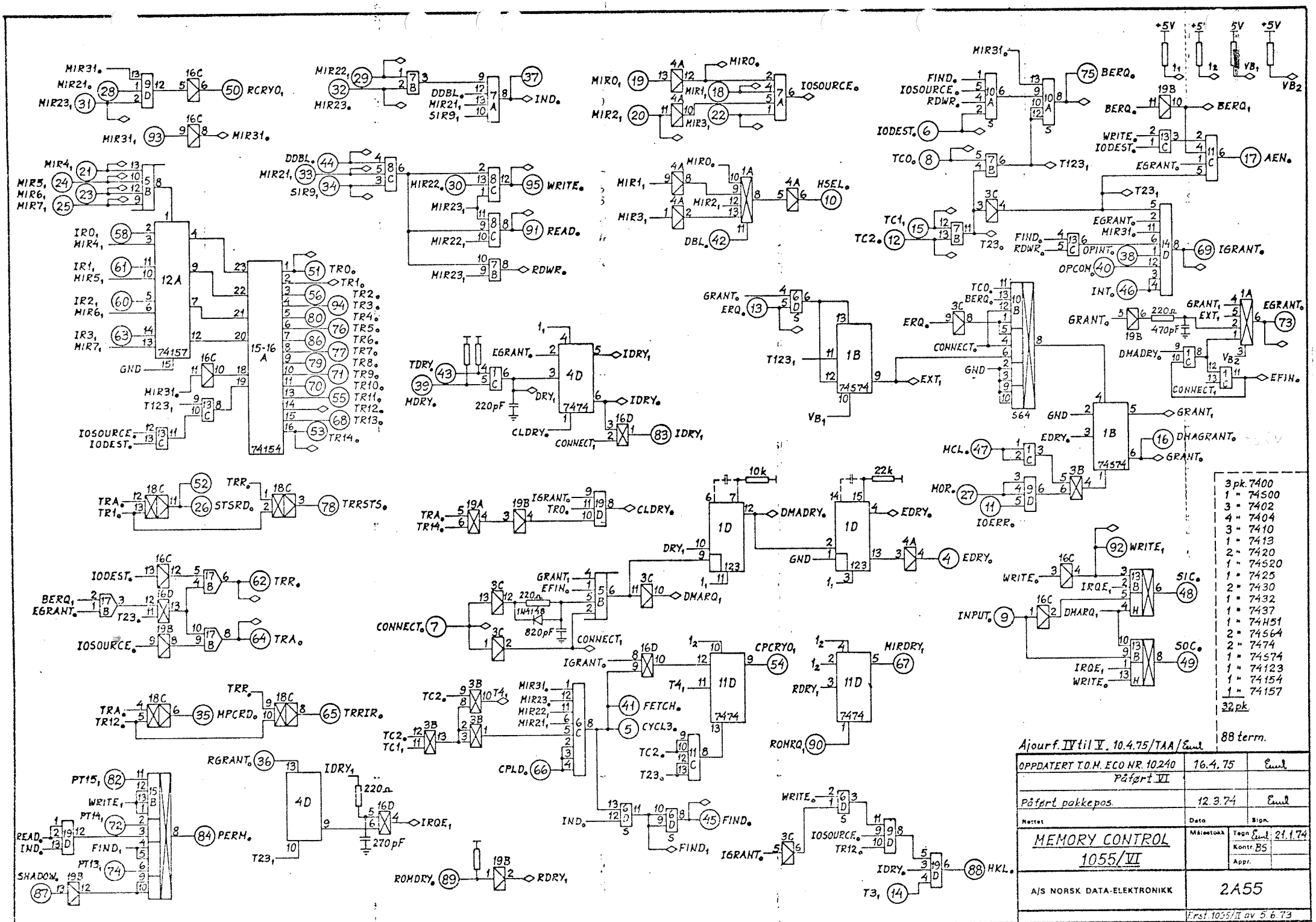


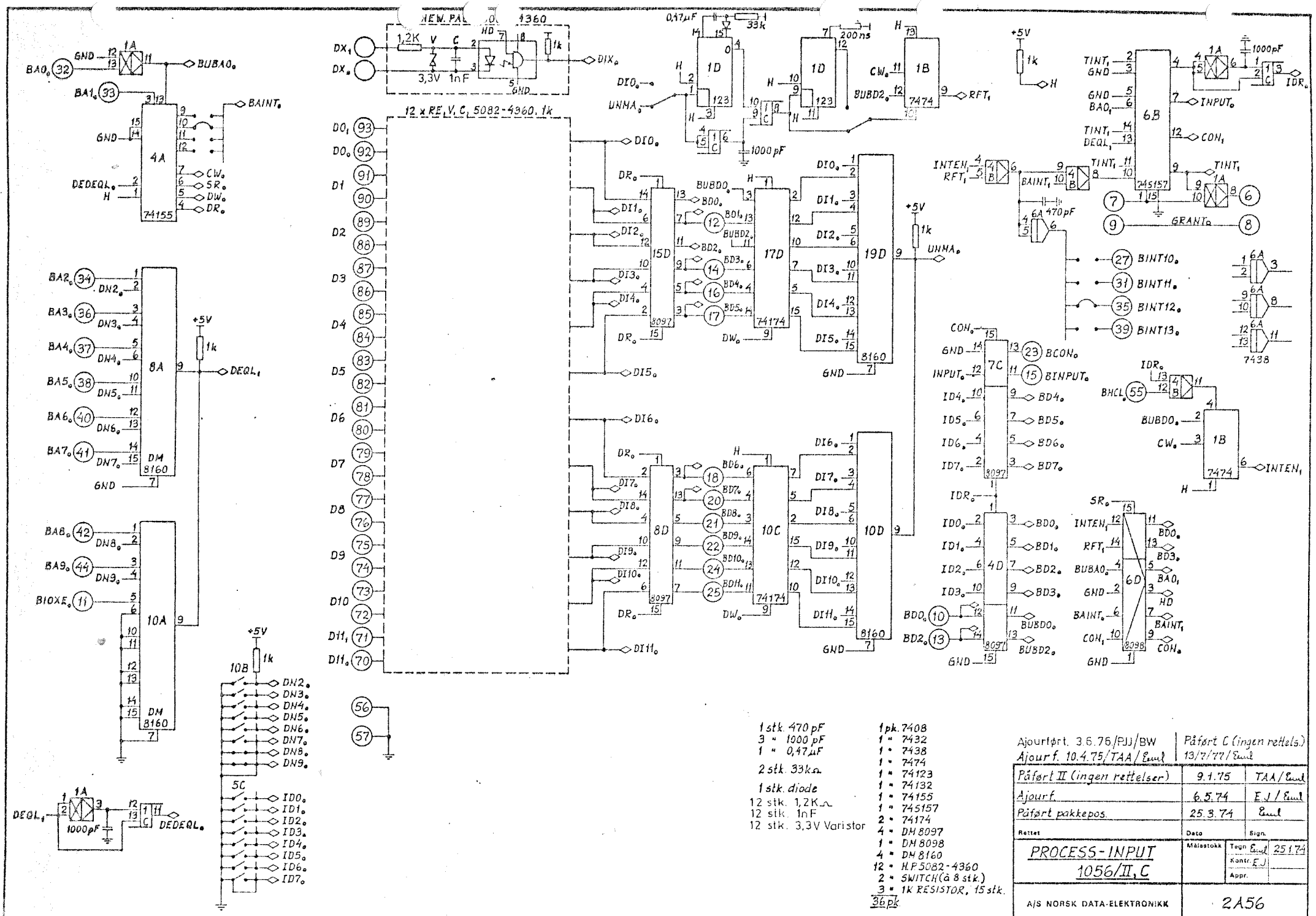
Pk pos. påført + div. rettelser	25.2.74.	J.H / BBO
Ajourført	3.12.73.	J.H / BBO.
Rettet	Dato	Sign.

WRITE CONTROL MODULE 1052	Målestokk	Tegn BEQ 9.8.73
		Kontr J.H.
		Appr.
A/S NORSK DATA-ELEKTRONIKK		2A52


SYSTEM CLEAR
BIT CLOCK
BCA1
BCB1
WRITE BIT
MODE 4
MODE 1
MODE 0
MODE 7
WES
WRITE ACTION
WRITE PROTECT VIOLATION
WRITE ENABLED
DRIVE FAIL
MODE 3
WCD0
WCD1
WDA1
WL
LEVEL
MODE 7
MODE 3
READL0
READL1
MODE3
TRANS
READ BIT
MODE 2
LCK1
SLT0
SLT1
READ FAIL DETECTED
FULL PLUS
HALF MINUS
THREE QUART
READ FAIL
MODE 4
MODE 5
MODE 6
REVERSE
RUN
FST
MAN REWIND
WEN0
WEN1
CIP0
CIP1
BEOT0
BEOT1
SIDEA0
SIDEA1
WES
DRIVE FAIL
BOTS
SIDEA
REV
FWD
BET0
BET1
RDA0
RDA1
READL0
READL1
VB4
RGT0
+5V
VB1
VB2
VB3
6 pk. 7400
1 " 7404
2 " 7408
1 " 7414
1 " 7428
2 " 7437
1 " 7474
1 " 7486
5 " 74123
1 " 74157
21 pk.
Påført pakkepos. 26.2.74 Eml
Ajourført og påført term.nr 28.11.73 J.H./Eml
DRIVE CONTROL MODULE
1053
A/S NORSK DATA-ELEKTRONIKK
2A53


SW1 (5L) 2.2K 10A STOP0
TC1 (12) SW10 (55) 74120 H21
SW5 (64) SWE1 SW60 (65) 12C SETADR0 74120 H21
SW12 (75) 14C 10A CONT0
SW2 (56) 1K SWE1 START0 SW20 (57) 74120 H21
SW8 (66) SWE1 SW80 (67) 14B LOAD0 74120 H21
SW3 (58) SWE1 SW30 (59) 12B SINST0 74120 H21
SW7 (68) TC1 SW70 (69) 2C DECP1 74120 H11
SW5 (60) SWE1 SW50 (61) 12C DEP1 74120 H21
SW9 (70) SWE1 SW90 (71) 14B RESTART0 74120 H21
SW4 (62) SWE1 SW40 (63) 12B ENT1 74120 H21
12A 2.2K H20
MCL0 SW12 (75) AUTOREST0
T13 (11) SINST0 CONT0 DEPOSIT0 SETADR0 RESTART0 AUTOREST0 LOAD0 16B 74114 SWR0 REST1 16A 8098 PB90 (89) PB110 (91) PB120 (92) PB80 (88)
0.47uF 10K 0.47uF 10K 2A 2A 123 123 TC1 2C 74120 TRAPAS0 PINT1 2.2K 14C H11
RUN0 SING0 RUN1 PINT1 NOOP0 18A 8097 PB110 SING1 PB100 PB120 CE0 (20)
RUN0 (80) SWL10 RUN1 (81) SWL20 LOADIND1 (82) SWL80 DECODE1 (83) SWL70
REST1 16A 8098 PB100 (90) PB150 (87) W170 (16) T130
SINST0 SING1 SSTOP0 (95) MCL0 RRUN0 SING0
2.2K W160 (15) NOOP0 18A 8097 T130 PB130 (93) PB140 (86)
8A START1 MCL START1 TP1 PCON0 CONT0 RUN1 PB90 PB9 STOP RRUN0 RUN0
DECP1 H11 4B 74107 DECODE0 (22) DECODE1
CPUR0 NOOP0 EXAM0 (30)
H11 PB80 PLS1 (14) PB81 4B 74107 TRAPAS0 PB90 TP11 PB91 10C 74107 SING0 PCON0 (5)
MCL0 PB120 TP10 8A 16C SWL100 (84) TEST1 (28)
ENT1 CPUR1 DEP CPUR0 DEPOSIT0 (6)
94
2B 2B 10K 10K TRAPAS0 123 123 SWE1 H11 H11
MCL0 SWR0
TP11 (PACS1) TP10 TP11 (PACS1) 8A PB130 10C 74107 GND TP11 PB81 LOADIND1
PFINT0 (45) (43) PFINT0 (44) (42)
POWFAIL0 (47) SW100 (73) 150 10µF 74A 34 MCL0 8A 43 MCL0 (to CPU) 46 40
CPUR0 EXM0 (8) 24 NOOP1 NOOP0
R8/150 (9) 21 CPUR0 R0/70 (10) 23 CPUR1
6 pk 7400
1 pk 7408
1 pk 7410
1 pk 7416
1 pk 7432
1 pk 7450
2 pk 74107
1 pk 74114
5 pk 74120
2 pk 74123
1 pk DM8097
1 pk DM8098
2 pk 2.2K BECKMAN
25 pk
2 stk 0,47µF
4 stk 10K
2 stk 1N4151
14.11.74. BBO
Rettet i flg. ECO 10-155 + påf/II
OPPDATERT T.O.M. ECO NR. 10.85 3.1.74. PB/BBO
Rettet 31.1.74 PB/BBO
Rettet 18.1.74 PB/BBO
Rettet Dato Sign.
PANEL CONTROL 2
1054/VI
Målestokk
Tegn BBO 7.12.73
Konstr PB
Godk.
A/S NORSK DATA-ELEKTRONIKK
2A54
Replaces 1030
() follows ECO for 1033/III 15.11.74/PB/Enl
Nr. på pk. bein i 16B endret 26.2.75. TAA/BBO
OPPDATERT T.O.M. ECO NR. 10-222. II t. VI. 20.1.75/TAA/Enl Ajourført. V t. VI. 24.2.75. TAA/BBO
5.4


MIR31, MIR21, MIR23, RCRY0, MIR31, MIR22, MIR23, DDBL, MIR21, SIR9, IND, MIR0, MIR1, MIR2, MIR3, IOSOURCE, MIR4, MIR5, MIR6, MIR7, IR0, IR1, IR2, IR3, 74157, 15-16 A, 74154, TR0, TR1, TR2, TR3, TR4, TR5, TR6, TR7, TR8, TR9, TR10, TR11, TR12, TR13, TR14, GND, IOSOURCE, IODEST, DDBL, MIR21, SIR9, MIR22, MIR23, WRITE, READ, RDWR, MIR1, MIR2, MIR3, DBL, HSEL, TDRY, MDRY, EGRANT, DRY1, CLDRY, CONNECT1, IDRY1, IDRY, GRANT, ERQ, T123, EXT1, VB1, 74574, TCO, BERQ, CONNECT, 564, GND, EDRY, MCL, MOR, IOERR, DMAGRANT, TRA, TR1, TRR, STSRD, TRRSTS, TR14, CLDRY, 10k, 22k, DMADRY, 74574, 3 pk. 7400, 1 " 74S00, 3 " 7402, 4 " 7404, 3 " 7410, 1 " 7413, 2 " 7420, 1 " 74S20, 1 " 7425, 2 " 7430, 1 " 7432, 1 " 7437, 1 " 74H51, 2 " 74S64, 2 " 7474, 1 " 74S74, 1 " 74123, 1 " 74154, 1 " 74157, 32 pk, 88 term., IODEST, BERQ, EGRANT, IOSOURCE, TRR, TRA, TR12, MPCRD, TRRIR, RGRANT, IDRY1, 220Ω, 270pF, IRQE1, PERM, PT15, PT14, PT13, WRITE1, FIND1, SHADOW, IGRANT, DMARQ1, CONNECT1, 1N4148, 820pF, CPCRYO, MIRDRY, RDRY1, ROMRQ, ROMDRY, RDRY1, TC2, TC1, CPLD, FETCH, CYCL3, T4, T23, IND, FIND, IRQE1, WRITE, SIC, SOC, INPUT, DMARQ1, IRQE1, WRITE, IGRANT, IOSOURCE, TR12, IDRY, T3, HKL, +5V, VB1, VB2, FIND, IOSOURCE, RDWR, IODEST, TCO, TC1, TC2, T123, T23, EGRANT, MIR31, OPINT, OPCON, INT, IGRANT, AEN, BERQ, BERQ1, WRITE, IODEST, GRANT, EXT1, 220Ω, 470pF, DMADRY, VB2, CONNECT1, EFIN, GRANT, EDRY, DMAGRANT, GRANT, MCL, EDRY, MOR, IOERR, Ajourf. IV til V. 10.4.75/TAA/Eml, OPPDATERT T.O.H. ECO NR. 10.240, 16.4.75, Eml, Påført VI, Påført pakkepos., 12.3.74, Eml, Rettet, Dato, Sign., MEMORY CONTROL, 1055/VI, Målestokk, Tegn Eml 21.1.74, Kontr BS, Appr., A/S NORSK DATA-ELEKTRONIKK, 2A55, Frst. 1055/II av 5.6.73



1 stk. 470 pF	1 pk. 7408
3 " 1000 pF	1 " 7432
1 " 0,47 µF	1 " 7438
	1 " 7474
2 stk. 33 kΩ	1 " 74123
	1 " 74132
1 stk. diode	1 " 74155
12 stk. 1,2K ~	1 " 745157
12 stk. 1nF	2 " 74174
12 stk. 3,3V Varistor	4 " DM 8097
	1 " DM 8098
	4 " DM 8160
	12 " H.P.5082-4360
	2 " SWITCH (à 8 stk.)
	3 " 1k RESISTOR, 15 stk.
	36 pk

Ajourført. 3.6.76/PJJ/BW
Ajour f. 10.4.75/TAA/Emil

Påført C (ingen rettels.)
13/7/77/Emil

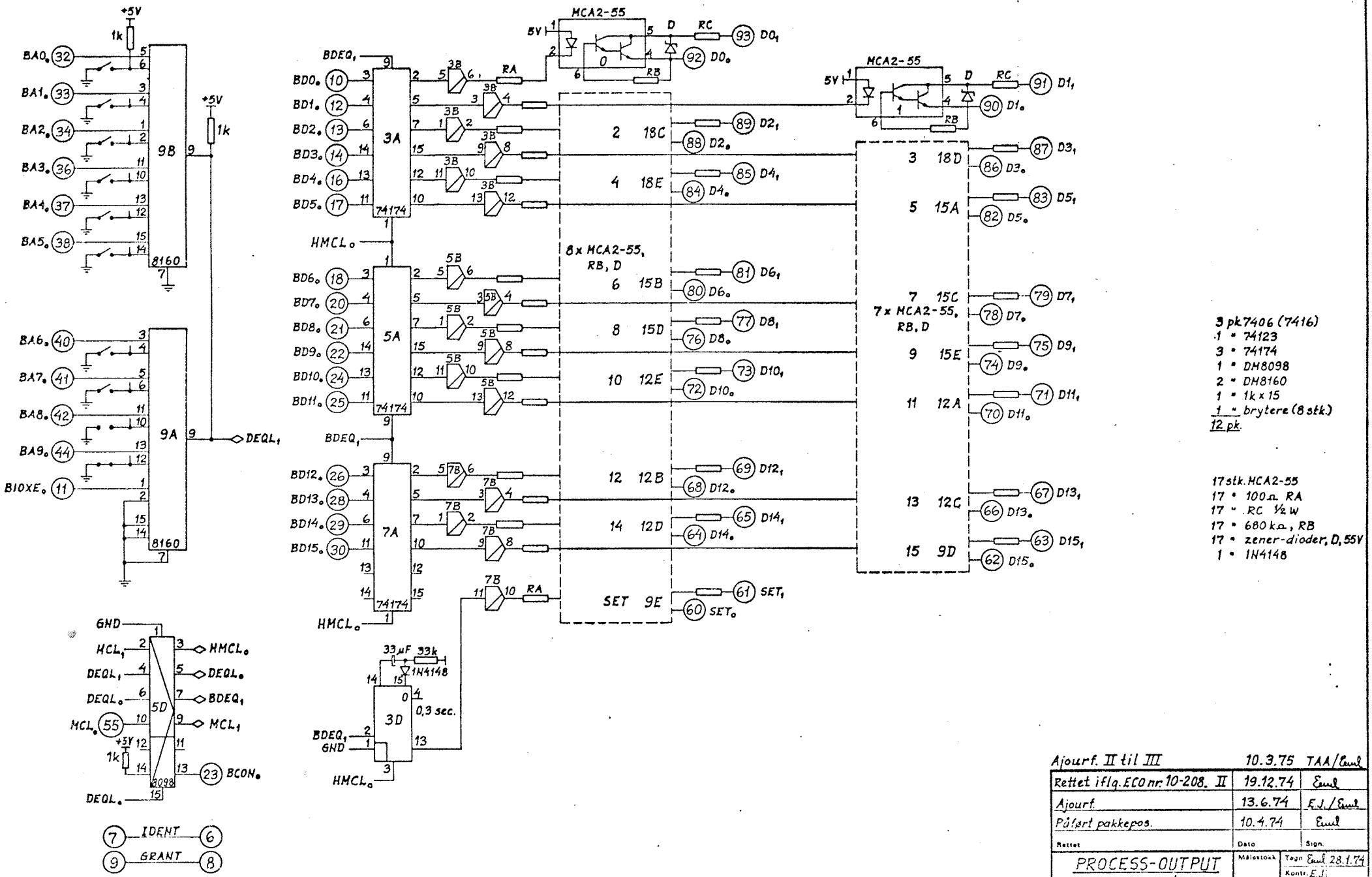
Påført II (ingen rettelser)	9.1.75	TAA/Emil
Ajour f.	6.5.74	EJ/Emil
Påført pakkepos.	25.3.74	Emil
Rettet	Dato	Sign.

PROCESS-INPUT
1056/II,C

Målestokk | Tegn. Emil 25174 | Kontr. EJ | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A56



3 pk 7406 (7416)
1 " 74123
3 " 74174
1 " DM8098
2 " DM8160
1 " 1k x 15
1 " brytere (8 stk.)
12 pk.

17 stk. MCA2-55
17 " 100 Ω RA
17 " RC ½ W
17 " 680 kΩ, RB
17 " zener-dioder, D, 55V
1 " 1N4148

Ajourf. II til III	10.3.75	TAA/Emil
Rettet iflg. ECO nr. 10-208. II	19.12.74	Emil
Ajourf.	13.6.74	E.J./Emil
Påført pakkepos.	10.4.74	Emil
Rettet	Dato	Sign.

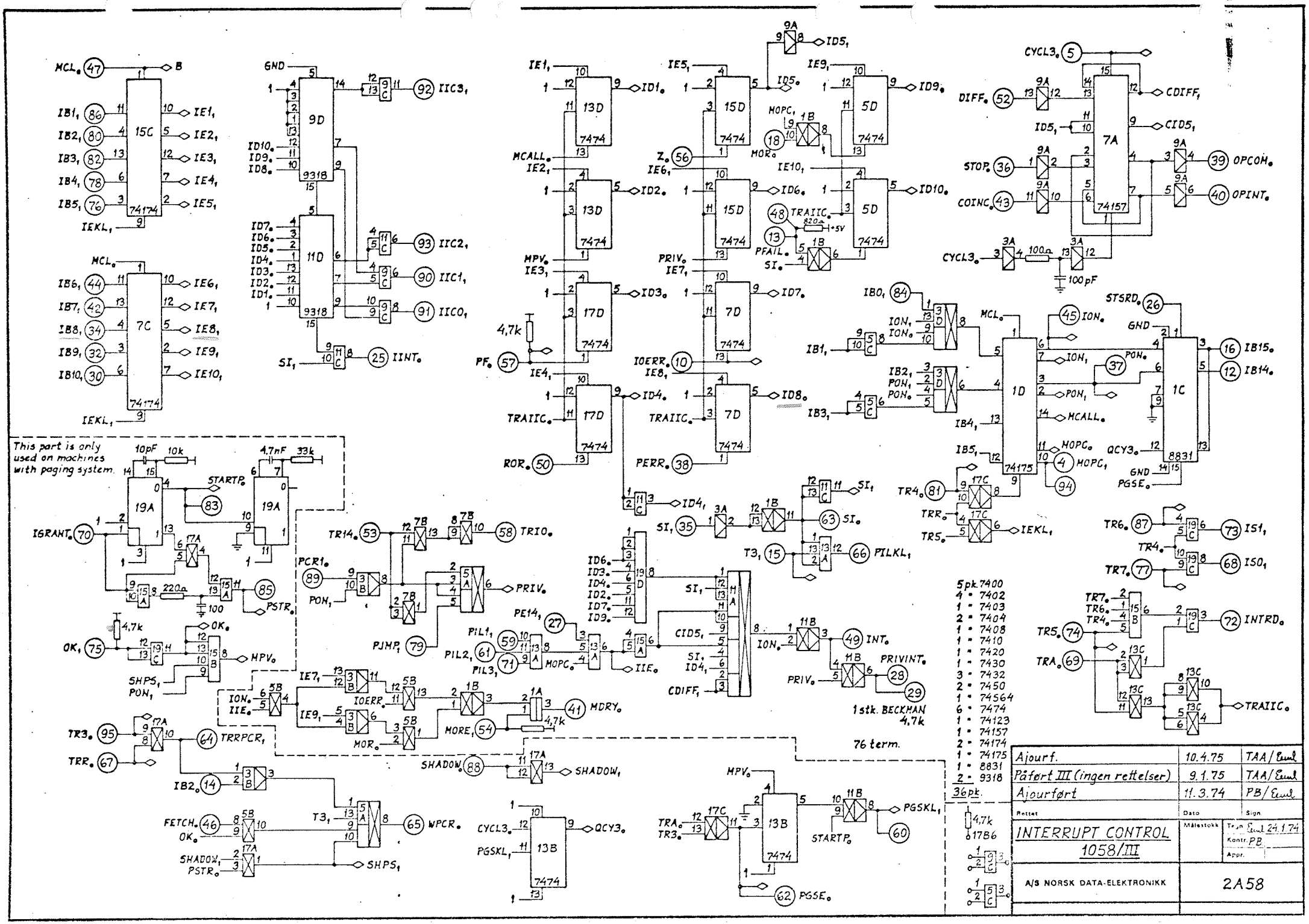
PROCESS-OUTPUT 1057/III

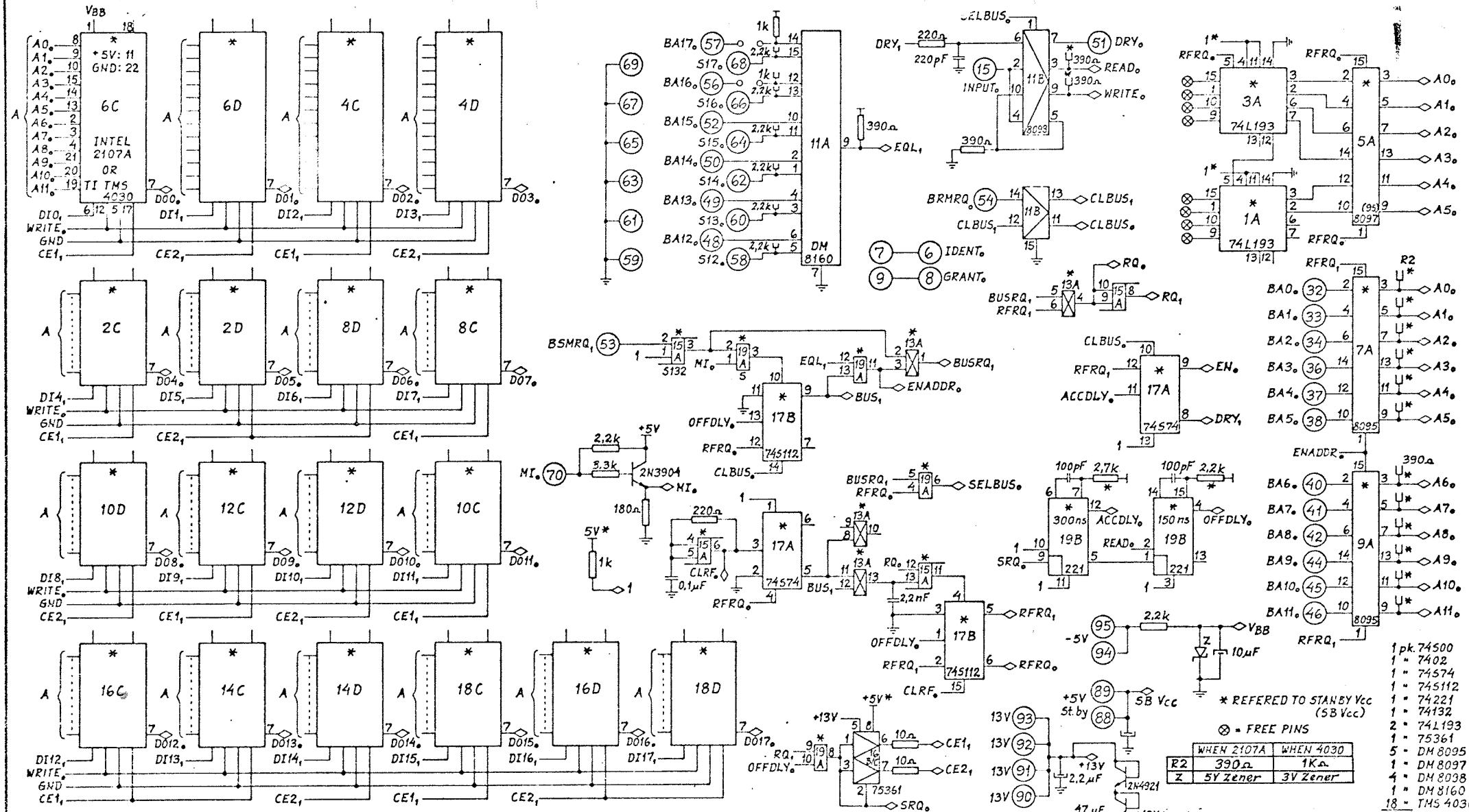
Målestokk

Tegn. Emil 28.1.74
Kontr. E.J.
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A57


This part is only used on machines with paging system.
1stk. BECKMAN 4,7k
76 term.
5pk 7400
4 = 7402
1 = 7403
2 = 7404
1 = 7408
1 = 7410
1 = 7420
1 = 7430
3 = 7432
2 = 7450
1 = 74564
6 = 7474
1 = 74123
1 = 74157
2 = 74174
1 = 74175
1 = 8831
2 = 9318
36pk
Ajourf. 10.4.75 TAA/Eml
Påført III (ingen rettelser) 9.1.75 TAA/Eml
Ajourført 11.3.74 PB/Eml
Rettet Dato Sign
INTERRUPT CONTROL
1058/III
Målestokk
Tegn. Eml 24.1.74
Kontr. PB
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A58



	WHEN 2107A	WHEN 4030
R2	390Ω	1KΩ
Z	5V Zener	3V Zener

1 pk. 74500
1 = 7402
1 = 74574
1 = 745112
1 = 74221
1 = 74132
2 = 74L193
1 = 75361
5 = DM 8095
1 = DM 8097
4 = DM 8098
1 = DM 8160
18 = TMS 4030
38 pk

OPPDATERT T.O.H. ECO NR.		
Ajourført og forandr f III t. IV	21.10.74	IB/Eml
Ajourført og forandr fra II til III	24.9.74	IB/Eml
Rettet	Dato	Sign.

4K RAM
1060/IV

Målestokk

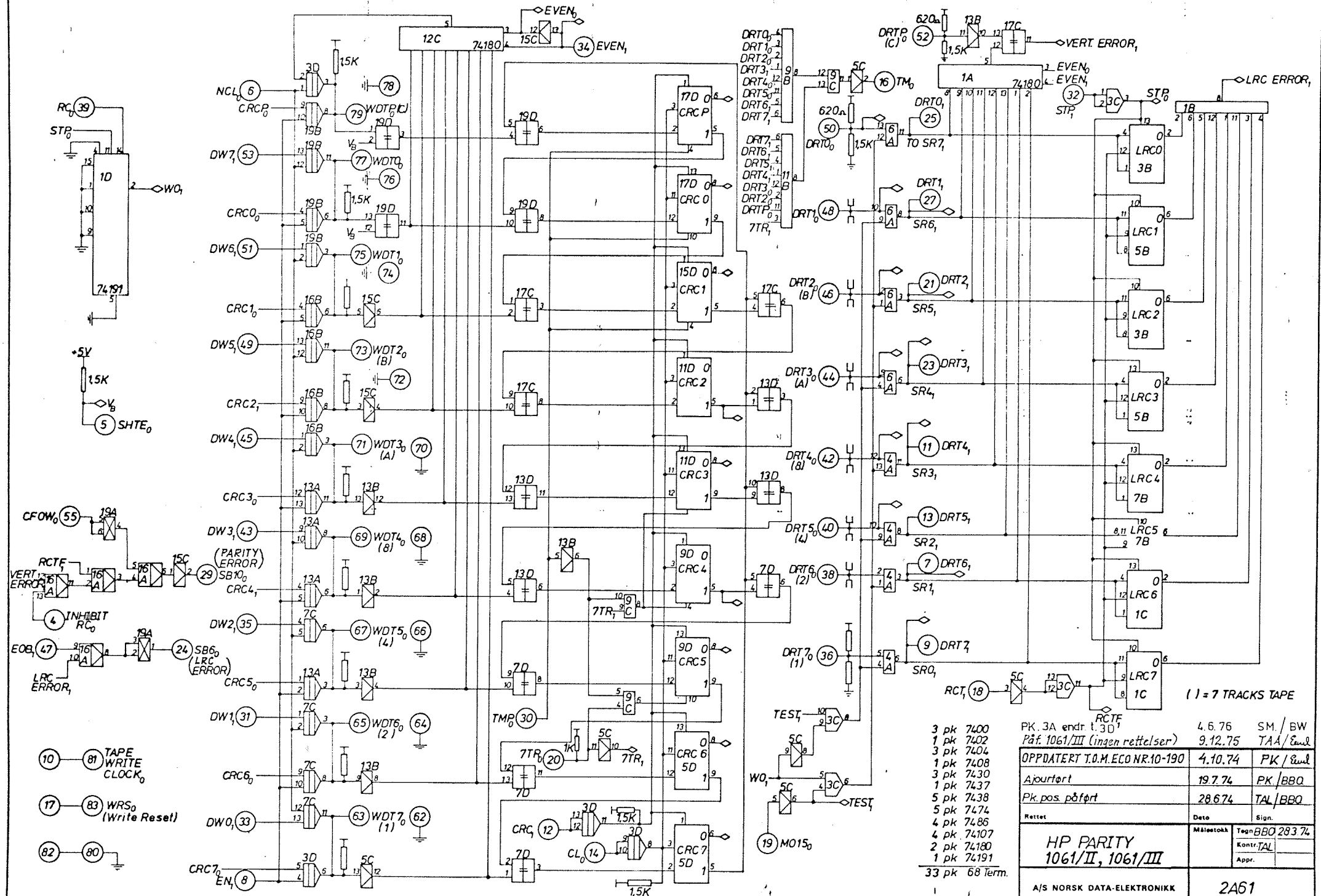
Tegn. Eml 5.6.74
Kontr. IB
Appr.

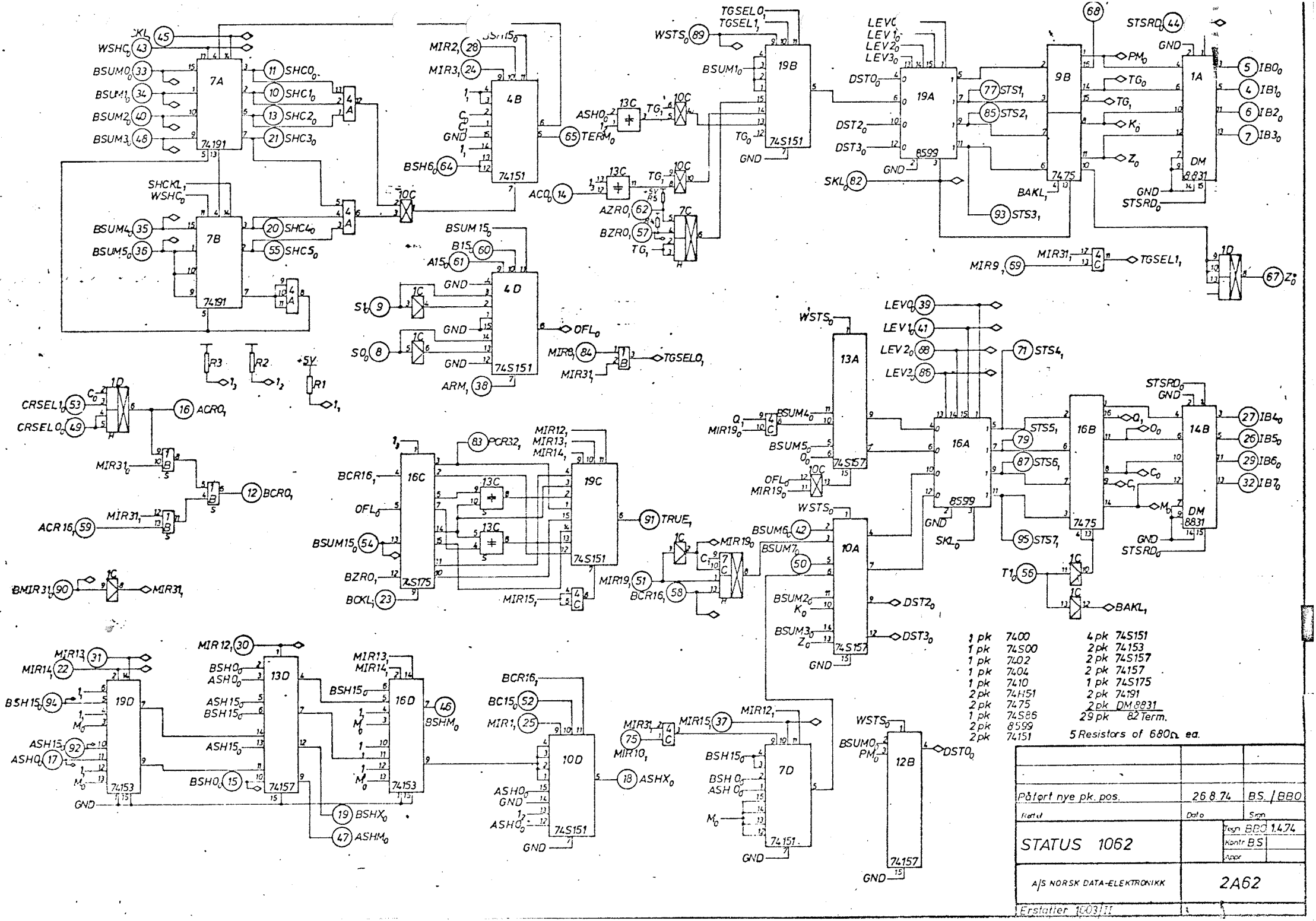
A/S NORSK DATA-ELEKTRONIKK

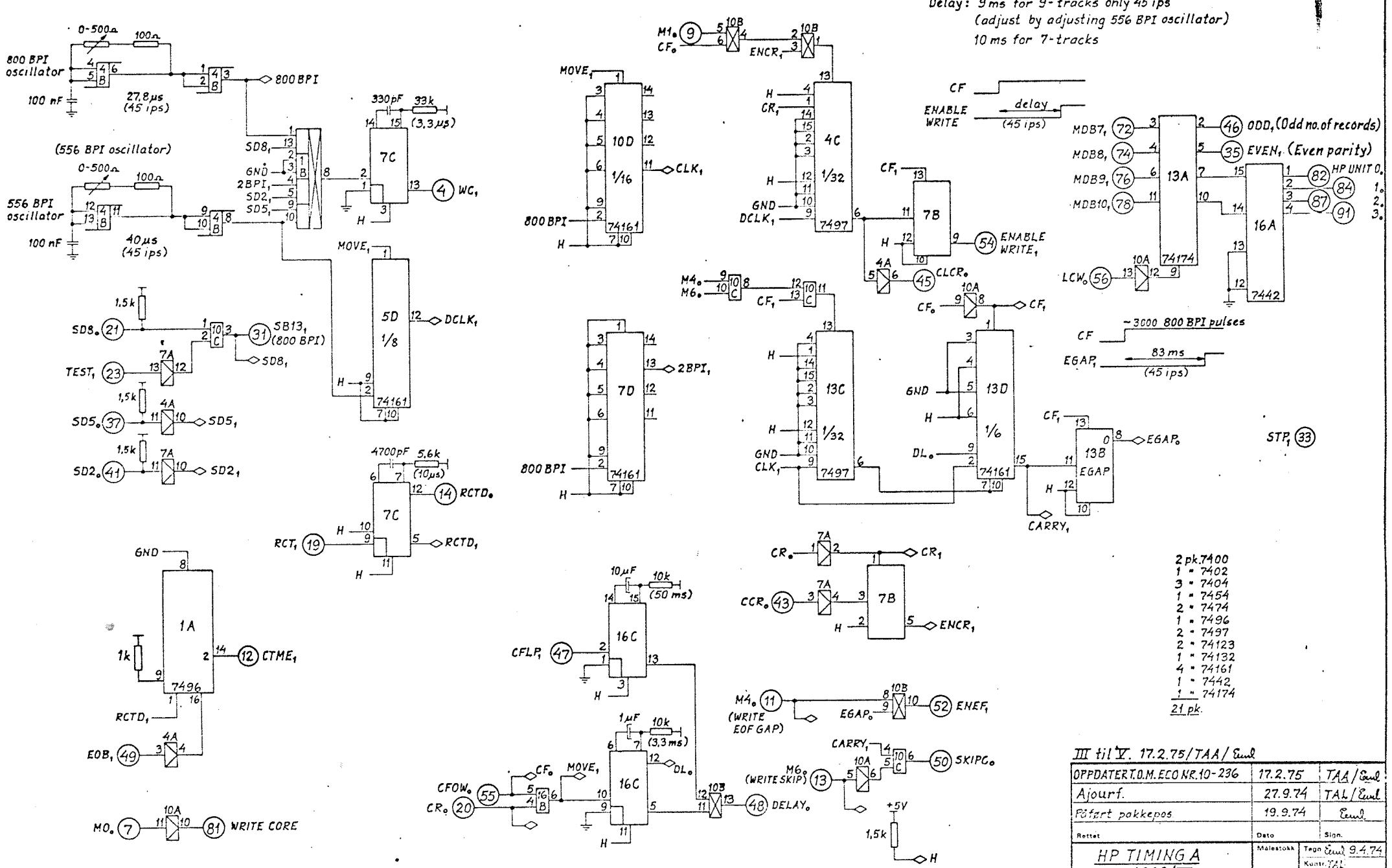
2A60

Ajourf. 18½-75/IB/Eml

1 stk. 3,3k + 6 stk. 1k erstatter av 1 stk. 899-1-R2,2k. 6.3.75/IB/Eml


12C 74180
15C
EVEN0
34 EVEN1
NCL0 6
CRCP0
DW7 53
CRC0 0
DW6 51
CRC1 0
DW5 49
CRC2 1
DW4 45
CRC3 0
DW3 43
CRC4 1
DW2 35
CRC5 0
DW1 31
CRC6 0
DW0 33
CRC7 0
EN1 8
RC0 39
STP0
1D 74191
WO1
+5V
1.5K
VB
5 SHTE0
CFOW0 55
RCTE1
VERT ERROR1
INHIBIT RC0
EOB1 47
LRC ERROR1
29 SB10 (PARITY ERROR)
24 SB60 (LRC ERROR)
10 81 TAPE WRITE CLOCK0
17 83 WRS0 (Write Reset)
82 80
78
79 WDTP0 (C)
77 WDT00
76
75 WDT10
74
73 WDT20 (B)
72
71 WDT30 (A)
70
69 WDT40 (8)
68
67 WDT50 (4)
66
65 WDT60 (2)
64
63 WDT70 (1)
62
17D CRCP
17D CRC0
15D CRC1
11D CRC2
11D CRC3
9D CRC4
9D CRC5
CRC6 5D
CRC7 5D
TMP0 30
7TR0 20
7TR1
CRC1 12
CL0 14
1K
1.5K
DRT0 4
DRT1 3
DRT2 2
DRT3 1
DRT4 12
DRT5 11
DRT6 5
DRT7 6
7TR1
620Ω
DRTP (C)0 52
VERT. ERROR1
1A 74180
EVEN0
EVEN1
32 STP1
STP0
LRC ERROR1
1B
TM0 16
DRT0 25
DRT10 50
TO SR7
DRT1 27
DRT10 48
SR6
DRT2 21
DRT2 (B) 46
SR5
DRT3 23
DRT3 (A) 44
SR4
DRT4 11
DRT4 (8) 42
SR3
DRT5 13
DRT5 (4) 40
SR2
DRT6 7
DRT6 (2) 38
SR1
DRT7 9
DRT7 (1) 36
SR0
LRC0 3B
LRC1 5B
LRC2 3B
LRC3 5B
LRC4 7B
LRC5 7B
LRC6 1C
LRC7 1C
RCT1 18
RCTE
TEST1
TEST0
WO1
MO150 19
( ) = 7 TRACKS TAPE
3 pk 7400
1 pk 7402
3 pk 7404
1 pk 7408
3 pk 7430
1 pk 7437
5 pk 7438
5 pk 7474
4 pk 7486
4 pk 74107
2 pk 74180
1 pk 74191
33 pk 68 Term.
PK. 3A endr. t. 3D 4.6.76 SM. / BW
Påf. 1061/III (ingen rettelser) 9.12.75 TAA / Emil
OPPDATERT T.O.M. ECO NR. 10-190 4.10.74 PK / Emil
Ajourført 19.7.74 PK / BBO
Pk. pos. påført 28.6.74 TAL / BBO
Rettet Dato Sign.
HP PARITY 1061/II, 1061/III
Målestokk
Tegn BBO 28.3.74
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A61


7A
7B
74191
4B
74151
4D
74S151
19B
19A
9B
7475
8599
1A
DM 8831
13A
16A
16B
14B
74S157
10A
16C
19C
74S175
13D
19D
74153
74157
16D
10D
7D
12B
R3
R2
R1
+5V
WSHC0
BSUM0 0
BSUM1 0
BSUM2 0
BSUM3 0
SHCKL1
BSUM4 0
BSUM5 0
MIR2 1
MIR3 1
BSH6 0
ACC0
TERM0
ASH0 0
AZR0 1
BZR0 1
WSTS0
BSUM1 0
TGSEL0 1
TGSEL1 1
DST0 0
DST2 0
DST3 0
SKL0
BAKL1
STS1 1
STS2 1
STS3 1
LEV0
LEV1 0
LEV2 0
LEV3 0
PM0
TG0
TG1
K0
Z0
IB0 0
IB1 0
IB2 0
IB3 0
STSRD0
GND
MIR9 1
MIR31 1
TGSEL1 1
Z0
B15
A15
S1
S0
OFL0
MIR8 1
MIR31 1
TGSEL0 1
ARM 1
CRSEL1 0
CRSEL0 0
ACR0 1
MIR31 0
BCR0 1
ACR16 1
BMIR31 0
MIR31 1
BCR16 1
OFL0
BSUM15 0
BZR0 1
BCKL1
PCR32 1
MIR12 1
MIR13 1
MIR14 1
TRUE1
MIR19 0
BCR16 1
MIR15 1
BSUM4 0
BSUM5 0
OFL0
MIR19 0
BSUM6 0
BSUM7 0
BSUM2 0
BSUM3 0
STS4 1
STS5 1
STS6 1
STS7 1
T1 0
BAKL1
IB4 0
IB5 0
IB6 0
IB7 0
MIR13 1
MIR14 1
BSH15 0
ASH15 0
ASH0 0
BSH0 0
BSHM0
BSHX0
ASHM0
BC15 0
MIR1 1
ASHX0
MIR10 1
BSH0 0
ASH0 0
DST0 0
1 pk 7400
1 pk 74S00
1 pk 7402
1 pk 7404
1 pk 7410
2 pk 74H51
2 pk 7475
1 pk 74S86
2 pk 8599
2 pk 74151
4 pk 74S151
2 pk 74153
2 pk 74S157
2 pk 74157
1 pk 74S175
2 pk 74191
2 pk DM 8831
29 pk 82 Term.
5 Resistors of 680Ω ea.
Påført nye pk. pos.
26 8 74
BS. / BBO
STATUS 1062
Tegn BBO 1.4.74
Kontr BS
A/S NORSK DATA-ELEKTRONIKK
2A62
Erstatter 1003 II



2 pk.7400
1 = 7402
3 = 7404
1 = 7454
2 = 7474
1 = 7496
2 = 7497
2 = 74123
1 = 74132
4 = 74161
1 = 7442
1 = 74174
21 pk.

III til V. 17.2.75/TAA/Eml

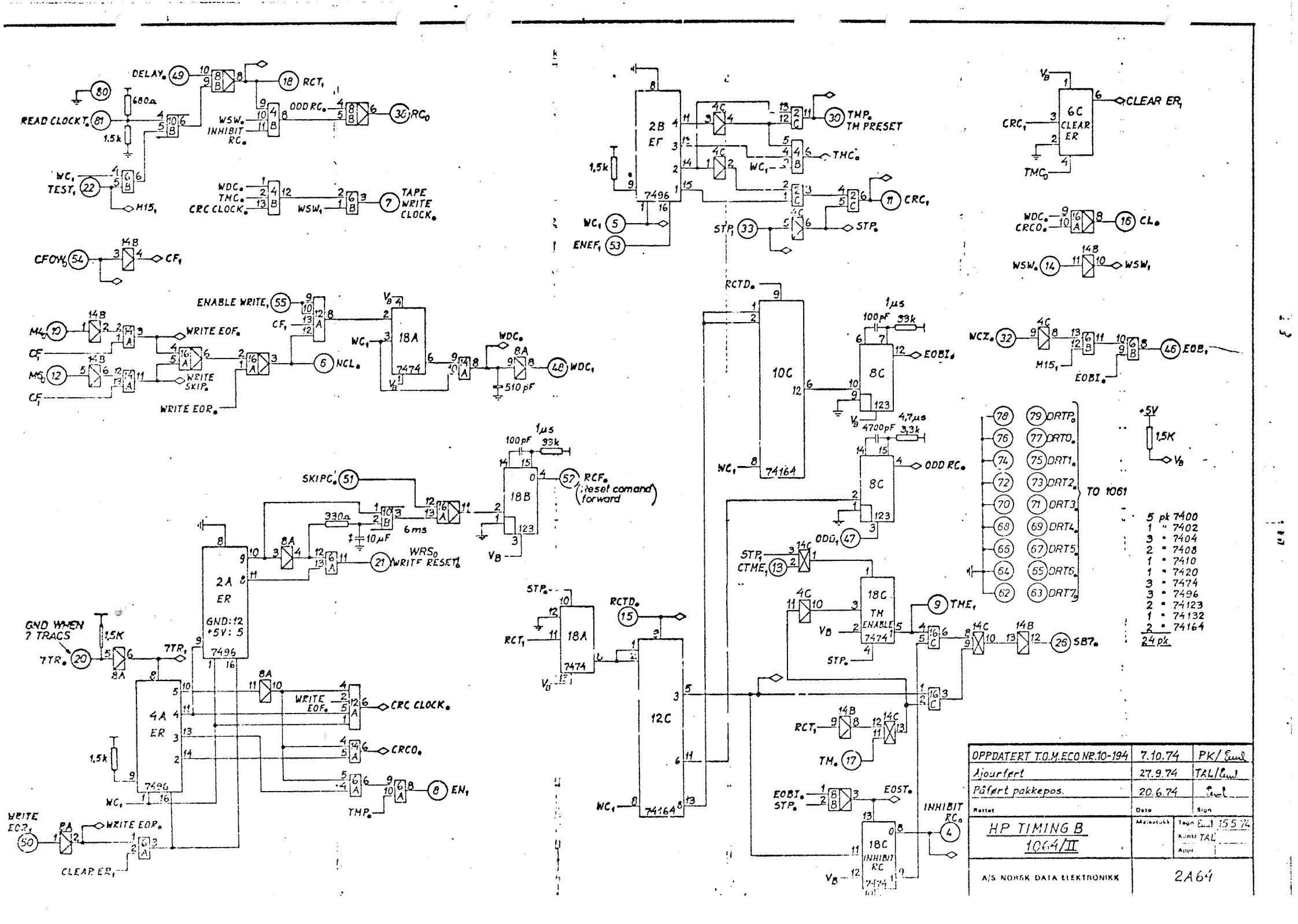
OPPDATERT.O.M. ECO NR. 10-236	17.2.75	TAA/Eml
Ajourf.	27.9.74	TAL/Eml
Påført pakkepos	19.9.74	Eml
Rettet	Dato	Sign.

HP TIMING A
1063/V

Målestokk | Tegn Eml 9.4.74 | Kontr. TAL | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A63



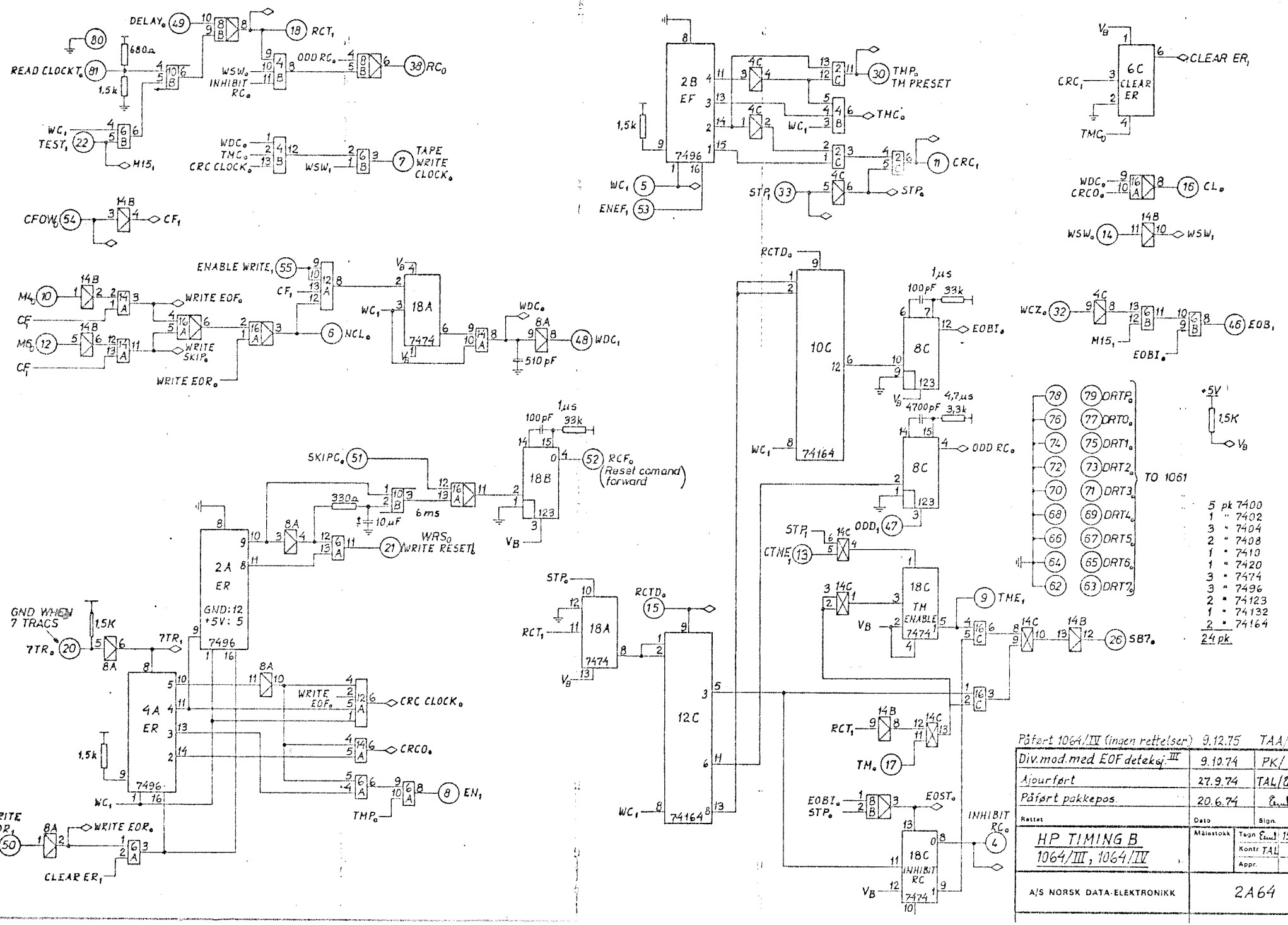
OPPDATERT T.O.M.ECO NR.10-194	7.10.74	PK/Emd
Ajourført	27.9.74	TAL/Emd
Påført pakkepos.	20.6.74	Emd
Rettet	Dato	Sign

HP TIMING B
1064/II

Målestokk | Tegn Emd 1557
Kontr TAL
Appr

A/S NORSK DATA ELEKTRONIKK

2A64



Påført 1064/III (ingen rettelser) 9.12.75 TAA/Emd

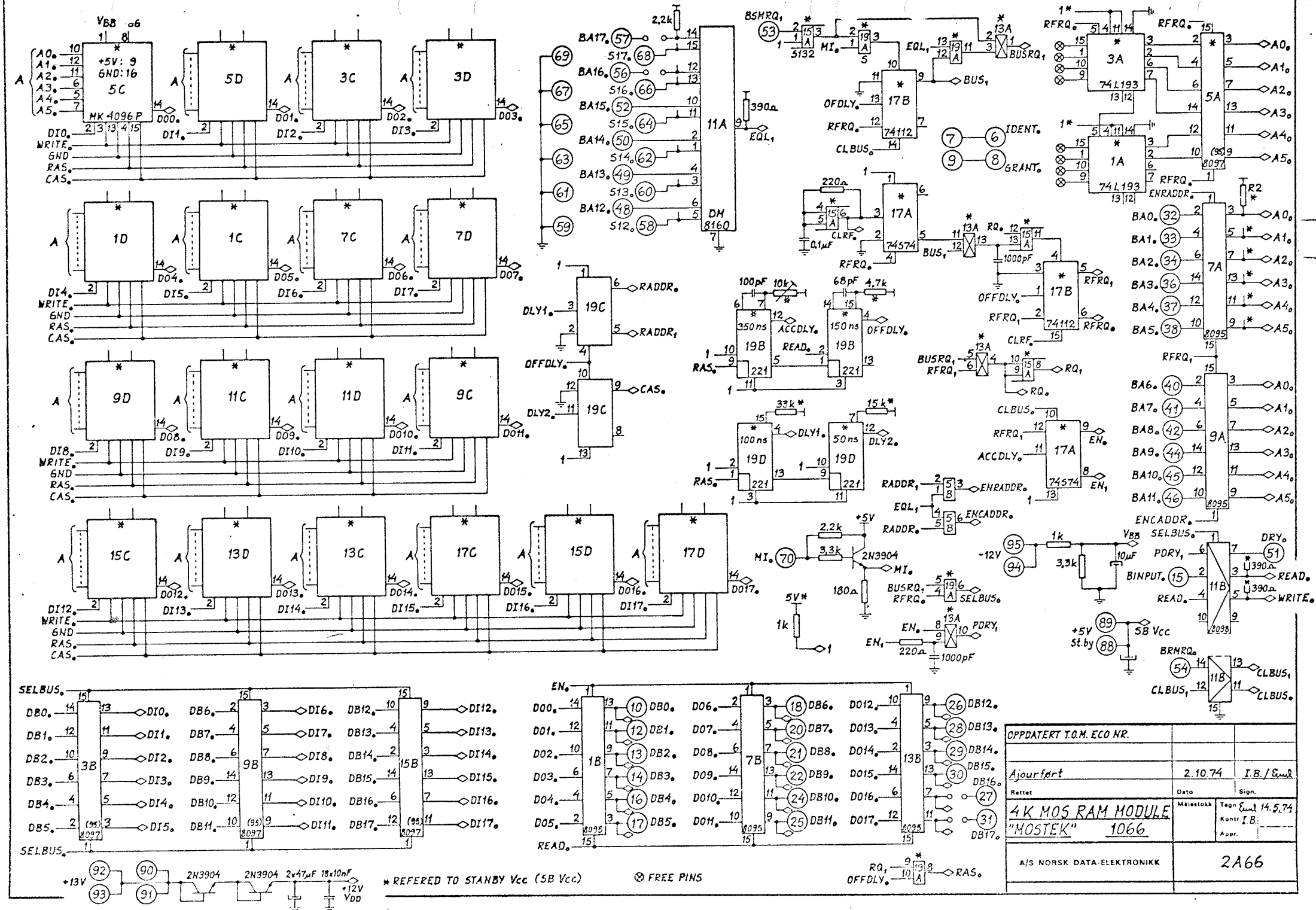
Div.mod.med EOF deteksj. III	9.10.74	PK/Emd
Ajourført	27.9.74	TAL/Emd
Påført pakkepos.	20.6.74	Emd
Rettet	Dato	Sign.

HP TIMING B
1064/III, 1064/IV

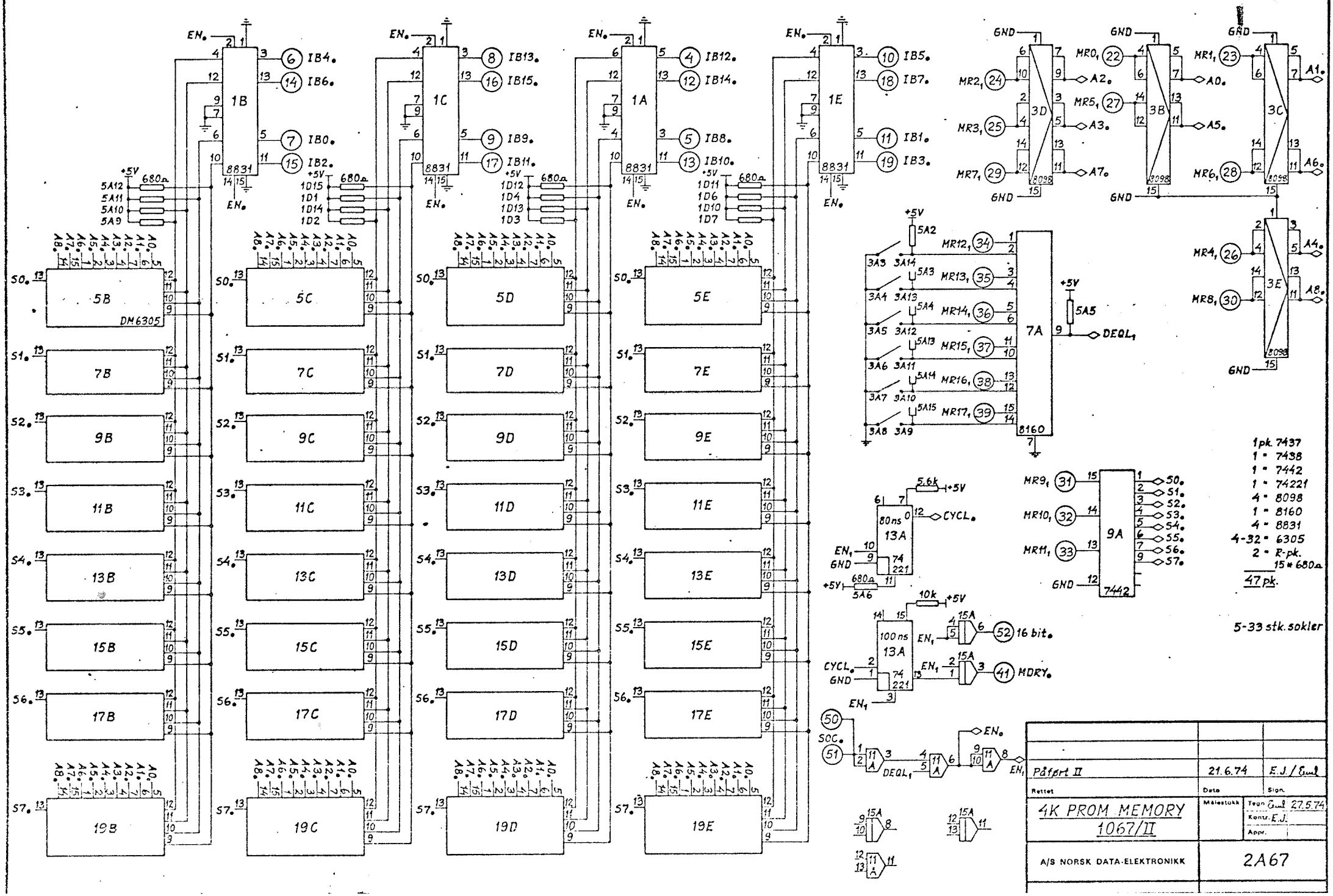
Målestokk | Tegn Emd 15 5 74 | Kontr TAL | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A64



OPPDATERT T.O.M. ECO NR.		
Ajourført	2.10.74	I.B./Emil
Rettet	Dato	Sign.
4K MOS RAM MODULE "MOSTEK" 1066	Målestokk	Tegn Emil 14.5.74 Konstr I.B. Appr.
A/S NORSK DATA-ELEKTRONIKK		2A66

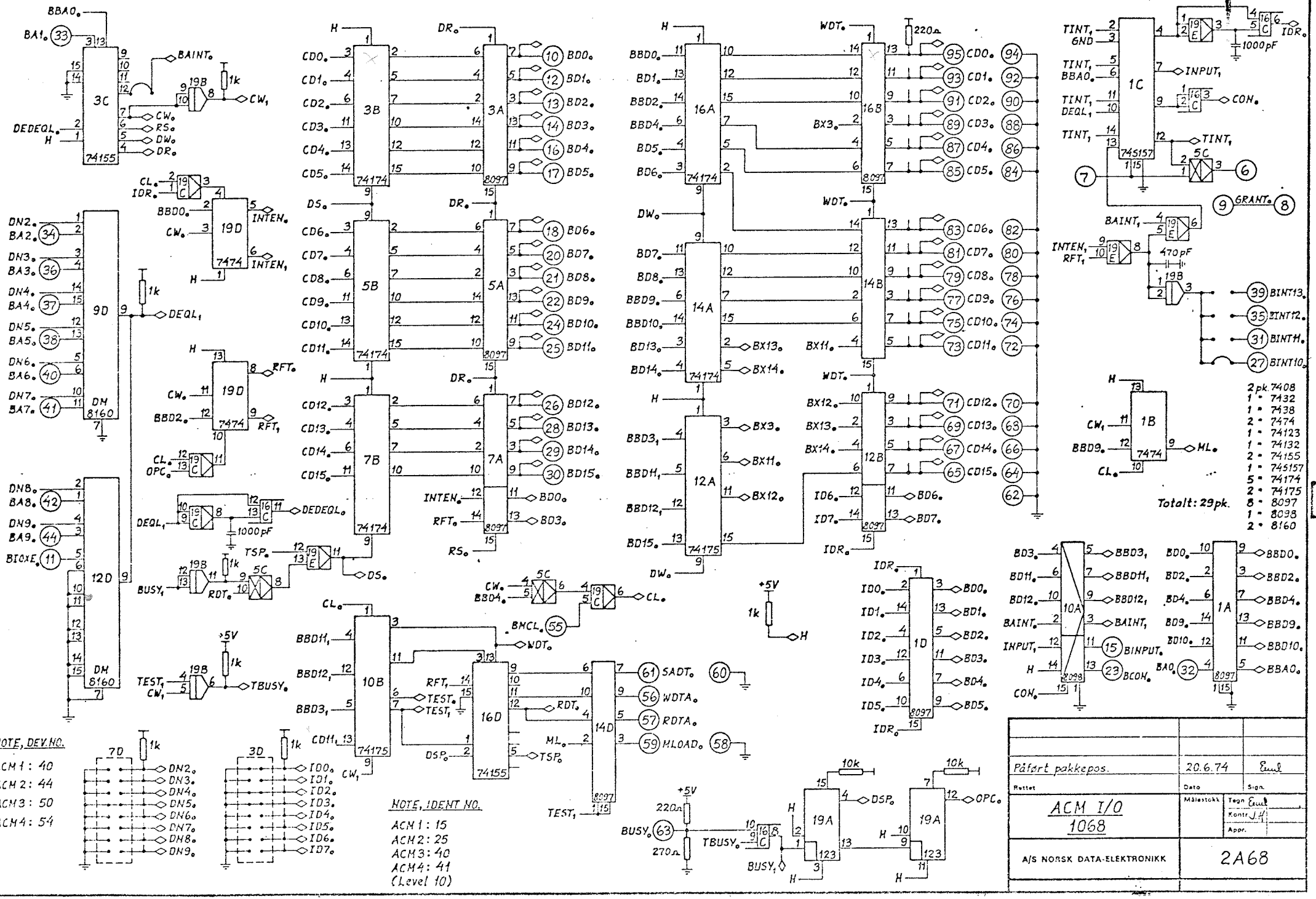


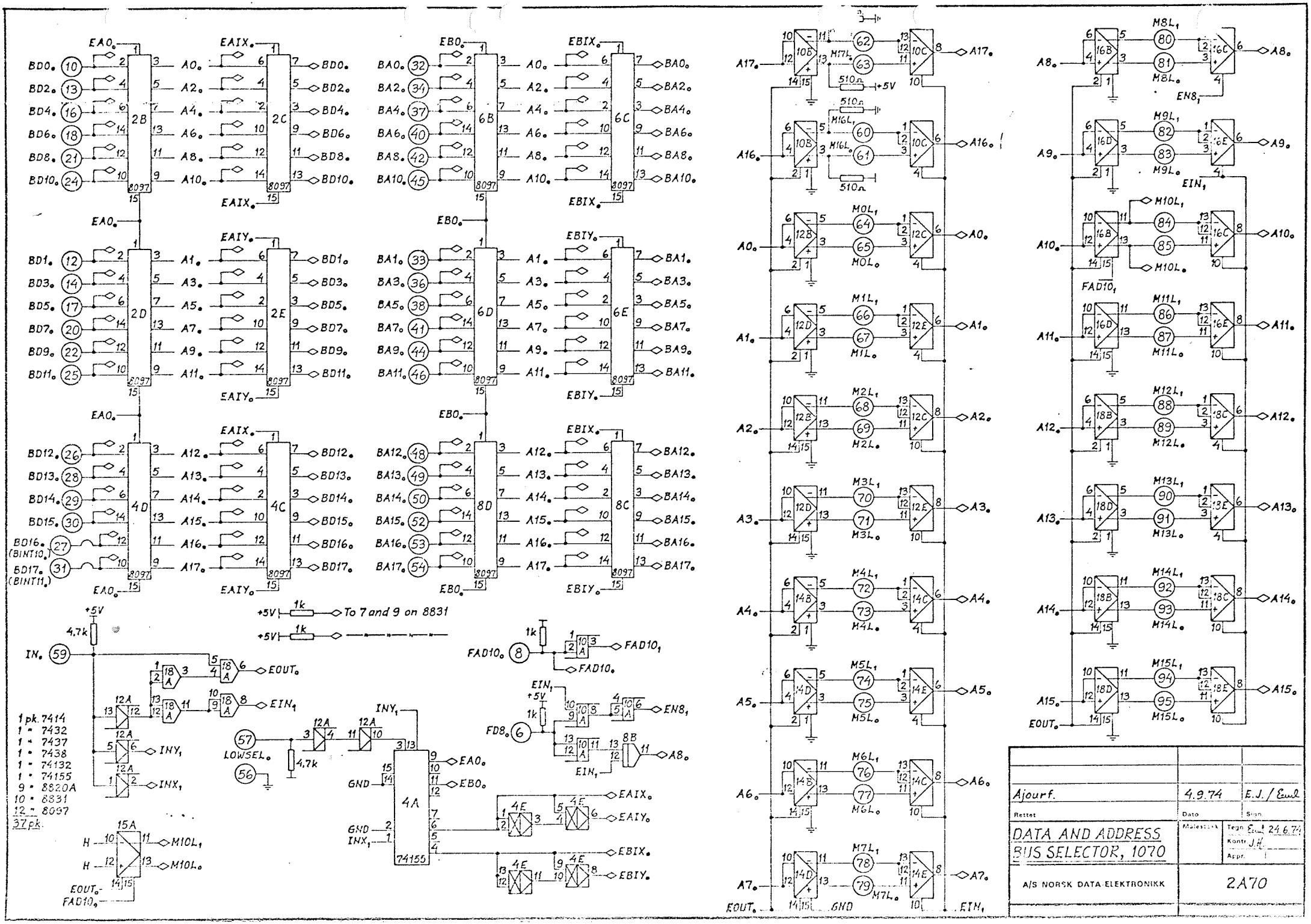
1pk 7437
1 = 7438
1 = 7442
1 = 74221
4 = 8098
1 = 8160
4 = 8831
4-32 = 6305
2 = R-pk.
15 × 680Ω

47 pk.

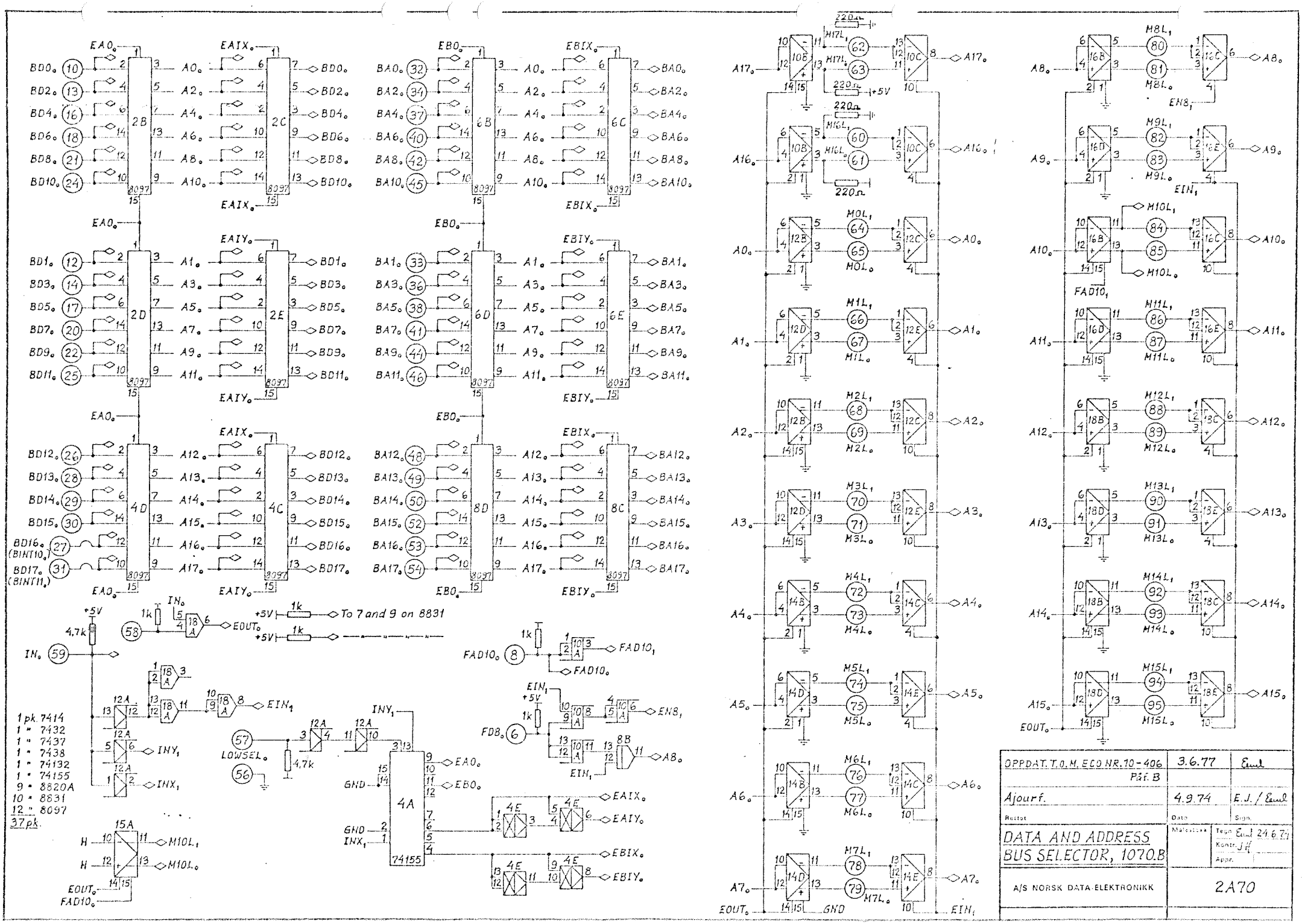
5-33 stk. sokler

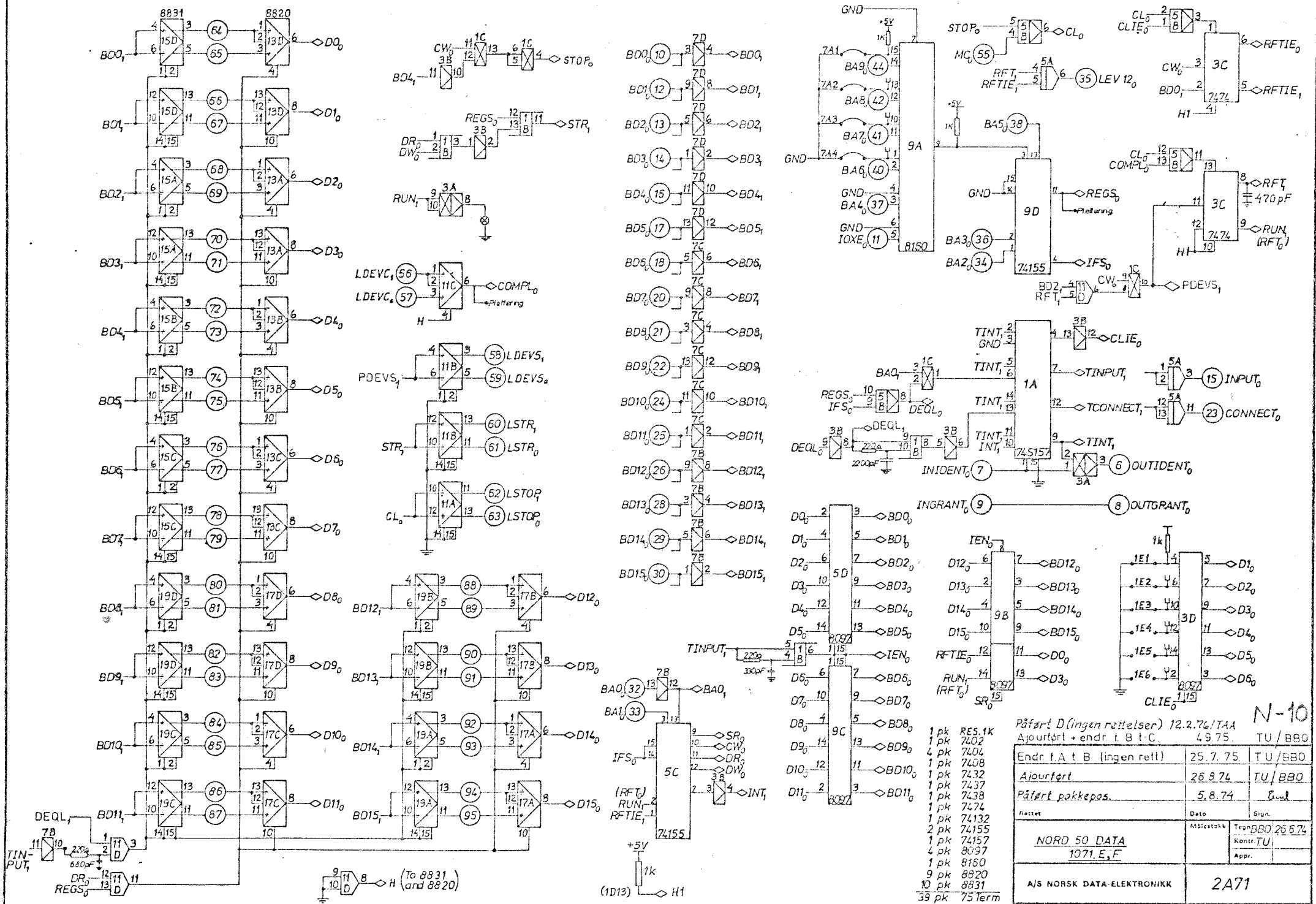
Påført II	21.6.74	E.J./Gmk
Rettet	Dato	Sign.
4K PROM MEMORY 1067/II	Målestokk	Tegn. Gmk 27.5.74 Kontr. E.J. Appr.
A/S NORSK DATA-ELEKTRONIKK		2A67

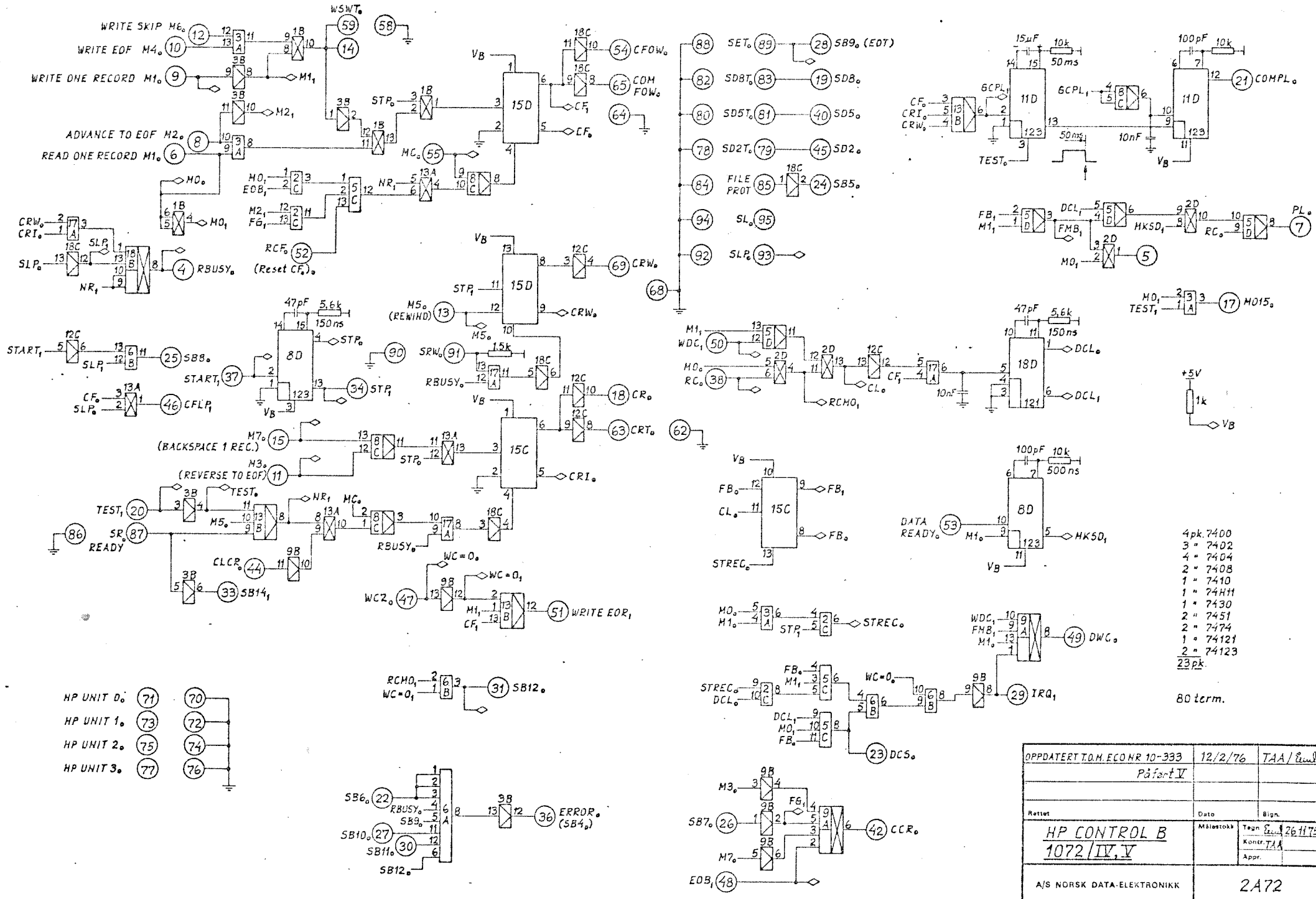

NOTE, DEV.NO.
ACM 1: 40
ACM 2: 44
ACM 3: 50
ACM 4: 54
NOTE, IDENT NO.
ACM 1: 15
ACM 2: 25
ACM 3: 40
ACM 4: 41
(Level 10)
2 pk. 7408
1 • 7432
1 • 7438
2 • 7474
1 • 74123
1 • 74132
2 • 74155
1 • 745157
5 • 74174
2 • 74175
8 • 8097
1 • 8098
2 • 8160
Totalt: 29pk.
Påført pakkepos.
20.6.74
Rettet
Dato
Sign.
ACM I/O
1068
Målestokk
Tegn
Kontr.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A68

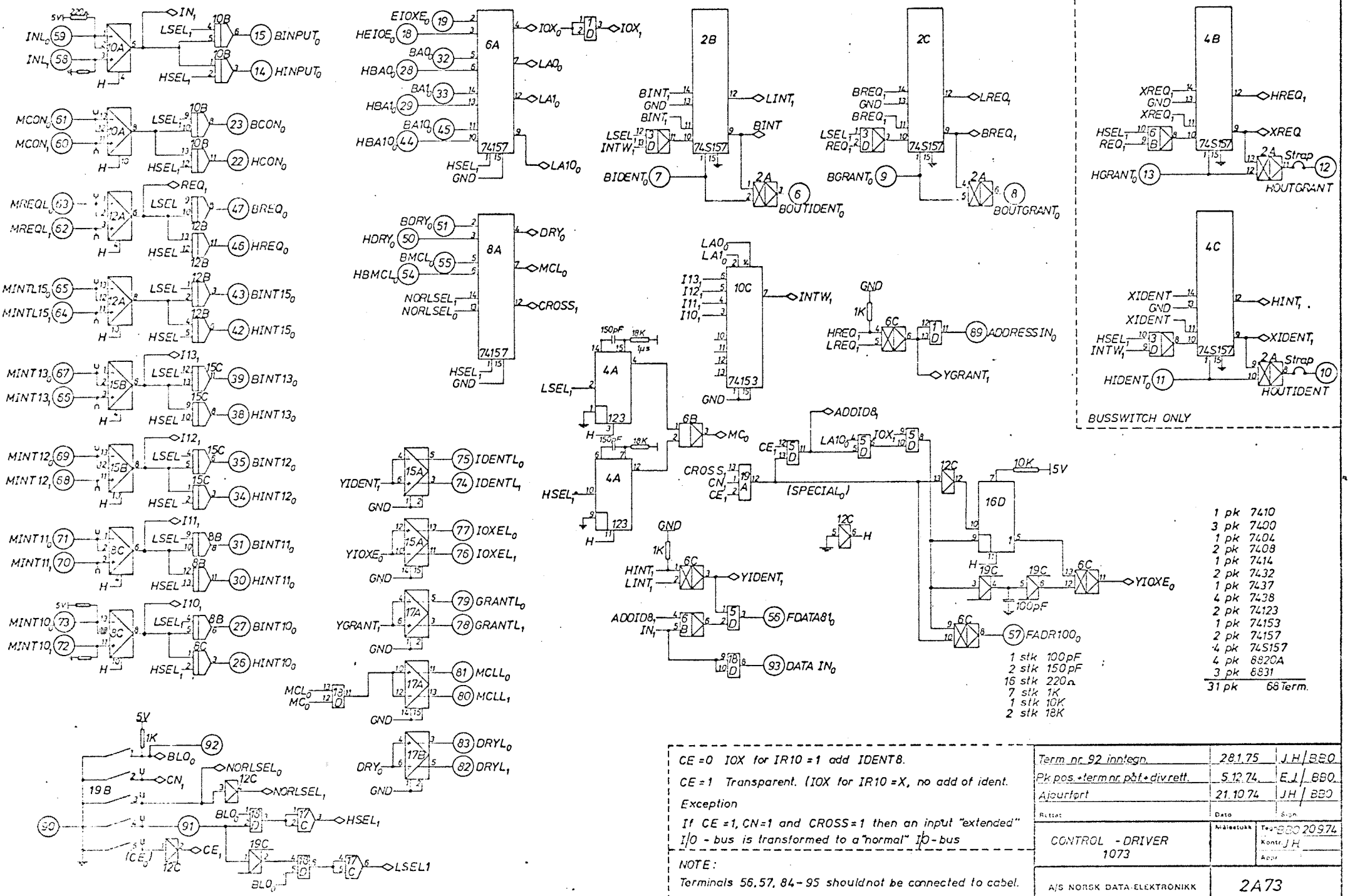


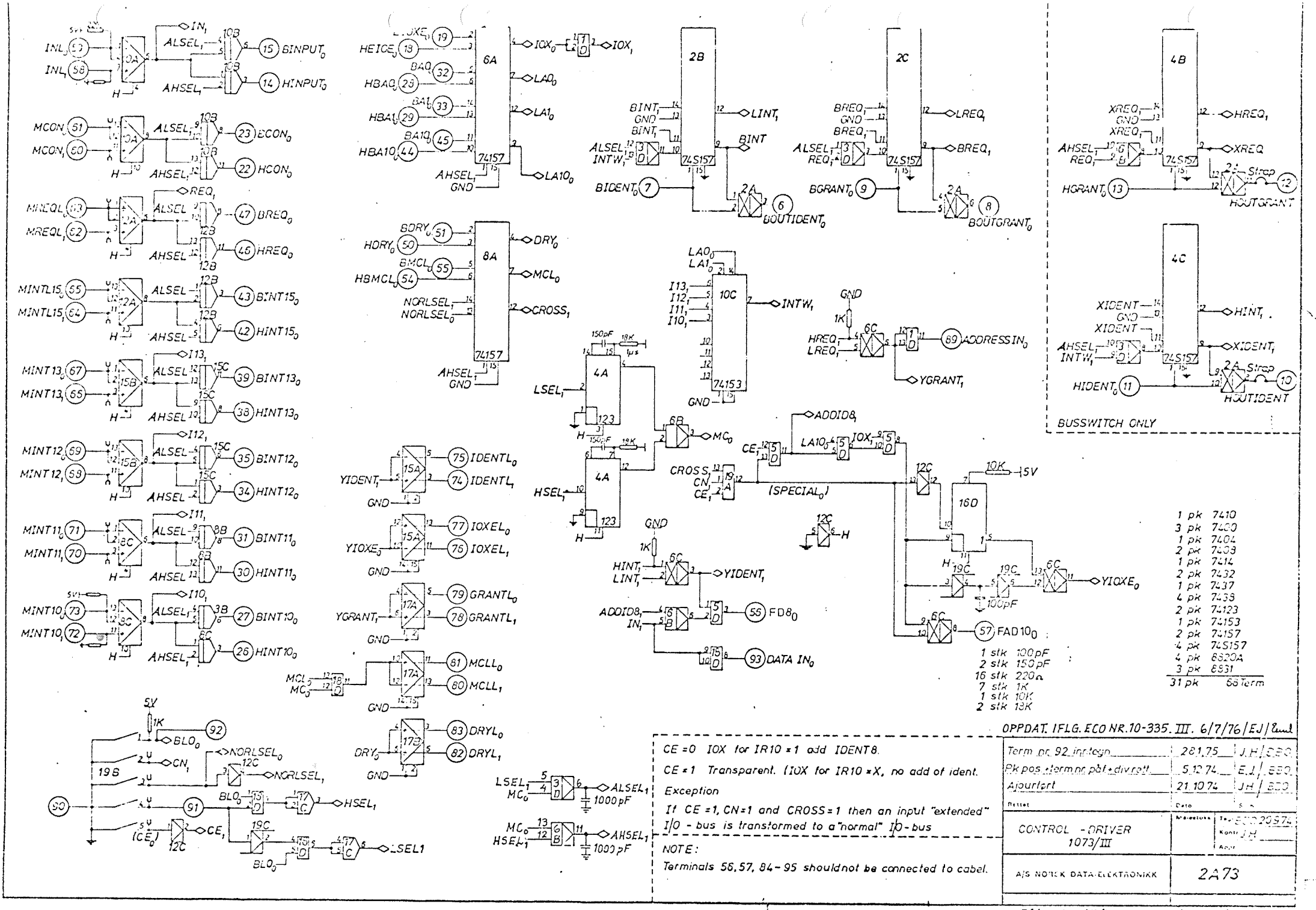
Ajourf.	4.9.74	E.J. / Eml
Rettet	Dato	Sign
DATA AND ADDRESS BUS SELECTOR, 1070	Målestokk	Tegn Eml 24.6.74 Kontr J.H. Appr.
A/S NORSK DATA ELEKTRONIKK		2A70

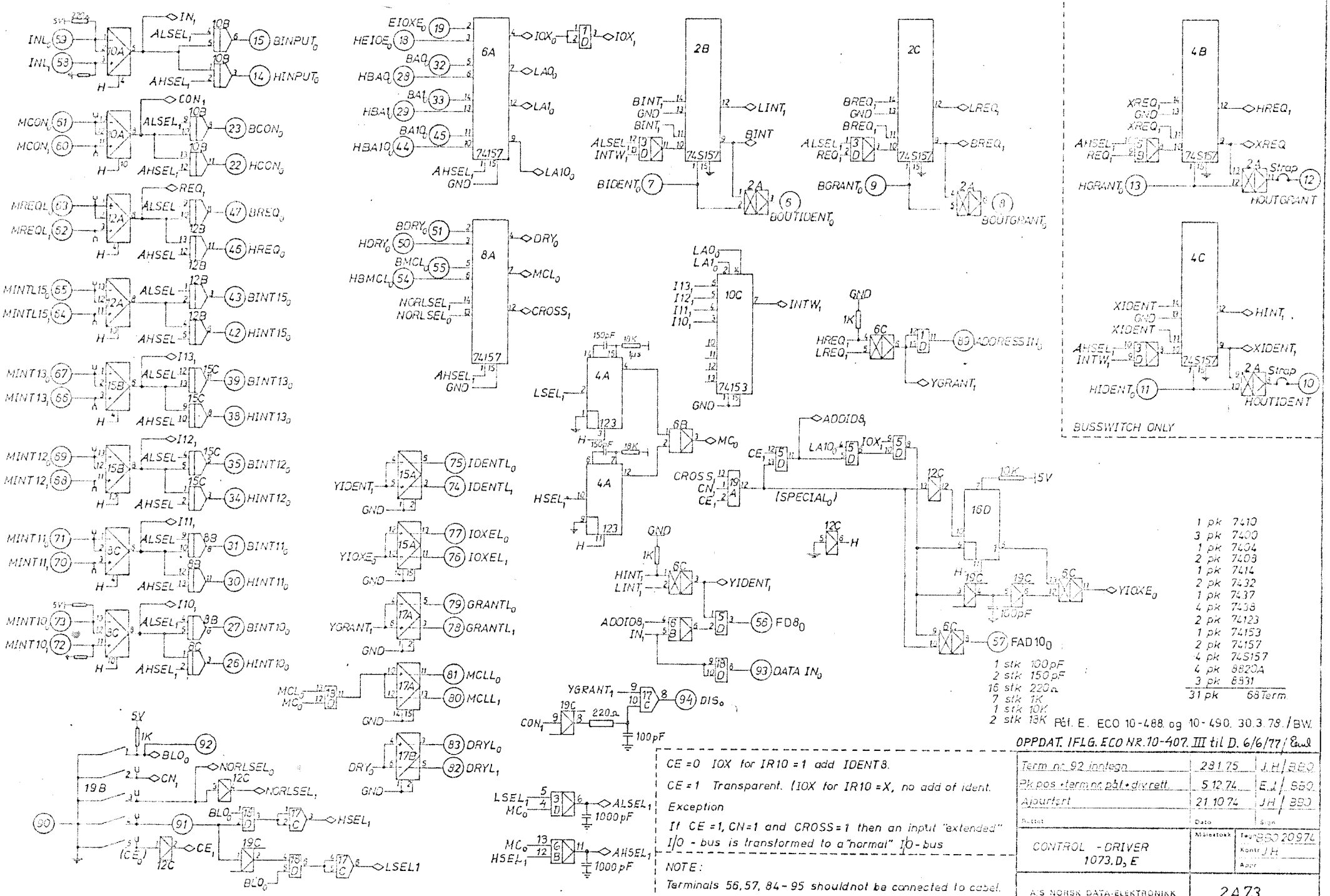

1 pk 7414
1 " 7432
1 " 7437
1 " 7438
1 " 74132
1 " 74155
9 " 8820A
10 " 8831
12 " 8097
37 pk
To 7 and 9 on 8831
DATA AND ADDRESS
BUS SELECTOR, 1070.B
A/S NORSK DATA-ELEKTRONIKK
2A70
OPPDAT. T.O.M. ECO NR. 10-406
3.6.77
Ajourf.
4.9.74
Tegn. Emil 24.6.71

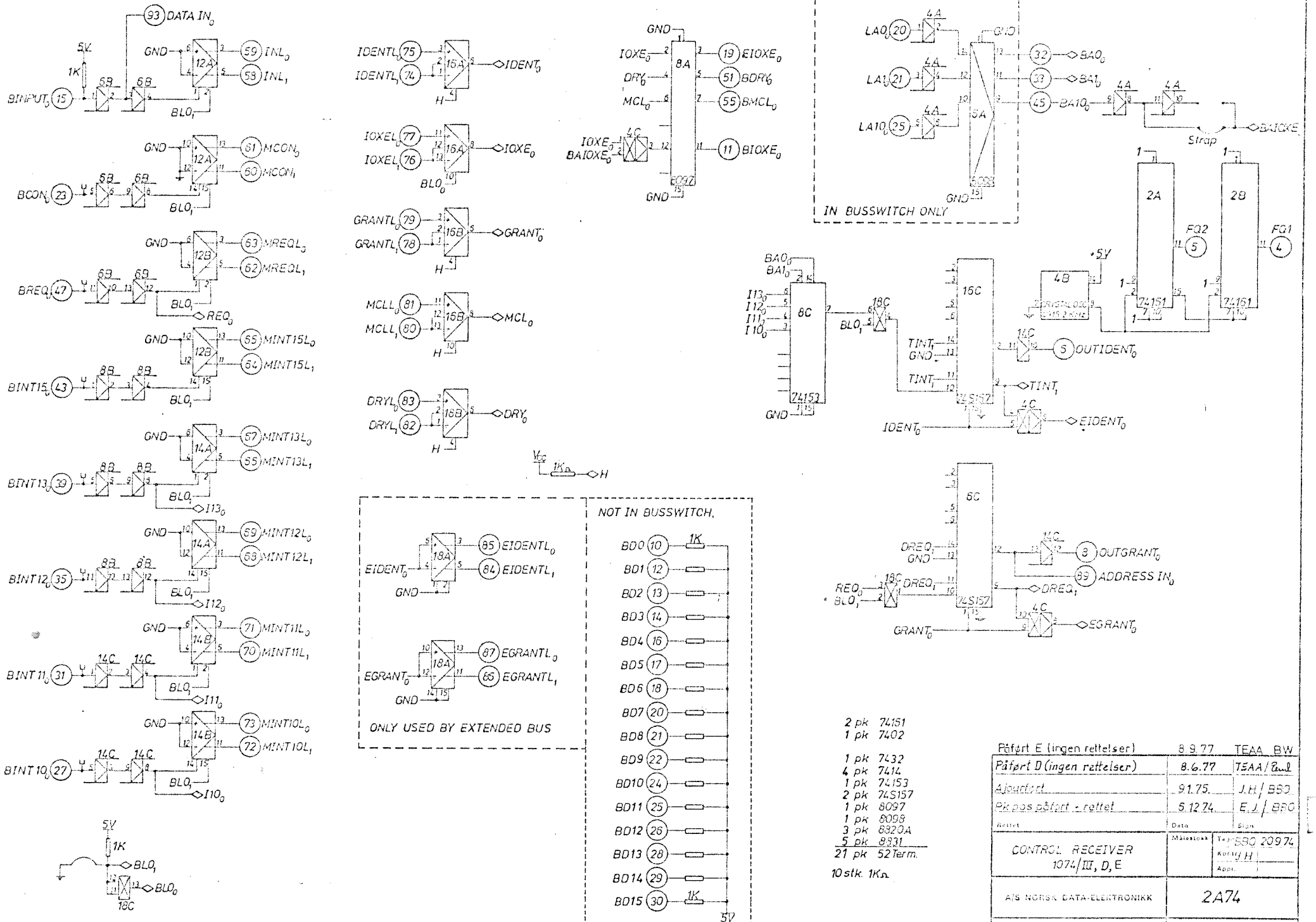

8831
8820
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
BD8
BD9
BD10
BD11
BD12
BD13
BD14
BD15
DO0
D1O
D2O
D3O
D4O
D5O
D6O
D7O
D8O
D9O
D10O
D11O
D12O
D13O
D14O
D15O
DEQL
TINPUT
DR
REGS
H
(To 8831 and 8820)
CW0
STOP0
BD4
REGS0
STR1
DR0
DW0
RUN1
LDEVC1
LDEVC2
COMPL0
Plotting
PDEVS1
LDEVS1
LDEVS2
LSTR1
LSTR0
STR1
CL0
LSTOP1
LSTOP0
BD12
BD13
BD14
BD15
BA0
BA1
IFS0
(RFT0)
RUN
RFTIE1
SR0
CW0
DR0
DW0
INT1
+5V
1k
(1D13)
H1
GND
BA9
BA8
BA7
BA6
BA4
IOXE0
9A
8150
STOP0
CL0
MC0
RFT1
RFTIE1
LEV 12
CL0
CLIE0
CW0
BD0
RFTIE0
RFTIE1
3C
7474
H1
BA5
BA3
BA2
9D
74155
REGS0
Plotting
IFS0
CL0
COMPL0
RFT1
470pF
RUN1
(RFT0)
PDEVS1
BD2
RFT1
TINT1
GND
TINT1
TINT1
INT1
1A
74S157
CLIE0
TINPUT1
TCONNECT1
TINT1
OUTIDENT0
INIDENT0
BA0
REGS0
IFS0
DEQL0
DEQL1
220Ω
200pF
INPUT0
CONNECT0
INGRANT0
OUTGRANT0
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
5D
9C
8097
IEN0
D12
D13
D14
D15
RFTIE0
RUN1
(RFT0)
SR0
9B
1k
1E1
1E2
1E3
1E4
1E5
1E6
3D
CLIE0
330pF
N-10
Påført D (ingen rettelser) 12.2.76/TAA
Ajourført + endr. f. B t. C. 49.75 TU/BBO
1 pk RES.1K
1 pk 7402
4 pk 7404
1 pk 7408
1 pk 7432
1 pk 7437
1 pk 7438
1 pk 7474
1 pk 74132
2 pk 74155
1 pk 74157
4 pk 8097
1 pk 8160
9 pk 8820
10 pk 8831
39 pk 75 Term
Endr. f. A t. B (ingen rett) 25.7.75 TU/BBO
Ajourført 26.8.74 TU/BBO
Påført pakkepos. 5.8.74 Emil
Rettet Dato Sign.
NORD 50 DATA
1071.E,F
Målestokk
Tegn. BBO 26.6.74
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A71
OPPDAT. IFLG. ECO NR. 50-19. PÅFØRT E. 31.5.76/TAA/Emil Uppdat. iflg. ECO 10-489 Påf. F. 28.4.78/BW

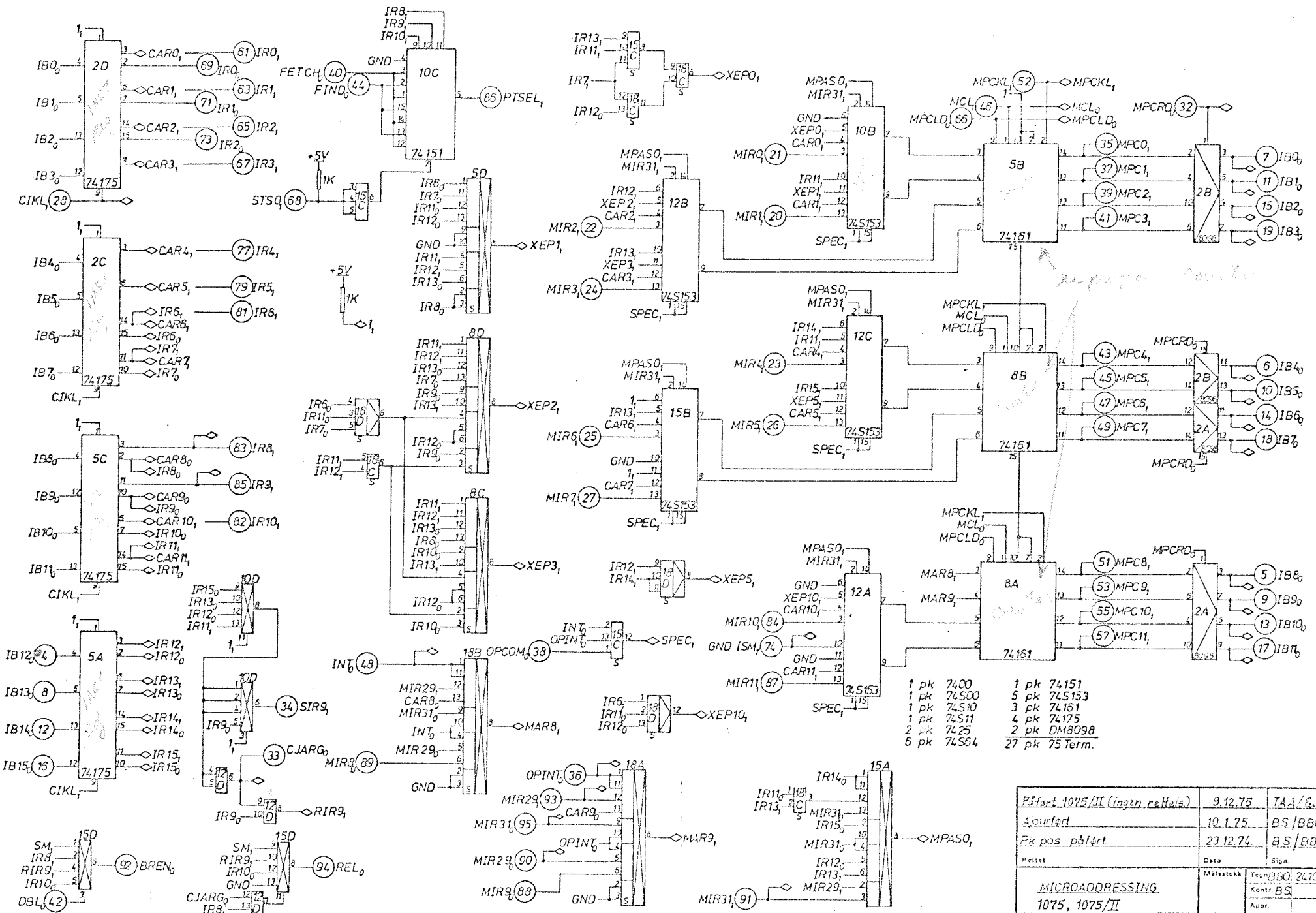

WRITE SKIP M6₀
WRITE EOF M4₀
WRITE ONE RECORD M1₀
ADVANCE TO EOF M2₀
READ ONE RECORD M1₀
WSWT₀
M1₁
M2₁
M0₀
M0₁
CRW₀
CRI₀
SLP₀
NR₁
RBUSY₀
(Reset CF₀)₀
STP₀
MC₀
EOB₁
FG₁
RCF₀
15D
CFOW₀
COM FOW₀
CF₁
CF₀
SET₀
SB9₀ (EOT)
SDBT₀
SD8₀
SD5T₀
SD5₀
SD2T₀
SD2₀
FILE PROT
SB5₀
SL₀
SLP₀
START₁
SB8₀
CFLP₁
8D
47pF
5.6k
150ns
M5₀ (REWIND)
SRW₀
1.5k
CRW₀
CR₀
CRT₀
15C
M7₀ (BACKSPACE 1 REC.)
M3₀ (REVERSE TO EOF)
TEST₀
TEST₁
SR₀
READY
CLCR₀
SB14₁
WC=0₀
WC=0₁
WC2₀
WRITE EOR₁
CRI₀
RCHO₁
SB12₀
HP UNIT 0₀
HP UNIT 1₀
HP UNIT 2₀
HP UNIT 3₀
SB6₀
SB10₀
SB11₀
ERROR₀ (SB4₀)
M1₁
WDC₁
RCHO₁
CL₀
18D
DCL₀
DCL₁
10nF
FB₀
FB₁
CL₀
STREC₀
DATA READY₀
HK5D₁
15µF
10k
50ms
6CPL₁
11D
COMPL₀
100pF
50ms
FB₁
M1₁
DCL₁
FMB₁
HK5D₁
RC₀
PL₀
MO₁
TEST₁
MO15₀
+5V
1k
STREC₀
DCL₀
WC=0₀
IRQ₁
WDC₁
FMB₁
M1₀
DWC₀
M3₀
FG₁
SB7₀
CCR₀
M7₀
EOB₁
4pk. 7400
3 " 7402
4 " 7404
2 " 7408
1 " 7410
1 " 74H11
1 " 7430
2 " 7451
2 " 7474
1 " 74121
2 " 74123
23 pk.
80 term.
OPPDATERT T.O.M. ECO NR 10-333
12/2/76
TAA/Emil
Påført V
Rettet
Dato
Sign.
HP CONTROL B
1072/IV, V
Målestokk
Tegn
Konstr. TAA
Appr.
26/11 75
A/S NORSK DATA-ELEKTRONIKK
2A72


BUSSWITCH ONLY
CE =0 IOX for IR10 =1 add IDENT8.
CE =1 Transparent. (IOX for IR10 =X, no add of ident.
Exception
If CE =1, CN=1 and CROSS=1 then an input "extended" I/O - bus is transformed to a "normal" I/O-bus
NOTE:
Terminals 56,57, 84-95 shouldnot be connected to cabel.
1 pk 7410
3 pk 7400
1 pk 7404
2 pk 7408
1 pk 7414
2 pk 7432
1 pk 7437
4 pk 7438
2 pk 74123
1 pk 74153
2 pk 74157
4 pk 74S157
4 pk 8820A
3 pk 8831
31 pk 68 Term.
1 stk 100pF
2 stk 150pF
16 stk 220Ω
7 stk 1K
1 stk 10K
2 stk 18K
Term nr 92 inntegn. 28.1.75 J.H/BBO
Pk pos.+term nr.påt+div.rett. 5.12.74 E.J/BBO
Ajourført 21.10.74 J.H/BBO
CONTROL - DRIVER 1073
A/S NORSK DATA-ELEKTRONIKK
2A73


CE =0 IOX for IR10 =1 add IDENT8.
CE =1 Transparent. (IOX for IR10 =X, no add of ident.
Exception
If CE =1, CN=1 and CROSS=1 then an input "extended" I/O - bus is transformed to a "normal" I/O - bus
NOTE:
Terminals 56, 57, 84 - 95 shouldnot be connected to cabel.
BUSSWITCH ONLY
1 pk 7410
3 pk 7400
1 pk 7404
2 pk 7408
1 pk 7414
2 pk 7432
1 pk 7437
4 pk 7438
2 pk 74123
1 pk 74153
2 pk 74157
4 pk 74S157
4 pk 8820A
3 pk 8831
31 pk 66 Term
1 stk 100pF
2 stk 150pF
16 stk 220Ω
7 stk 1K
1 stk 10K
2 stk 18K
OPPDAT. IFLG. ECO NR.10-335. III. 6/7/76/EJ/
Term pr. 92 inntegn 28.1.75 J.H/EBO
Pk pos.-term pr pbl.-diverse 5.12.74 E.J./EBO
Ajourført 21.10.74 J.H./EBO
Rettet Dato Sign
CONTROL - DRIVER 1073/III
Tegn. EBO 29.9.74
Kontr. J.H.
Godkj.
A/S NORSK DATA-ELEKTRONIKK
2A73
Påført 1073/II (ingen rettelser) 3.12.75/TAR/


BUSSWITCH ONLY
CE =0 IOX for IR10 =1 add IDENT8.
CE =1 Transparent. (IOX for IR10 =X, no add of ident.
Exception
If CE =1, CN=1 and CROSS=1 then an input "extended" I/O - bus is transformed to a "normal" IO-bus
NOTE:
Terminals 56, 57, 84-95 shouldnot be connected to cabel.
1 pk 7410
3 pk 7400
1 pk 7404
2 pk 7408
1 pk 7414
2 pk 7432
1 pk 7437
4 pk 7438
2 pk 74123
1 pk 74153
2 pk 74157
4 pk 74S157
4 pk 8820A
3 pk 8831
31 pk 68Term
1 stk 100pF
2 stk 150pF
16 stk 220Ω
7 stk 1K
1 stk 10K
2 stk 18K
Pål. E. ECO 10-488 og 10-490. 30.3.78./BW.
OPPDAT. IFLG. ECO NR. 10-407. III til D. 6/6/77/ Emil
Term nr. 92 innlegg 28.1.75 J.H./BSO
Pk pos. term nr. påf. diverse 5.12.74 E.J./BSO
Ajourført 21.10.74 J.H./BSO
Dato Sign
CONTROL - DRIVER
1073. D, E
Tegn BSO 20.9.74
Konstr J.H.
A/S NORSK DATA-ELEKTRONIKK
2A73
Påført 1073/II (inner-tabellen) 9.12.75/TAA/E...


DATA IN0
INL0
INL1
BINPUT0
BLO1
MCON0
MCON1
BCON0
MREQL0
MREQL1
BREQ0
REQ0
MINT15L0
MINT15L1
BINT15
MINT13L0
MINT13L1
BINT13
I130
MINT12L0
MINT12L1
BINT12
I120
MINT11L0
MINT11L1
BINT11
I110
MINT10L0
MINT10L1
BINT10
I100
5V
1K
BLO0
18C
IDENT0
IOXE0
GRANT0
MCL0
DRY0
H
EIDENTL0
EIDENTL1
EGRANTL0
EGRANTL1
ONLY USED BY EXTENDED BUS
NOT IN BUSWITCH.
BD00
BD01
BD02
BD03
BD04
BD05
BD06
BD07
BD08
BD09
BD10
BD11
BD12
BD13
BD14
BD15
EIOXE0
BDRY0
BMCL0
BIOXE0
IN BUSWITCH ONLY
LA00
LA01
LA010
BA00
BA01
BA010
BAIOXE0
Strap
OUTIDENT0
TINT1
EIDENT0
OUTGRANT0
ADDRESS IN0
DREQ1
EGRANT0
2A
2B
FQ2
FQ1
74151
74S157
2 pk 74151
1 pk 7402
1 pk 7432
4 pk 7414
1 pk 74153
2 pk 74S157
1 pk 8097
1 pk 8098
3 pk 8320A
5 pk 8331
21 pk 52Term.
10 stk. 1KΩ
Påført E (ingen rettelser) 8.9.77 TEAA BW
Påført D (ingen rettelser) 8.6.77 TEAA/BWL
Ajourført 91.75. J.H/BSO
Pk pos påført - rettet 5.12.74. E.J/BRO
Rettet Dato Sign
CONTROL RECEIVER 1074/III, D, E
Målestokk
Tegn BSO 20974
Kontr H
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A74



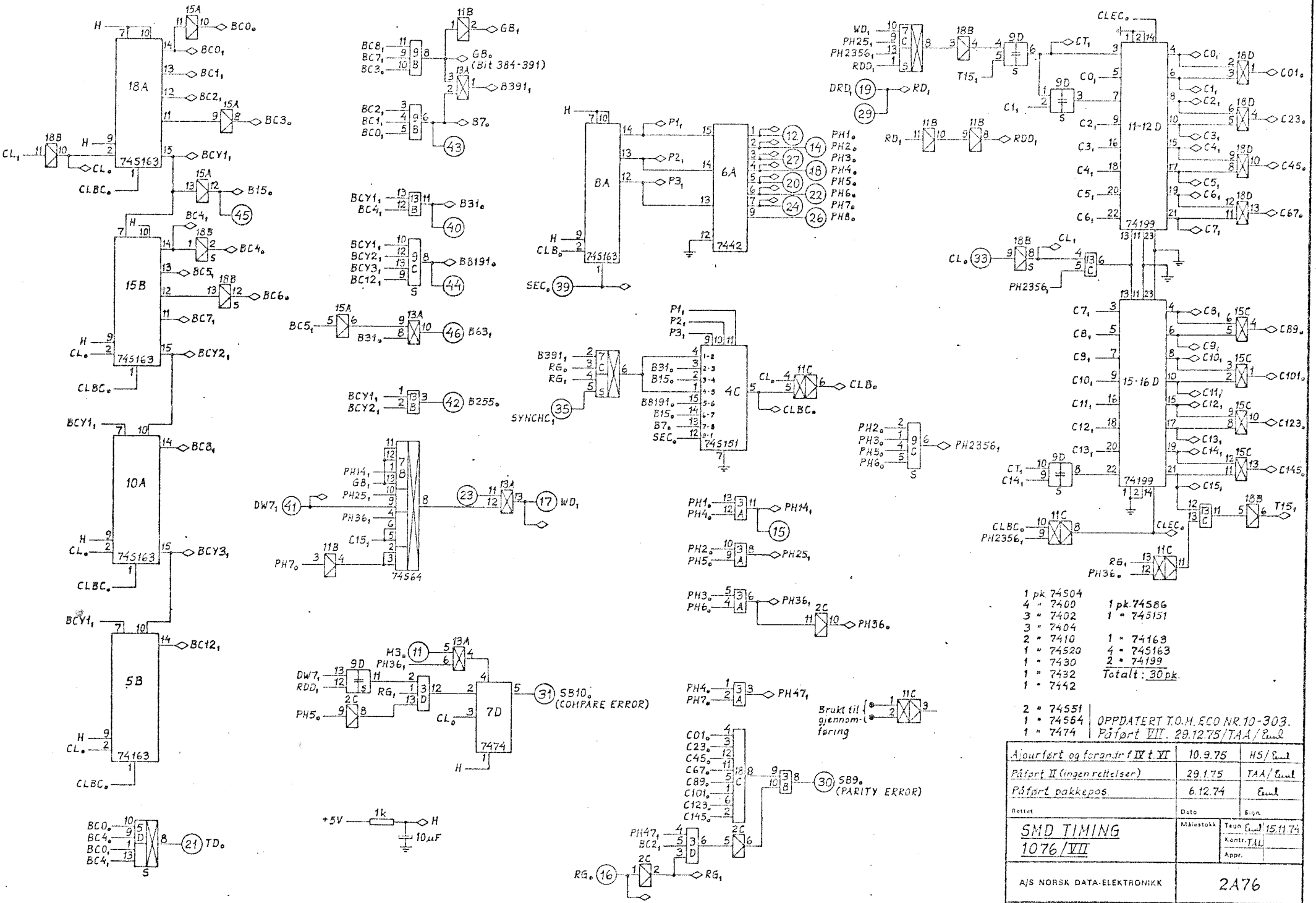
Påført 1075/II (ingen retteis)	9.12.75	TAA/Gwk
Kontrollert	12.1.75	BS/BBO
Pk pos. påført	23.12.74	BS/BBO
Rettet	Dato	Sign.

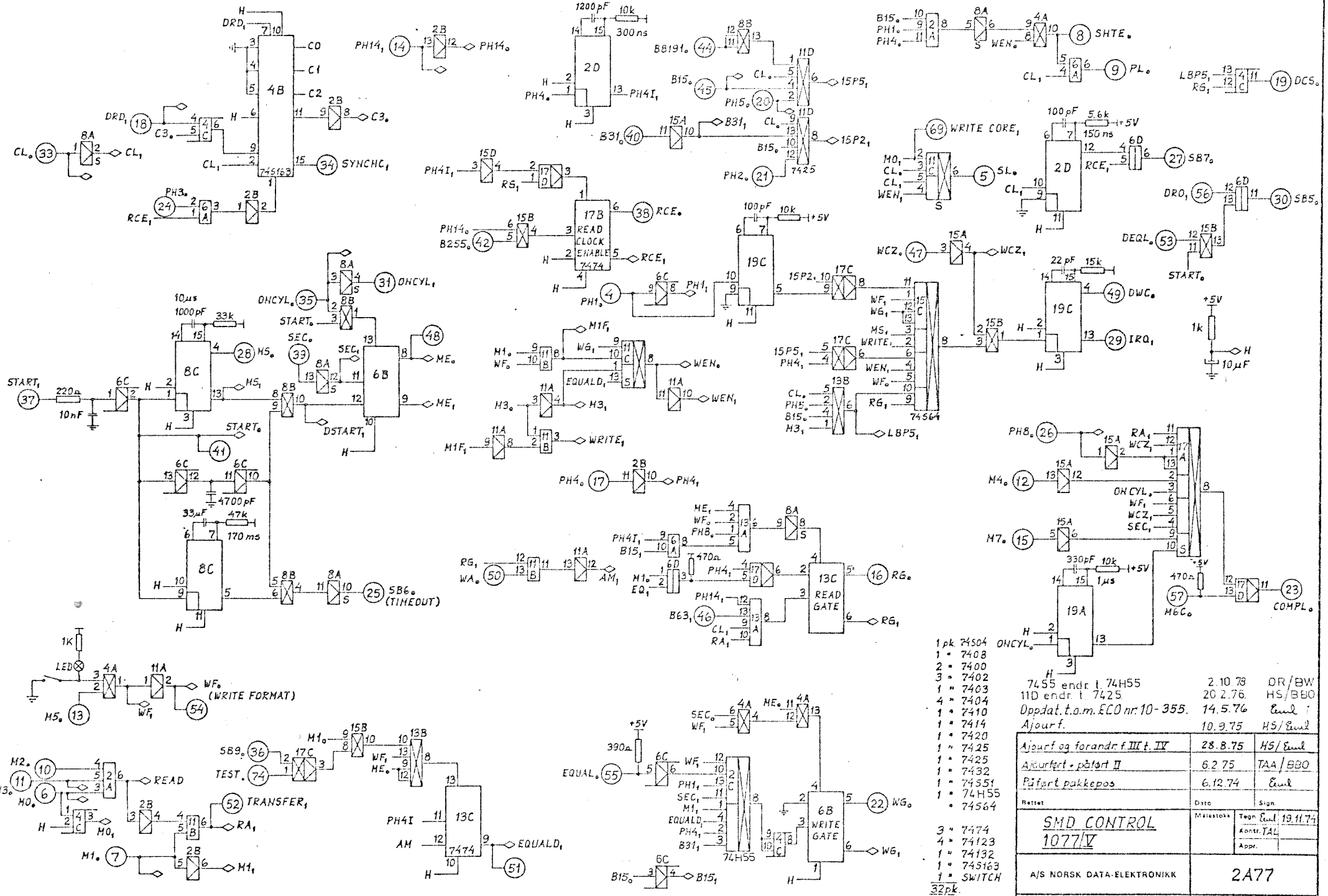
MICROADDRESSING
1075, 1075/II

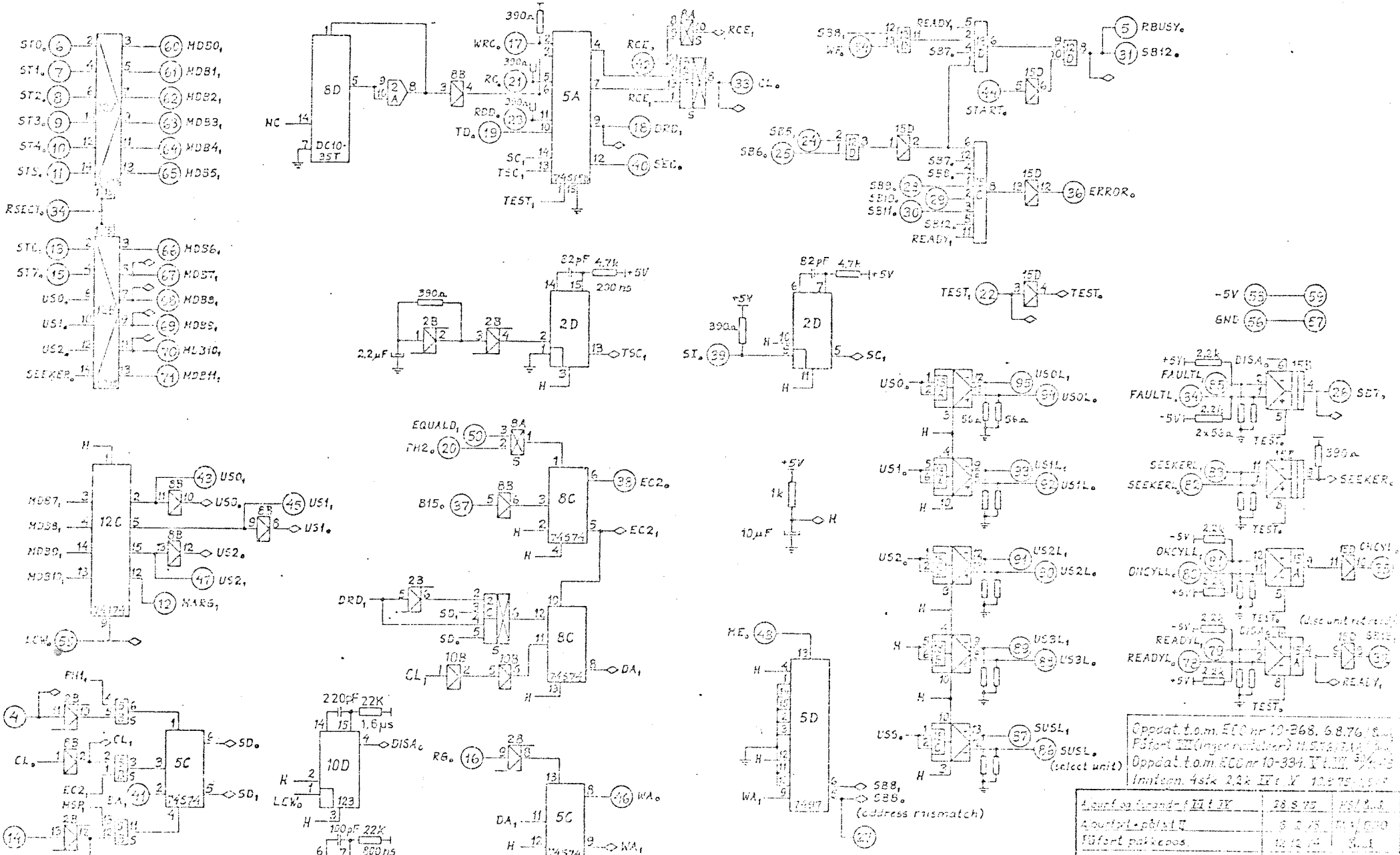
Målestokk | Tegnet BBO 24.10.74 | Kontr. BS | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A75


18A
15B
10A
5B
BA
6A
7442
4C
745151
11-12 D
15-16 D
74199
745163
74163
74564
7D
7474
2C
74504
7400
7402
7404
7410
74520
7430
7432
7442
74551
74564
7474
74586
745151
74163
745163
74199
OPPDATERT T.O.H. ECO NR. 10-303.
Påført VII. 29.12.75/TAA/Eml
Ajourført og forandr. f. IV t. VI 10.9.75 HS/Eml
Påført II (ingen rettelser) 29.1.75 TAA/Eml
Påført pakkepos. 6.12.74 Eml
SMD TIMING
1076/VII
A/S NORSK DATA-ELEKTRONIKK
2A76


4B
2D
17B READ CLOCK ENABLE 7474
19C
8C
6B
13C READ GATE
6B WRITE GATE
19A
74564
74H55
7425
74S163
SHTE.
PL.
DCS.
WRITE CORE
SL.
SB7.
SB5.
DWC.
IRQ.
RCE.
WCZ.
SYNCHC.
ONCYL.
START.
DSTART.
ME.
HS.
5B6. (TIMEDOUT)
WF. (WRITE FORMAT)
READ
TRANSFER
RG.
WG.
COMPL.
M6C.
1 pk. 74S04
1 " 7408
2 " 7400
3 " 7402
1 " 7403
4 " 7404
1 " 7410
1 " 7414
1 " 7420
1 " 7425
1 " 7425
1 " 7432
1 " 74S51
1 " 74H55
" 74564
3 " 7474
4 " 74123
1 " 74132
1 " 74S163
1 " SWITCH
32pk.
7455 endr. t. 74H55 2.10.78 DR/BW
11D endr. t. 7425 20.2.76. HS/BBO
Oppdat. t.o.m. ECO nr. 10-355. 14.5.76 Emil
Ajour f. 10.9.75 HS/Emil
Ajour og forandr. f. III t. IV 28.8.75 HS/Emil
Ajourført - påført II 6.2.75 TAA/BBO
Påført pakkepos 6.12.74 Emil
Rettet Dato Sign.
SMD CONTROL 1077/V
Målestokk
Tegn. Emil 19.11.74
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A77



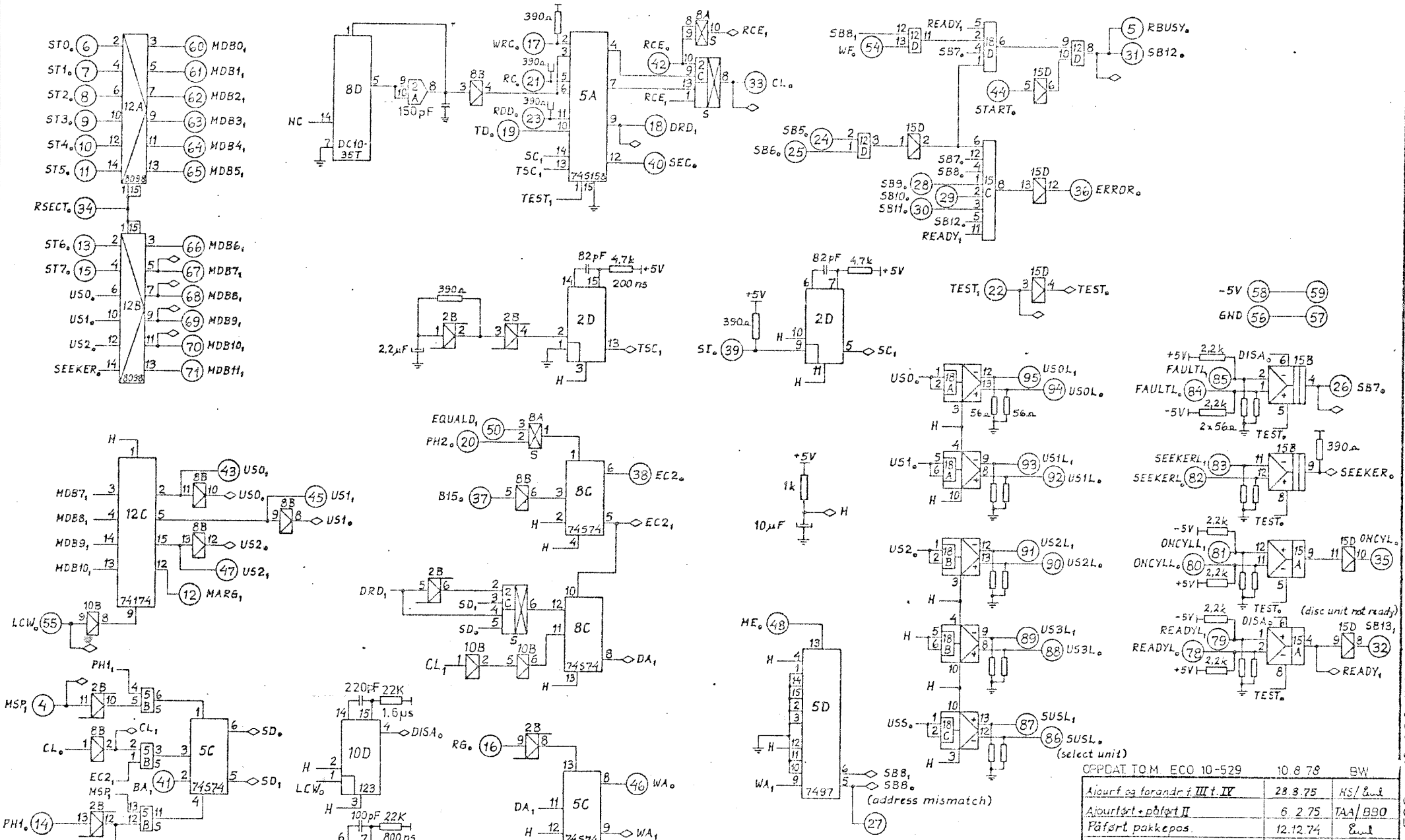
Oppdat. t.o.m. ECO nr 10-368, 6.8.76 [illegible]
Fjernet III (linjer modifisert) H.Strø 21.10.76
Oppdat. t.o.m. ECO nr 10-334. I, II, III, V, VI, IX
Innsatt 4stk 2.2k IV, X 13.9.77 [illegible]

Konstruert/tegnet	28.8.75	[illegible]
Kontrollert	5.2.75	[illegible]
Godkjent	12.12.75	[illegible]
Endret	[illegible]	[illegible]

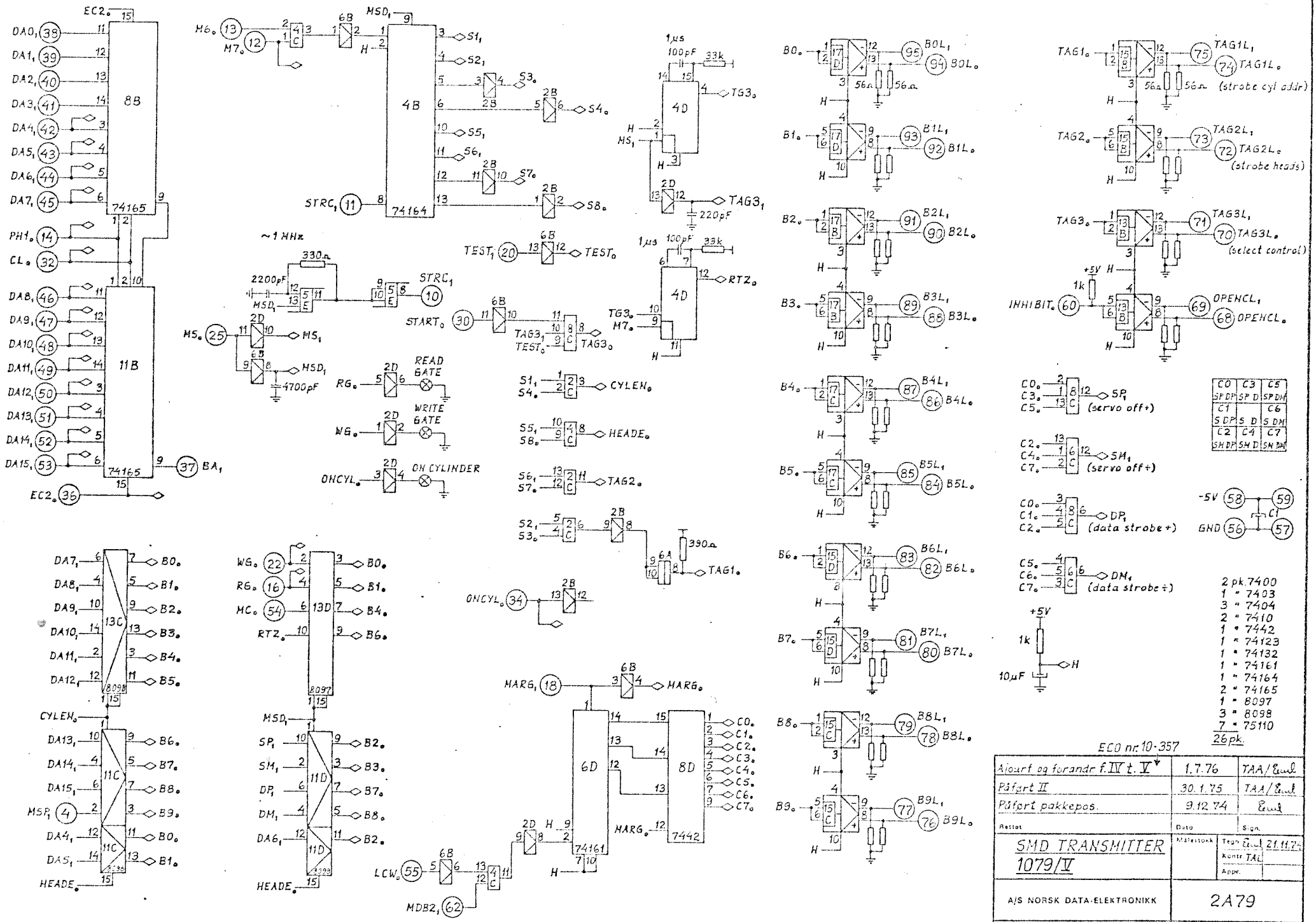
SMD RECEIVER
10 kort VI, VII, IX

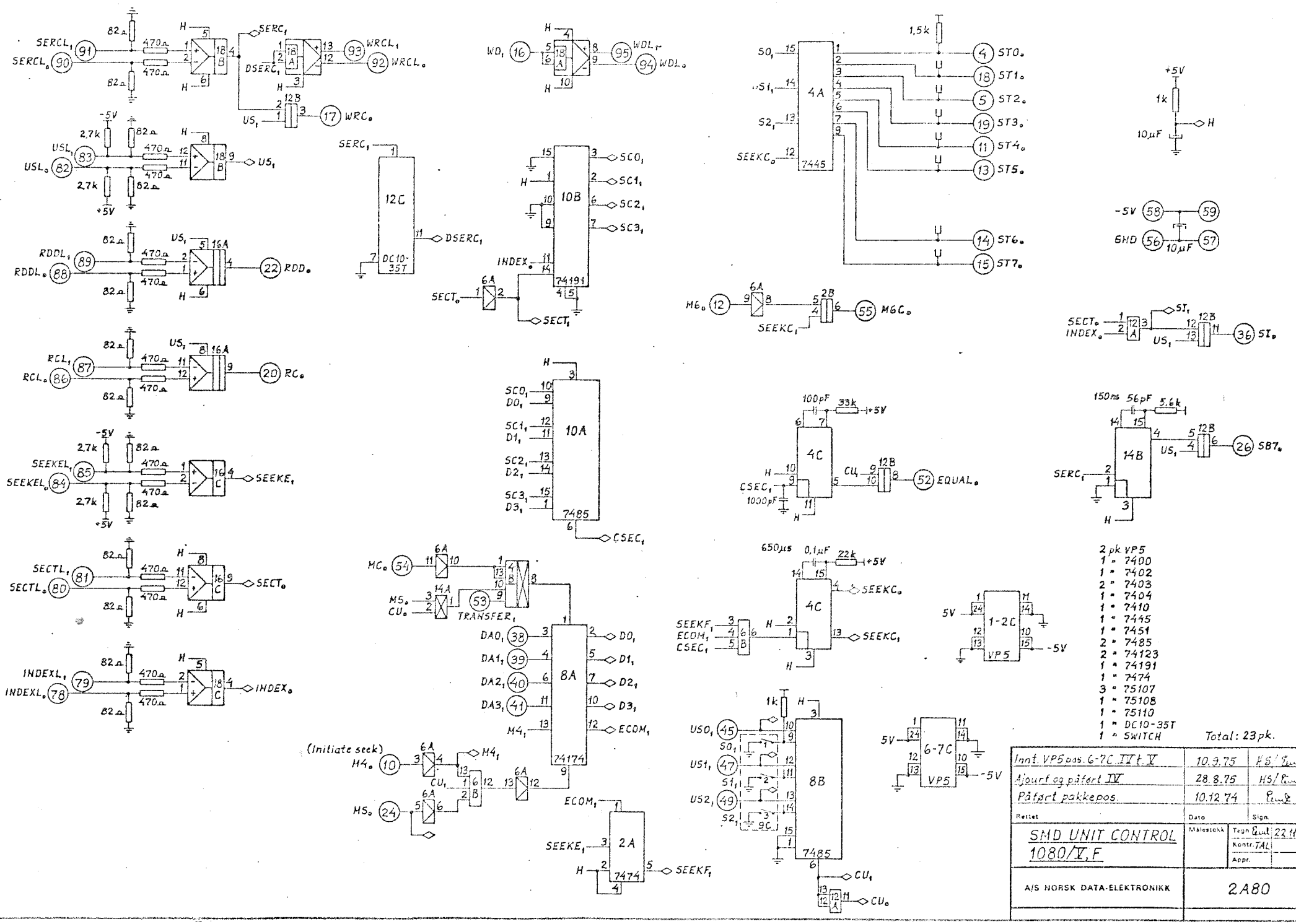
A/S NORSK DATA ELEKTRONIKK

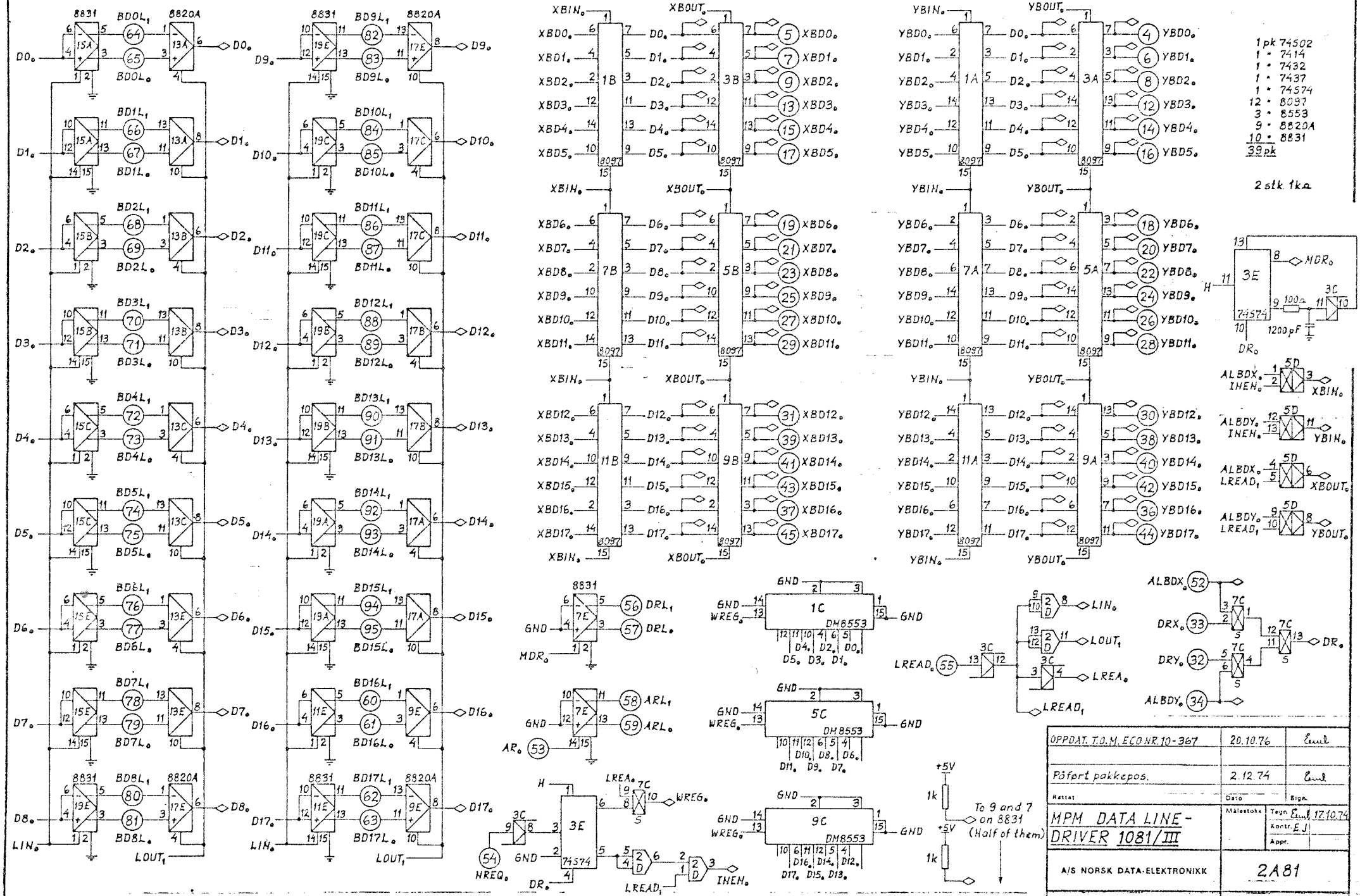
2173



Oppdat. t.o.m. ECO nr. 10-388. H til J. 22/12/77/Emil


EC2₀
DA0₁ (38)
DA1₁ (39)
DA2₁ (40)
DA3₁ (41)
DA4₁ (42)
DA5₁ (43)
DA6₁ (44)
DA7₁ (45)
8B
74165
PH1₀ (14)
CL₀ (32)
DA8₁ (46)
DA9₁ (47)
DA10₁ (48)
DA11₁ (49)
DA12₁ (50)
DA13₁ (51)
DA14₁ (52)
DA15₁ (53)
11B
74165
BA₁ (37)
EC2₀ (36)
M6₀ (13)
M7₀ (12)
6B
MSD₁
4B
74164
STRC₁ (11)
S1₁
S2₁
S3₀
S4₀
S5₁
S6₁
S7₀
S8₀
~1 MHz
330Ω
2200pF
MSD₁
STRC₁ (10)
MS₀ (25)
2D
MS₁
MSD₁
4700pF
TEST₁ (20)
TEST₀
START₀ (30)
TAG3₁
TEST₀
TAG3₀
RG₀
READ GATE
WG₀
WRITE GATE
ONCYL₀
ON CYLINDER
1µs
100pF
33k
4D
TG3₀
MS₁
220pF
TAG3₁
TG3₀
M7₀
RTZ₀
S1₁
S4₀
CYLEN₀
S5₁
S8₀
HEADE₀
S6₁
S7₀
TAG2₀
S2₁
S3₀
390Ω
TAG1₀
ONCYL₀ (34)
B0₀
B1₀
B2₀
B3₀
B4₀
B5₀
B6₀
B7₀
B8₀
B9₀
B0L₁ (95)
B0L₀ (94)
B1L₁ (93)
B1L₀ (92)
B2L₁ (91)
B2L₀ (90)
B3L₁ (89)
B3L₀ (88)
B4L₁ (87)
B4L₀ (86)
B5L₁ (85)
B5L₀ (84)
B6L₁ (83)
B6L₀ (82)
B7L₁ (81)
B7L₀ (80)
B8L₁ (79)
B8L₀ (78)
B9L₁ (77)
B9L₀ (76)
56Ω
TAG1₀
TAG1L₁ (75)
TAG1L₀ (74)
(strobe cyl addr)
TAG2₀
TAG2L₁ (73)
TAG2L₀ (72)
(strobe heads)
TAG3₀
TAG3L₁ (71)
TAG3L₀ (70)
(select control)
+5V
1k
INHIBIT₀ (60)
OPENCL₁ (69)
OPENCL₀ (68)
C0₀
C3₀
C5₀
5P₁
(servo off+)
C2₀
C4₀
C7₀
5M₁
(servo off+)
C0₀
C1₀
C2₀
DP₁
(data strobe+)
C5₀
C6₀
C7₀
DM₁
(data strobe÷)
+5V
1k
10µF
-5V (58)
GND (56)
C1
(59)
(57)
C0 C3 C5
SP DP SP D SP DM
C1 C6
S DP S D S DM
C2 C4 C7
SM DP SM D SN DM
2 pk. 7400
1 = 7403
3 = 7404
2 = 7410
1 = 7442
1 = 74123
1 = 74132
1 = 74161
1 = 74164
2 = 74165
1 = 8097
3 = 8098
7 = 75110
26 pk.
DA7₁
DA8₁
DA9₁
DA10₁
DA11₁
DA12₁
13C
8098
B0₀
B1₀
B2₀
B3₀
B4₀
B5₀
CYLEN₀
DA13₁
DA14₁
DA15₁
11C
B6₀
B7₀
B8₀
MSP₁ (4)
DA4₁
DA5₁
11C
B9₀
B0₀
B1₀
HEADE₀
WG₀ (22)
RG₀ (16)
MC₀ (54)
RTZ₀ (10)
13D
8097
B0₀
B1₀
B4₀
B6₀
MSD₁
SP₁
SM₁
DP₁
DM₁
DA6₁
11D
B2₀
B3₀
B7₀
B8₀
B2₀
8098
HEADE₀
MARG₁ (18)
MARG₀
6D
74161
8D
7442
CO₀
C1₀
C2₀
C3₀
C4₀
C5₀
C6₀
C7₀
MARG₀
LCW₀ (55)
MDB2₁ (62)
ECO nr. 10-357
Ajourf og forandr f. IV t. V
1.7.76
TAA/Emil
Påført II
30.1.75
TAA/Emil
Påført pakkepos.
9.12.74
Emil
Rettet
Dato
Sign.
SMD TRANSMITTER
1079/V
Målestokk
Tegn Emil 21.11.74
Kontr TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A79


SERCL1 (91)
SERCL0 (90)
82Ω
470Ω
SERC1
DSERC1
WRCL1 (93)
WRCL0 (92)
US1
WRC0 (17)
USL1 (83)
USL0 (82)
2,7k
-5V
+5V
US1
RDDL1 (89)
RDDL0 (88)
RDD0 (22)
RCL1 (87)
RCL0 (86)
RC0 (20)
SEEKEL1 (85)
SEEKEL0 (84)
SEEKE1
SECTL1 (81)
SECTL0 (80)
SECT0
INDEXL1 (79)
INDEXL0 (78)
INDEX0
WD1 (16)
WDL1 (95)
WDL0 (94)
12C
DC10-35T
DSERC1
10B
SCO1
SC11
SC21
SC31
INDEX0
74191
SECT0
SECT1
10A
7485
CSEC1
MC0 (54)
MS0
CU0
TRANSFER1
8A
DA0 (38)
DA1 (39)
DA2 (40)
DA3 (41)
M41
74174
DO1
D11
D21
D31
ECOM1
(Initiate seek)
M40 (10)
M41
MS0 (24)
CU1
2A
SEEKE1
7474
SEEKF1
SO1 (15)
US1 (14)
S21 (13)
SEEKC0 (12)
4A
7445
1,5k
ST0 (4)
ST1 (18)
ST2 (5)
ST3 (19)
ST4 (11)
ST5 (13)
ST6 (14)
ST7 (15)
M60 (12)
SEEKC1
M6C0 (55)
1k
10µF
-5V (58)
GND (56)
(59)
(57)
SECT0
INDEX0
SI1
US1
SI0 (36)
100pF
33k
4C
CSEC1
1000pF
CU1
EQUAL0 (52)
150ns
56pF
5,6k
14B
SERC1
SB70 (26)
650µs
0,1µF
22k
SEEKF1
ECOM1
CSEC1
SEEKC0
SEEKC1
8B
USO1 (45)
US11 (47)
US21 (49)
SO1
S11
S21
7485
CU1
CU0
1-2C
VP5
6-7C
2 pk VP5
1 " 7400
1 " 7402
2 " 7403
1 " 7404
1 " 7410
1 " 7445
1 " 7451
2 " 7485
2 " 74123
1 " 74191
1 " 7474
3 " 75107
1 " 75108
1 " 75110
1 " DC10-35T
1 " SWITCH
Total: 23pk.
Innf. VP5 pos. 6-7C IV t. V
10.9.75
HS/Emd
Ajourf og påført IV
28.8.75
HS/Emd
Påført pakkepos.
10.12.74
Emd
Rettet
Dato
Sign.
SMD UNIT CONTROL
1080/V,F
Målestokk
Tegn Emd 22.11.74
Kontr. TAL
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A80
Påført F (ingen rettelser) 8/6/77/TEAA/Emd



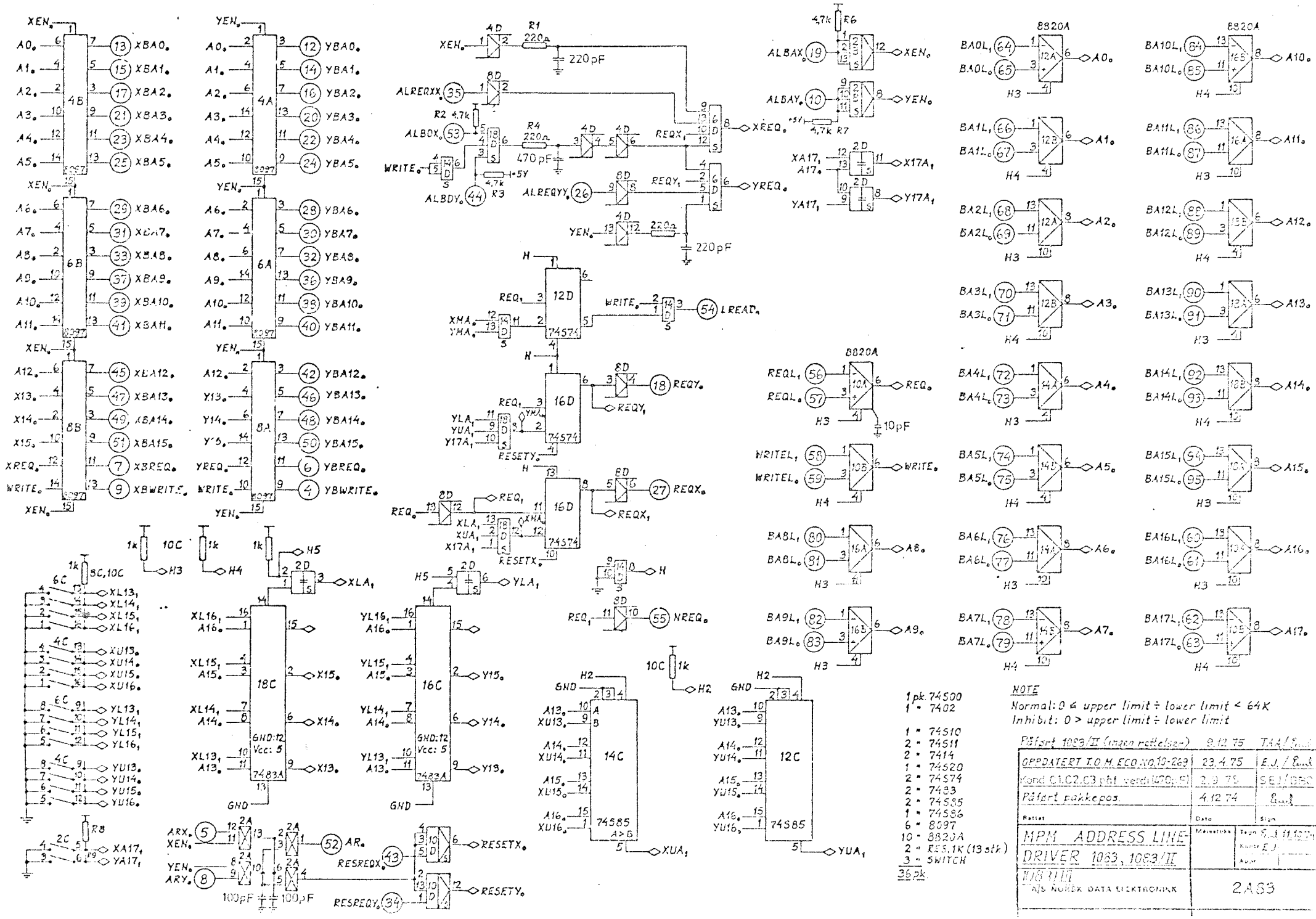
OPPDAT. T.O.M. ECO NR. 10-367	20.10.76	Emil
Påført pakkepos.	2.12.74	Emil
Rettet	Dato	Sign.

MPM DATA LINE - DRIVER 1081/III

Målestokk | Tegn Emil 17.10.74 | Kontr. E J | Appr.

A/S NORSK DATA-ELEKTRONIKK

2A81



1 pk. 74500
1 " 7402

1 " 74S10
2 " 74S11
2 " 7414
1 " 74S20
2 " 74S74
2 " 7483
2 " 74S85
1 " 74S86
6 " 8097
10 " 8820A
2 " RES. 1K (13 stk)
3 " SWITCH
36 pk

NOTE
Normal: 0 ≤ upper limit ÷ lower limit < 64K
Inhibit: 0 > upper limit ÷ lower limit

Påført 1083/II (mapro rettelser) 9.12.75 TAA/Sund

OPPDATERT TO M.ECO NO.10-269	23.4.75	EJ / Sund
Kond. C1,C2,C3 økt verdi (470pF)	3.9.75	SEJ/BBO
Påført pakkepos.	4.12.74	Sund
Rettet	Dato	Sign

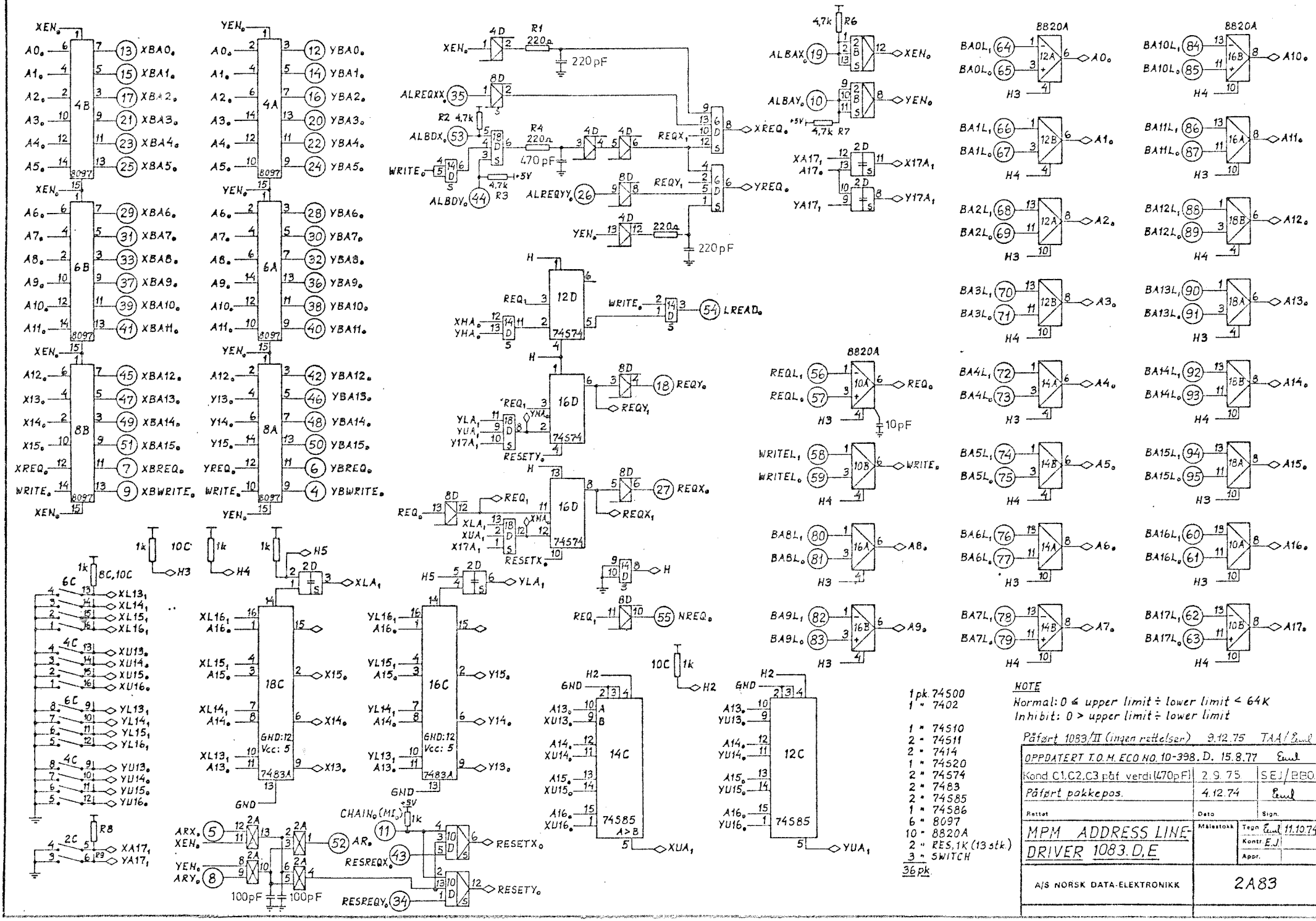
MPM ADDRESS LINE DRIVER 1063, 1083/II

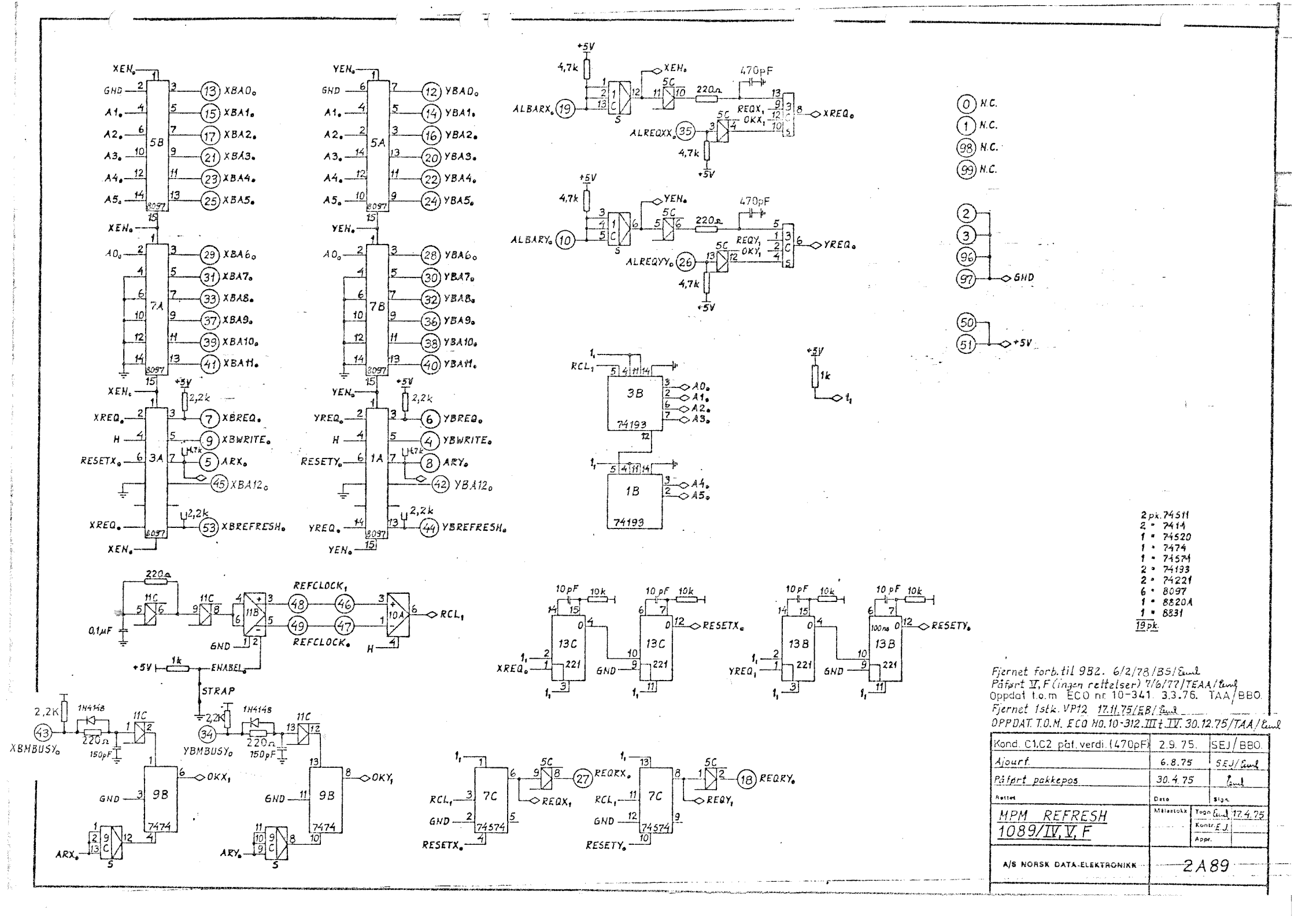
Maskinkode | Tegn. Nr. 11.1974 | Konstr. EJ | Kontr.

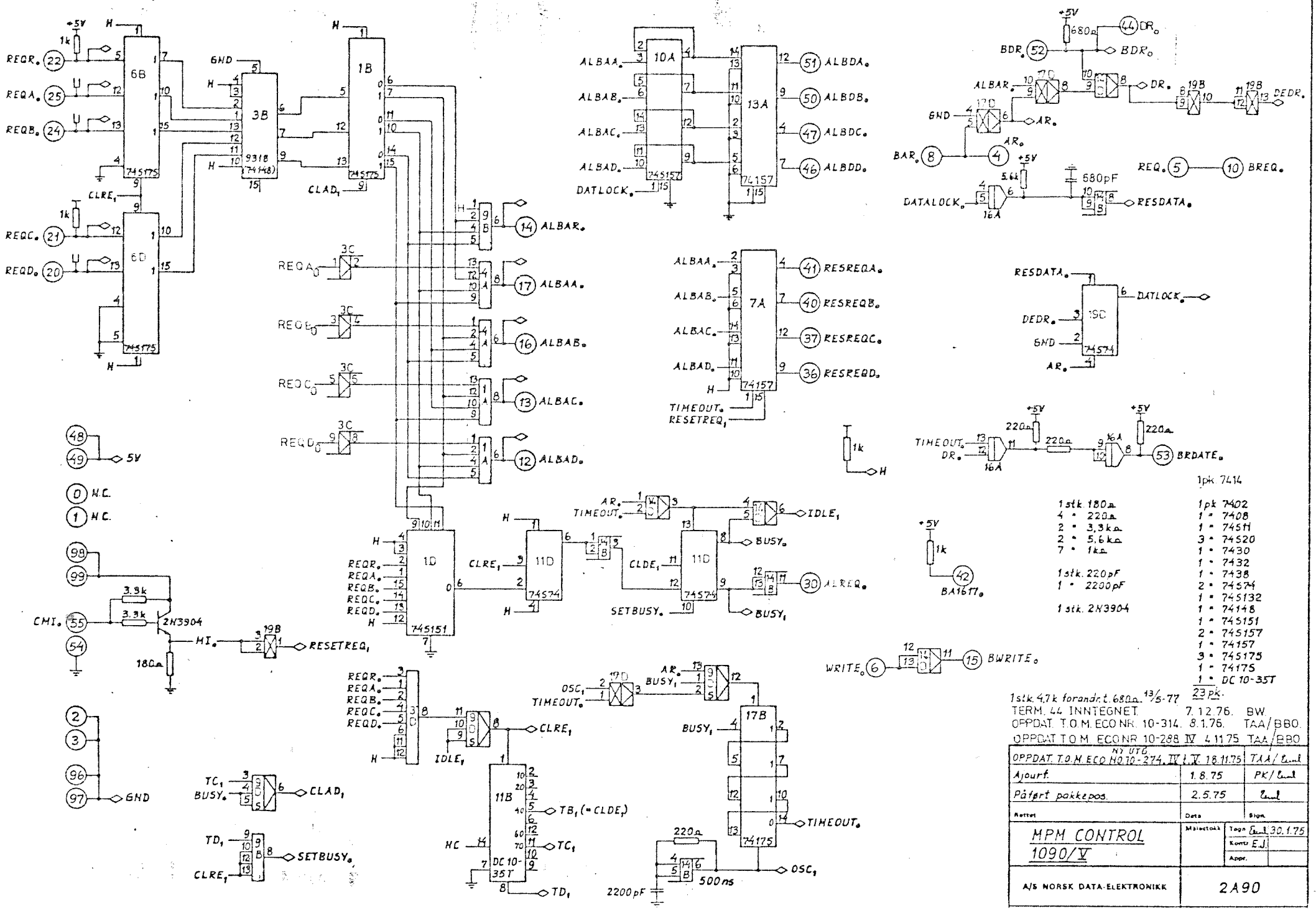
A/S NORSK DATA ELEKTRONIKK

2A83

CONTRACT NO. F10 215 AT NTH


4B
6B
8B
4A
6A
8A
8097
12D
74574
16D
74574
18C
16C
7483A
14C
74585
A>B
12C
74585
8820A
NOTE
Normal: 0 ≤ upper limit ÷ lower limit < 64K
Inhibit: 0 > upper limit ÷ lower limit
Påført 1083/II (ingen rettelser) 9.12.75 TAA/Eml
OPPDATERT T.O.M. ECO NO. 10-398. D. 15.8.77 Eml
Kond C1,C2,C3 påf verdi (470pF) 2.9.75 SEJ/BBO
Påført pakkepos. 4.12.74 Eml
MPM ADDRESS LINE-DRIVER 1083.D,E
A/S NORSK DATA-ELEKTRONIKK
2A83
Oppdat. t.o.m. ECO 10-528, pat. E.I/28.7.78/TEAA/BW


XEN
GND
A1
A2
A3
A4
A5
5B
XBA0
XBA1
XBA2
XBA3
XBA4
XBA5
5A
YBA0
YBA1
YBA2
YBA3
YBA4
YBA5
A0
7A
7B
XBA6
XBA7
XBA8
XBA9
XBA10
XBA11
YBA6
YBA7
YBA8
YBA9
YBA10
YBA11
8097
+5V
2,2k
XREQ
YREQ
H
RESETX
RESETY
3A
1A
XBREQ
YBREQ
XBWRITE
YBWRITE
ARX
ARY
4,7k
XBA12
YBA12
XBREFRESH
YBREFRESH
ALBARX
ALBARY
ALREQXX
ALREQYY
5C
220n
470pF
REQX1
REQY1
OKX1
OKY1
XREQ0
YREQ0
3B
1B
74193
RCL1
1k
N.C.
GND
50
51
2 pk. 74S11
2 " 7414
1 " 74S20
1 " 7474
1 " 74S74
2 " 74193
2 " 74221
6 " 8097
1 " 8820A
1 " 8831
19 pk.
0,1µF
REFCLOCK1
REFCLOCK0
11C
11B
10A
ENABEL
STRAP
XBMBUSY
YBMBUSY
1N4148
150pF
9B
7474
OKX1
OKY1
10pF
10k
13C
221
RESETX
RESETY
100ns
13B
7C
74S74
REQX
REQY
RCL1
Fjernet forb. til 982. 6/2/78/BS/Emil
Påført V, F (ingen rettelser) 7/6/77/TEAA/Emil
Oppdat t.o.m ECO nr 10-341. 3.3.76. TAA/BBO.
Fjernet fstk. VP12 17.11.75/EB/Emil
OPPDAT. T.O.M. ECO NO. 10-312. III t. IV. 30.12.75/TAA/Emil
Kond. C1,C2 påt. verdi. (470pF) 2.9.75. SEJ/BBO.
Ajourf. 6.8.75 SEJ/Emil
Påført pakkepos 30.4.75 Emil
Rettet Dato Sign.
MPM REFRESH
1089/IV, V, F
Målestokk
Tegn Emil 17.4.75
Kontr. E.J.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A89



1pk 7414

1 stk 180Ω	1pk 7402
4 • 220Ω	1 • 7408
2 • 3,3kΩ	1 • 7451
2 • 5,6kΩ	3 • 74520
7 • 1kΩ	1 • 7430
	1 • 7432
1 stk. 220pF	1 • 7438
1 • 2200pF	2 • 74574
	1 • 745132
1 stk. 2N3904	1 • 74148
	1 • 745151
	2 • 745157
	1 • 74157
	3 • 745175
	1 • 74175
	1 • DC 10-35T

23 pk.

1 stk. 4,7k forandret t. 680Ω. 13/5-77
TERM. 44 INNTEGNET 7.12.76. BW
OPPDAT. T.O.M. ECO NR. 10-314. 8.1.76. TAA/BBO.
OPPDAT T.O.M. ECO NR 10-288 IV 4.11.75 TAA/BBO

NY UTG. OPPDAT. T.O.M. ECO NO. 10-274. IV i V. 18.11.75		TAA/Emil
Ajourf.	1.8.75	PK/Emil
Påført pakkepos.	2.5.75	Emil
Rettet	Dato	Sign.

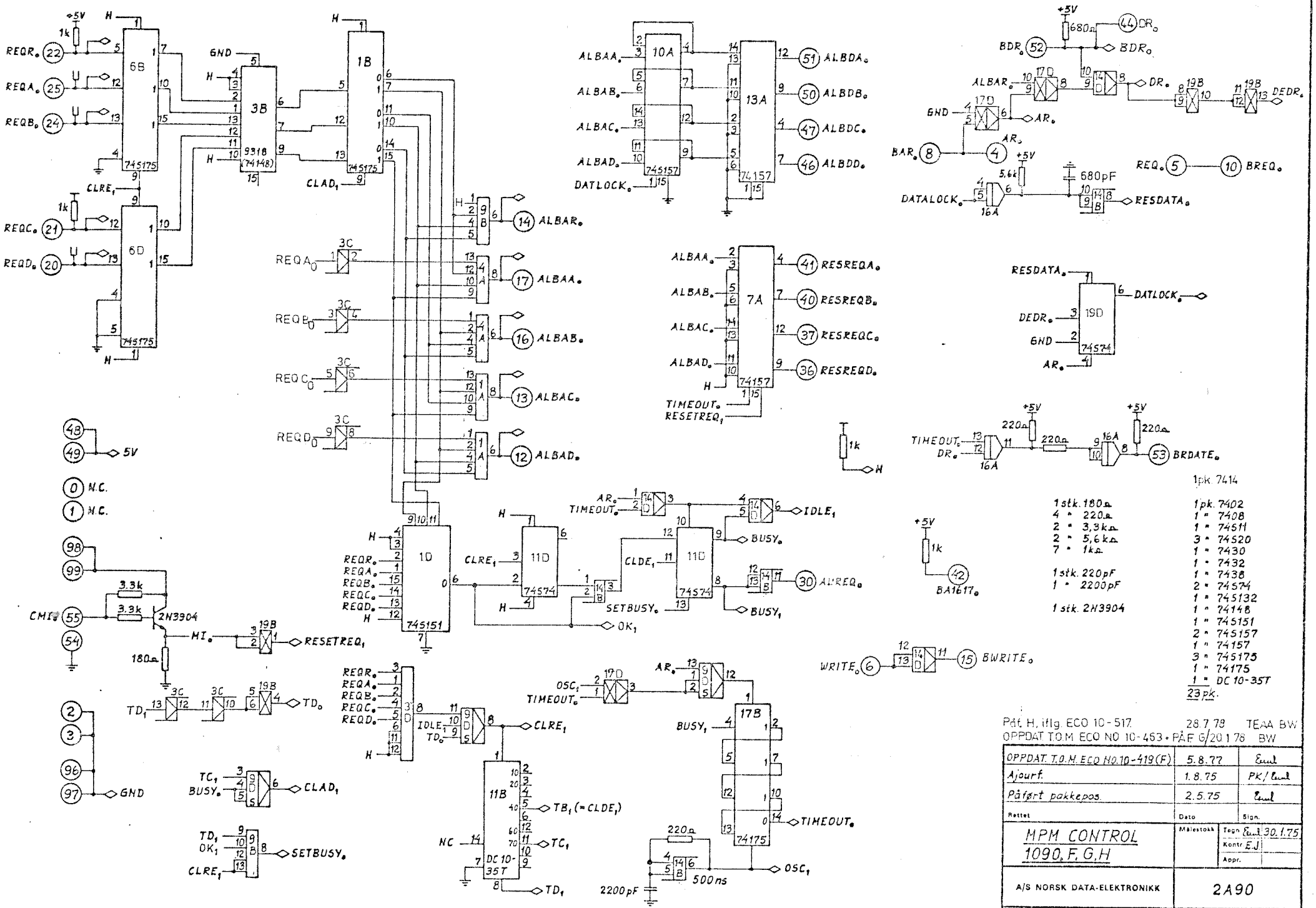
MPM CONTROL 1090/V

Målestokk

Tegn Emil 30.1.75
Kontr E.J.
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A90



1pk 7414

1 stk. 180Ω
4 " 220Ω
2 " 3,3kΩ
2 " 5,6kΩ
7 " 1kΩ

1 stk. 220pF
1 " 2200pF

1 stk. 2N3904

1pk. 7402
1 " 7408
1 " 74S11
3 " 74S20
1 " 7430
1 " 7432
1 " 7438
2 " 74S74
1 " 74S132
1 " 74148
1 " 74S151
2 " 74S157
1 " 74157
3 " 74S175
1 " 74175
1 " DC 10-35T
23 pk.

Påf. H, iflg. ECO 10-517 28.7.78 TEAA BW
OPPDAT. T.O.M. ECO NO 10-453 + PÅF 6/20 1.78 BW

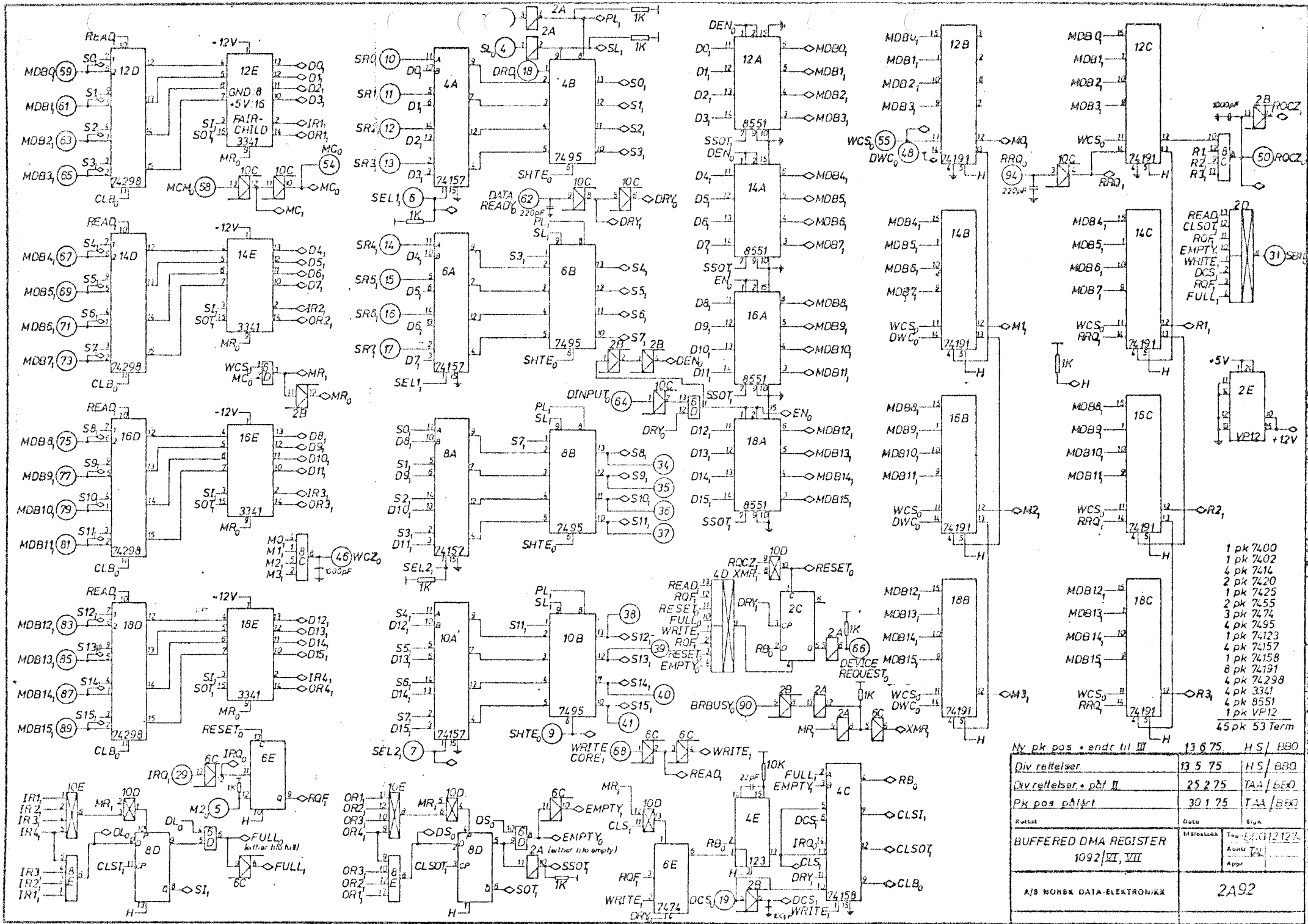
OPPDAT. T.O.M. ECO NO 10-419 (F)	5.8.77	Emil
Ajourf.	1.8.75	PK/Emil
Påført pakkepos.	2.5.75	Emil
Rettet	Dato	Sign.

MPM CONTROL
1090, F, G, H

Målestokk
Tegn Emil 30.1.75
Kontr E.J
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A90



Ny pk pos + endr til III 13 6 75 HS/BBO

Div rettelser	13 5 75	HS/BBO
Div rettelser + pkt II	25 2 75	TAA/BBO
Pk pos påført	30 1 75	TAA/BBO
Rettet	Dato	Sign.

BUFFERED DMA REGISTER 1092/VI, VII

Målestokk: Tegn. BBO 12 12 74
Kontr. TAA
Appr.

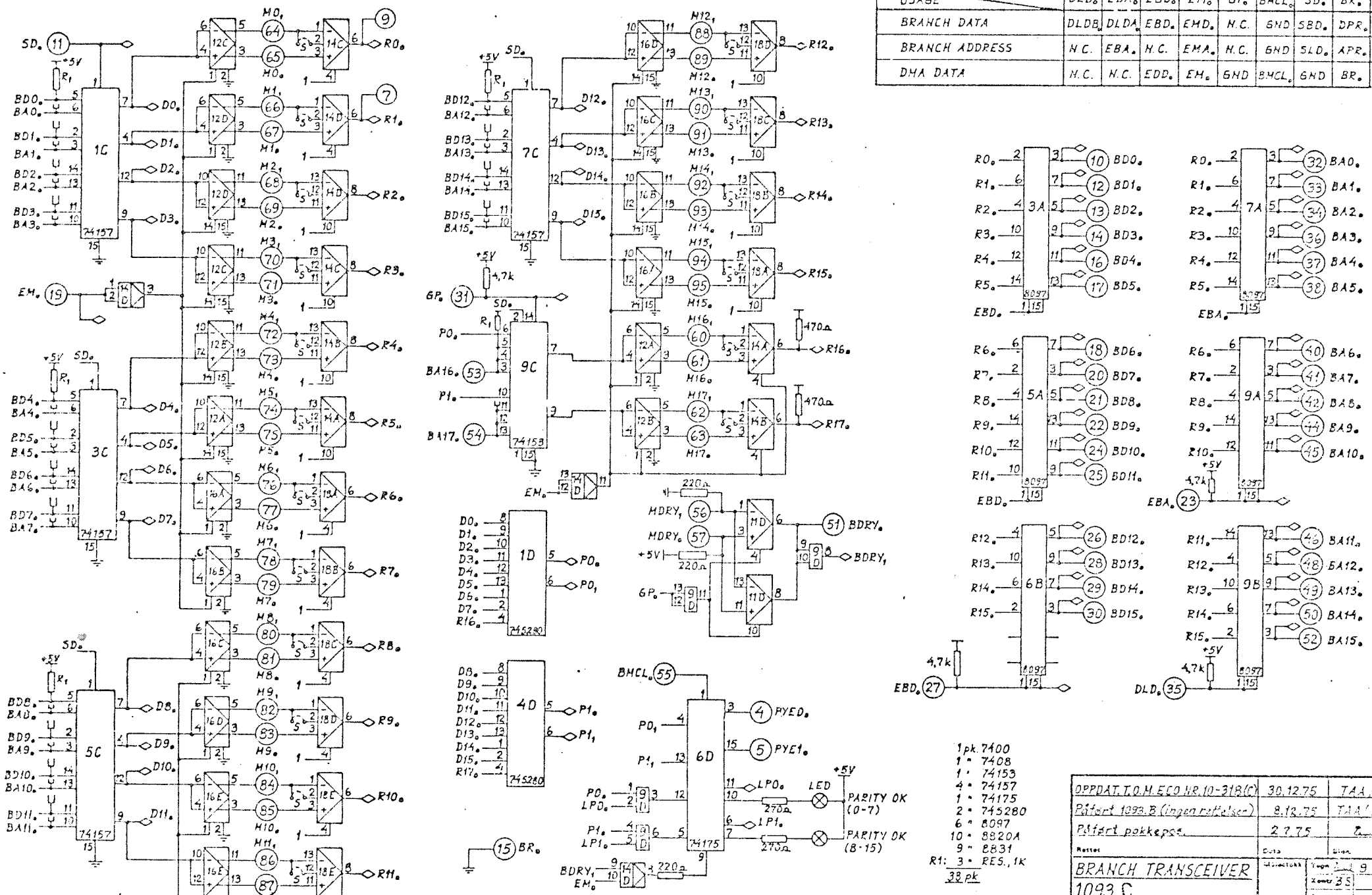
A/S NORSK DATA-ELEKTRONIKK

2A92

Endr. til 3052/A ingen rett. 7.7.76 TAA/BW Innganger på 74298 pkr endr. 19.11.76. HS/BW

Påført III (ingen rettelser) 30/3/77/KK

Oppdat T.O.M. ECO nr 10-305 pkt VI/V 19.12.75/BBO
Oppdat T.O.M. ECO nr 10-365 pkt I - 6/2-78 KVE/BW



ACTUAL SIGNAL NAMES

USAGE \ SIGNAL	DLD0	EBA0	EBD0	EM0	GP0	BMCL0	SD0	BR0	BDRY0
BRANCH DATA	DLDB0	DLDA0	EBD0	EMD0	N.C.	GND	SBD0	DPR0	N.C.
BRANCH ADDRESS	N.C.	EBA0	N.C.	EMA0	N.C.	GND	SLD0	APR0	N.C.
DMA DATA	N.C.	N.C.	EDD0	EM0	GND	BMCL0	GND	BR0	BDRY0

NOTE: WHEN USED FOR DMA DATA, PREFERABLY REMOVE R1 AND INSERT STRAPS 5.

1 pk. 7400
1 " 7408
1 " 74153
4 " 74157
1 " 74175
2 " 745280
6 " 8097
10 " 8820A
9 " 8831
R1: 3 " RES., 1k
38 pk

85 term.

OPPDAT. T.O.M. ECO NR. 10-318(C)	30.12.75	TAA
Påført 1093.B (ingen retteser)	8.12.75	TAA
Påført pakkepse.	2.7.75	E.J.
Rettet	Dato	Sign.
BRANCH TRANSCEIVER 1093.C	Tegn. 9.6.75	
A/S NORSK DATA-ELEKTRONIKK	2A93	

NOTE: WHEN USED FOR DMA DATA, PREFERABLY REMOVE R_1 AND INSERT STRAPS S.

ACTUAL SIGNAL NAMES

USAGE \ SIGNAL	DLD$_0$	EBA$_0$	EBD$_0$	EM$_0$	GP$_0$	BNCL	SD$_0$	BR$_0$	BDRY
BRANCH DATA	DLDB	DLDA	EBD$_0$	EMD$_0$	N.C.	GND	SBD$_0$	DPR$_0$	N.C.
BRANCH ADDRESS	N.C.	EBA$_0$	N.C.	EMA$_0$	N.C.	GND	SLD$_0$	APR$_0$	N.C.
DMA DATA	N.C.	N.C.	EDD$_0$	EM$_0$	GND	BNCL	GND	BR$_0$	BDRY

1pk 7408
1 " 7474
1 " 74132
1 " 74153
4 " 74157
2 " 745280
6 " 8097
10 " 8820A
9 " 8551
R1: 3 " RES., 1K
33 pk

85 term.

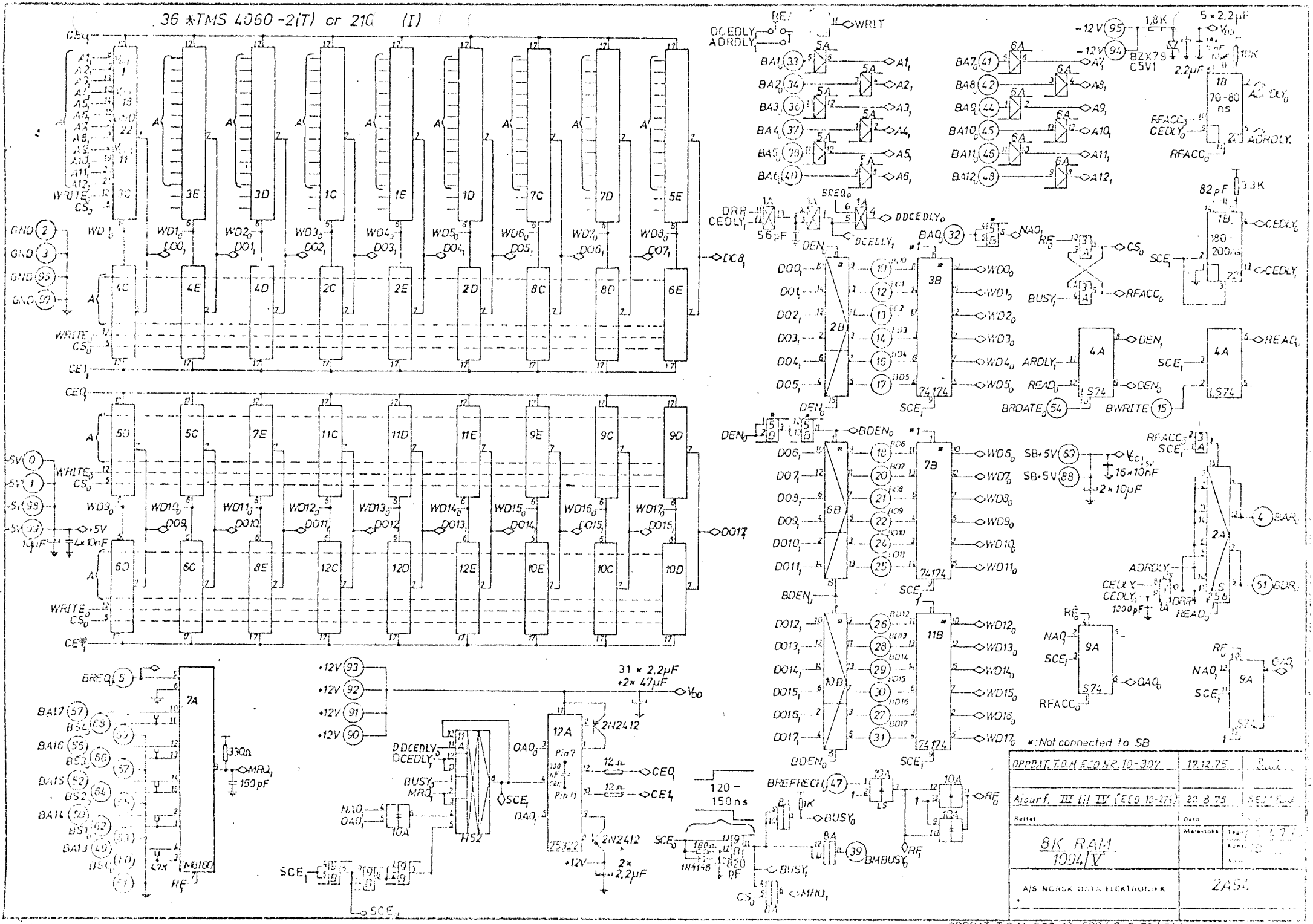
Pdf.G iflg. ECO 10-516. 28.7.78. TEA/BW

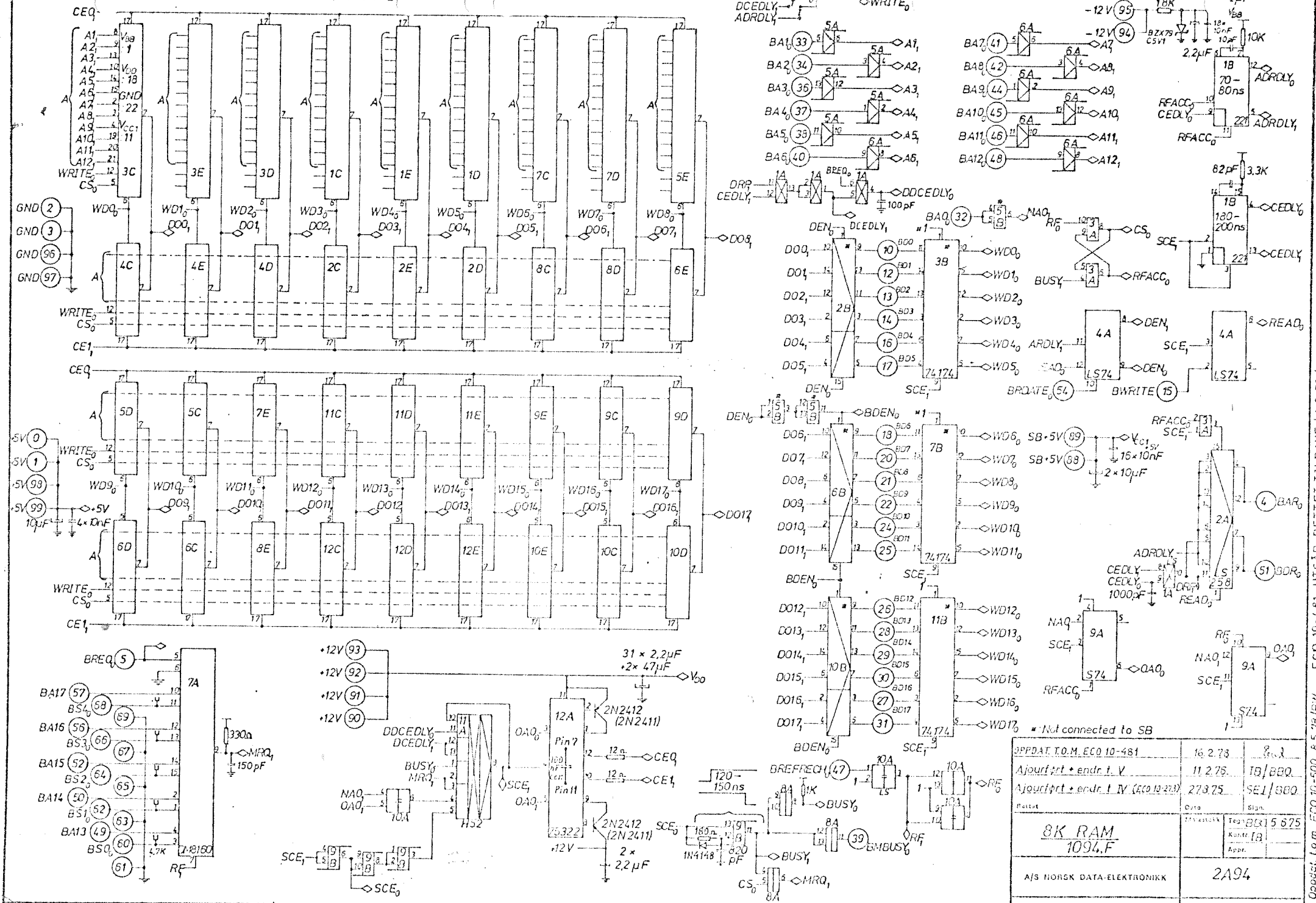
DPPDAT. T.O.M. ECO 10-454. F	17.10.77	Eml
:4 and :5 are swapped	3.7.78	Eml

BRANCH TRANSCEIVER
1093.E,F,G

A/S NORSK DATA ELEKTRONIKK

2A93


36 * TMS 4060 -2(T) or 210 (I)
3E
3D
1C
1E
1D
7C
7D
5E
4C
4E
4D
2C
2E
2D
8C
8D
6E
5D
5C
7E
11C
11D
11E
9E
9C
9D
6D
6C
8E
12C
12D
12E
10E
10C
10D
7A
12A
2B
3B
6B
7B
10B
11B
4A
9A
2A
1B
18
70-80 ns
180-200ns
74174
LS74
LS86
LS32
75322
2N2412
MB100
BREQ0
WRIT
BA1 (33)
BA2 (34)
BA3 (36)
BA4 (37)
BA5 (38)
BA6 (40)
BA7 (41)
BA8 (42)
BA9 (44)
BA10 (45)
BA11 (46)
BA12 (48)
-12V (95)
-12V (94)
BZX79 C5V1
1,8K
5 × 2,2µF
2,2µF
18K
82pF
3,3K
DRP
CEDLY1
56pF
DDCEDLY0
BAQ0 (32)
NAQ0
RFACC0
BUSY1
CS0
DEN0
BDEN0
BWRITE (15)
BRDATE (54)
SB+5V (89)
SB+5V (88)
16 × 10nF
2 × 10µF
1200pF
RFACC1
SCE1
ADRDY1
CEDLY1
READ0
BAR
BDR
NAQ1
QAQ0
BREFREQ1 (47)
1K
BUSY0
BMBUSY0
RF1
120 - 150 ns
1N4148
820 pF
MRQ1
+12V (93)
+12V (92)
+12V (91)
+12V (90)
31 × 2,2µF
2× 47µF
VDD
2N2412
2 × 2,2µF
330Ω
150pF
47k
DDCEDLY1
DCEDLY1
ARDLY1
BUSY1
MRQ1
OAQ0
SCE1
NAQ0
OAQ1
SCE0
CEQ1
CE1
WRITE0
CS0
WD0 - WD17
DO0 - DO15
DO17
GND (2)
GND (3)
GND (93)
GND (92)
5V (0)
5V (1)
5V (99)
5V (98)
1µF
4 × 10nF
* : Not connected to SB
OPPDAT. T.O.M ECO 10-307
12.12.75
Alourf. III til IV (ECO 10-276)
23.8.75
8K RAM
1094/V
A/S NORSK DATA-ELEKTRONIKK
2A54
OPPDAT. T.O.M ECO 10-500 / 8.5.78 / BW



OPPDAT. T.O.M. ECO 10-481	16.2.78	[illegible]
Ajourført + endr. t. V	11.2.76.	TB/BBO
Ajourført + endr. t. IV (ECO 10-223)	27.8.75	SE1/BBO
Rettet	Dato	Sign.

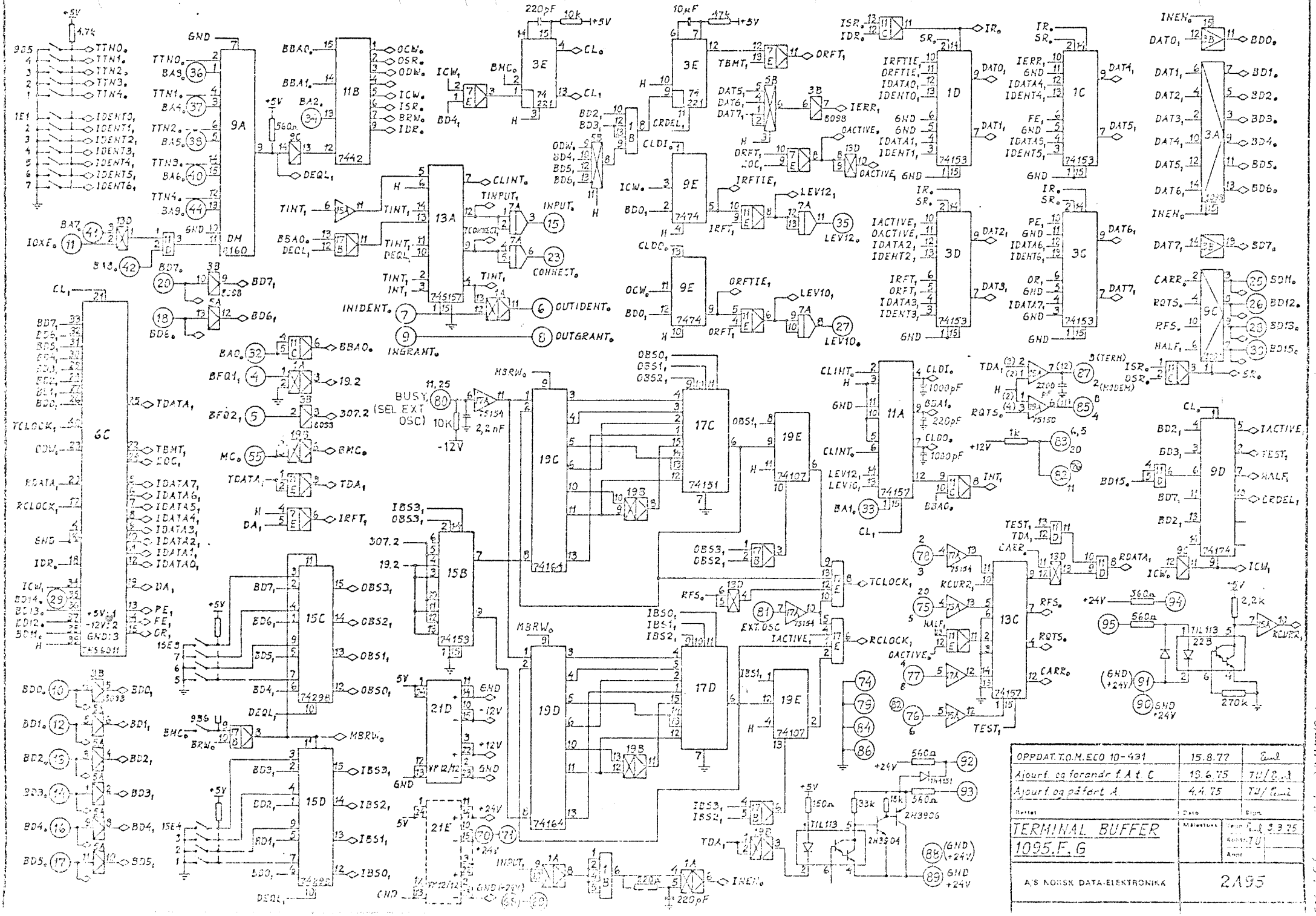
8K RAM
1094.F

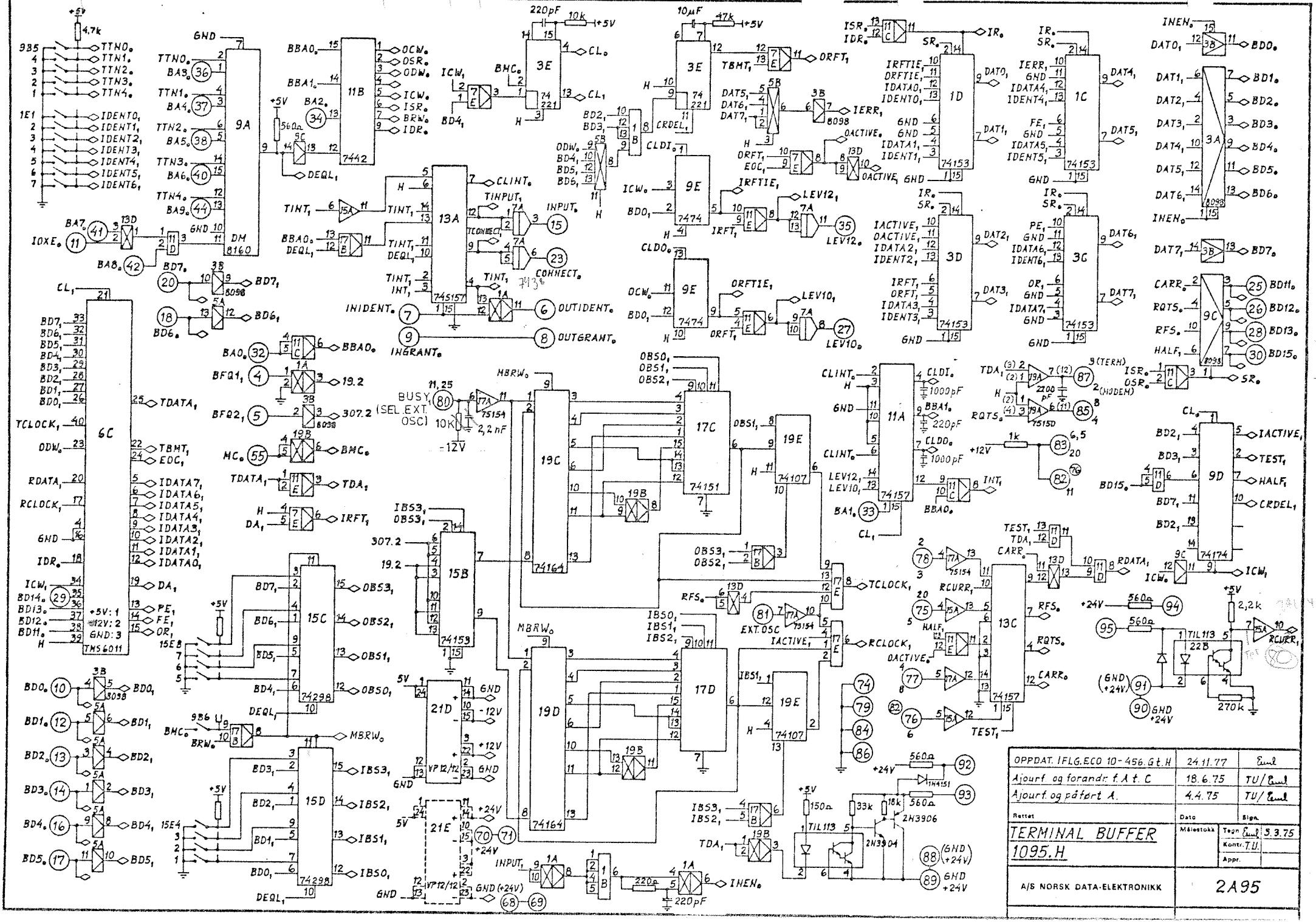
Tegn. BOX 5 675
Kontr. TB
Appr.

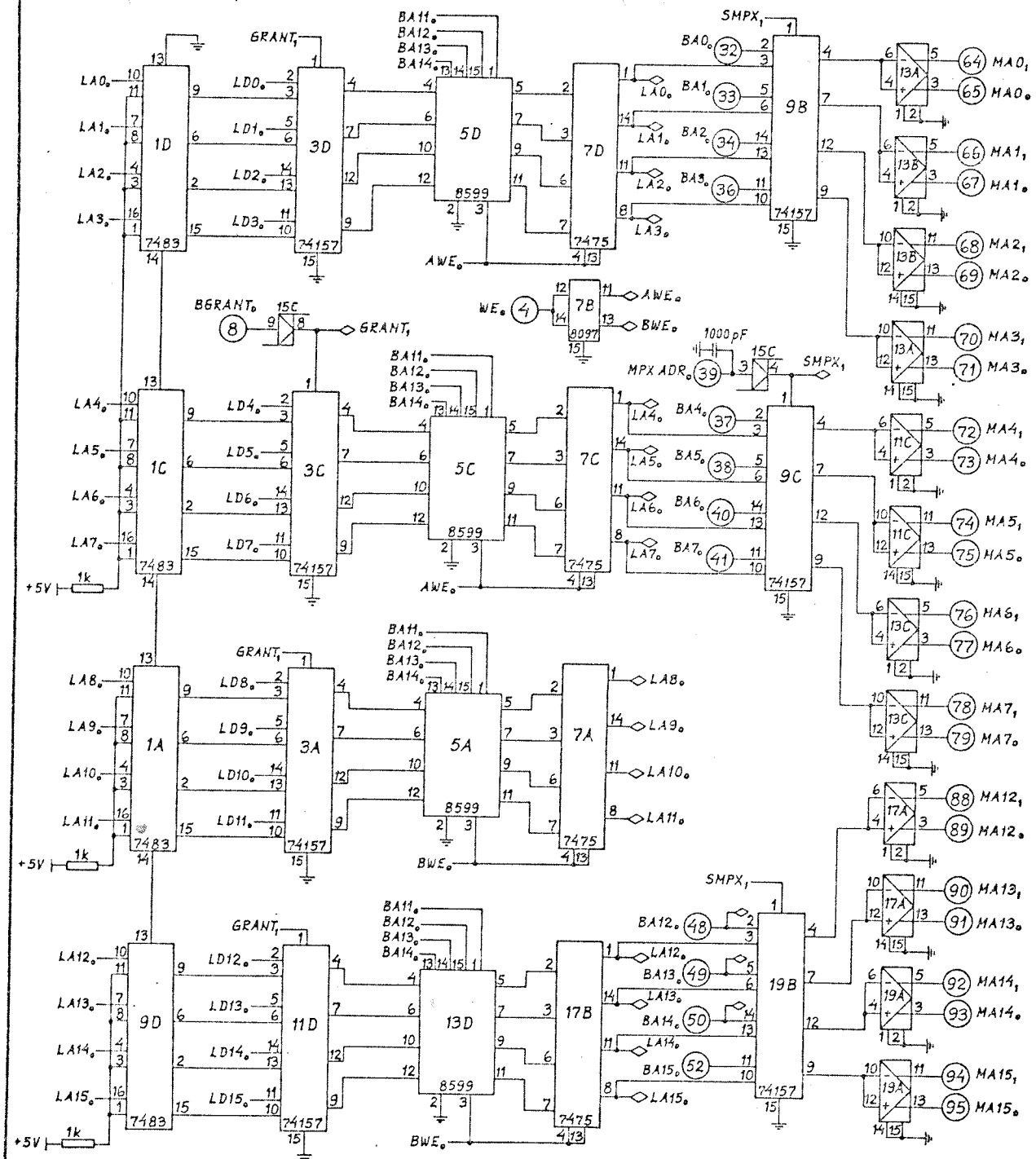
A/S NORSK DATA-ELEKTRONIKK

2A94

Oppdat. t.o.m. ECO 10-500 6.5.78/BW ECO 10-481 UTGÅR, RETTET TILBAKE / 4.9.78 / BW


TERMINAL BUFFER
1095.F.G
A/S NORSK DATA-ELEKTRONIKA
2A95
OPPDAT. T.O.M. ECO 10-431
15.8.77
Ajourf. og forandr. f. A, t. C
15.6.75
Ajourf. og påført A
4.4.75
9.9.75
9A
11B
3E
9E
13A
74157
74153
74174
74107
74164
74161
74159
74298
74151
74154


TERMINAL BUFFER
1095.H
A/S NORSK DATA-ELEKTRONIKK
2A95
OPPDAT. IFLG.ECO 10-456. G.t.H 24.11.77
Ajourf. og forandr. f. A.t. C 18.6.75 TU/
Ajourf. og påført A. 4.4.75 TU/
Rettet Dato Sign.
Målestokk
Tegn 5.3.75
Kontr. T.U.
Appr.
9A
11B
3E
10µF
47k
220pF
10k
+5V
4.7k
560Ω
GND
DM 8160
7442
74221
74153
74157
74174
74164
74107
74151
74298
74154
74159
74LS157
74LS74
TMS6011
8098
8038
8039
8T26
75154
75150
75451
75452
2N3906
2N3904
TIL113
1N4151
270k
2,2k
33k
18k
150Ω
220Ω
2200pF
1000pF
2,2nF
10K
-12V
+12V
+24V
GND +24V
11.25 BUSY (SEL.EXT OSC)
EXT.OSC
TERM
MODEM



OPPDAT T.O.M. ECO NR. 10-316. B til C. 30.12.75/TAA/Emil

Påført B	22.9.75	BS/Emil
Forandr pinnenr på 7483	5.9.75	TAA/Emil
Påført pakkepos.	17.6.75	Emil
Rettet	Dato	Sign.

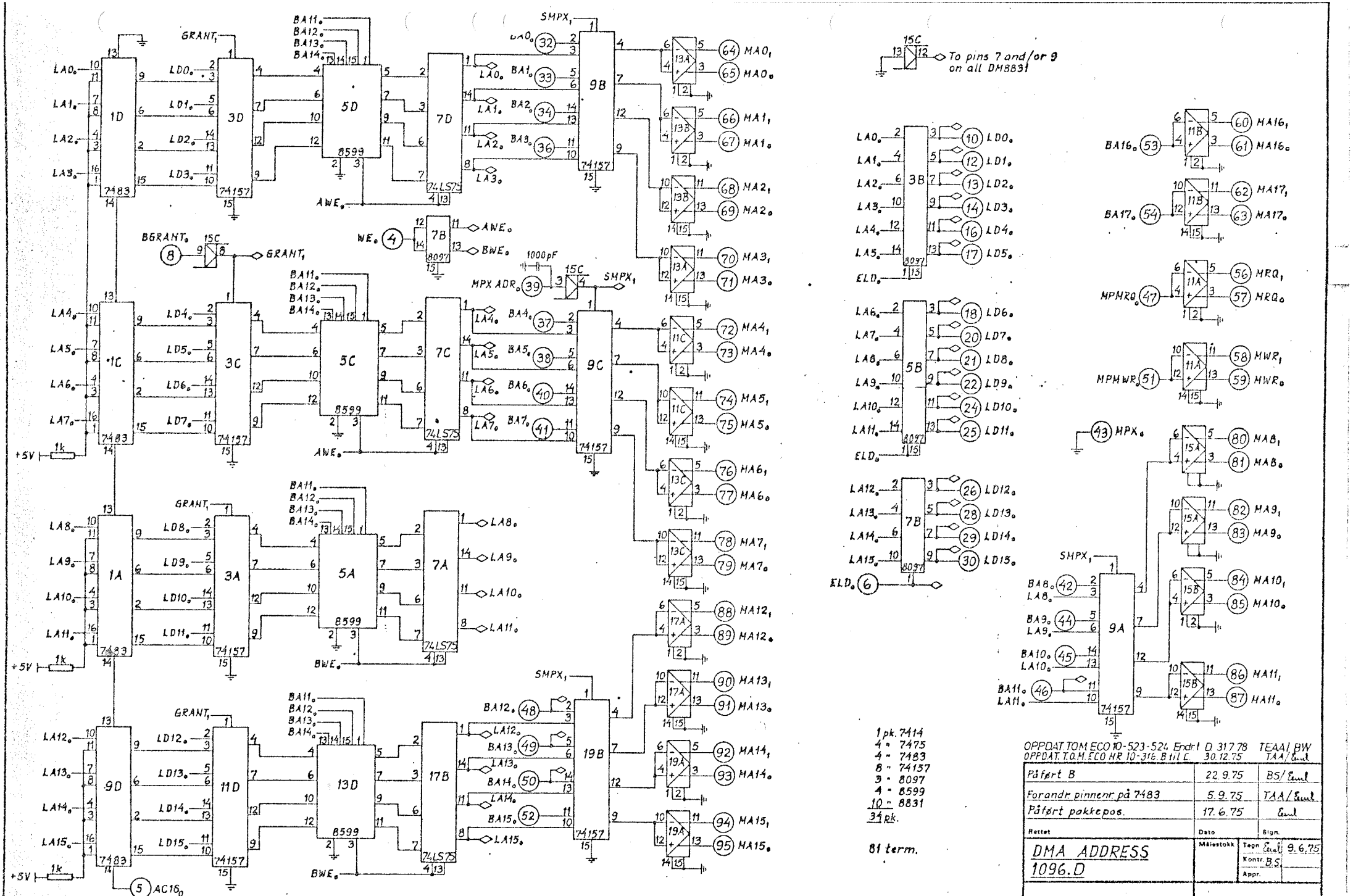
DMA ADDRESS
1096.C

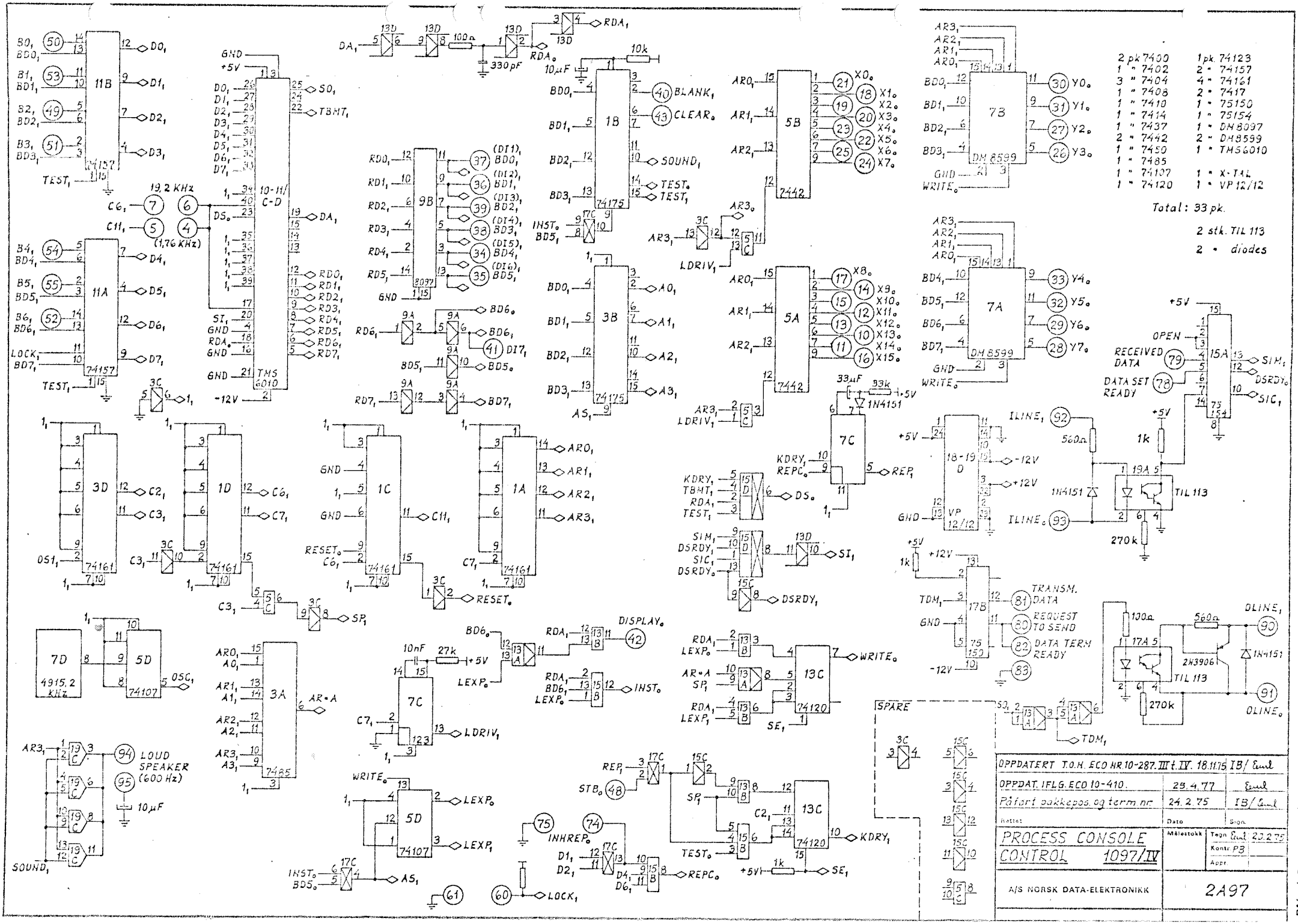
Målestokk

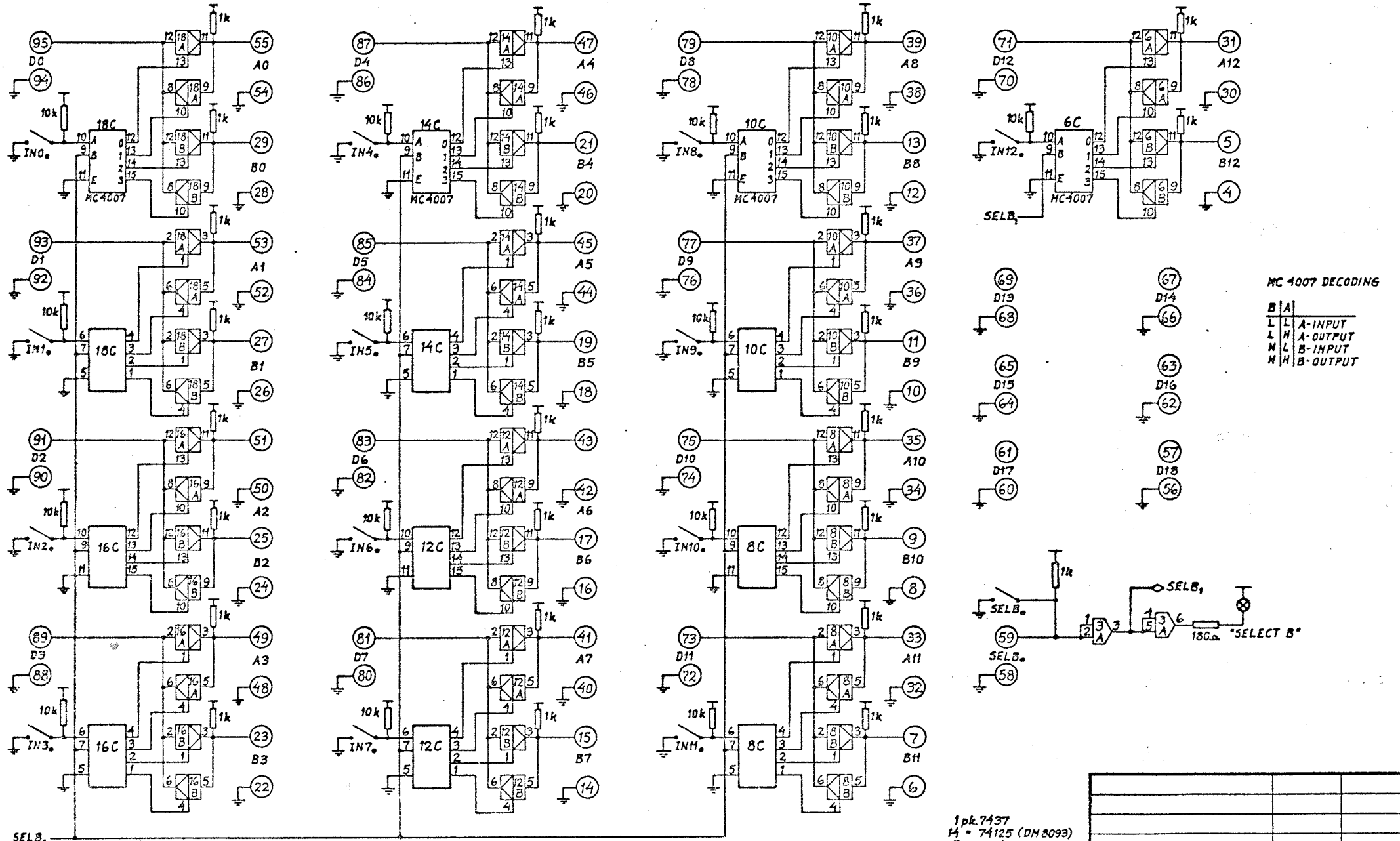
Tegn Emil 9.6.75
Kontr. BS
Appr.

A/S NORSK DATA-ELEKTRONIKK

2A96


GRANT1
BA11o
BA12o
BA13o
BA14o
SMPX1
LA0o
LA1o
LA2o
LA3o
LD0o
LD1o
LD2o
LD3o
1D
3D
5D
7D
9B
7483
74157
8599
74LS75
AWEo
MA0o
MA1o
MA2o
MA3o
BGRANTo
15C
WEo
7B
8097
BWEo
1000pF
MPX ADRo
LA4o
LA5o
LA6o
LA7o
LD4o
LD5o
LD6o
LD7o
1C
3C
5C
7C
9C
MA4o
MA5o
MA6o
MA7o
+5V
1k
LA8o
LA9o
LA10o
LA11o
LD8o
LD9o
LD10o
LD11o
1A
3A
5A
7A
MA12o
MA13o
LA12o
LA13o
LA14o
LA15o
LD12o
LD13o
LD14o
LD15o
9D
11D
13D
17B
19B
MA14o
MA15o
AC16o
To pins 7 and/or 9 on all DM8831
3B
5B
7B
ELDo
11B
11A
MA16o
MA17o
MRQo
MWRo
MPXo
9A
15A
15B
MA8o
MA9o
MA10o
MA11o
1 pk. 7414
4 " 7475
4 " 7483
8 " 74157
3 " 8097
4 " 8599
10 " 8831
34 pk.
81 term.
OPPDAT TOM ECO 10-523-524 Endrt D 317.78 TEAA/BW
OPPDAT. T.O.M. ECO HR 10-316. B til C. 30.12.75 TAA/Emil
Påført B 22.9.75 BS/Emil
Forandr. pinnenr. på 7483 5.9.75 TAA/Emil
Påført pakkepos. 17.6.75 Emil
Rettet Dato Sign.
DMA ADDRESS
1096.D
Målestokk
Tegn Emil 9.6.75
Kontr. B.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
2A96


11B
11A
74157
10-11/C-D
TMS 6010
9B
8097
3B
74175
1B
74175
5B
7442
7B
DM 8599
5A
7442
7A
DM 8599
3D
1D
74161
1C
1A
7C
7D
4915.2 kHz
5D
74107
3A
74285
7C
5D
74107
13C
74120
17B
75 150
18-19 D
VP 12/12
15A
75 154
19A
TIL 113
17A
2N3906
TIL 113
1N4151
1N4151
SPARE
2 pk 7400
1 " 7402
3 " 7404
1 " 7408
1 " 7410
1 " 7414
1 " 7437
2 " 7442
1 " 7450
1 " 7485
1 " 74107
1 " 74120
1 pk. 74123
2 " 74157
4 " 74161
2 " 7417
1 " 75150
1 " 75154
1 " DM8097
2 " DM8599
1 " TMS6010
1 " X-TAL
1 " VP 12/12
Total: 33 pk.
2 stk. TIL 113
2 " diodes
OPPDATERT T.O.H. ECO NR 10-287. III t. IV. 18.11.75 IB/ Emil
OPPDAT. IFLG. ECO 10-410. 25.4.77 Emil
Påført pakkepos. og term nr 24.2.75 IB/ Emil
PROCESS CONSOLE CONTROL 1097/IV
A/S NORSK DATA-ELEKTRONIKK
2A97
Påført III. 24.5.75/TAA/Emil



1 pk. 7437
14 • 74125 (DM 8093)
7 • MC4007
2 • SWITCH
24 pk.

Rettet	Dato	Sign.

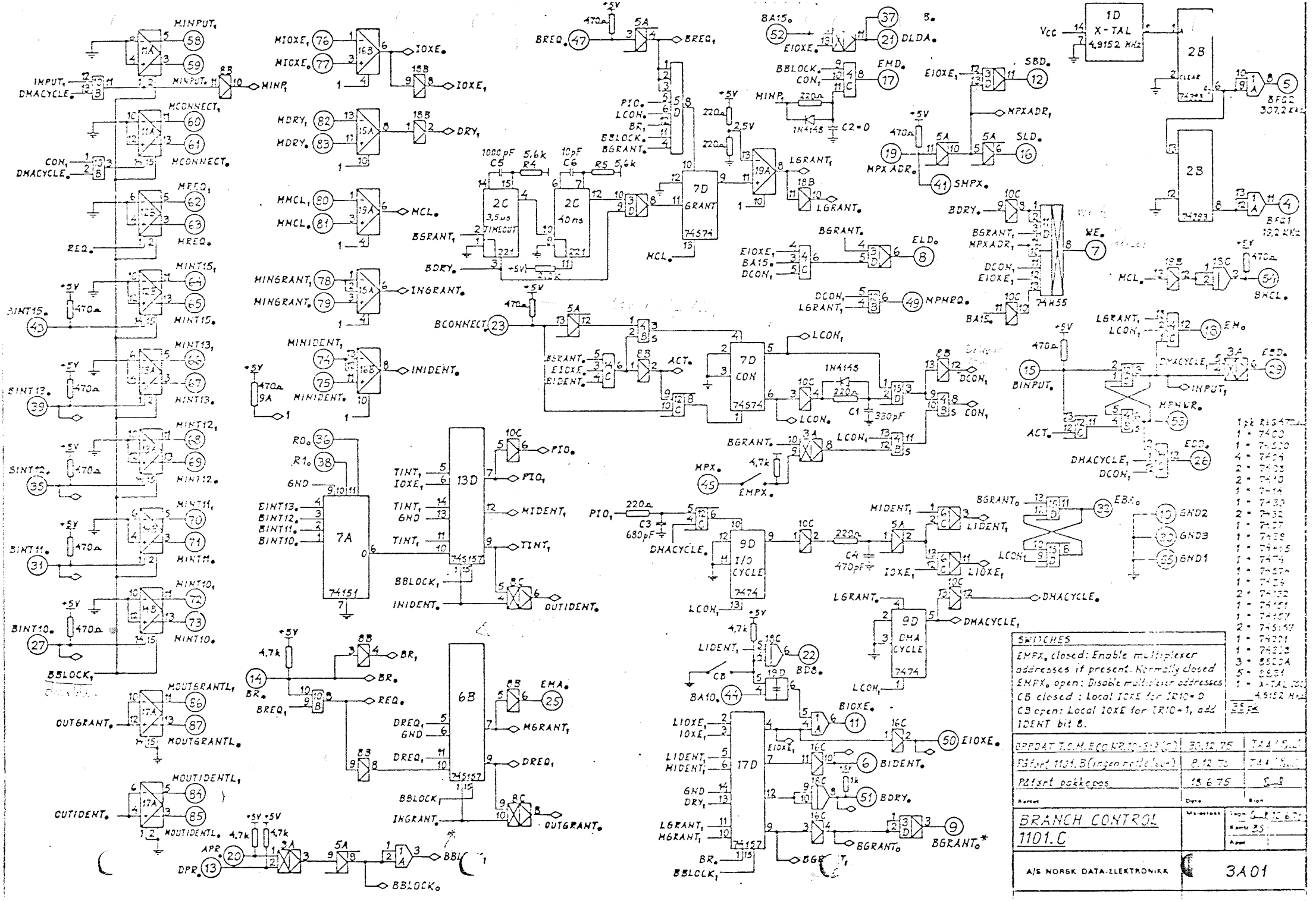
DEVICE SWITCH
1098

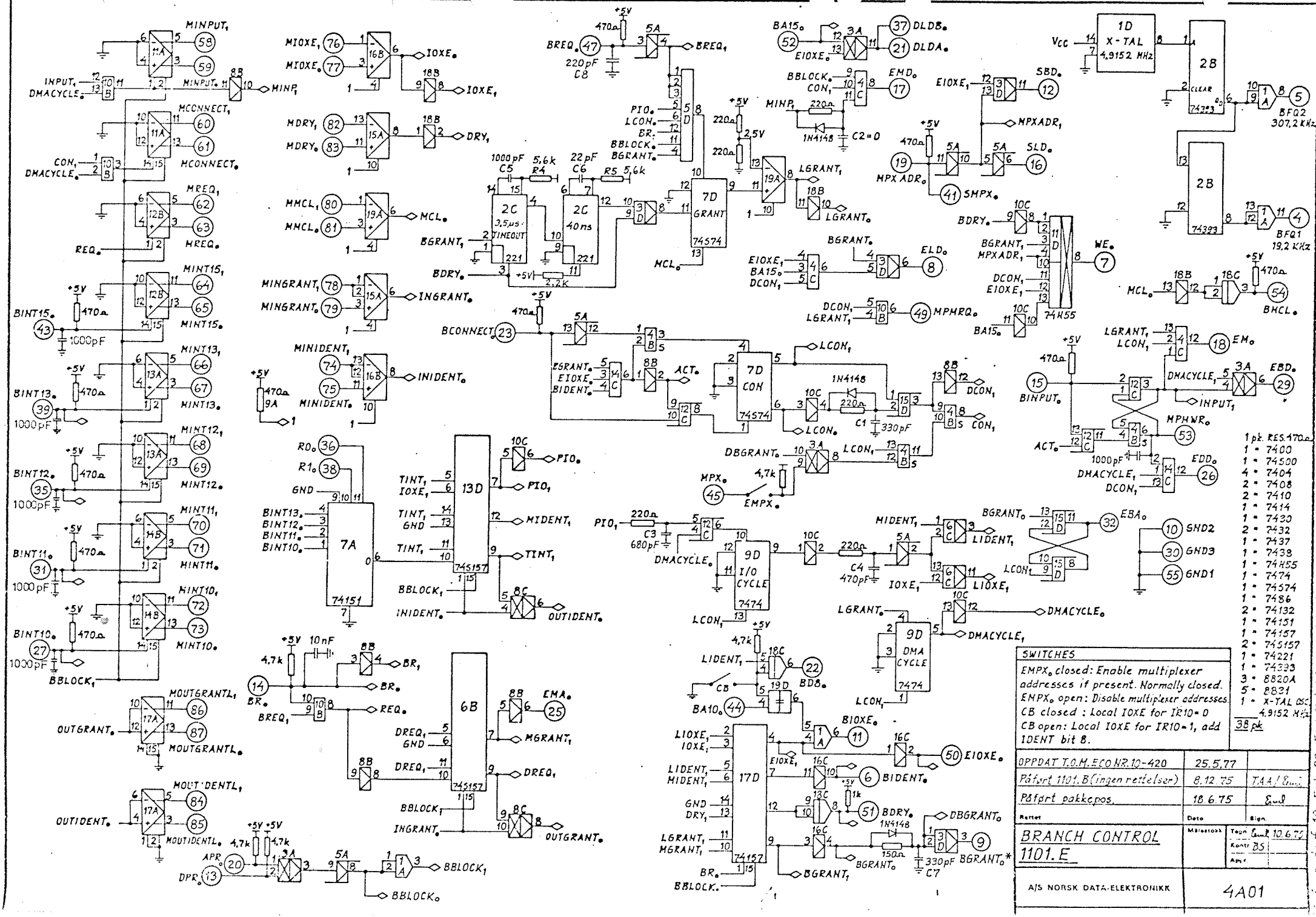
Målestokk

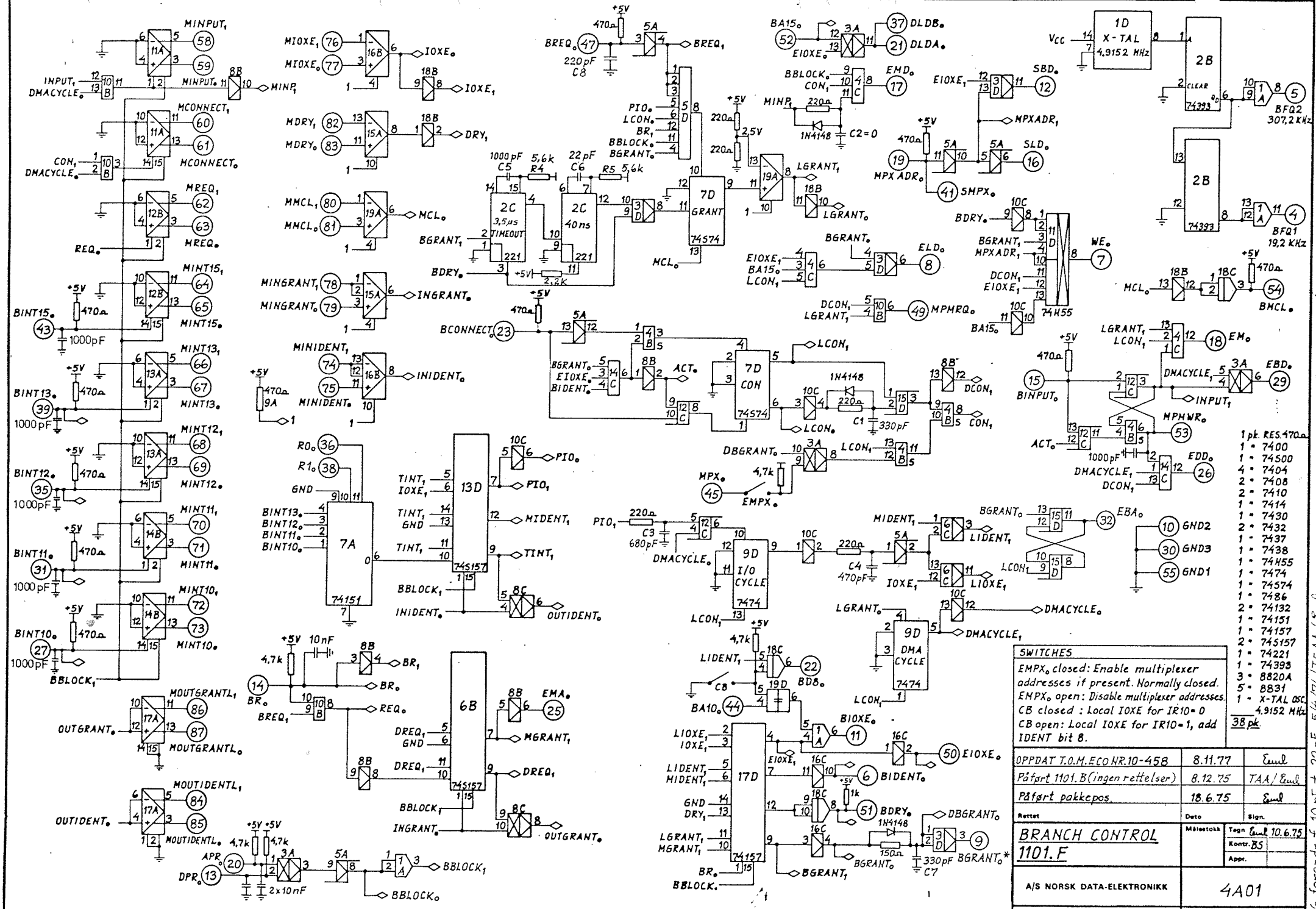
Tegn. Emil 14.4.75
Kontr. PJ
Appr.

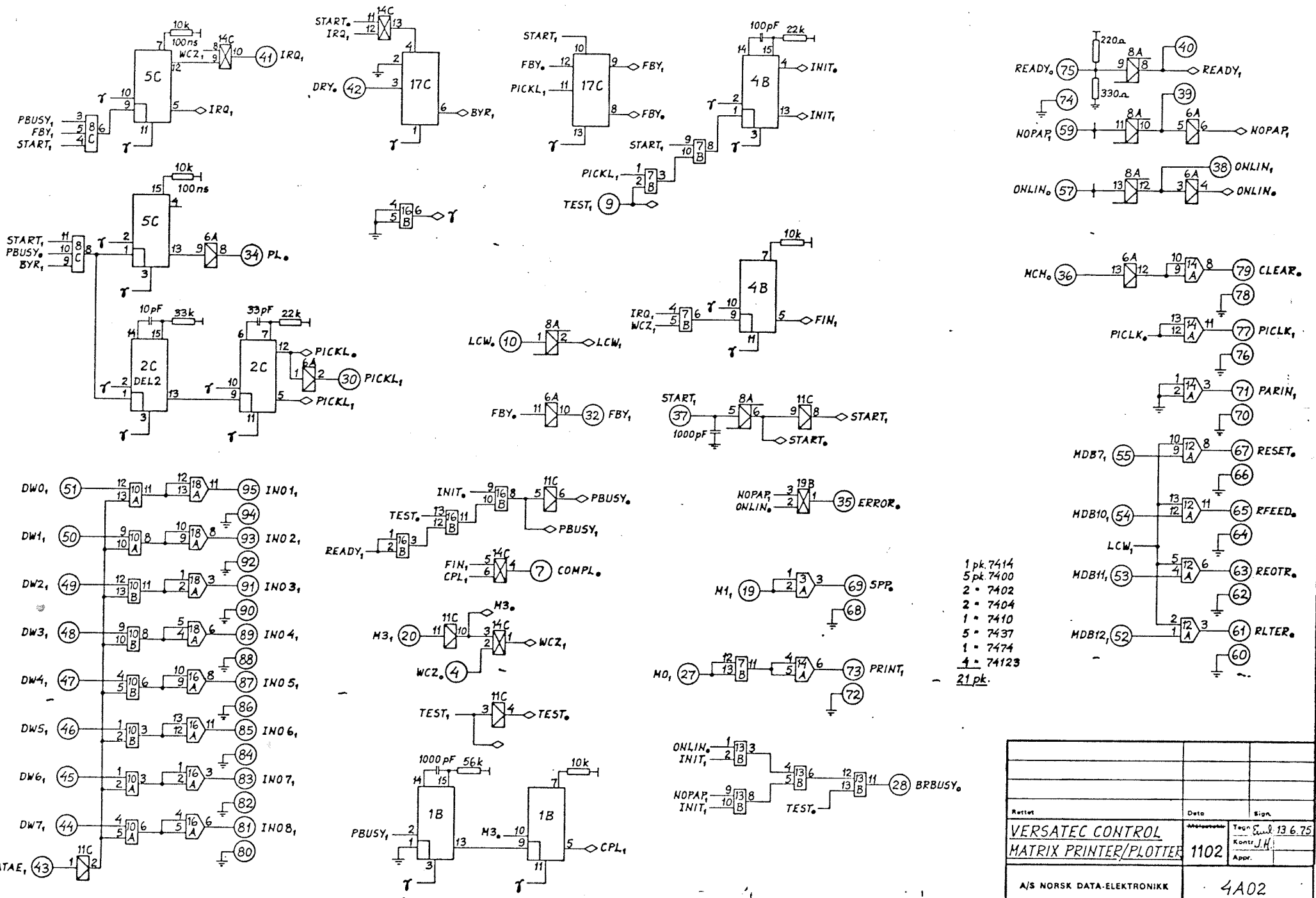
A/S NORSK DATA-ELEKTRONIKK

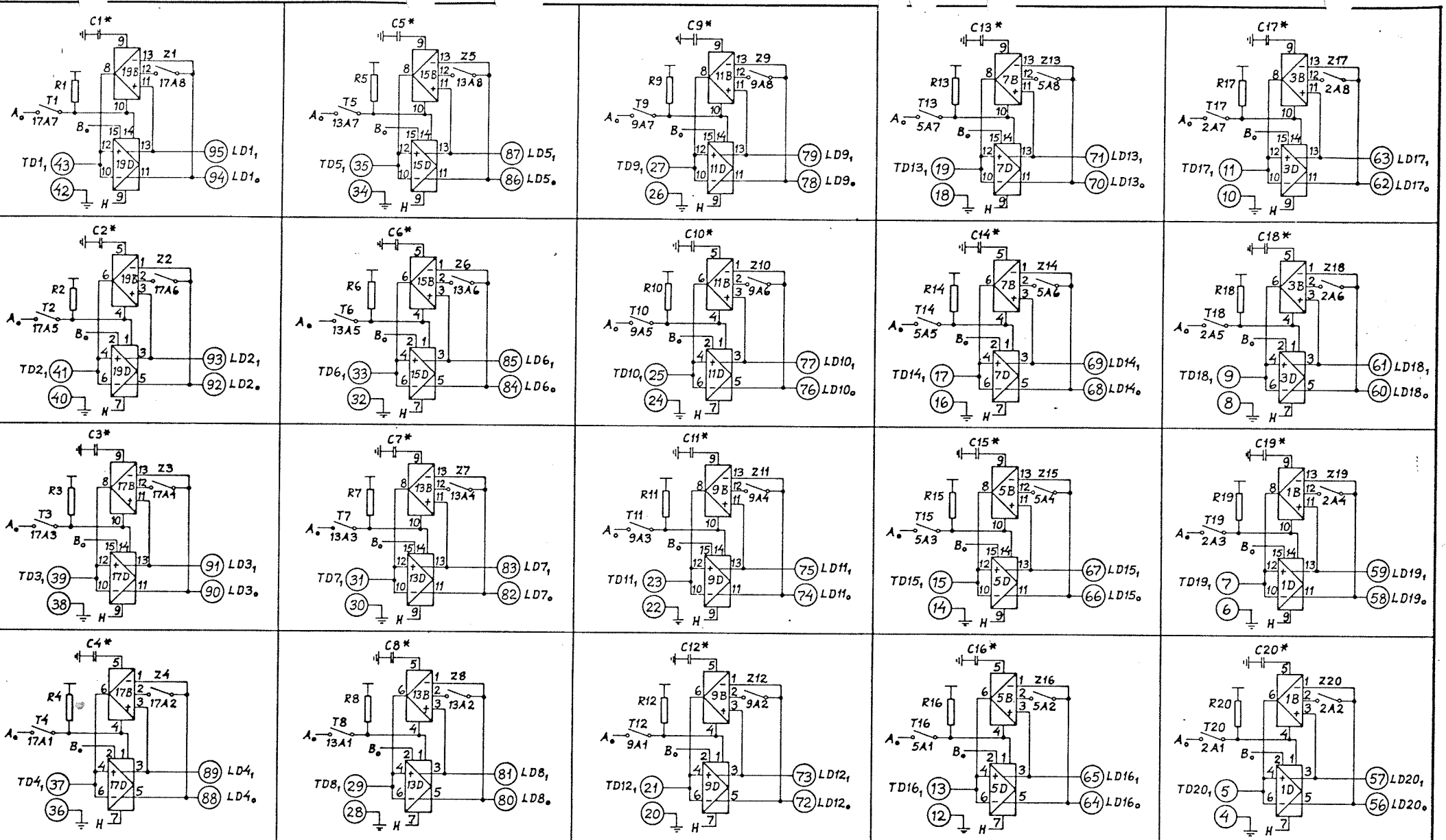
2A98

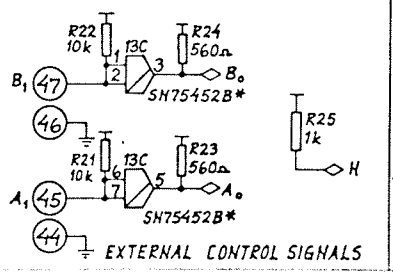

MINPUT
MIOXE
MDRY
MMCL
MINGRANT
MINIDENT
INPUT
DMACYCLE
CON
MCONNECT
MREQ
REQ
MINT15
MINT13
MINT12
MINT11
MINT10
BINT15
BINT13
BINT12
BINT11
BINT10
BBLOCK
MOUTGRANTL
OUTGRANT
MOUTIDENTL
OUTIDENT
IOXE
DRY
MCL
INGRANT
INIDENT
BREQ
PIO
LCON
BR
BBLOCK
BGRANT
1000pF
C5
5,6k
R4
10pF
C6
R5
5,6k
2C
3,5µs
TIMEOUT
40ms
7D
GRANT
74S74
LGRANT
BSRANT
BDRY
MCL
BCONNECT
7D
CON
74S74
ACT
EIOXE
BIDENT
LCON
1N4148
C1
330pF
DCON
BA15
ELD
MPHRQ
MPX ADR
SMPX
SBD
SLD
EMD
BA15
DLDA
EIOXE
BBLOCK
CON
HINP
C2=D
1N4148
1D
X-TAL
4,9152 MHz
2B
74293
BFG2
307,2 kHz
BFG1
19,2 kHz
74LS55
WE
BMCL
EMA
EBD
MPHER
EBR
GND2
GND3
GND1
BINPUT
INPUT
DMACYCLE
DCON
7A
74151
13D
74S157
TINT
IOXE
GND
EINT13
EINT12
EINT11
EINT10
RO
R1
PIO
MIDENT
TINT
BBLOCK
INIDENT
OUTIDENT
6B
74S157
DREQ
GND
MGRANT
DREQ
BBLOCK
INGRANT
OUTGRANT
BR
REQ
BREQ
EMA
APR
DPR
BBLOCK
9A
9D
I/O
CYCLE
7474
9D
DMA
CYCLE
7474
LIDENT
CB
BA10
BDS
BIOXE
17D
74S157
LIOXE
IOXE
LIDENT
MIDENT
GND
DRY
LGRANT
MGRANT
BR
BBLOCK
BIDENT
BDRY
BGRANT
BGRANT*
EIOXE
LIDENT
LCON
DMACYCLE
LGRANT
DMACYCLE
BGRANT
4,7k
EMPX
SWITCHES
EMPX closed: Enable multiplexer addresses if present. Normally closed
EMPX open: Disable multiplexer addresses
CB closed: Local IOXE for TR10=0
CB open: Local IOXE for TR10=1, add IDENT bit 8.
BRANCH CONTROL
1101.C
A/S NORSK DATA-ELEKTRONIKK
3A01


SWITCHES
EMPX₀ closed: Enable multiplexer addresses if present. Normally closed.
EMPX₀ open: Disable multiplexer addresses.
CB closed: Local IOXE for IR10=0
CB open: Local IOXE for IR10=1, add IDENT bit 8.
1 pk. RES. 470Ω
1 • 7400
1 • 74500
4 • 7404
2 • 7408
2 • 7410
1 • 7414
1 • 7430
2 • 7432
1 • 7437
1 • 7438
1 • 74H55
1 • 7474
1 • 74574
1 • 7486
2 • 74132
1 • 74151
1 • 74157
2 • 745157
1 • 74221
1 • 74393
3 • 8820A
5 • 8831
1 • X-TAL 4.9152 MHz
38 pk
OPPDAT T.O.M. ECO NR. 10-420 25.5.77
Påført 1101.B (ingen rettelser) 8.12.75 TAA / Sund
Påført pakkepos. 18.6.75 Sund
Rettet Dato Sign.
BRANCH CONTROL
1101.E
Målestokk
Tegn Sund 10.6.75
Kontr BS
Appr
A/S NORSK DATA-ELEKTRONIKK
4A01
Tolerance ± 10 pF ± 22 pF, 5%/1%/TEAA/Sund


SWITCHES
EMPX₀ closed: Enable multiplexer addresses if present. Normally closed.
EMPX₀ open: Disable multiplexer addresses.
CB closed : Local IOXE for IR10=0
CB open: Local IOXE for IR10=1, add IDENT bit 8.
1 pk. RES. 470Ω
1 • 7400
1 • 74500
4 • 7404
2 • 7408
2 • 7410
1 • 7414
1 • 7430
2 • 7432
1 • 7437
1 • 7438
1 • 74H55
1 • 7474
1 • 74574
1 • 7486
2 • 74132
1 • 74151
1 • 74157
2 • 745157
1 • 74221
1 • 74393
3 • 8820A
5 • 8831
1 • X-TAL osc. 4.9152 MHz
38 pk.
OPPDAT T.O.M. ECO NR. 10-458 8.11.77 Eml
Påført 1101.B (ingen rettelser) 8.12.75 TAA/Eml
Påført pakkepos. 18.6.75 Eml
Rettet Dato Sign.
BRANCH CONTROL
1101.F
Målestokk
Tegn Eml 10.6.75
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A01
16 forandr. f. 10 pF 1. 22 pF. 5/4/76/TEAA/Eml


5C
10k
100ns
WCZ,
IRQ,
PBUSY,
FBY,
START,
14C
START,
IRQ,
17C
DRY,
BYR,
START,
FBY,
PICKL,
17C
FBY,
FBY,
100pF
22k
4B
INIT,
INIT,
START,
PICKL,
TEST,
7B
220Ω
330Ω
8A
READY,
READY,
NOPAP,
NOPAP,
ONLIN,
ONLIN,
6A
5C
START,
PBUSY,
BYR,
PL,
10pF
33k
33pF
22k
2C
DEL2
2C
PICKL,
PICKL,
PICKL,
4B
IRQ,
WCZ,
FIN,
LCW,
LCW,
MCM,
CLEAR,
PICLK,
PICLK,
PARIN,
FBY,
FBY,
START,
1000pF
START,
START,
MDB7,
RESET,
MDB10,
RFEED,
LCW,
MDB11,
REOTR,
MDB12,
RLTER,
DW0,
DW1,
DW2,
DW3,
DW4,
DW5,
DW6,
DW7,
DATAE,
INO1,
INO2,
INO3,
INO4,
INO5,
INO6,
INO7,
INO8,
INIT,
TEST,
READY,
PBUSY,
PBUSY,
FIN,
CPL,
COMPL,
M3,
M3,
WCZ,
WCZ,
TEST,
TEST,
1000pF
56k
10k
1B
1B
PBUSY,
M3,
CPL,
NOPAP,
ONLIN,
ERROR,
M1,
SPP,
MO,
PRINT,
ONLIN,
INIT,
NOPAP,
INIT,
TEST,
BRBUSY,
1 pk. 7414
5 pk. 7400
2 • 7402
2 • 7404
1 • 7410
5 • 7437
1 • 7474
4 • 74123
21 pk.
Rettet
Dato
Sign.
VERSATEC CONTROL
MATRIX PRINTER/PLOTTER
1102
Tegn Emil 13.6.75
Kontr J.H.
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A02





TRANSMIT-SWITCH (T) OPEN:
ALWAYS RECEIVE (A,B HAS NO EFFECT)

TRANSMIT-SWITCH (T) CLOSED:

A	B	
1	1	TRANSMIT (NORMAL CASE)
1	0	DISABLE
0	1	RECEIVE
0	0	RECEIVE

TERMINATE-SWITCH (Z) SHOULD BE CLOSED IN THE RECEIVE END ONLY

SWITCH-SETTING

* C1-20 ARE NORMALLY NOT MOUNTED. (NEEDED ONLY UNDER HEAVY NOISE CONDITION)

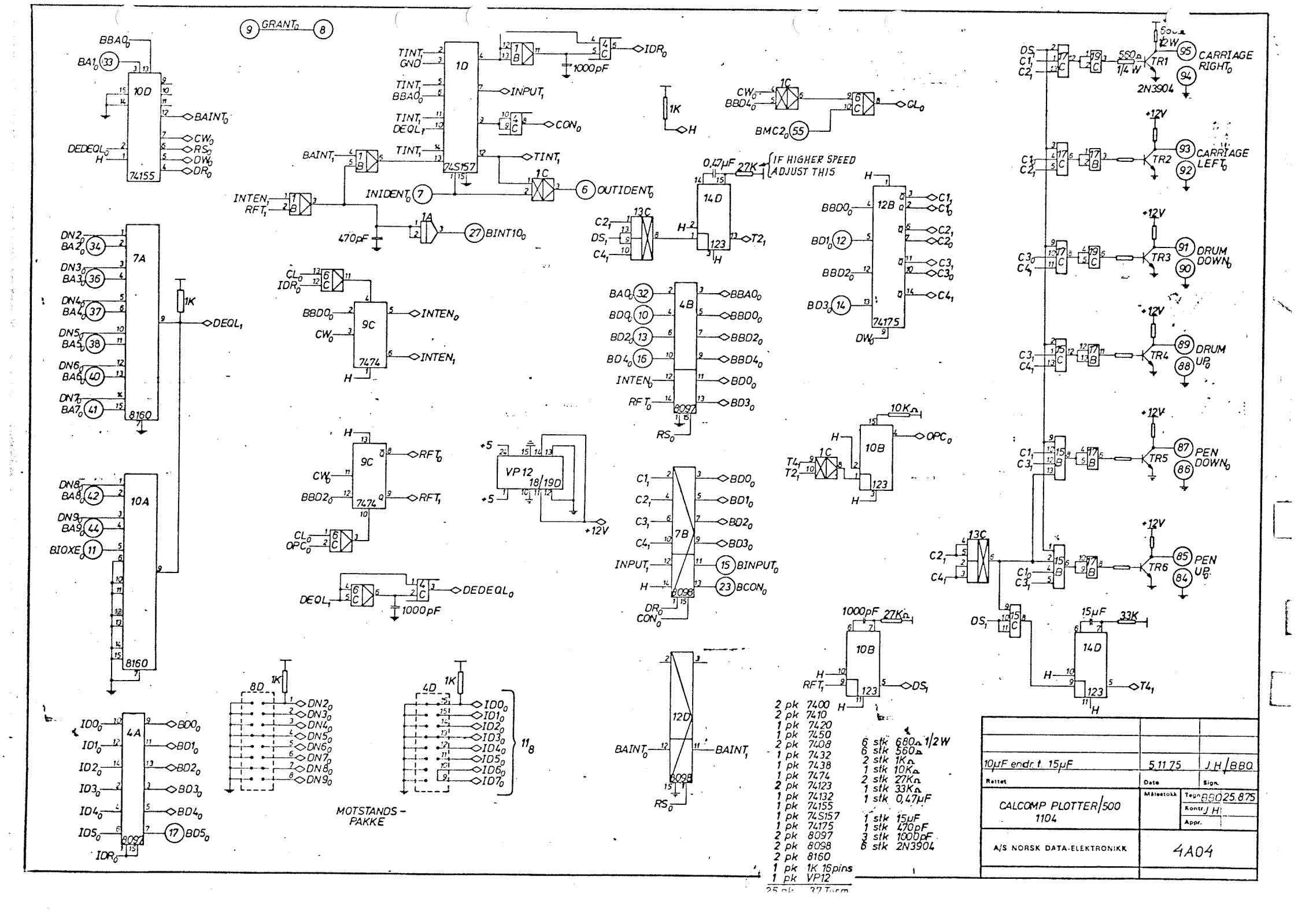
* SN75452B AND R24-R25 ARE NORMALLY NOT MOUNTED. (A0 IS GROUNDED BY STRAP X. B0 IS GROUNDED BY STRAP Y. IN ROW 2 ON CARD).

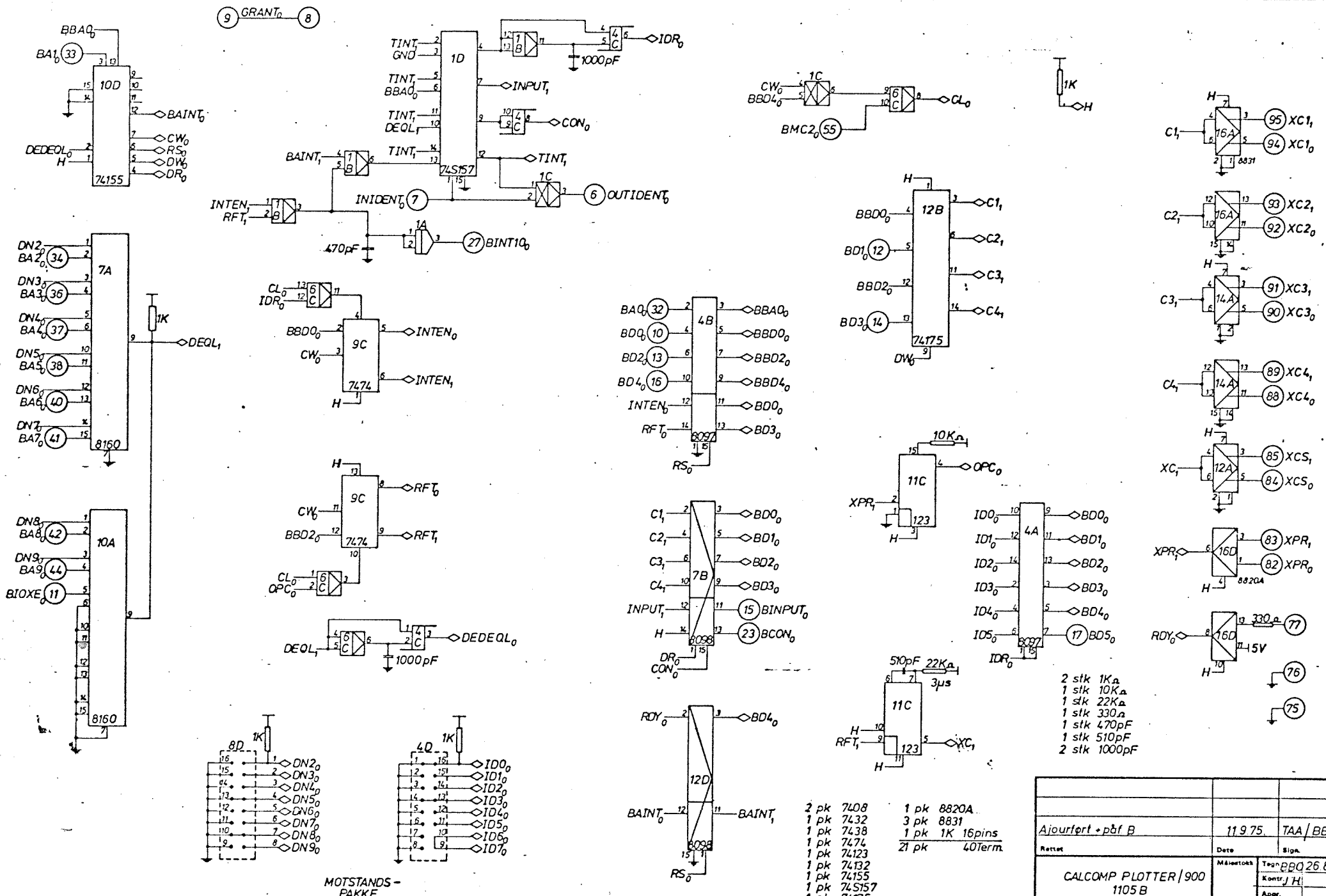
NOTE!

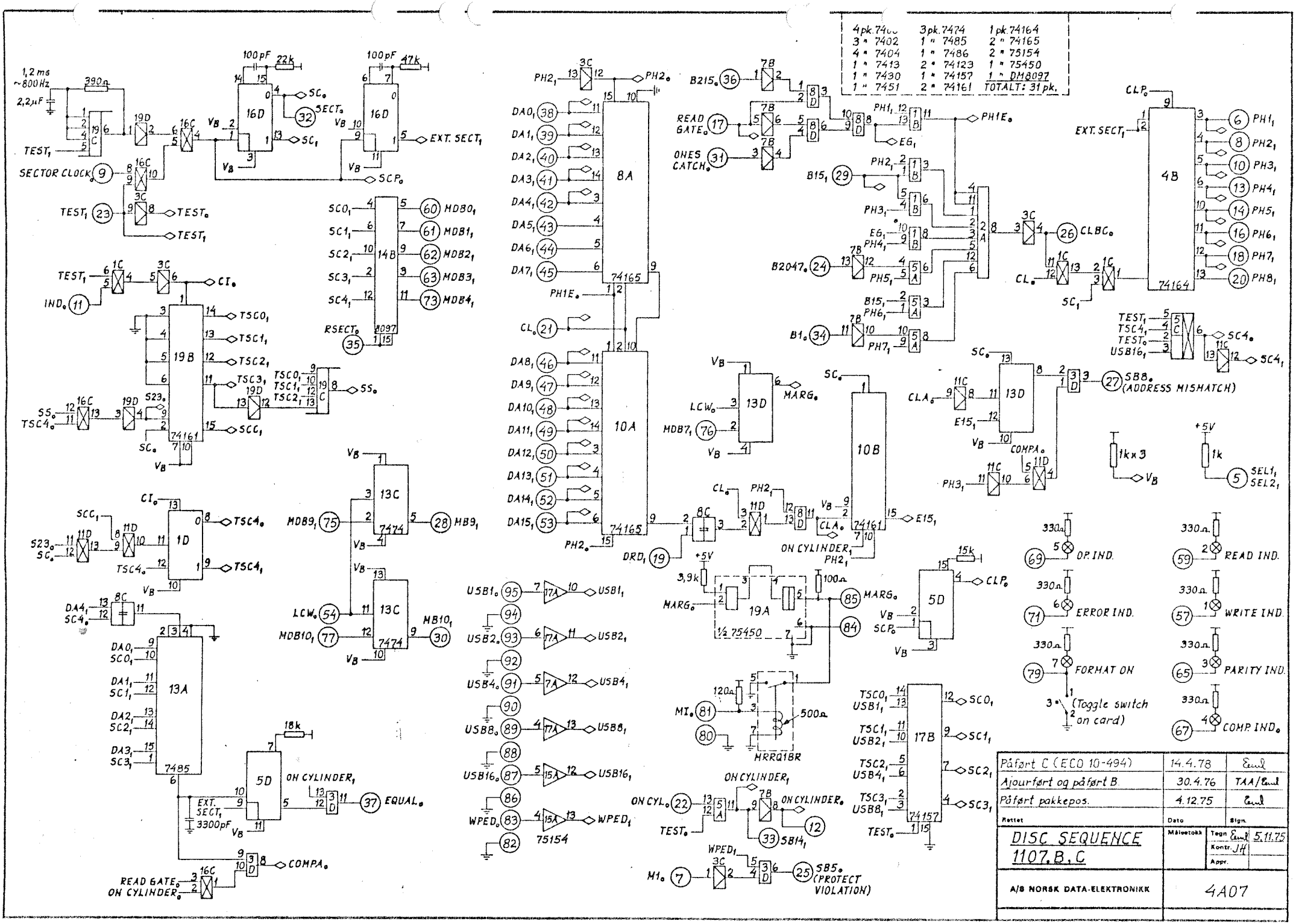
10 pk. DM8820A
10 • DM8831
1 • SN75452B
5 • SWITCH
26 pk.

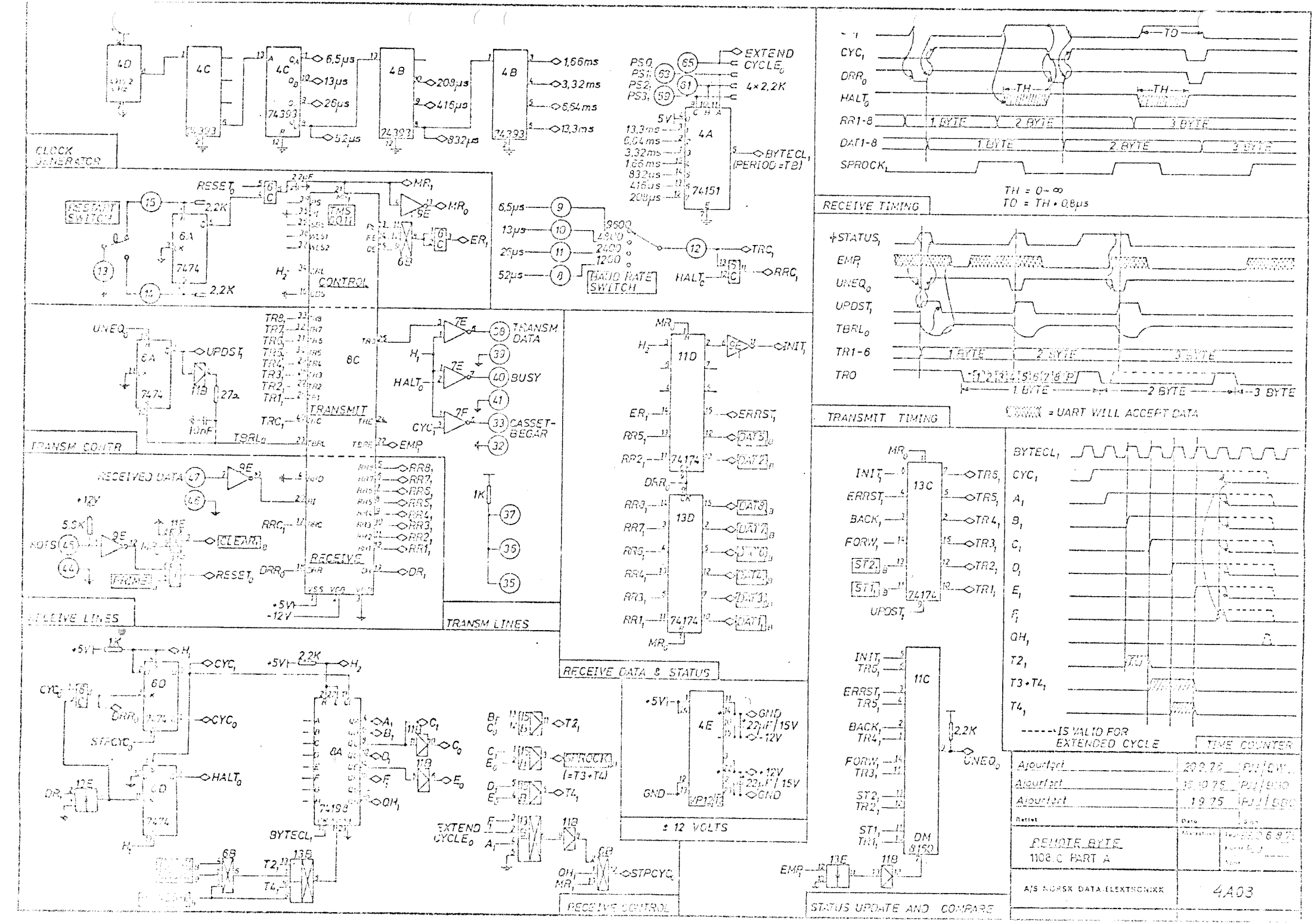
22 stk. 10kΩ (R1-R22)
1 • 1kΩ (R25)
2 • 560Ω (R23-R24)
20 • capacitors (C1-C20)

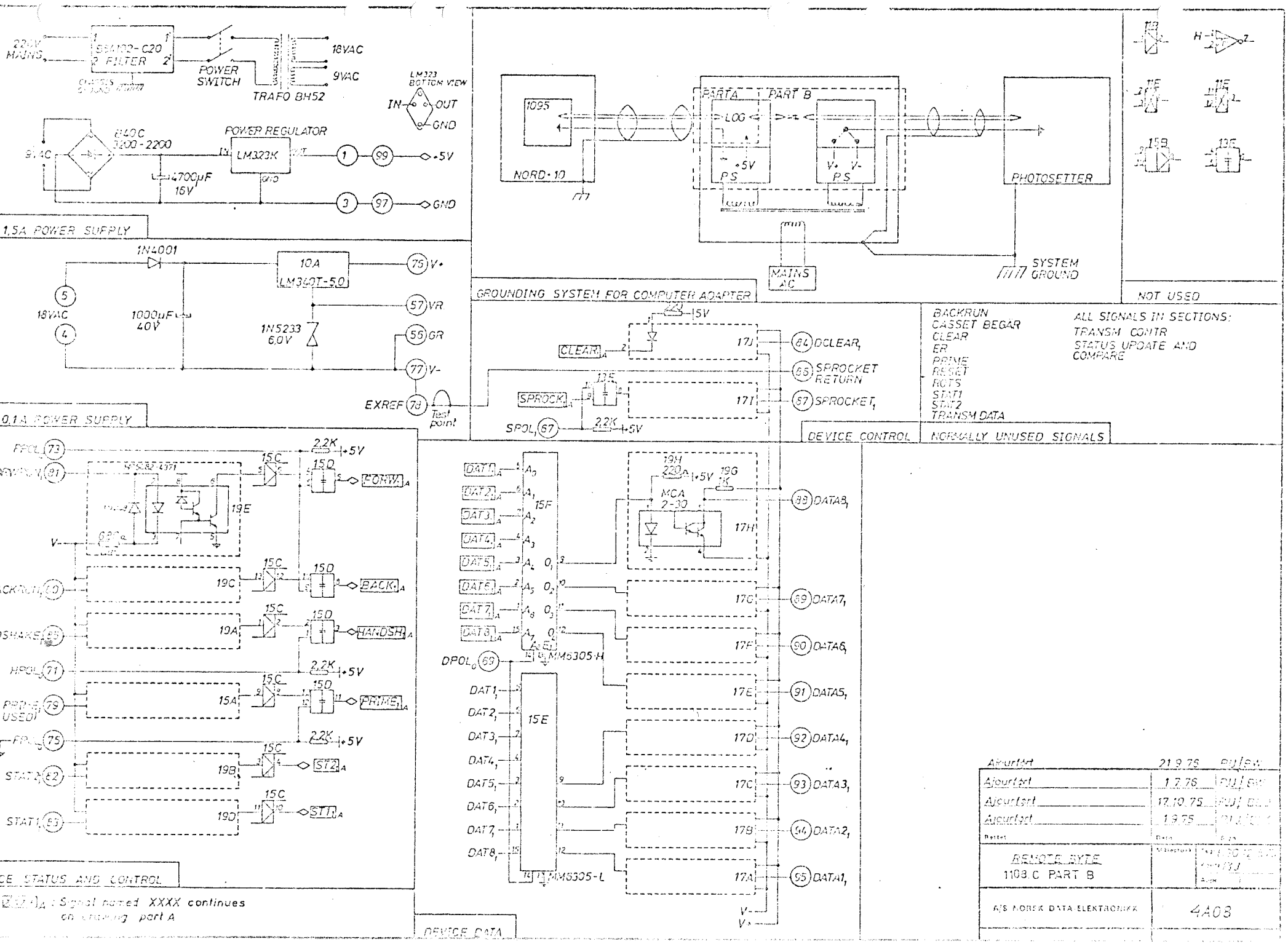
Rettet	Dato	Sign.
DIFF. TRANCEIVER 1103	Målestokk	Tegn. 17.6.75
		Kontr. PJJ
		Appr.
A/S NORSK DATA-ELEKTRONIKK		4A03

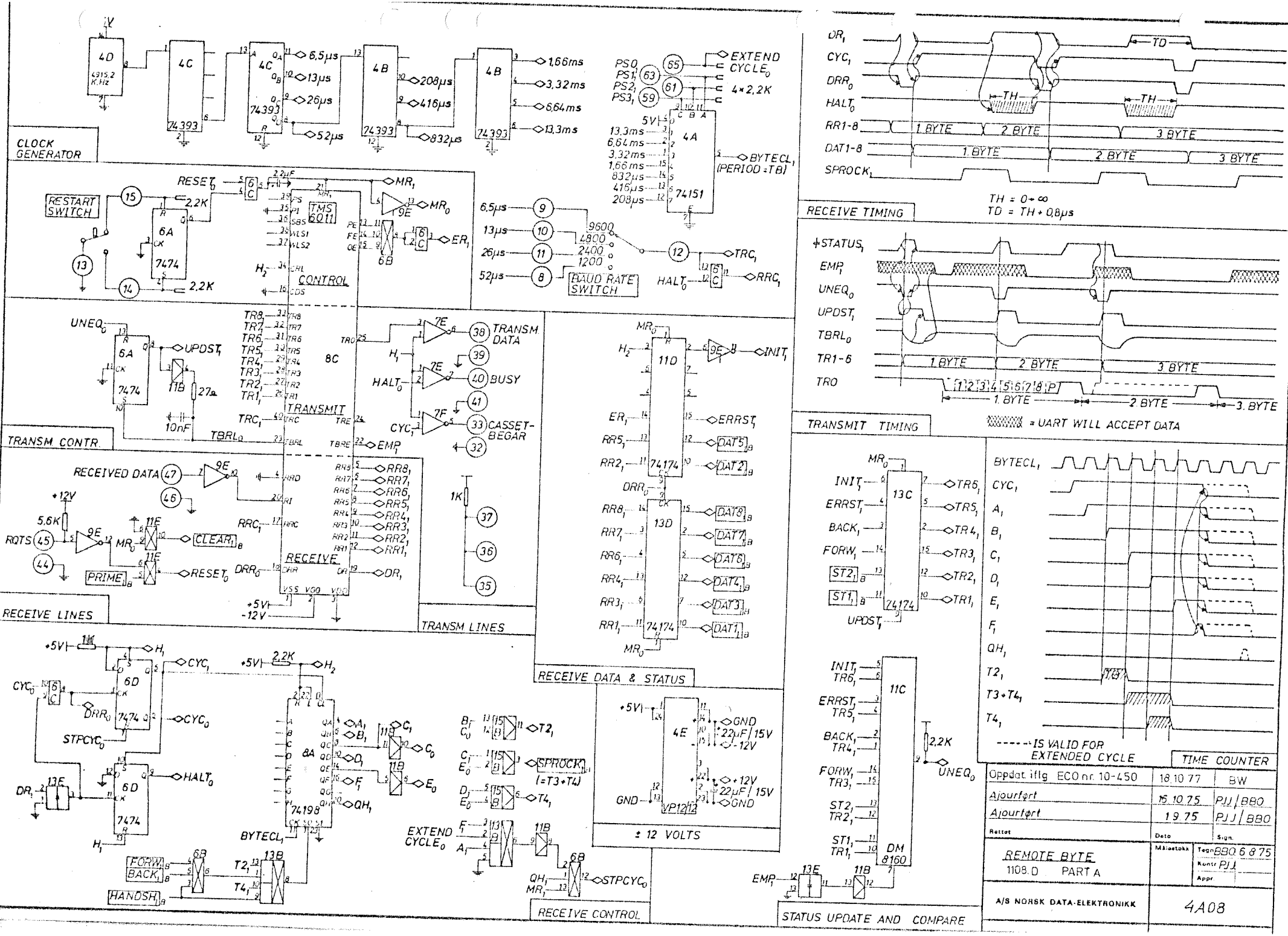

GRANT0
BBAQ0
BA1 0 (33)
10D
74155
BAINT0
DEDEQL0
H
CW0
RS0
DW0
DR0
TINT1
GND
TINT1
BBAQ0
TINT1
DEQL1
1D
74S157
INPUT1
CON0
TINT1
IDR0
1000pF
1K
H
CW0
BBD40
BMC2 0 (55)
1C
GL0
BAINT1
INTEN1
RFT1
INIDENT0 (7)
470pF
1A
27 BINT10
1C
6 OUTIDENT0
0,47µF
27K
IF HIGHER SPEED ADJUST THIS
C21
DS1
C41
13C
H
14D
123
T21
DN2
BA2 0 (34)
DN3
BA3 0 (36)
DN4
BA4 0 (37)
DN5
BA5 0 (38)
DN6
BA6 0 (40)
DN7
BA7 0 (41)
7A
8160
1K
DEQL1
CL0
IDR0
BBD0 0
CW0
9C
7474
INTEN0
INTEN1
H
BAQ 0 (32)
BDQ 0 (10)
BD2 0 (13)
BD4 0 (16)
INTEN0
RFT0
4B
8097
BBAQ0
BBD0 0
BBD2 0
BBD4 0
BD0 0
BD3 0
RS0
H
12B
BBD0 0
BD1 0 (12)
BBD2 0
BD3 0 (14)
74175
DW0
C1 1
C1 0
C2 1
C2 0
C3 1
C3 0
C4 1
H
9C
CW0
BBD2 0
7474
RFT0
RFT1
CL0
QPC0
+5
VP 12
18/19D
+12V
+5
10K Ω
H
T4 1
T2 1
1C
10B
123
H
OPC0
C11
C21
7B
C31
C41
INPUT1
H
8098
DR0
CON0
BD0 0
BD1 0
BD2 0
BD3 0
15 BINPUT0
23 BCON0
DEQL1
CL0
QPC0
1000pF
DEDEQL0
DN8
BA8 0 (42)
DN9
BA9 0 (44)
BIOXE 0 (11)
10A
8160
ID0 0
ID1 0
ID2 0
ID3 0
ID4 0
ID5 0
4A
8098
IDR0
BD0 0
BD1 0
BD2 0
BD3 0
BD4 0
17 BD5 0
8D
1K
DN2 0
DN3 0
DN4 0
DN5 0
DN6 0
DN7 0
DN8 0
DN9 0
4D
1K
ID0 0
ID1 0
ID2 0
ID3 0
ID4 0
ID5 0
ID6 0
ID7 0
11 8
MOTSTANDS-PAKKE
12D
BAINT0
BAINT1
8098
RS0
1000pF 27KΩ
10B
H
RFT1
123
H
DS1
15µF 33K
14D
H
123
T41
H
DS1
C21
C41
13C
C11
C31
DS1
C11
C21
C31
500Ω 2W
580Ω 1/4 W
TR1
2N3904
95 CARRIAGE RIGHT0
94
+12V
TR2
93 CARRIAGE LEFT0
92
TR3
91 DRUM DOWN0
90
TR4
89 DRUM UP0
88
TR5
87 PEN DOWN0
86
TR6
85 PEN UP0
84
2 pk 7400
2 pk 7410
1 pk 7420
1 pk 7450
2 pk 7408
1 pk 7432
1 pk 7438
1 pk 7474
2 pk 74123
1 pk 74132
1 pk 74155
1 pk 74S157
1 pk 74175
2 pk 8097
2 pk 8098
2 pk 8160
1 pk 1K 16pins
1 pk VP12
6 stk 680Ω 1/2W
6 stk 560Ω
2 stk 1KΩ
1 stk 10KΩ
2 stk 27KΩ
1 stk 33KΩ
1 stk 0,47µF
1 stk 15µF
1 stk 470pF
3 stk 1000pF
6 stk 2N3904
10µF endr t. 15µF
5.11.75
J.H/BBQ
Rettet
Dato
Sign.
CALCOMP PLOTTER/500
1104
Måleetokk
Tegn. BBQ 25.8.75
Kontr. J.H.
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A04

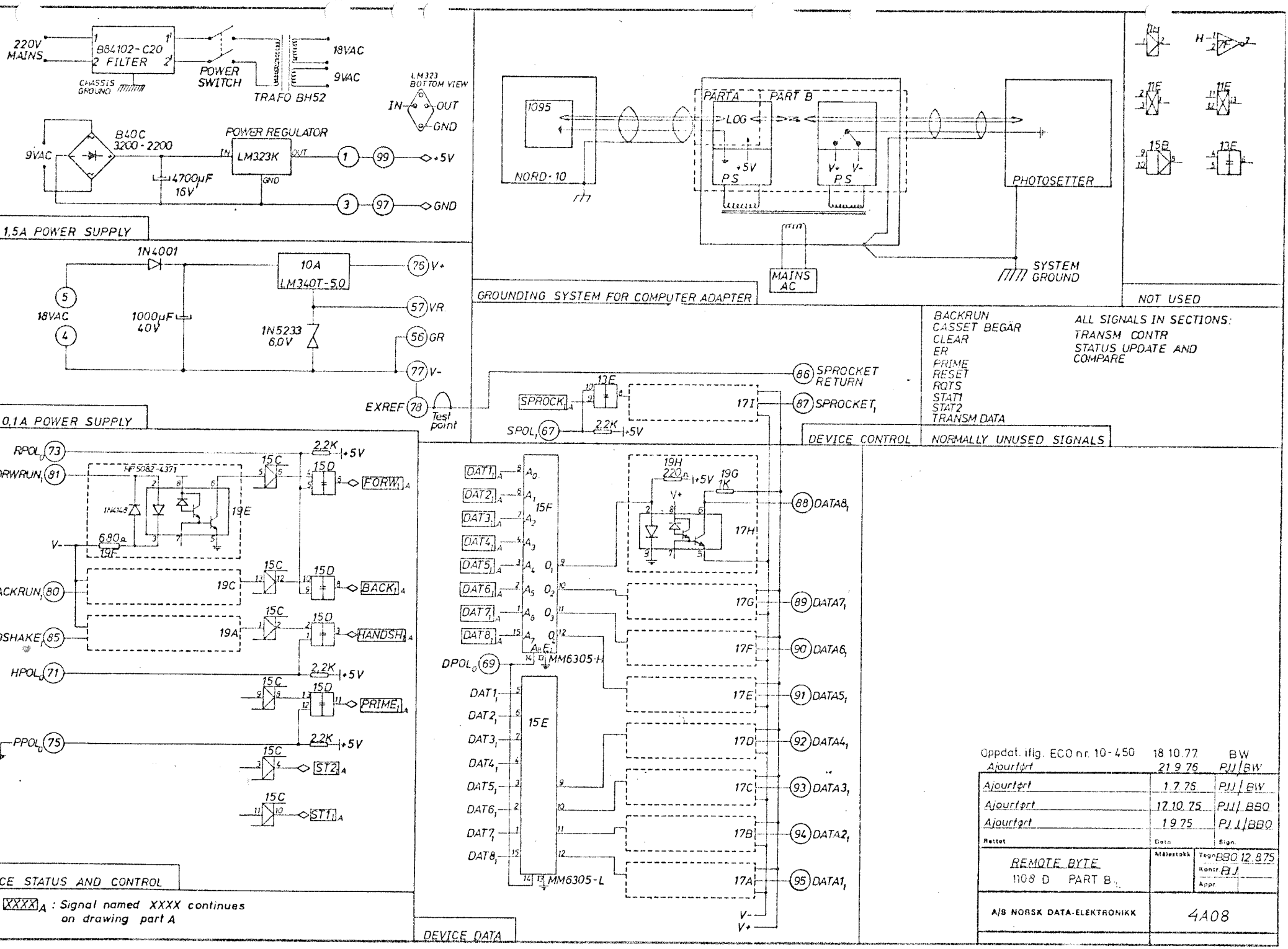

GRANT0
BBAQ0
BA1 0 (33)
10D
74155
BAINT0
CW0
RS0
DW0
DR0
DEDEQL0
H
TINT1
GND
BBAQ0
DEQL1
74S157
INPUT1
CON0
IDR0
1000pF
BAINT1
INTEN1
RFT1
INIDENT0 (7)
OUTIDENT0 (6)
470pF
27 BINT10
1C
CW0
BBD40
BMC20 (55)
CL0
1K
DN2 0 BA2 0 (34)
DN3 0 BA3 0 (36)
DN4 0 BA4 0 (37)
DN5 0 BA5 0 (38)
DN6 0 BA6 0 (40)
DN7 0 BA7 0 (41)
7A
8160
DEQL1
DN8 0 BA8 0 (42)
DN9 0 BA9 0 (44)
BIOXE0 (11)
10A
8160
IDR0
9C
7474
BBD00
CW0
INTEN0
INTEN1
RFT0
RFT1
BBD20
CL0
OPC0
DEQL1
1000pF
DEDEQL0
8D
40
DN20
DN30
DN40
DN50
DN60
DN70
DN80
DN90
ID00
ID10
ID20
ID30
ID40
ID50
ID60
ID70
MOTSTANDS-
PAKKE
BA0 (32)
BD0 (10)
BD2 (13)
BD4 (16)
INTEN0
RFT0
4B
8097
RS0
BBA0
BBD0
BBD2
BBD4
BD0
BD3
C1
C2
C3
C4
INPUT1
H
DR0
CON0
7B
8098
15 BINPUT0
23 BCON0
ROY0
12D
BAINT0
BAINT1
RS0
BD40
BB00
BD10
BD20
BD30
12B
74175
DW0
10K
11C
XPR1
OPC0
123
H
ID00
ID10
ID20
ID30
ID40
ID50
4A
8097
17 BD50
IDR0
510pF
22K
3µs
11C
RFT1
XC1
123
H
2 stk 1K
1 stk 10K
1 stk 22K
1 stk 330
1 stk 470pF
1 stk 510pF
2 stk 1000pF
1K
H
C1
16A
8831
95 XC10
94 XC11
C2
93 XC20
92 XC21
C3
14A
91 XC30
90 XC31
C4
89 XC40
88 XC41
XC
12A
85 XCS0
84 XCS1
XPR1
16D
8820A
83 XPR0
82 XPR1
ROY0
330
77
5V
76
75
2 pk 7408
1 pk 7432
1 pk 7438
1 pk 7474
1 pk 74123
1 pk 74132
1 pk 74155
1 pk 74S157
1 pk 74175
2 pk 8097
2 pk 8098
2 pk 8160
1 pk 8820A
3 pk 8831
1 pk 1K 16pins
21 pk 40Term.
Ajourført + påf B
11 9 75
TAA/BBO
Rettet
Dato
Sign.
CALCOMP PLOTTER/900
1105 B
Målestokk
Tegn BBO 26 8 75
Kontr. / H
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A05


4pk. 74xx
3 " 7402
4 " 7404
1 " 7413
1 " 7430
1 " 7451
3pk. 7474
1 " 7485
1 " 7486
2 " 74123
1 " 74157
2 " 74161
1 pk. 74164
2 " 74165
2 " 75154
1 " 75450
1 " DM8097
TOTALT: 31 pk.
DISC SEQUENCE
1107.B.C
A/B NORSK DATA-ELEKTRONIKK
4A07
Påført C (ECO 10-494) 14.4.78 Emil
Ajourført og påført B 30.4.76 TAA/Emil
Påført pakkepos. 4.12.75 Emil
Rettet Dato Sign.
Målestokk
Tegn Emil 5.11.75
Kontr. JH
Appr.
1,2 ms ~800Hz
2,2µF
390Ω
TEST1
SECTOR CLOCK
TEST1 (23)
TEST0
TEST1
100pF 22k
100pF 47k
16D
SC0
SECT0
SC1
EXT. SECT1
SCP0
VB
74165
74161
74164
74157
74B097
74154
7485
7474
75450
8A
10A
10B
13A
13B
13C
13D
14B
17A
17B
19A
19B
1D
4B
5D
PH2
PH1E
CL
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
DA8
DA9
DA10
DA11
DA12
DA13
DA14
DA15
DRD1 (19)
B215 (36)
READ GATE0 (17)
ONES CATCH0 (31)
B15 (29)
B2047 (24)
B1 (34)
PH1
PH2
PH3
PH4
PH5
PH6
PH7
EG1
CLBC0 (26)
SC1
EXT. SECT1
CLP0
PH1 (6)
PH2 (8)
PH3 (10)
PH4 (13)
PH5 (14)
PH6 (16)
PH7 (18)
PH8 (20)
TEST1
TSC41
TEST0
USB161
SC4
SC41
SC0 (13)
CLA0
E151
COMPA0
SB8 (27)
(ADDRESS MISMATCH)
1kx3
VB
+5V
1k
SEL1 (5)
SEL2
CI0
SCC1
S230
SC0
TSC40
TSC41
SS0
S5
TSC40
SCC1
IND0 (11)
TSC00
TSC01
TSC02
TSC03
MDB0 (60)
MDB1 (61)
MDB2 (62)
MDB3 (63)
MDB4 (73)
RSECT0 (35)
MDB9 (75)
MB9 (28)
LCW0 (54)
MDB10 (77)
MB10 (30)
MDB7 (76)
MARG0
USB10 (95)
USB20 (93)
USB40 (91)
USB80 (89)
USB160 (87)
WPED0 (83)
(82)
USB11
USB21
USB41
USB81
USB161
WPED1
MI0 (81)
(80)
MRRQ18R
120Ω
500Ω
100Ω
3,9k
ON CYLINDER1
ON CYL0 (22)
ON CYLINDER0
SB14 (33)
(12)
M10 (7)
SB5 (25)
(PROTECT VIOLATION)
TEST0
DA40
SC40
DA00
SC01
DA10
SC11
DA20
SC21
DA30
SC31
EXT. SECT1
3300pF
EQUAL0 (37)
COMPA0
READ GATE0
ON CYLINDER0
18k
15k
330Ω
OP. IND. (69)
ERROR IND. (71)
FORMAT ON (79)
(Toggle switch on card)
READ IND. (59)
WRITE IND. (57)
PARITY IND. (65)
COMP. IND0 (67)
MARG0 (85)
(84)
CLP0
SCP0
ON CYLINDER1
PH2
CL0 (21)
PH1E0
DA00 (38)
DA10 (39)
DA20 (40)
DA30 (41)
DA40 (42)
DA50 (43)
DA60 (44)
DA70 (45)
DA80 (46)
DA90 (47)
DA100 (48)
DA110 (49)
DA120 (50)
DA130 (51)
DA140 (52)
DA150 (53)
PH20
DRD1 (19)
SC0
CL0
PH21
CLA0
ON CYLINDER1
PH21
E151
MARG0
MDB7 (76)
LCW0
VB
SC0
CLA0
E151
COMPA0
PH31
SB8 (27)
(ADDRESS MISMATCH)
1kx3
+5V
1k
SEL1 (5)
SEL2
TSC00
USB11
TSC01
USB21
TSC02
USB41
TSC03
USB81
TEST0
SC01
SC11
SC21
SC31
TSC00
TSC01
TSC02
TSC03
TSC40
TSC41
SS0
S5
TSC40
SCC1
IND0 (11)
TEST1
CI0
SC0
SCC1
S230
TSC40
TSC40
TSC41
DA40
SC40
DA00
SC01
DA10
SC11
DA20
SC21
DA30
SC31
EXT. SECT1
3300pF
EQUAL0 (37)
COMPA0
READ GATE0
ON CYLINDER0
18k
ON CYLINDER1
MDB9 (75)
MB9 (28)
LCW0 (54)
MDB10 (77)
MB10 (30)
USB10 (95)
USB20 (93)
USB40 (91)
USB80 (89)
USB160 (87)
WPED0 (83)
(82)
USB11
USB21
USB41
USB81
USB161
WPED1
MI0 (81)
(80)
MRRQ18R
120Ω
500Ω
100Ω
3,9k
MARG0
MARG0 (85)
(84)
15k
CLP0
SCP0
ON CYL0 (22)
ON CYLINDER0
SB14 (33)
(12)
M10 (7)
SB5 (25)
(PROTECT VIOLATION)
TEST0
330Ω
OP. IND. (69)
ERROR IND. (71)
FORMAT ON (79)
(Toggle switch on card)
READ IND. (59)
WRITE IND. (57)
PARITY IND. (65)
COMP. IND0 (67)


CLOCK GENERATOR
RESET SWITCH
CONTROL
BAUD RATE SWITCH
TRANSMIT
TRANSM CONTR
TRANSM DATA
BUSY
CASSETT-BEGAR
RECEIVE
RECEIVE LINES
TRANSM LINES
RECEIVE DATA & STATUS
± 12 VOLTS
RECEIVE CONTROL
STATUS UPDATE AND COMPARE
RECEIVE TIMING
TRANSMIT TIMING
TIME COUNTER
= UART WILL ACCEPT DATA
IS VALID FOR EXTENDED CYCLE
REMOTE BYTE
1108.C PART A
A/S NORSK DATA ELEKTRONIKK
4A03


220V MAINS
SSA102-C20 FILTER
POWER SWITCH
TRAFO BH52
18VAC
9VAC
LM323 BOTTOM VIEW
IN
OUT
GND
9VAC
BAC 2200
POWER REGULATOR
LM323K
4700µF 16V
+5V
GND
5V - 1.5A POWER SUPPLY
1N4001
10A LM340T-5.0
18VAC
1000µF 40V
1N5233 6.0V
V+
VR
GR
V-
EXREF
Test point
12V - 0.1A POWER SUPPLY
NORD-10
PART A
PART B
LOG
+5V P.S
V+ V- P.S
PHOTOSETTER
MAINS AC
SYSTEM GROUND
GROUNDING SYSTEM FOR COMPUTER ADAPTER
NOT USED
CLEAR
SPROCK
SPOL
2.2K
+5V
DCLEAR
SPROCKET RETURN
SPROCKET
BACKRUN
CASSET BEGAR
CLEAR
ER
PRIME
RESET
RQTS
STAT1
STAT2
TRANSM DATA
ALL SIGNALS IN SECTIONS:
TRANSM CONTR
STATUS UPDATE AND COMPARE
DEVICE CONTROL
NORMALLY UNUSED SIGNALS
FPOL
FORWARD
BACKRUN
HANDSHAKE
HPOL
PRIME (NOT USED)
FR
STAT2
STAT1
MCA 2-30
DAT0
DAT1
DAT2
DAT3
DAT4
DAT5
DAT6
DAT7
DAT8
DPOL
MM6305-H
MM6305-L
DATA8
DATA7
DATA6
DATA5
DATA4
DATA3
DATA2
DATA1
DEVICE STATUS AND CONTROL
DEVICE DATA
(XXXX)A : Signal named XXXX continues on sheet part A
Ajourført 21.9.76 Rutan
Ajourført 1.2.76 PUJ/BIO
Ajourført 17.10.75 PUJ/BIO
Ajourført 1.9.75 PUJ/BIO
Rettet Dato Sign.
REMOTE SITE 1108.C PART B
Tegningsnr. 4A08
A/S NORSK DATA ELEKTRONIKK


CLOCK GENERATOR
RESTART SWITCH
RESET
CONTROL
TMS 6011
BAUD RATE SWITCH
EXTEND CYCLE
4×2,2K
74151
BYTECL1 (PERIOD = T8)
TRANSM CONTR
TRANSMIT
TRANSM DATA
BUSY
CASSET-BEGAR
RECEIVE LINES
RECEIVE
TRANSM LINES
RECEIVE DATA & STATUS
± 12 VOLTS
RECEIVE CONTROL
STATUS UPDATE AND COMPARE
RECEIVE TIMING
TH = 0 → ∞
TD = TH + 0,8µs
TRANSMIT TIMING
= UART WILL ACCEPT DATA
IS VALID FOR EXTENDED CYCLE
TIME COUNTER
REMOTE BYTE
1108.0 PART A
A/S NORSK DATA-ELEKTRONIKK
4A08



Oppdat. iflg. ECO nr. 10-450 18.10.77 BW

Ajourført 21.9.76 PJJ/BW

Ajourført	1.7.75	PJJ / BW
Ajourført	17.10.75	PJJ / BBO
Ajourført	1.9.75	PJJ / BBO
Rettet	Dato	Sign.

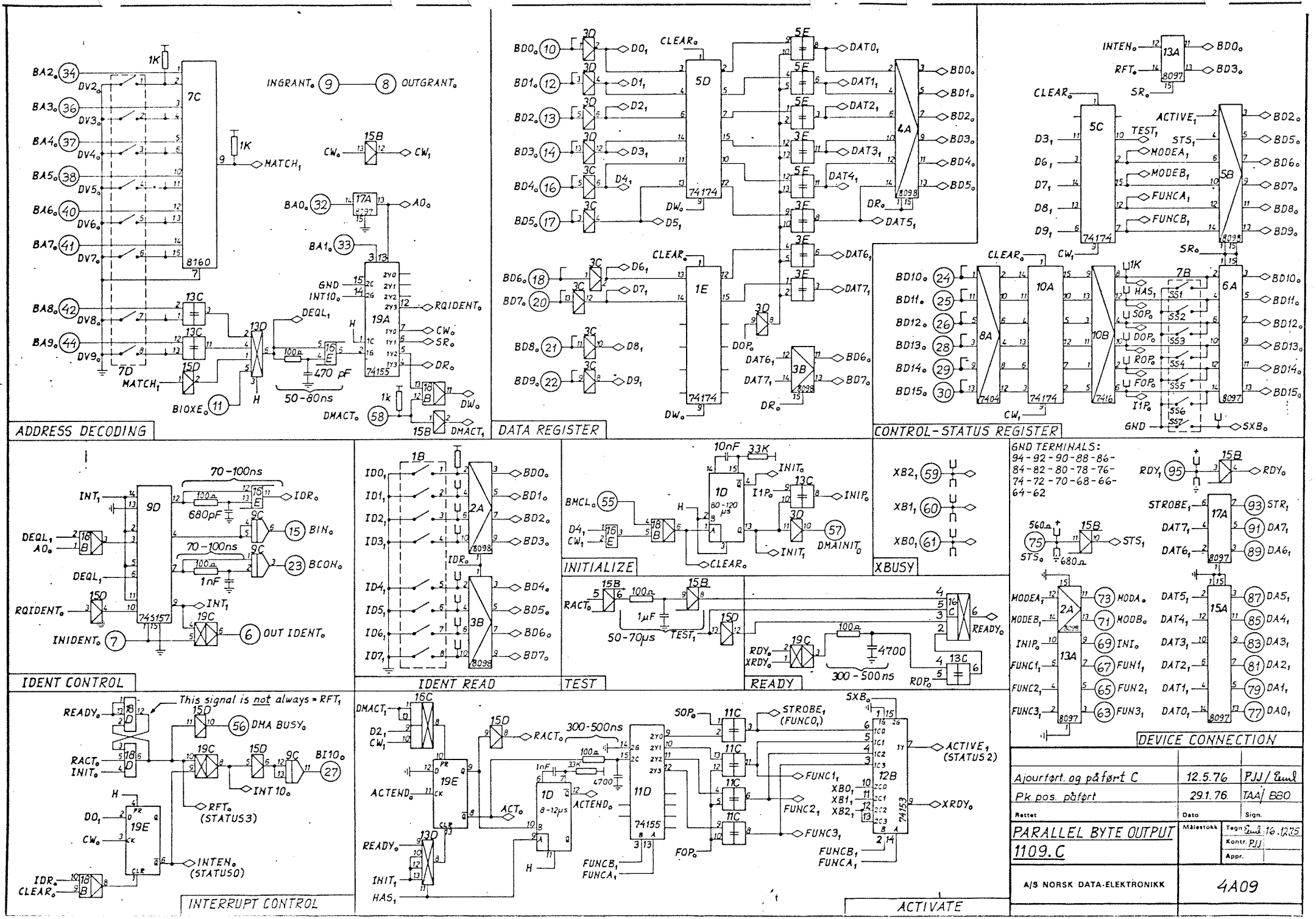
REMOTE BYTE
1108 D PART B

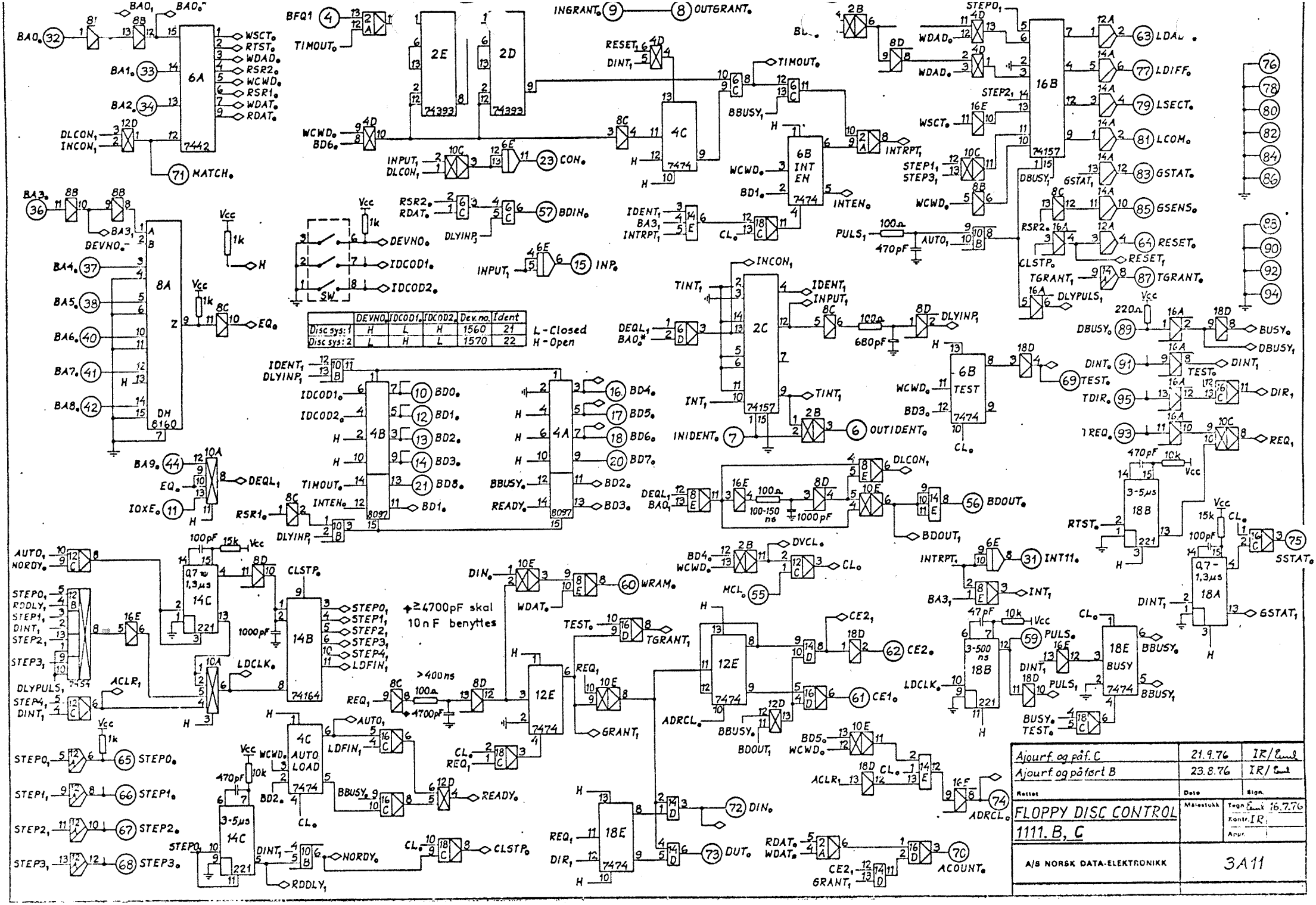
Målestokk

Tegn BBO 12.8.75
Kontr BJ
Appr

A/S NORSK DATA-ELEKTRONIKK

4A08


ADDRESS DECODING
DATA REGISTER
CONTROL-STATUS REGISTER
IDENT CONTROL
IDENT READ
INITIALIZE
XBUSY
TEST
READY
DEVICE CONNECTION
INTERRUPT CONTROL
ACTIVATE
GND TERMINALS:
94-92-90-88-86-
84-82-80-78-76-
74-72-70-68-66-
64-62
This signal is not always = RFT1
PARALLEL BYTE OUTPUT
1109.C
A/S NORSK DATA-ELEKTRONIKK
4A09
Ajourført og påført C
12.5.76
PJJ / Emil
Pk pos. påført
29.1.76.
TAA/ B60
Rettet
Dato
Sign.
Målestokk
Tegn Emil 16.12.75
Kontr. PJJ
Appr.



	DEVNO	IDCOD1	IDCOD2	Dev.no.	Ident
Disc sys: 1	H	L	H	1560	21
Disc sys: 2	L	H	L	1570	22

L - Closed
H - Open

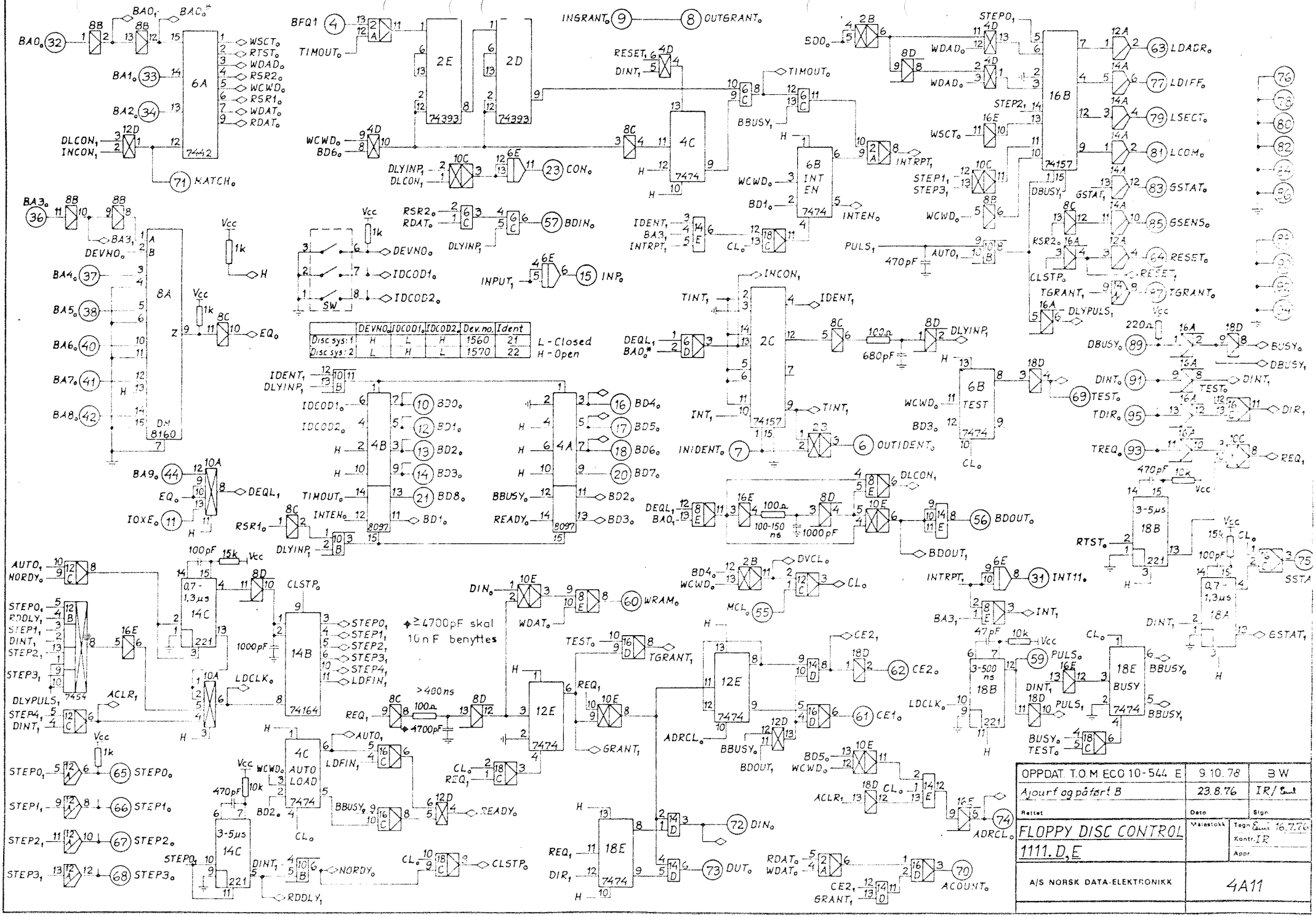
Ajourf. og påf. C	21.9.76	IR/Emil
Ajourf og påført B	23.8.76	IR/Emil
Rettet	Dato	Sign.

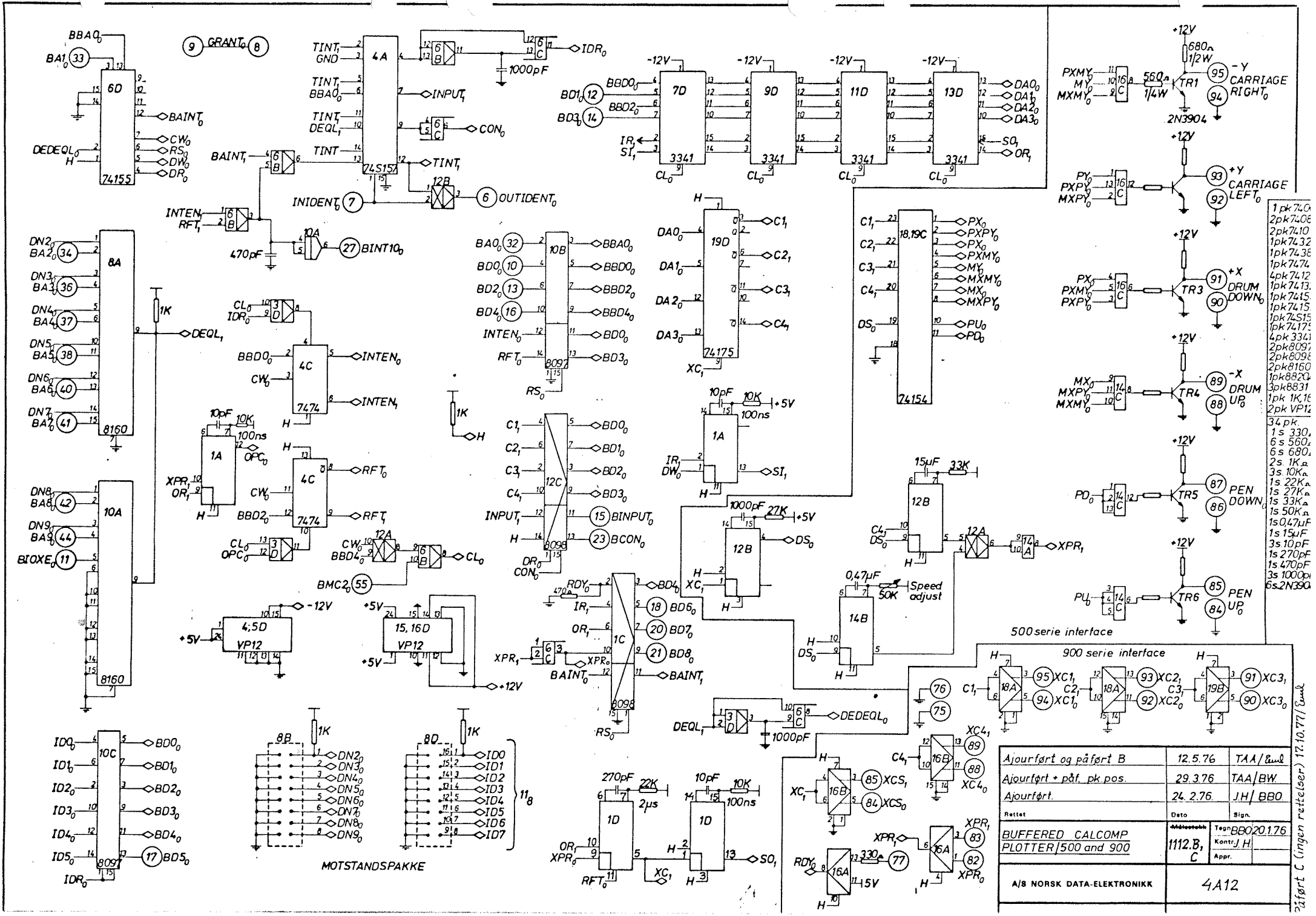
FLOPPY DISC CONTROL
1111. B, C

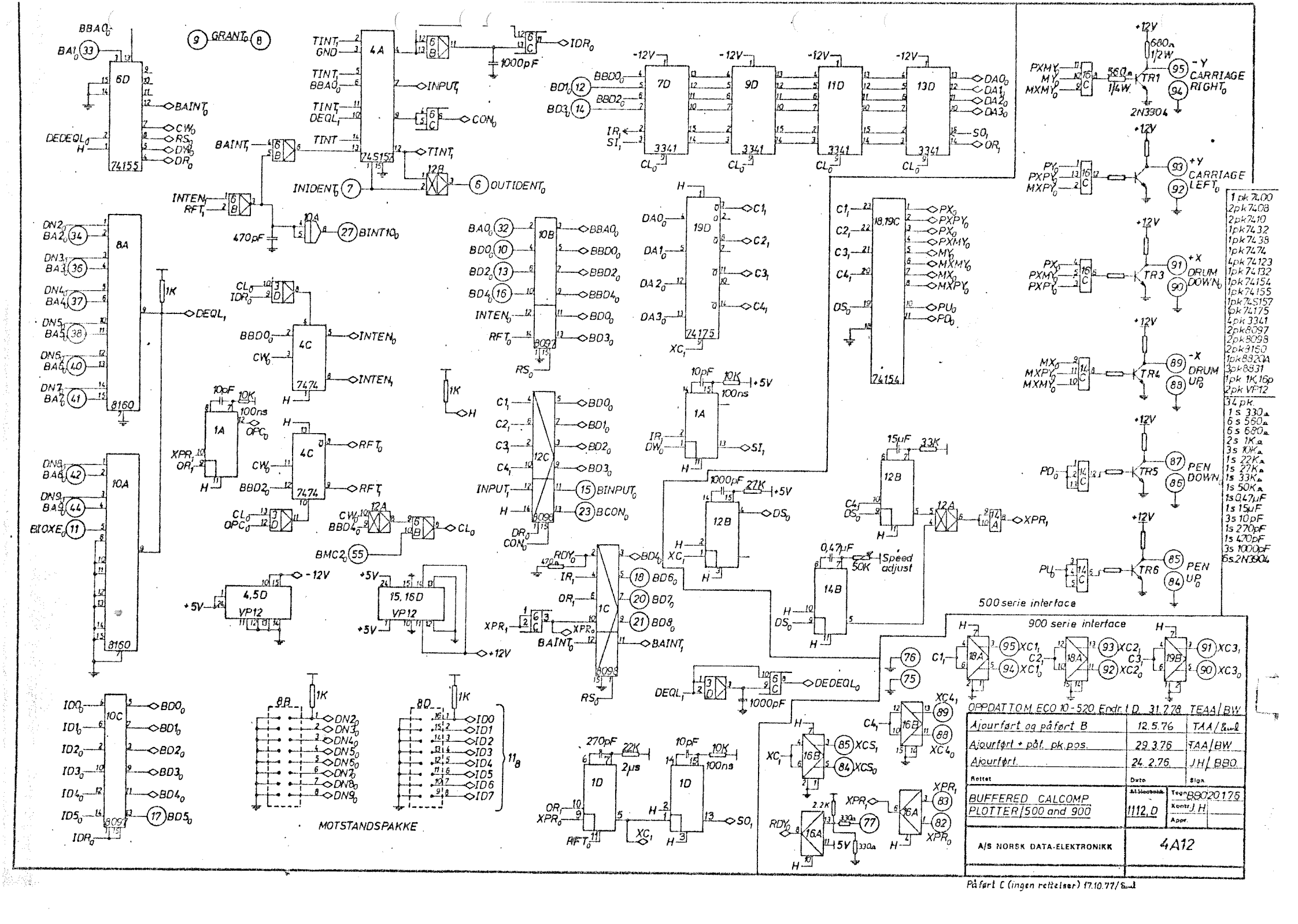
Malestokk | Tegn. Emil 16.7.76 | Kontr. IR | Appr.

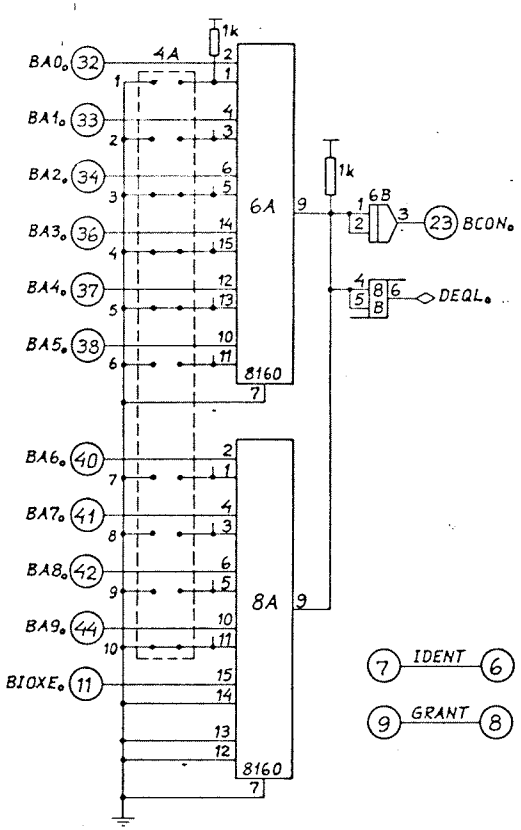
A/S NORSK DATA-ELEKTRONIKK

3A11

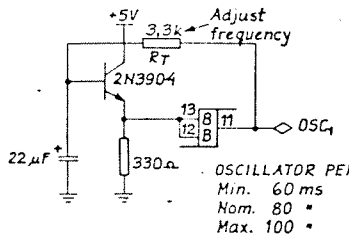

6A
74S42
2E
2D
74393
4C
74S74
6B INT EN
74S74
16B
74157
8A
DM 8160
2C
74157
4B
4A
8097
6B TEST
74S74
14C
74164
14B
4C AUTO LOAD
74S74
12E
74S74
18E BUSY
74S74
18B
74S21
18A
74S21
16A
74S21
3-5µs
0,7-1,3µs
3-500 ns
100-150 ns
>400ns
≥4700pF skal 10nF benyttes
DEVNO IDCOD1 IDCOD2 Dev.no. Ident
Disc sys: 1 H L H 1560 21
Disc sys: 2 L H L 1570 22
L - Closed
H - Open
OPPDAT. T.O.M ECO 10-544 E 9.10.78 BW
Ajourf og påført B 23.8.76 IR/
Rettet Dato Sign
FLOPPY DISC CONTROL
1111. D,E
Målestokk
Tegn 16.7.76
Kontr. IR
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A11


MOTSTANDSPAKKE
500serie interface
900 serie interface
BUFFERED CALCOMP PLOTTER/500 and 900
A/S NORSK DATA-ELEKTRONIKK
4A12
1112.B, C
Ajourført og påført B 12.5.76 TAA/Emil
Ajourført + påf. pk pos. 29.3.76 TAA/BW
Ajourført. 24.2.76 J.H/BBO
Rettet Dato Sign.
Tegn BBO 20.1.76
Kontr. J.H
Appr.
påført C (ingen rettelser) 17.10.77/Emil


500 serie interface
900 serie interface
MOTSTANDSPAKKE
OPPDATTOM ECO 10-520 Endr.ID 31.278 TEAA/BW
Ajourført og påført B 12.5.76 TAA/BW
Ajourført + påt. pk.pos. 29.3.76 TAA/BW
Ajourført. 24.2.76 JH/BBO
BUFFERED CALCOMP PLOTTER/500 and 900
1112.D
A/S NORSK DATA-ELEKTRONIKK
4A12
Påført C (ingen rettelser) 17.10.77/BW

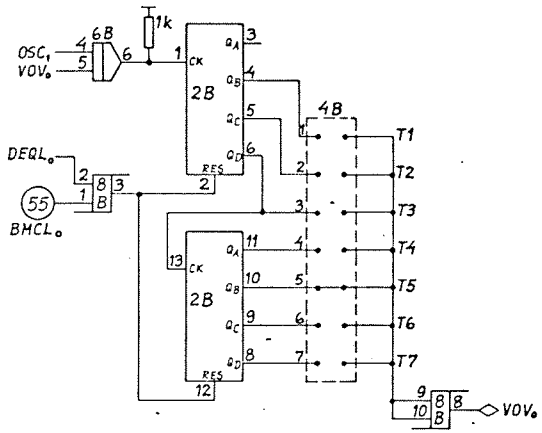


DEVICE SELECT

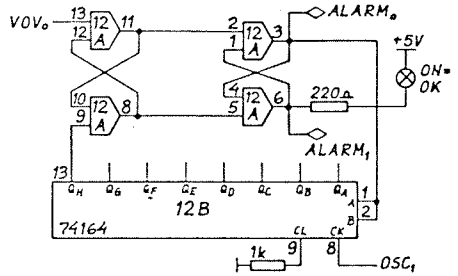


OSCILLATOR PERIOD:
Min. 60 ms
Nom. 80 "
Max. 100 "

12,5 HZ OSCILLATOR

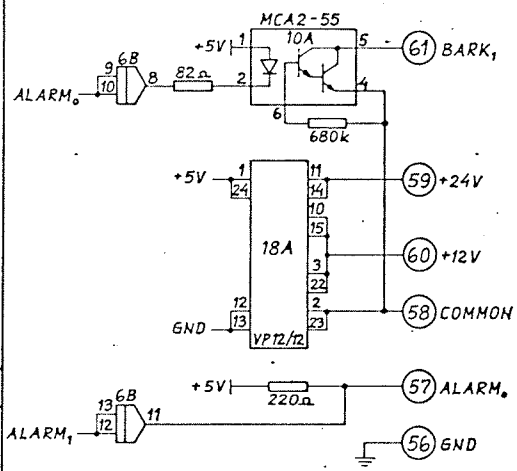


- WATCH TIME GENERATE



SHIFT REGISTER DELAY:
Min. 7·60 = 420 ms
Nom. 7,5·80 = 600 "
Max. 8·100 = 800 "

LOGIC PROBE



DEVICE CONNECTION

NO.	SIGNAL	POL.	PLUG BERG	PLUG BURNDY
18	BARK	1	61	NN
	+12V		60	RR
19	+24V		59	SS
	COMMON		58	UU
20	ALARM	0	57	TT
	GND		56	VV

PLUG LIST

WATCH TIME 2,5 s
DEVICE NUMBER 1034

FIXED STRAPS

1 PK. 7437
1 " 7438
1 " 74132
1 " 74164
1 " 74393
2 " 8160
1 " MCA 2-55
1 " VP 12/12
1 " 1K RESISTOR

10 PK.

WATCH TIMES IN ms.

	MIN	NOM.	MAX.
T1	60	120	200
T2	180	280	400
T3	420	600	800
T4	900	1240	1600
T5	1860	2520	3200
T6	3780	5080	6400
T7	7620	10200	12800

SPECIFICATIONS

THE LOGIC PROBE DETECTS SHORT HIGH AND LOW VOV-SIGNALS AND STRETCH THEM FOR DELAY TIME OF SHIFT REGISTER.

HENCE BOTH ALARM- AND OK- STATES HAVE MINIMUM DURATION 0,4 SEC. AND MAXIMUM ∞

MAX. VOLTAGE : TERMINALS 58 - 61 : -1V, +50V
MAX. CURRENT : TERMINAL 61 : -50 mA
MIN. LOAD RESIST : TERMINALS 61 - 59 : 700 Ω
MIN. LOAD RESIST : TERMINALS 61 - 60 : 300 Ω

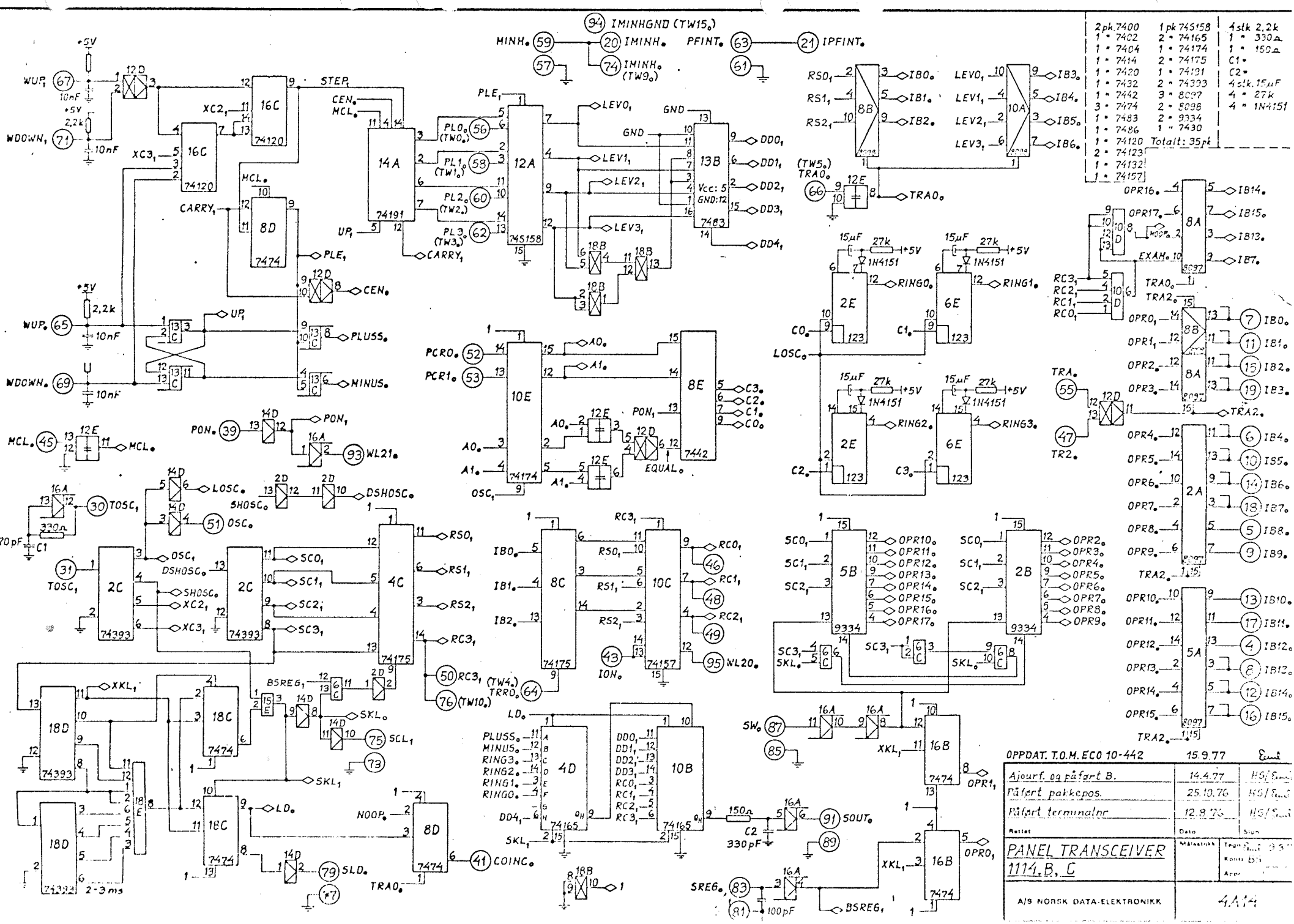
DESCRIPTION

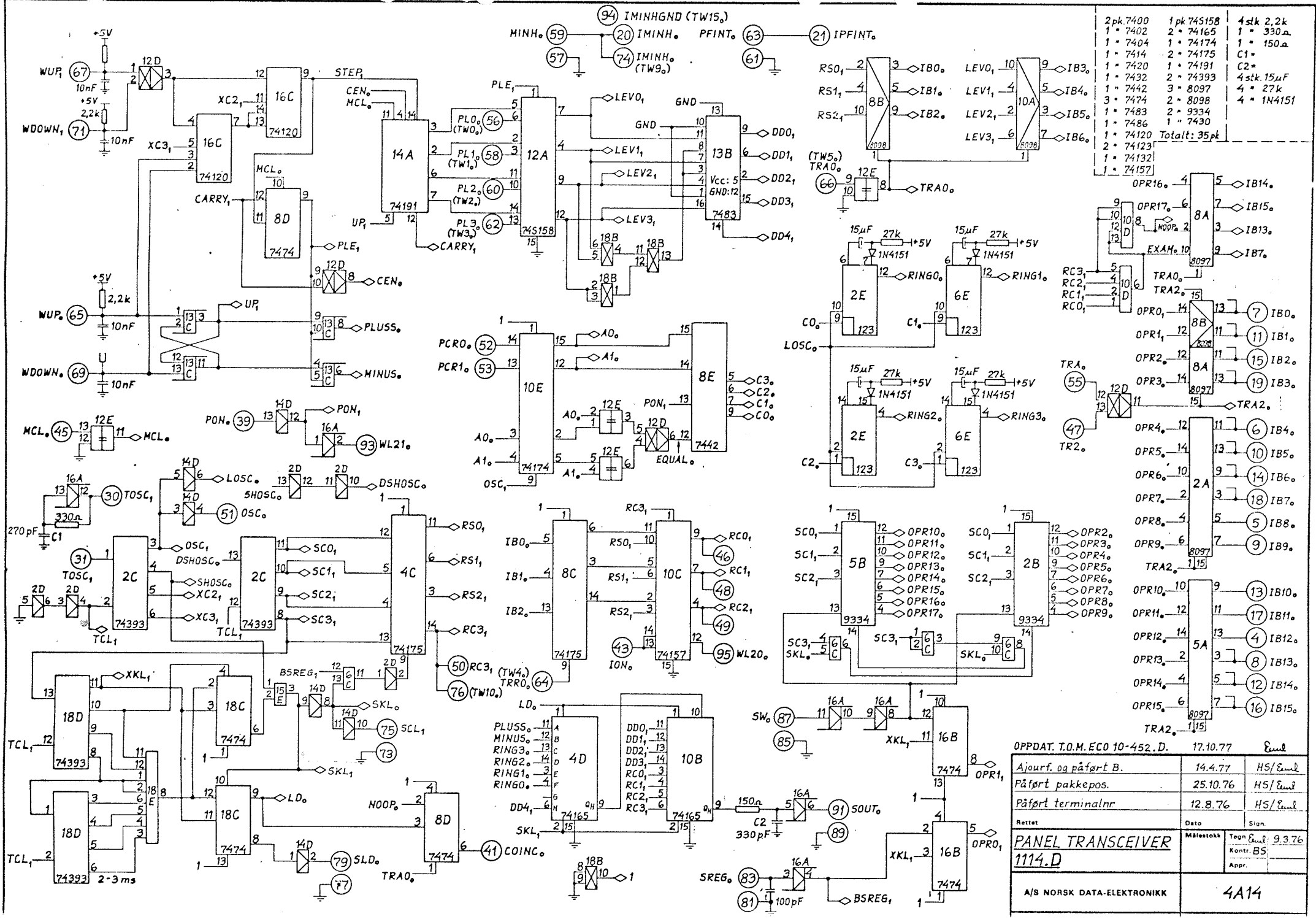
THE OSCILLATOR INCREMENTS THE COUNTER TRYING TO REACH THE SELECTED WATCH TIME.

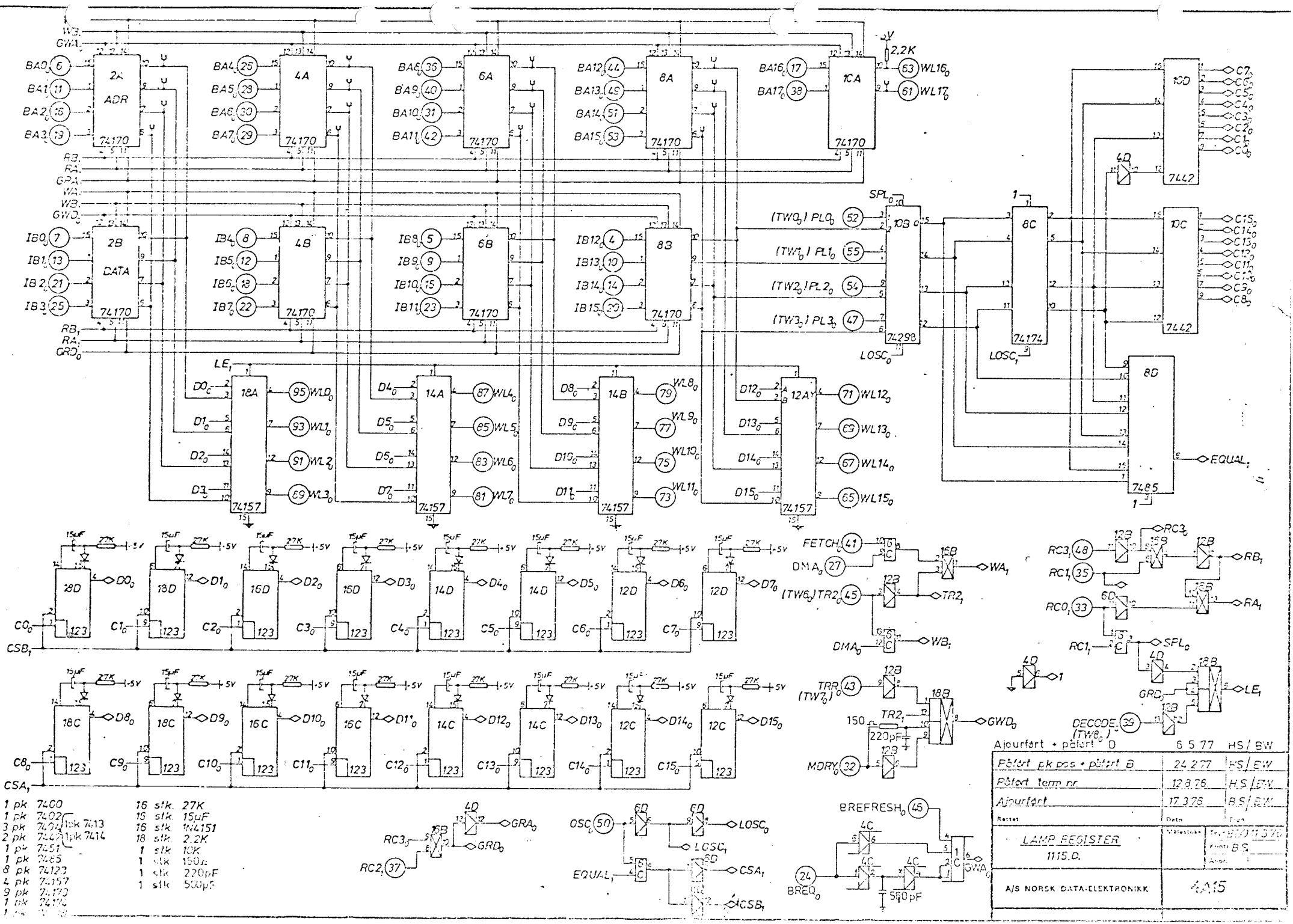
IOX 1034 OR MASTER CLEAR CLEARS THE COUNTER.

IF WATCH TIME IS REACHED AND LASTS MORE THAN 20 ms ALARM LASTS FOR 0,4 sec. OR MORE. IF IOX 1034 OCCURS ONCE, WHEN ALARM IS ON, THE ALARM SIGNAL DISAPPEARS FOR MINIMUM 0,4 sec. MAXIMUM DEPENDS ON SELECTED WATCHTIME.

Rettet	Dato	Sign.
WATCHDOG 1113	Målestokk	Tegn Gunl 11.2.76
		Kontr. PJJ
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A13	


2pk.7400
1 * 7402
1 * 7404
1 * 7414
1 * 7420
1 * 7432
1 * 7442
3 * 7474
1 * 7483
1 * 7486
1 * 74120
2 * 74123
1 * 74132
1 * 74157
1 pk 745158
2 * 74165
1 * 74174
2 * 74175
1 * 74191
2 * 74393
1 * 8097
2 * 9334
4 stk 2,2k
1 * 330Ω
1 * 150Ω
C1 =
C2 =
4 stk. 15µF
4 * 27k
4 * 1N4151
OPPDAT. T.O.M. ECO 10-442 15.9.77
Ajourf. og påført B. 14.4.77
Påført pakkepos. 25.10.76
Påført terminalnr 12.8.76
PANEL TRANSCEIVER 1114. B, C
A/S NORSK DATA-ELEKTRONIKK
4.4.14


2 pk. 7400
1 • 7402
1 • 7404
1 • 7414
1 • 7420
1 • 7432
1 • 7442
3 • 7474
1 • 7483
1 • 7486
1 • 74120
2 • 74123
1 • 74132
1 • 74157
1 pk 745158
2 • 74165
1 • 74174
2 • 74175
1 • 74191
2 • 74393
3 • 8097
2 • 8098
1 • 9334
1 • 7430
Totalt: 35 pk
4 stk 2,2k
1 • 330Ω
1 • 150Ω
C1 =
C2 =
4 stk. 15µF
4 • 27k
4 • 1N4151
OPPDAT. T.O.M. ECO 10-452.D. 17.10.77
Ajourf. og påført B. 14.4.77 HS/Eml
Påført pakkepos. 25.10.76 HS/Eml
Påført terminalnr 12.8.76 HS/Eml
Rettet Dato Sign.
PANEL TRANSCEIVER 1114.D
Målestokk
Tegn Eml 9.3.76
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A14



1 pk	7400	16 stk.	27K
1 pk	7402	15 stk.	15µF
3 pk	7404 (1pk 7413)	16 stk.	1N4151
2 pk	7408 (1pk 7414)	18 stk.	2.2K
1 pk	7451	1 stk.	10K
1 pk	7465	1 stk.	150Ω
8 pk	74123	1 stk.	220pF
4 pk	74157	1 stk.	560pF
9 pk	7470		
1 pk	7474		

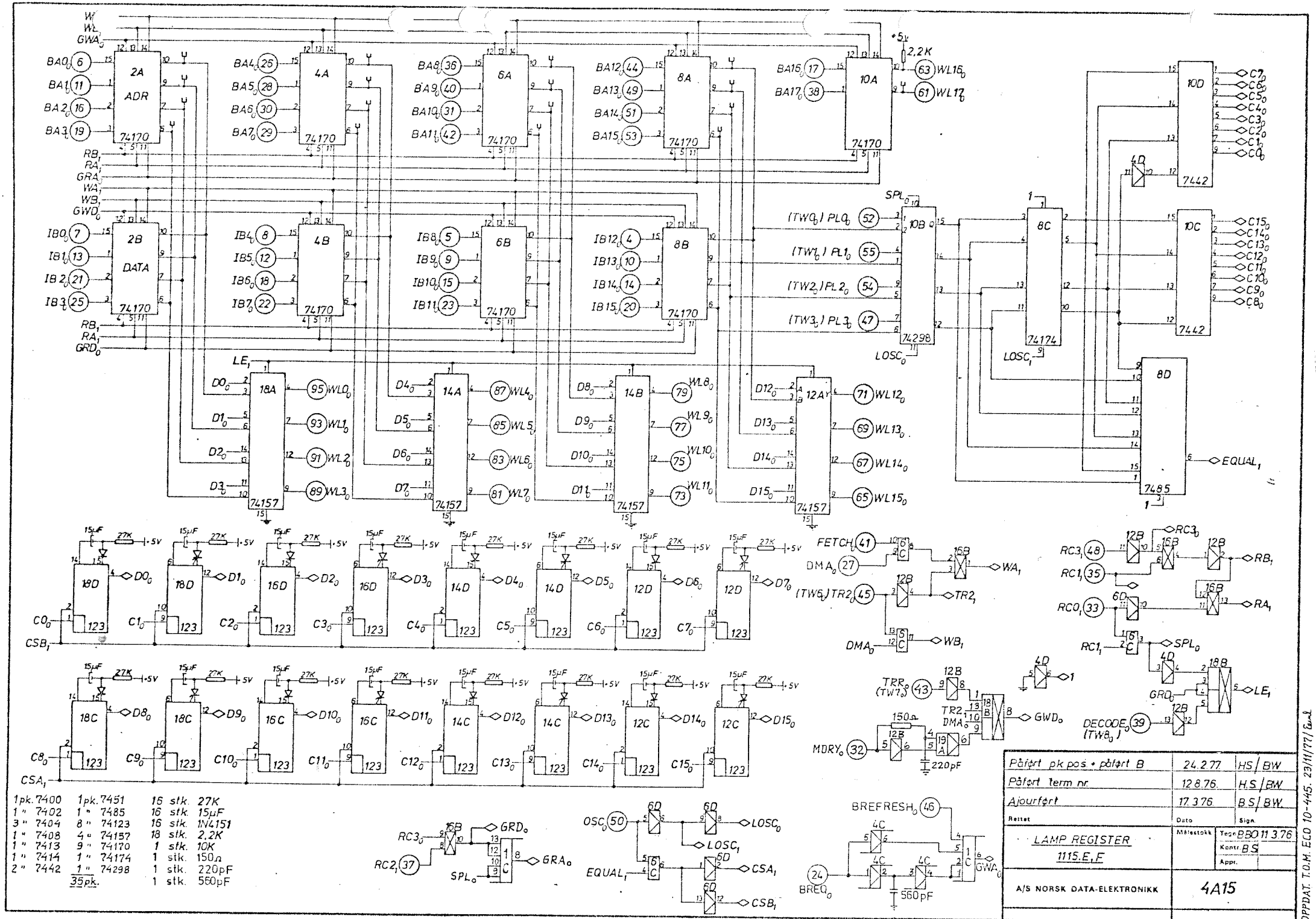
Ajourført + påført D	6.5.77	HS/BW
Påført pk pos + påført B	24.2.77	HS/BW
Påført term nr	12.8.76	HS/BW
Ajourført	17.3.76	BS/BW
Rettet	Dato	Sign

LAMP REGISTER
1115.D.

Tegnenr. 13.570
Kontr. BS

A/S NORSK DATA-ELEKTRONIKK

4A15



1pk. 7400	1pk. 7451	16 stk. 27K
1 " 7402	1 " 7485	16 stk. 15µF
3 " 7404	8 " 74123	16 stk. 1N4151
1 " 7408	4 " 74157	18 stk. 2,2K
1 " 7413	9 " 74170	1 stk. 10K
1 " 7414	1 " 74174	1 stk. 150Ω
2 " 7442	1 " 74298	1 stk. 220pF
	35pk.	1 stk. 560pF

Påført pk pos + påført B	24.2.77	HS / BW
Påført term nr.	12.8.76.	H.S / BW
Ajourført	17.3.76	B.S / B.W.
Rettet	Dato	Sign.

LAMP REGISTER
1115.E,F

Målestokk

Tegn BBO 11.3.76
Kontr B.S
Appr.

A/S NORSK DATA-ELEKTRONIKK

4A15

Påført F (ingen rettelser) 14.4.78 / Emd

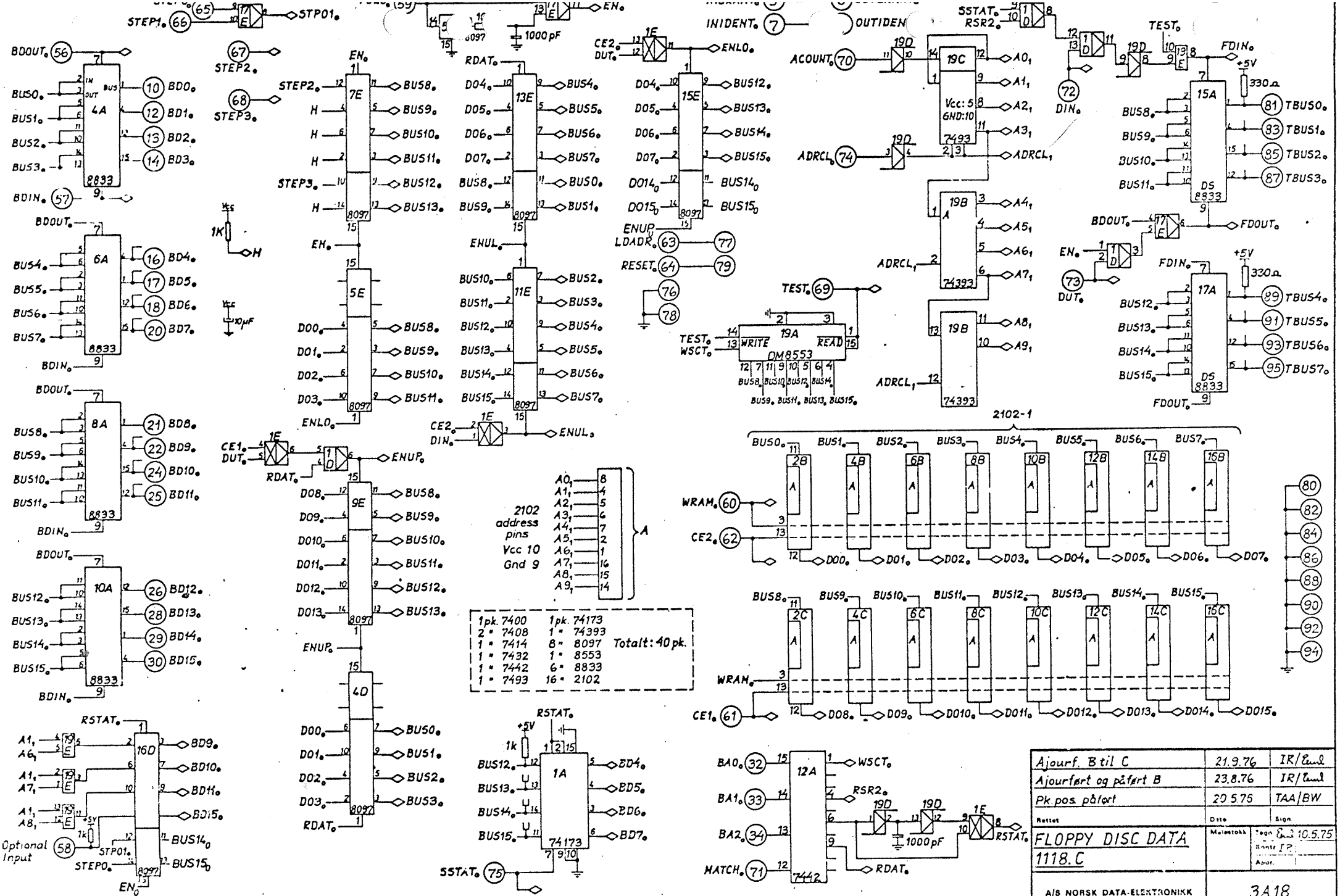
OPPDAT. T.O.M. ECO 10-445. 23/11/77 / Emd

1	1117	
2	1118	FLOPPY DISC DATA
3	1119	TRANSCEIVER ADDRESS
4	1120	TIME CONTROL
5	1121	PANEL CONTROL
6	1122	4 AS. CURRENT LOOP
7	1123-24	PERTEC CONTROL A-B
8	1125	TRANSCEIVER DATA
9	1126	CACHE.
10	1127	TRANSCEIVER CONTROL
11	1128	
12	1129	DMA REGISTERS
13	1130	CDC LINEPRINTER INTERFACE
14	1131	
15	1132	32 K RAM
16	1133-34	ECC POLYNOM-CONTROL
17	1135	SMD TIMING
18	1136	MPM REFRESH
19	1137-39	PS-DATA OUT-IN-CONTROL
20	1140	TALLY MATRIX PRINTER
21	1141	HDLC DATA
22	1142-46	BIG MPM ADDR-DATA-CTRL-X-Y-LOG-I/O-INTERFACE
23	1147	2 ASYNC V.24
24	1148	2 K PROM
25	1149	
26	1150	
27	1151	HDLC DMA CONTROL
28	1152	
29	1153	DMA ADDRESS EXTENDER
30	1154	SMD TRANSMITTER
31	1155	BUS CONTROL

DORAS Art.nr. 59207

1	1156	SMD UNIT CONTROL
2	1157	BIG MPM Y-LOG
3	1158	DR11-INTERFACE
4	1159	16 K RAM
5	1160	8 K STATIC RAM
6	1161	CHANNEL ADDR. DRIVER
7	1162	CHANNEL DATA DRIVER
8	1163-65	MEMORY ADDRESS-DATA-MULTIPLEXER REC.
9	1166	PROGRAMMABLE CLOCK
10	1167	4 K PROM
11	1168	DIFFERENTIAL DMA ADAPTOR
12	1169	DUAL TELEX CONTROL
13	1170	EXTERNAL DATA
14	1171	DMA CHANNEL DATA
15	1172	UNIVERSAL DMA INTERFACE
16	1173	TELEX TRANSMITTER
17	1181	HDLC DATA
18	1183	MEMORY ADDRESS RECEIVER
19	1184	MEMORY DATA
20	1185	BUS CONTROL
21		
22	1210	CLOCK AND PARITY
23	1211-1213	MPM DATA-ADDRESS LINE DRIVER
24	1301-06,1314	NORDCOM
25		
26	1907	OPERATOR'S PANEL
27		
28		
29		
30		
31		

DORAB Art.nr. 59207



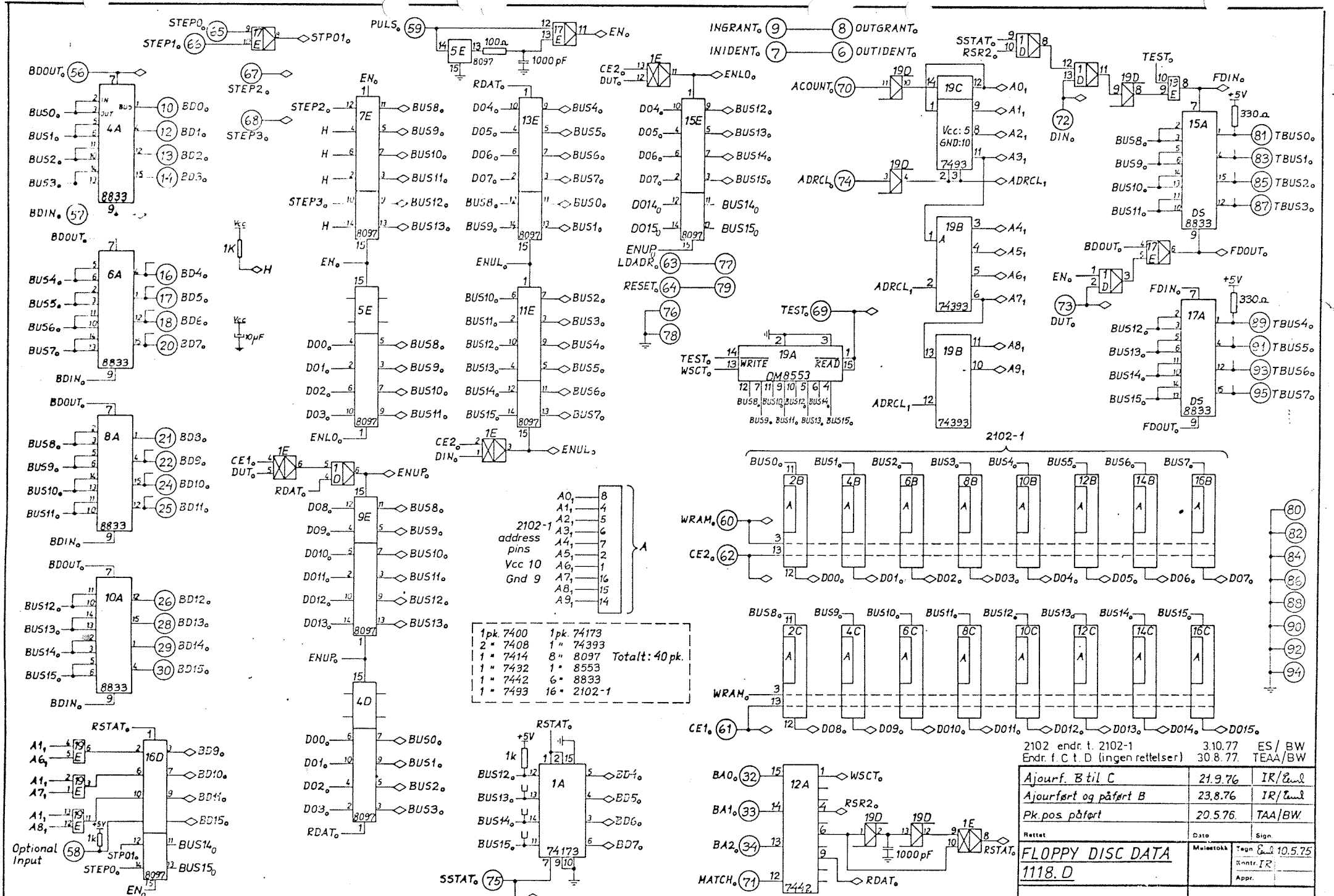
Ajourf. B til C	21.9.76	IR/Emil
Ajourført og påført B	23.8.76	IR/Emil
Pk.pos. påført	20.5.75	TAA/BW
Rettet	Dato	Sign

FLOPPY DISC DATA
1118.C

Malestokk | Tegn. Emil 10.5.75 | Kontr. TP | Godk.

A/S NORSK DATA-ELEKTRONIKK

3.A18



2102 endr. t. 2102-1 3.10.77 ES/BW
Endr. f. C t. D (ingen rettelser) 30.8.77. TEAA/BW

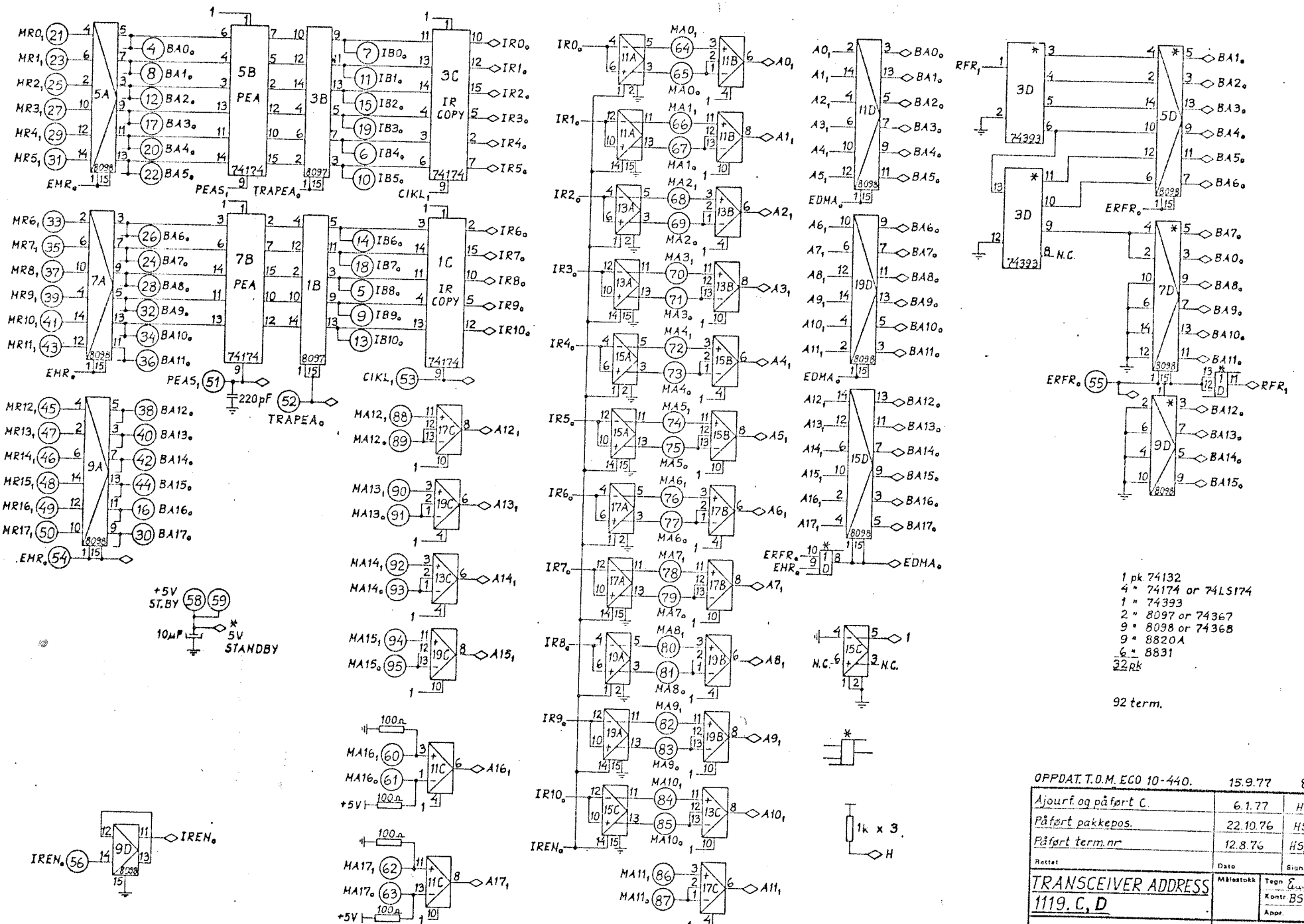
Ajourf. B til C	21.9.76	IR/Emil
Ajourført og påført B	23.8.76	IR/Emil
Pk.pos. påført	20.5.76.	TAA/BW
Rettet	Dato	Sign.

FLOPPY DISC DATA
1118. D

Målestokk | Tegn. Emil 10.5.75 | Kontr. IR | Appr.

A/S NORSK DATA-ELEKTRONIKK

4A18



Pin 7 and/or 9 on 8831 to H

1 pk 74132
4 " 74174 or 74LS174
1 " 74393
2 " 8097 or 74367
9 " 8098 or 74368
9 " 8820A
6 " 8831
32 pk

92 term.

OPPDAT. T.O.M. ECO 10-440. 15.9.77 Emil

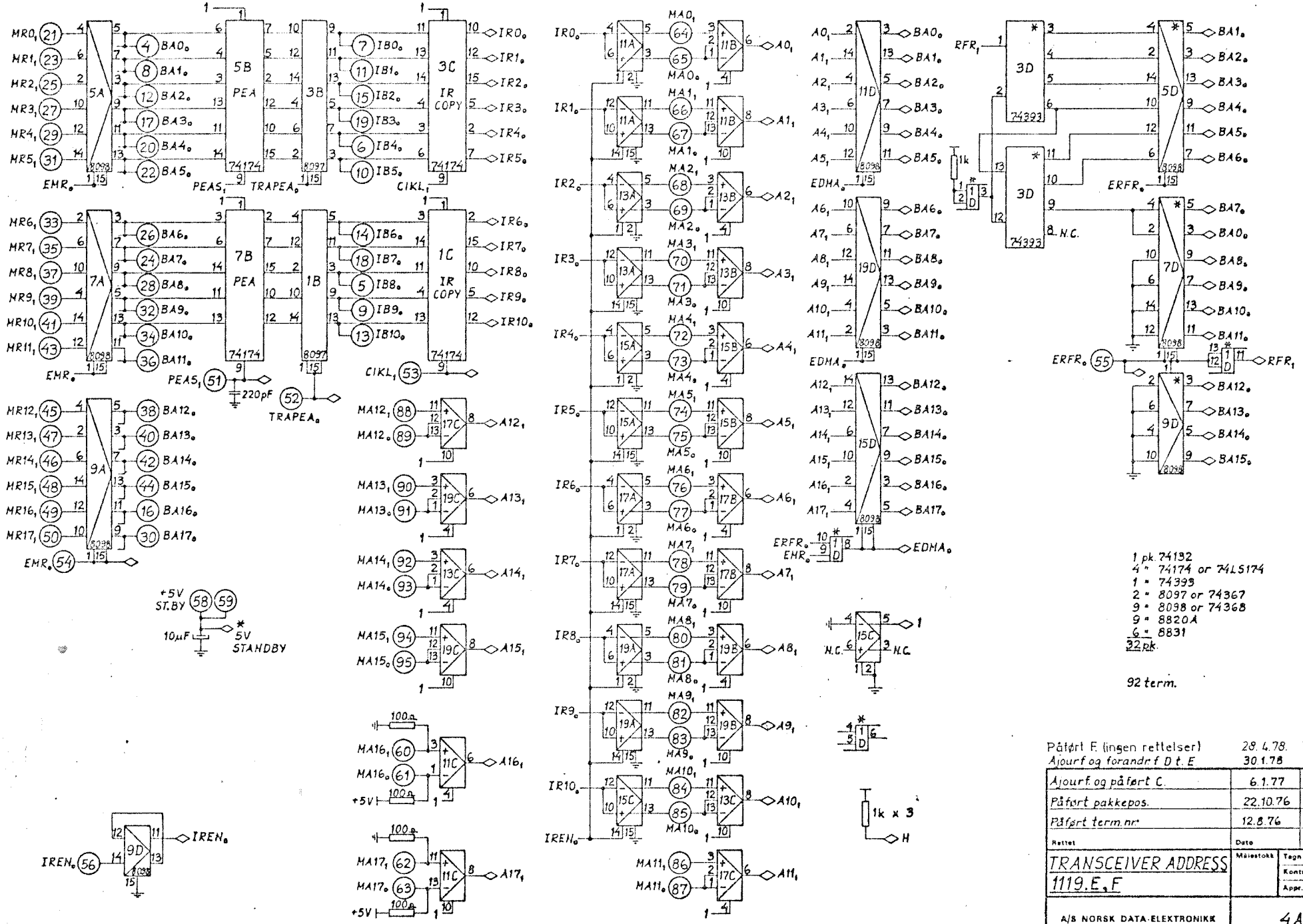
Ajourf. og påført C.	6.1.77	HS/Emil
Påført pakkepos.	22.10.76	HS/Emil
Påført term.nr	12.8.76	HS/Emil
Rettet	Dato	Sign.

TRANSCEIVER ADDRESS
1119. C, D

Målestokk
Tegn Emil 14.5.76
Kontr. BS
Appr.

A/S NORSK DATA-ELEKTRONIKK

4A19



Påført F (ingen rettelser)	28.4.78.	TEAA/BW
Ajourf og forandr f D t. E	30.1.78	HS/Emil
Ajourf. og påført C.	6.1.77	HS/Emil
Påført pakkepos.	22.10.76	HS/Emil
Påført term.nr.	12.8.76	HS/Emil
Rettet	Dato	Sign.

TRANSCEIVER ADDRESS
1119.E.F

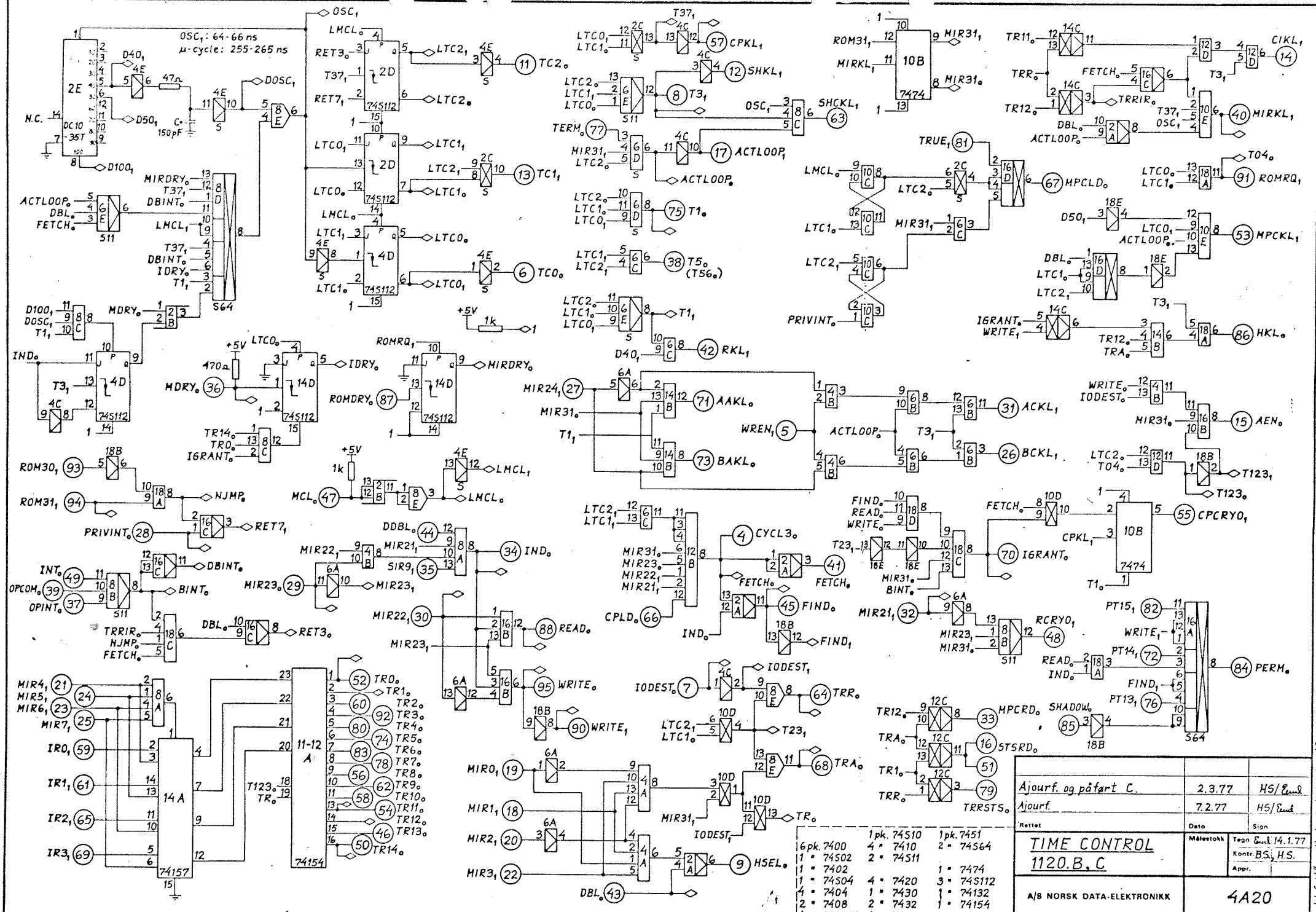
Målestokk

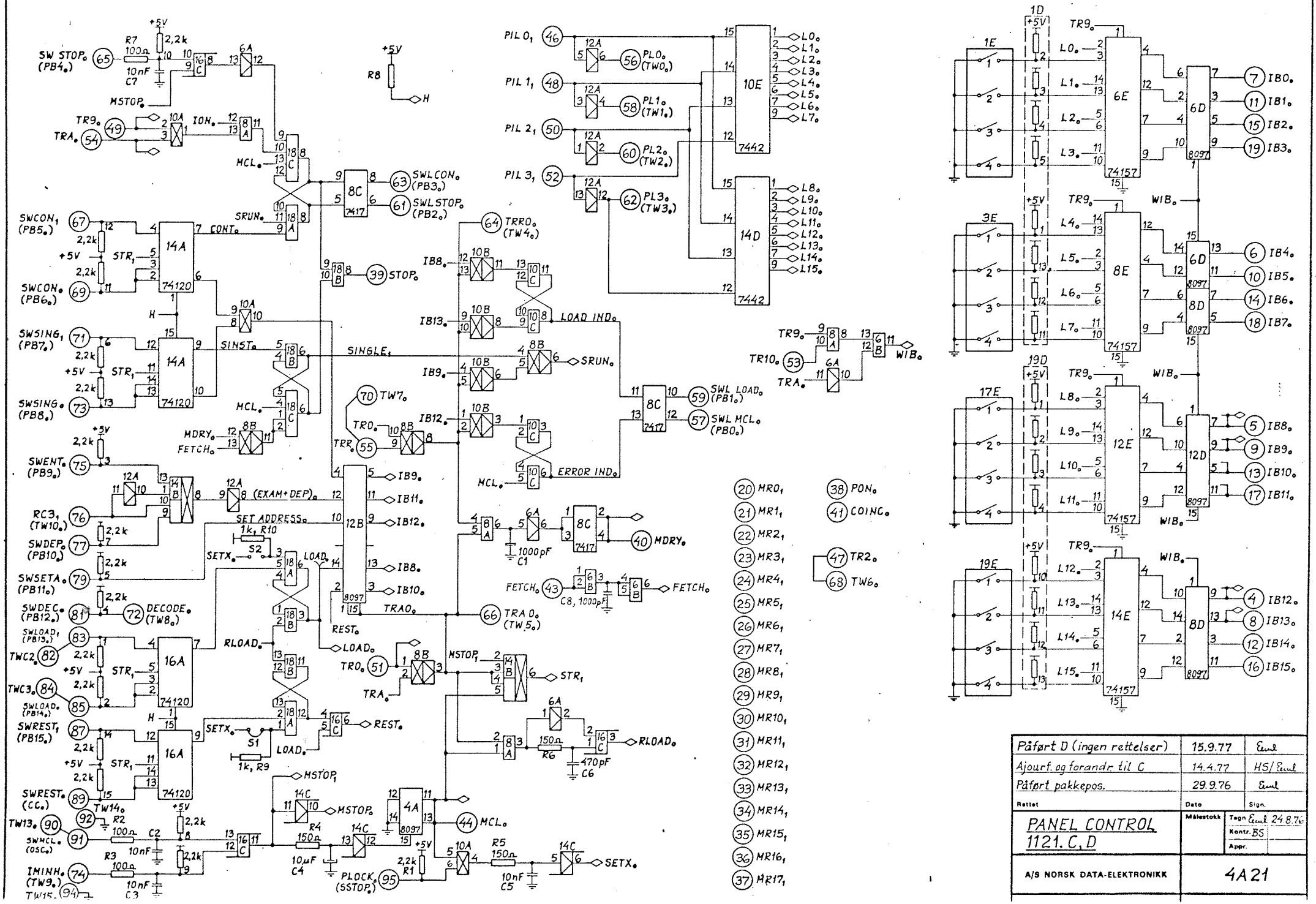
Tegn Emil 14.5.76
Kontr. BS
Appr.

A/S NORSK DATA-ELEKTRONIKK

4A19

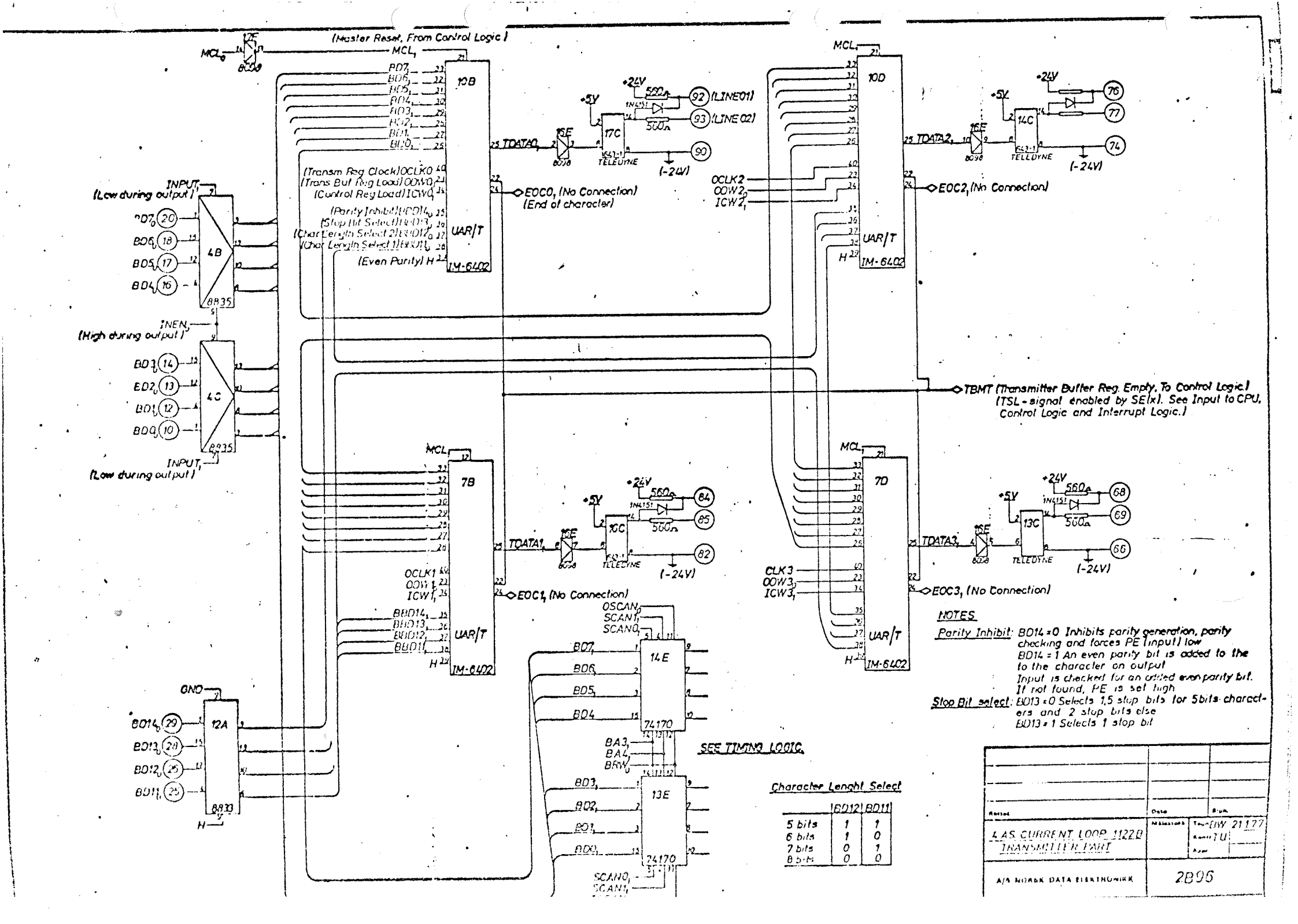
OPPDAT. T.O.M. ECO 10-440. 15/9/77/Emil


OSC1: 64-66 ns
µ-cycle: 255-265 ns
2E
DC10-35T
N.C.
D40
D50
D100
47Ω
150pF
DOSC
OSC
LMCL
RET3
RET7
T37
2D
74S112
LTC2
LTC1
LTC0
4D
TC2
TC1
TC0
MIRDRY
DBINT
IDRY
T1
S64
ACTLOOP
DBL
FETCH
511
MDRY
IND
T3
IGRANT
TR14
TRO
ROMRQ
ROMDRY
470Ω
1k
+5V
ROM30
ROM31
NJMP
PRIVINT
RET7
INT
OPCOM
OPINT
BINT
TRRIR
NJMP
FETCH
DBL
RET3
MIR22
MIR21
SIR9
MIR23
DDBL
IND
READ
WRITE
MIR4
MIR5
MIR6
MIR7
IR0
IR1
IR2
IR3
14A
74157
11-12
A
74154
TR0
TR1
TR2
TR3
TR4
TR5
TR6
TR7
TR8
TR9
TR10
TR11
TR12
TR13
TR14
LTCO
CPKL
SHKL
T3
SHCKL
TERM
ACTLOOP
T1
T5
(T56)
RKL
MIR24
AAKL
MIR31
T1
BAKL
WREN
ACKL
BCKL
AEN
T123
CYCL3
FETCH
FIND
CPLD
IND
IODEST
TRR
TRA
HSEL
ROM31
MIRKL
10B
7474
TR11
TR12
TRR
TRRIR
CIKL
MIRKL
TRUE
LMCL
LTC2
LTC1
MIR31
PRIVINT
MPCLD
D50
ACTLOOP
MPCKL
DBL
IGRANT
WRITE
TR12
TRA
HKL
ROMRQ
WRITE
IODEST
FIND
READ
IND
FETCH
T23
MIR31
BINT
MIR21
MIR23
CPKL
IGRANT
CPCRYO
T1
RCRYO
PT15
PT14
PT13
WRITE
READ
IND
FIND
PERM
SHADOW
MPCRDO
STSRDO
TRRSTSO
Ajourf. og påført C.
2.3.77
HS/Emil
Ajourf.
7.2.77
HS/Emil
Rettet
Dato
Sign.
TIME CONTROL
1120.B, C
Målestokk
Tegn Emil 14.1.77
Kontr. B.S., H.S.
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A20
6 pk. 7400
1 " 74S02
1 " 7402
1 " 74S04
4 " 7404
2 " 7408
1 " DC10-35
1 pk. 74S10
4 " 7410
2 " 74S11
2 " 7420
1 " 7430
2 " 7432
1 " 7437
1 pk. 7451
2 " 74S64
1 " 7474
3 " 74S112
1 " 74132
1 " 74154
1 " 74157
rykkfeil" rettet. 16/3/77/HS/Emil



Påført D (ingen rettelser)	15.9.77	Eml
Ajourf. og forandr. til C	14.4.77	HS/Eml
Påført pakkepos.	29.9.76	Eml
Rettet	Dato	Sign.

PANEL CONTROL 1121. C, D	Målestokk	Tegn Eml 24.8.76 Kontr. BS Appr.
A/S NORSK DATA-ELEKTRONIKK	4A21	



NOTES

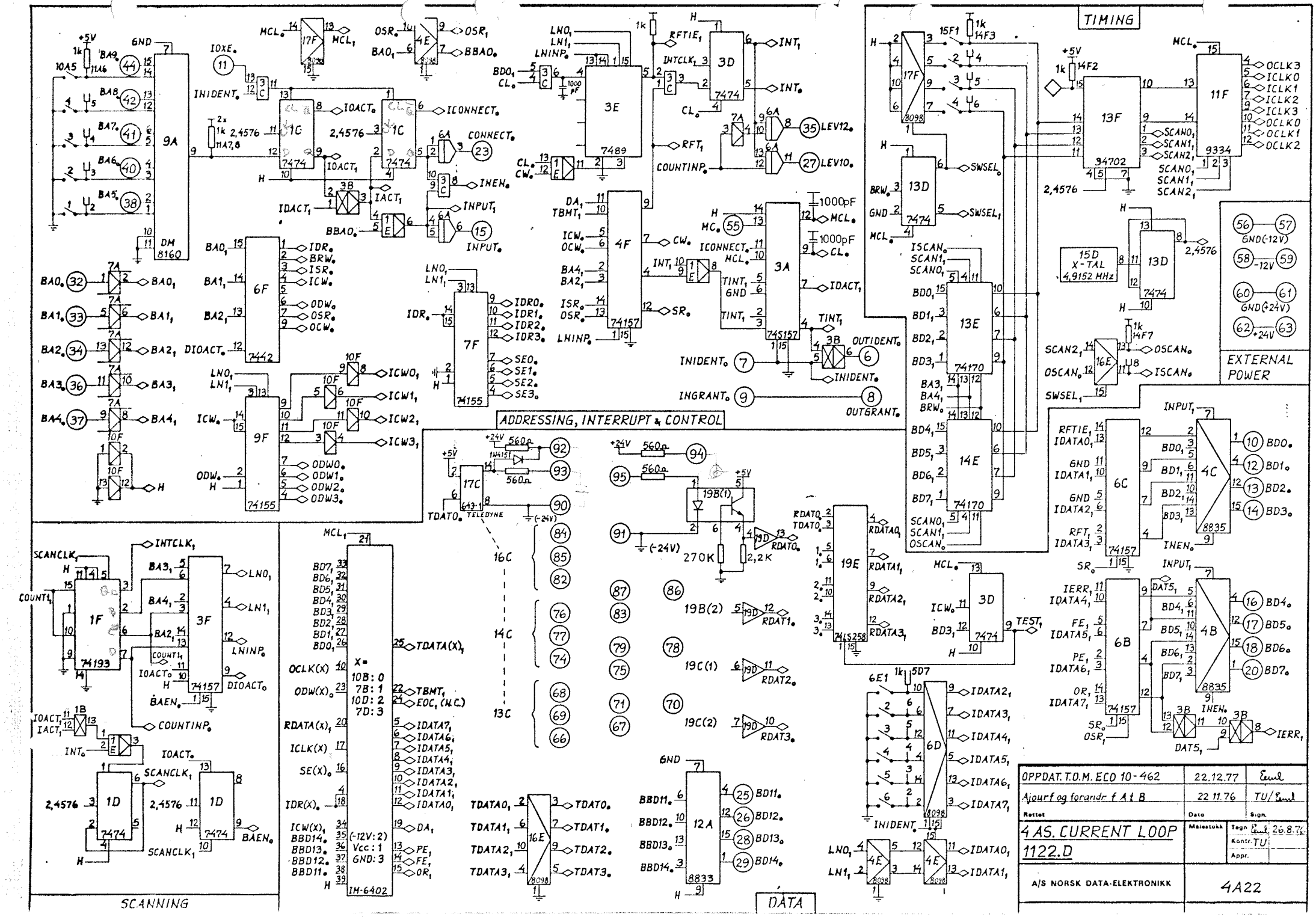
Parity Inhibit: BD14 = 0 Inhibits parity generation, parity checking and forces PE (input) low
BD14 = 1 An even parity bit is added to the to the character on output
Input is checked for an added even parity bit. If not found, PE is set high

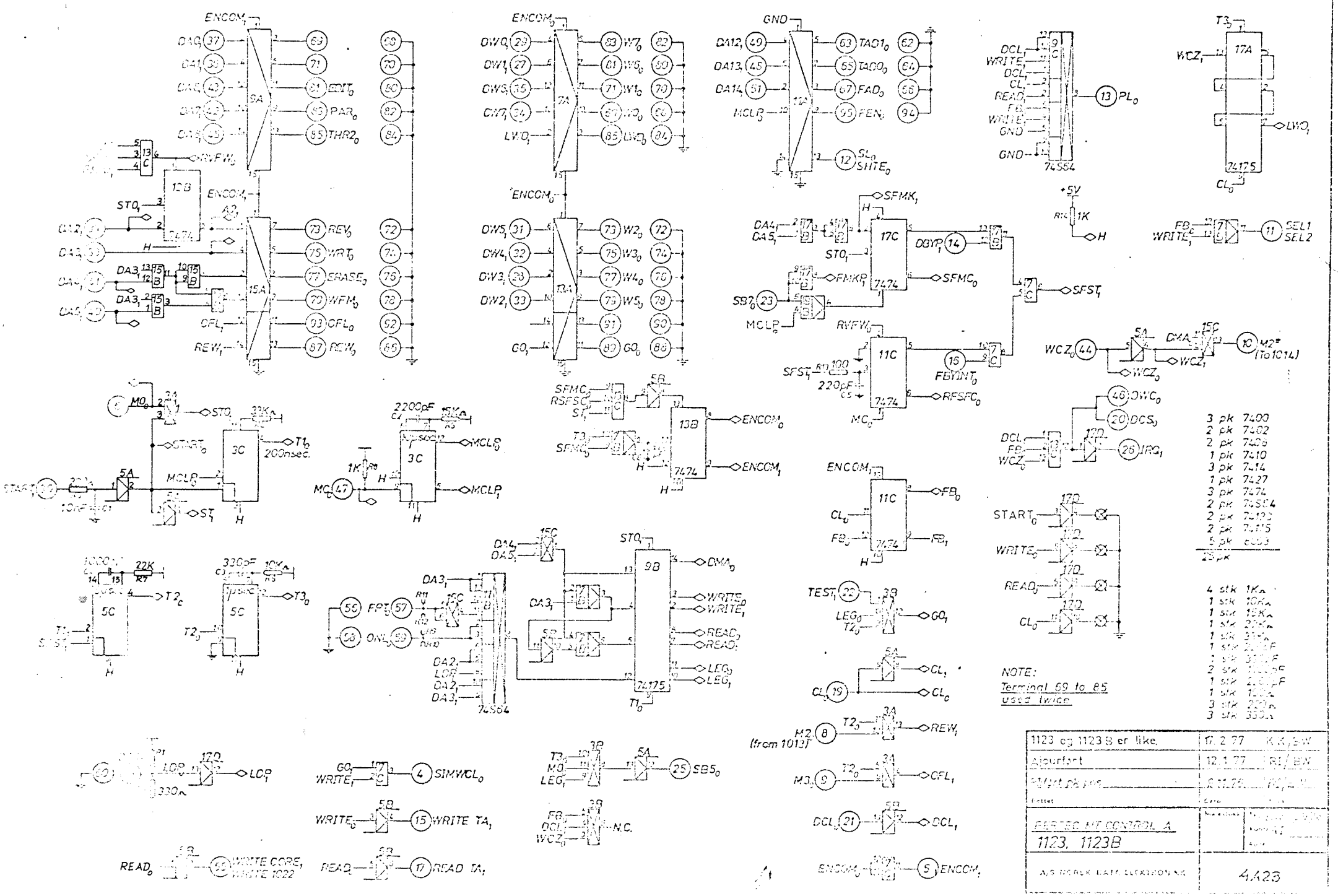
Stop Bit select: BD13 = 0 Selects 1,5 stop bits for 5bits characters and 2 stop bits else
BD13 = 1 Selects 1 stop bit

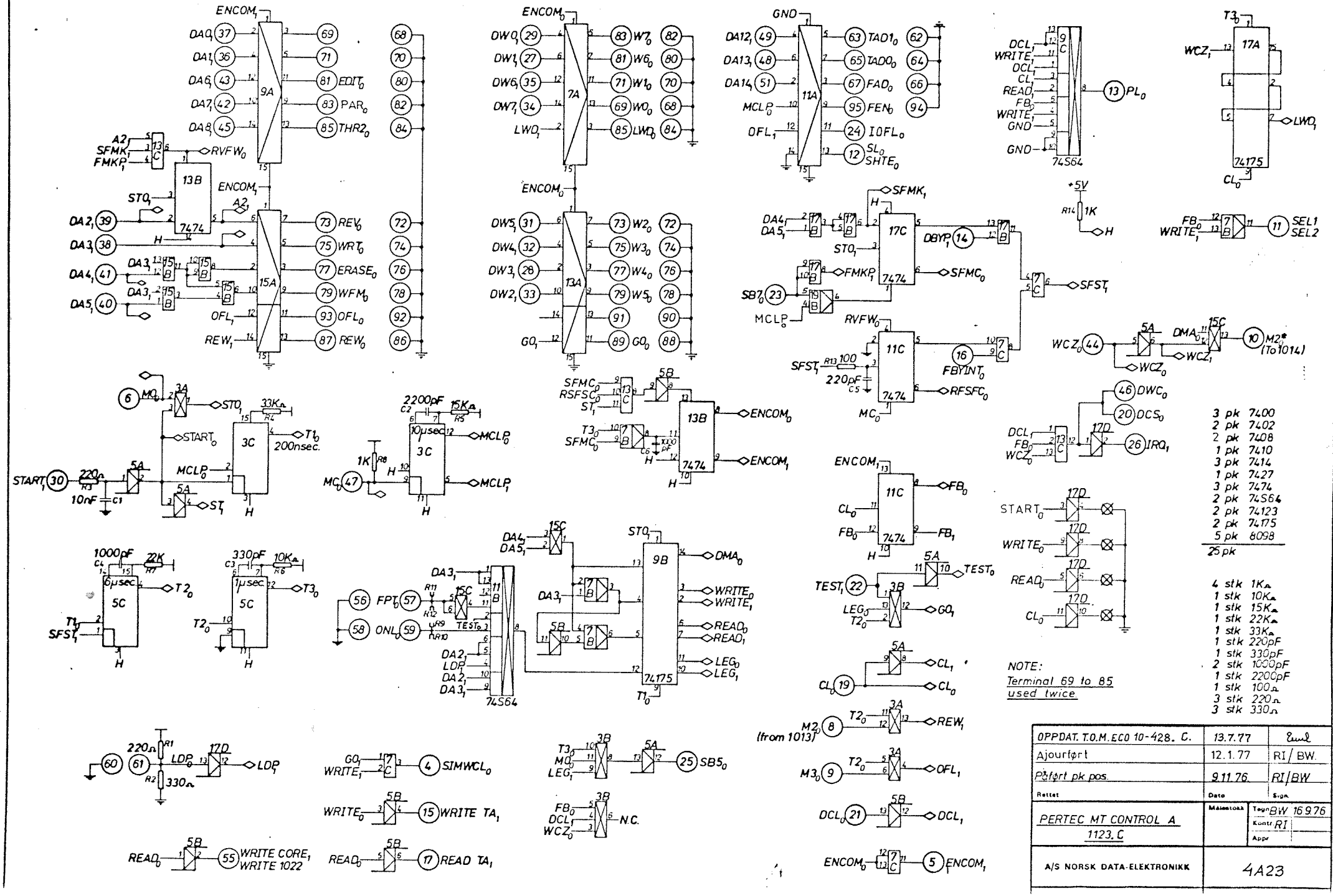
Character Lenght Select

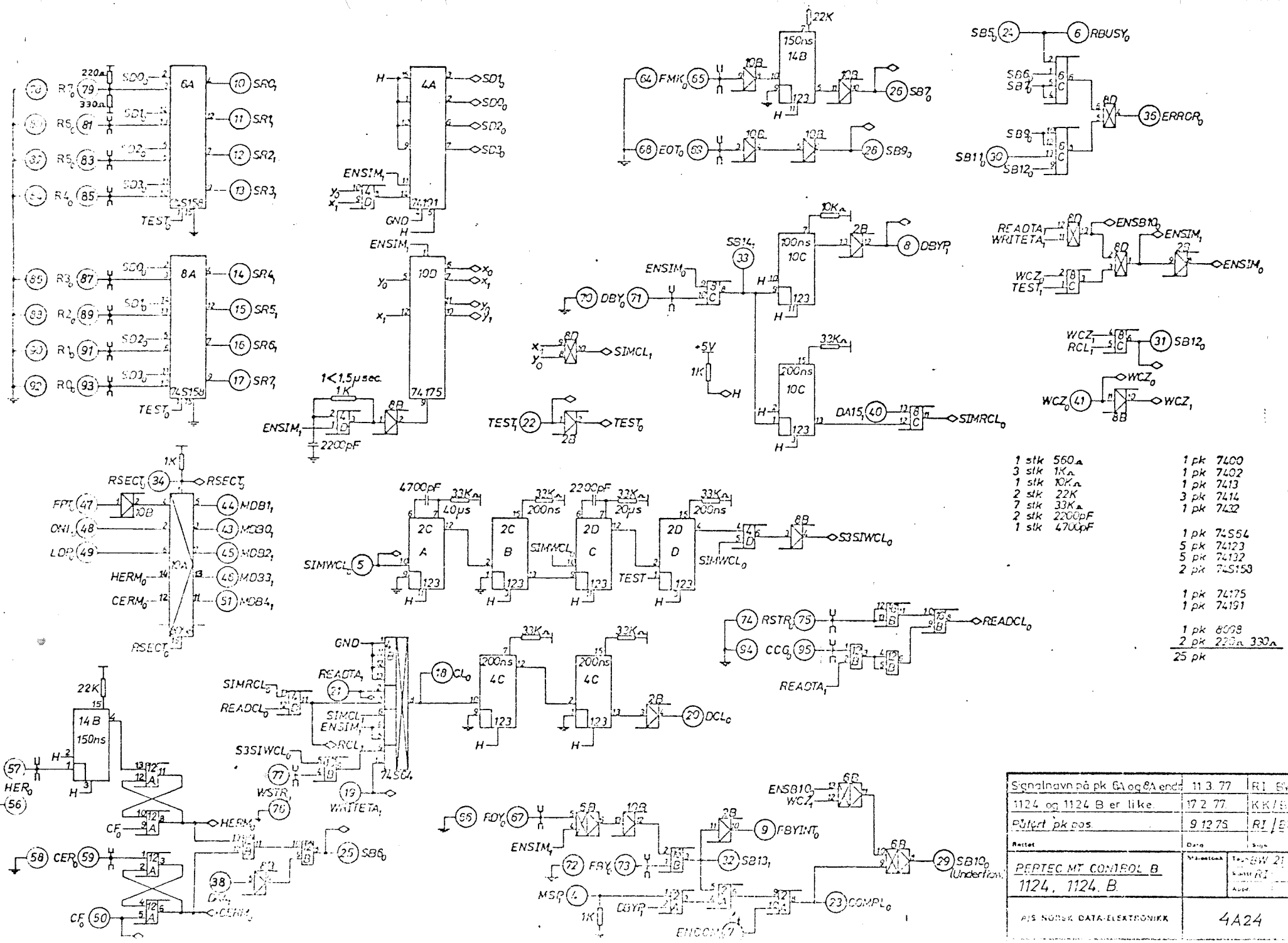
	BD12	BD11
5 bits	1	1
6 bits	1	0
7 bits	0	1
8 bits	0	0

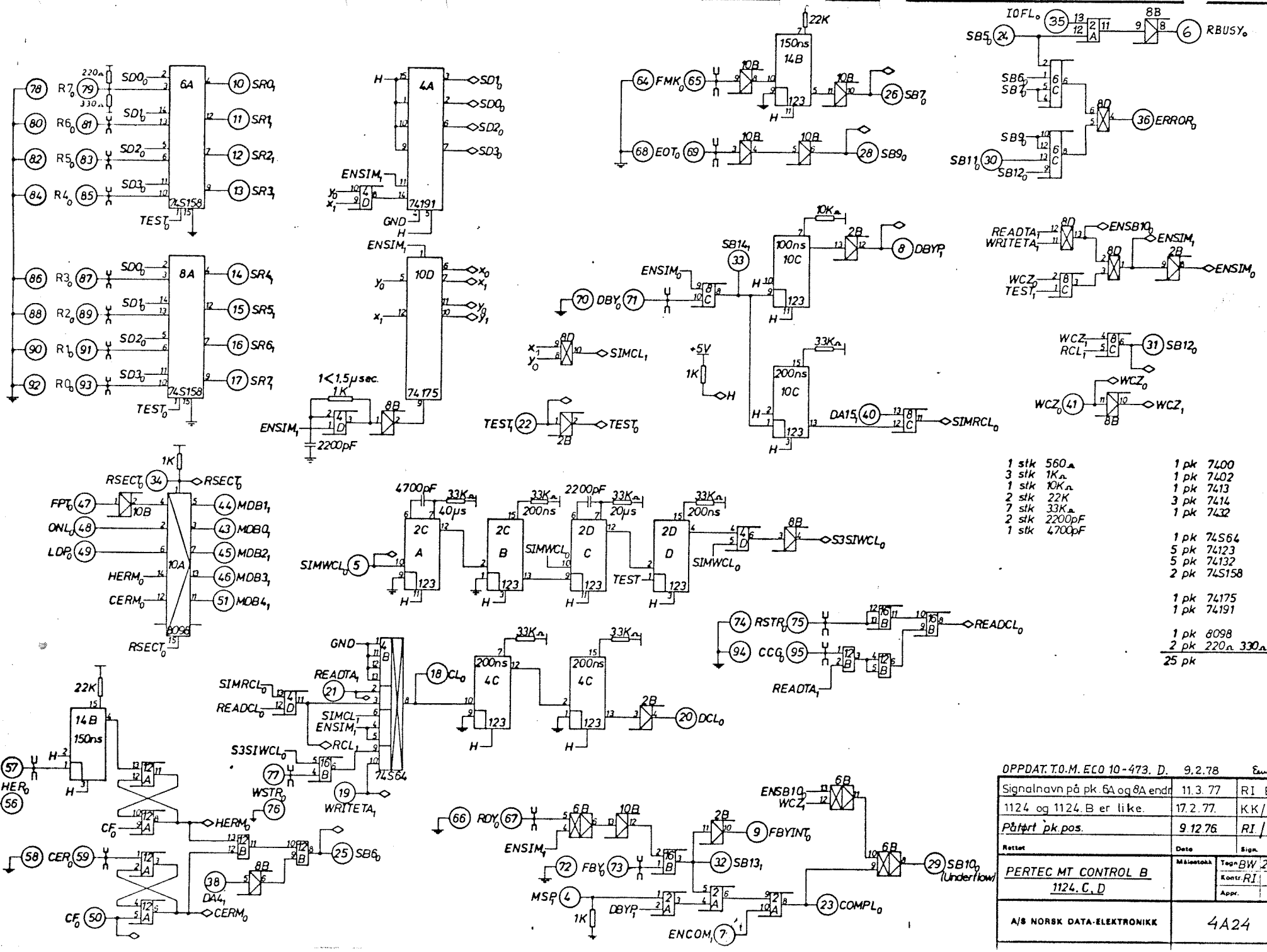
Revised	Date	Sign
4 AS CURRENT LOOP 1122B TRANSMITTER PART	Mikrotronik	Tegn DW 21177 Kontr TU Appr
A/S NORSK DATA ELEKTRONIKK	2896	


TIMING
ADDRESSING, INTERRUPT & CONTROL
EXTERNAL POWER
SCANNING
DATA
9A
DM 8160
1C
3E
3D
4F
3A
7F
9F
6F
1F
3F
1D
17F
13F
11F
13D
14E
6C
4C
6B
4B
19E
6D
12A
16E
17C
16C
14C
13C
IM-6402
15D X-TAL 4,9152 MHz
A/S NORSK DATA-ELEKTRONIKK
4A22
4 AS. CURRENT LOOP
1122.D
OPPDAT. T.O.M. ECO 10-462
22.12.77
Ajourf og forandr f A t B
22.11.76
Rettet
Dato
Sign.
Målestokk
Tegn
Kontr.
Appr.


ENCOM1
DA0 37
DA1 36
DA2 42
DA7 43
DA8 45
9A
69
71
EDIT0
PAR0
THR20
RVFW0
10B
STO1
DA2 34
DA3 33
DA4 31
DA5 32
DA3 13 12 B
DA3 2 3 B
15A
CFL1
REW1
REV0
WRT0
ERASE0
WFM0
CFL0
REW0
72
74
76
78
52
65
ENCOM0
DW0 29
DW1 27
DW3 25
DW7 24
LW0 1
7A
83
81
71
87
85
W7 0
W5 0
W1 0
W0 0
LW3 0
82
80
70
86
84
DW5 31
DW4 32
DW3 28
DW2 33
12A
73
75
77
79
91
90
W2 0
W3 0
W4 0
W5 0
GO1
GO0
72
74
76
78
50
88
GND
DA12 49
DA13 48
DA14 51
MCLP 10
12A
63
65
67
95
12
TAD1 0
TAD0 0
FAD0
FEN0
SL0
SHIFE0
62
64
66
94
DCL
WRITE
DCL
CL
READ
FB
WRITE
GND
GND
74S64
13
PL0
T3
WCZ1
17A
LW0 1
74175
CL0
+5V
R14 1K
H
FB
WRITE1
11
SEL1
SEL2
SFMK1
H
DA4
DA5
17C
DBYP1 14
STO1
FMKP1
7474
SFMC0
SB2 23
MCLP0
SFST1
RVFW0
11C
SFST1 R12 100
220pF C5
7474
15
FBTINT0
RFSFC0
MC0
WCZ0 44
5A
DMA1
15C
M2* (To 1014)
WCZ0
WCZ1
46
DWC0
20
DCS0
DCL
FB
WCZ0
17D
25
IRQ1
3 pk 7400
2 pk 7402
2 pk 7403
1 pk 7410
3 pk 7414
1 pk 7427
3 pk 7474
2 pk 74S04
2 pk 74123
2 pk 74175
5 pk 8T03
25 pk
4 stk 1K
1 stk 10K
1 stk 15K
1 stk 22K
1 stk 33K
1 stk 220pF
1 stk 330pF
2 stk 1000pF
1 stk 2200pF
1 stk 1µF
3 stk 220n
3 stk 330n
M0 24
STO
33K R1
START 22
200nsec.
T1 0
START0
5A
MCLP0
ST1
H
1000pF C1
10nF C2
3C
2200pF C4
15K R5
MCLP1
MC 47
1K R6
SFMC
RSFSC
ST1
T3
SFMC0
5B
13B
7474
ENCOM0
ENCOM1
ENCOM1
11C
CL0
FB0
7474
FB1
H
TEST1 22
LEG0
T20
3B
GO1
5A
CL1
CL 19
CL0
M2 (from 1013) 8
T20
3A
REW1
M3 9
T20
3A
CFL1
DCL 21
5B
DCL1
ENCOM0
5
ENCOM1
START0
WRITE0
READ0
CL0
17D
17D
17D
17D
NOTE:
Terminal 69 to 85
used twice
5C
T2C
330pF C3
10K R3
5C
T30
T2 0
H
DA4
DA5
15C
DA3 1
56 FPT 57
58 ONL 59
R11 1K
R12 1K
15C
DA3
5B
DA2
LOP
DA2
DA3
74S64
STO
9B
DMA0
WRITE0
WRITE1
READ0
READ1
LEG0
LEG1
74175
T10
22
LOP
17D
LOP1
330n
GO
WRITE1
5B
4
SIMWCL0
WRITE0
5B
15
WRITE TA1
READ0
5B
17
READ TA1
T3
M0
LEG1
3B
5A
25
SB50
FB
DCL
WCZ0
3B
N.C.
READ0
5B
66
WRITE CORE, WRITE 1022
1123 og 1123 B er like.
17. 2. 77
K.K./B.W.
Ajourført
12. 1. 77
R.I./B.W.
8. 11. 78
M.K./B.W.
RESTEC MT CONTROL A
1123, 1123B
A/S REGNECENTRALEN
4A23


ENCOM1
DA0 37
DA1 36
DA6 43
DA7 42
DA8 45
9A
69
71
81 EDIT0
83 PAR0
85 THR20
68
70
80
82
84
A2
SFMK
FMKP1
13C
RVFW0
13B
STO1
7474
ENCOM1
A2
DA2 39
DA3 38
DA4 41
DA5 40
H
DA3
15B
15A
73 REV0
75 WRT0
77 ERASE0
79 WFM0
OFL1
REW1
93 OFL0
87 REW0
72
74
76
78
92
86
ENCOM0
DW0 29
DW1 27
DW6 35
DW7 34
LWO1
7A
83 W70
81 W60
71 W10
69 WO0
85 LWO0
82
80
70
68
84
ENCOM0
DW5 31
DW4 32
DW3 28
DW2 33
13A
73 W20
75 W30
77 W40
79 W50
91
89 GO0
72
74
76
78
90
88
GND
DA12 49
DA13 48
DA14 51
MCLP0
OFL1
11A
63 TAD10
65 TAD00
67 FAD0
95 FEN0
24 IOFL0
12 SL0 SHTE0
62
64
66
94
DCL1
WRITE1
DCL1
CL1
READ1
FB0
WRITE1
GND
GND
9C
13 PL0
74S64
T30
WCZ1
17A
LWO1
74175
CL0
+5V
R14 1K
H
FB0
WRITE1
7B
11 SEL1
SEL2
SFMK1
H
DA4
DA5
17C
DBYP1 14
STO1
FMKP1
7474
SFMC0
SB70 23
MCLP0
SFST1
RVFW0
11C
SFST1
R13 100
220pF C5
16 FBYINT0
RFSFC0
MC0
WCZ0 44
5A
DMA0
15C
10 M20* (To1014)
WCZ0
WCZ1
46 DWC0
20 DCS0
DCL1
FB0
WCZ0
17D
26 IRQ1
6 MC0
3A
STO1
33KΩ R4
START0
3C
T10 200nsec.
MCLP0
5A
ST1
H
START1 30
220Ω R3
10nF C1
5A
2200pF C2
15KΩ R5
10μsec
3C
MCLP0
1K R8
MC0 47
H
MCLP1
SFMC0
RSFSC0
ST1
5B
13B
ENCOM0
T30
SFMC0
C6 100pF
7474
ENCOM1
H
ENCOM1
11C
FB0
CL0
FB0
7474
FB1
H
5A
TEST0
TEST1 22
3B
LEG0
T20
GO1
5A
CL1
CL0 19
CL0
M20 (from 1013) 8
T20
3A
REW1
M30 9
T20
OFL1
DCL0 21
5B
DCL1
ENCOM0
12 13 7C 11
5 ENCOM1
1000pF C4
22K R7
6μsec
5C
T20
T1
SFST1
H
330pF C3
10KΩ R6
1μsec
5C
T30
T20
H
56 FPT0
58 ONL0
57
59
R11 1KΩ
R9 TR10
15C
TEST0
DA31
DA21
LDP
DA21
DA31
74S64
DA4
DA5
15C
DA31
5B
STO1
9B
74175
T10
DMA0
WRITE0
WRITE1
READ0
READ1
LEG0
LEG1
60
61
220Ω R1
R2 330Ω
LDP0
17D
LDP1
GO1
WRITE1
4 SIMWCL0
WRITE0
5B
15 WRITE TA1
READ0
5B
17 READ TA1
READ0
5B
55 WRITE CORE, WRITE 1022
T30
MO1
LEG1
3B
5A
25 SB50
FB0
DCL1
WCZ0
3B
N.C.
START0
17D
WRITE0
17D
READ0
17D
CL0
17D
NOTE:
Terminal 69 to 85
used twice.
3 pk 7400
2 pk 7402
2 pk 7408
1 pk 7410
3 pk 7414
1 pk 7427
3 pk 7474
2 pk 74S64
2 pk 74123
2 pk 74175
5 pk 8098
25 pk
4 stk 1KΩ
1 stk 10KΩ
1 stk 15KΩ
1 stk 22KΩ
1 stk 33KΩ
1 stk 220pF
1 stk 330pF
2 stk 1000pF
1 stk 2200pF
1 stk 100Ω
3 stk 220Ω
3 stk 330Ω
OPPDAT. T.O.M. ECO 10-428. C.
13.7.77
Ajourført
12.1.77
RI / BW.
Påført pk pos.
9.11.76.
RI/BW
Rettet
Dato
Sign.
PERTEC MT CONTROL A
1123. C
Målestokk
Tegn BW 16976
Kontr RI
Appr
A/S NORSK DATA-ELEKTRONIKK
4A23


6A
4A
8A
10D
74175
74S158
14B
150ns
4C
200ns
10C
100ns
2C
2D
74S64
74123
74S32
74S51
74S04
SIMRCL
SIMWCL
SIMCL
ENSIM
READETA
WRITETA
RCL
WSTR
RSTR
CCG
FMK
EOT
RBUSY
ERROR
DBY
DBYP
FBY
FBYINT
SB13
SB14
SB10
SB12
DA15
COMPL
ENDCOM
MSP
HER
CER
CF
RSECT
FPT
ONI
LOP
HERM
CERM
MDB1
MDB0
MDB2
MDB3
MDB4
TEST
SIMRCL
READCL
S3SIWCL
WCZ
RCL
ENSB10
READETA
WRITETA
WCZ
TEST
1 stk 560Ω
3 stk 1KΩ
1 stk 10KΩ
2 stk 22K
7 stk 33KΩ
2 stk 2200pF
1 stk 4700pF
1 pk 7400
1 pk 7402
1 pk 7413
3 pk 7414
1 pk 7432
1 pk 74S64
5 pk 74123
5 pk 74132
2 pk 74S158
1 pk 74175
1 pk 74191
1 pk 8098
2 pk 220Ω 330Ω
25 pk
Signalnavn på pk 6A og 8A endr. 11 3 77 RI BW
1124 og 1124 B er like. 17 2 77 KK/BW
Rettet pk pos. 9 12 75 RI / BW
Rettet Dato Sign
PERTEC MT CONTROL B
1124. 1124. B
P/S NORSK DATA-ELEKTRONIKK
4A24



1 stk	560Ω
3 stk	1KΩ
1 stk	10KΩ
2 stk	22K
7 stk	33KΩ
2 stk	2200pF
1 stk	4700pF

1 pk	7400
1 pk	7402
1 pk	7413
3 pk	7414
1 pk	7432
1 pk	74S64
5 pk	74123
5 pk	74132
2 pk	74S158
1 pk	74175
1 pk	74191
1 pk	8098
2 pk	220Ω 330Ω
25 pk	

OPPDAT. T.O.M. ECO 10-473. D. 9.2.78 Emil

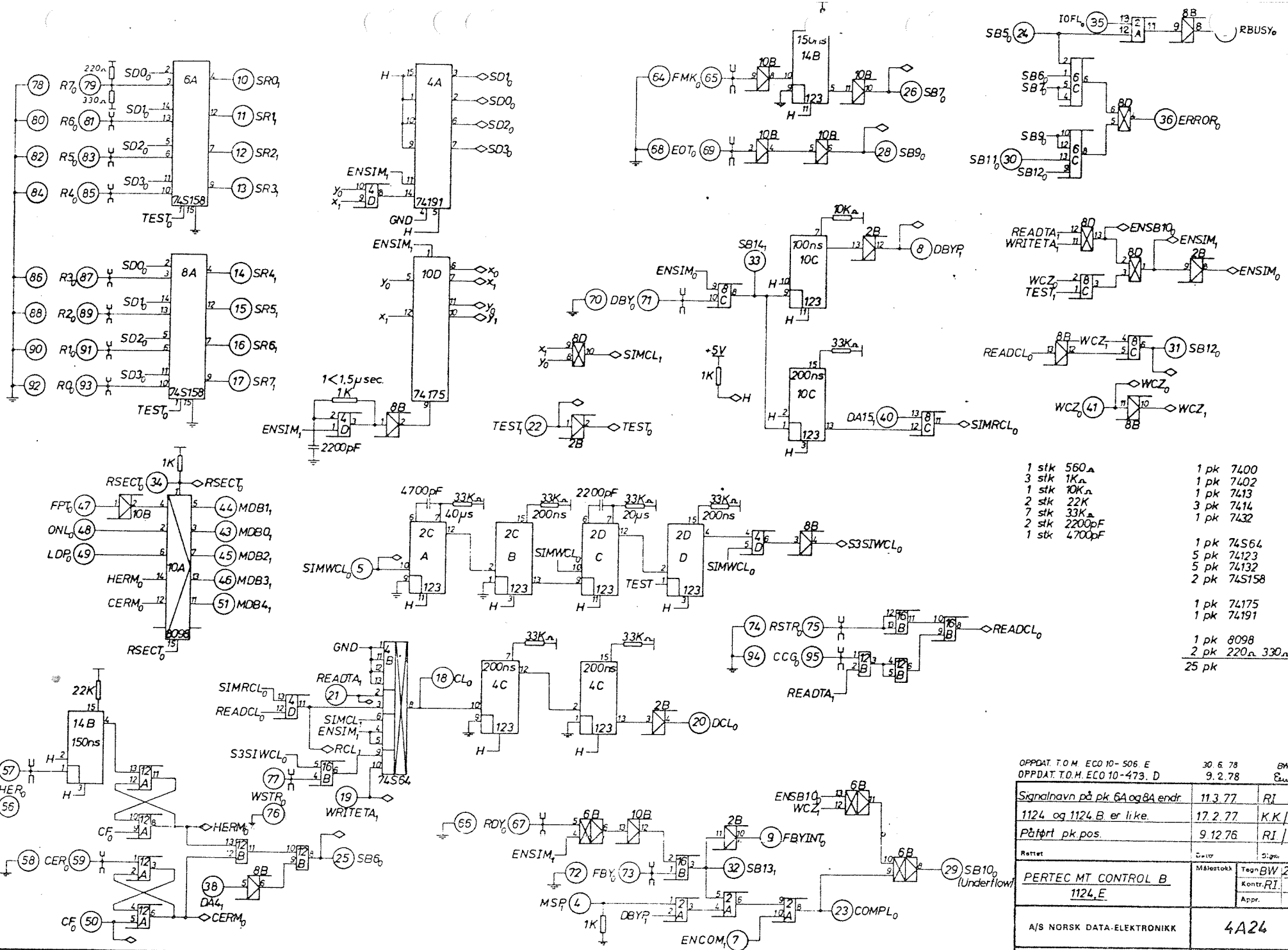
Signalnavn på pk. 6A og 8A endr	11.3.77	RI BW
1124 og 1124.B er like.	17.2.77.	KK/BW
Påført pk.pos.	9.12.76	RI / BW
Rettet	Dato	Sign.

PERTEC MT CONTROL B
1124. C, D

Målestokk | Tegn. BW 21976 | Kontr. RI | Appr.

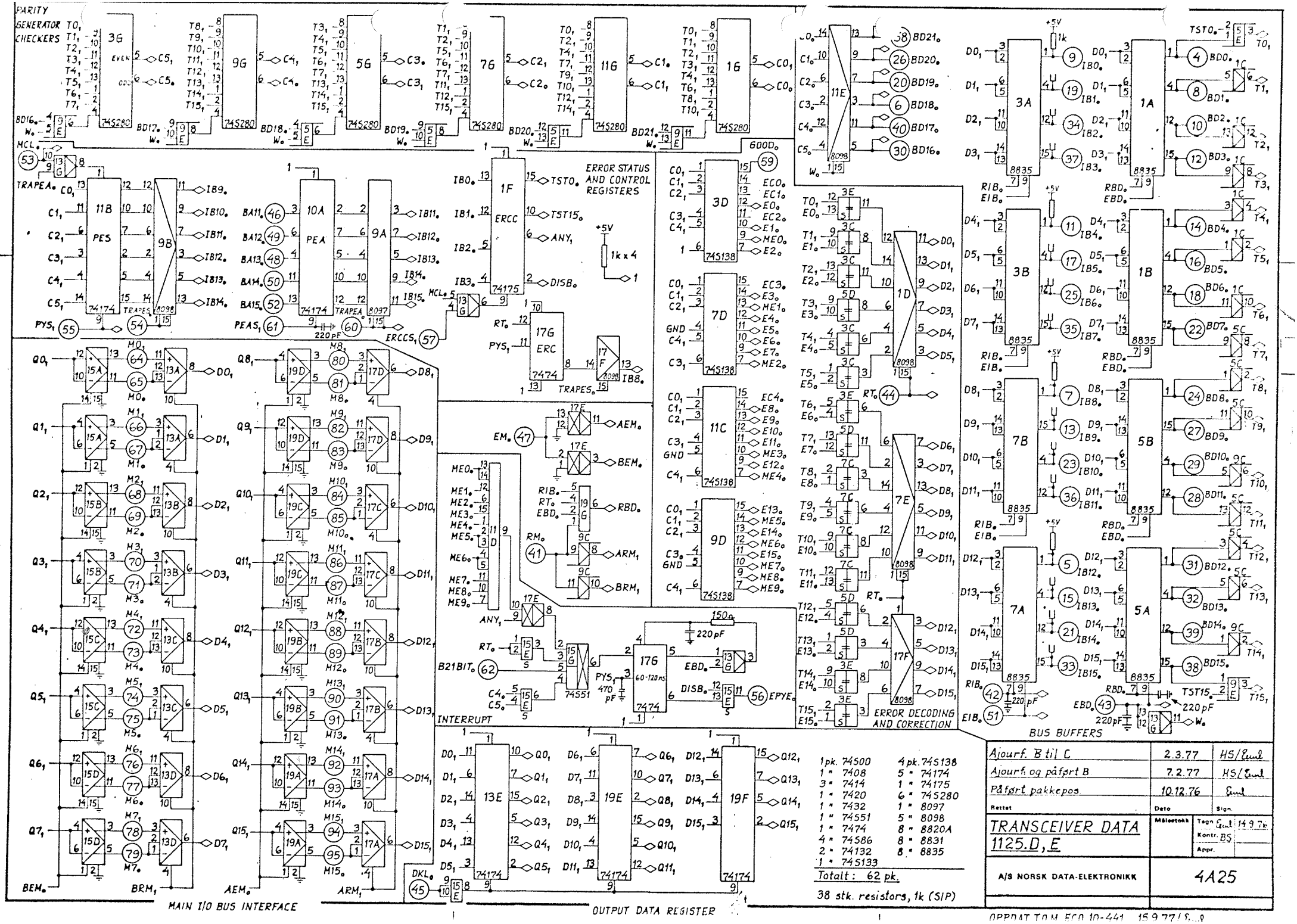
A/S NORSK DATA-ELEKTRONIKK

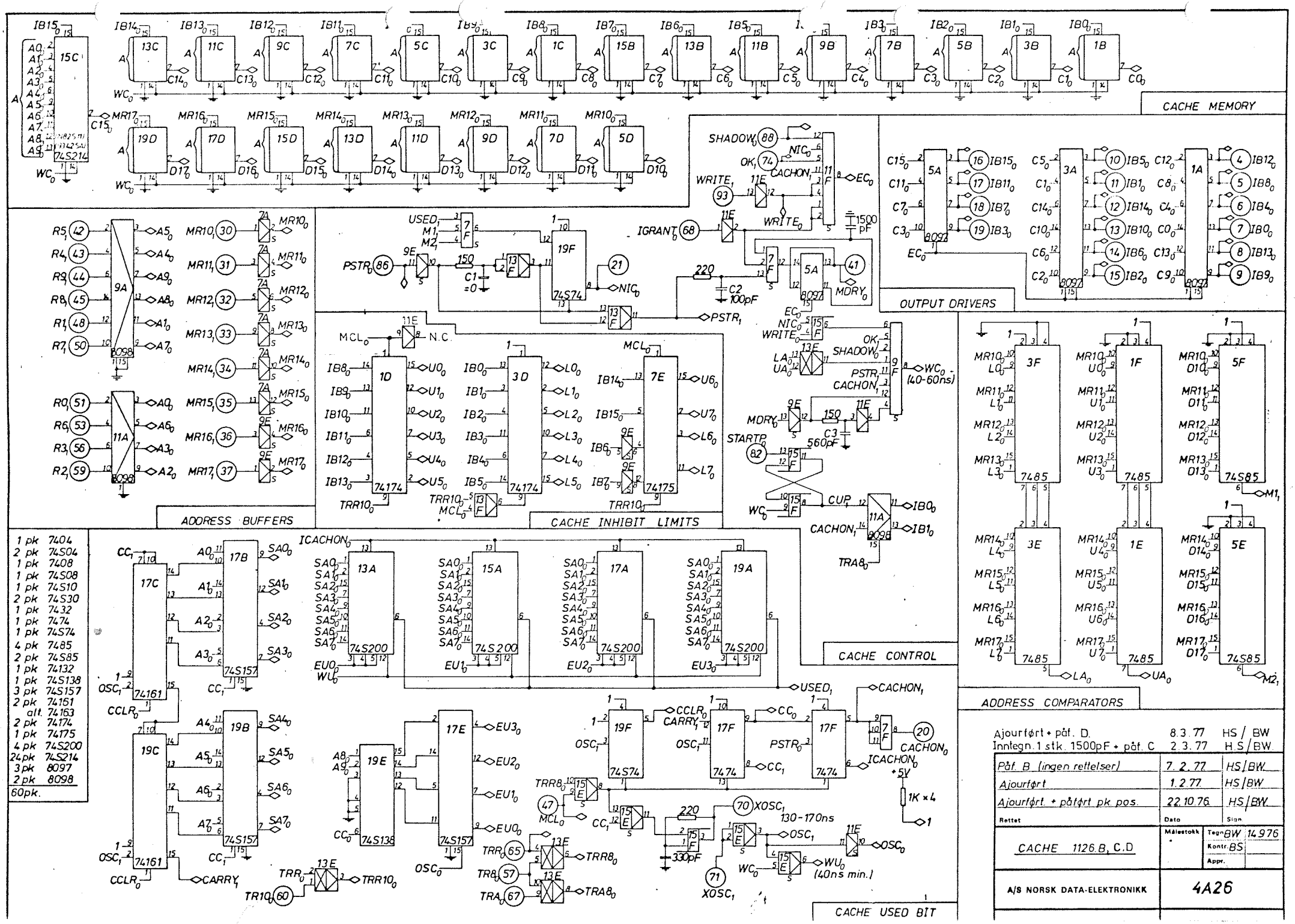
4A24

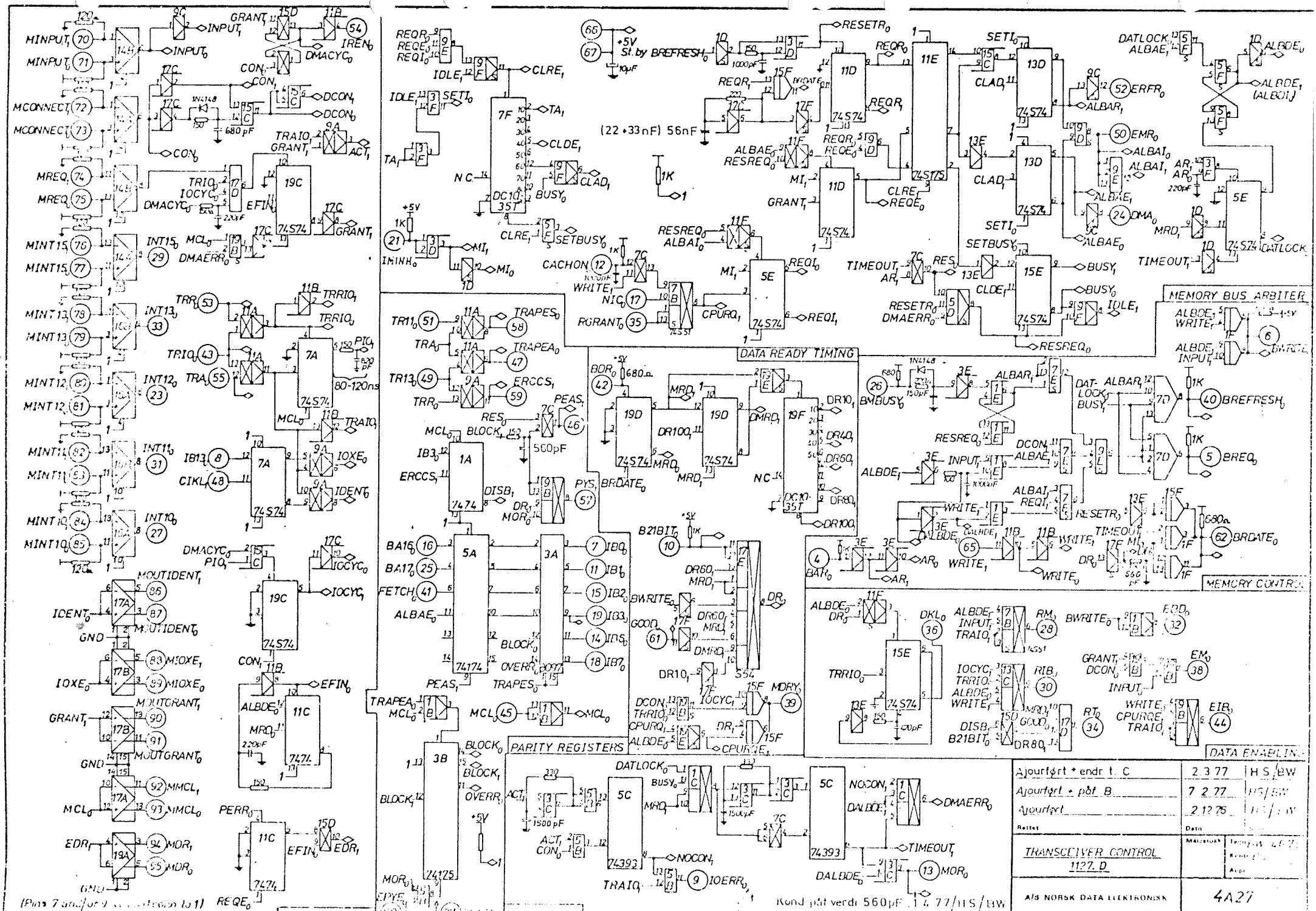


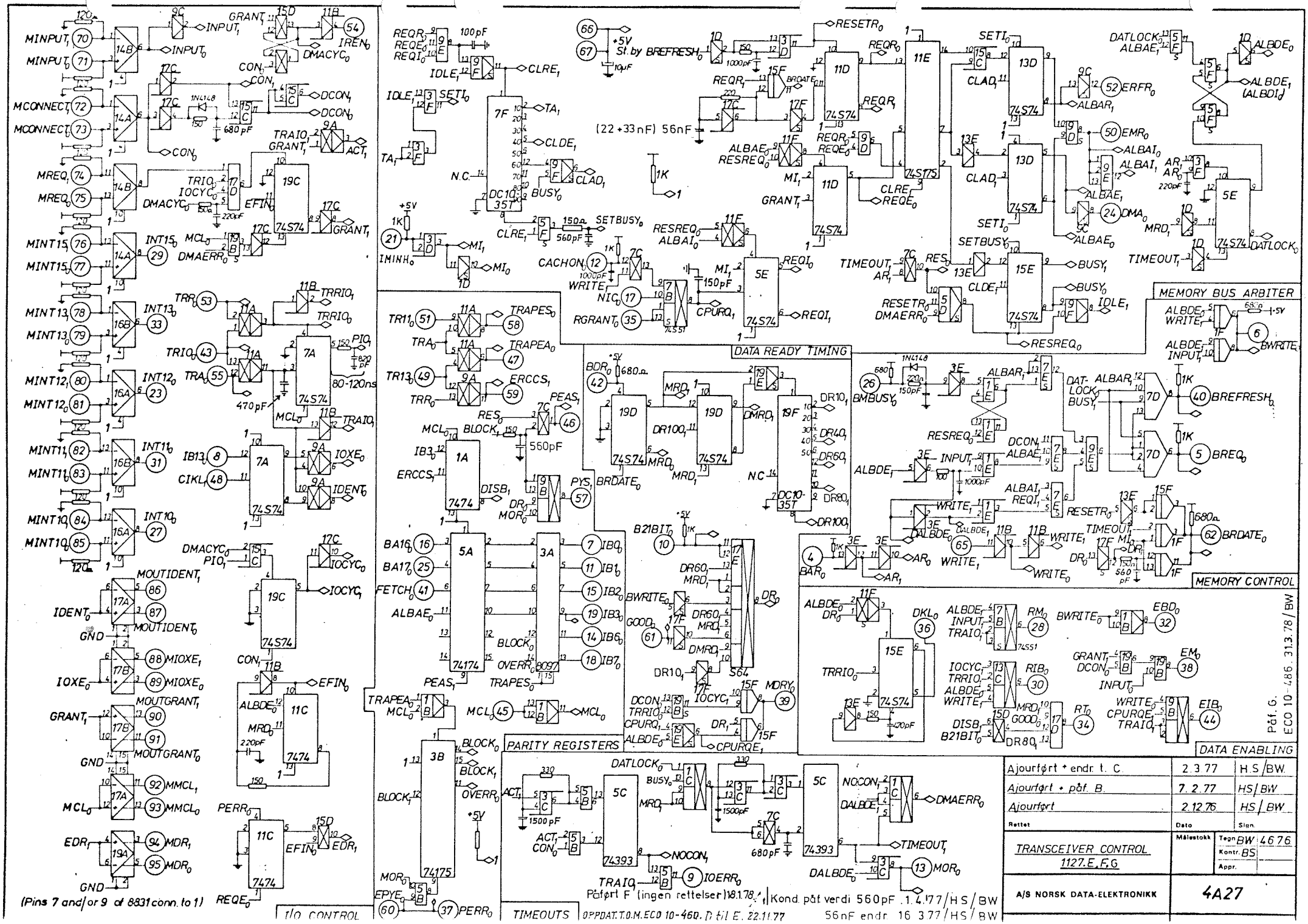
OPPDAT. T.O.M. ECO 10-506. E	30. 6. 78	BW
OPPDAT. T.O.M. ECO 10-473. D	9. 2. 78	BW
Signalnavn på pk 6A og 8A endr.	11.3.77	RI BW
1124 og 1124.B er like.	17.2.77	K.K./BW
Påført pk.pos.	9.12.76	RI / BW
Rettet	Dato	Sign.

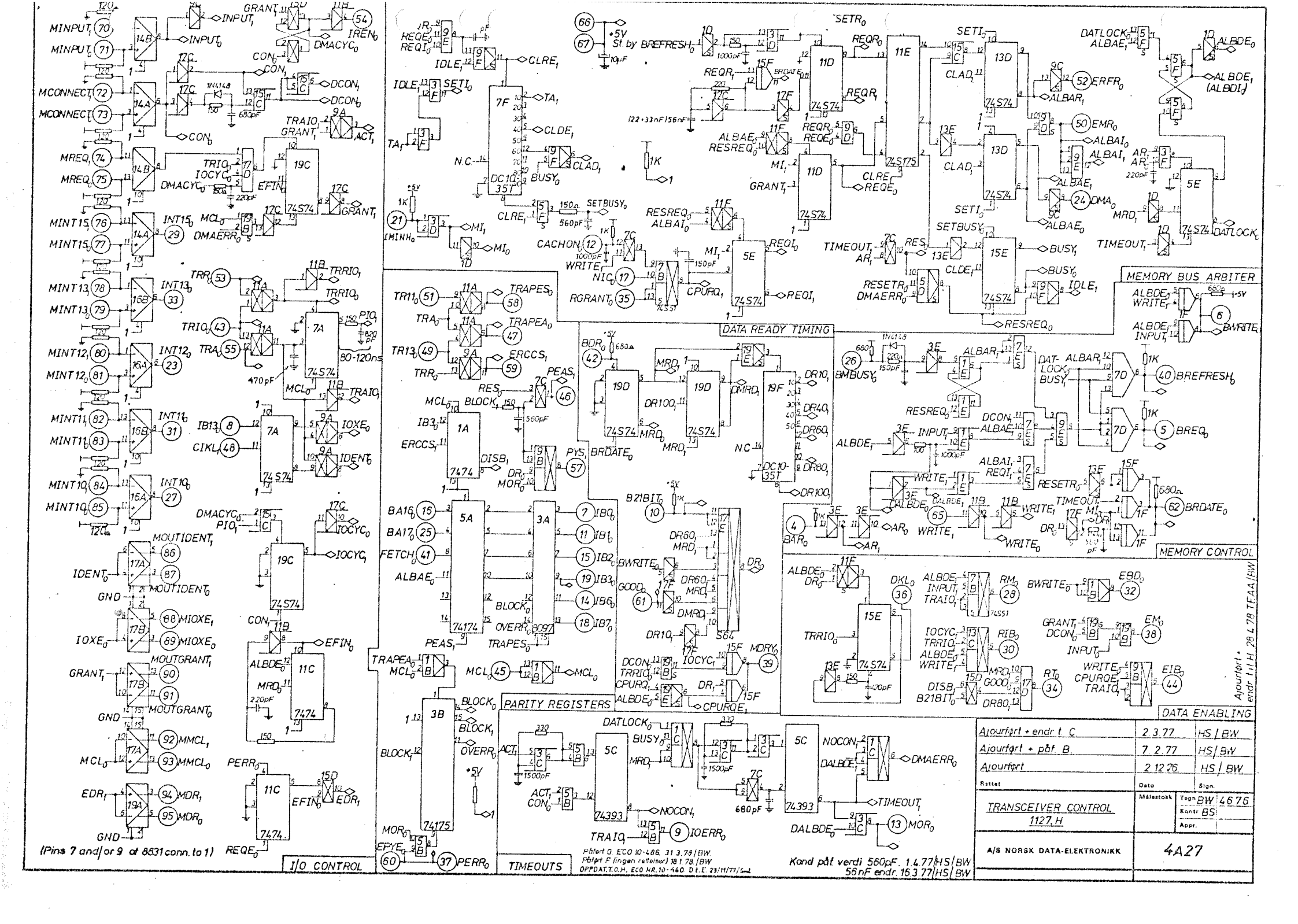
PERTEC MT CONTROL B 1124,E	Målestokk	Tegn. BW 21.9.76
		Kontr. RI
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A24	

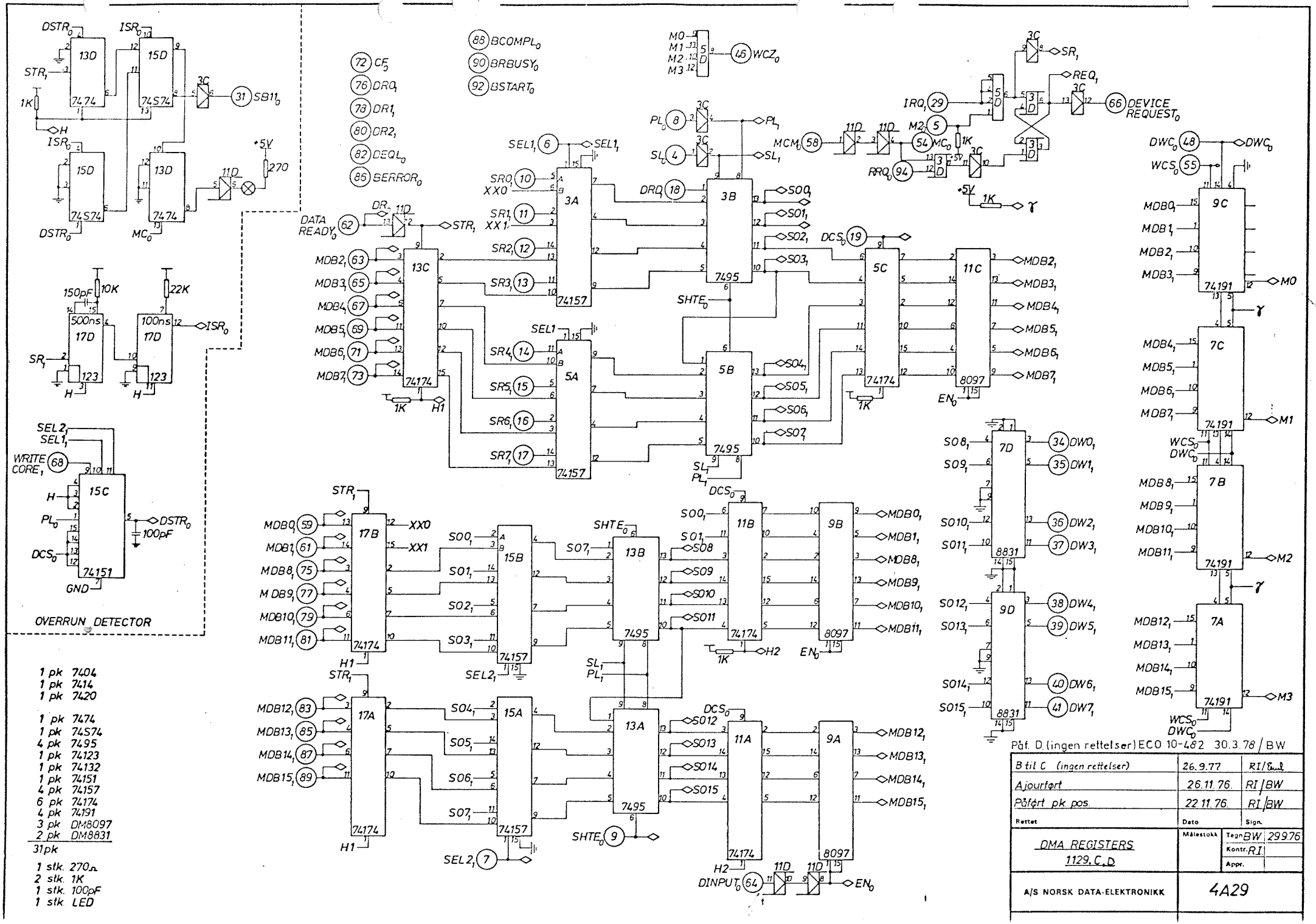

PARITY GENERATOR CHECKERS
3G
9G
5G
7G
11G
1G
745280
EVEN
ODD
ERROR STATUS AND CONTROL REGISTERS
1F
ERCC
74175
17G
ERC
7474
11B
PES
74174
9B
8098
10A
PEA
9A
8097
TRAPEA
TRAPES
220pF
1k x 4
+5V
3D
7D
11C
9D
745138
1A
1B
5A
5B
7A
7B
3A
3B
8835
1C
5C
1D
7E
17F
8098
11E
8098
BUS BUFFERS
ERROR DECODING AND CORRECTION
15A
15B
15C
15D
13A
13B
13C
13D
19D
19C
19B
19A
17D
17C
17B
17A
MAIN I/O BUS INTERFACE
INTERRUPT
17E
74551
176
60-120ns
7474
470 pF
150Ω
220 pF
13E
19E
19F
74174
OUTPUT DATA REGISTER
1 pk. 74500
1 " 7408
3 " 7414
1 " 7420
1 " 7432
1 " 74551
1 " 7474
4 " 74586
2 " 74132
1 " 74S133
4 pk. 74S138
5 " 74174
1 " 74175
6 " 74S280
1 " 8097
5 " 8098
8 " 8820A
8 " 8831
8 " 8835
Totalt : 62 pk.
38 stk. resistors, 1k (SIP)
Ajourf. B til C 2.3.77 HS/Emil
Ajourf. og påført B 7.2.77 HS/Emil
Påført pakkepos. 10.12.76 Emil
Rettet Dato Sign
TRANSCEIVER DATA 1125.D,E
Målestokk
Tegn. Emil 14.9.76
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A25
OPPDAT TOM ECO 10-441 15.9.77/S...0

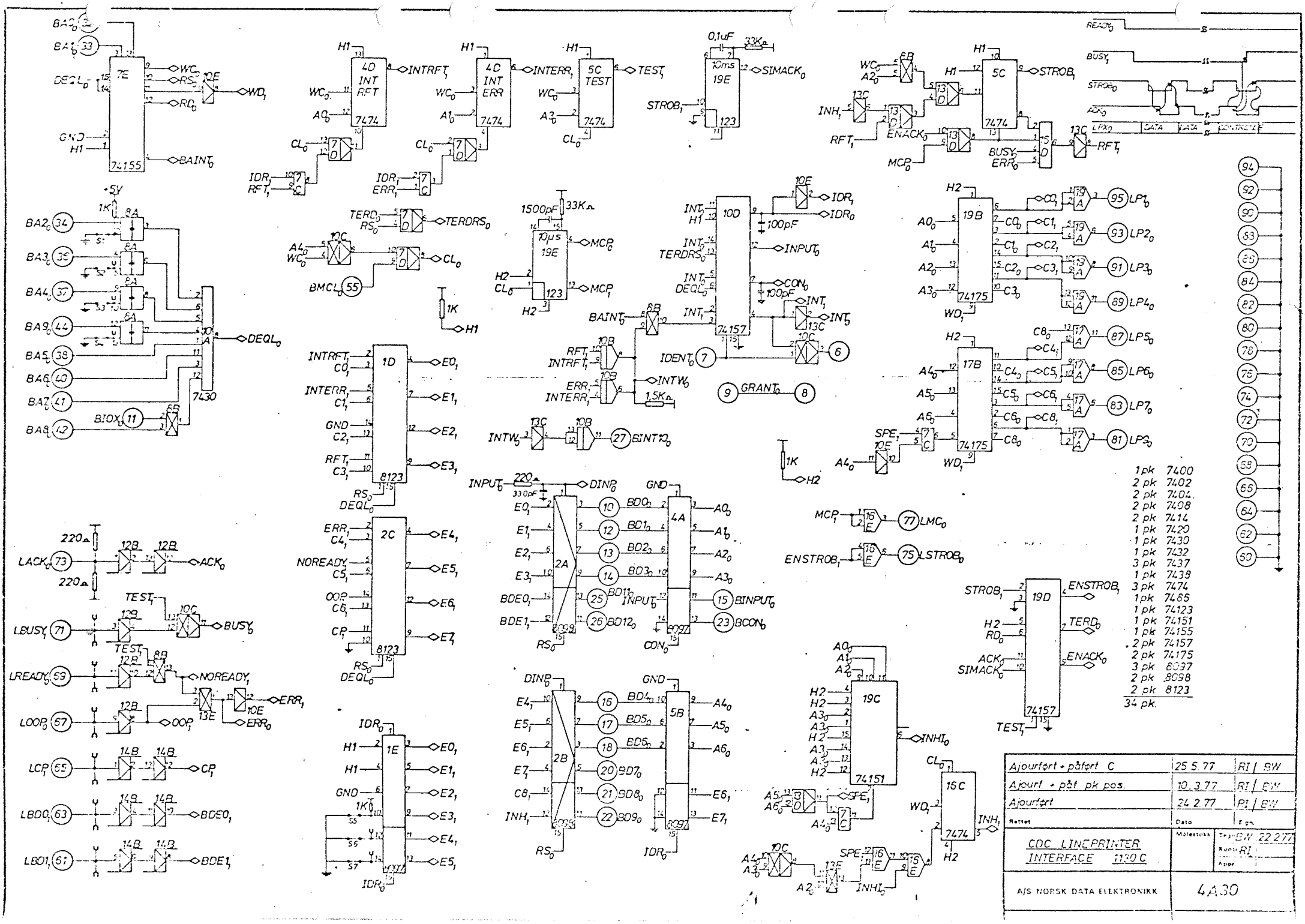

CACHE MEMORY
OUTPUT DRIVERS
ADDRESS BUFFERS
CACHE INHIBIT LIMITS
CACHE CONTROL
ADDRESS COMPARATORS
CACHE USED BIT
1 pk 7404
2 pk 74S04
1 pk 7408
1 pk 74S08
1 pk 74S10
2 pk 74S30
1 pk 7432
1 pk 7474
1 pk 74S74
4 pk 7485
2 pk 74S85
1 pk 74132
1 pk 74S138
3 pk 74S157
2 pk 74161
alt. 74163
2 pk 74174
1 pk 74175
4 pk 74S200
24pk 74S214
3pk 8097
2pk 8098
60pk.
Ajourført + pål. D. 8.3.77 HS / BW
Innlegn. 1 stk. 1500pF + pål. C 2.3.77 H.S / BW
Pål. B (ingen rettelser) 7.2.77 HS/BW
Ajourført 1.2.77 HS/BW
Ajourført. + påført pk. pos. 22.10.76 HS/BW
Rettet Dato Sign
CACHE 1126.B, C.D
Målestokk
Tegn BW 14.976
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A26

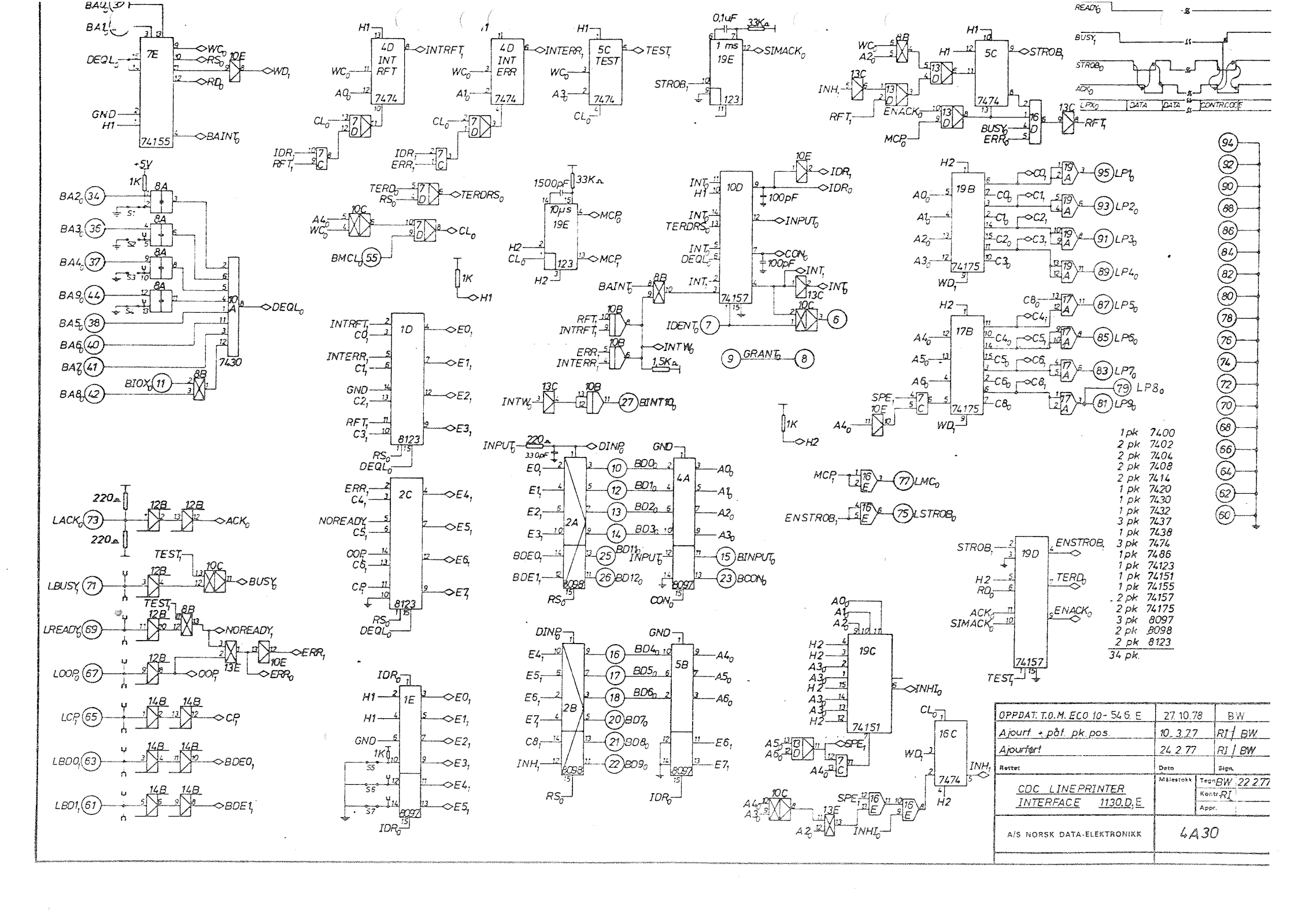

MEMORY BUS ARBITER
DATA READY TIMING
PARITY REGISTERS
MEMORY CONTROL
DATA ENABLING
TRANSCEIVER CONTROL
1127. D
A/S NORSK DATA ELEKTRONIKK
4A27
Ajourført + endr. t. C
2 3 77
HS/BW
Ajourført + pst. B
7 2 77
HS/BW
Ajourført
2 12 76
Rettet
Dato
Kond. pst verdi 560pF 14 77/HS/BW

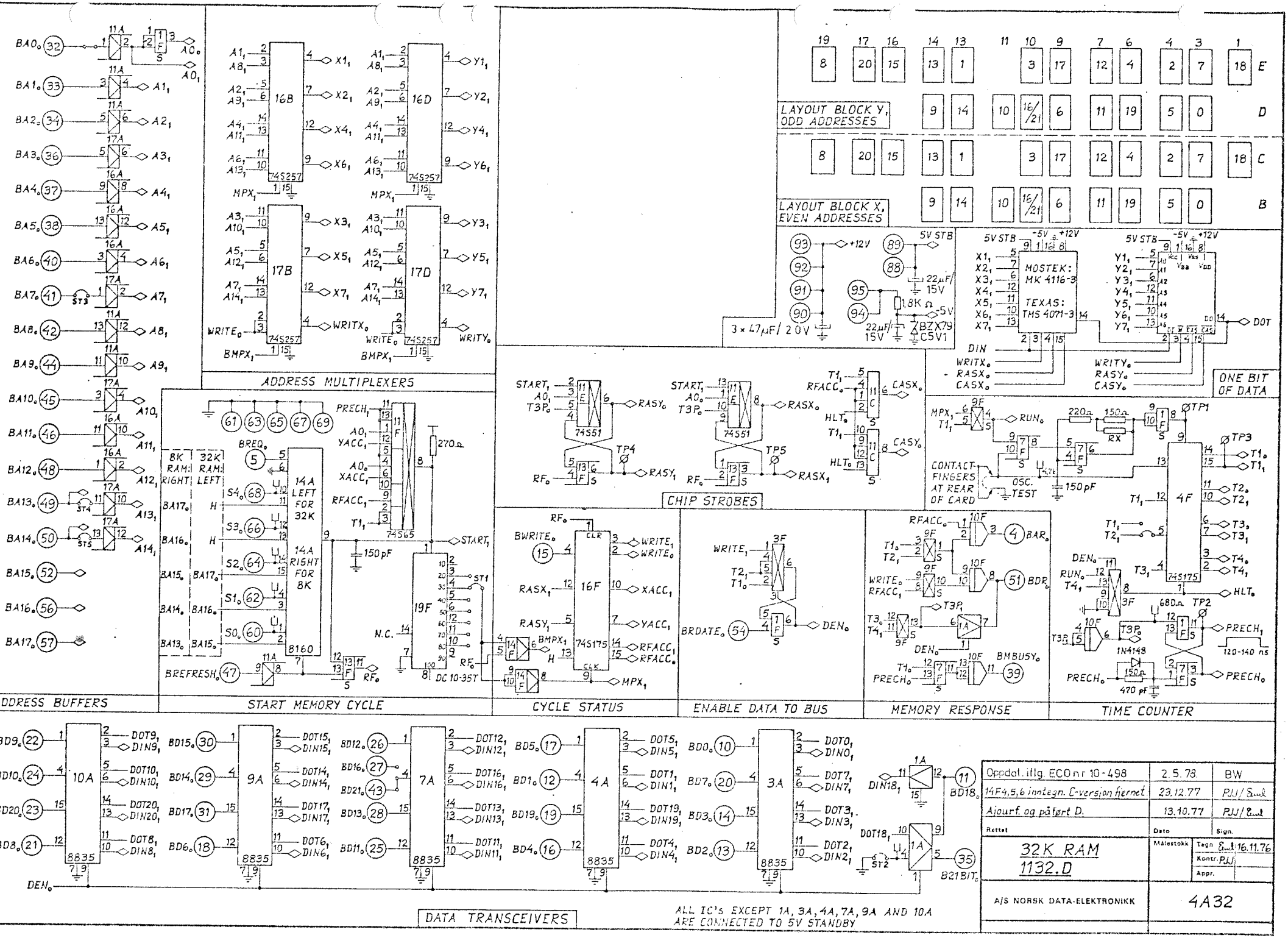

MEMORY BUS ARBITER
DATA READY TIMING
MEMORY CONTROL
PARITY REGISTERS
DATA ENABLING
I/O CONTROL
TIMEOUTS
(Pins 7 and/or 9 of 8831 conn. to 1)
Påført F (ingen rettelser) 18.1.78 / Kond. på t verdi 560pF 1.4.77/HS/BW
OPPDAT. T.O.M. ECO 10-460. D til E. 22.11.77
56nF endr 16.3.77/HS/BW
Ajourført + endr. t. C 2.3.77 HS/BW
Ajourført + påf. B 7.2.77 HS/BW
Ajourført 2.12.76 HS/BW
Rettet Dato Sign
TRANSCEIVER CONTROL
1127.E.F.G
Målestokk
Tegn BW 4676
Kontr. BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A27
Påf. G. ECO 10-486. 31.3.78/BW

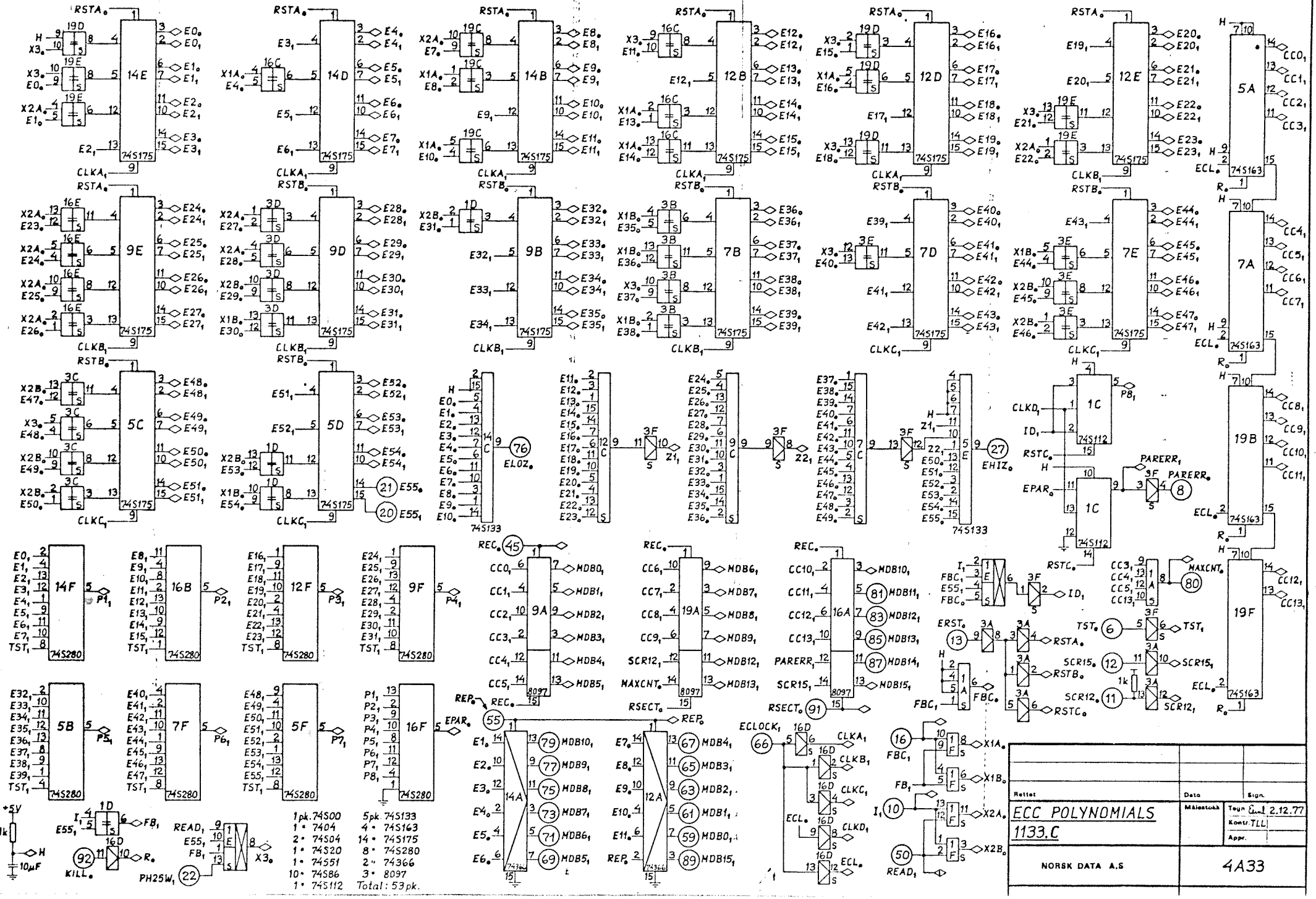

I/O CONTROL
TIMEOUTS
PARITY REGISTERS
DATA READY TIMING
MEMORY BUS ARBITER
MEMORY CONTROL
DATA ENABLING
TRANSCEIVER CONTROL
1127.H
A/S NORSK DATA-ELEKTRONIKK
4A27
(Pins 7 and/or 9 of 8831 conn. to 1)

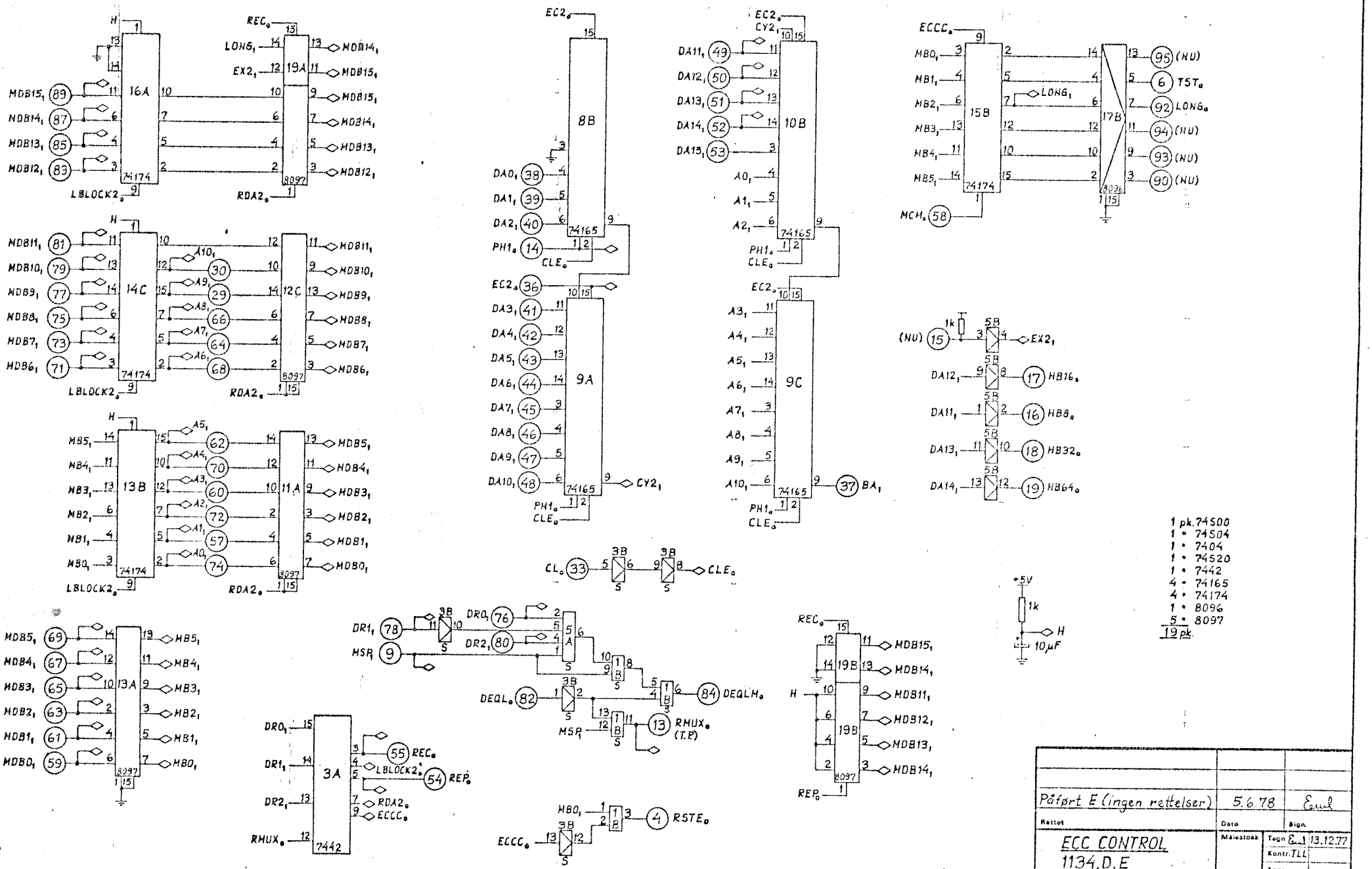

OVERRUN DETECTOR
1 pk 7404
1 pk 7414
1 pk 7420
1 pk 7474
1 pk 74S74
4 pk 7495
1 pk 74123
1 pk 74132
1 pk 74151
4 pk 74157
6 pk 74174
4 pk 74191
3 pk DM8097
2 pk DM8831
31pk
1 stk. 270Ω
2 stk. 1K
1 stk. 100pF
1 stk. LED
Påf. D.(ingen rettelser) ECO 10-482 30.3.78 / BW
B til C (ingen rettelser) 26.9.77 RI/Sml
Ajourført 26.11.76. RI/BW
Påført pk. pos. 22.11.76. RI/BW
Rettet Dato Sign.
DMA REGISTERS
1129.C.D
Målestokk
Tegn BW 29.9.76
Kontr. RI
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A29


74155
74175
74157
74174
74151
8123
8097
8098
7430
1pk 7400
2 pk 7402
2 pk 7404
2 pk 7408
2 pk 7414
1 pk 7420
1 pk 7430
1 pk 7432
3 pk 7437
1 pk 7438
3 pk 7474
1 pk 7486
1 pk 74123
1 pk 74151
1 pk 74155
2 pk 74157
2 pk 74175
3 pk 8097
2 pk 8098
2 pk 8123
34 pk.
Ajourført - påført C 25 5 77 RI / SW
Ajourf - påt pk pos. 10 3 77 RI / SW
Ajourført 24 2 77 PI / SW
Rettet Dato Sign
CDC LINEPRINTER INTERFACE 1130 C
Tegn SW 22 2 77
Kontr RI
A/S NORSK DATA ELEKTRONIKK
4A30


7E
74155
4D INT RFT
7474
4D INT ERR
5C TEST
1 ms 19E
123
5C
10D
74157
19B
74175
17B
19C
74151
19D
1E
8123
2C
1D
2A
4A
2B
5B
16C
7474
10µs 19E
1pk 7400
2 pk 7402
2 pk 7404
2 pk 7408
2 pk 7414
1 pk 7420
1 pk 7430
1 pk 7432
3 pk 7437
1 pk 7438
3 pk 7474
1pk 7486
1 pk 74123
1 pk 74151
1 pk 74155
2 pk 74157
2 pk 74175
3 pk 8097
2 pk 8098
2 pk 8123
34 pk.
OPPDAT. T.O.M. ECO 10-546. E
27.10.78
B.W
Ajourf. + påf. pk pos.
10.3.77
RI / BW
Ajourført
24.2.77
RI / BW
Retter
Dato
Sign.
CDC LINEPRINTER INTERFACE 1130.D.E
Målestokk
Tegn BW 22.2.77
Kontr. RI
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A30

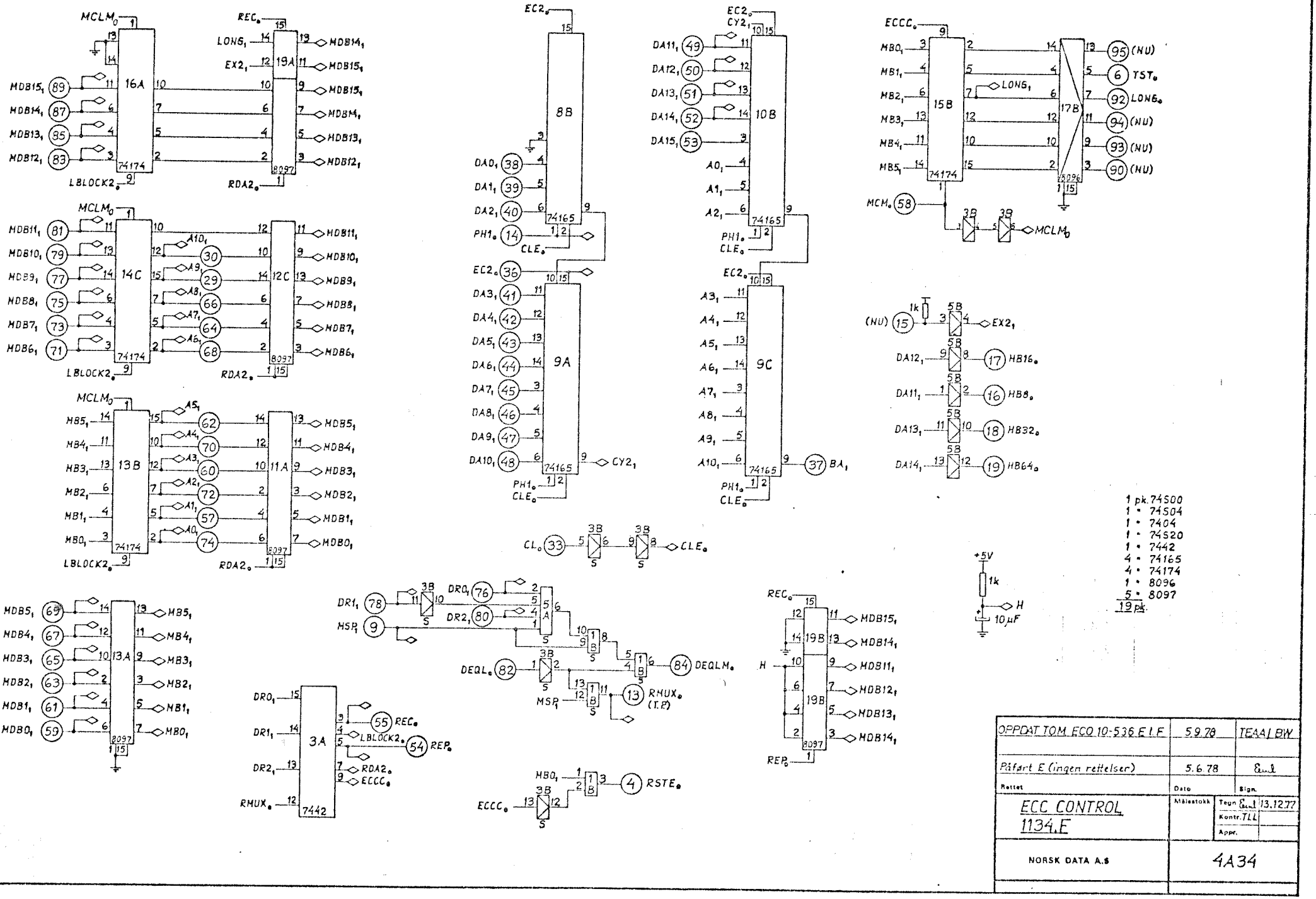

ADDRESS MULTIPLEXERS
LAYOUT BLOCK Y, ODD ADDRESSES
LAYOUT BLOCK X, EVEN ADDRESSES
CHIP STROBES
ONE BIT OF DATA
ADDRESS BUFFERS
START MEMORY CYCLE
CYCLE STATUS
ENABLE DATA TO BUS
MEMORY RESPONSE
TIME COUNTER
DATA TRANSCEIVERS
ALL IC's EXCEPT 1A, 3A, 4A, 7A, 9A AND 10A ARE CONNECTED TO 5V STANDBY
32K RAM
1132.D
A/S NORSK DATA-ELEKTRONIKK
4A32
Oppdat. iflg. ECO nr 10-498
14F4,5,6 inntegn. C-versjon hjernet
Ajourf. og påført D.
Rettet
Dato
Sign.
Målestokk
Tegn
Kontr.
Appr.

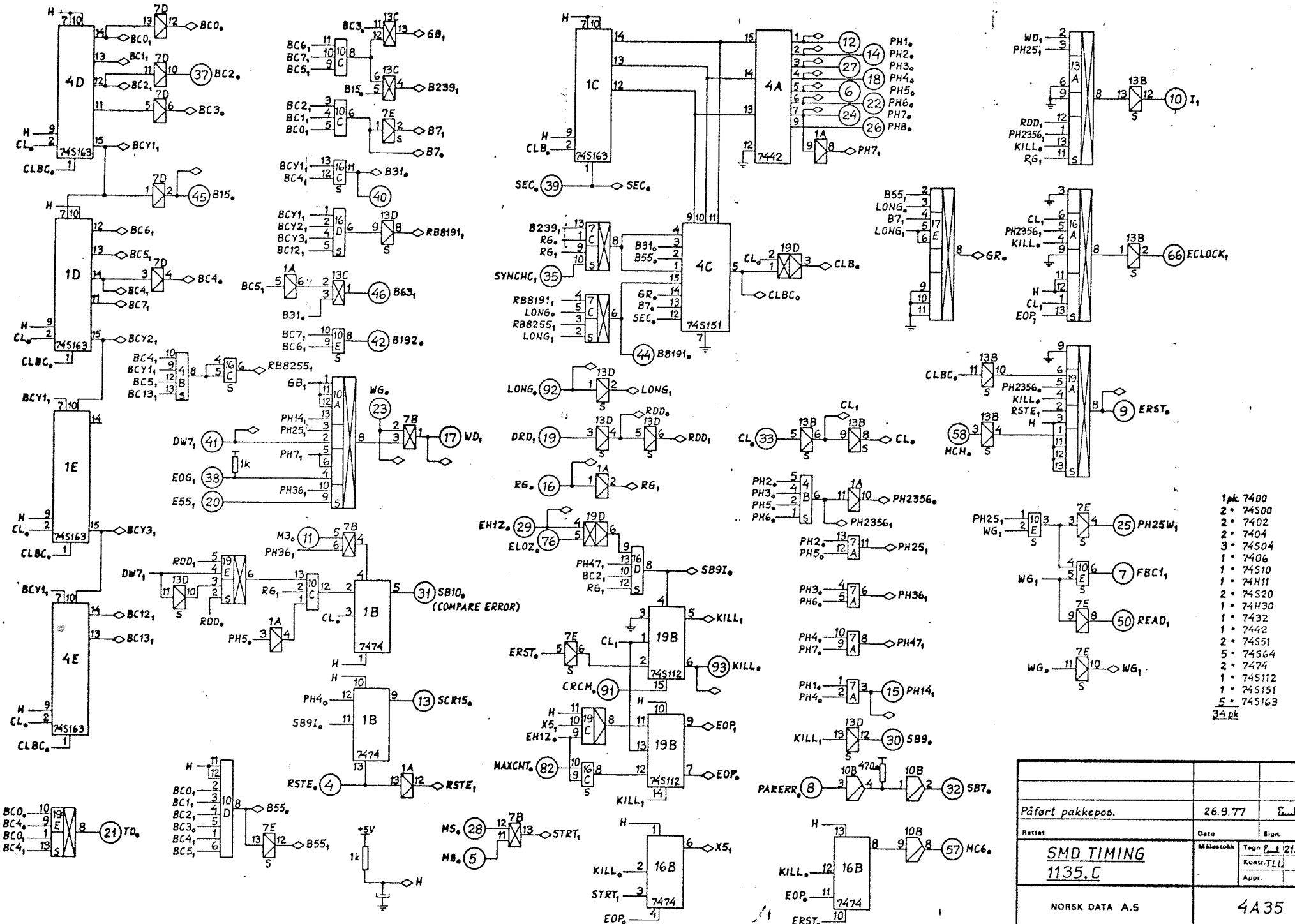

14E
14D
14B
12B
12D
12E
5A
9E
9D
9B
7B
7D
7E
7A
5C
5D
14F
16B
12F
9F
5B
7F
5F
16F
19B
19F
1C
9A
19A
16A
14A
12A
74S175
74S163
74S280
74S133
74S112
8097
74S366
RSTA
RSTB
CLKA
CLKB
CLKC
RSTC
CLKD
ECL
EHIZ
EL0Z
PAREERR
MAXCNT
ERST
TST
SCR15
SCR12
RSECT
REC
EPAR
KILL
PH25W
READ
FBC
FB
ECLOCK
1 pk. 74S00
1 • 7404
2 • 74S04
1 • 74S20
1 • 74S51
10 • 74S86
1 • 74S112
5 pk. 74S133
4 • 74S163
14 • 74S175
8 • 74S280
2 • 74366
3 • 8097
Total: 53 pk.
ECC POLYNOMIALS
1133.C
NORSK DATA A.S
4A33
Tegn
Kontr. TLL
Appr.
Dato
Sign.
Målestokk
2.12.77

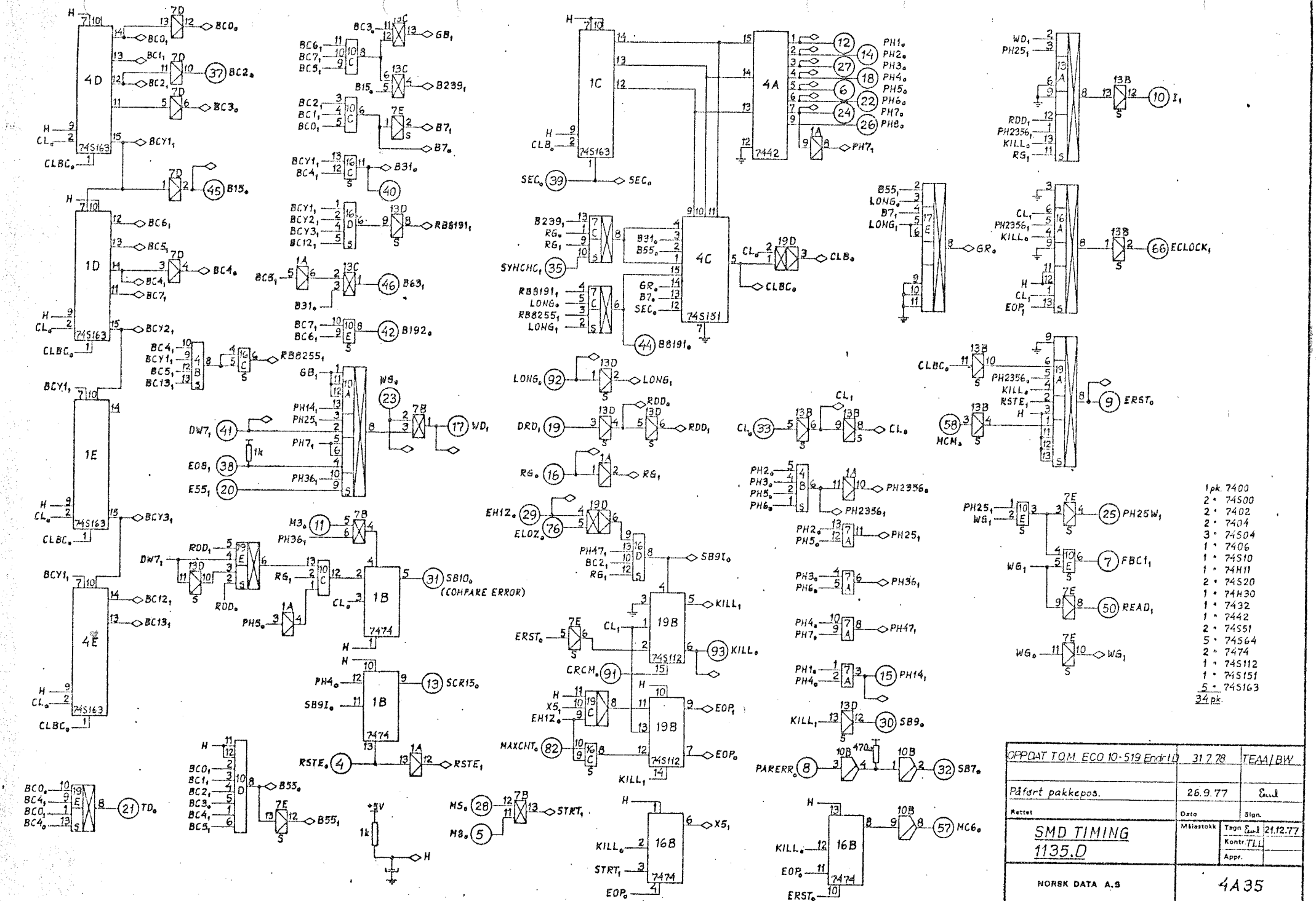


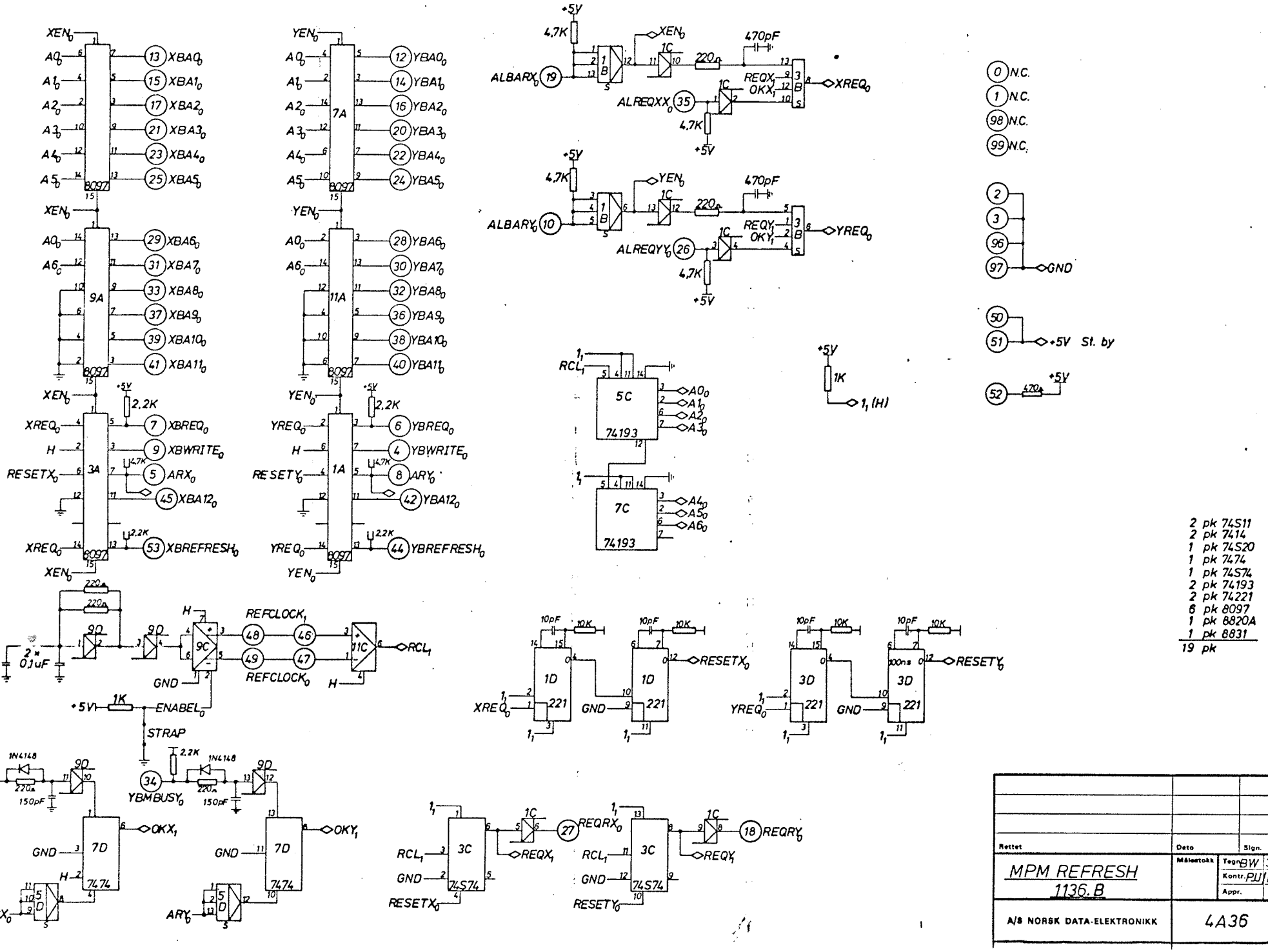
1 pk. 74S00
1 • 74S04
1 • 7404
1 • 74S20
1 • 7442
4 • 74165
4 • 74174
1 • 8096
5 • 8097
19 pk

Påført E (ingen rettelser)	5.6.78	Emil
Rettet	Dato	Sign.
ECC CONTROL 1134, D, E	Målestokk	Tegn E.J 13.12.77
		Kontr. TLL
		Appr.
NORSK DATA A.S	4A34	


MCLM0
REC0
LONG1
EX21
MDB151
MDB141
MDB131
MDB121
MDB15 (89)
MDB14 (87)
MDB13 (85)
MDB12 (83)
16A
19A
74174
8097
LBLOCK20
RDA20
MDB111
MDB101
MDB91
MDB81
MDB71
MDB61
MDB11 (81)
MDB10 (79)
MDB9 (77)
MDB8 (75)
MDB7 (73)
MDB6 (71)
14C
12C
A10 (30)
A9 (29)
A8 (66)
A7 (64)
A6 (68)
MDB5 (69)
MDB4 (67)
MDB3 (65)
MDB2 (63)
MDB1 (61)
MDB0 (59)
13B
11A
A5 (62)
A4 (70)
A3 (60)
A2 (72)
A1 (57)
A0 (74)
MDB51
MDB41
MDB31
MDB21
MDB11
MDB01
13A
EC20
8B
DA0 (38)
DA1 (39)
DA2 (40)
PH10 (14)
CLE0
74165
EC2 (36)
9A
DA3 (41)
DA4 (42)
DA5 (43)
DA6 (44)
DA7 (45)
DA8 (46)
DA9 (47)
DA10 (48)
CY21
CY20
DA11 (49)
DA12 (50)
DA13 (51)
DA14 (52)
DA15 (53)
10B
A0
A1
A2
9C
A3
A4
A5
A6
A7
A8
A9
A10
BA1 (37)
ECCC0
MB01
MB11
MB21
MB31
MB41
MB51
15B
17B
LONG1
MCM0 (58)
3B
MCLM0
(NU) (95)
TST0 (6)
LONG0 (92)
(NU) (94)
(NU) (93)
(NU) (90)
1k
(NU) (15)
EX21
DA121
DA111
DA131
DA141
5B
HB160 (17)
HB80 (16)
HB320 (18)
HB640 (19)
CL0 (33)
CLE0
DR1 (78)
MSR1 (9)
DRQ0 (76)
DR21 (80)
DEQL0 (82)
DEQLM0 (84)
RMUX0 (T.P) (13)
MSR1
3A
DRQ0
DR11
DR21
RMUX0
REC0 (55)
LBLOCK20
REP0 (54)
RDA20
ECCC0
7442
MB01
ECCC0
RSTE0 (4)
+5V
1k
H
10µF
19B
REC0
REP0
MDB151
MDB141
MDB111
MDB121
MDB131
MDB141
1 pk. 74S00
1 • 74S04
1 • 7404
1 • 74S20
1 • 7442
4 • 74165
4 • 74174
1 • 8096
5 • 8097
19 pk
OPPDAT TOM ECO 10-536 E I F
5.9.78
TEAA/BW
Påført E (ingen rettelser)
5.6.78
Rettet
Dato
Sign.
ECC CONTROL
1134.E
Nibleatokk
Tegn
13.12.77
Kontr. TLL
Appr.
NORSK DATA A.S
4A34

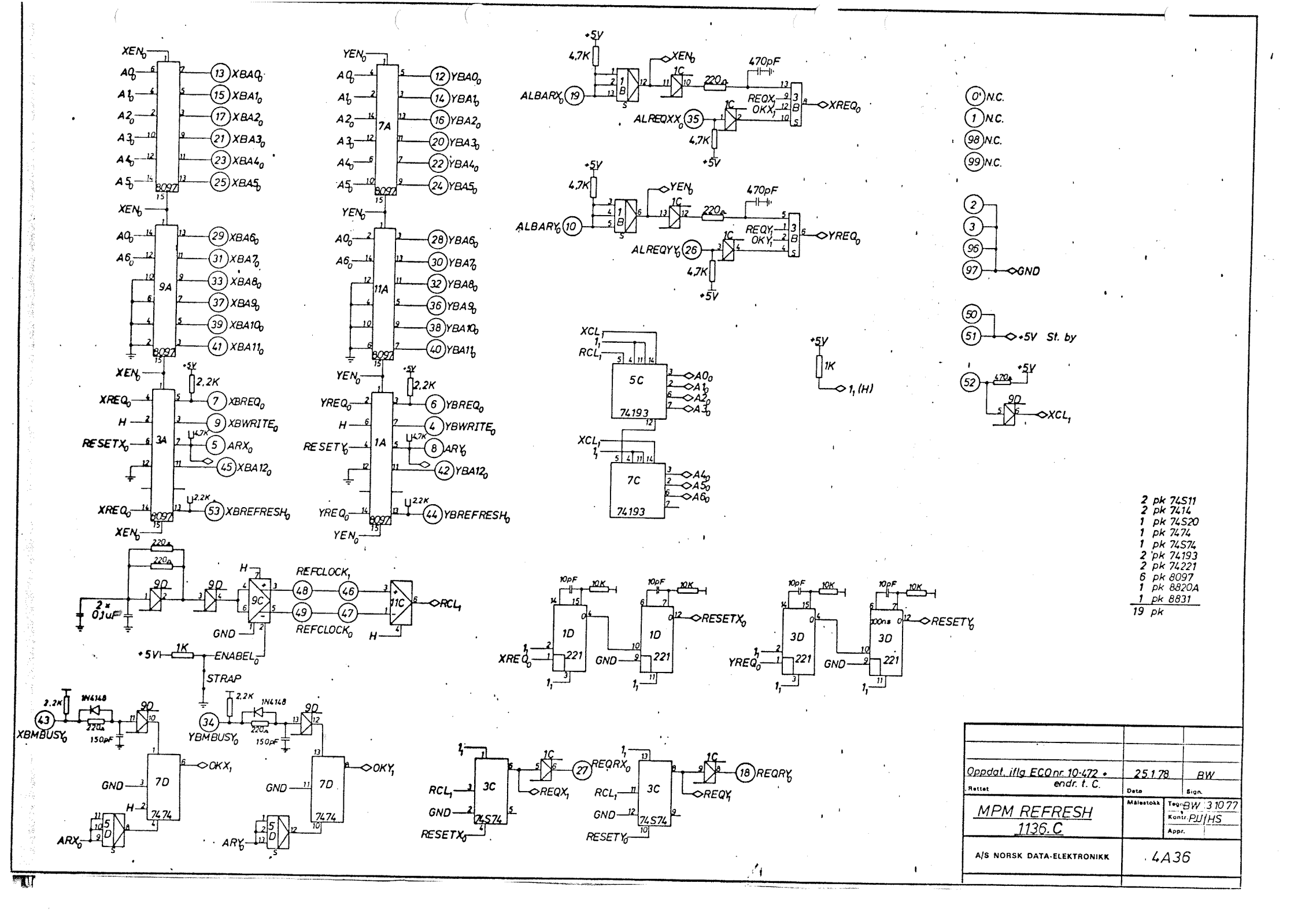

4D
1D
1E
4E
1C
4A
4C
1B
19B
16B
74S163
74S151
74S112
7442
7474
BC0
BC1
BC2
BC3
BC4
BC5
BC6
BC7
BCY1
BCY2
BCY3
BC12
BC13
B15
B31
B63
B192
B239
B8191
RB8191
RB8255
SEC
SYNCHC
LONG
RDD
DRD
RG
EH1Z
ELOZ
PH1
PH2
PH3
PH4
PH5
PH6
PH7
PH8
PH14
PH25
PH36
PH47
PH2356
WD
DW7
EOG
E55
WG
SB9I
SB10
(COMPARE ERROR)
SCR15
RSTE
KILL
EOP
MAXCNT
EH12
X5
SB9
SB7
PARERR
MC6
MS
MB
STRT
+5V
1k
GR
CL
CLB
CLBC
B55
LONG
B7
ECLOCK
I
ERST
CRCM
MCM
PH25W
FBC1
READ
TD
SB91
SB84
SB85
SB86
SB87
SB88
SB89
SB90
SB92
SB93
SB94
SB95
SB96
SB97
SB98
SB99
SB100
SB101
SB102
SB103
SB104
SB105
SB106
SB107
SB108
SB109
SB110
SB111
SB112
SB113
SB114
SB115
SB116
SB117
SB118
SB119
SB120
SB121
SB122
SB123
SB124
SB125
SB126
SB127
SB128
SB129
SB130
SB131


4D
74S163
1D
74S163
1E
74S163
4E
74S163
1C
74S163
4A
7442
4C
74S151
1B
7474
1B
7474
19B
74S112
19B
74S112
16B
7474
16B
7474
BCO,
BC1,
BC2,
BC3,
BC4,
BC5,
BC6,
BC7,
BCY1,
BCY2,
BCY3,
BC12,
BC13,
CLBC,
CLB,
SEC,
PH1,
PH2,
PH3,
PH4,
PH5,
PH6,
PH7,
PH8,
WD,
PH25,
RDD,
PH2356,
KILL,
RG,
B55,
LONG,
B7,
CL,
EOP,
H
I,
GR,
ECLOCK,
ERST,
HCM,
RSTE,
PH25W,
WG,
FBC1,
READ,
B239,
B31,
B63,
B192,
RB8191,
RB8255,
GB,
DW7,
EO8,
E55,
PH14,
PH7,
PH36,
WG,
EH12,
ELOZ,
DRD,
LONG,
SYNCHC,
SEC,
CRCM,
MAXCHT,
PAREER,
MS,
M8,
M3,
SB91,
SB10,
(COMPARE ERROR)
SCR15,
RSTE,
TD,
B55,
STRT,
X5,
KILL,
EOP,
ERST,
MC6,
SB7,
SB9,
PH14,
PH36,
PH47,
PH25,
PH2356,
+5V
1k
1pk. 7400
2 " 74S00
2 " 7402
2 " 7404
3 " 74S04
1 " 7406
1 " 74S10
1 " 74H11
2 " 74S20
1 " 74H30
1 " 7432
1 " 7442
2 " 74S51
5 " 74S64
2 " 7474
1 " 74S112
1 " 74S151
5 " 74S163
34 pk.
OPPDAT TOM ECO 10-519 Endr ID 31 7 78 TEAA/BW
Påført pakkepos. 26.9.77 Emil
Rettet Dato Sign.
SMD TIMING
1135.D
Målestokk
Tegn Emil 21.12.77
Kontr TLL
Appr.
NORSK DATA A.S
4A35

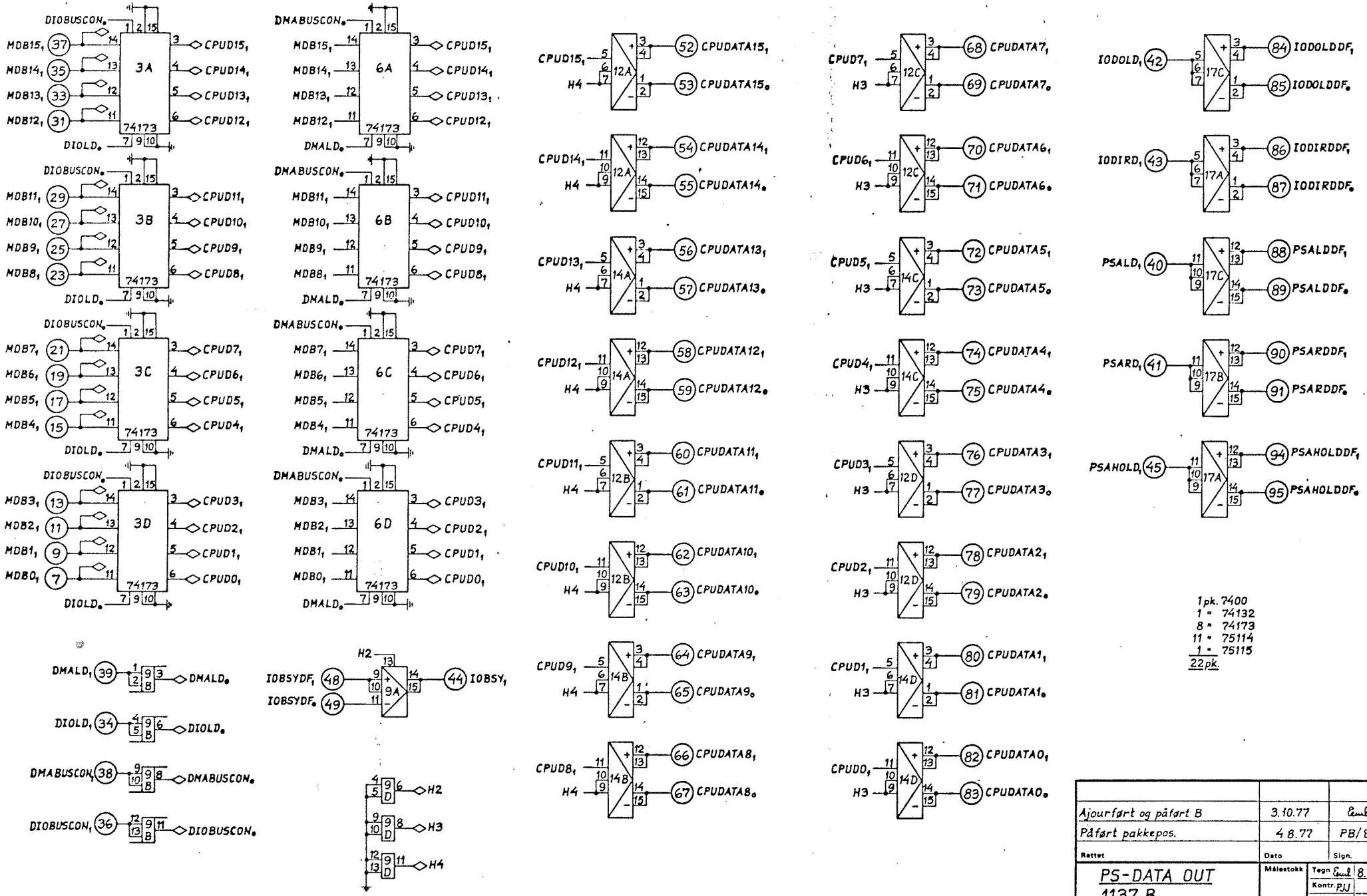


2 pk	74S11
2 pk	7414
1 pk	74S20
1 pk	7474
1 pk	74S74
2 pk	74193
2 pk	74221
6 pk	8097
1 pk	8820A
1 pk	8831
19 pk	

Rettet	Dato	Sign.
MPM REFRESH 1136.B	Målestokk	Tegn. BW 3 10 77 Kontr. PJJ HS Appr.
A/S NORSK DATA-ELEKTRONIKK	4A36	



Oppdat. iflg ECO nr. 10-472 • endr. t. C.	25.1.78	BW
Rettet	Dato	Sign.
MPM REFRESH 1136.C	Målestokk	Tegn. BW 3.10.77 Kontr. PJJ/HS Appr.
A/S NORSK DATA-ELEKTRONIKK	4A36	



1pk. 7400
1 = 74132
8 = 74173
11 = 75114
1 = 75115
22pk.

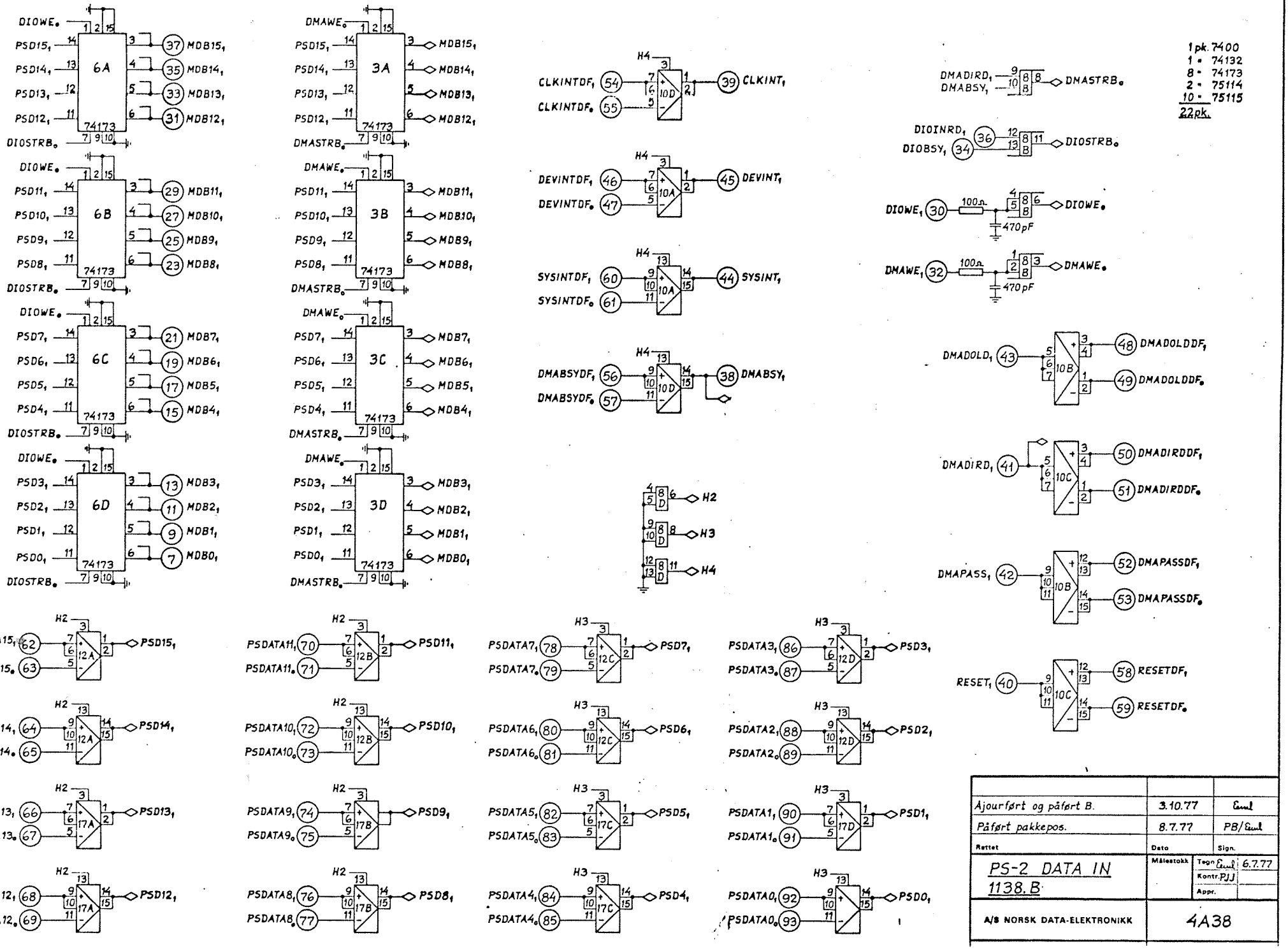
Ajourført og påført B	3.10.77	Emil
Påført pakkepos.	4.8.77	PB/Emil
Rettet	Dato	Sign.

PS-DATA OUT
1137.B

Målestokk | Tegn Emil 8.7.77 | Kontr. PJJ | Appr.

A/S NORSK DATA-ELEKTRONIKK

4A37



1 pk. 7400
1 = 74132
8 = 74173
2 = 75114
10 = 75115
22 pk.

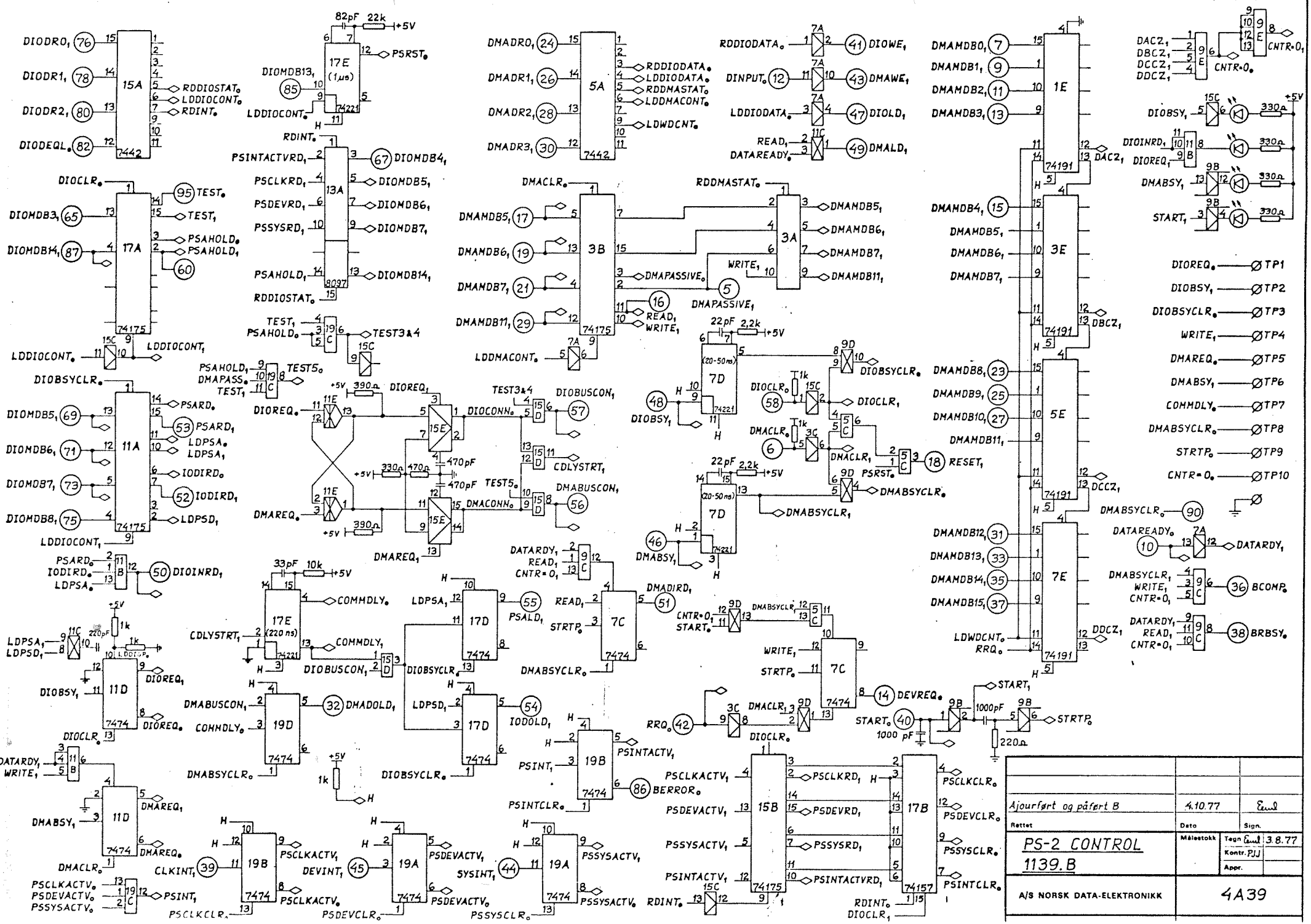
Ajourført og påført B.	3.10.77	Emil
Påført pakkepos.	8.7.77	PB/Emil
Rettet	Dato	Sign.

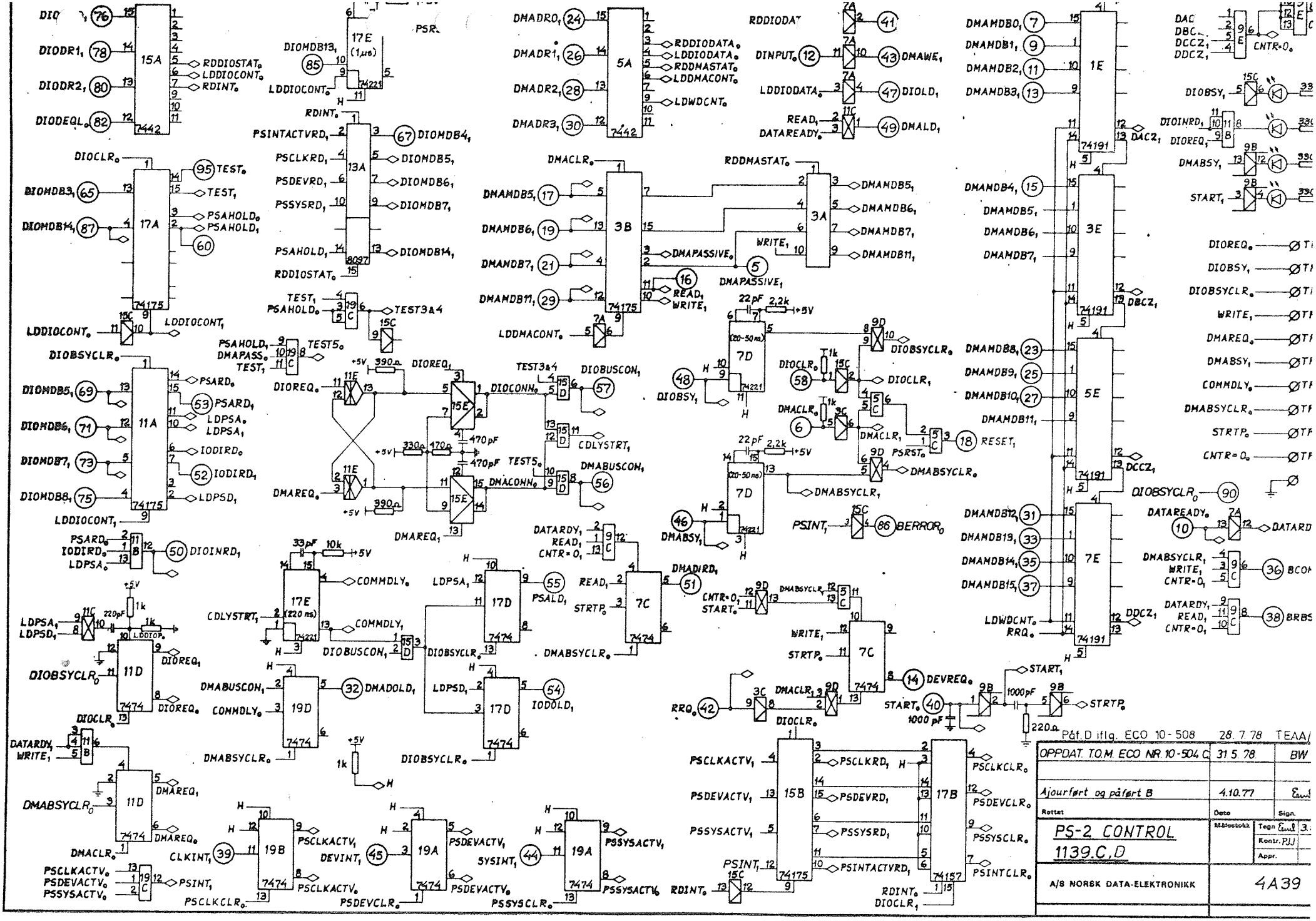
PS-2 DATA IN
1138.B

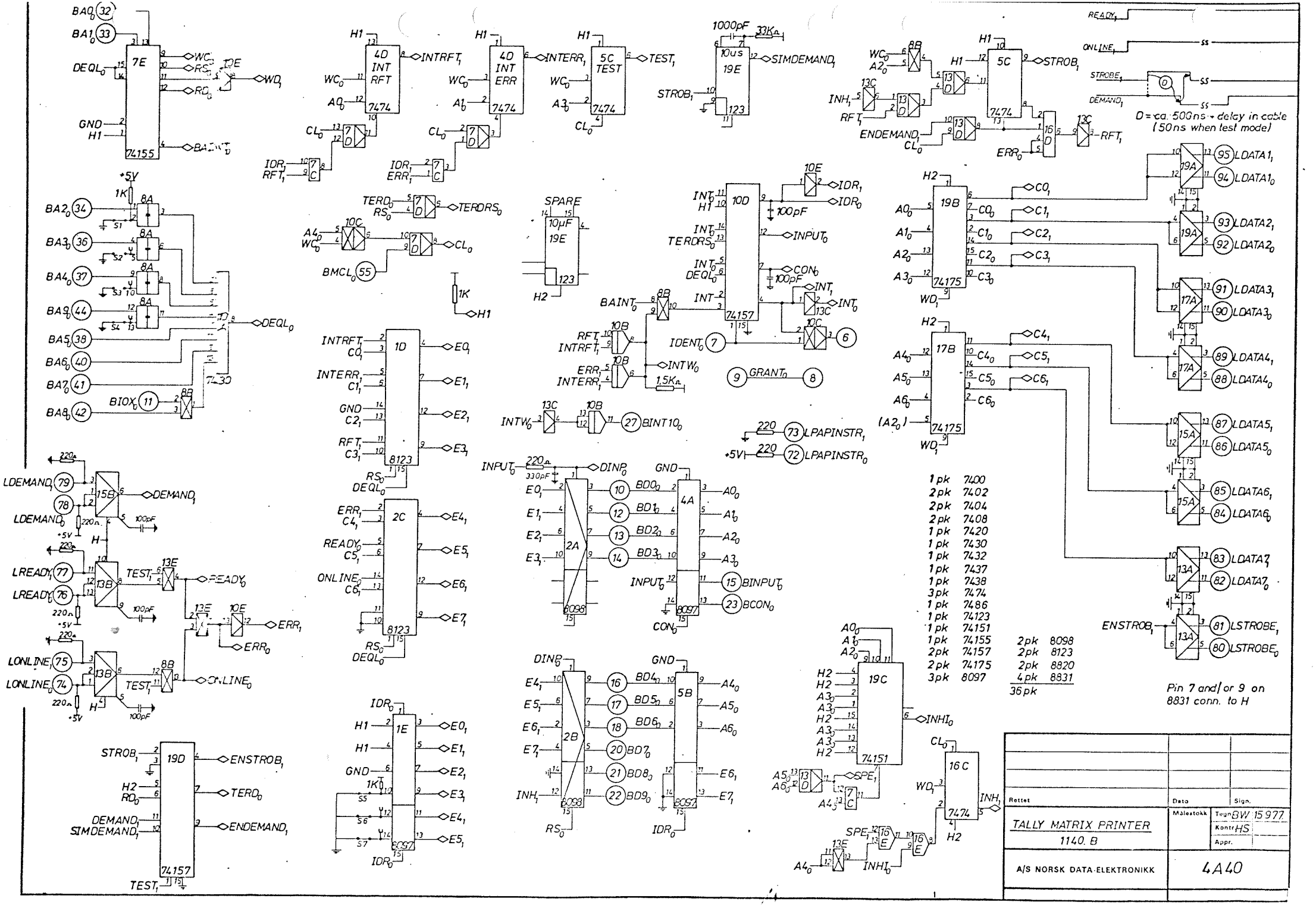
Målestokk | Tegn Emil 6.7.77 | Kontr. PJJ | Appr.

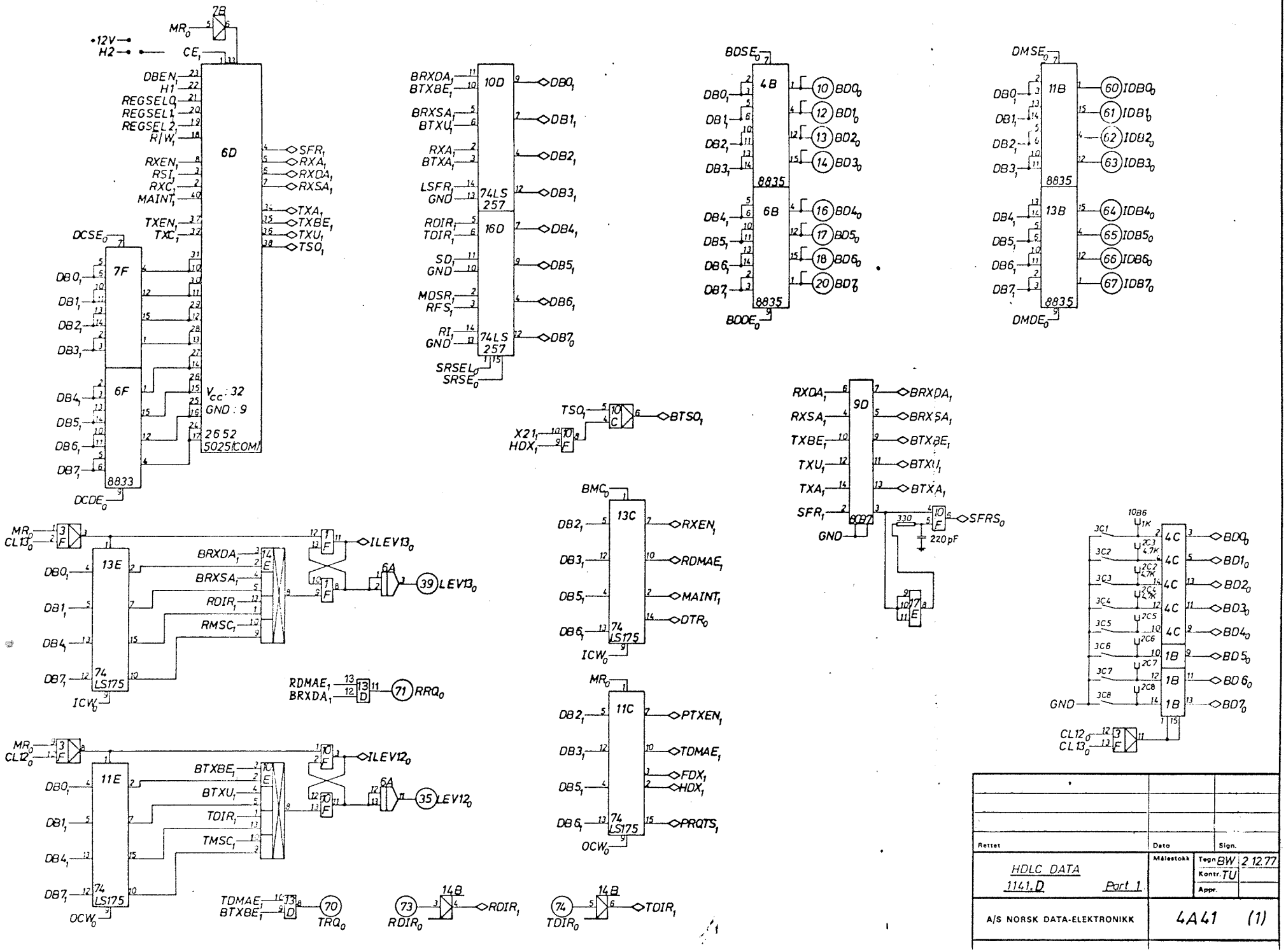
A/S NORSK DATA-ELEKTRONIKK

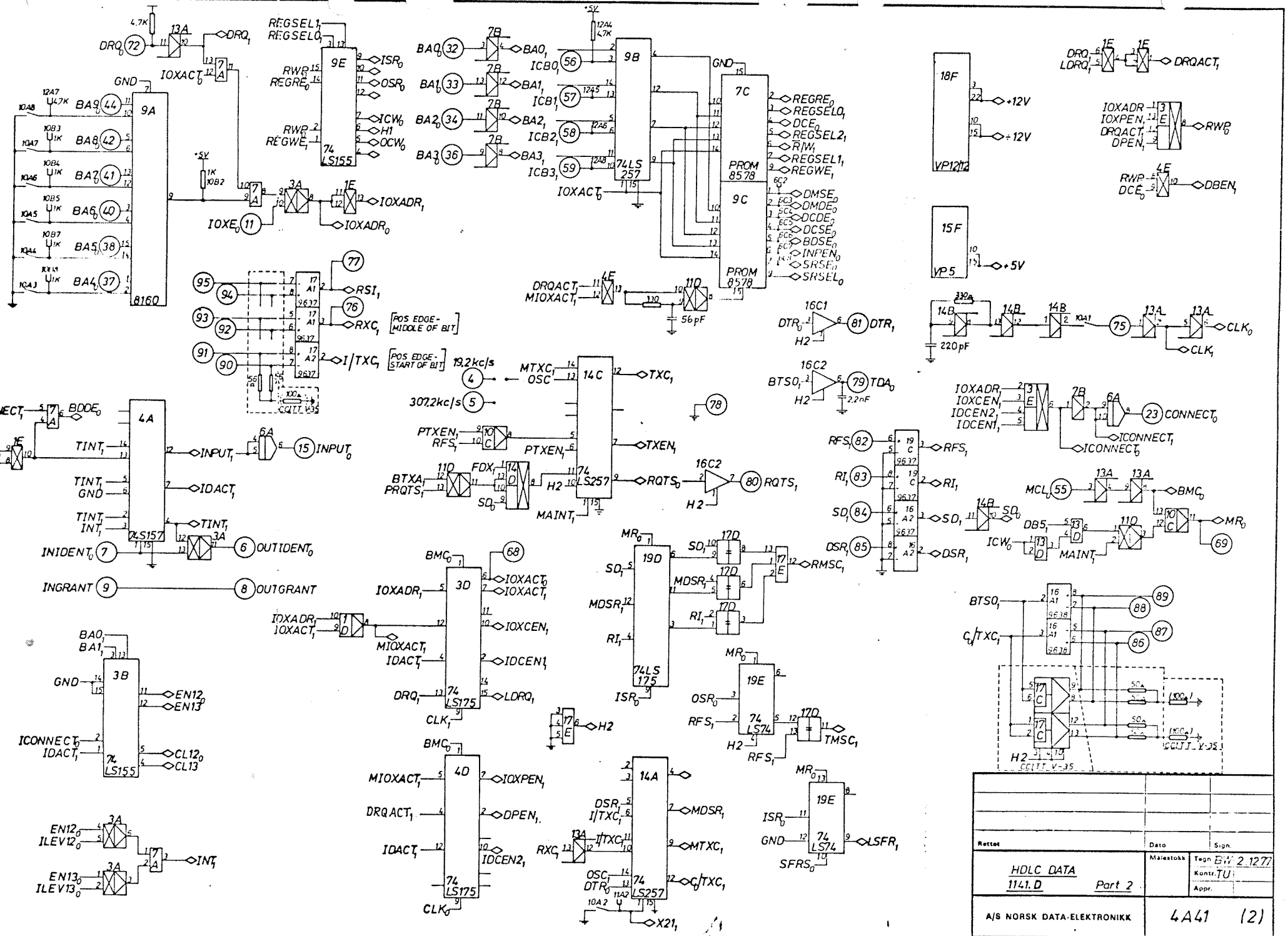
4A38

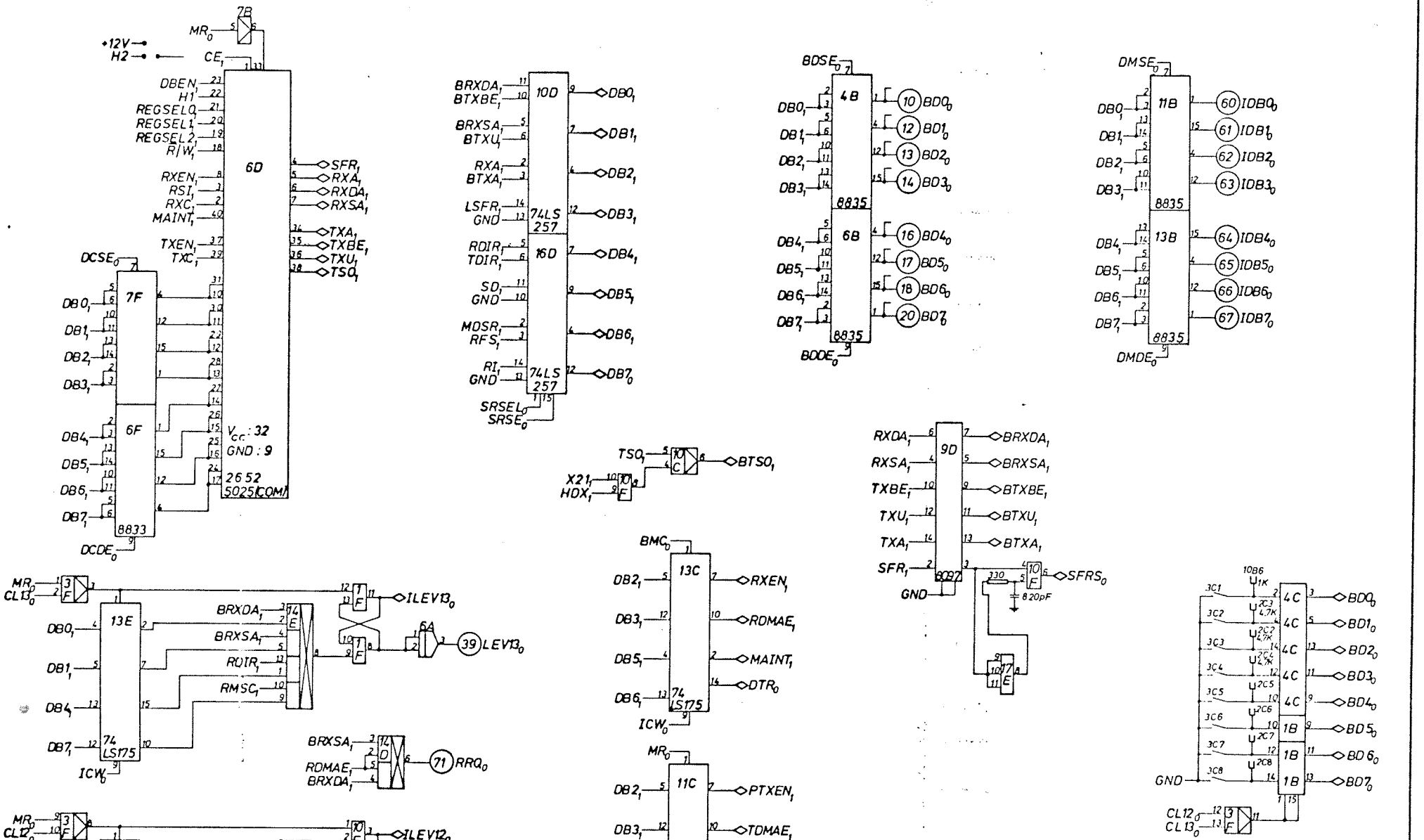

PS-2 CONTROL
1139.B
A/S NORSK DATA-ELEKTRONIKK
4A39
Ajourført og påført B
4.10.77
Rettet
Dato
Sign.
Tegn
3.8.77
Kontr.
Appr.
Målestokk
DIODR0,
DIODR1,
DIODR2,
DIODEQL,
DIOCLR,
DIOMDB3,
DIOMDB14,
LDDIOCONT,
DIOBSYCLR,
DIOMDB5,
DIOMDB6,
DIOMDB7,
DIOMDB8,
PSARD,
IODIRD,
LDPSA,
LDPSD,
DIOBSY,
DATARDY,
WRITE,
DMABSY,
DMACLR,
PSCLKACTV,
PSDEVACTV,
PSSYSACTV,
PSINT,
PSCLKCLR,
PSDEVCLR,
PSSYSCLR,
CLKINT,
DEVINT,
SYSINT,
RDDIOSTAT,
LDDIOCONT,
RDINT,
PSRST,
DIOMDB13,
LDDIOCONT,
PSINTACTVRD,
PSCLKRD,
PSDEVRD,
PSSYSRD,
PSAHOLD,
RDDIOSTAT,
TEST,
PSAHOLD,
TEST3&4,
PSAHOLD,
DMAPASS,
TEST,
TEST5,
DIOREQ,
DMAREQ,
DMADR0,
DMADR1,
DMADR2,
DMADR3,
LDWDCNT,
DMACLR,
DMAMDB5,
DMAMDB6,
DMAMDB7,
DMAMDB11,
LDDMACONT,
RDDMASTAT,
WRITE,
DMAPASSIVE,
READ,
WRITE,
RDDIODATA,
DINPUT,
LDDIODATA,
READ,
DATAREADY,
DIOWE,
DMAWE,
DIOLD,
DMALD,
DMAMDB0,
DMAMDB1,
DMAMDB2,
DMAMDB3,
DMAMDB4,
DMAMDB5,
DMAMDB6,
DMAMDB7,
DMAMDB8,
DMAMDB9,
DMAMDB10,
DMAMDB11,
DMAMDB12,
DMAMDB13,
DMAMDB14,
DMAMDB15,
LDWDCNT,
RRQ,
DACZ,
DBCZ,
DCCZ,
DDCZ,
CNTR=0,
DIOBSY,
DIOINRD,
DIOREQ,
DMABSY,
START,
DIOREQ TP1
DIOBSY TP2
DIOBSYCLR TP3
WRITE TP4
DMAREQ TP5
DMABSY TP6
COMMDLY TP7
DMABSYCLR TP8
STRTP TP9
CNTR=0 TP10
DMABSYCLR,
DATAREADY,
DMABSYCLR,
WRITE,
CNTR=0,
DATARDY,
READ,
CNTR=0,
BCOMP,
BRBSY,
RESET,
DMABSYCLR,
DIOCLR,
DIOBSYCLR,
DMACLR,
DMACLR,
PSRST,
DMABSYCLR,
DMABSYCLR,
DIOBUSCON,
DMABUSCON,
CDLYSTRT,
DIOBUSCON,
DMABUSCON,
COMMDLY,
COMMDLY,
DIOBSYCLR,
DMABSYCLR,
LDPSA,
LDPSD,
DIOBSYCLR,
DMABSYCLR,
DIOCONN,
DMACONN,
TEST3&4,
TEST5,
DIOBUSCON,
CDLYSTRT,
DMABUSCON,
DMADIRD,
PSALD,
IODOLD,
DMADOLD,
PSINTACTV,
BERROR,
PSINTCLR,
PSCLKACTV,
PSDEVACTV,
PSSYSACTV,
PSCLKRD,
PSDEVRD,
PSSYSRD,
PSINTACTVRD,
RDINT,
DIOCLR,
PSCLKCLR,
PSDEVCLR,
PSSYSCLR,
PSINTCLR,
DEVREQ,
START,
STRTP,
RESET,
DMAPASSIVE,
DMAPASSIVE,
READ,
WRITE,
DIOREQ,
DIOBSY,
DMABSY,
DMAREQ,
DMACLR,
DIOCLR,
DIOBSYCLR,
DMABSYCLR,
DMABSYCLR,
DATARDY,
BCOMP,
BRBSY,
DACZ,
DBCZ,
DCCZ,
DDCZ,
CNTR=0,
DIOBSY,
DIOINRD,
DIOREQ,
DMABSY,
START,
15A
7442
17E
(1 µs)
74221
13A
8097
17A
74175
11A
74175
5A
7442
3B
74175
3A
74175
7D
(20-50 ns)
74221
11E
15E
11D
7474
19D
7474
17D
7474
7C
7474
19B
7474
19A
7474
19B
7474
17E
(220 ns)
74221
15B
74175
17B
74157
1E
74191
3E
74191
5E
74191
7E
74191
82pF
22k
+5V
330Ω
390Ω
470pF
33pF
10k
1k
220pF
1000pF
2,2k
22pF
1000 pF
220Ω


PS-2 CONTROL
1139.C,D
A/S NORSK DATA-ELEKTRONIKK
4A39
OPPDAT. T.O.M. ECO NR. 10-504 C
31.5.78
BW
Påf. D iflg. ECO 10-508
28.7.78
TEAA/
Ajourført og påført B
4.10.77
Rettet
Dato
Sign.
Målestokk
Tegn.
Kontr. PJJ
Appr.
15A
17E
5A
RDDIODAT
DINPUT
LDDIODATA
READ
DATAREADY
1E
DAC
DBC
DCCZ
DDCZ
CNTR=0
DIOBSY
DIOINRD
DIOREQ
DMABSY
START
17A
13A
3B
3A
3E
5E
7E
74191
DACZ
DBCZ
DCCZ
DDCZ
DIOCLR
TEST
PSAHOLD
LDDIOCONT
PSINTACTVRD
PSCLKRD
PSDEVRD
PSSYSRD
RDDIOSTAT
DMACLR
DMAMDB5
DMAMDB6
DMAMDB7
DMAMDB11
DMAPASSIVE
WRITE
RDDMASTAT
DMAMDB4
DMAMDB8
DMAMDB9
DMAMDB10
DMAMDB12
DMAMDB13
DMAMDB14
DMAMDB15
LDWDCNT
RRQ
11A
DIOMDB5
DIOMDB6
DIOMDB7
DIOMDB8
PSARD
IODIRD
LDPSD
LDPSA
11E
15E
DIOREQ
DMAREQ
DIOCONN
DMACONN
DIOBUSCON
CDLYSTRT
TEST5
TEST3&4
DMABUSCON
7D
DIOBSYCLR
DMABSYCLR
PSRST
RESET
BERROR
PSINT
DMABSY
11D
DIOBSYCLR
DIOCLR
DATARDY
WRITE
DMABSYCLR
DMACLR
17E
COMMDLY
CDLYSTRT
DIOBUSCON
19D
DMABUSCON
COMMDLY
DMABSYCLR
17D
LDPSA
LDPSD
DIOBSYCLR
7C
DMADIRD
CNTR=0
START
STRTP
DMABSYCLR
7C
WRITE
STRTP
RRQ
DMACLR
DEVREQ
START
STRTP
DIOCLR
15B
17B
PSCLKACTV
PSDEVACTV
PSSYSACTV
PSINT
RDINT
PSCLKRD
PSDEVRD
PSSYSRD
PSINTACTVRD
PSCLKCLR
PSDEVCLR
PSSYSCLR
PSINTCLR
DIOCLR
19B
19A
19A
CLKINT
DEVINT
SYSINT
PSCLKACTV
PSDEVACTV
PSSYSACTV
PSCLKCLR
PSDEVCLR
PSSYSCLR
DIOREQ
DIOBSY
DIOBSYCLR
WRITE
DMAREQ
DMABSY
COMMDLY
DMABSYCLR
STRTP
CNTR=0
DATARDY
READ
CNTR=0
DATARD
BCOM
BRBS

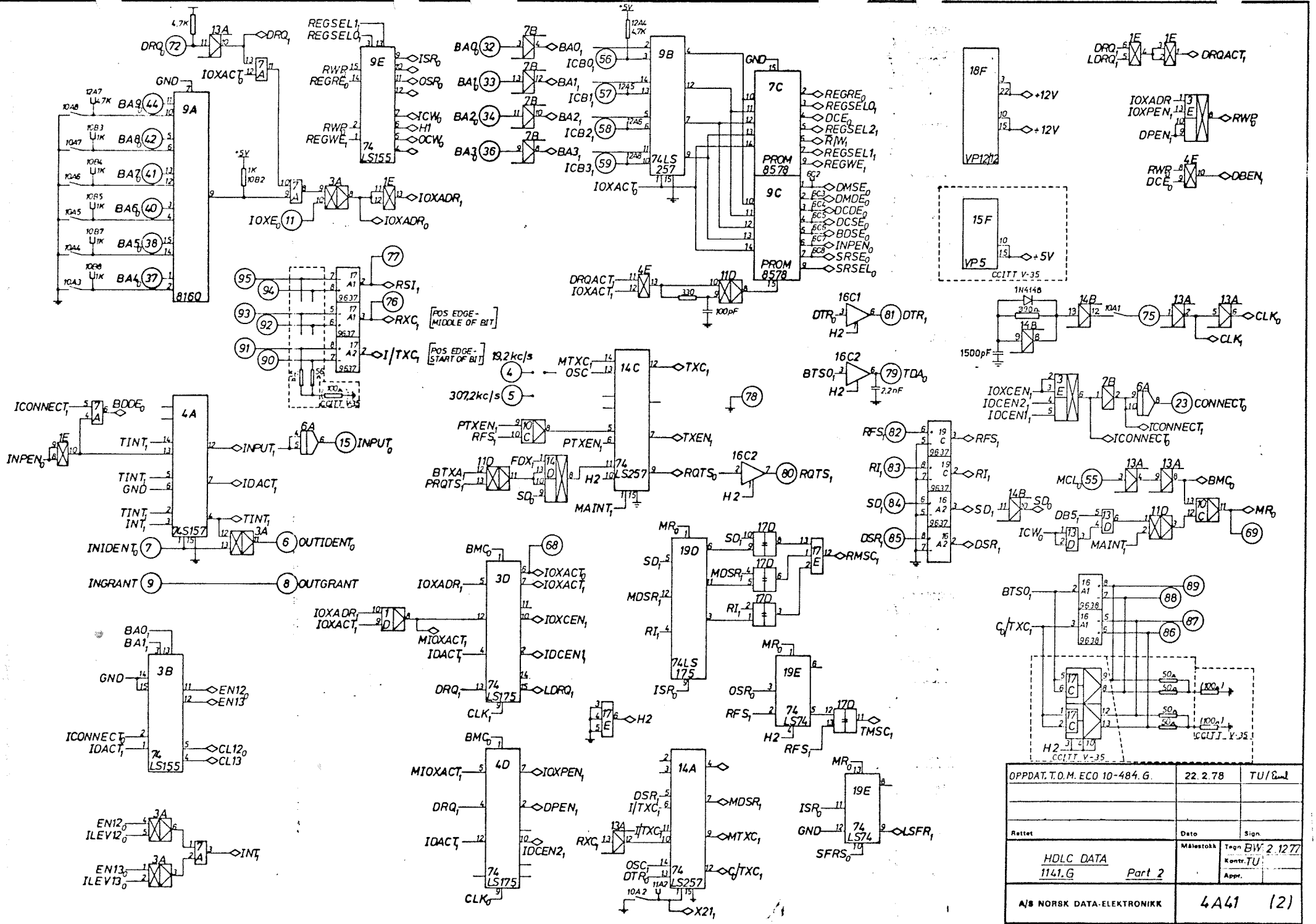

7E
74155
4D INT RFT
7474
4D INT ERR
5C TEST
100ns 19E
123
8B
13C
5C
74174
16D
READY1
ONLINE1
STROBE1
DEMAND1
D = ca. 500ns + delay in cable (50ns when test mode)
+5V
1K
8A
7430
10C
7D
10μF 19E
10D
100pF
74157
10B
1.5KΩ
10E
19B
74175
17B
19A
17A
15A
13A
LDATA1
LDATA2
LDATA3
LDATA4
LDATA5
LDATA6
LDATA7
LSTROBE
ENSTROBE1
15B
13B
13E
10E
LDEMAND
LREADY
LONLINE
8123
2C
READY0
ONLINE0
ERR1
ERR0
8098
2A
4A
8097
2B
5B
6097
19C
74151
16C
7474
SPE1
INHI0
19D
74157
ENSTROB1
TERO0
ENDEMAND1
1 pk 7400
2 pk 7402
2 pk 7404
2 pk 7408
1 pk 7420
1 pk 7430
1 pk 7432
1 pk 7437
1 pk 7438
3 pk 7474
1 pk 7486
1 pk 74123
1 pk 74151
1 pk 74155
2 pk 74157
2 pk 74175
3 pk 8097
2 pk 8098
2 pk 8123
2 pk 8820
4 pk 8831
36 pk
Pin 7 and/or 9 on 8831 conn. to H
TALLY MATRIX PRINTER
1140. B
A/S NORSK DATA-ELEKTRONIKK
4A40

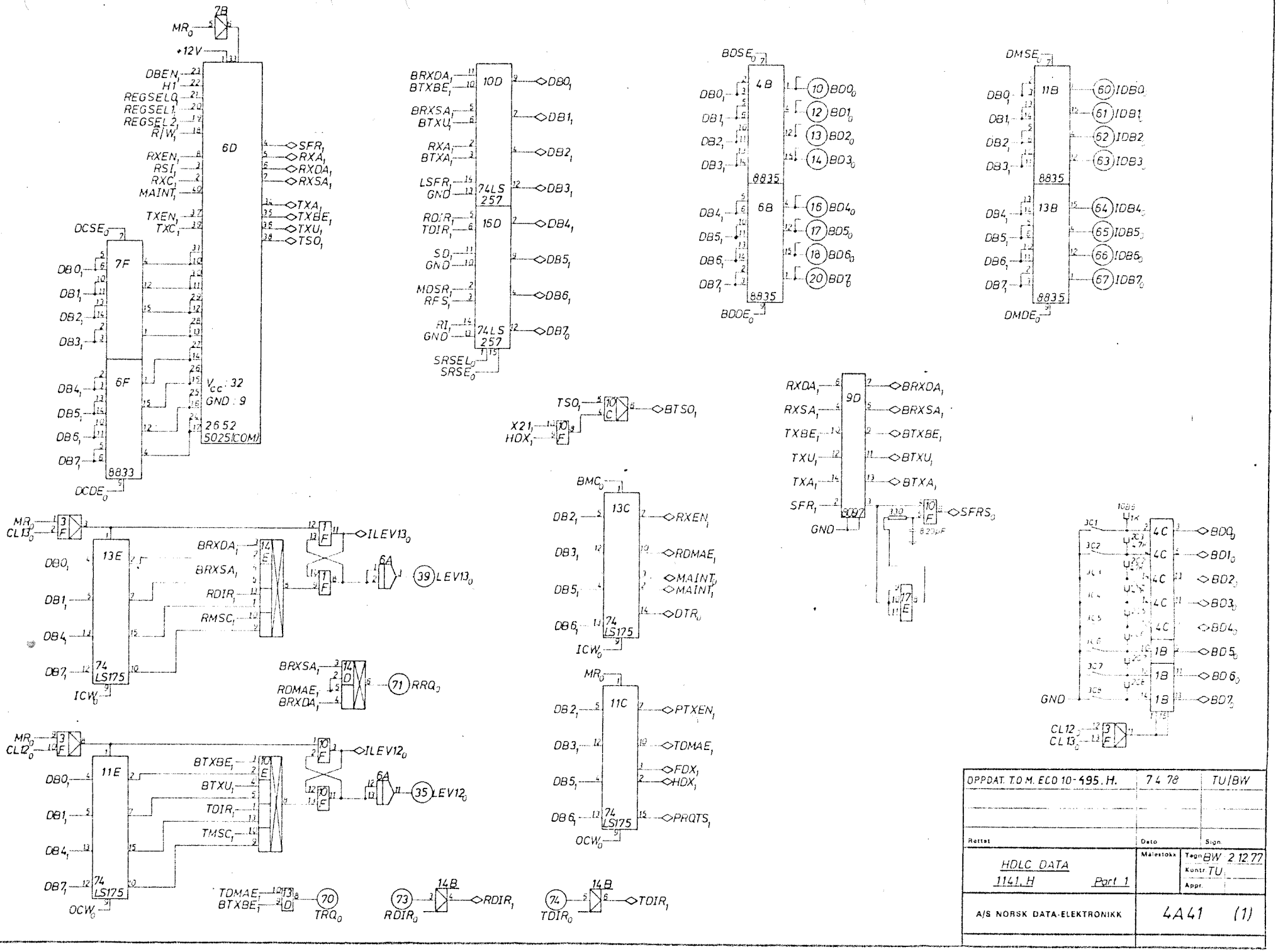

6D
7F
6F
8833
DCDE0
DCSE0
V_CC : 32
GND : 9
2652
5025/COM/
7B
MR0
+12V
H2
CE1
DBEN1
H1
REGSEL0
REGSEL1
REGSEL2
R/W1
RXEN1
RSI1
RXC1
MAINT1
TXEN1
TXC1
SFR1
RXA1
RXDA1
RXSA1
TXA1
TXBE1
TXU1
TSO1
10D
16D
74LS 257
SRSEL0
SRSE0
BRXDA1
BTXBE1
BRXSA1
BTXU1
RXA1
BTXA1
LSFR1
GND
ROIR1
TOIR1
SD1
MDSR1
RFS1
RI1
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
4B
6B
8835
BDOE0
BDOSE0
BD0
BD1
BD2
BD3
BD4
BD5
BD6
BD7
11B
13B
DMDE0
DMSE0
IDB0
IDB1
IDB2
IDB3
IDB4
IDB5
IDB6
IDB7
9D
BRXDA1
BRXSA1
BTXBE1
BTXU1
BTXA1
SFR1
GND
330
220pF
SFRS0
TSO1
X21
HDX1
BTSO1
13E
74 LS175
ICW0
MR0
CL13 0
BRXDA1
BRXSA1
RDIR1
RMSC1
DB0 1
DB1 1
DB4 1
DB7 1
ILEV13 0
LEV13 0
RDMAE1
BRXDA1
RRQ0
13C
BMC0
DB2 1
DB3 1
DB5 1
DB6 1
RXEN1
RDMAE1
MAINT1
DTR0
11E
CL12 0
BTXBE1
BTXU1
TDIR1
TMSC1
ILEV12 0
LEV12 0
TDMAE1
BTXBE1
TRQ0
11C
PTXEN1
TDMAE1
FDX1
HDX1
PRQTS1
14B
RDIR0
RDIR1
TDIR0
TDIR1
10B6
4C
1B
3C1
3C2
3C3
3C4
3C5
3C6
3C7
3C8
2C1
2C2
2C3
2C4
2C5
2C6
2C7
2C8
1K
4.7K
CL12 0
CL13 0
Rettet
Dato
Sign.
HDLC DATA
1141.D
Part 1
Målestokk
Tegn BW 2 12 77
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A41 (1)


9A
8160
9E
74 LS155
9B
7C
PROM 8578
9C
PROM 8578
74LS 257
4A
74LS157
14C
74 LS257
19D
74LS 175
19E
74 LS74
3B
74 LS155
3D
74 LS175
4D
74 LS175
14A
74 LS257
18F
VP12
15F
VP 5
16C1
16C2
DRQACT
IOXACT
IOXADR
IOXPEN
DPEN
DBEN
RWP
DCE
CONNECT
ICONNECT
BDDE
INPUT
IDACT
TINT
OUTIDENT
INGRANT
OUTGRANT
RSI
RXC
I/TXC
TXC
TXEN
RQTS
MAINT
DTR
TDA
CLK
RMSC
TMSC
LSFR
MDSR
MTXC
LDRQ
IDCEN2
IOXCEN
DMSE
DMDE
OCDE
DCSE
BDSE
INPEN
SRSE
SRSEL
REGRE
REGSEL2
R/W
REGWE
ISR
OSR
ICW
H1
OCW
IOXE
DRQ
BA0
BA1
BA2
BA3
BA4
BA5
BA6
BA7
BA8
BA9
ICB0
ICB1
ICB2
ICB3
BMC
MCL
DB5
MR
SD
RFS
RI
DSR
BTSO
C/TXC
EN12
EN13
CL12
CL13
ILEV12
ILEV13
INT
H2
+12V
-12V
+5V
56pF
220pF
2.2nF
19,2kc/s
307,2kc/s
CCITT V-35
Rettet
Dato
Sign.
HDLC DATA
1141.D
Part 2
Målestokk
Tegn BW 2.1277
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A41 (2)

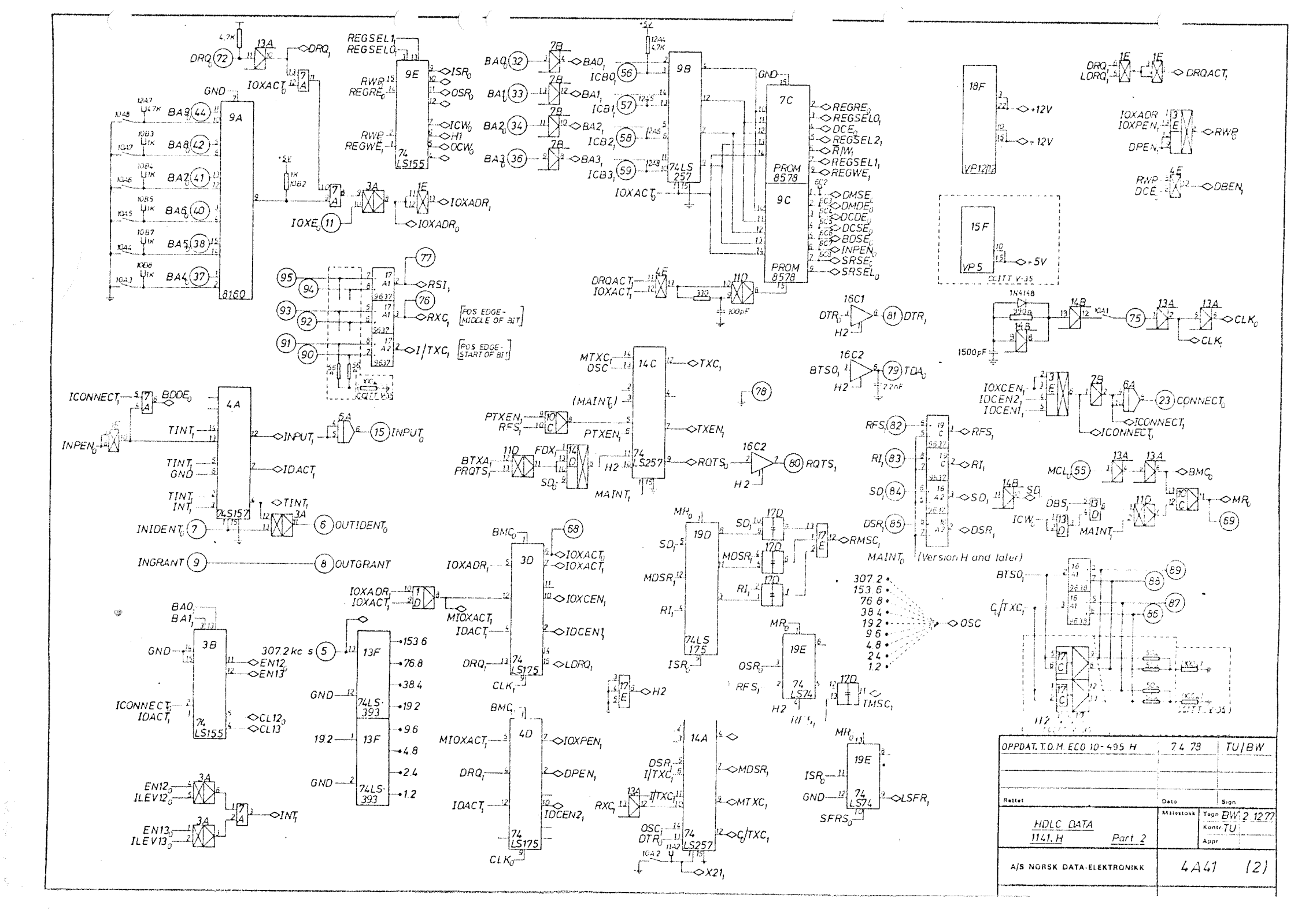


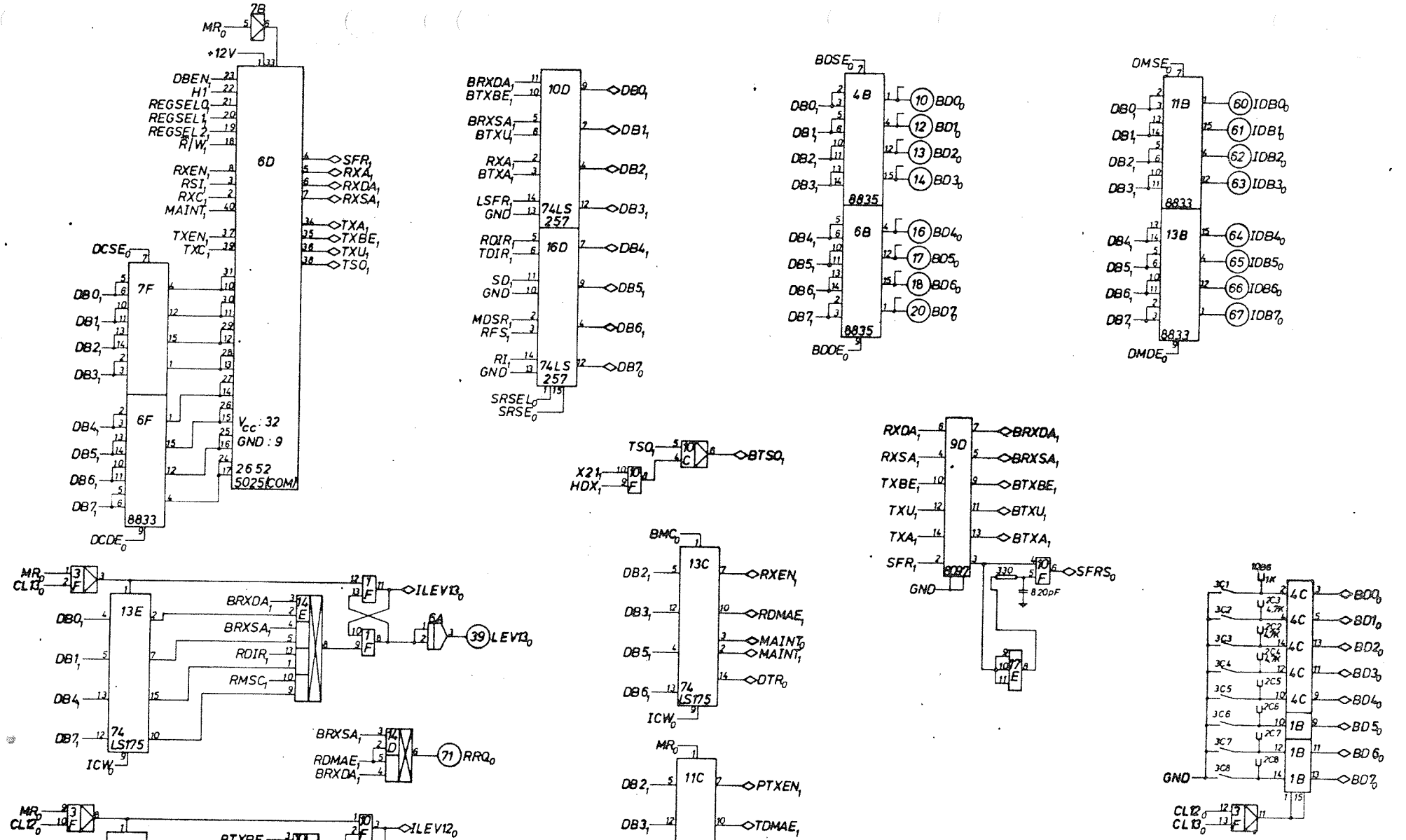
OPPDAT. T.O.M. ECO 10-484. G.	22.2.78	TU/Emil
Rettet	Dato	Sign
HDLC DATA 1141.G Part 1	Målestokk	Tegn BW 2.12.77 Kontr. TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A41 (1)	


9A
8160
9E
74 LS155
7C
9C
PROM 8578
9B
74LS 257
4A
74 LS157
14C
74 LS257
3B
74 LS155
3D
74 LS175
4D
74 LS175
19D
74LS 175
19E
74 LS74
14A
74 LS257
18F
VP12/12
15F
VP5
CCITT V-35
DRQ0
IOXACT0
REGSEL1
REGSEL0
RWP
REGRE0
REGWE1
ISR0
OSR0
ICW0
H1
OCW0
BA0 32
BA1 33
BA2 34
BA3 36
ICB0 56
ICB1 57
ICB2 58
ICB3 59
IOXACT1
REGRE0
REGSEL0
DCE0
REGSEL2
R/W1
REGSEL1
REGWE1
DMSE0
DMDE0
DCDE0
DCSE0
BOSE0
INPEN0
SRSE0
SRSEL0
DRQACT1
IOXACT1
IOXE0 11
IOXADR1
IOXADR0
BA0 44
BA1 42
BA2 41
BA3 40
BA4 37
BA5 38
BA6 40
BA7 41
BA8 42
BA9 44
RSI1
RXC1
I/TXC1
POS EDGE - MIDDLE OF BIT
POS EDGE - START OF BIT
13,2kc/s
307,2kc/s
MTXC1
OSC
TXC1
TXEN1
PTXEN1
RFS1
RQTS0
MAINT1
BTXA1
PRQTS1
FOX1
SQ0
H2
ICONNECT1
BODE0
INPEN0
TINT1
GND
INT1
IDACT1
INPUT1
INPUT0
INIDENT0
OUTIDENT0
INGRANT 9
OUTGRANT 8
BMC0
IOXADR1
IOXACT1
MIOXACT1
IDACT1
DRQ1
CLK1
IOXCEN1
IDCEN1
LDRQ1
IOXACT0
IOXACT1
MIOXACT1
DRQ1
IDACT1
CLK0
IDCEN2
DPEN1
MR0
SQ1
MDSR1
RI1
ISR0
RMSC1
OSR0
RFS1
TMSC1
H2
DSR1
I/TXC1
RXC1
OSC1
DTR0
X21
MTXC1
C/TXC1
ISR0
GND
LSFR1
SFRS0
EN12
EN13
CL12
CL13
ICONNECT0
IDACT1
EN12
ILEV12
EN13
ILEV13
INT1
DRQ0
IOXACT0
DRQACT1
IOXACT1
330
100pF
DTR0
H2
DTR1
BTSO1
TDA0
2,2nF
1N4148
200n
1500pF
CLK0
CLK1
RFS1
RI1
SD1
DSR1
IOXCEN1
IDCEN2
IDCEN1
ICONNECT1
ICONNECT0
CONNECT0
MCL0
BMC0
DB51
ICW0
MAINT1
MR0
BTSO1
C/TXC1
50Ω
100Ω
CCITT V-35
DRQ0
LDRQ1
DRQACT1
IOXADR1
IOXPEN1
RWP
DPEN1
DCE0
DBEN1
+12V
+5V
OPPDAT. T.O.M. ECO 10-484.G.
22.2.78
TU/Emil
Rettet
Dato
Sign.
HDLC DATA
1141.G
Part 2
Målestokk
Tegn BW 2.1277
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A41 (2)

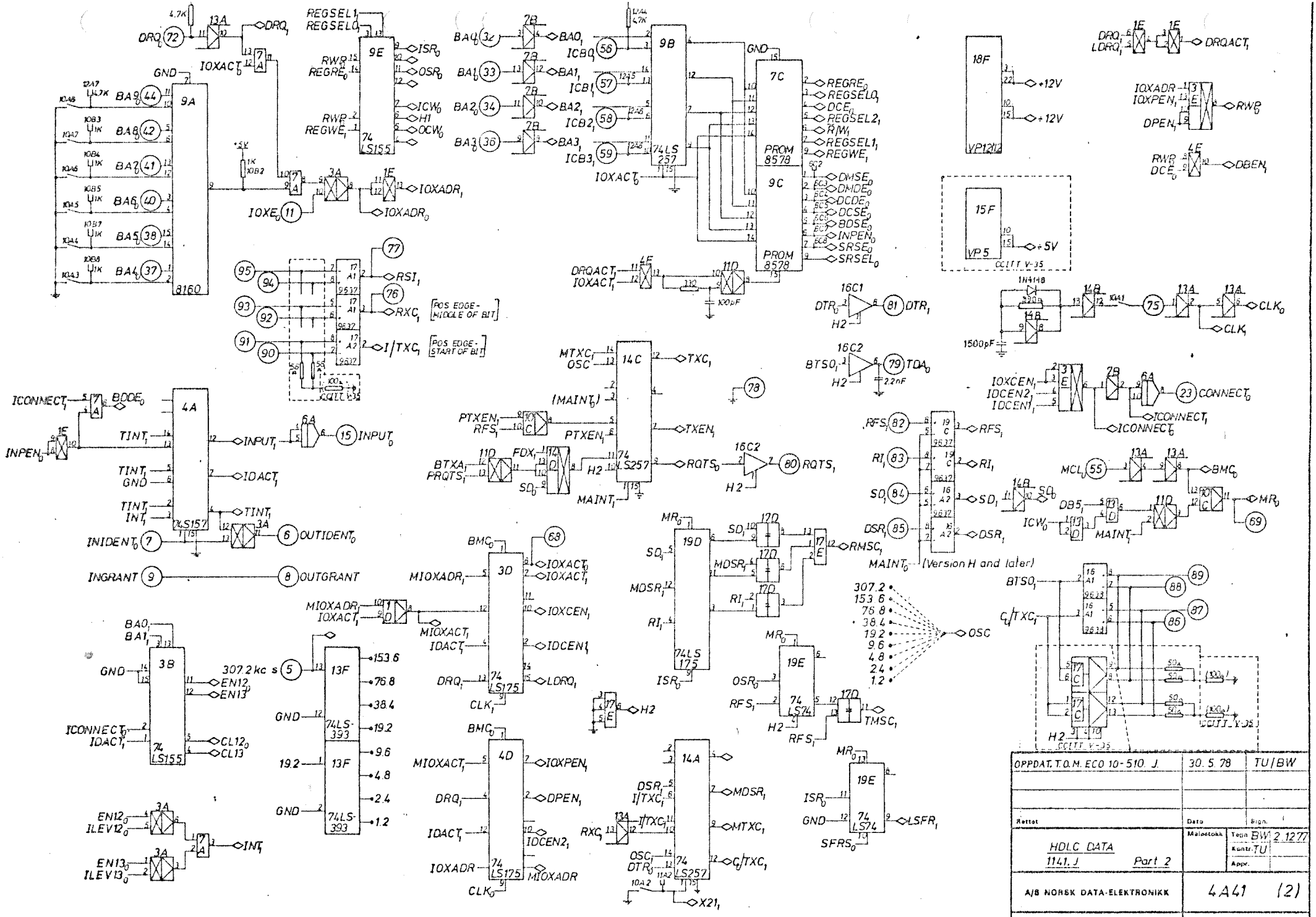


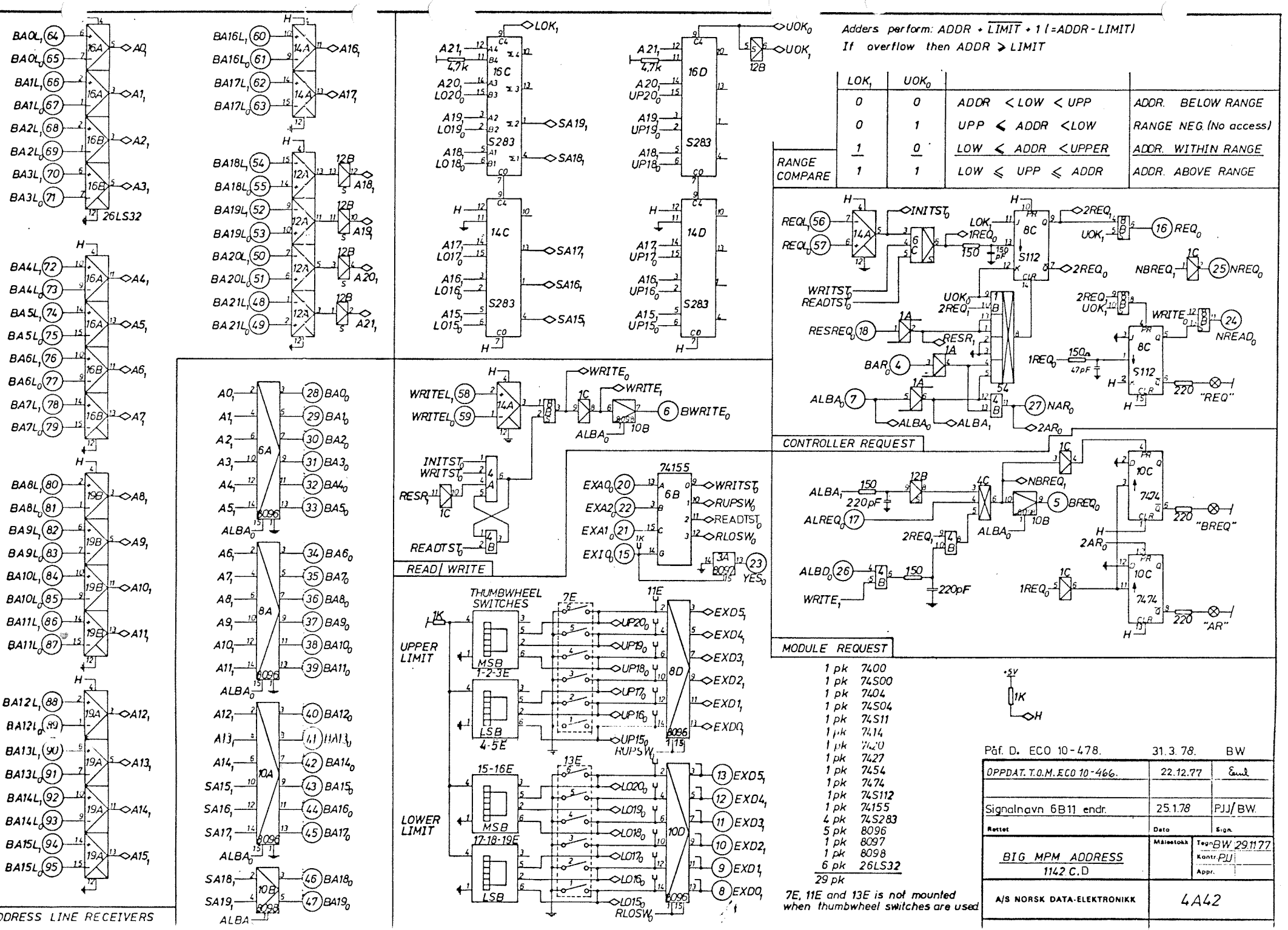
OPPDAT. T.O.M. ECO 10-495.H.	7 4 78	TU/BW
Rettet	Dato	Sign.
HDLC DATA 1141.H Part 1	Målestokk	Tegn BW 2 12 77 Kontr TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A41 (1)	

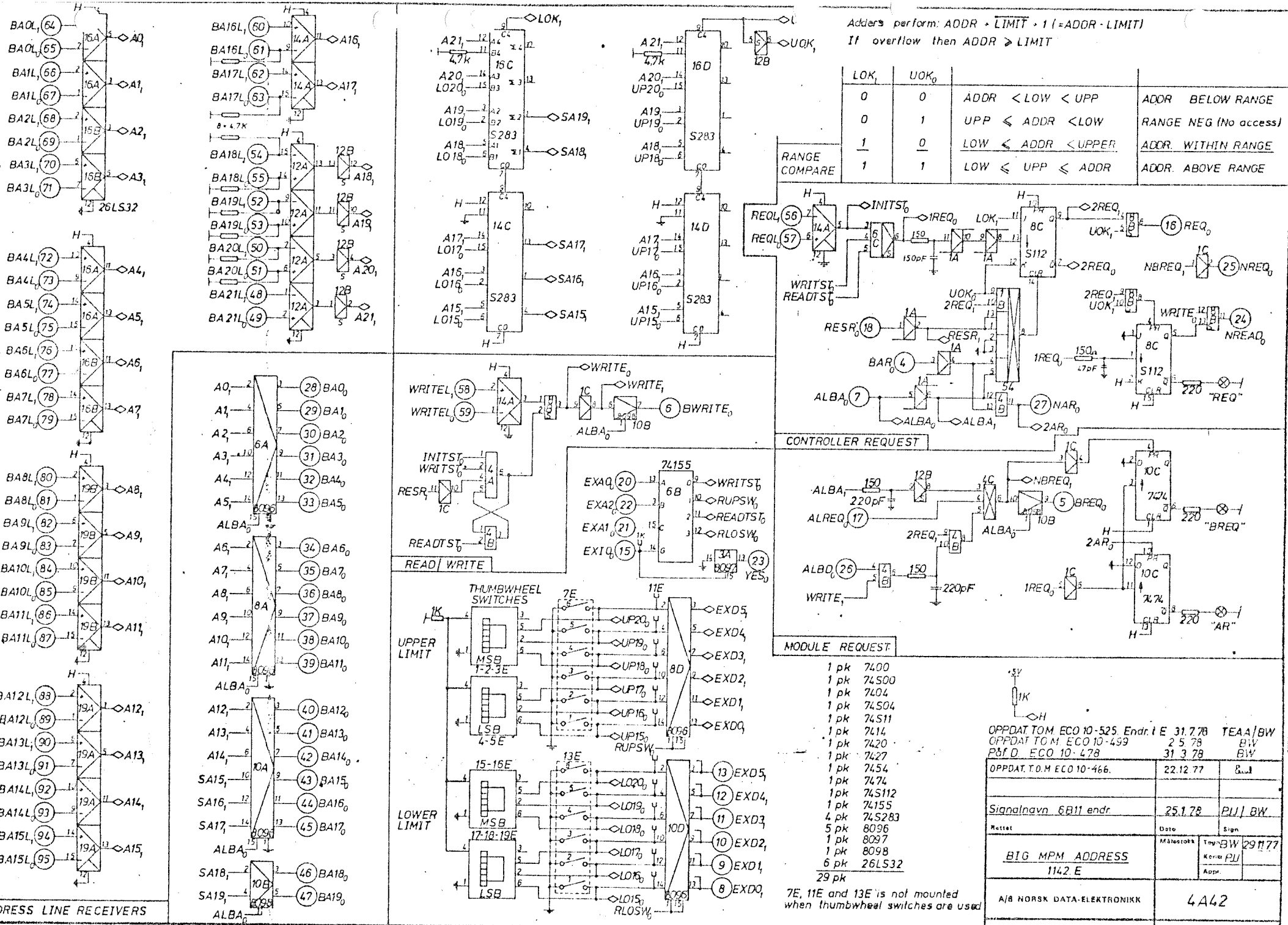

DRQ0 (72)
IOXACT0
DRQ1
REGSEL1
REGSEL0
RWR
REGRE0
RWP
REGWE1
9E
74 LS155
ISR0
OSR0
ICW0
H1
OCW0
GND
9A
8160
BA3 (44)
BA8 (42)
BA7 (41)
BA6 (40)
BA5 (38)
BA4 (37)
IOXE0 (11)
3A
1E
IOXADR1
IOXADR0
BA0 (32)
BA1 (33)
BA2 (34)
BA3 (36)
7B
BA0
BA1
BA2
BA3
ICB0 (56)
ICB1 (57)
ICB2 (58)
ICB3 (59)
9B
74LS 257
IOXACT0
7C
PROM 8578
9C
PROM 8578
REGRE0
REGSEL0
DCE0
REGSEL2
R/W1
REGSEL1
REGWE1
DMSE0
DMDE0
DCDE0
DCSE0
BDSE0
INPEN0
SRSE0
SRSEL0
18F
VP12/7
+12V
-12V
15F
VP 5
-5V
CCITT V.35
DRQ1
LDRQ1
1E
DRQACT1
IOXADR
IOXPEN1
DPEN1
RWP0
DCE0
4E
DBEN1
DRQACT1
IOXACT1
4E
110
330
100pF
95
94
93
92
91
90
77
76
RSI1
RXC1
I/TXC1
[POS EDGE - MIDDLE OF BIT]
[POS EDGE - START OF BIT]
96S37
17 A1
17 A2
56
22
XC
CCITT V.35
16C1
DTR0
H2
81
DTR1
16C2
BTS01
79
TDA0
22nF
1N4148
1500pF
14B
13A
75
CLK0
CLK1
ICONNECT1
IDACT1
BODE0
4A
INPEN0
TINT1
GND
INT1
74S157
INPUT1
IDACT1
TINT1
INIDENT0 (7)
6A
15
INPUT0
6
OUTIDENT0
INGRANT (9)
8
OUTGRANT
MTXC1
OSC1
(MAINT0)
14C
74 LS257
TXC1
TXEN1
RQTS0
MAINT1
PTXEN1
RFS1
BTXA1
PRQTS1
FDX1
SQ0
11D
78
16C2
H2
80
RQTS1
IOXCEN1
IDCEN2
IDCEN1
7B
6A
23
CONNECT0
ICONNECT1
ICONNECT0
RFS (82)
RI1 (83)
SD1 (84)
DSR1 (85)
19 C
16 A2
RFS1
RI1
SD1
DSR1
MAINT0
(Version H and later)
MCL0 (55)
13A
BMC0
OBS1
ICW0
14B
SQ
MAINT1
11D
MR0
69
BMC0
68
30
74 LS175
IOXACT0
IOXACT1
IOXCEN1
IDCEN1
LDRQ1
CLK1
IOXADR1
IOXACT1
1D
MIOXACT1
IDACT1
DRQ1
MR0
19D
74LS 175
SD1
MDSR1
RI1
ISR0
17D
RMSC1
TMSC1
H2
19E
74 LS74
OSR0
RFS1
307.2
153.6
76.8
38.4
19.2
9.6
4.8
2.4
1.2
OSC
BTSO1
C/TXC1
88
87
86
85
112
BA0
BA1
GND
3B
74 LS155
307.2kc s (5)
EN12
EN13
ICONNECT1
IDACT1
CL12
CL13
13F
74LS 393
153.6
76.8
38.4
19.2
9.6
4.8
2.4
1.2
EN12
ILEV12
EN13
ILEV13
3A
INT1
BMC1
40
74 LS175
MIOXACT1
DRQ1
IDACT1
IOXPEN1
DPEN1
IDCEN2
CLK0
14A
DSR1
I/TXC1
RXC1
MDSR1
MTXC1
OSC1
DTR0
74 LS257
C/TXC1
X21
19E
74 LS74
ISR0
GND
LSFR1
SFRS0
OPPDAT. T.O.M. ECO 10-495 H
7 4 78
TU/BW
Rettet
Dato
Sign
HDLC DATA
1141. H
Part 2
Tegn BW 2 1277
Kontr TU
Appr
A/S NORSK DATA-ELEKTRONIKK
4A41 (2)

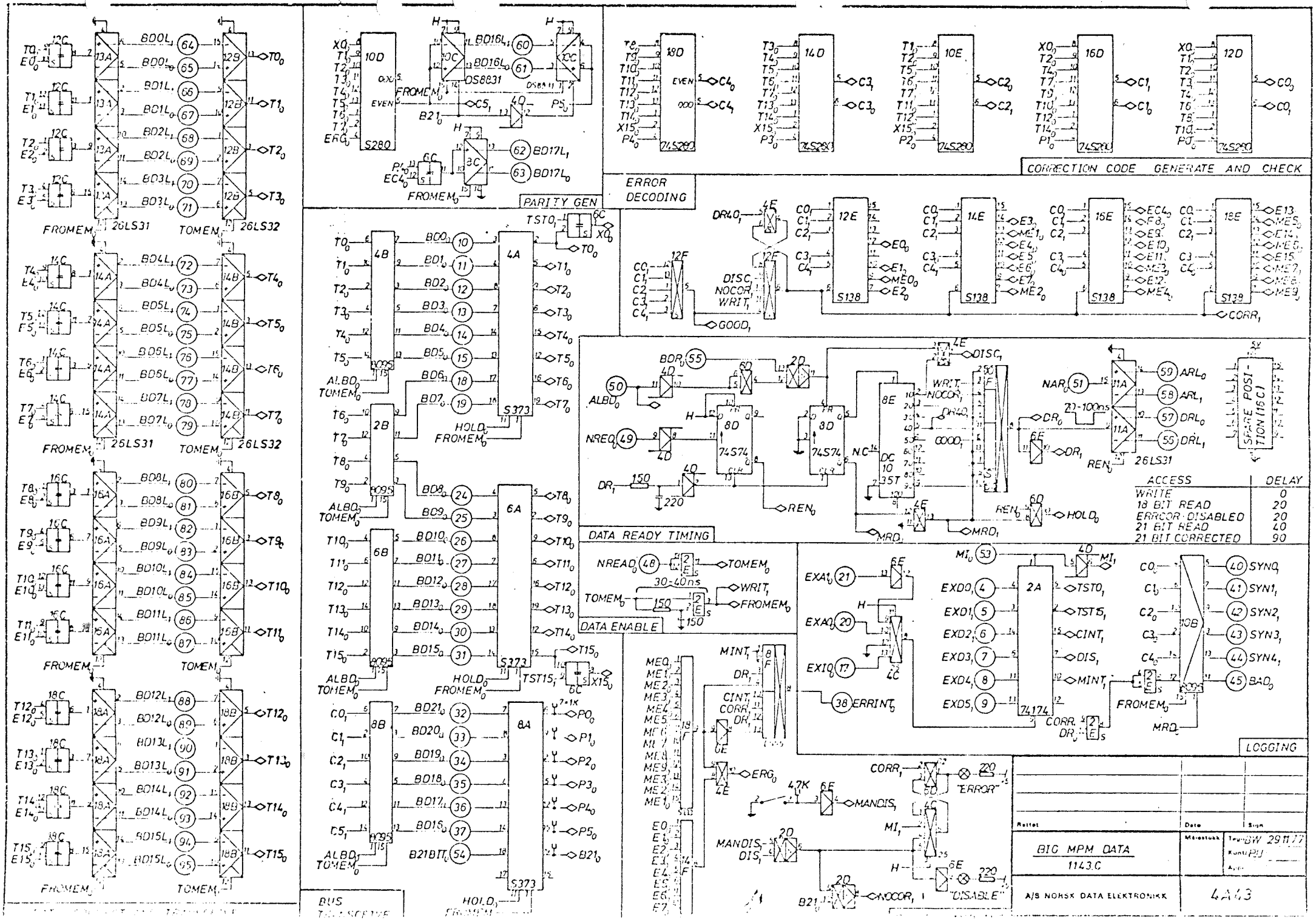


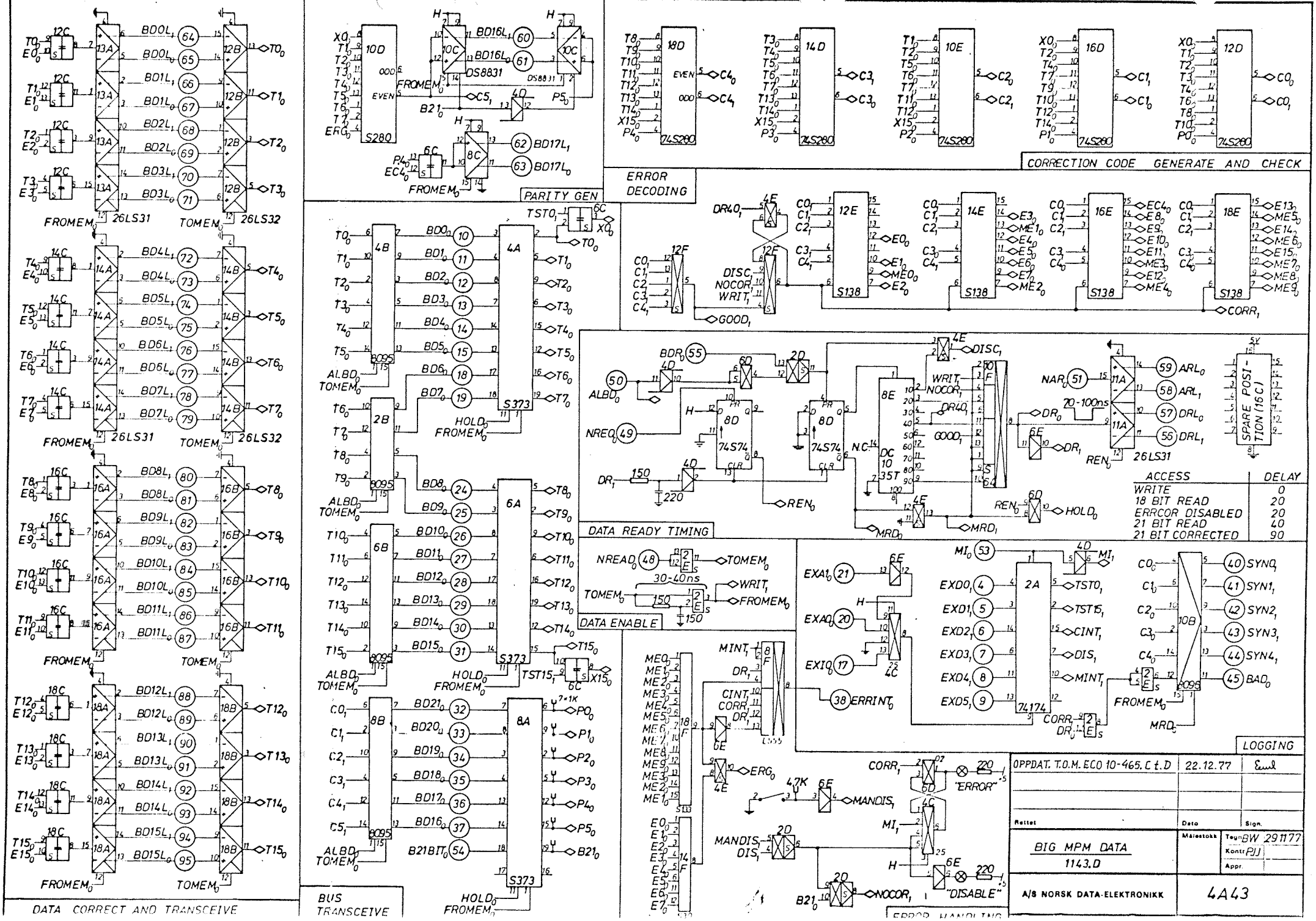
OPPDAT. I.O.M. ECO 10-510. J.	30 5.78	TU/BW
Rettet	Dato	Sign.
HDLC DATA 1141.J Part 1	Målestokk	Tegn BW 2.12.77 Kontr. TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A41 (1)	

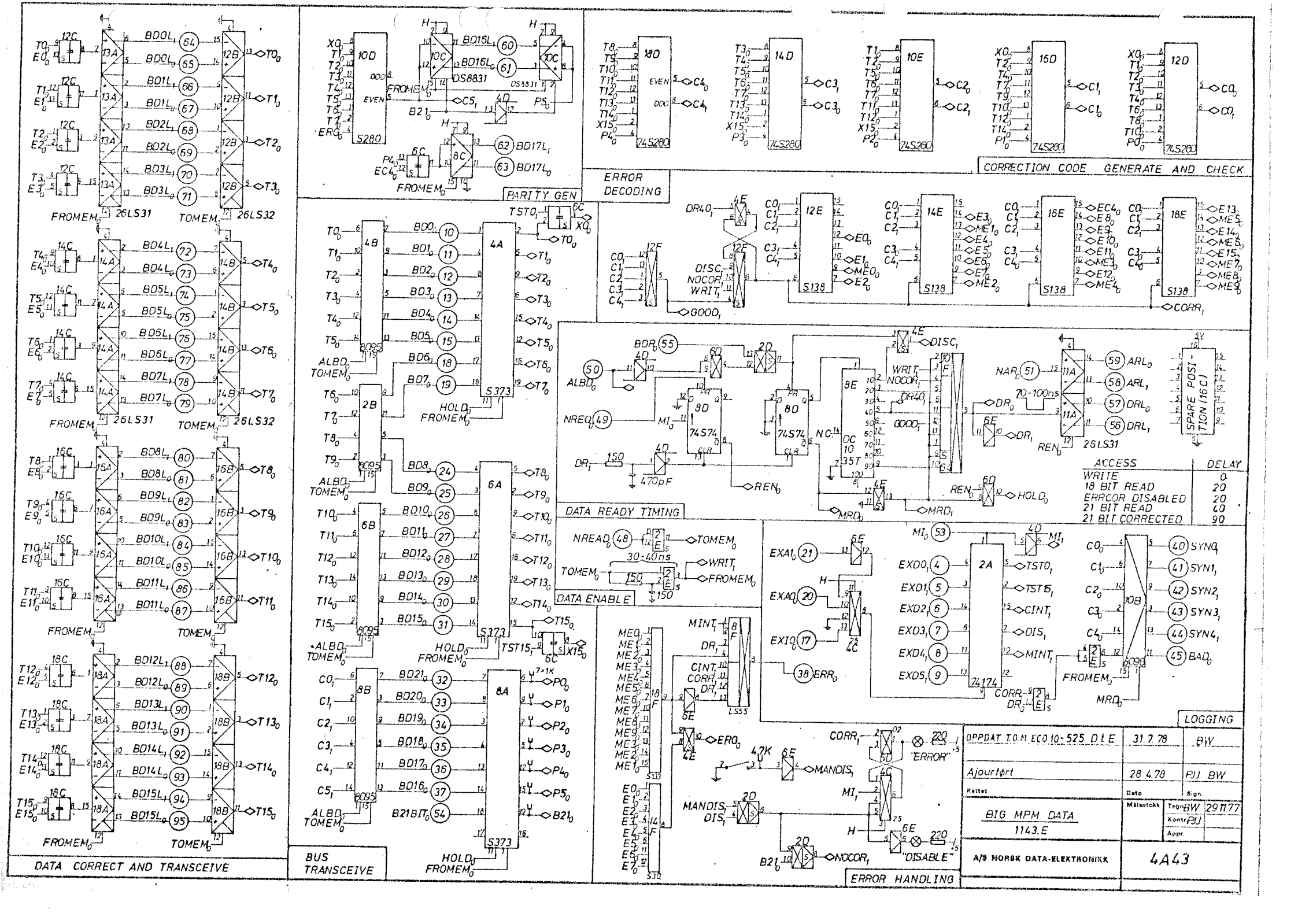

9A
8160
9E
74 LS155
7C
9C
PROM 8578
9B
74LS 257
18F
VP12
15F
VP5
16C1
16C2
14C
74 LS257
4A
74LS157
3D
4D
74 LS175
19D
74LS 175
19E
74 LS74
14A
3B
74 LS155
13F
74LS 393
3A
14B
1N4148
1500pF
2.2nF
100pF
(Version H and later)
307.2 kc s
153.6
76.8
38.4
19.2
9.6
4.8
2.4
1.2
OSC
OPPDAT. T.O.M. ECO 10-510. J.
30. 5. 78
TU/BW
Rettet
Dato
Sign.
HDLC DATA
1141. J
Part 2
Målestokk
Tegn BW 2 1277
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A41 (2)

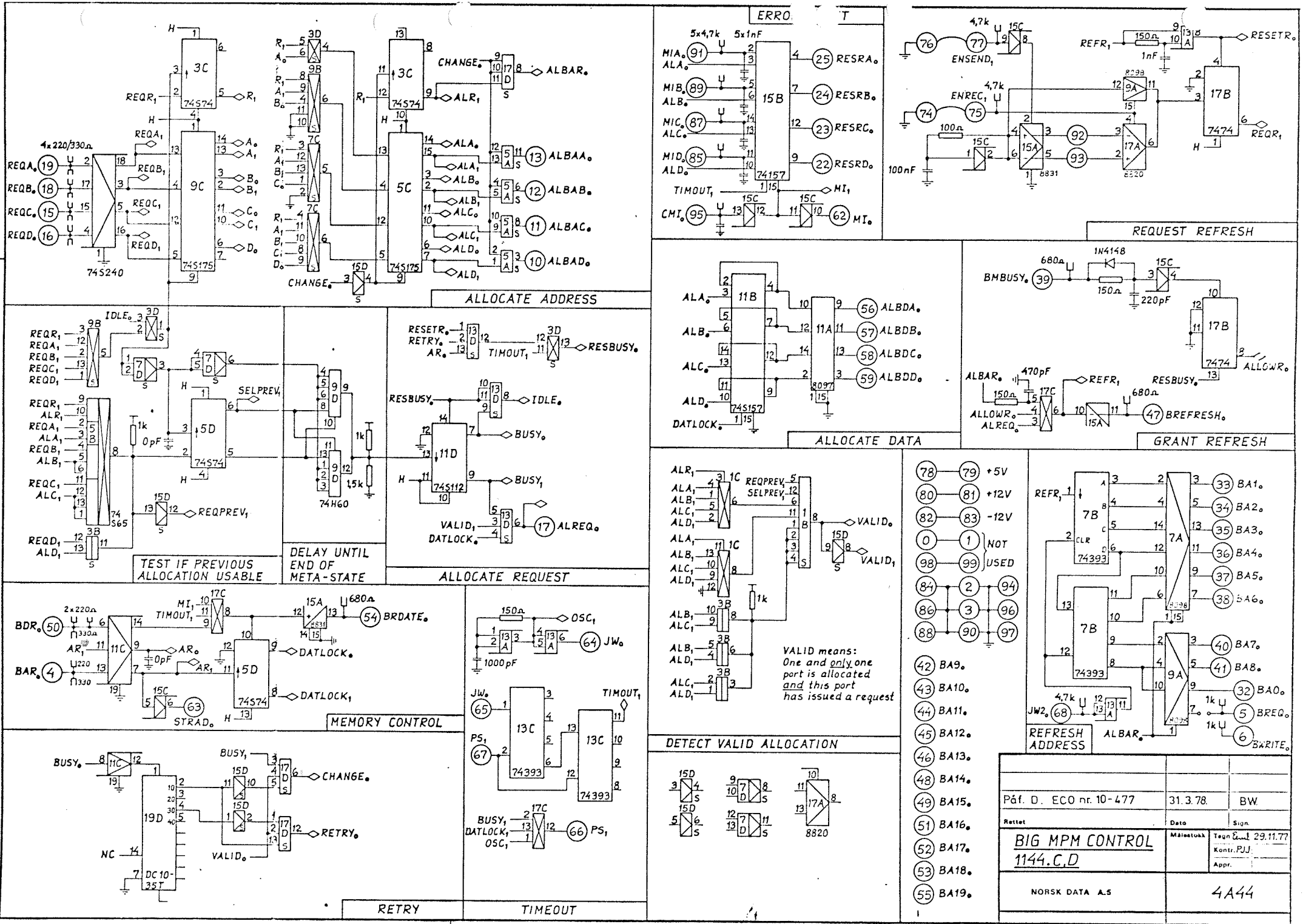

Adders perform: ADDR + $\overline{LIMIT}$ + 1 (=ADDR - LIMIT)
If overflow then ADDR ≥ LIMIT
LOK1 UOK0 UOK1
0 0 ADDR < LOW < UPP ADDR. BELOW RANGE
0 1 UPP ≤ ADDR < LOW RANGE NEG. (No access)
1 0 LOW ≤ ADDR < UPPER ADDR. WITHIN RANGE
RANGE COMPARE
1 1 LOW ≤ UPP ≤ ADDR ADDR. ABOVE RANGE
26LS32
ADDRESS LINE RECEIVERS
CONTROLLER REQUEST
MODULE REQUEST
READ/ WRITE
THUMBWHEEL SWITCHES
UPPER LIMIT
LOWER LIMIT
7E and 11E and 13E is not mounted when thumbwheel switches are used
1 pk 7400
1 pk 74S00
1 pk 7404
1 pk 74S04
1 pk 74S11
1 pk 7414
1 pk 7420
1 pk 7427
1 pk 7454
1 pk 7474
1 pk 74S112
1 pk 74155
4 pk 74S283
5 pk 8096
1 pk 8097
1 pk 8098
6 pk 26LS32
29 pk
Påf. D. ECO 10-478. 31.3.78. BW
OPPDAT. T.O.M. ECO 10-466. 22.12.77 Emil
Signalnavn 6B11 endr. 25.1.78 PJJ/BW
Rettet Dato Sign.
BIG MPM ADDRESS 1142 C.D
Målestokk
Tegn BW 29.11.77
Kontr. PJJ
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A42

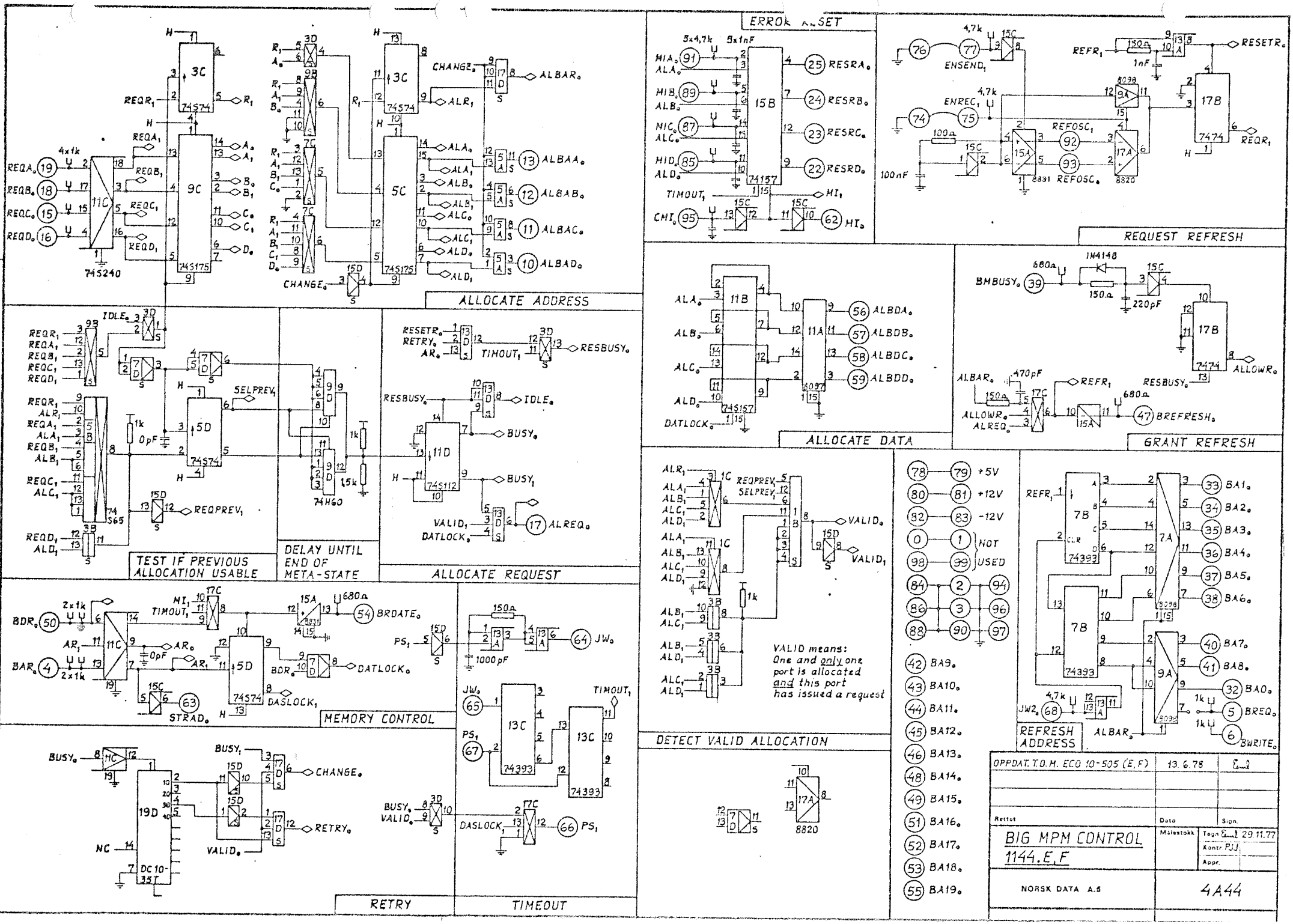

Adders perform: ADDR + $\overline{LIMIT}$ + 1 (=ADDR - LIMIT)
If overflow then ADDR ≥ LIMIT
LOK1 UOK1
RANGE COMPARE
LOK1 UOK0
0 0 ADDR < LOW < UPP — ADDR BELOW RANGE
0 1 UPP ≤ ADDR < LOW — RANGE NEG (No access)
1 0 LOW ≤ ADDR < UPPER — ADDR WITHIN RANGE
1 1 LOW ≤ UPP ≤ ADDR — ADDR ABOVE RANGE
ADDRESS LINE RECEIVERS
READ/WRITE
CONTROLLER REQUEST
MODULE REQUEST
THUMBWHEEL SWITCHES
UPPER LIMIT
LOWER LIMIT
1 pk 7400
1 pk 7500
1 pk 7404
1 pk 7504
1 pk 7511
1 pk 7414
1 pk 7420
1 pk 7427
1 pk 7454
1 pk 7474
1 pk 74S112
1 pk 74155
4 pk 74S283
5 pk 8096
1 pk 8097
1 pk 8098
6 pk 26LS32
29 pk
7E, 11E and 13E is not mounted when thumbwheel switches are used
OPPDAT TOM ECO 10-525. Endr i E 31.7.78 TEAA/BW
OPPDAT TO M. ECO 10-499 2 5 78 BW
P&f D ECO 10-478 31 3 78 BW
OPPDAT. T.O.M ECO 10-466. 22.12.77
Signalnavn 6B11 endr 25.1.78 PLJ/BW
Rettet Dato Sign
BIG MPM ADDRESS 1142 E
Målestokk
Tegn BW 29/11/77
Konstr PLJ
Godkj
A/S NORSK DATA-ELEKTRONIKK
4A42

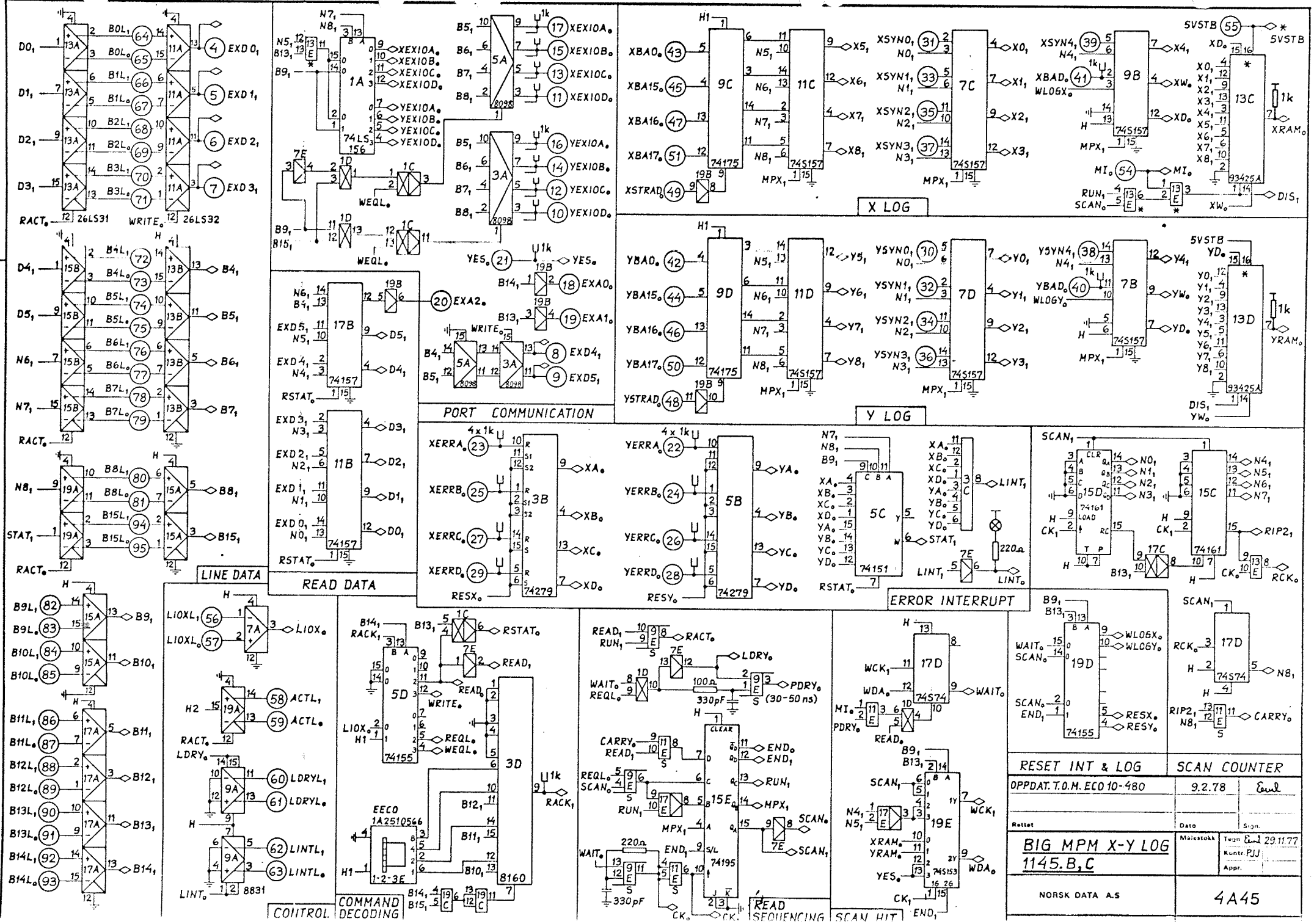

CORRECTION CODE GENERATE AND CHECK
ERROR DECODING
PARITY GEN
DATA READY TIMING
DATA ENABLE
BUS
LOGGING
26LS31
26LS32
FROMEM
TOMEM
S280
S373
S138
74S280
74S74
74174
ACCESS
DELAY
WRITE 0
18 BIT READ 20
ERRCOR DISABLED 20
21 BIT READ 40
21 BIT CORRECTED 90
SPARE POSITION (16C)
"ERROR"
"DISABLE"
BIG MPM DATA
1143.C
A/S NORSK DATA ELEKTRONIKK
4A43

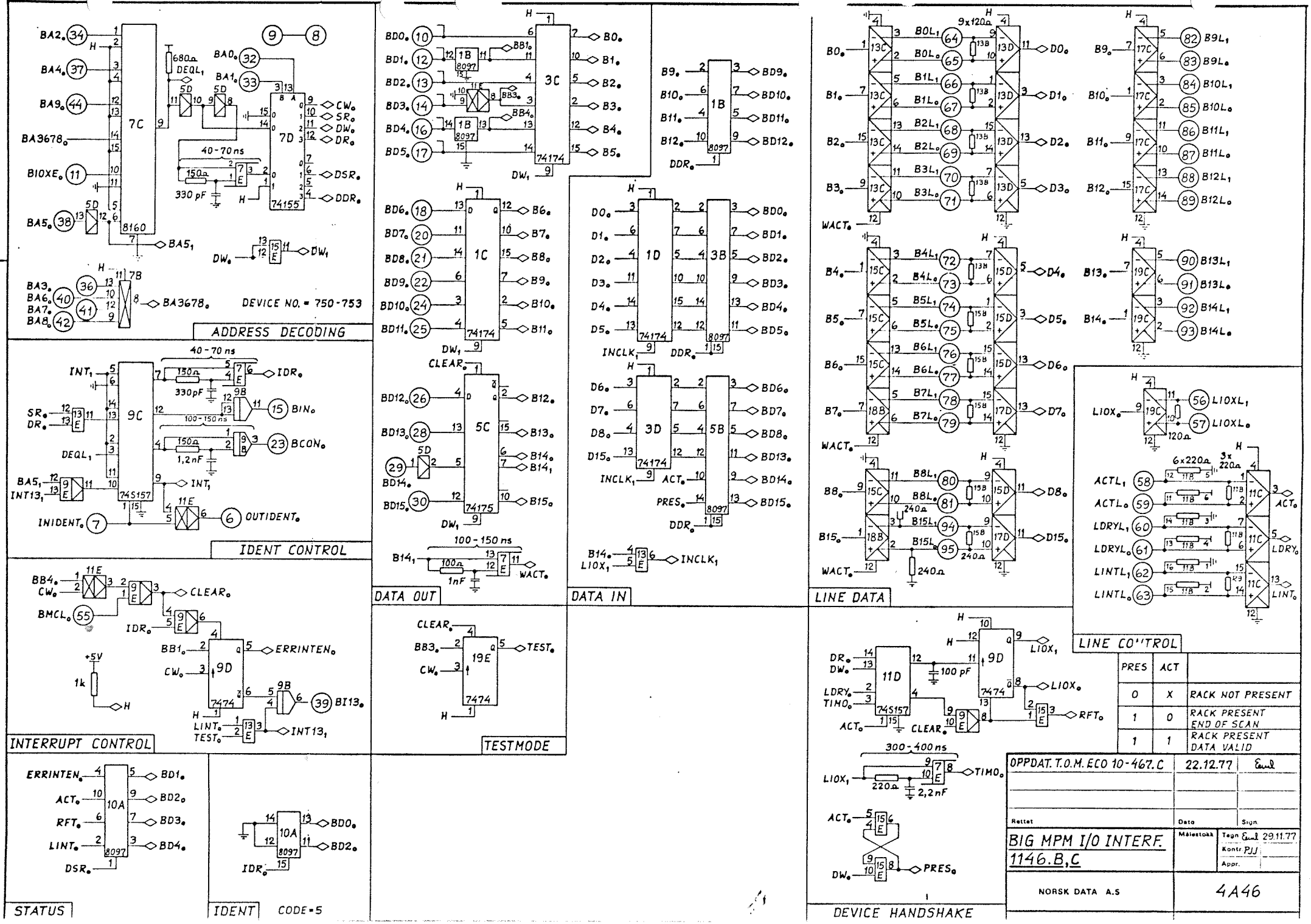

DATA CORRECT AND TRANSCEIVE
BUS TRANSCEIVE
PARITY GEN
ERROR DECODING
CORRECTION CODE GENERATE AND CHECK
DATA READY TIMING
DATA ENABLE
ERROR HANDLING
LOGGING
FROMEM
TOMEM
26LS31
26LS32
S280
S373
S138
74S280
74S74
74174
ACCESS
DELAY
WRITE 0
18 BIT READ 20
ERRCOR DISABLED 20
21 BIT READ 40
21 BIT CORRECTED 90
SPARE POSITION (16 C)
"ERROR"
"DISABLE"
OPPDAT. T.O.M. ECO 10-465. C t.D
22.12.77
BIG MPM DATA
1143.D
A/S NORSK DATA-ELEKTRONIKK
4A43

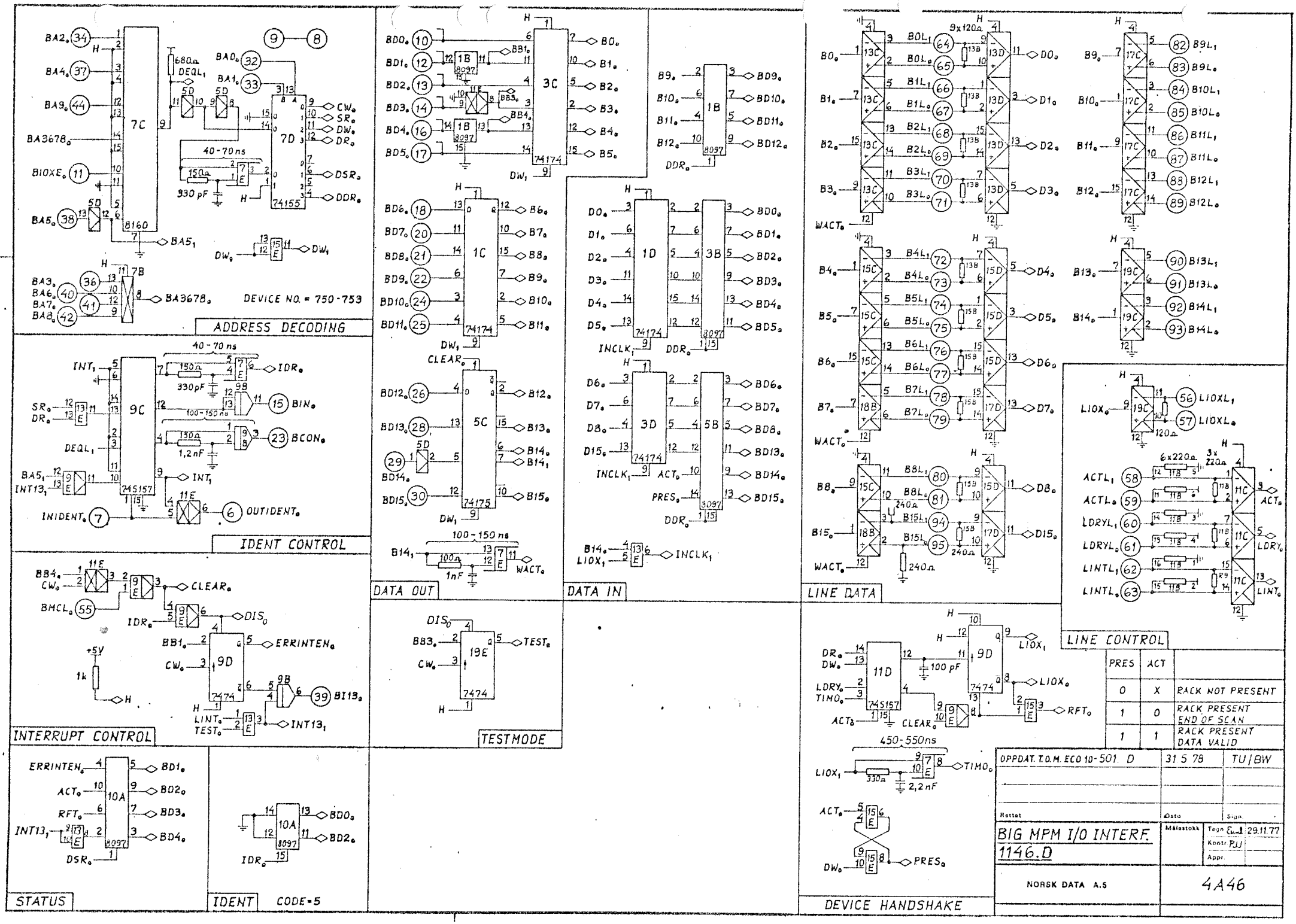

DATA CORRECT AND TRANSCEIVE
BUS TRANSCEIVE
PARITY GEN
ERROR DECODING
CORRECTION CODE GENERATE AND CHECK
DATA READY TIMING
DATA ENABLE
ERROR HANDLING
LOGGING
SPARE POST-TION (16 C)
ACCESS
DELAY
WRITE 0
18 BIT READ 20
ERRCOR DISABLED 20
21 BIT READ 40
21 BIT CORRECTED 90
OPPDAT. T.O.M. ECO 10-525 D.I.E
31.7.78
BW
Ajourført
28.4.78
PJJ BW
BIG MPM DATA
1143.E
A/S NORSK DATA-ELEKTRONIKK
4A43

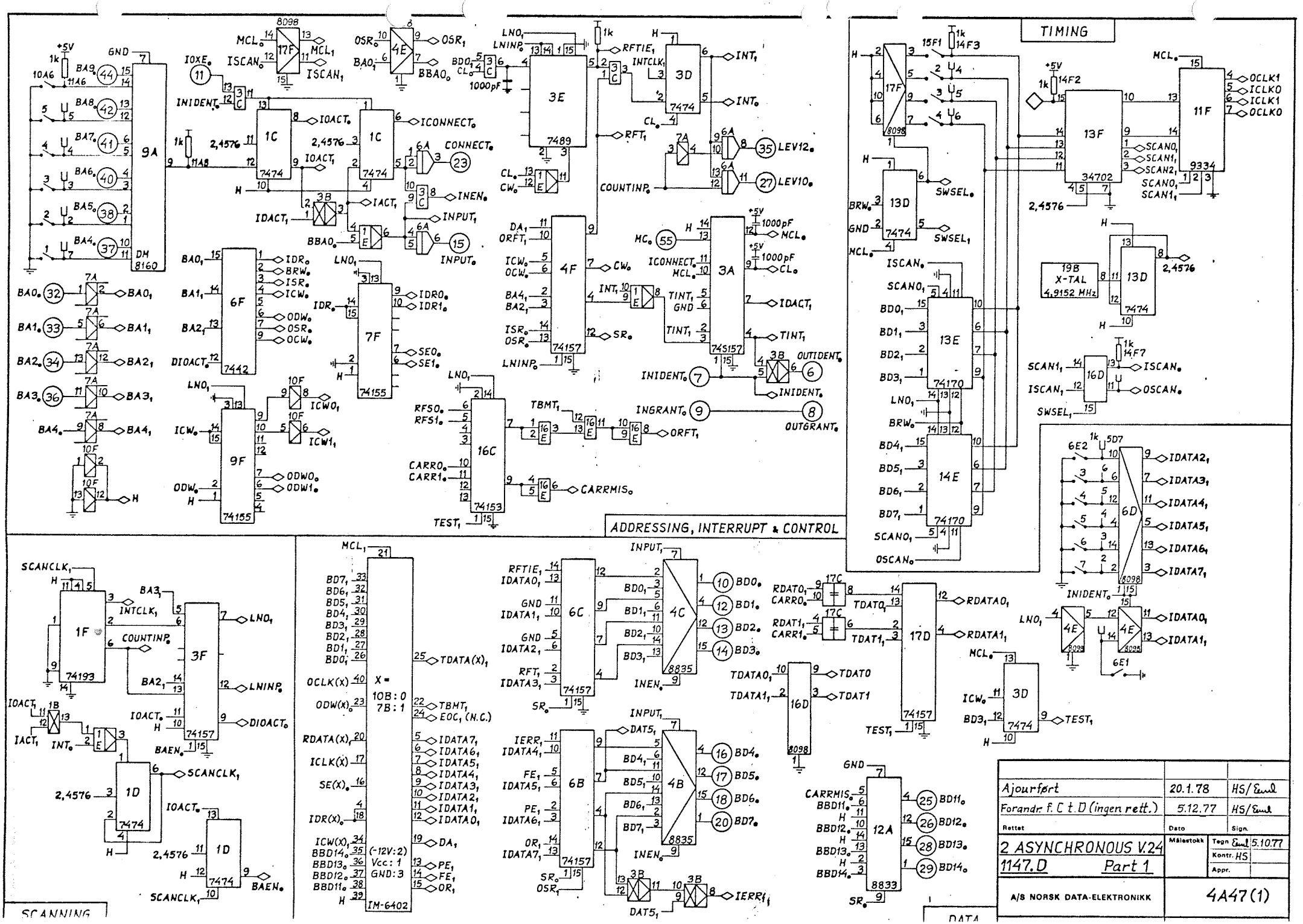

ALLOCATE ADDRESS
ERRO T
REQUEST REFRESH
TEST IF PREVIOUS ALLOCATION USABLE
DELAY UNTIL END OF META-STATE
ALLOCATE REQUEST
ALLOCATE DATA
GRANT REFRESH
MEMORY CONTROL
DETECT VALID ALLOCATION
REFRESH ADDRESS
RETRY
TIMEOUT
VALID means: One and only one port is allocated and this port has issued a request
BIG MPM CONTROL
1144.C,D
Påf. D. ECO nr. 10-477
31.3.78
BW
NORSK DATA A.S
4A44

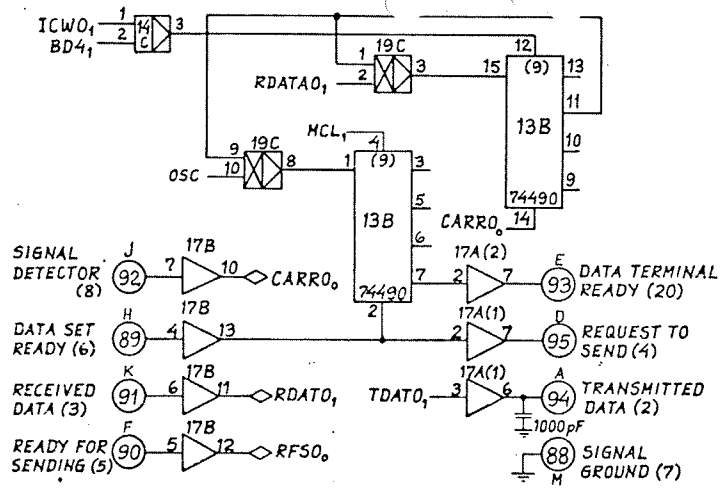

ALLOCATE ADDRESS
ERROR RESET
REQUEST REFRESH
ALLOCATE DATA
GRANT REFRESH
TEST IF PREVIOUS ALLOCATION USABLE
DELAY UNTIL END OF META-STATE
ALLOCATE REQUEST
MEMORY CONTROL
DETECT VALID ALLOCATION
REFRESH ADDRESS
RETRY
TIMEOUT
VALID means: One and only one port is allocated and this port has issued a request
OPPDAT. T.O.M. ECO 10-505 (E,F)
13.6.78
BIG MPM CONTROL 1144.E,F
NORSK DATA A.S
4A44

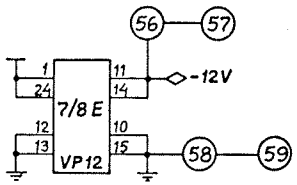

LINE DATA
READ DATA
PORT COMMUNICATION
X LOG
Y LOG
ERROR INTERRUPT
CONTROL
COMMAND DECODING
READ SEQUENCING
SCAN HIT
RESET INT & LOG
SCAN COUNTER
OPPDAT. T.O.M. ECO 10-480
9.2.78
Rettet
Dato
Sign.
BIG MPM X-Y LOG
1145.B,C
Malestokk
Tegn. 29.11.77
Kontr. PJJ
Appr.
NORSK DATA A.S
4A45

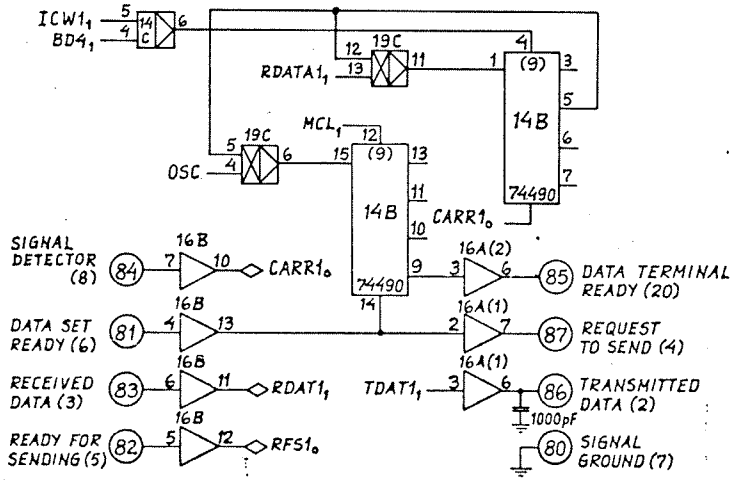

ADDRESS DECODING
DEVICE NO. = 750-753
IDENT CONTROL
INTERRUPT CONTROL
STATUS
IDENT
CODE=5
DATA OUT
DATA IN
TESTMODE
LINE DATA
LINE CONTROL
DEVICE HANDSHAKE
PRES | ACT
0 | X | RACK NOT PRESENT
1 | 0 | RACK PRESENT END OF SCAN
1 | 1 | RACK PRESENT DATA VALID
OPPDAT. T.O.M. ECO 10-467.C
22.12.77
BIG MPM I/O INTERF.
1146.B,C
NORSK DATA A.S
4A46

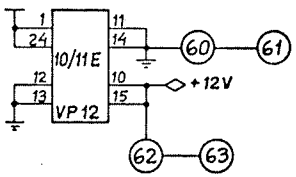

ADDRESS DECODING
DEVICE NO. = 750-753
IDENT CONTROL
DATA OUT
DATA IN
LINE DATA
LINE CONTROL
INTERRUPT CONTROL
TESTMODE
STATUS
IDENT CODE=5
DEVICE HANDSHAKE
PRES ACT
0 X RACK NOT PRESENT
1 0 RACK PRESENT END OF SCAN
1 1 RACK PRESENT DATA VALID
OPPDAT. T.O.M. ECO 10-501. D 31 5 78 TU/BW
Rettet Dato Sign
BIG MPM I/O INTERF.
1146.D
Målestokk
Tegn. 29.11.77
Kontr. PJJ
Appr.
NORSK DATA A.S
4A46

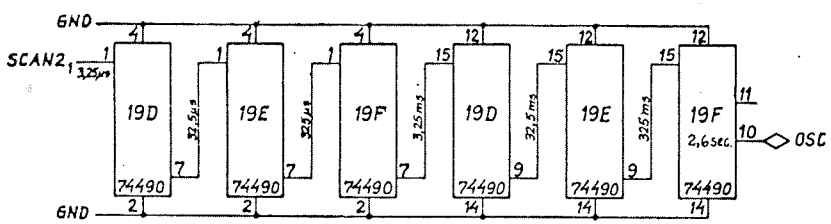

TIMING
ADDRESSING, INTERRUPT & CONTROL
SCANNING
DATA
9A
1C
3E
3D
4F
3A
6F
7F
9F
16C
1F
3F
1D
6C
4C
6B
4B
17D
16D
12A
13F
11F
13E
14E
13D
16D
6D
4E
IM-6402
A/S NORSK DATA-ELEKTRONIKK
2 ASYNCHRONOUS V.24
1147.D Part 1
4A47(1)
Ajourført 20.1.78 HS/Emd
Forandr. f. C t. D (ingen rett.) 5.12.77 HS/Emd
Rettet Dato Sign.
Målestokk
Tegn Emd 5.10.77
Kontr. HS
Appr.



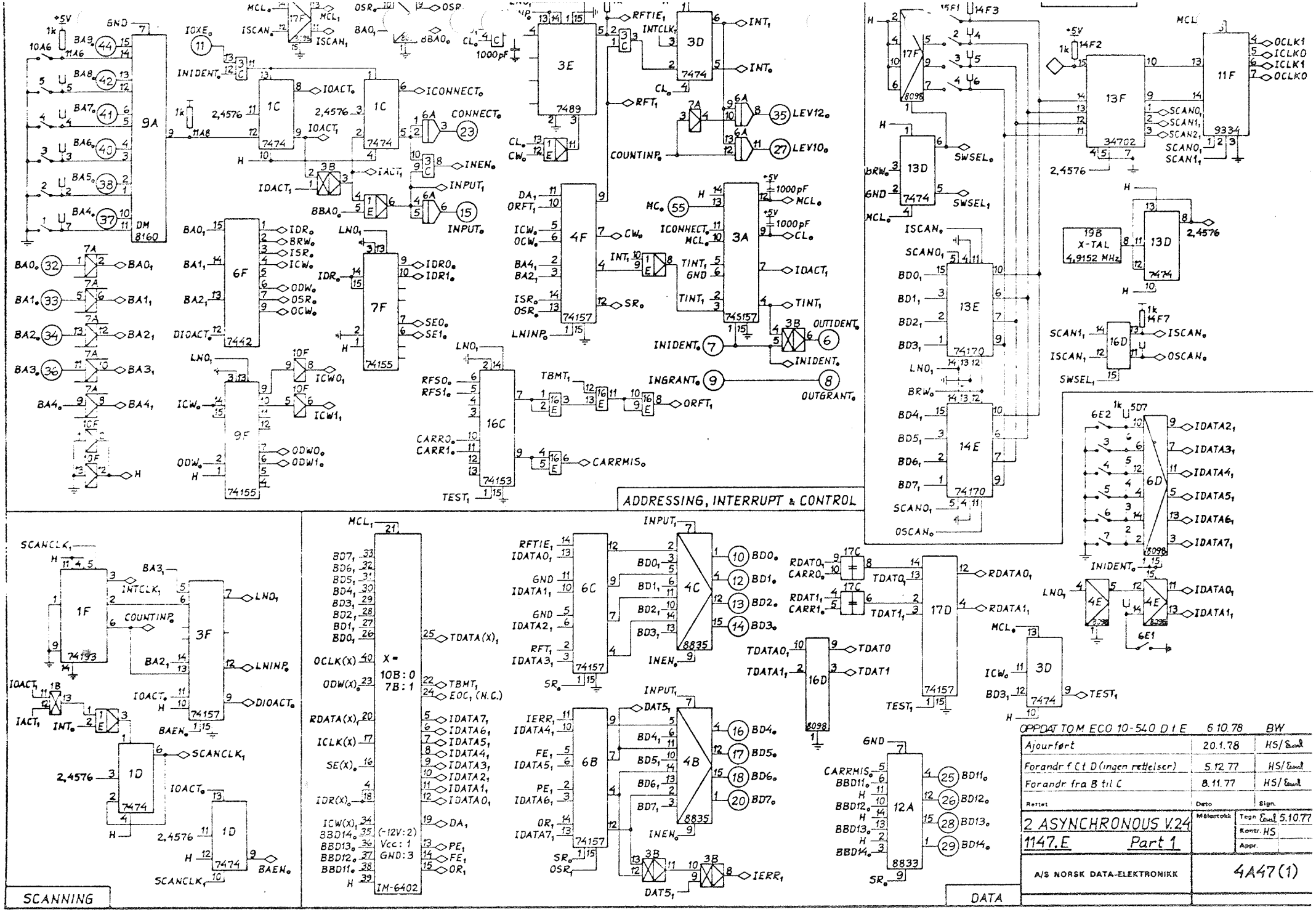


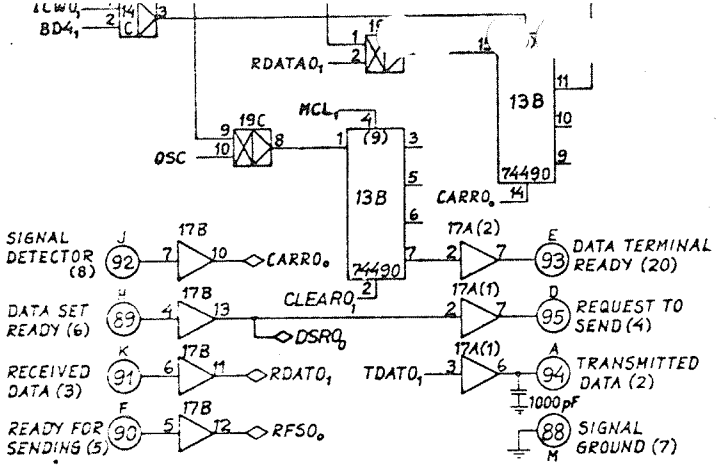


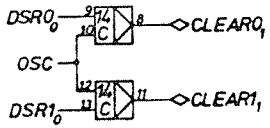


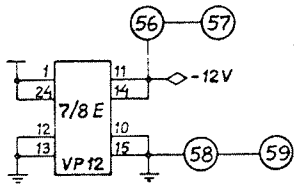


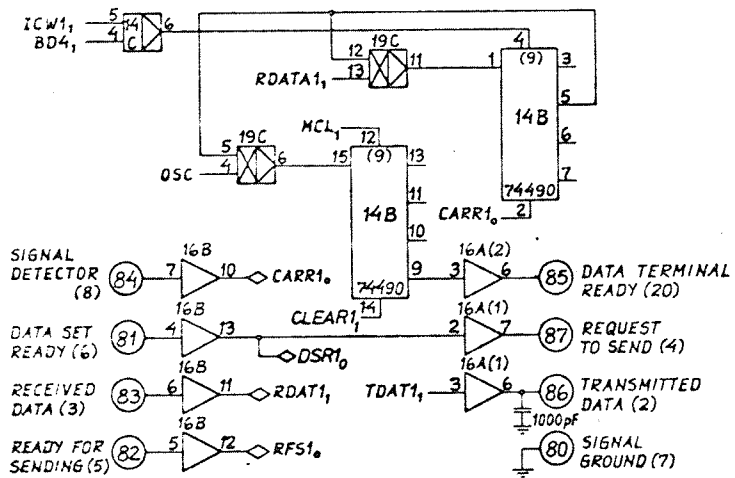
Forandr. f. C t. D (ingen rett.)	5.12.77	HS/Emil
Rettet	Dato	Sign.
2 ASYNCHRONOUS V.24 1147.D Part 2	Målestokk	Tegn Emil 5.10.77 Kontr. HS Appr.
A/S NORSK DATA-ELEKTRONIKK		4A47 (2)

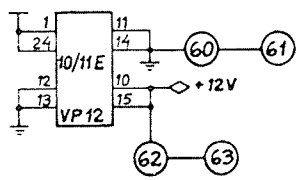

ADDRESSING, INTERRUPT & CONTROL
SCANNING
DATA
9A
DM 8160
1C
7474
3E
7489
3D
7474
4F
74157
3A
74157
6F
7442
7F
74155
9F
74155
16C
74153
13D
7474
13E
74170
14E
74170
11F
9334
13F
34702
16D
6D
4E
4C
8835
6C
4B
6B
17D
74157
16D
8098
12A
8833
1F
74193
3F
74157
1D
7474
IM-6402
X = 10B:0 7B:1
(-12V:2) Vcc:1 GND:3
OPPDAT TOM ECO 10-540 D I E 6.10.78 BW
Ajourført 20.1.78 HS/Emil
Forandr f C t D (ingen rettelser) 5.12.77 HS/Emil
Forandr fra B til C 8.11.77 HS/Emil
2 ASYNCHRONOUS V.24
1147.E Part 1
A/S NORSK DATA-ELEKTRONIKK
4A47(1)

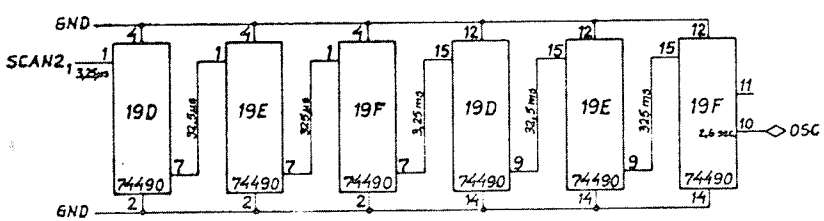












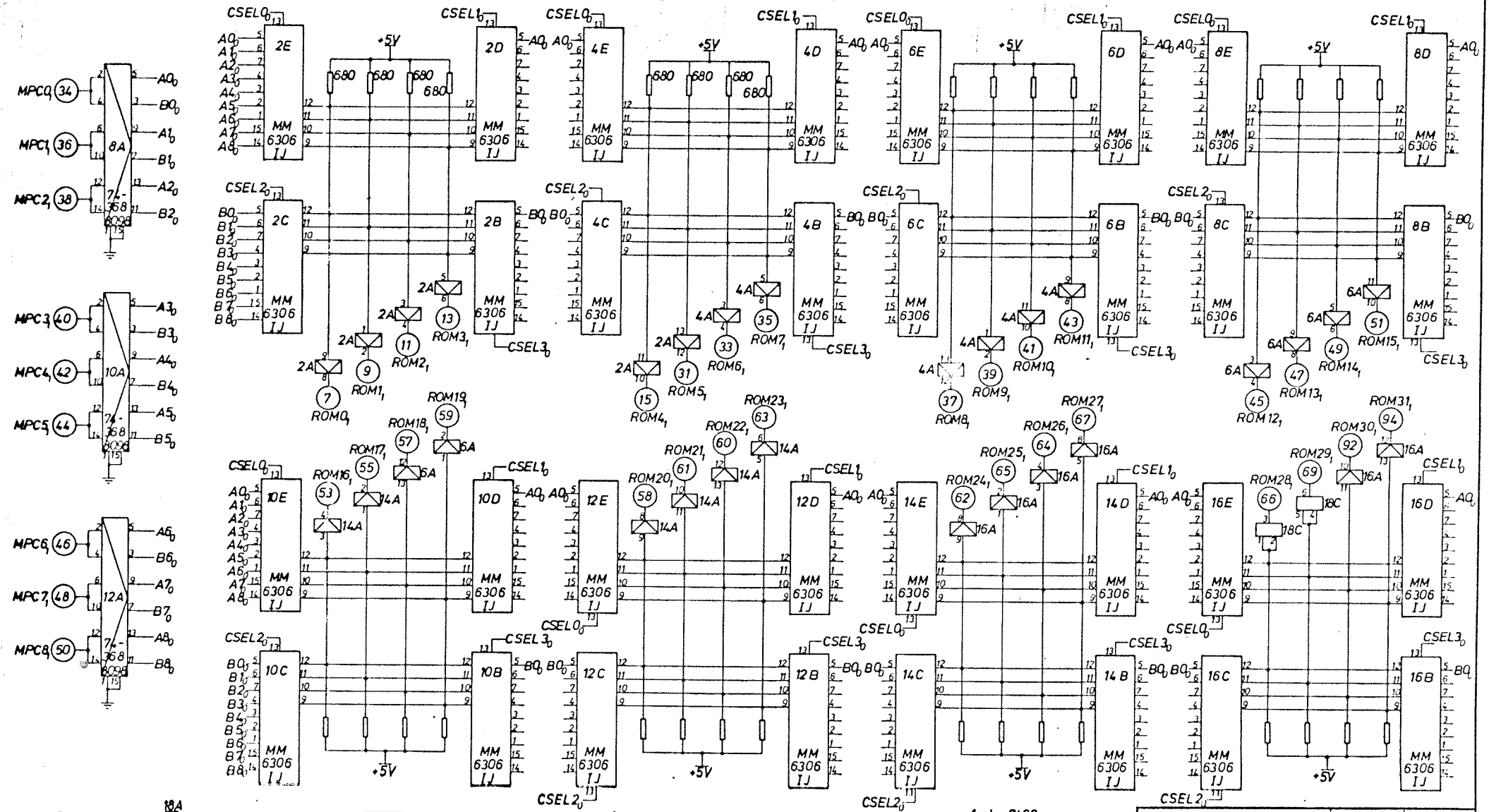
OPPDAT TOM ECO10-540 D t E	6.10.78	BW
Forandr f C t D (ingen rettelser)	5.12.77	HS/Emil
Ajourf. og forandr fra B til C	9.11.77	HS/Emil
Rettet	Dato	Sign.

2 ASYNCHRONOUS V.24
1147.E Part 2

Målestokk | Tegn Emil 5.10.77 | Kontr. HS | Appr.

A/S NORSK DATA-ELEKTRONIKK

4A47 (2)

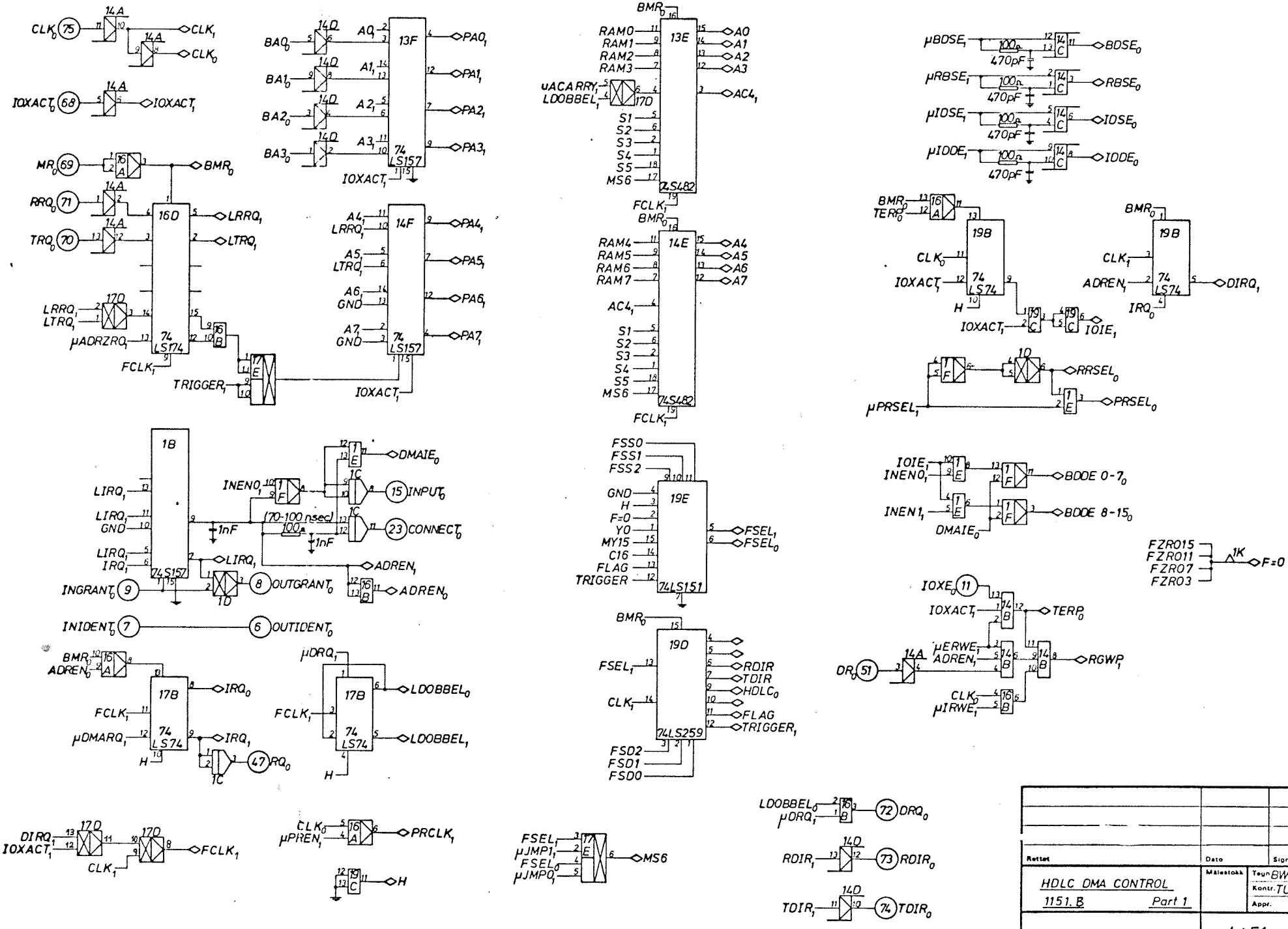


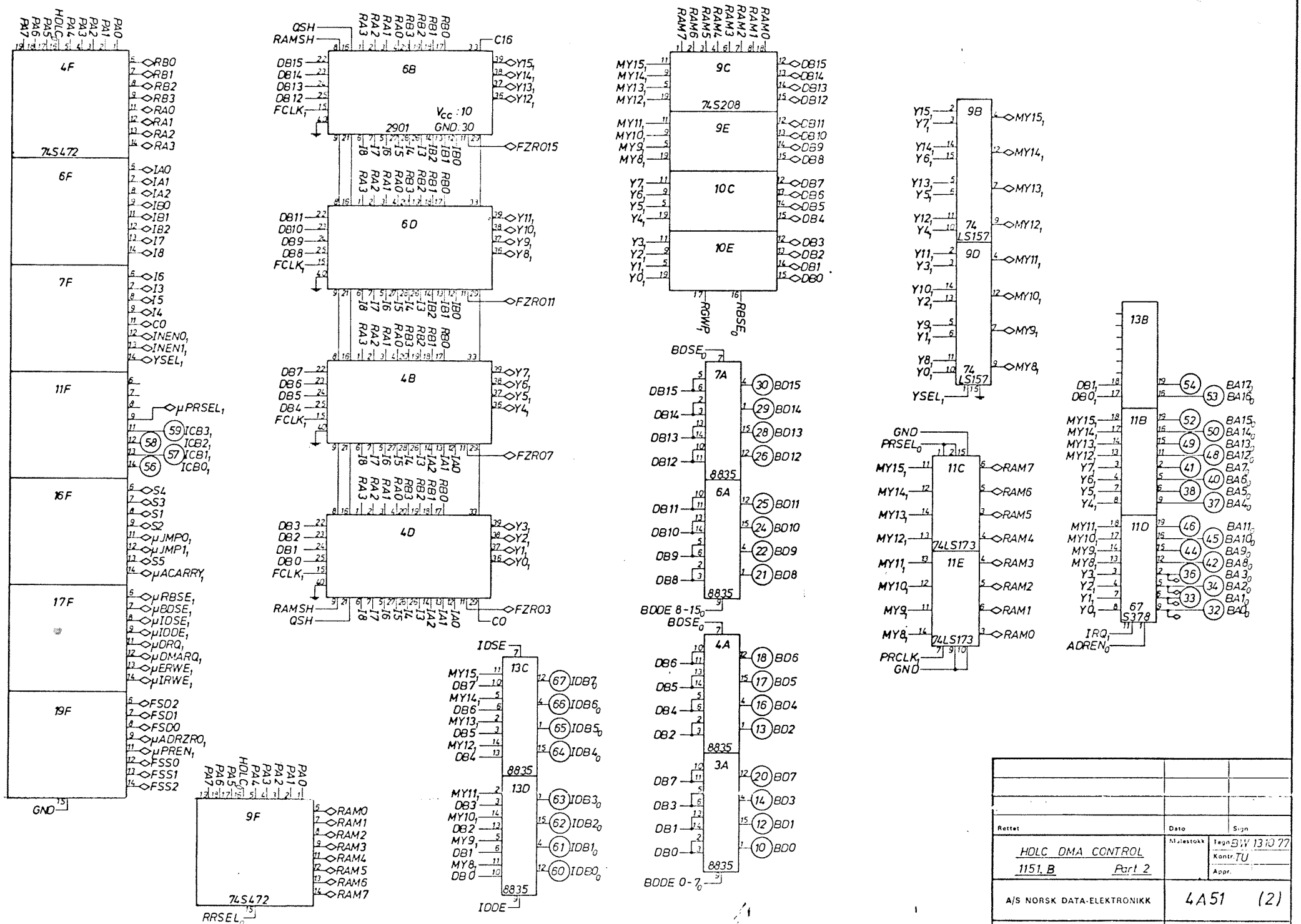
1 pk 7400
1 pk 74S00
6 pk 7404
1 pk 74123
1 pk 74S138
3 pk 74368 (8098)
32 pk MM6306

45 pk

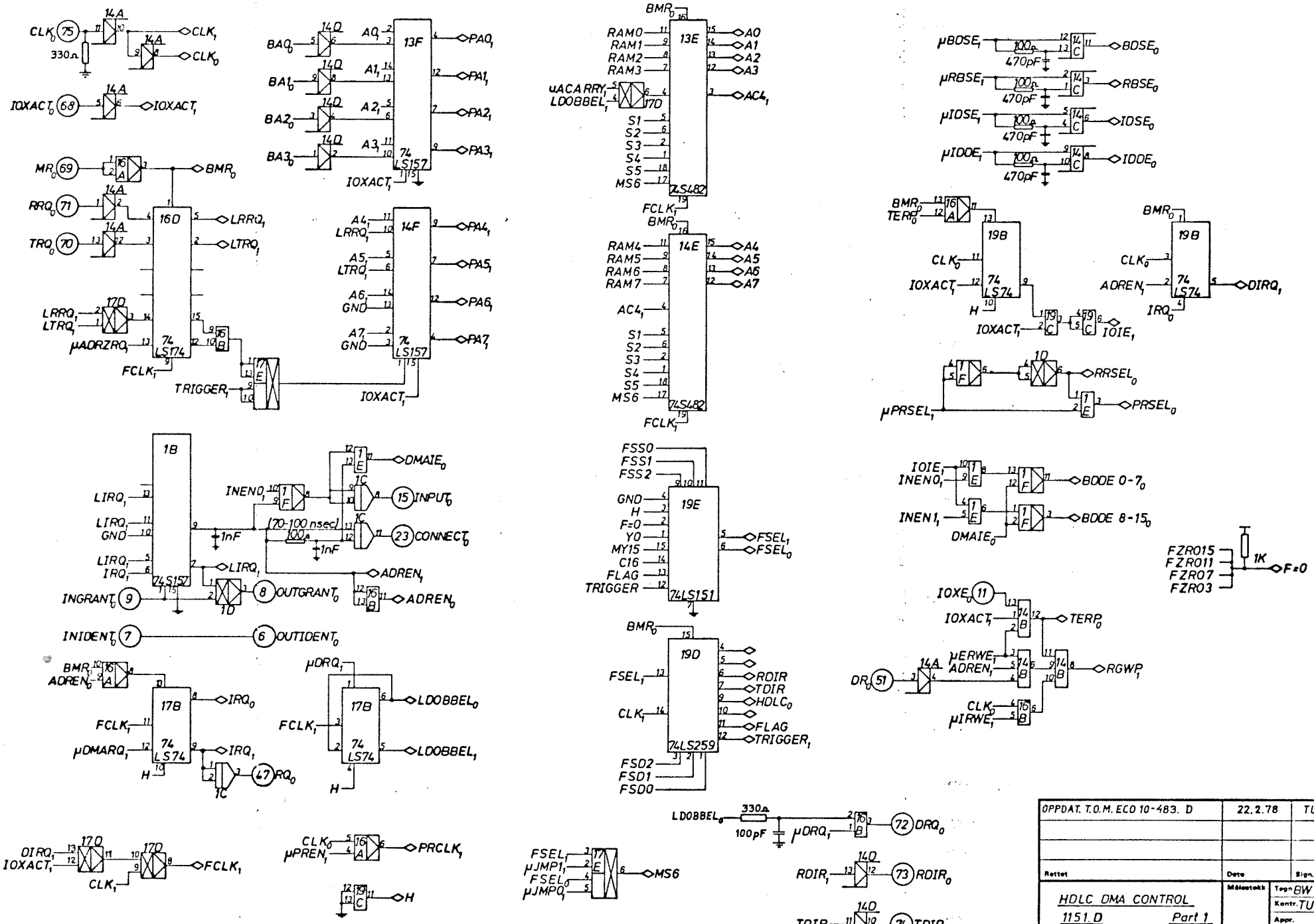
680Ω resistors are Bourns 4308R-101-680 single in line resistor networks.

Rettet	Dato	Sign
2 K PROM 1148 B	Målestokk	Tegn BW 13 12 77 Kontr. PHE Appr.
A/S NORSK DATA-ELEKTRONIKK		4A48

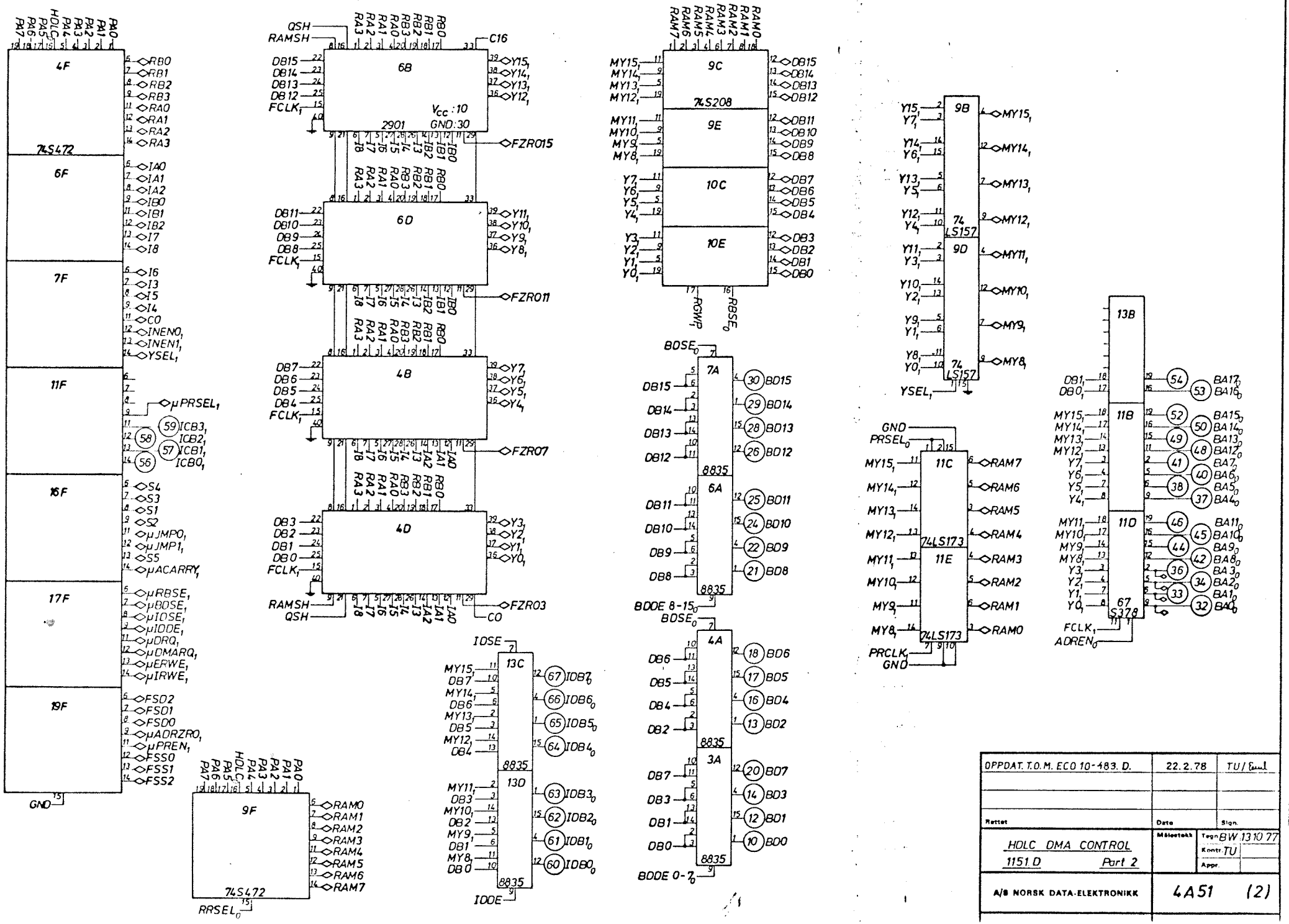

CLK0 (75)
IOXACT0 (68)
MR0 (69)
RRQ0 (71)
TRQ0 (70)
BA0
BA1
BA2
BA3
IOXACT1
BMR0
LRRQ1
LTRQ1
μADRZRQ1
FCLK1
TRIGGER1
A4
A5
A6
A7
GND
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
RAM0
RAM1
RAM2
RAM3
uACARRY1
LDOBBEL1
S1
S2
S3
S4
S5
MS6
A0
A1
A2
A3
AC4
RAM4
RAM5
RAM6
RAM7
H
F=0
Y0
MY15
C16
FLAG
TRIGGER
FSS0
FSS1
FSS2
FSEL1
FSEL0
FSD2
FSD1
FSD0
μBDSE1
μRBSE1
μIDSE1
μIDDE1
470pF
BDSE0
RBSE0
IDSE0
IDDE0
TERP0
CLK1
ADREN1
DIRQ1
IRQ0
IOIE1
RRSEL0
μPRSEL1
PRSEL0
IOIE1
INEN0
INEN1
DMAIE0
BDDE 0-7
BDDE 8-15
FZR015
FZR011
FZR07
FZR03
1K
IOXE0 (11)
IOXACT1
μERWE1
ADREN1
DR0 (51)
CLK0
μIRWE1
TERP0
RGWP1
LIRQ1
GND
IRQ1
INEN0
70-100 nsec
100Ω
1nF
DMAIE0
INPUT0 (15)
CONNECT0 (23)
ADREN1
ADREN0
INGRANT0 (9)
OUTGRANT0 (8)
INIDENT0 (7)
OUTIDENT0 (6)
BMR0
ADREN0
IRQ0
FCLK1
μDMARQ1
IRQ1
RQ0 (47)
μDRQ1
LDOBBEL0
LDOBBEL1
DIRQ1
IOXACT1
CLK1
FCLK1
CLK0
μPREN1
PRCLK1
H
FSEL1
μJMP1
FSEL0
μJMP0
MS6
LDOBBEL0
μDRQ1
DRQ0 (72)
RDIR1
RDIR0 (73)
TDIR1
TDIR0 (74)
FSD2
FSD1
FSD0
FSEL1
FSEL0
RDIR
TDIR
HDLC0
FLAG
TRIGGER1
Rettet
Dato
Sign.
HDLC DMA CONTROL
1151. B
Part 1
Målestokk
Tegn. BW 11.10.77
Kontr. TU
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A51 (1)



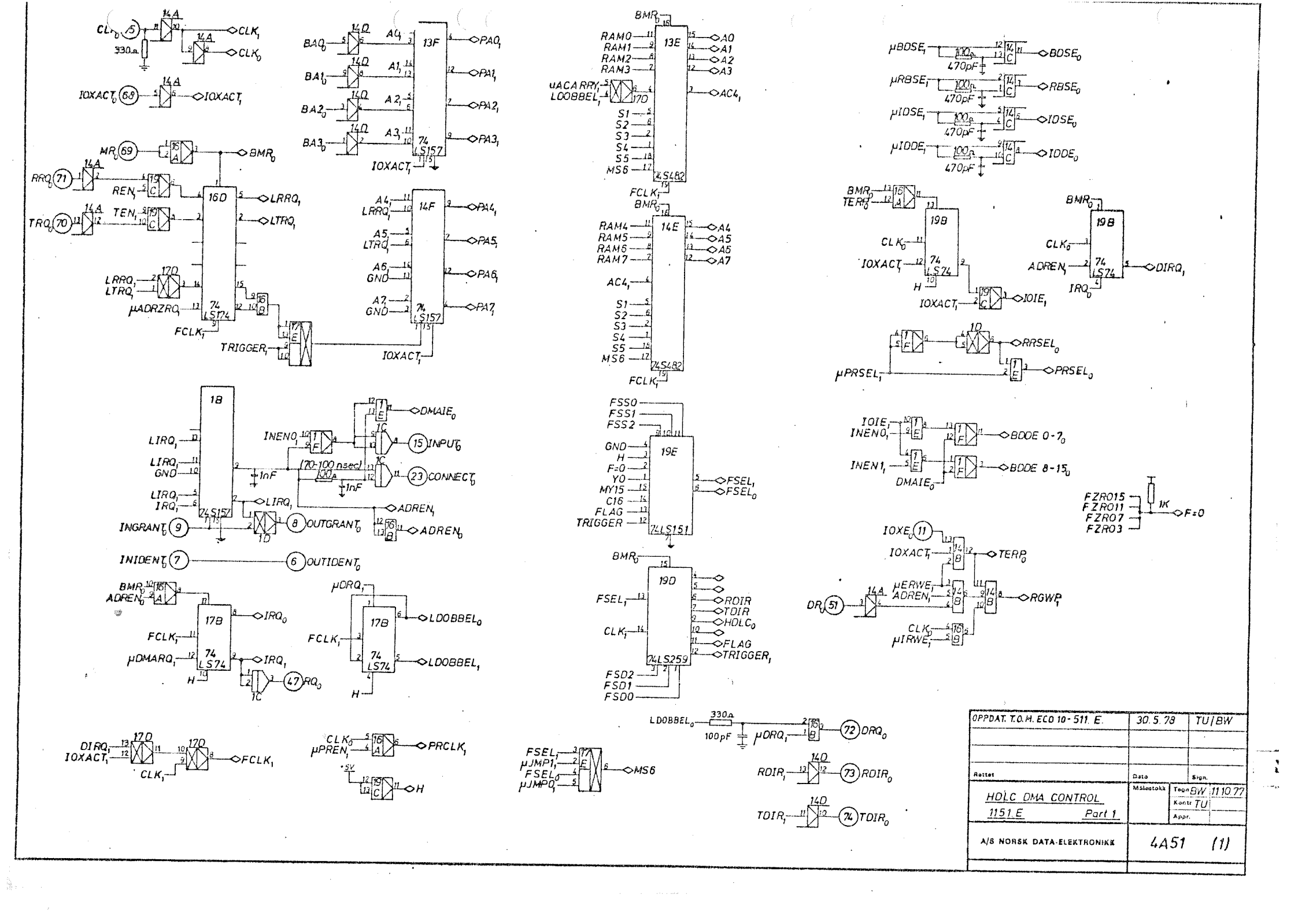
Rettet	Dato	Sign
HOLC DMA CONTROL 1151. B Part 2	Målestokk	Tegn SW 13.10.77 Kontr. TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A51 (2)	



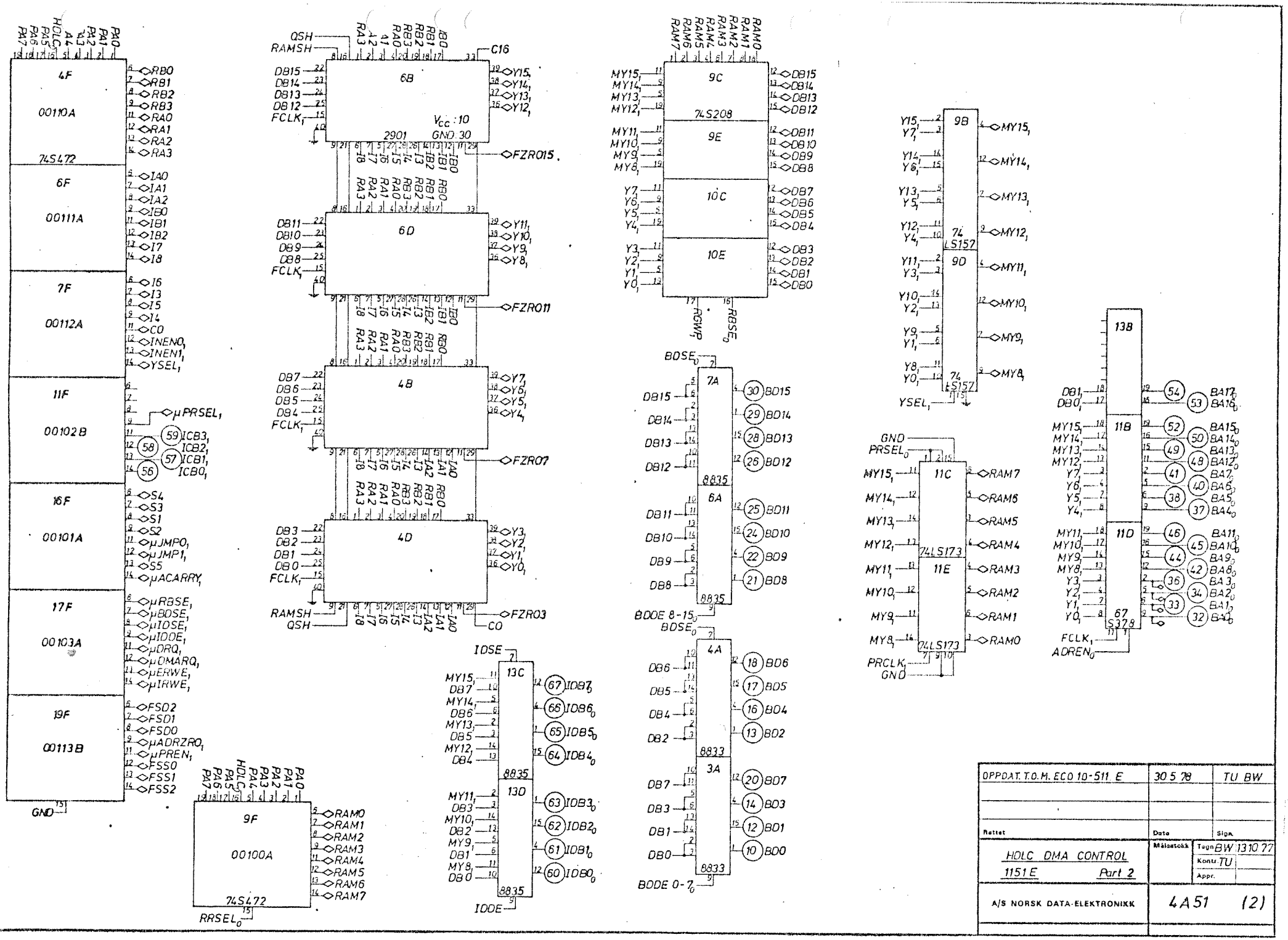
OPPDAT. T.O.M. ECO 10-483. D	22.2.78	TU/
Rettet	Dato	Sign.
HDLC DMA CONTROL	Målestokk	Tegn BW 11.10.77
1151.D Part 1		Kontr. TU
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A51 (1)	



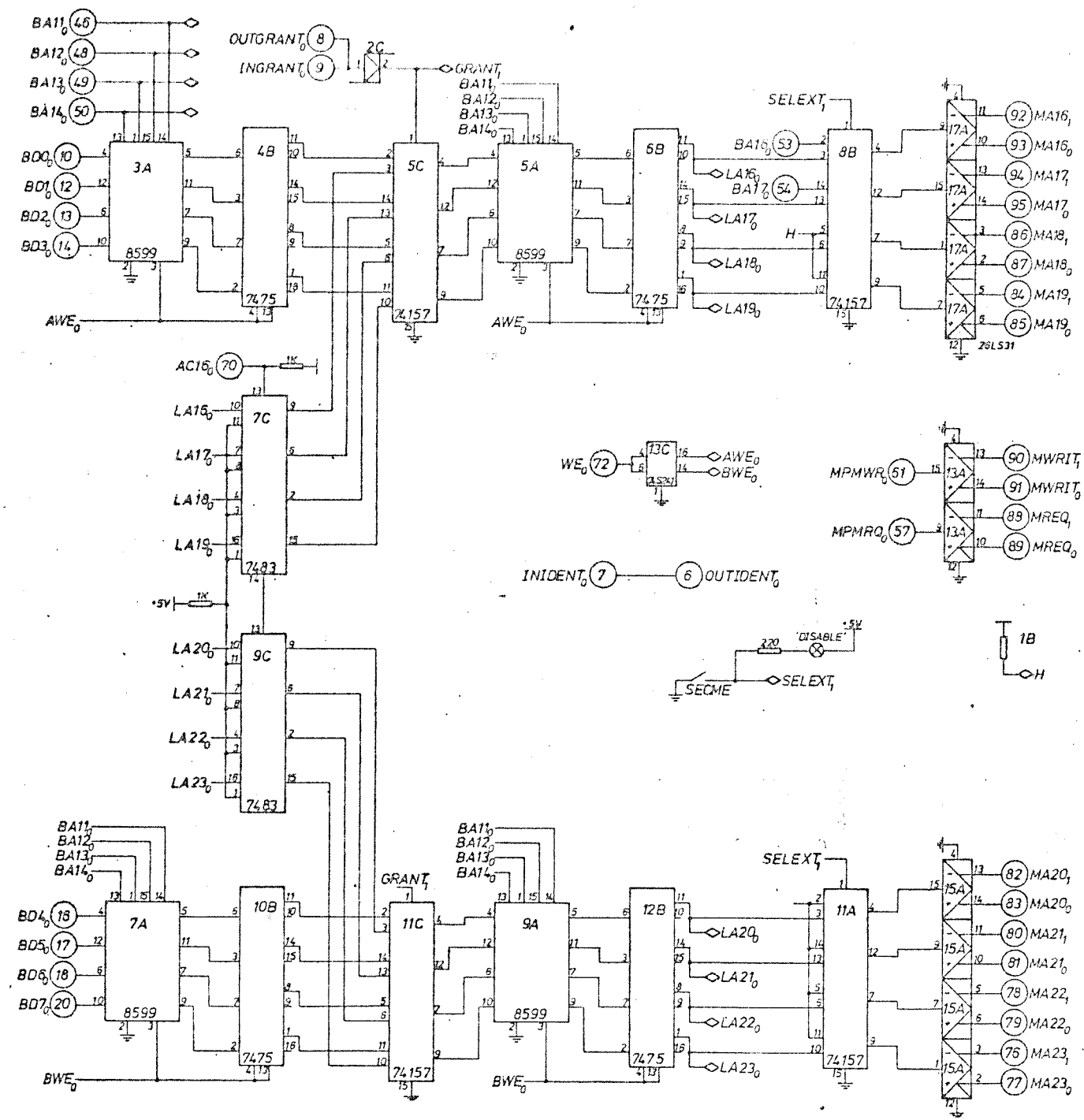
OPPDAT. T.O.M. ECO 10-483. D.	22.2.78	TU/
Rettet	Dato	Sign.
HDLC DMA CONTROL 1151 D Part 2	Målestokk	Tegn BW 13 10 77 Kontr. TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A51 (2)	



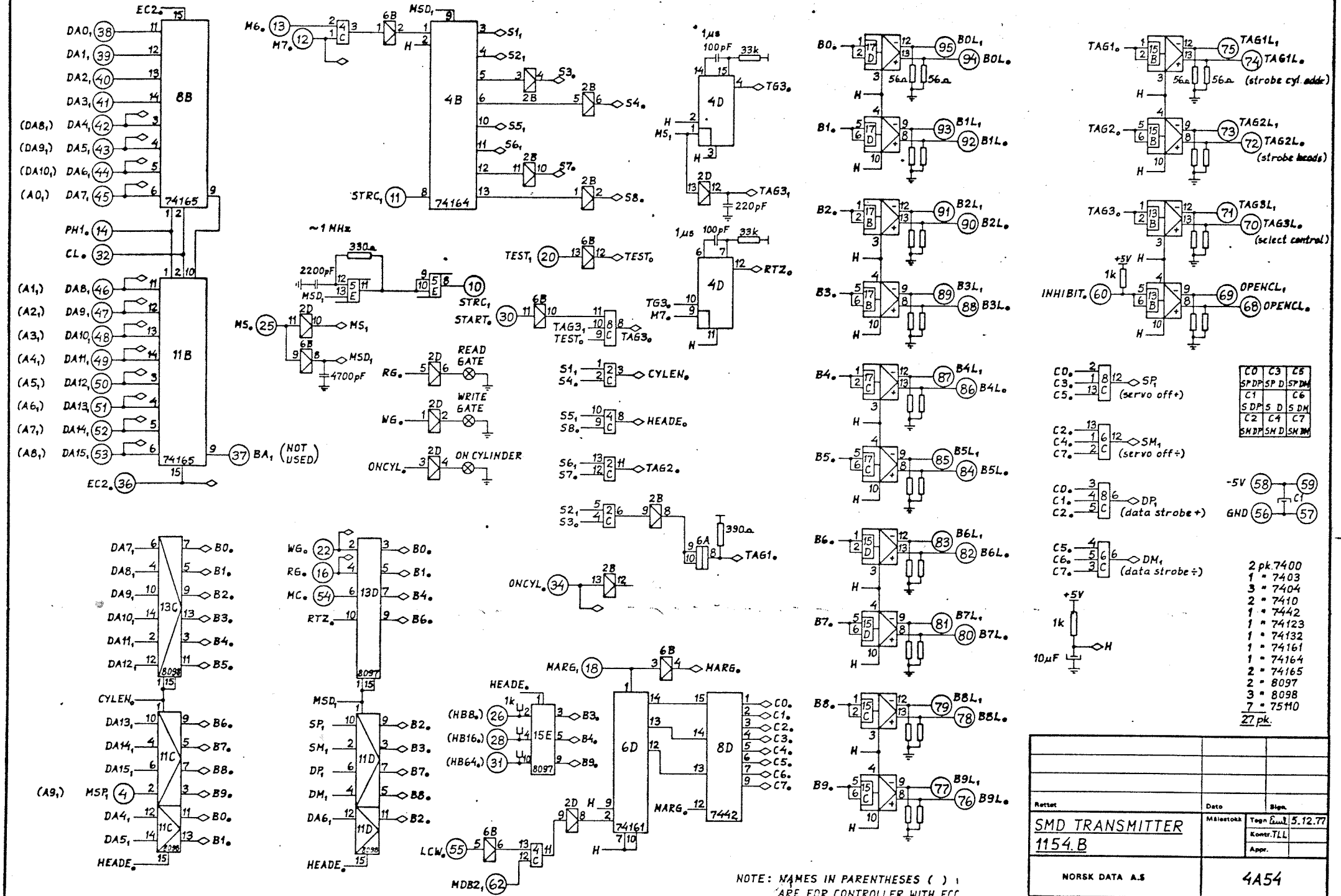
OPPDAT. T.O.H. ECO 10-511. E.	30.5.78	TU/BW
Rettet	Dato	Sign.
HDLC DMA CONTROL	Målestokk	Tegn BW 11.10.77
1151 E Part 1		Kontr TU
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A51 (1)	

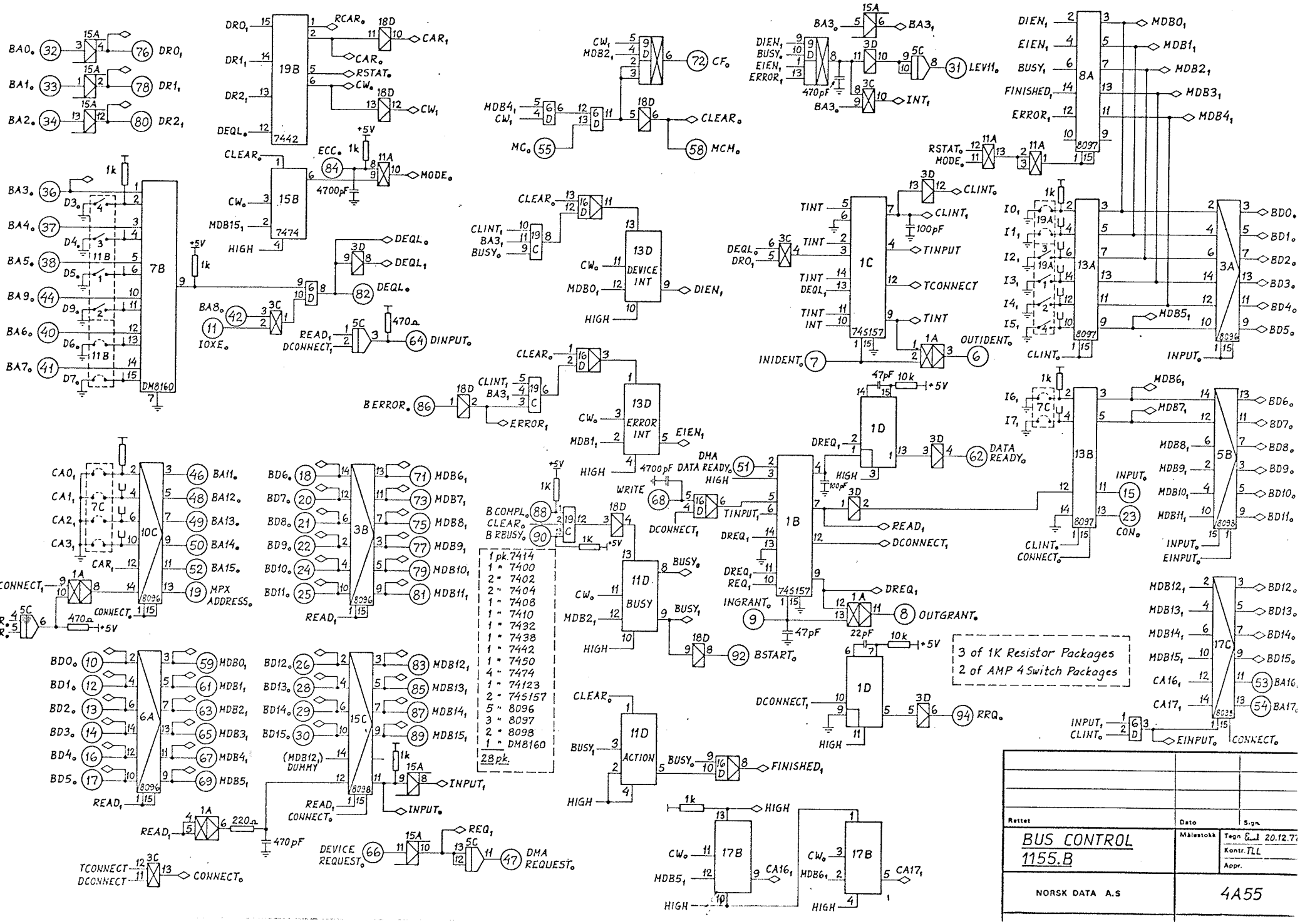


OPPDAT. T.O.M. ECO 10-511 E	30 5 78	TU BW
Rettet	Dato	Sign.
HOLC DMA CONTROL 1151 E Part 2	Målestokk	Tegn BW 13 10 77 Kontr. TU Appr.
A/S NORSK DATA-ELEKTRONIKK	4A51 (2)	

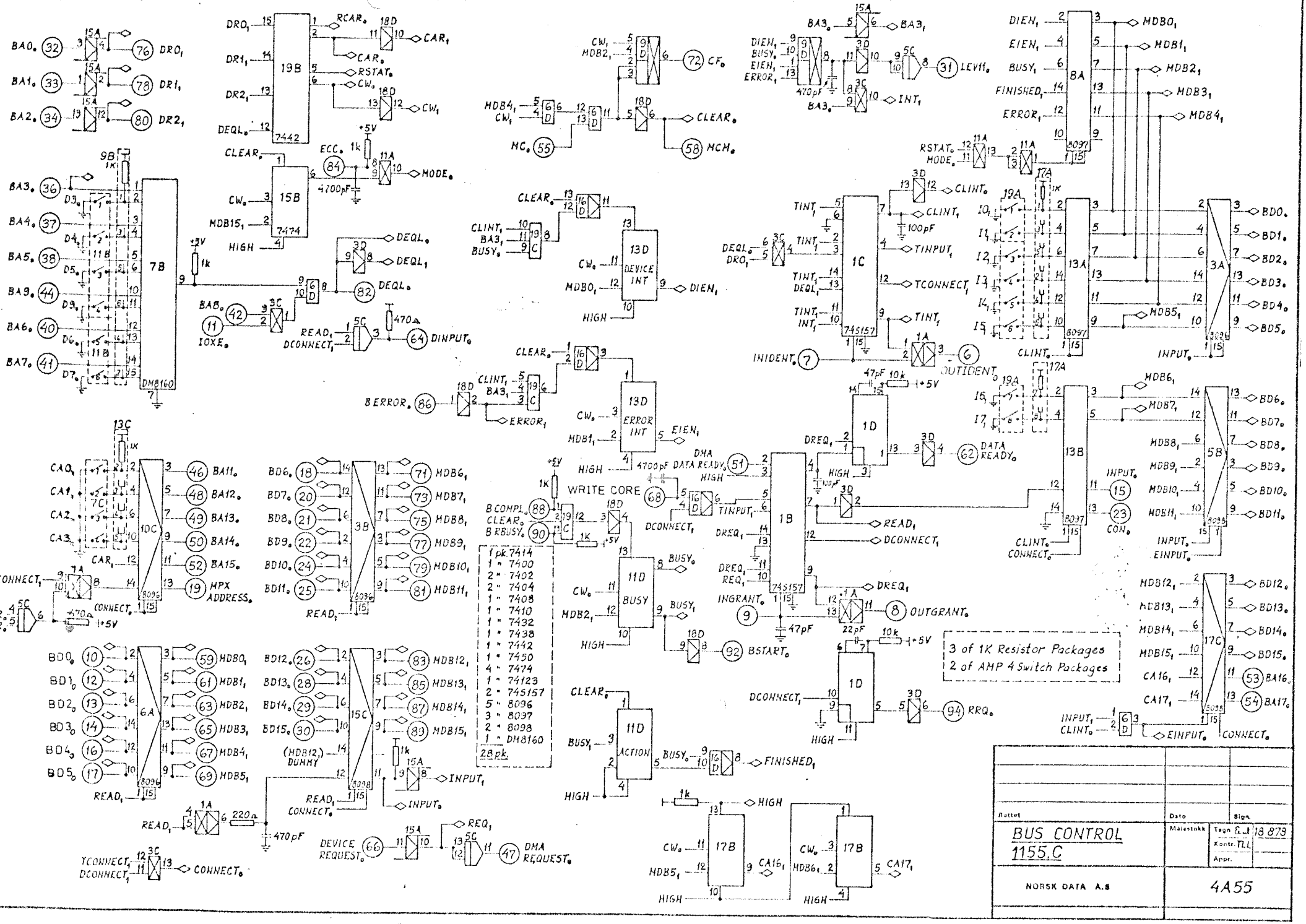


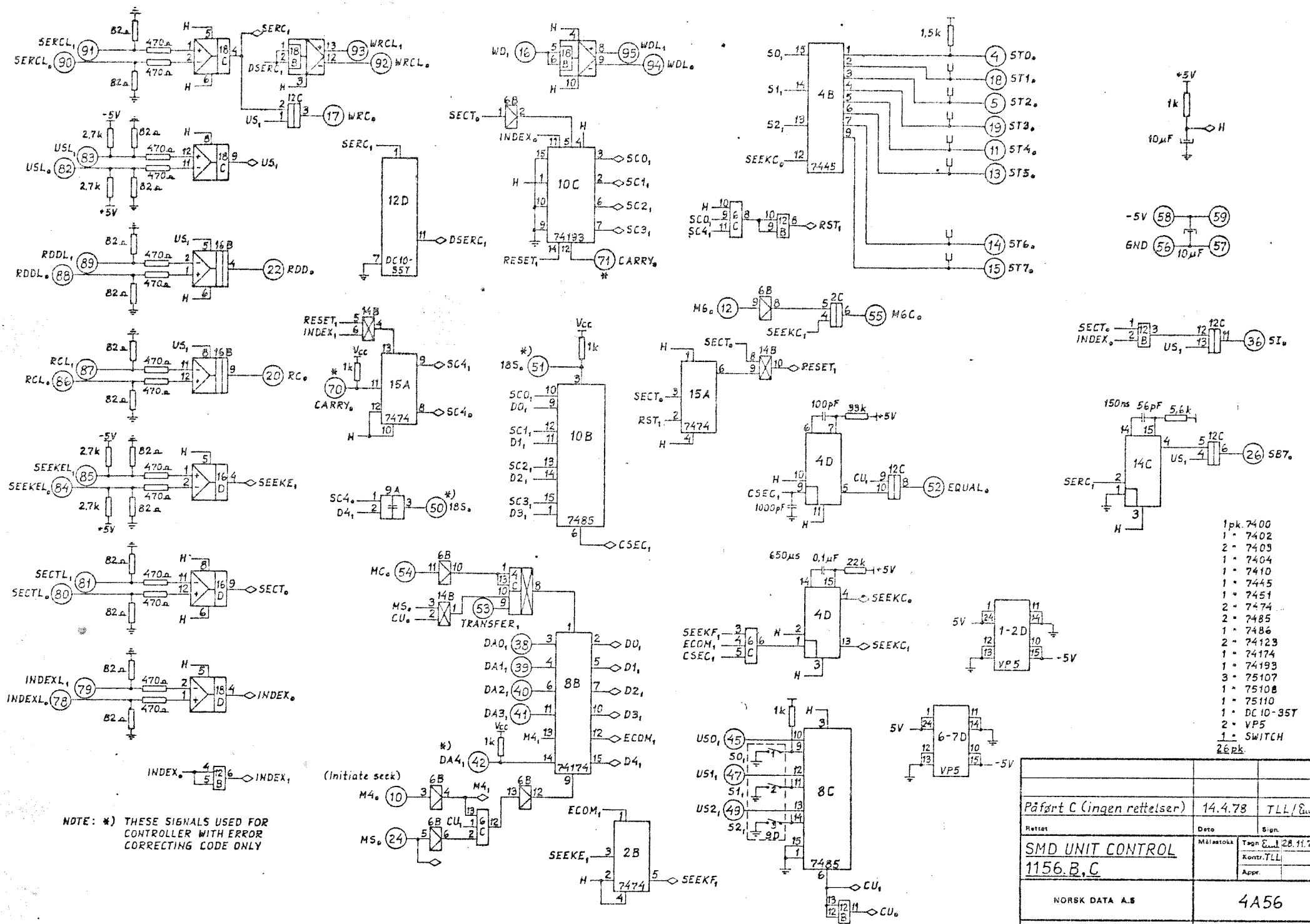
ADDED VERS. C (NO CORRECTIONS)	26.9.78	SEJ/ Emil
Rettet	Dato	Sign.
DMA ADDRESS EXTENDER 1153.C,D	Målestokk	Tegn BW 27.7.78 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK	4A53	


8B
74165
11B
74165
4B
74164
4D
4D
74161
6D
8D
7442
13C
11C
11C
13D
11D
11D
15E
8097
8098
8098
8097
2098
2098
~1 MHz
1µs
100pF
33k
2200pF
4700pF
330Ω
220pF
390Ω
56Ω
1k
10µF
+5V
-5V
GND
READ GATE
WRITE GATE
ON CYLINDER
(NOT USED)
(strobe cyl. addr)
(strobe heads)
(select control)
(servo off+)
(servo off÷)
(data strobe+)
(data strobe÷)
EC2
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
DA8
DA9
DA10
DA11
DA12
DA13
DA14
DA15
PH1
CL
BA
STRC
START
TEST
MS
MSD
RG
WG
ONCYL
CYLEN
HEADE
TAG1
TAG2
TAG3
RTZ
MARG
INHIBIT
OPENCL
SP
SM
DP
DM
MC
LCW
MDB2
S1
S2
S3
S4
S5
S6
S7
S8
B0
B1
B2
B3
B4
B5
B6
B7
B8
B9
TAG1L
TAG2L
TAG3L
B0L
B1L
B2L
B3L
B4L
B5L
B6L
B7L
B8L
B9L
CO
C1
C2
C3
C4
C5
C6
C7
HB8
HB16
HB64
MSP
NOTE: NAMES IN PARENTHESES ( ) ARE FOR CONTROLLER WITH ECC.
2 pk.7400
1 = 7403
3 = 7404
2 = 7410
1 = 7442
1 = 74123
1 = 74132
1 = 74161
1 = 74164
2 = 74165
2 = 8097
3 = 8098
7 = 75110
27 pk.
Rettet
Dato
Sign.
SMD TRANSMITTER
1154.B
Målestokk
Tegn. Emil
5.12.77
Konstr. TLL
Appr.
NORSK DATA A.S
4A54



Rettet	Dato	Sign.
BUS CONTROL 1155.B	Målestokk	Tegn. Emil 20.12.7 Kontr. TLL Appr.
NORSK DATA A.S		4A55

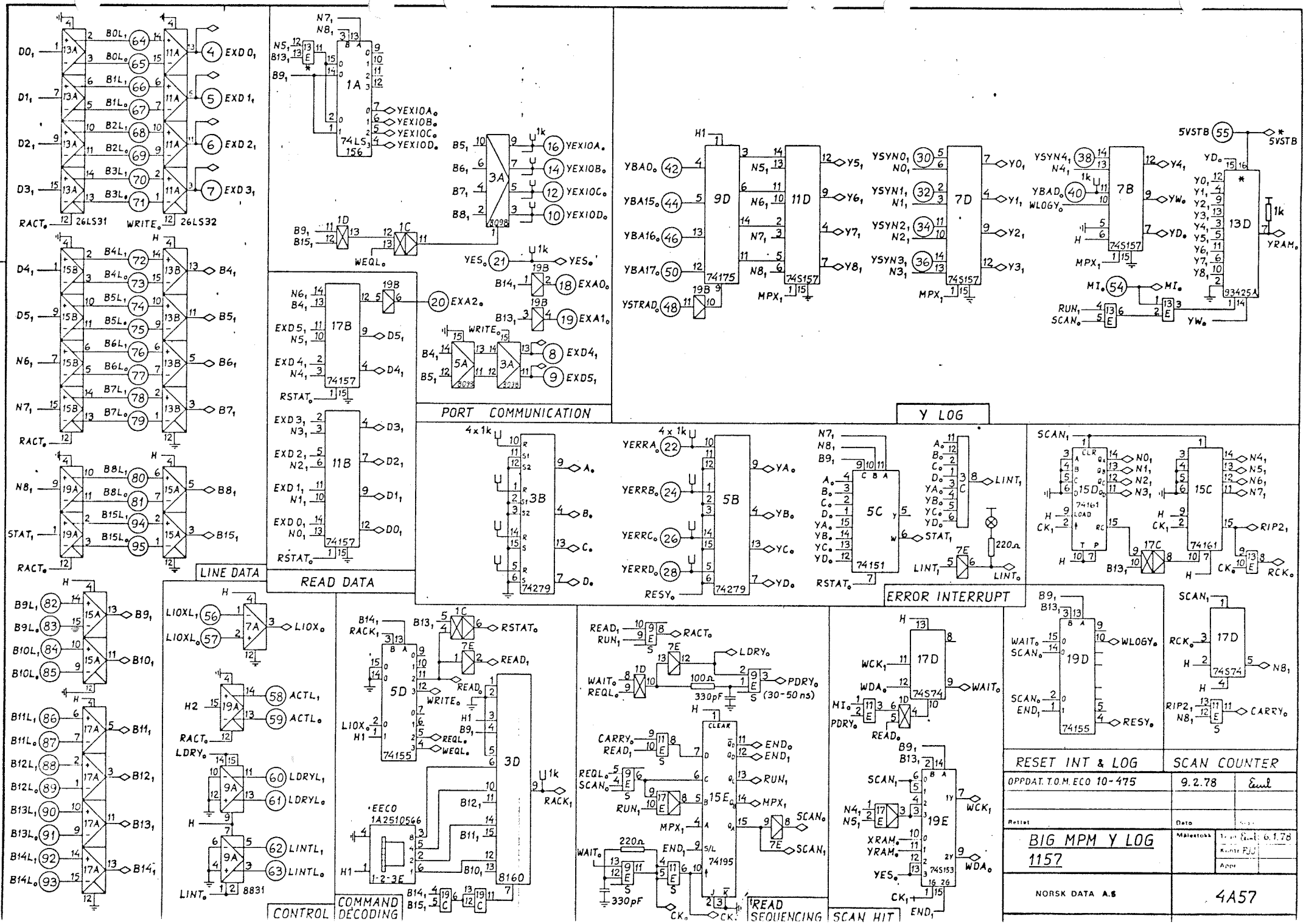

BA0 (32) 15A 76 DR0
BA1 (33) 15A 78 DR1
BA2 (34) 15A 80 DR2
DR0 DR1 DR2 DEQL 19B 7442 RCAR CAR RSTAT CW CAR1 CW1
CLEAR 15B 7474 CW MDB15 HIGH
ECC (84) 1k +5V 4700pF 11A MODE
9B 1K BA3 (36) BA4 (37) BA5 (38) BA9 (44) BA6 (40) BA7 (41) D3 D4 D5 D6 D7 11B 7B DM8160
DEQL 3D DEQL1 DEQL (82)
BA8 (42) 3C IOXE (11) READ1 DCONNECT1 5C 470Ω DINPUT (64)
CW1 MDB2 (72) CF
MDB4 CW1 MC (55) 6D 18D CLEAR (58) MCH
CLEAR CLINT1 BA3 BUSY 16D 13D DEVICE INT DIEN HIGH
BERROR (86) 18D ERROR1 13D ERROR INT EIEN
BA3 (31) LEV11 INT1 3D 5C 3C 470pF
DIEN1 BUSY1 EIEN1 ERROR1 8A 8097 MDB0 MDB1 MDB2 MDB3 MDB4 MDB5
RSTAT MODE 11A 17A
CLINT 3D 100pF TINPUT TCONNECT TINT 1C 74S157 INIDENT (7) 1A OUTIDENT (6)
DEQL DRQ1 3C
19A 1K I0 I1 I2 I3 I4 I5 13A 3A BD0 BD1 BD2 BD3 BD4 BD5
CA0 CA1 CA2 CA3 7C 13C 1K 10C 8096 BA11 BA12 BA13 BA14 BA15 MPX ADDRESS (19)
CAR1 7A DCONNECT CONNECT RCAR CAR 5C 470Ω +5V
BD6 BD7 BD8 BD9 BD10 BD11 (18) (20) (21) (22) (24) (25) 3B 8096 READ1 MDB6 MDB7 MDB8 MDB9 MDB10 MDB11 (71) (73) (75) (77) (79) (81)
1 pk 7414
1 " 7400
2 " 7402
2 " 7404
1 " 7408
1 " 7410
1 " 7432
1 " 7438
1 " 7442
1 " 7450
4 " 7474
1 " 74123
2 " 745157
5 " 8096
3 " 8097
2 " 8098
1 " DM8160
28 pk.
BD0 BD1 BD2 BD3 BD4 BD5 (10) (12) (13) (14) (16) (17) 6A 8096 READ1 59 61 63 65 67 69 MDB0 MDB1 MDB2 MDB3 MDB4 MDB5
BD12 BD13 BD14 BD15 (MDB12) DUMMY (26) (28) (29) (30) 15C 8098 READ1 CONNECT1 1k 15A INPUT1 INPUT 83 85 87 89 MDB12 MDB13 MDB14 MDB15
1A 220Ω 470pF TCONNECT1 DCONNECT1 3C CONNECT
DEVICE REQUEST (66) 15A REQ1 5C DMA REQUEST (47)
B COMPL (88) B RBUSY (90) CLEAR 1K +5V WRITE CORE (68) 16D 18D 4700pF DCONNECT1
11D BUSY CW MDB2 HIGH BUSY1 BSTART (92)
11D ACTION BUSY FINISHED1
DMA DATA READY (51) 1B 74S157 INGRANT (9) 47pF DREQ1 DREQ REQ1 1A 22pF OUTGRANT (8)
1D 47pF 10k +5V DREQ1 3D DATA READY (62) 100pF TINPUT READ1 DCONNECT
1D 10k +5V DCONNECT 3D RRQ (94)
3 of 1K Resistor Packages
2 of AMP 4 Switch Packages
13B 8097 INPUT (15) CON (23) CLINT CONNECT
MDB6 MDB7 MDB8 MDB9 MDB10 MDB11 5B 8098 BD6 BD7 BD8 BD9 BD10 BD11
MDB12 MDB13 MDB14 MDB15 CA16 CA17 17C 8098 BD12 BD13 BD14 BD15 BA16 (53) BA17 (54)
INPUT CLINT 6D EINPUT CONNECT
1k HIGH 17B CW MDB5 CA16 MDB6 CA17
Rettet Dato Sign.
Målestokk Tegn. Godkj. 18 879
Kontr. TLL
Appr.
BUS CONTROL
1155.C
NORSK DATA A.S
4A55

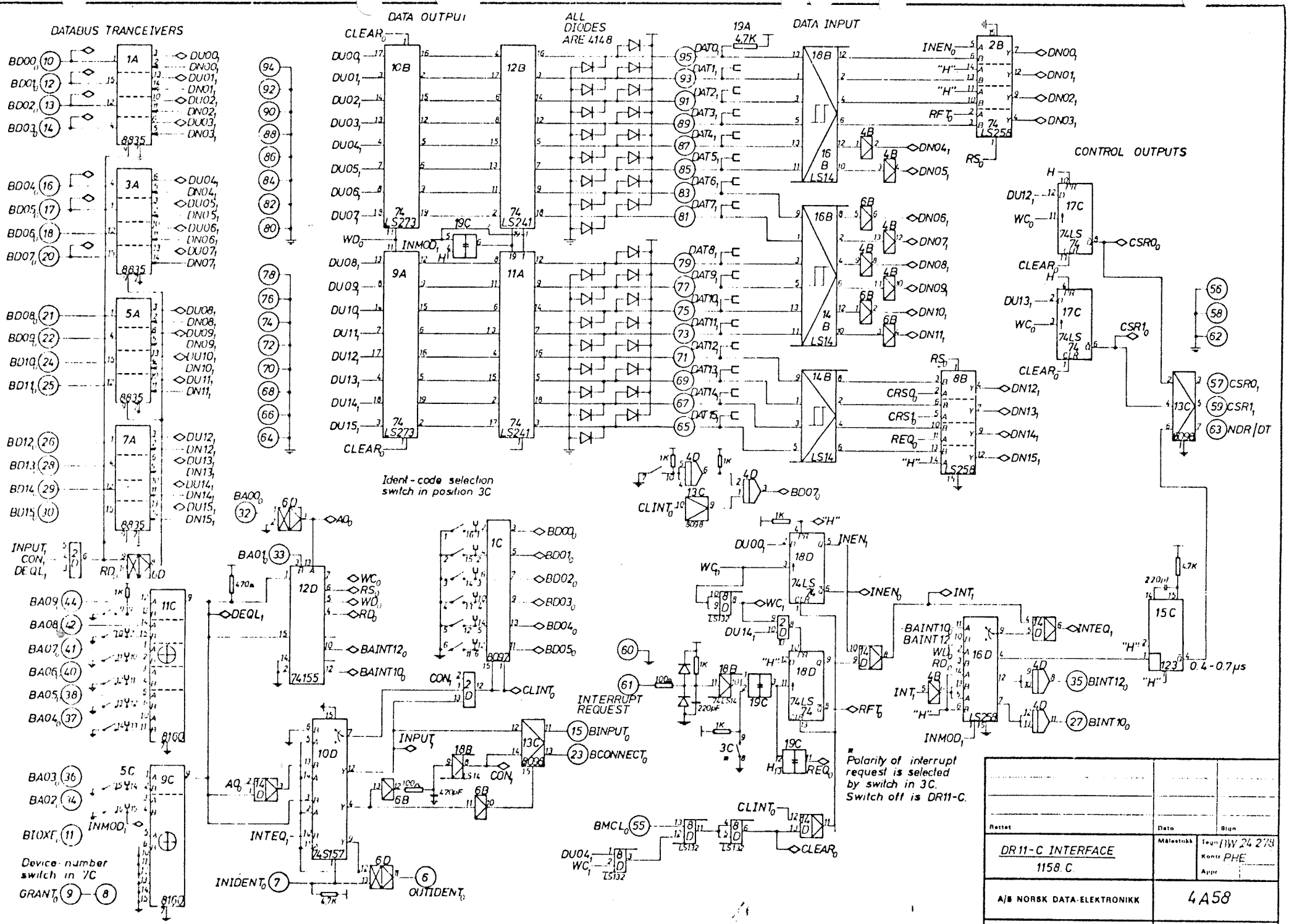


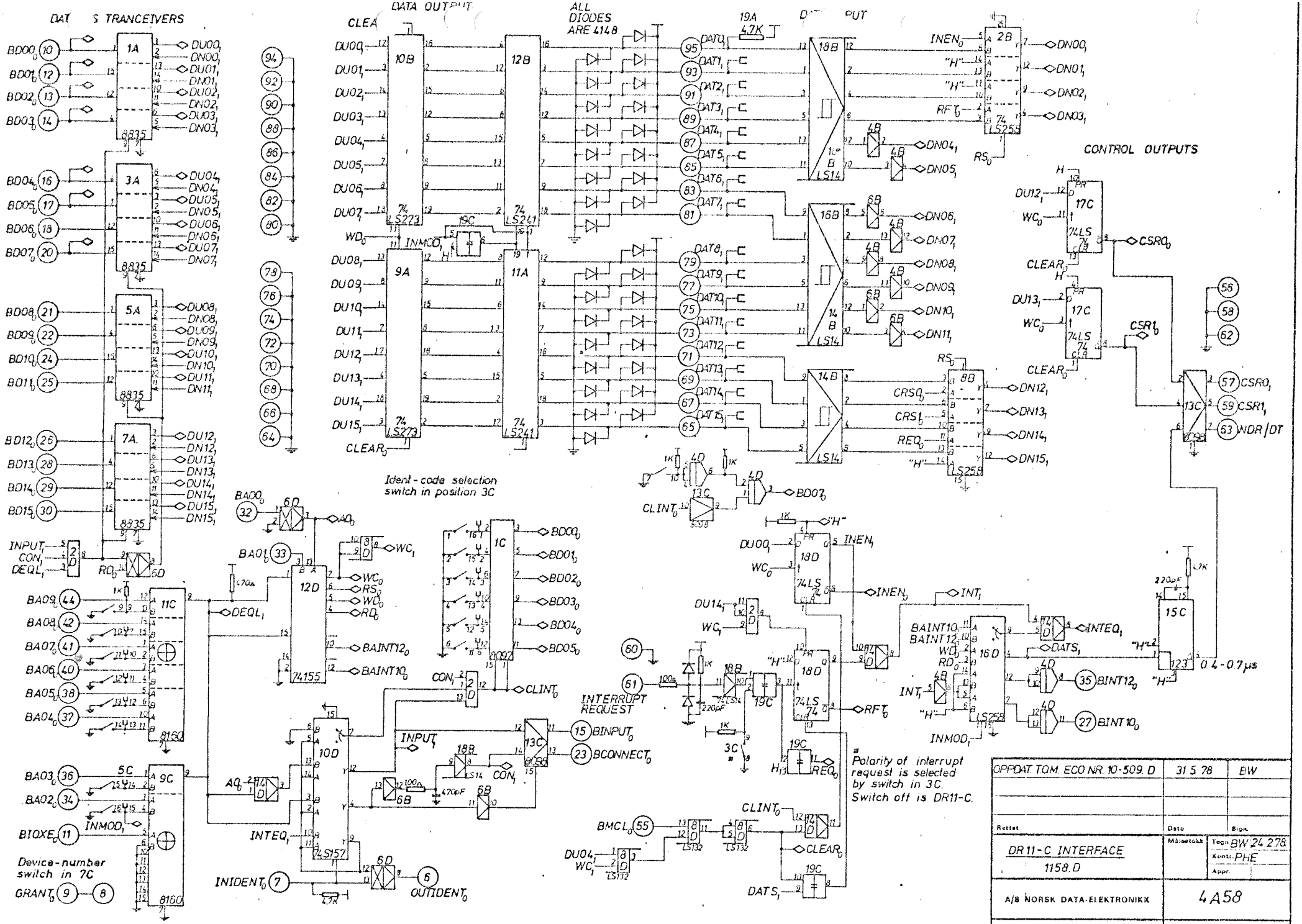
NOTE: *) THESE SIGNALS USED FOR CONTROLLER WITH ERROR CORRECTING CODE ONLY

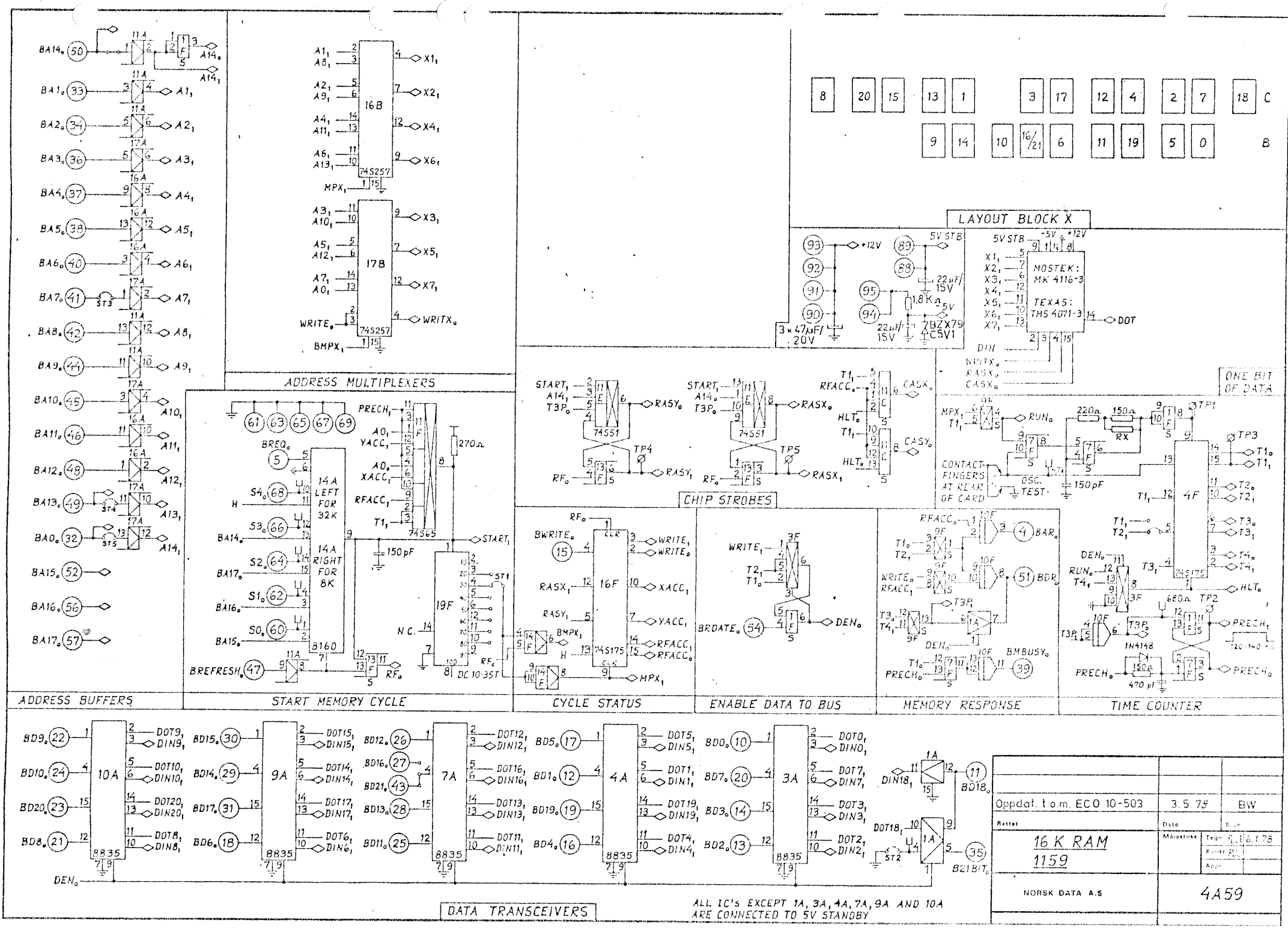
1 pk.	7400
1 "	7402
2 "	7403
1 "	7404
1 "	7410
1 "	7445
1 "	7451
2 "	7474
2 "	7485
1 "	7486
2 "	74123
1 "	74174
1 "	74193
3 "	75107
1 "	75108
1 "	75110
1 "	DC10-35T
2 "	VP5
1 "	SWITCH
26 pk.	

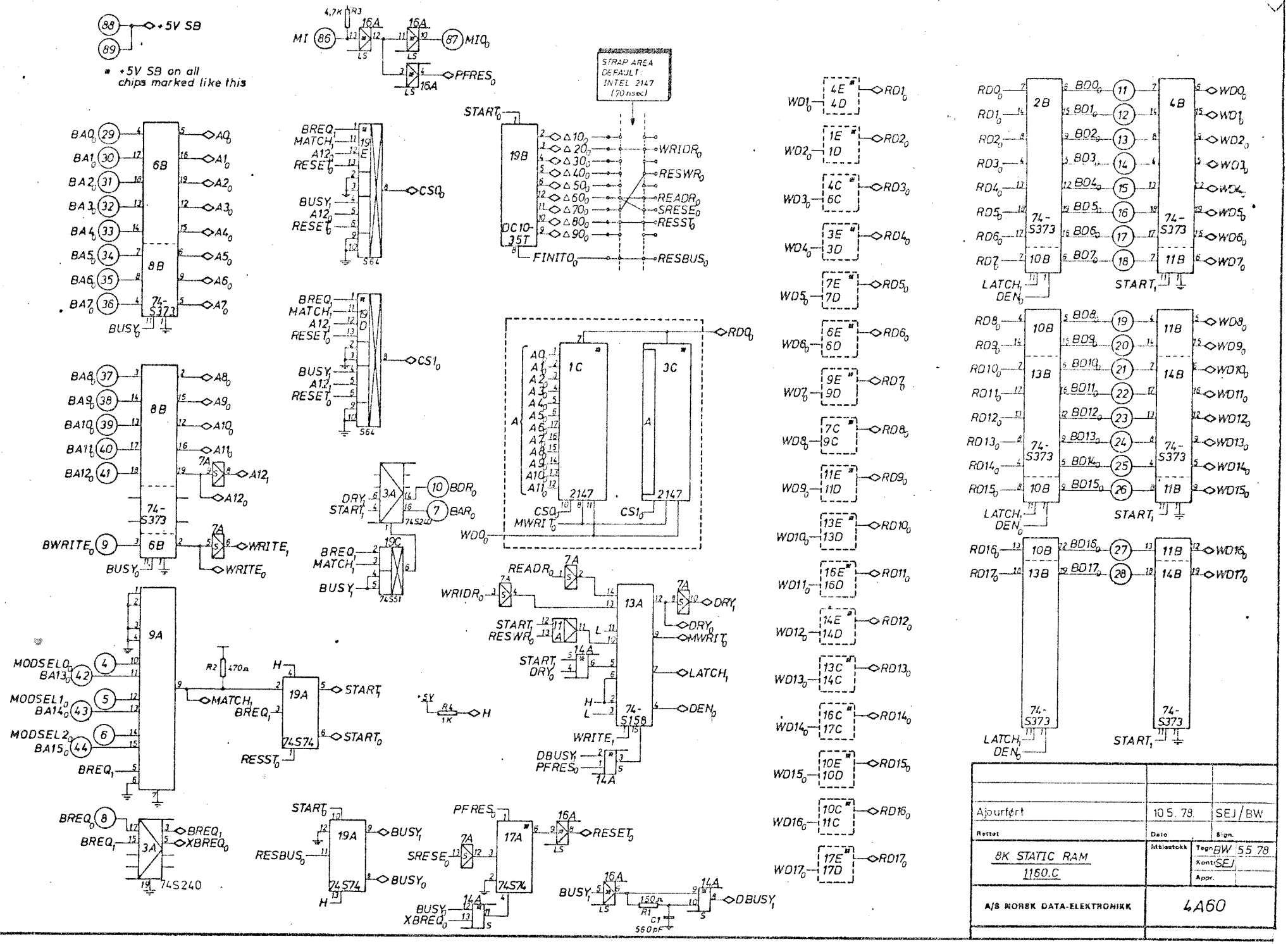
Påført C (ingen rettelser)	14.4.78	TLL/Emd
Rettet	Dato	Sign.
SMD UNIT CONTROL 1156.B,C	Målestokk	Tegn Emd 28.11.77 Kontr. TLL Appr.
NORSK DATA A.S	4A56	


PORT COMMUNICATION
Y LOG
LINE DATA
READ DATA
ERROR INTERRUPT
CONTROL
COMMAND DECODING
READ SEQUENCING
SCAN HIT
RESET INT & LOG
SCAN COUNTER
BIG MPM Y LOG
1157
OPPDAT. T.O.M. ECO 10-475
9.2.78
NORSK DATA A.S
4A57

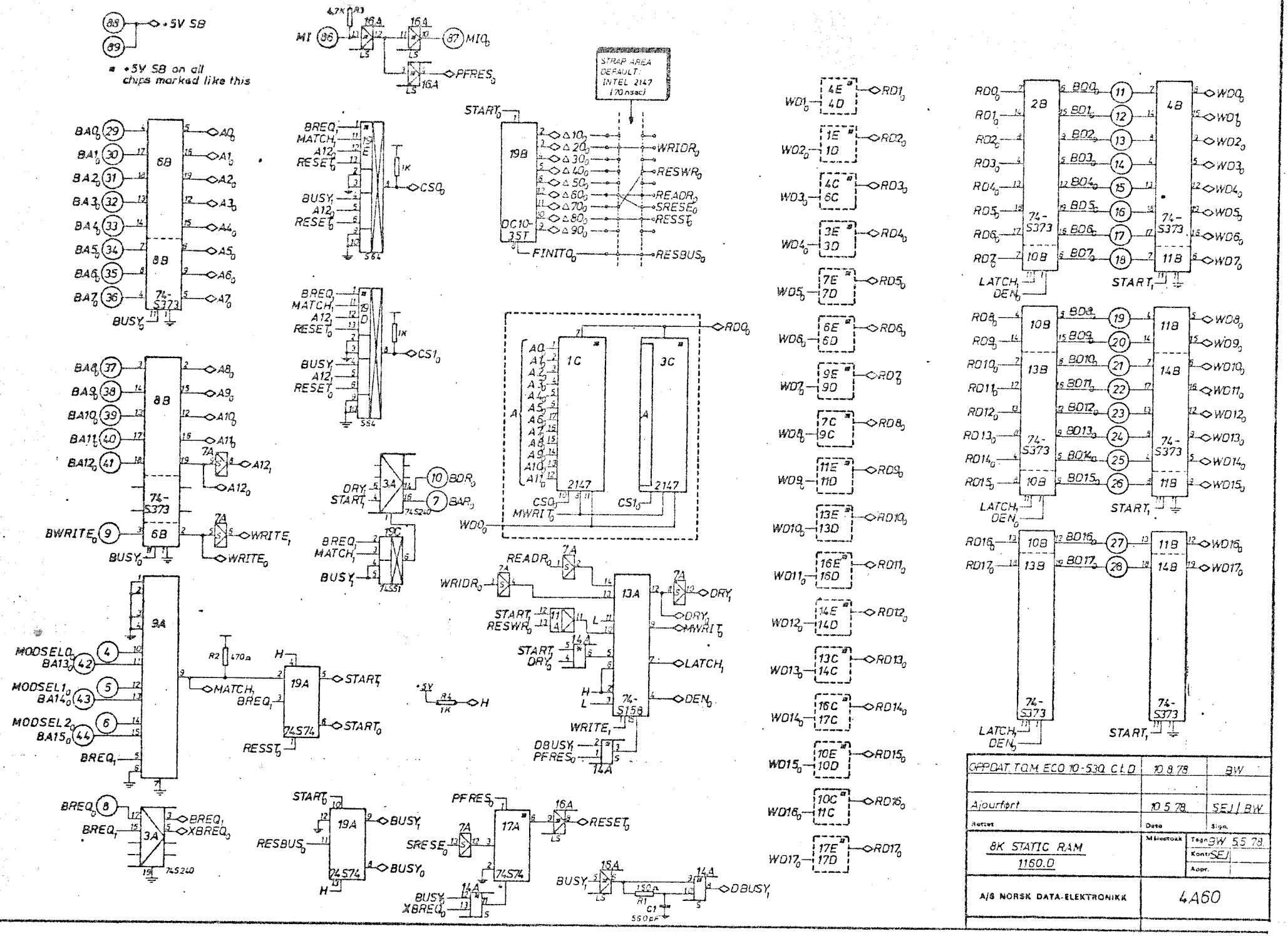

DATABUS TRANCEIVERS
DATA OUTPUT
ALL DIODES ARE 4148
DATA INPUT
CONTROL OUTPUTS
Ident-code selection switch in position 3C
INTERRUPT REQUEST
Device-number switch in 7C
Polarity of interrupt request is selected by switch in 3C. Switch off is DR11-C.
0.4-0.7µs
DR11-C INTERFACE
1158.C
A/S NORSK DATA-ELEKTRONIKK
4A58


DATA TRANSCEIVERS
DATA OUTPUT
ALL DIODES ARE 4148
CONTROL OUTPUTS
Ident - code selection switch in position 3C
Device-number switch in 7C
INTERRUPT REQUEST
Polarity of interrupt request is selected by switch in 3C. Switch off is DR11-C.
0.4 - 0.7 µs
OPPDAT. TOM. ECO NR. 10-509. D
31.5.78
BW
DR11-C INTERFACE
1158. D
A/S NORSK DATA-ELEKTRONIKK
4A58

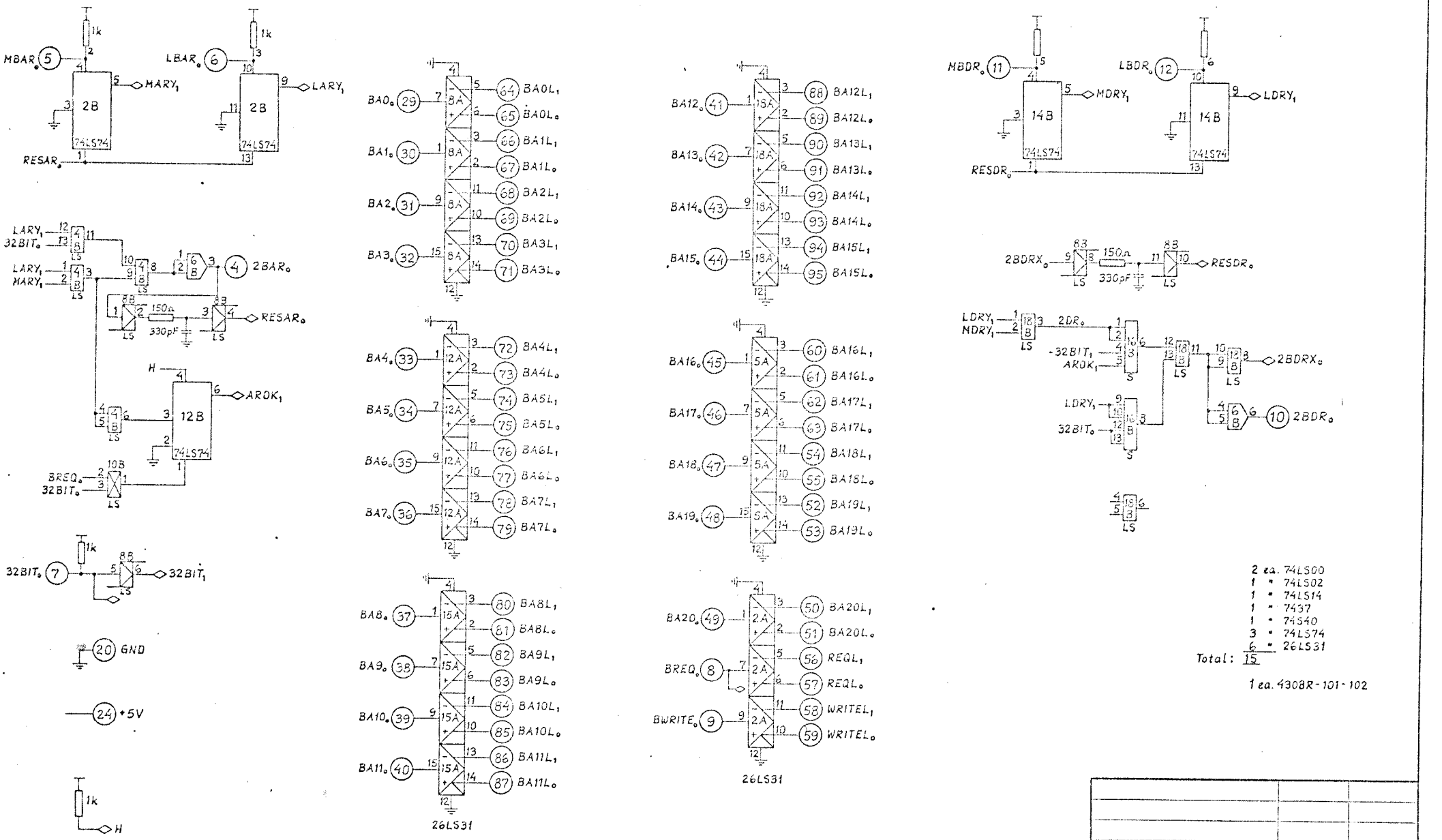

LAYOUT BLOCK X
8 20 15 13 1 3 17 12 4 2 7 18 C
9 14 10 16/21 6 11 19 5 0 B
ADDRESS MULTIPLEXERS
16B
17B
74S257
ONE BIT OF DATA
MOSTEK: MK 4116-3
TEXAS: THS 4071-3
DOT
DIN
WRITX
RASX
CASX
CHIP STROBES
74S51
74S65
74S175
ADDRESS BUFFERS
START MEMORY CYCLE
CYCLE STATUS
ENABLE DATA TO BUS
MEMORY RESPONSE
TIME COUNTER
DATA TRANSCEIVERS
10A
9A
7A
4A
3A
8835
ALL IC'S EXCEPT 1A, 3A, 4A, 7A, 9A AND 10A ARE CONNECTED TO 5V STANDBY
Oppdat. t.o.m. ECO 10-503
3.5.79
BW
16 K RAM
1159
NORSK DATA A.S
4A59



Ajourført	10.5.78	SEJ/BW
Rettet	Dato	Sign.
8K STATIC RAM 1160.C	Målestokk	Tegn BW 55 78 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK	4A60	



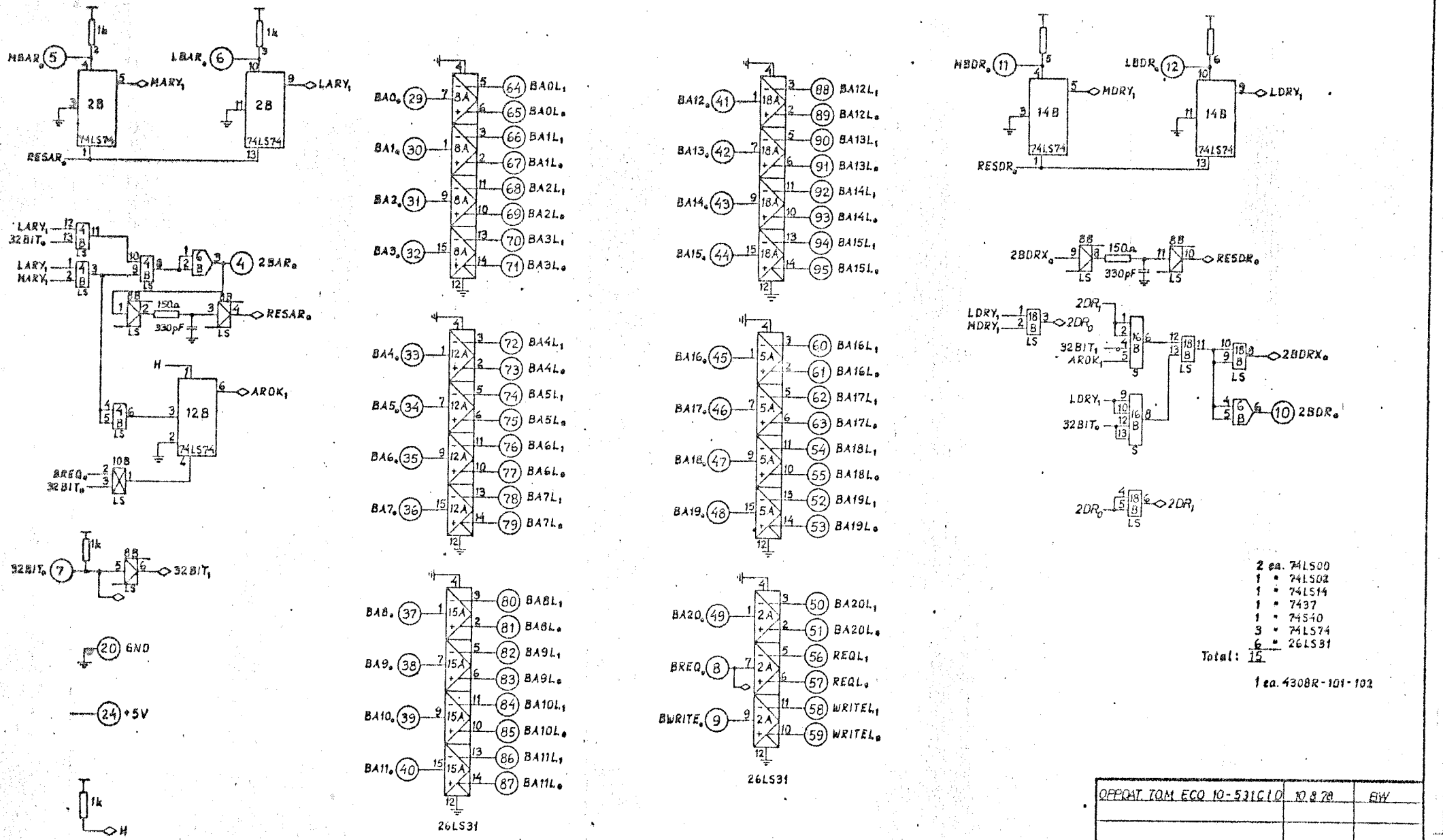
OPPDAT. TQM ECO 10-530 C.L.D	10.8.78	BW
Ajourført	10.5.78	SEJ / BW
Rettet	Dato	Sign.
8K STATIC RAM 1160.0	Målestokk	Tegn BW 5.5.78
		Kontr SEJ
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A60	



2 ea. 74LS00
1 " 74LS02
1 " 74LS14
1 " 7437
1 " 74S40
3 " 74LS74
6 " 26LS31
Total: 15

1 ea. 4308R-101-102

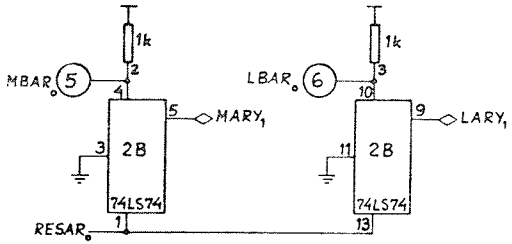
Rettet	Dato	Sign
CHANNEL ADDR. DRIVER 1161.C	Målestokk	Tegn [illegible] 18.4.78 Kontr SEJ Appr
NORSK DATA A.S	4A61	

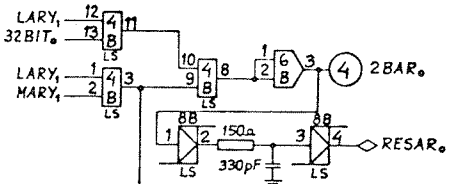


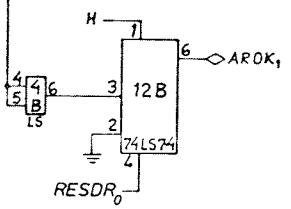
2 ea. 74LS00
1 " 74LS02
1 " 74LS14
1 " 7437
1 " 74S40
3 " 74LS74
6 " 26LS31
Total: 15

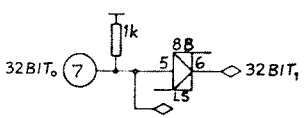
1 ea. 4308R-101-102

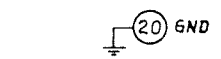
OPPDAT. TOM ECO 10-531C1D	10.8.78	EW
Rettet	Dato	Sign.
CHANNEL ADDR. DRIVER 1161.D	Tegnet G.J. 18.4.78 Kontr SEJ Appr.	
NORSK DATA A.S	4A61	

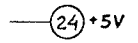


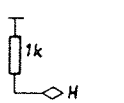


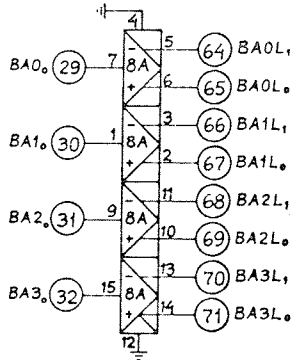


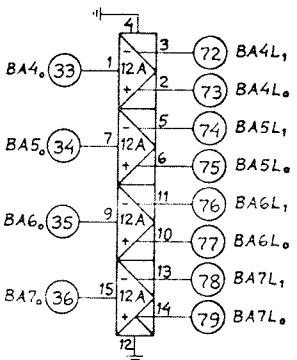


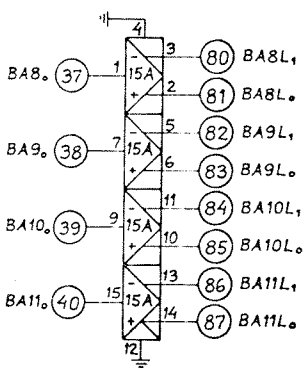




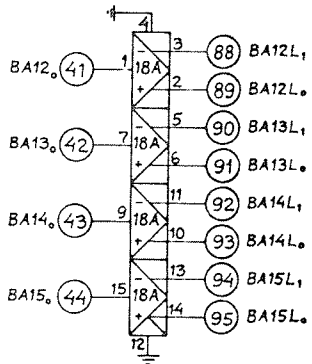


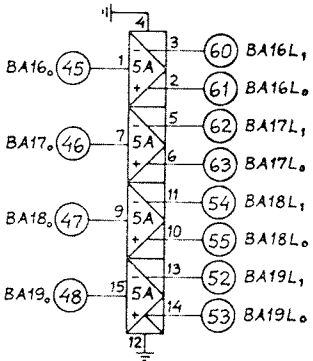


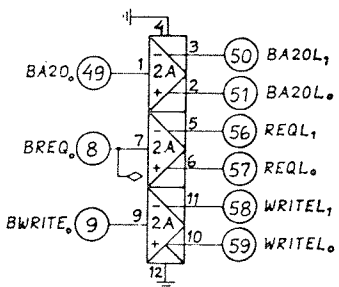




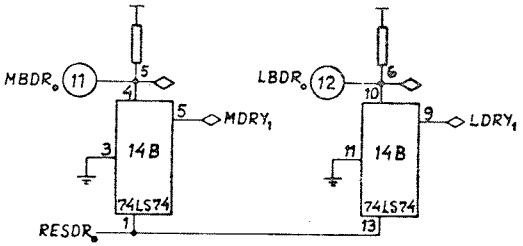
26LS31

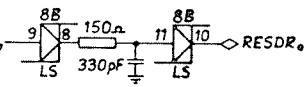


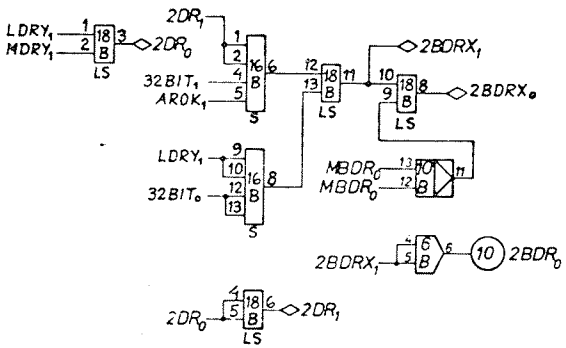




26LS31



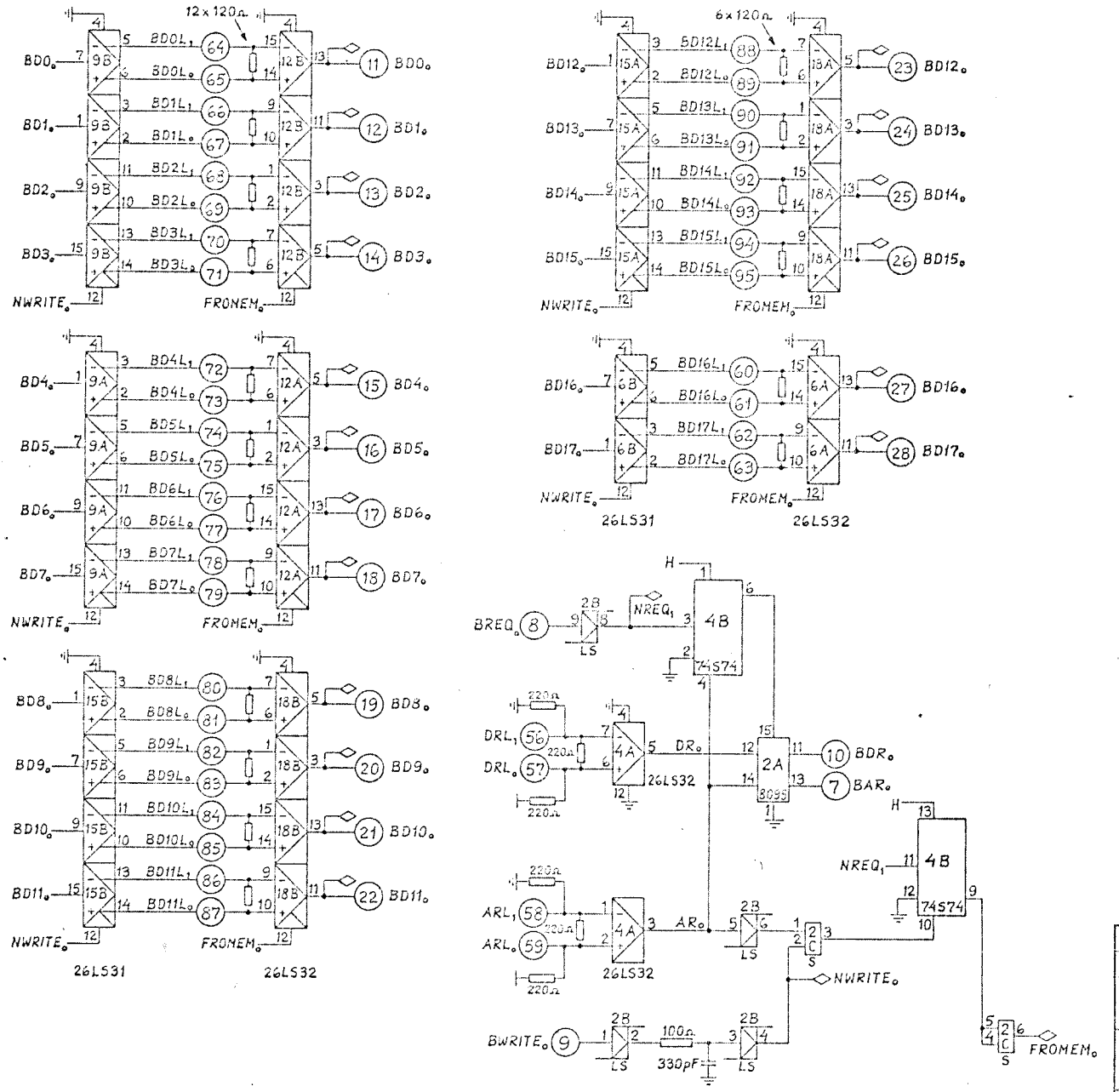




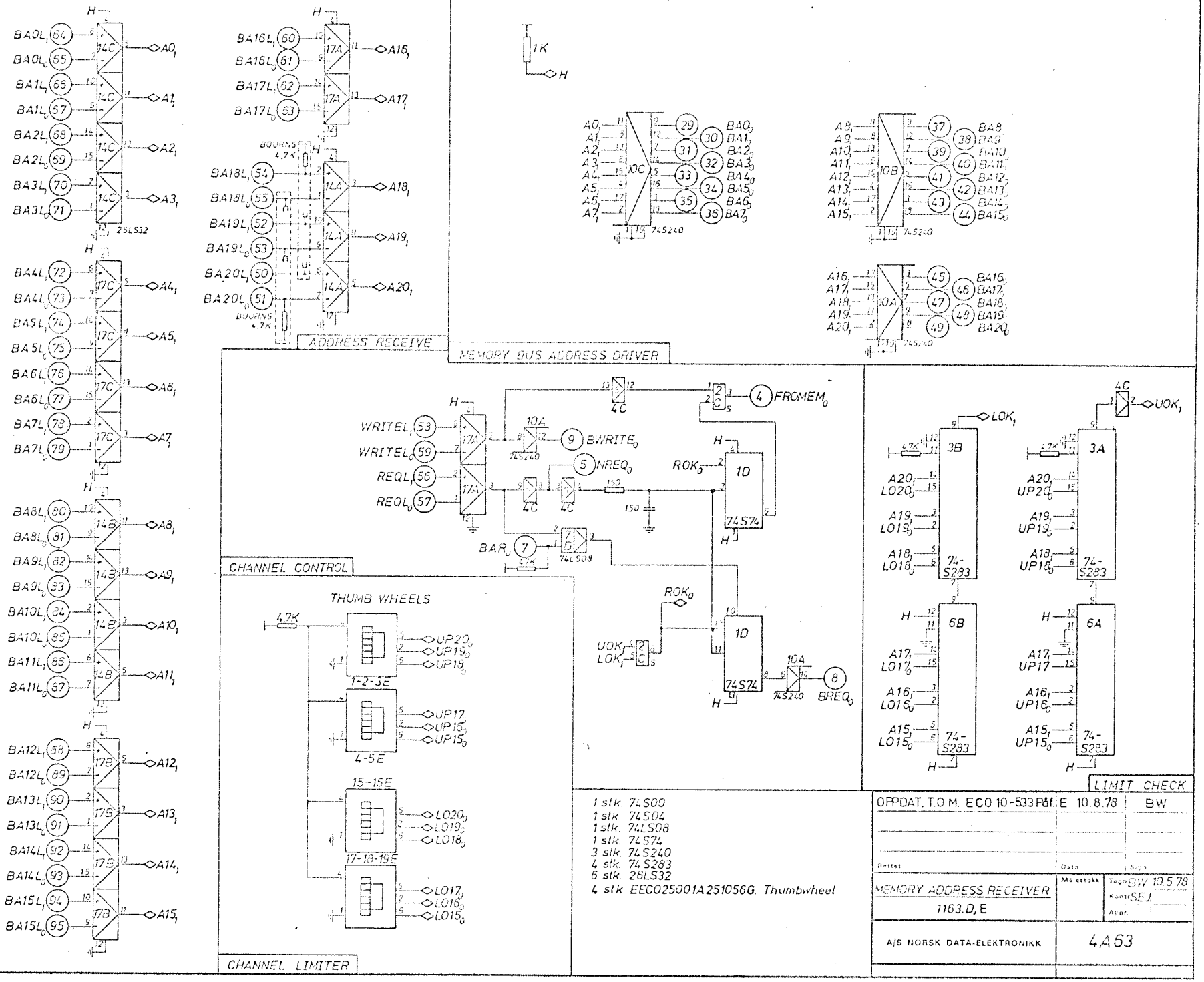
2 ea. 74LS00
1 " 74LS02
1 " 74LS14
1 " 7437
1 " 74S40
3 " 74LS74
6 " 26LS31
Total: 15

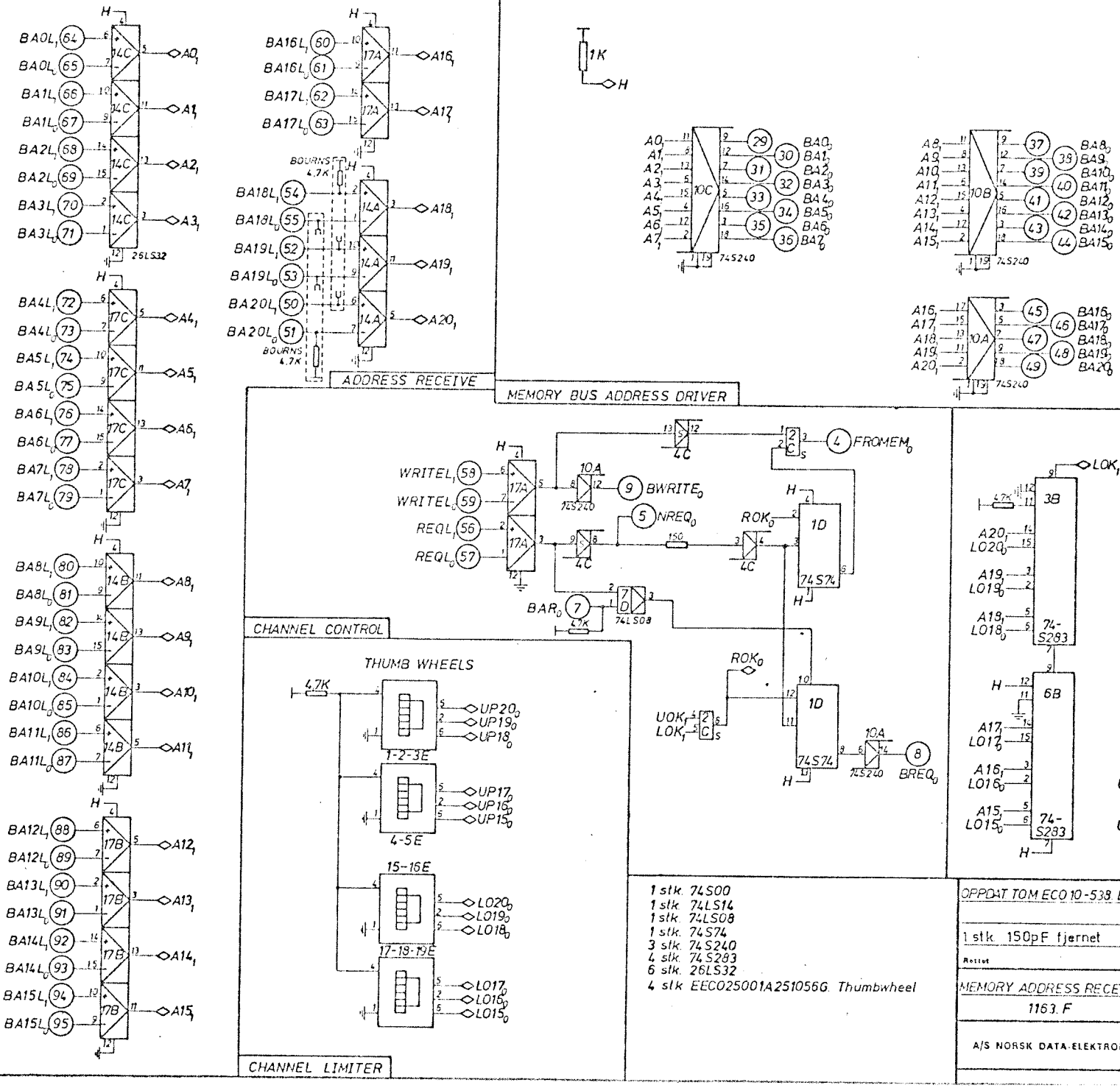
1 ea. 4308R-101-102

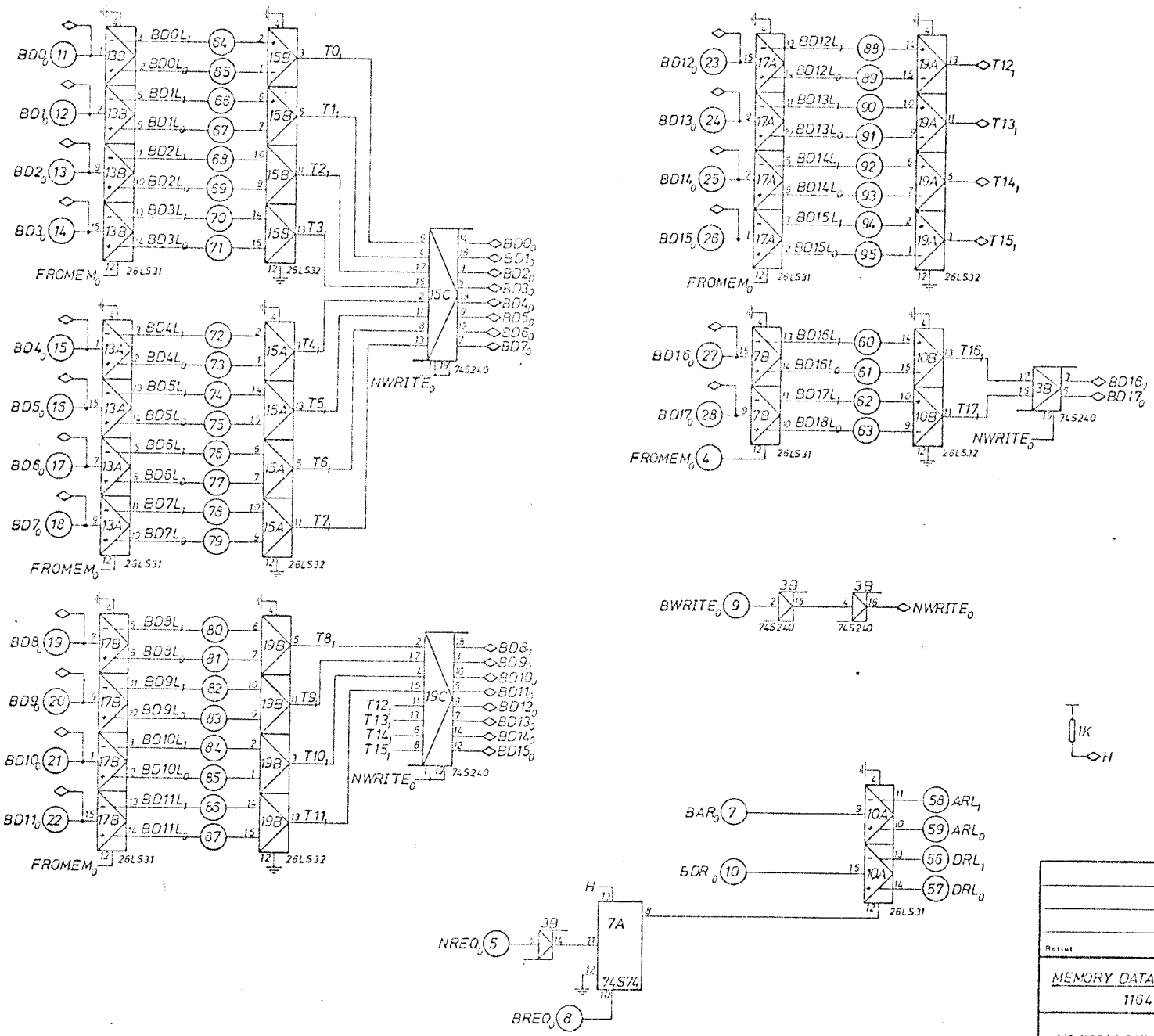
OPPDAT TOM ECO 10-539.01 E	2710.78	BW
Rettet	Dato	Sign.
CHANNEL ADDR. DRIVER 1161, E	Målestokk	Tegn Gmd 18.4.78 Kontr SEJ Appr.
NORSK DATA A.S		4A61



Rettet	Dato	Sign.
CHANNEL DATA DRIVER 1162.D	Målestokk	Tegn. E... 10.5.78 Kontr. SEJ Appr.
NORSK DATA A.S	4A62	

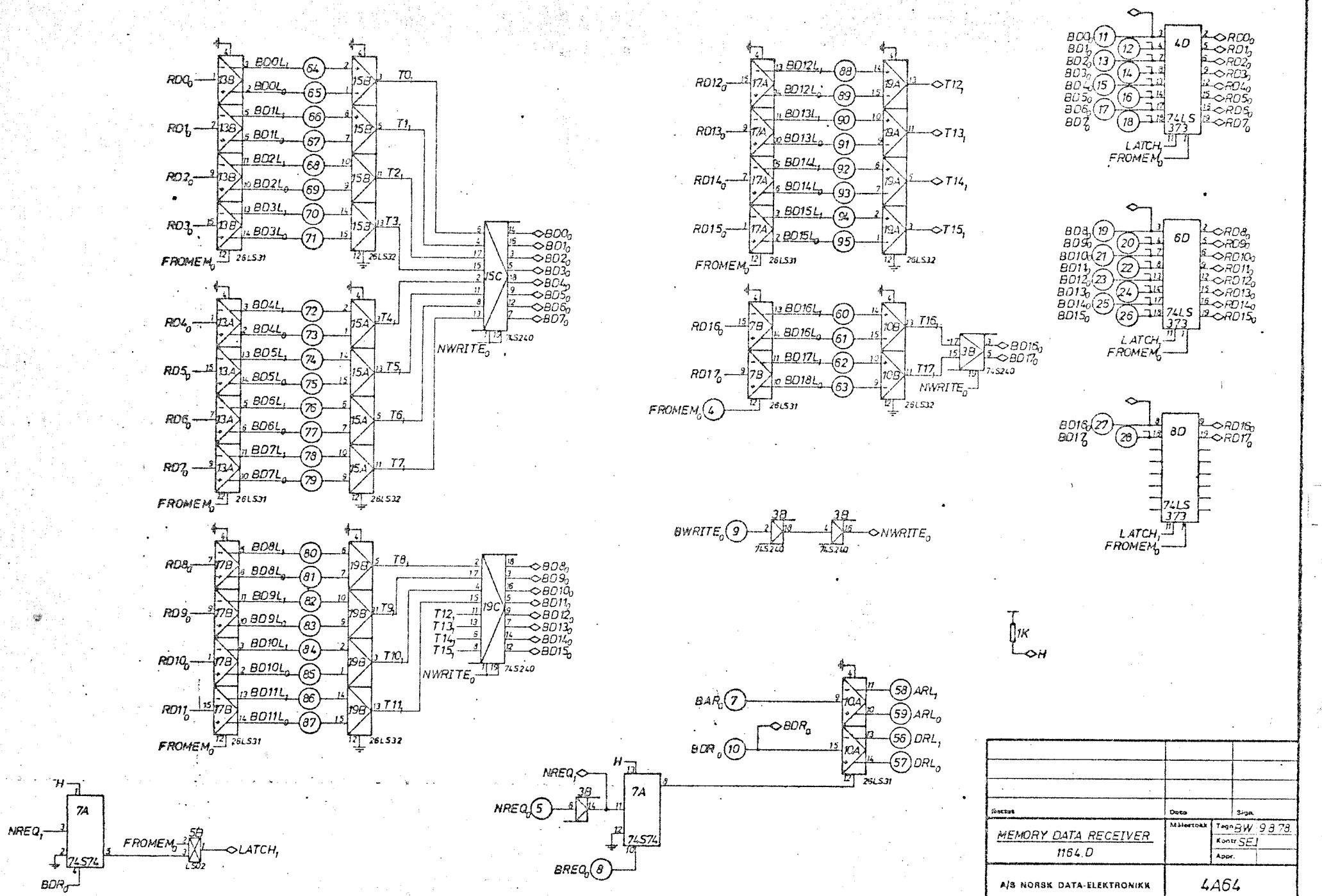

ADDRESS RECEIVE
MEMORY BUS ADDRESS DRIVER
CHANNEL CONTROL
CHANNEL LIMITER
LIMIT CHECK
THUMB WHEELS
1K
4.7K
BOURNS 4.7K
26LS32
74S240
74S74
74LS08
74LS09
74S283
1 stk 74S00
1 stk 74S04
1 stk 74LS08
1 stk 74S74
3 stk 74S240
4 stk 74S283
6 stk 26LS32
4 stk EEC025001A251056G. Thumbwheel
OPPDAT. T.O.M. ECO 10-533 Påf. E 10.8.78 BW
Rettet Dato Sign
MEMORY ADDRESS RECEIVER
1163.D,E
Målestokk
Tegn BW 10.5.78
Kontr SEJ
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A63


ADDRESS RECEIVE
MEMORY BUS ADDRESS DRIVER
CHANNEL CONTROL
CHANNEL LIMITER
THUMB WHEELS
1K
4.7K
BOURNS 4.7K
26LS32
74S240
74S74
74LS08
74-S283
150
1 stk. 74S00
1 stk. 74LS14
1 stk. 74LS08
1 stk. 74S74
3 stk. 74S240
4 stk. 74S283
6 stk. 26LS32
4 stk EEC025001A251056G. Thumbwheel
OPPDAT TOM ECO 10-538 E.I.F 6.9.78 BW
1 stk. 150pF fjernet 20.9.78 SEJ/BW
Rettet Dato Sign
MEMORY ADDRESS RECEIVER
1163. F
Målestokk
Tegn BW 10.5.78
Kontr SEJ
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A63

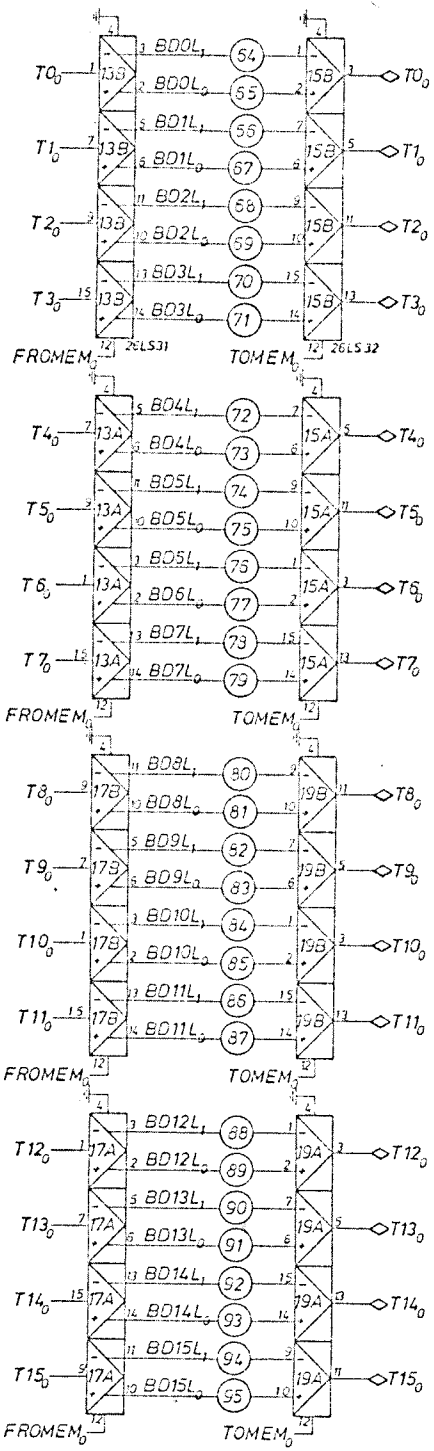


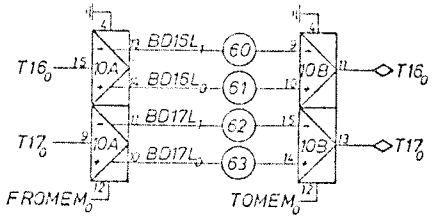
1 ea 74LS74
3 ea 74S240
6 ea 26LS31
5 ea 26LS32

Rettet	Dato	Sign
MEMORY DATA RECEIVER 1164.C	Målestokk	Tegn BW 10 5 78 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK	4A64	

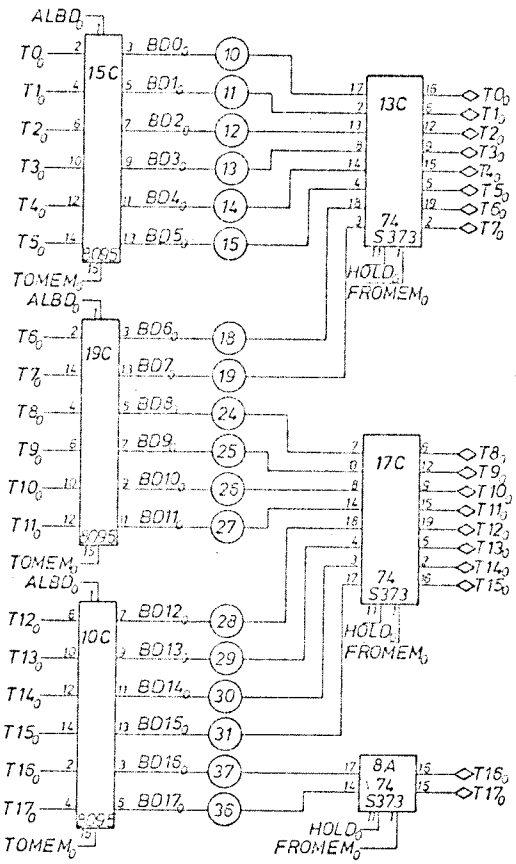


Rettet	Dato	Sign.
MEMORY DATA RECEIVER 1164.D	Målestokk	Tegn BW 9378 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK	4A64	

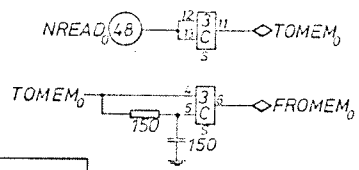




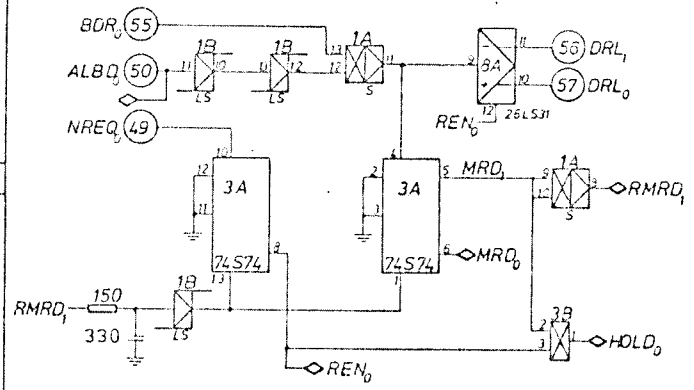
DATA TRANSCEIVE

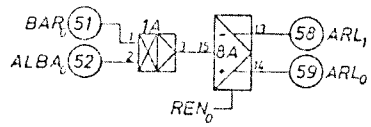


BUS TRANCEIVE



DATA ENABLE

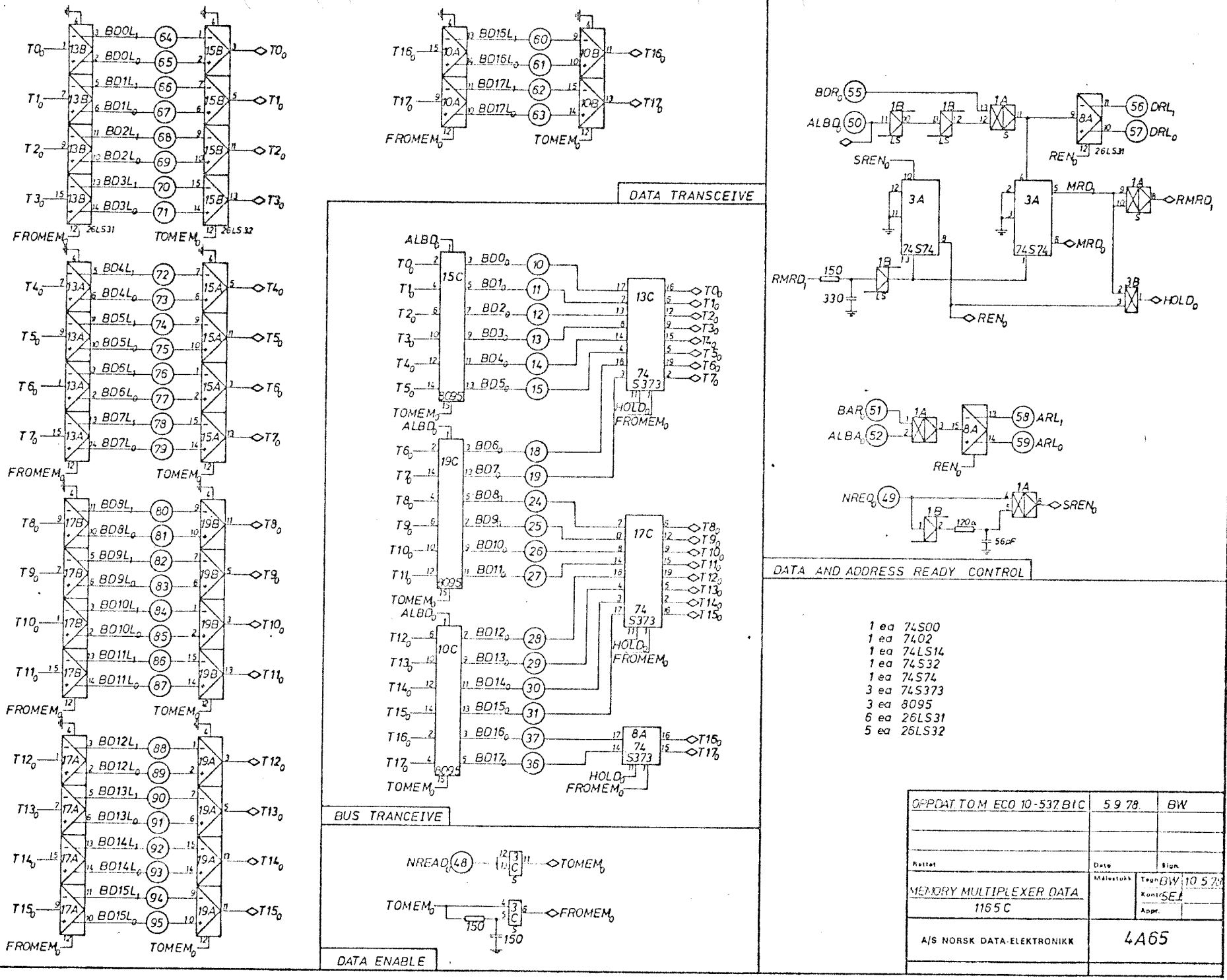


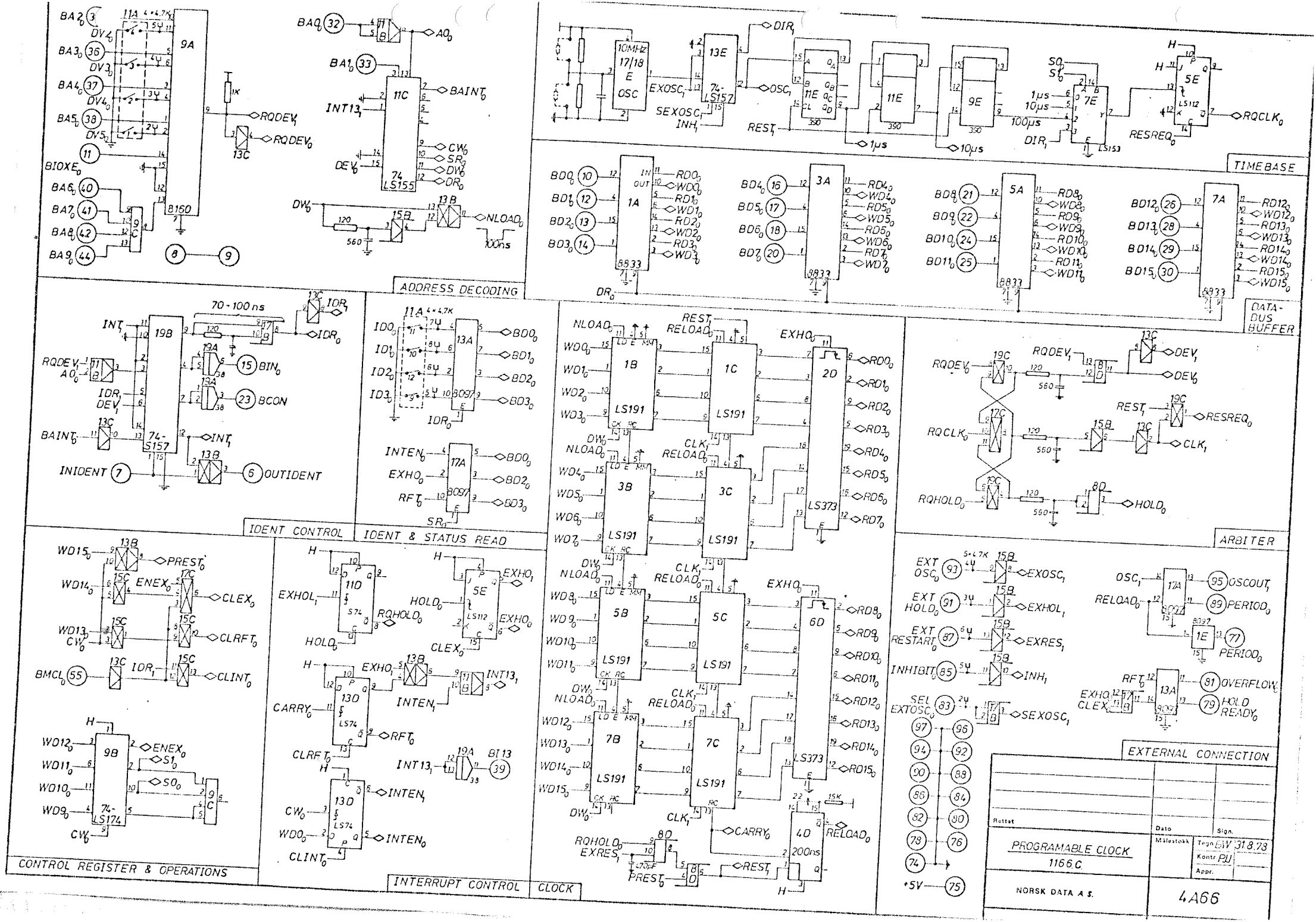


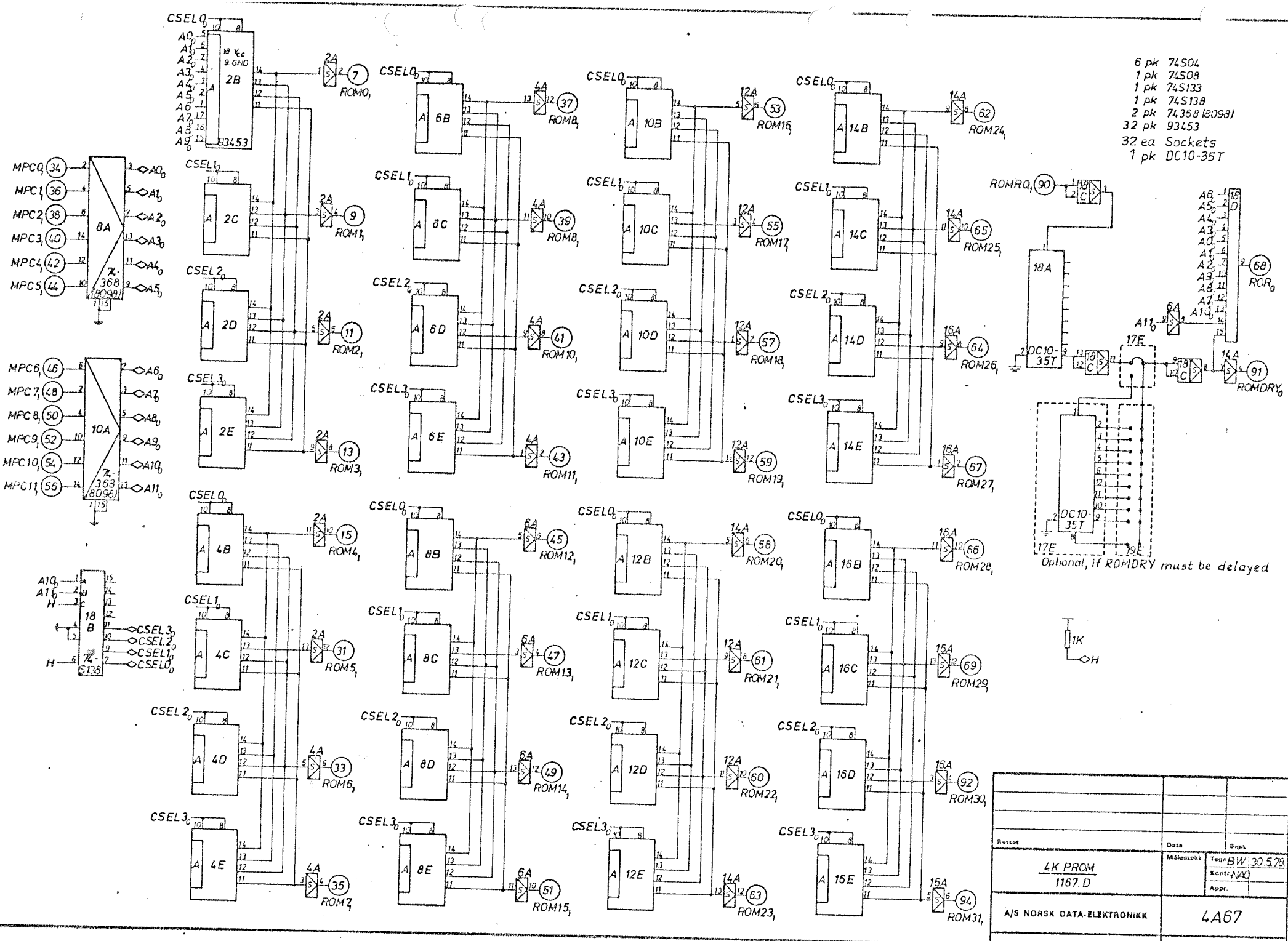
DATA AND ADDRESS READY CONTROL

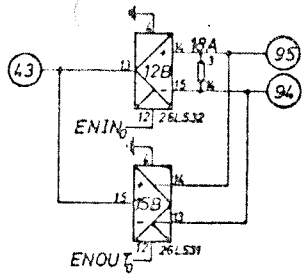
1 ea 74S00
1 ea 7402
1 ea 74LS14
1 ea 74S32
1 ea 74S74
3 ea 74S373
3 ea 8095
6 ea 26LS31
5 ea 26LS32

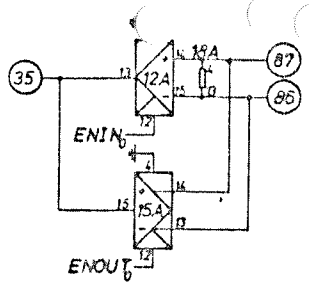
OPPDAT. T.OM ECO 10-532	10.8.78	BW
Rettet	Dato	Sign.
MEMORY MULTIPLEXER DATA 1165 B	Målestokk	Tegn BW 10.5.78 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK		4A65

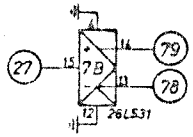

DATA TRANSCEIVE
BUS TRANCEIVE
DATA ENABLE
DATA AND ADDRESS READY CONTROL
FROMEM0
TOMEM0
HOLD0
REN0
SREN0
RMRQ1
MRQ0
HOLD0
NREAD0
NREQ0
ALBA0
ALBQ0
BAR0
BDR0
1 ea 74S00
1 ea 7402
1 ea 74LS14
1 ea 74S32
1 ea 74S74
3 ea 74S373
3 ea 8095
6 ea 26LS31
5 ea 26LS32
OPPDAT TOM ECO 10-537 BIC
5 9 78
BW
Rettet
Dato
Sign.
MEMORY MULTIPLEXER DATA
1165 C
Målestokk
Tegn BW 10 5 78
Kontr SEA
Appr.
A/S NORSK DATA-ELEKTRONIKK
4A65

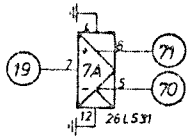

ADDRESS DECODING
TIMEBASE
DATA-BUS BUFFER
IDENT CONTROL
IDENT & STATUS READ
ARBITER
CONTROL REGISTER & OPERATIONS
INTERRUPT CONTROL
CLOCK
EXTERNAL CONNECTION
PROGRAMABLE CLOCK
1166.C
NORSK DATA A.S.
4A66
Rettet
Dato
Sign.
Målestokk
Tegn. BV 31.8.78
Kontr. PJ
Appr.
10MHz
17/18 E OSC
74-LS157
74-LS155
74-LS174
74-LS373
74-LS153
74-LS112
8097
8833
70-100 ns
+5V

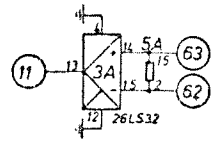

6 pk 74S04
1 pk 74S08
1 pk 74S133
1 pk 74S138
2 pk 74368 (8098)
32 pk 93453
32 ea Sockets
1 pk DC10-35T
MPC0 MPC1 MPC2 MPC3 MPC4 MPC5
MPC6 MPC7 MPC8 MPC9 MPC10 MPC11
A10 A11 H
CSEL0 CSEL1 CSEL2 CSEL3
ROM0 ROM1 ROM2 ROM3 ROM4 ROM5 ROM6 ROM7
ROM8 ROM9 ROM10 ROM11 ROM12 ROM13 ROM14 ROM15
ROM16 ROM17 ROM18 ROM19 ROM20 ROM21 ROM22 ROM23
ROM24 ROM25 ROM26 ROM27 ROM28 ROM29 ROM30 ROM31
ROMRQ0 ROR0 ROMDRY0
1K
Optional, if ROMDRY must be delayed
4K PROM
1167.D
A/S NORSK DATA-ELEKTRONIKK
4A67

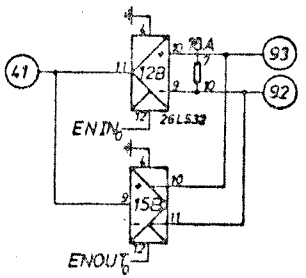


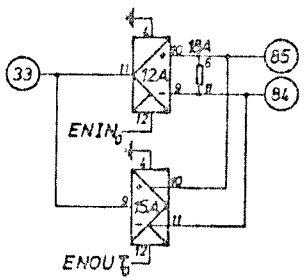


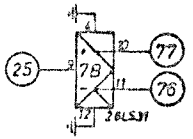


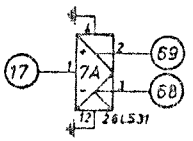


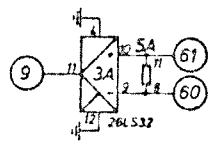


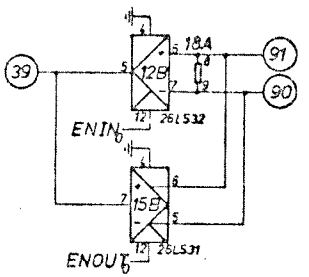


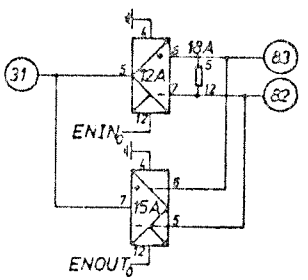


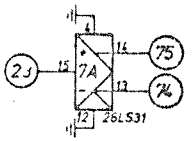


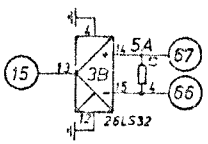


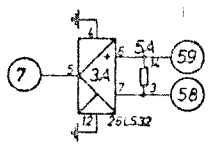


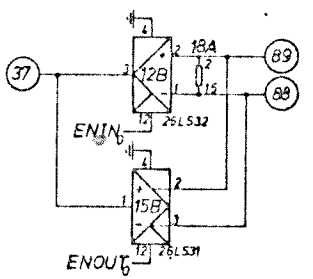


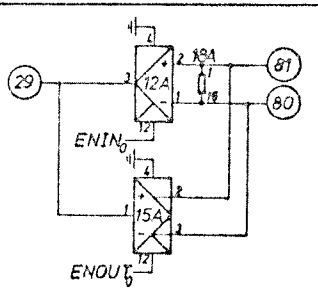


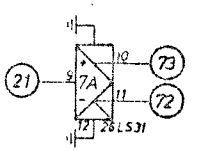


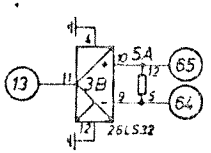


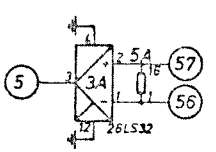


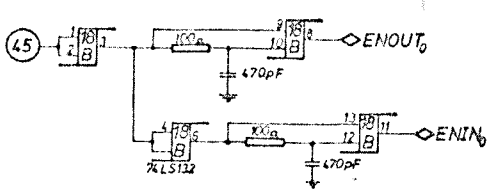






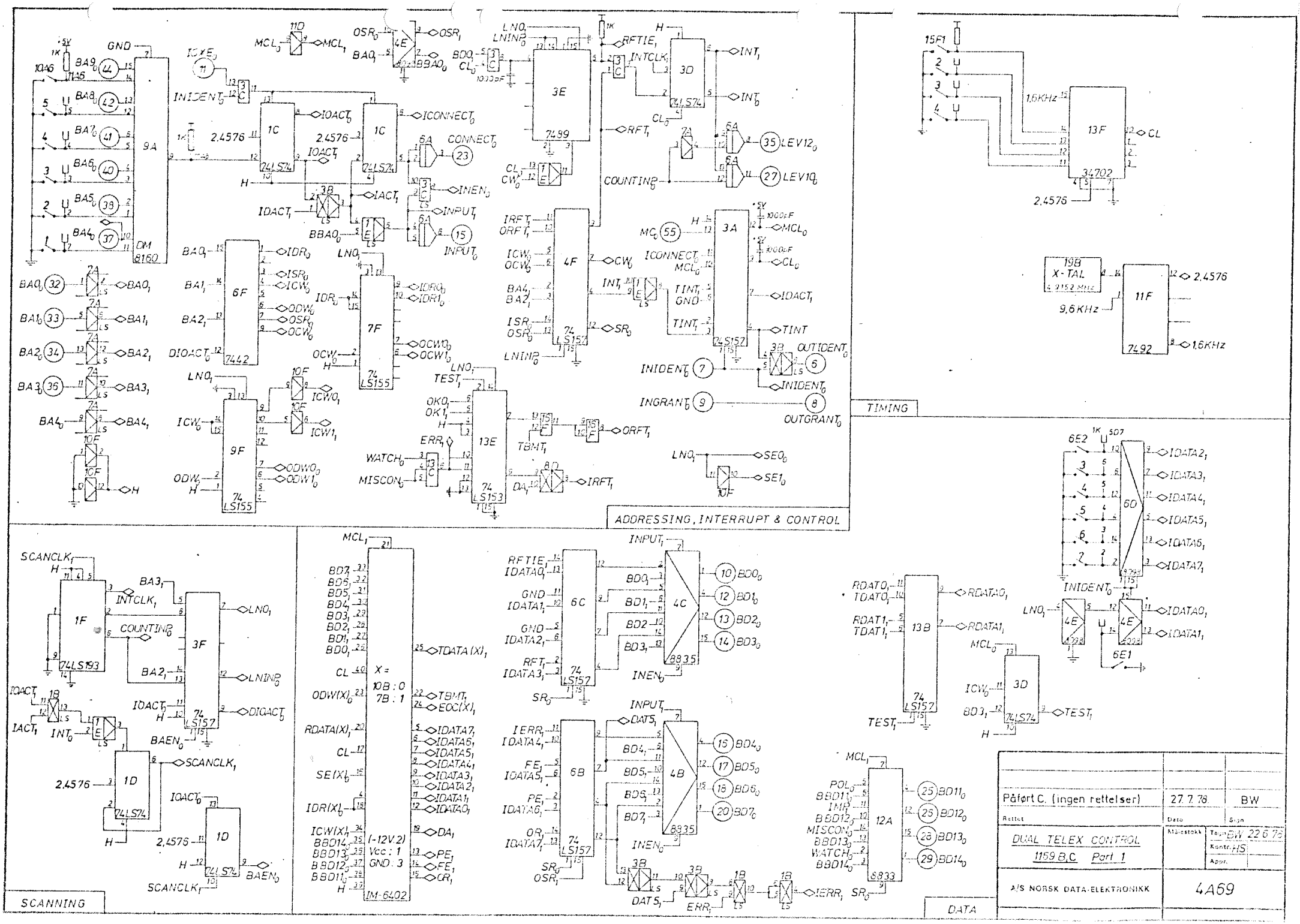






1 pk 74LS132
4 pk 26LS31
4 pk 26LS32
2 pk 4116R-001-121
11 pk

Rettet	Dato	Sign.
DIFFERENTIAL DMA ADOPTOR	Målestokk	Tegn BW 15.8.78
1168		Kontr. PIJ
		Appr.
A/S NORSK DATA-ELEKTRONIKK	4A68	


ADDRESSING, INTERRUPT & CONTROL
TIMING
SCANNING
DATA
DM 8160
74LS74
74LS157
74LS153
74LS155
74LS193
74LS04
74LS00
74LS32
74LS86
74LS08
74LS10
74LS20
74LS27
74LS30
74LS51
74LS86
74LS93
74LS125
74LS126
74LS132
74LS138
74LS139
74LS151
74LS152
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74LS154
74LS155
74LS157
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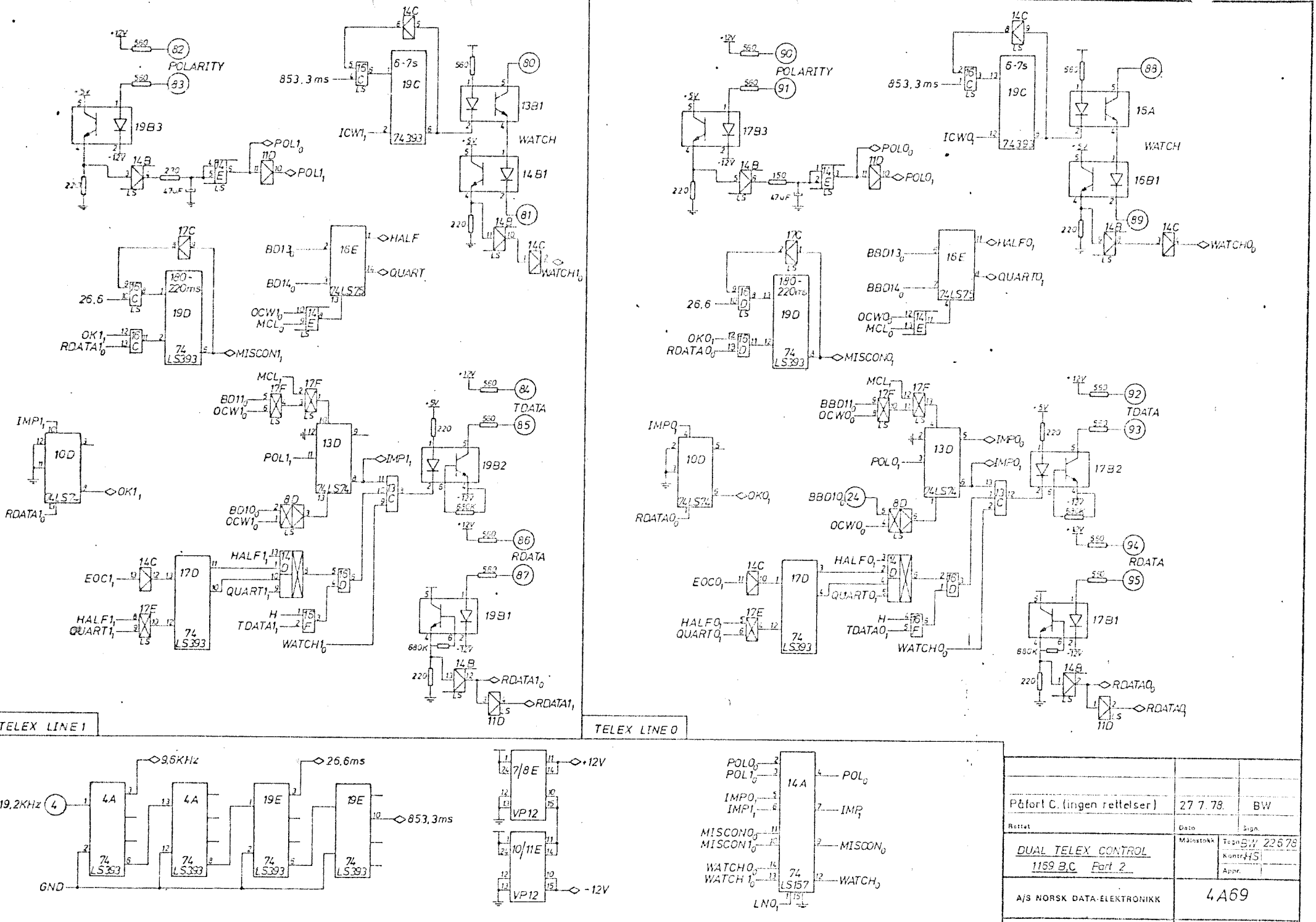
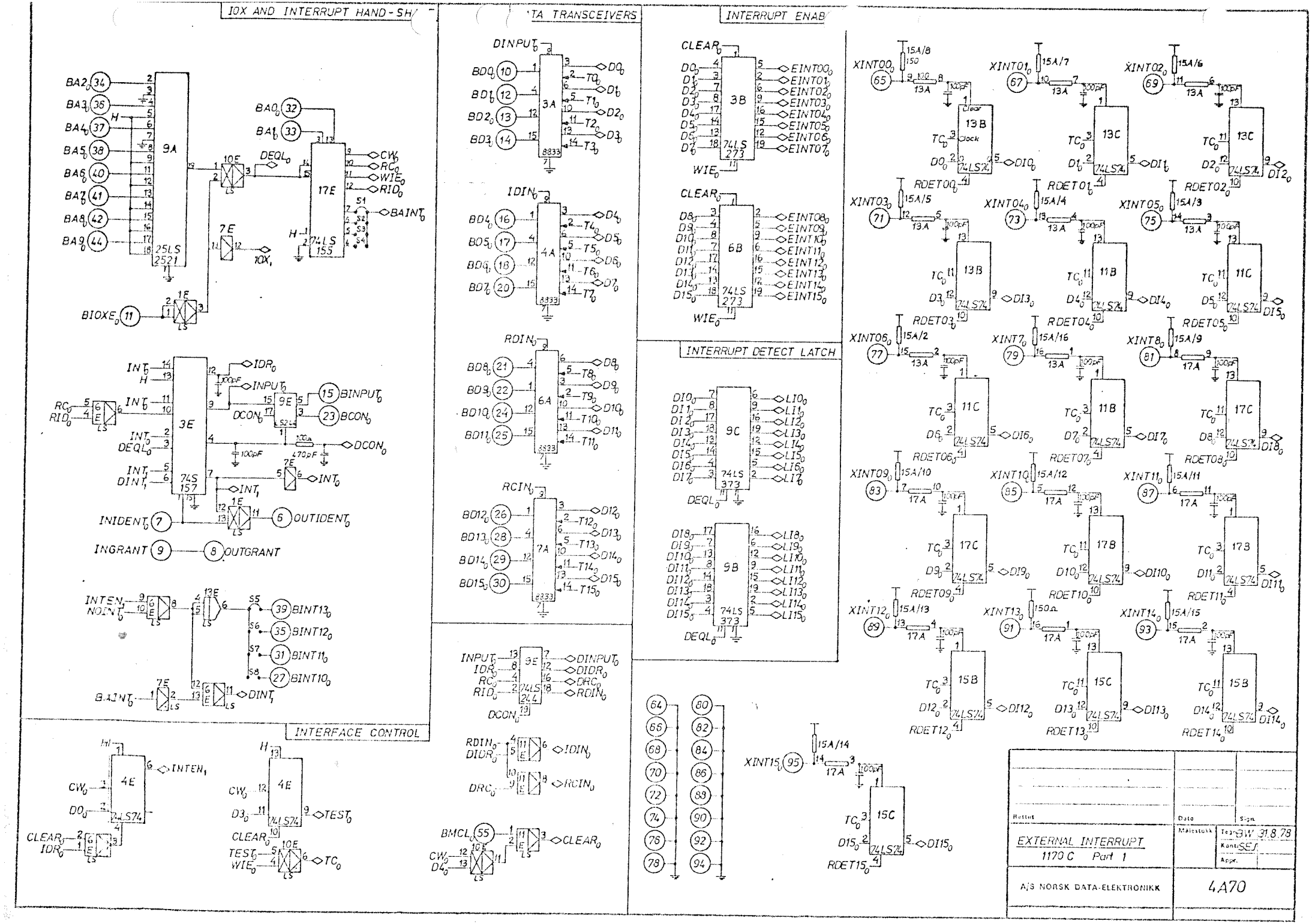
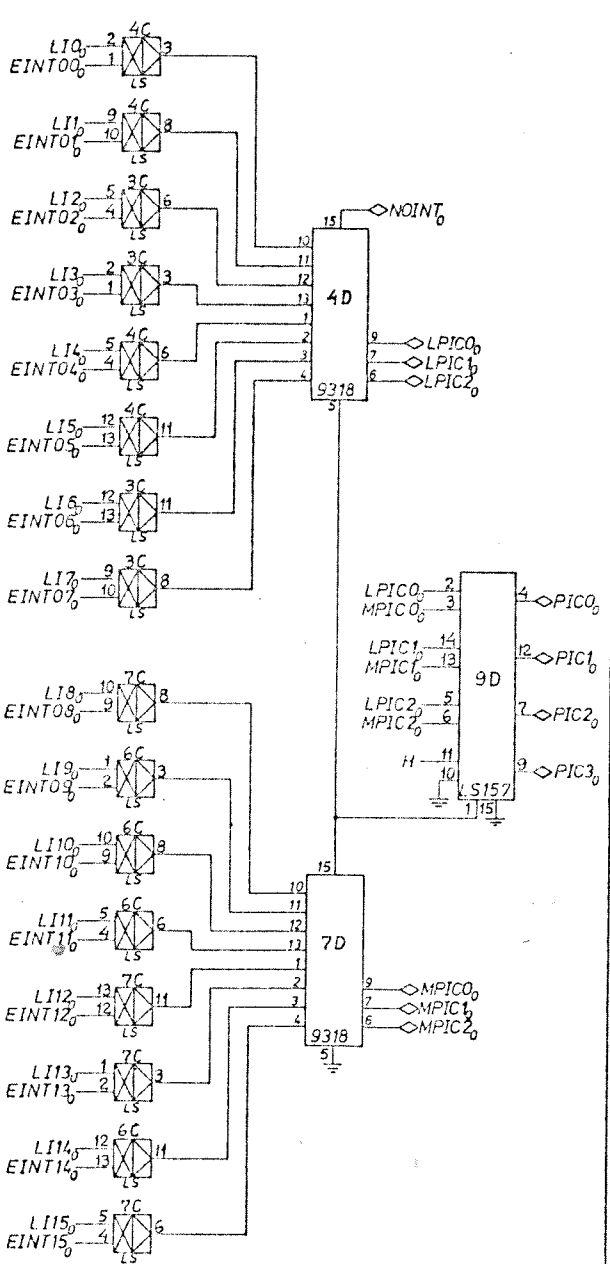
74LS458
74LS459
74LS460
74LS461
74LS462
74LS463

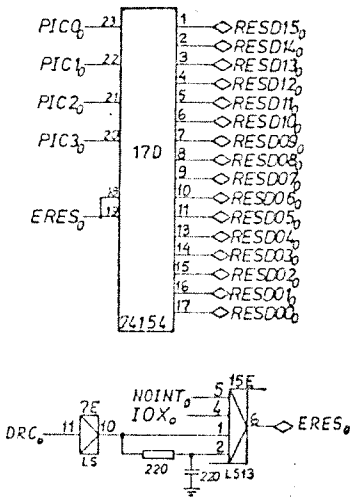

TELEX LINE 1
TELEX LINE 0
POLARITY
TDATA
RDATA
WATCH
HALF
QUART
MISCON
9,6KHz
26,6ms
853,3ms
19,2KHz
GND
+12V
-12V
Påført C. (ingen rettelser)
27 7 78
BW
DUAL TELEX CONTROL
1169 B,C Part 2
A/S NORSK DATA-ELEKTRONIKK
4A69

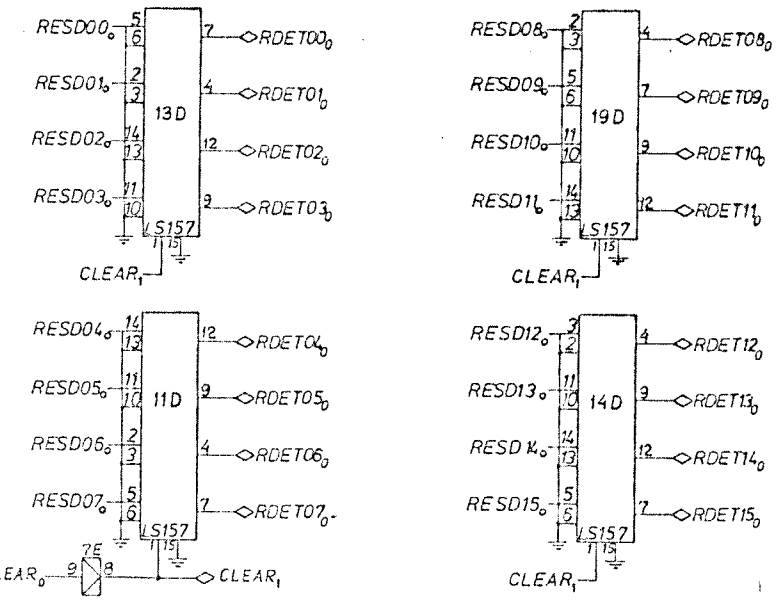

IOX AND INTERRUPT HAND-SH
TA TRANSCEIVERS
INTERRUPT ENAB
INTERRUPT DETECT LATCH
INTERFACE CONTROL
9A
25LS 2521
10E
DEQL0
17E
74LS 155
CW0
RC0
WIE0
RID0
BAINT0
7E
10X1
1E
BIOXE0
BA2 0 (34)
BA3 0 (36)
BA4 0 (37)
BA5 0 (38)
BA6 0 (40)
BA7 0 (41)
BA8 0 (42)
BA9 0 (44)
BA0 0 (32)
BA1 0 (33)
3E
74S 157
INT0
INT1
DEQL0
DIN1
IDR0
INPUT0
DCON0
INIDENT0 (7)
OUTIDENT0 (6)
INGRANT (9)
OUTGRANT (8)
9E
74LS 373
BINPUT0 (15)
BCON0 (23)
DCON0
INT0
INT1
9E
74LS 244
INTEN
NOIN0
BINT13 0 (39)
BINT12 0 (35)
BINT11 0 (31)
BINT10 0 (27)
DINT1
4E
74LS74
INTEN1
TEST0
TC0
IDR0
CLEAR0
WIE0
DINPUT0
BD0 0 (10)
BD1 0 (12)
BD2 0 (13)
BD3 0 (14)
3A
8833
DO0
T0
D1 0
T1 0
D2 0
T2 0
D3 0
T3 0
IDIN0
BD4 0 (16)
BD5 0 (17)
BD6 0 (18)
BD7 0 (20)
4A
D4 0
T4 0
D5 0
T5 0
D6 0
T6 0
D7 0
T7 0
RDIN0
BD8 0 (21)
BD9 0 (22)
BD10 0 (24)
BD11 0 (25)
6A
D8 0
T8 0
D9 0
T9 0
D10 0
T10 0
D11 0
T11 0
RCIN0
BD12 0 (26)
BD13 0 (28)
BD14 0 (29)
BD15 0 (30)
7A
D12 0
T12 0
D13 0
T13 0
D14 0
T14 0
D15 0
T15 0
INPUT0
IDR0
RC0
RID0
DCON0
9E
74LS 244
DINPUT0
DIDR0
DRC0
RDIN0
DIDR0
IDIN0
DRC0
RCIN0
BMCL0 (55)
CW0
D4 0
10E
CLEAR0
CLEAR0
WIE0
3B
74LS 273
EINT00 0
EINT01 0
EINT02 0
EINT03 0
EINT04 0
EINT05 0
EINT06 0
EINT07 0
6B
EINT08 0
EINT09 0
EINT10 0
EINT11 0
EINT12 0
EINT13 0
EINT14 0
EINT15 0
9C
74LS 373
LI0 0
LI1 0
LI2 0
LI3 0
LI4 0
LI5 0
LI6 0
LI7 0
DEQL0
9B
74LS 373
LI8 0
LI9 0
LI10 0
LI11 0
LI12 0
LI13 0
LI14 0
LI15 0
DEQL0
XINT00 0 (65)
XINT01 0 (67)
XINT02 0 (69)
XINT03 0 (71)
XINT04 0 (73)
XINT05 0 (75)
XINT06 0 (77)
XINT7 0 (79)
XINT8 0 (81)
XINT09 0 (83)
XINT10 0 (85)
XINT11 0 (87)
XINT12 0 (89)
XINT13 0 (91)
XINT14 0 (93)
XINT15 0 (95)
13B
13C
11B
11C
17C
17B
15B
15C
74LS74
RDET00 0
RDET01 0
RDET02 0
RDET03 0
RDET04 0
RDET05 0
RDET06 0
RDET07 0
RDET08 0
RDET09 0
RDET10 0
RDET11 0
RDET12 0
RDET13 0
RDET14 0
RDET15 0
DI0 0
DI1 0
DI2 0
DI3 0
DI4 0
DI5 0
DI6 0
DI7 0
DI8 0
DI9 0
DI10 0
DI11 0
DI12 0
DI13 0
DI14 0
DI15 0
64
66
68
70
72
74
76
78
80
82
84
86
89
90
92
94
EXTERNAL INTERRUPT
1170 C Part 1
A/S NORSK DATA-ELEKTRONIKK
4A70
Date
Sign.
Milestone
Tegn. SW 31.8.78
Kontr. SEJ
Appr.

PRIORITY ENCODER

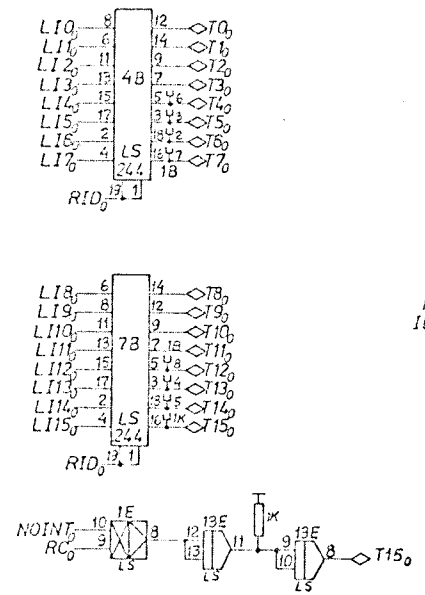
INTERRUPT RESET







DATA ENABLE



1 ea	74LS04
2 ea	74LS08
1 ea	74LS13
6 ea	74LS32
1 ea	74LS38
9 ea	74LS74
1 ea	74154
1 ea	74S157
5 ea	74LS157
3 ea	74LS241
1 ea	74LS257
2 ea	74LS273
2 ea	74LS373
1 ea	25LS2521
4 ea	DM8833
2 ea	DM9318
1 ea	74LS155

Rettet	Dato	Sign.

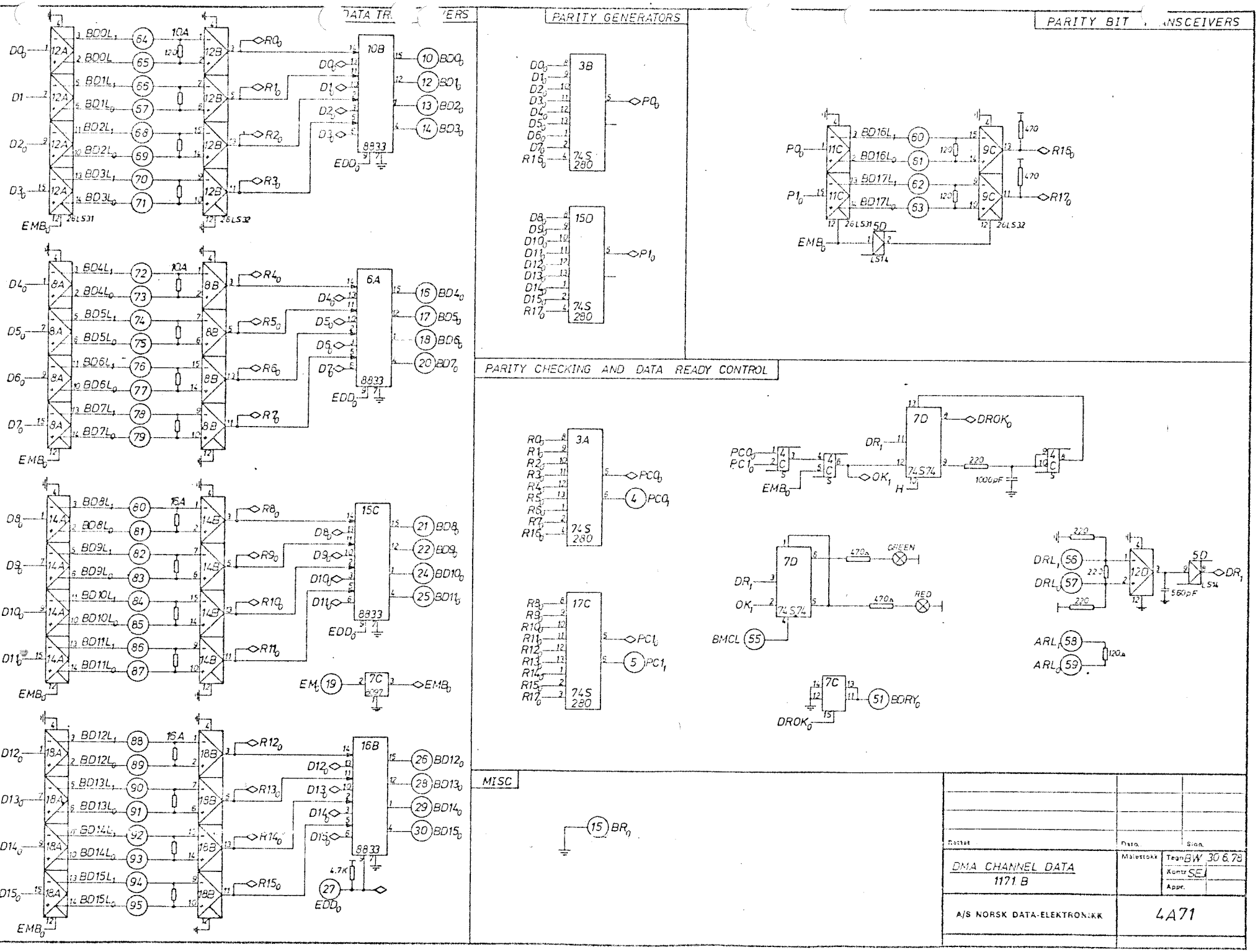
EXTERNAL INTERRUPT
1170C Part 2

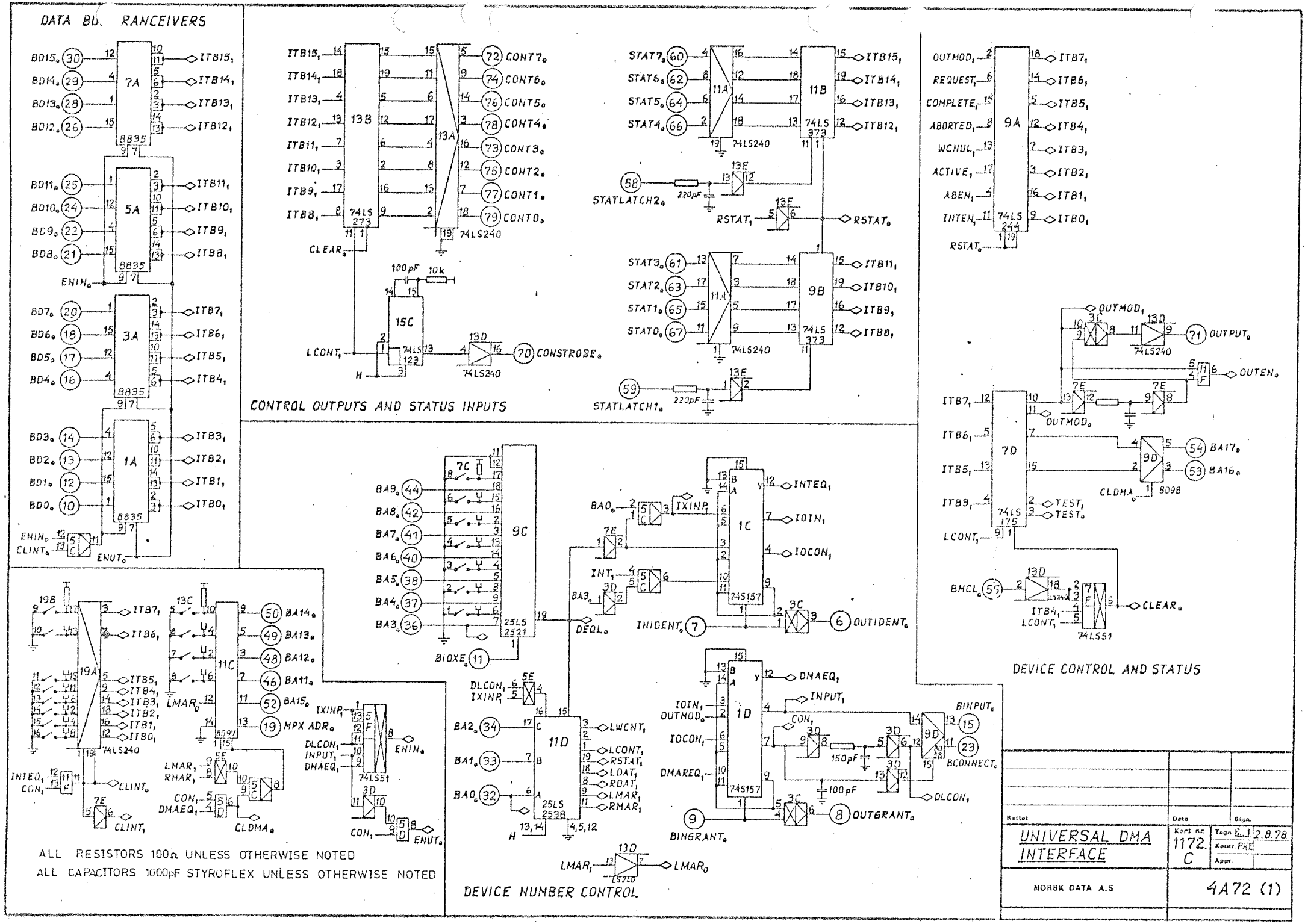
Målestokk

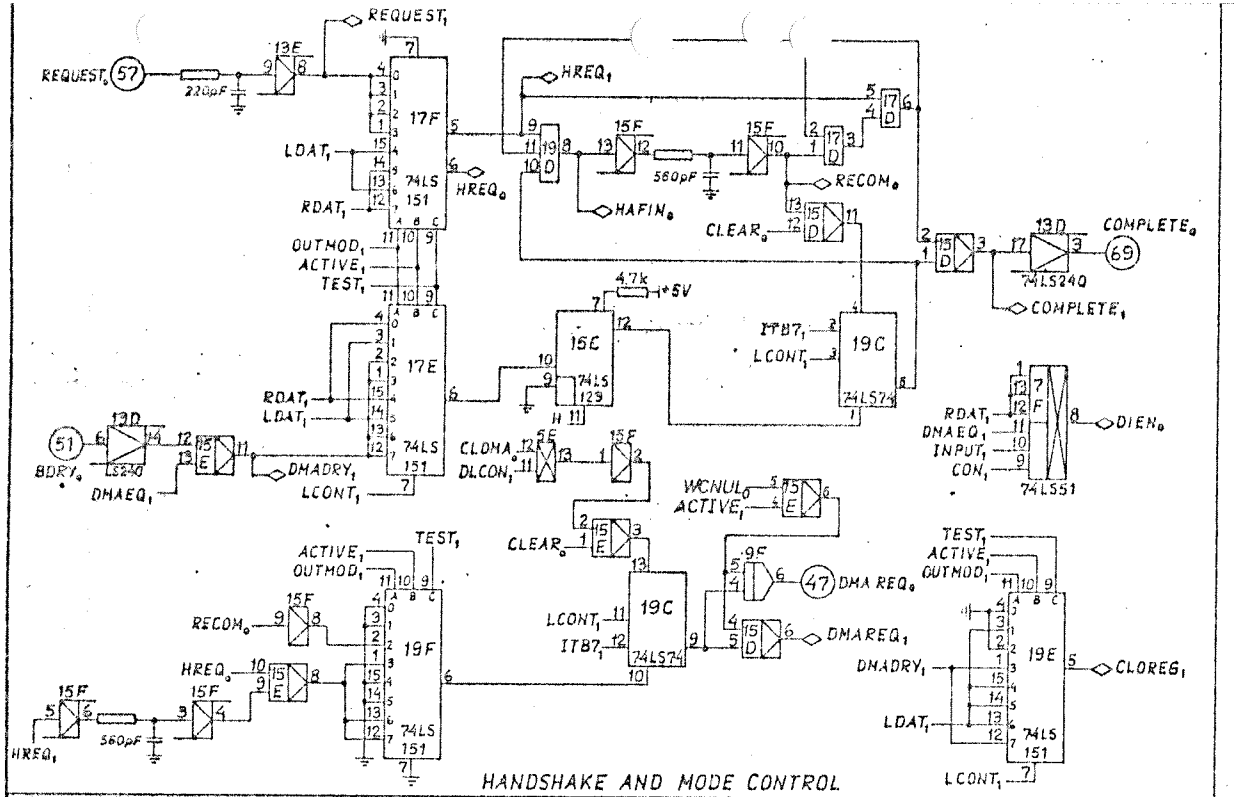
Tegn BW 31.8.78
Kontr SEJ
Appr.

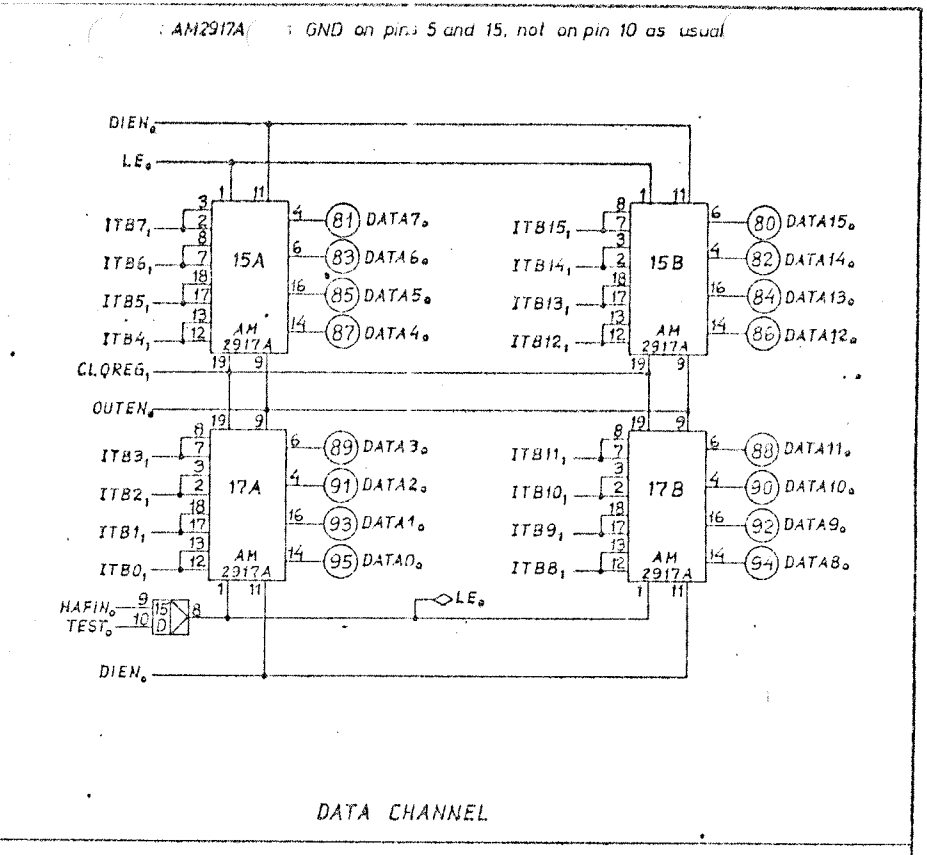
A/S NORSK DATA-ELEKTRONIKK

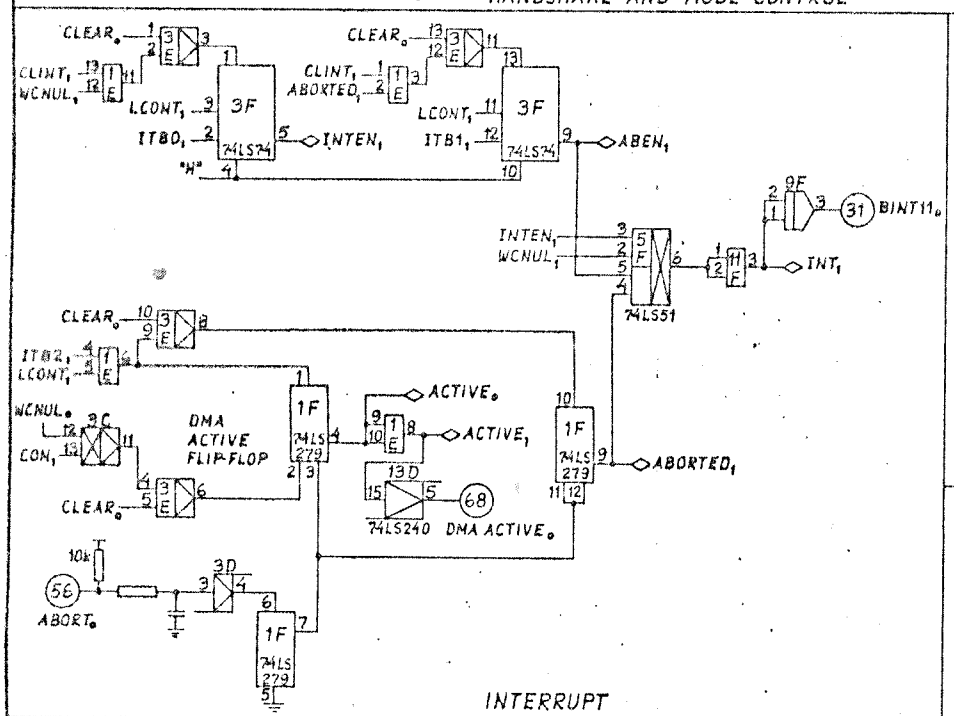
4A70

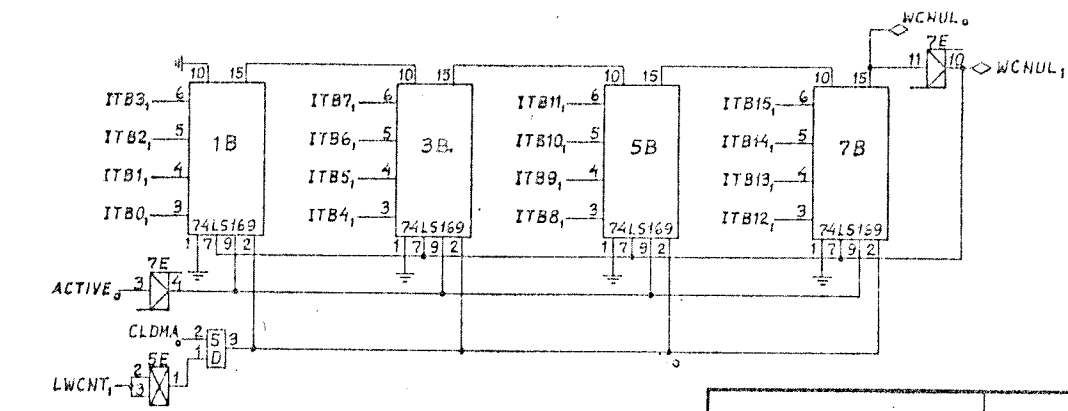

DATA TR... ...ERS
PARITY GENERATORS
PARITY BIT ... ...NSCEIVERS
PARITY CHECKING AND DATA READY CONTROL
MISC
DMA CHANNEL DATA
1171 B
A/S NORSK DATA-ELEKTRONIKK
4A71
Tegn BW 30.6.78
Kontr SE
Appr.
Dato
Sign.
Rettet
Malestokk


DATA BD. RANCEIVERS
CONTROL OUTPUTS AND STATUS INPUTS
DEVICE CONTROL AND STATUS
DEVICE NUMBER CONTROL
ALL RESISTORS 100Ω UNLESS OTHERWISE NOTED
ALL CAPACITORS 1000pF STYROFLEX UNLESS OTHERWISE NOTED
UNIVERSAL DMA INTERFACE
1172. C
NORSK DATA A.S
4A72 (1)



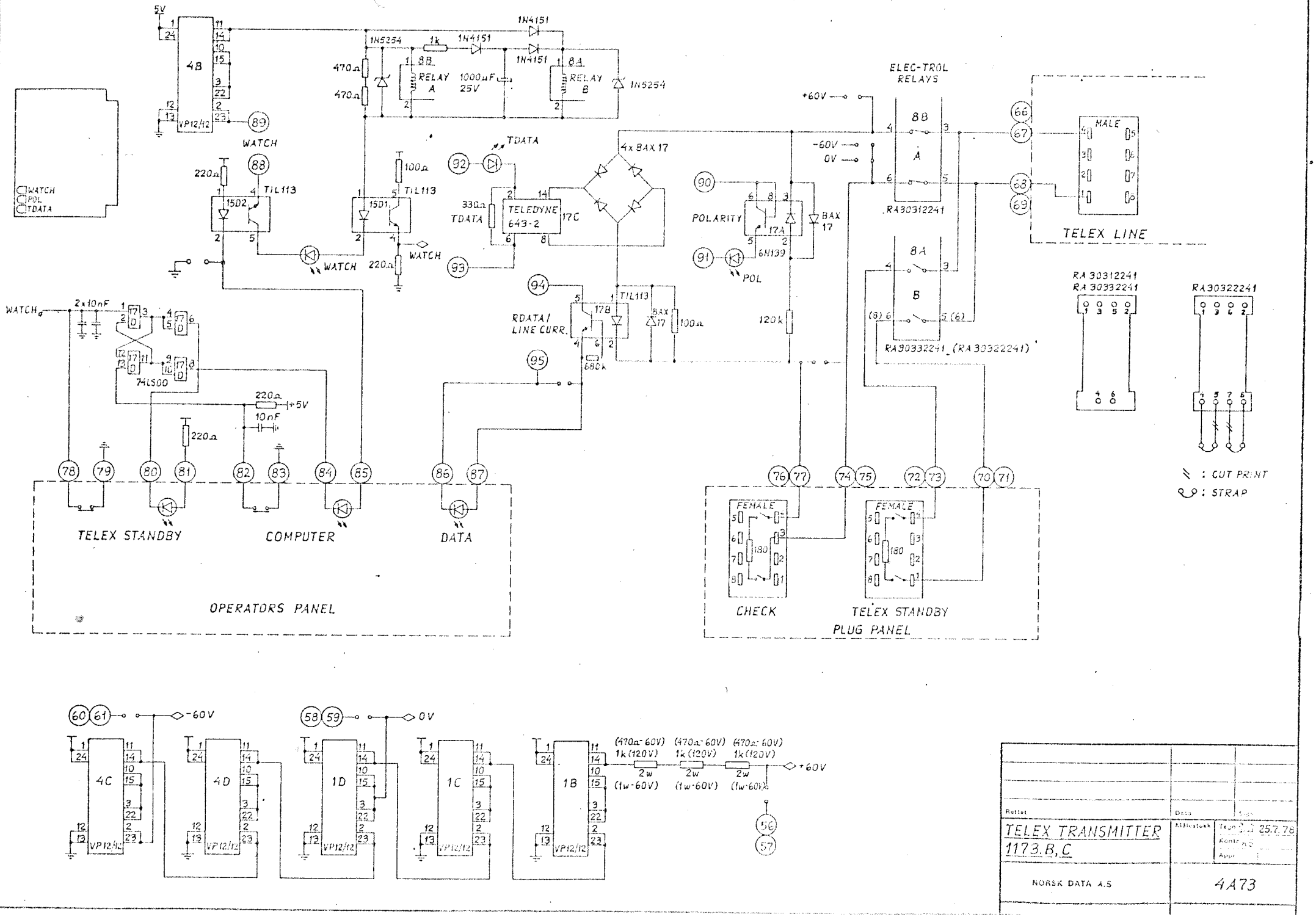




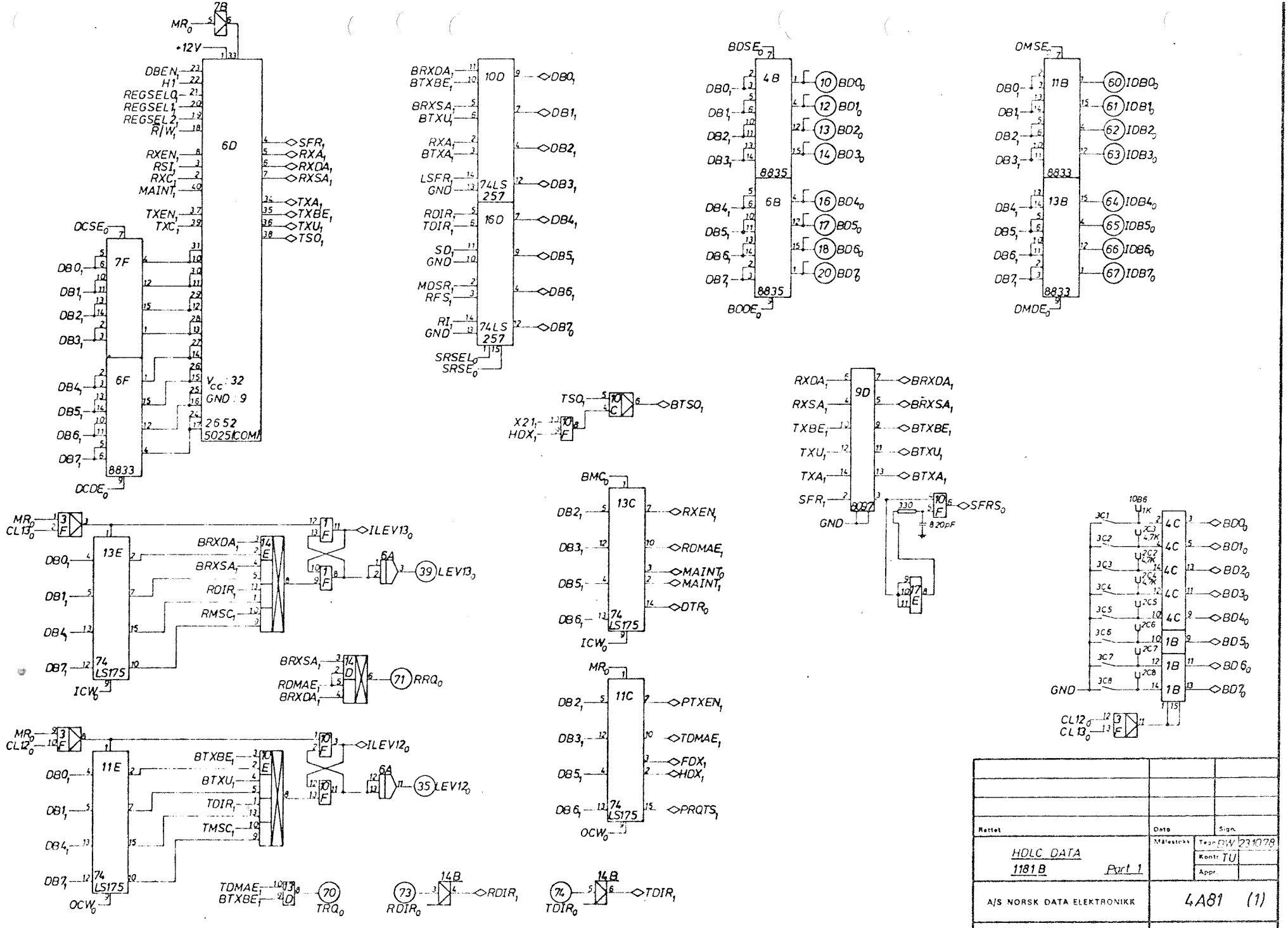


ALL RESISTORS 100 Ω UNLESS OTHERWISE NOTED.
ALL CAPACITORS 1000 PF STYROFLEX UNLESS OTHERWISE NOTED

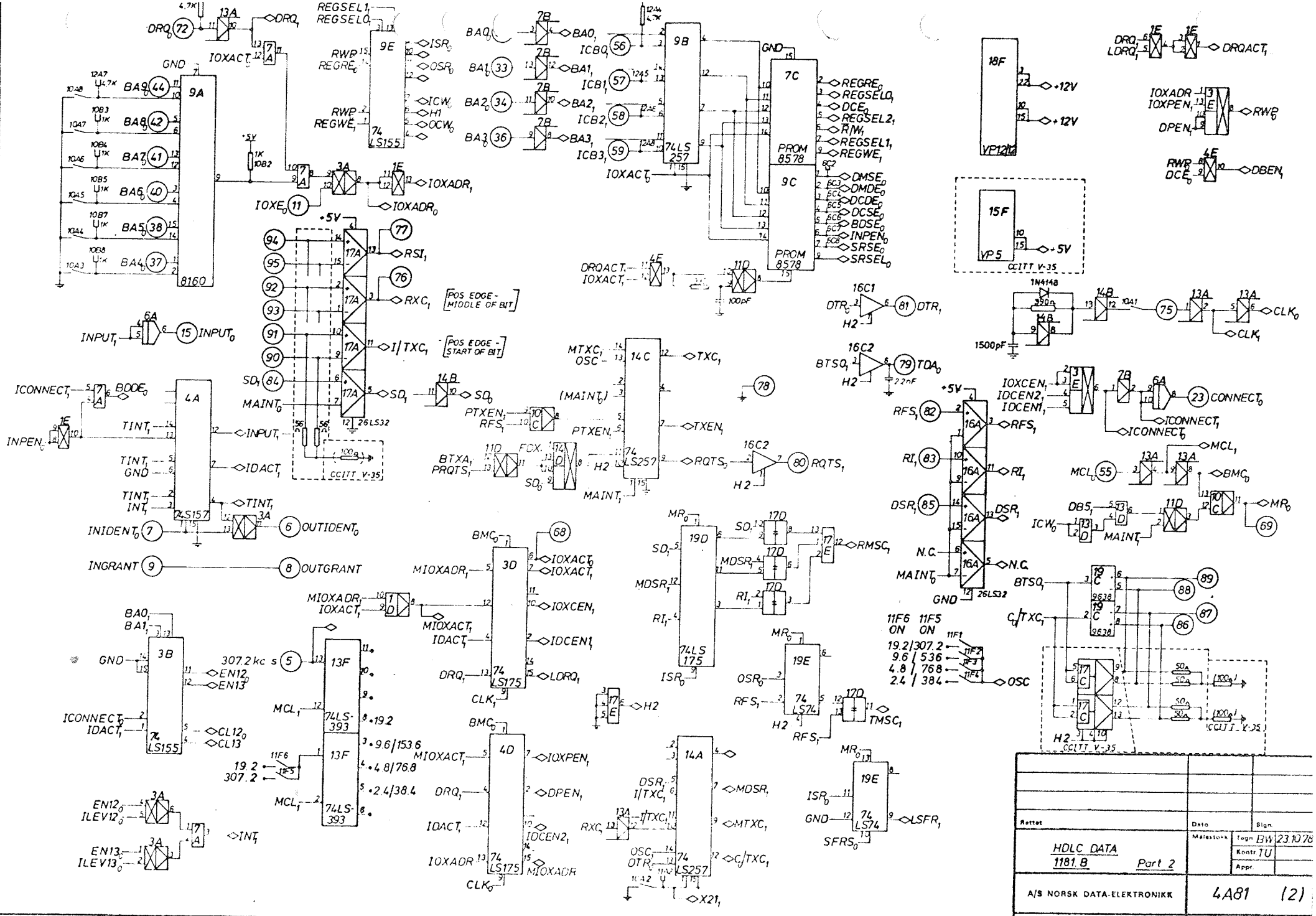
Rettet	Dato	Sign.
UNIVERSAL DMA INTERFACE	Kort nr. 1172 C	Tegn. 8.1.2879 Kontr. PHE Appr.
NORSK DATA A.S	4A72 (2)	

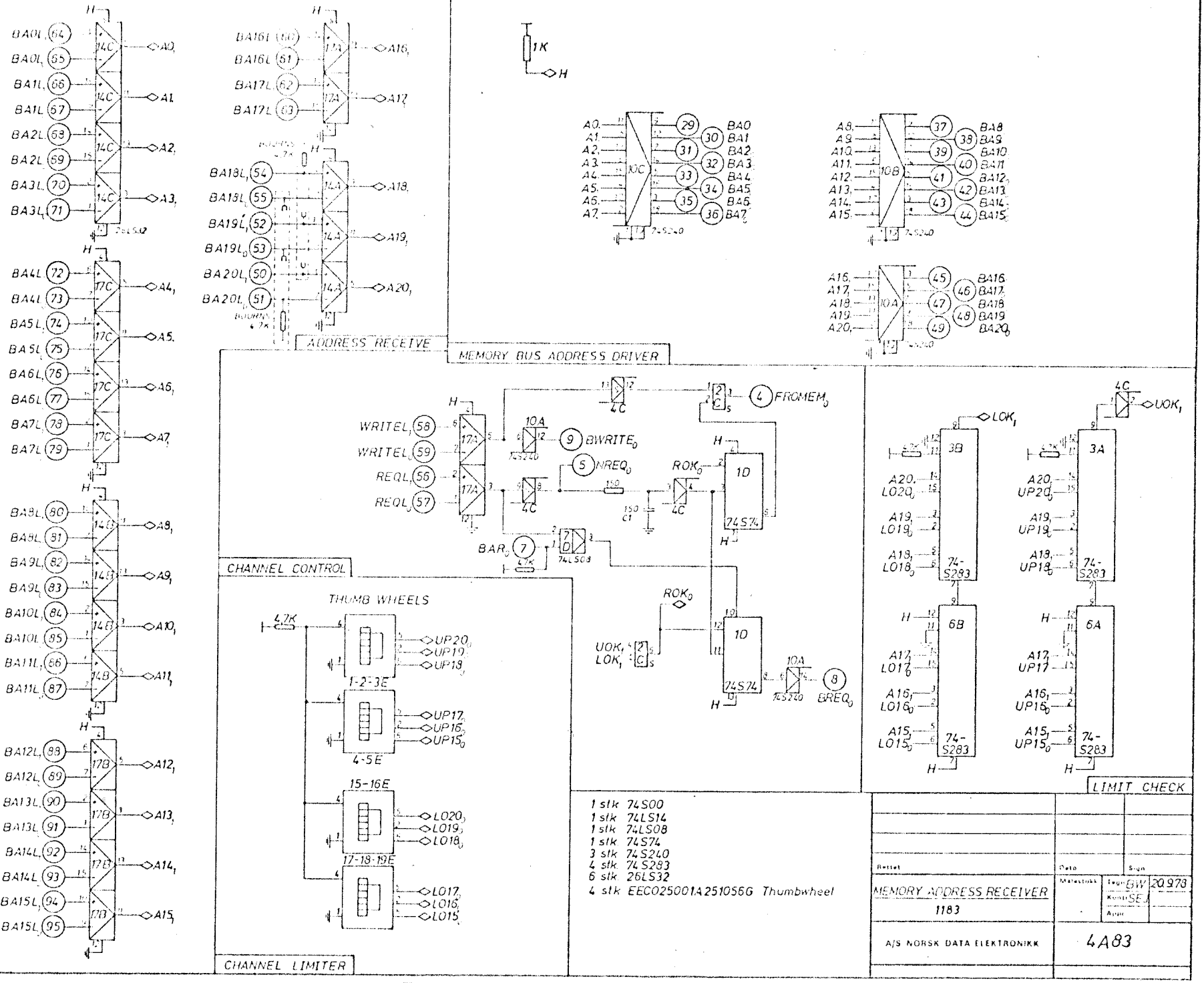


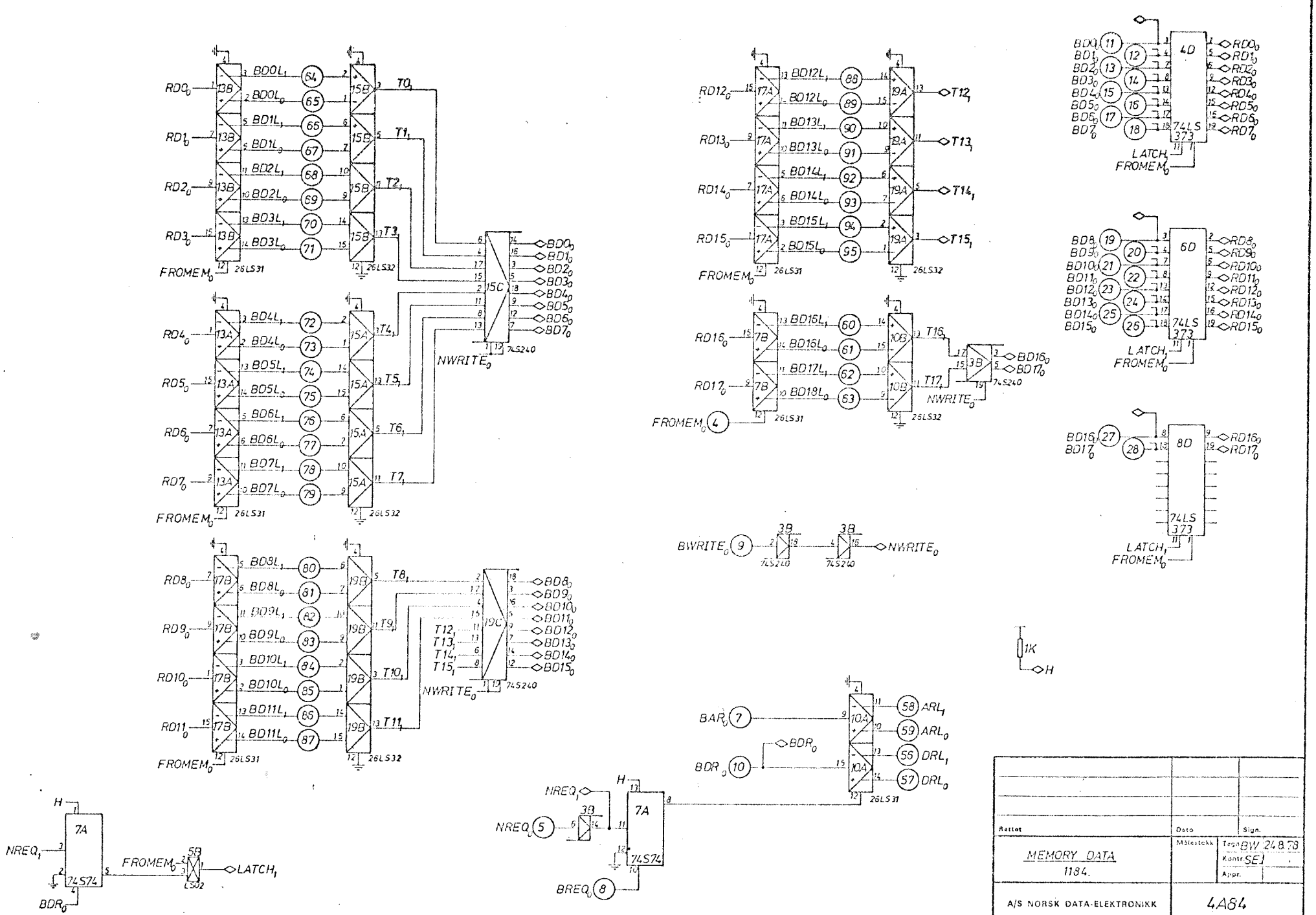
Rettet	Dato	Sign
TELEX TRANSMITTER 1173.B,C	Kvalitetskontr.	Tegn. J.J. 25.7.78 Kontr. Godk. Appr.
NORSK DATA A.S		4A73



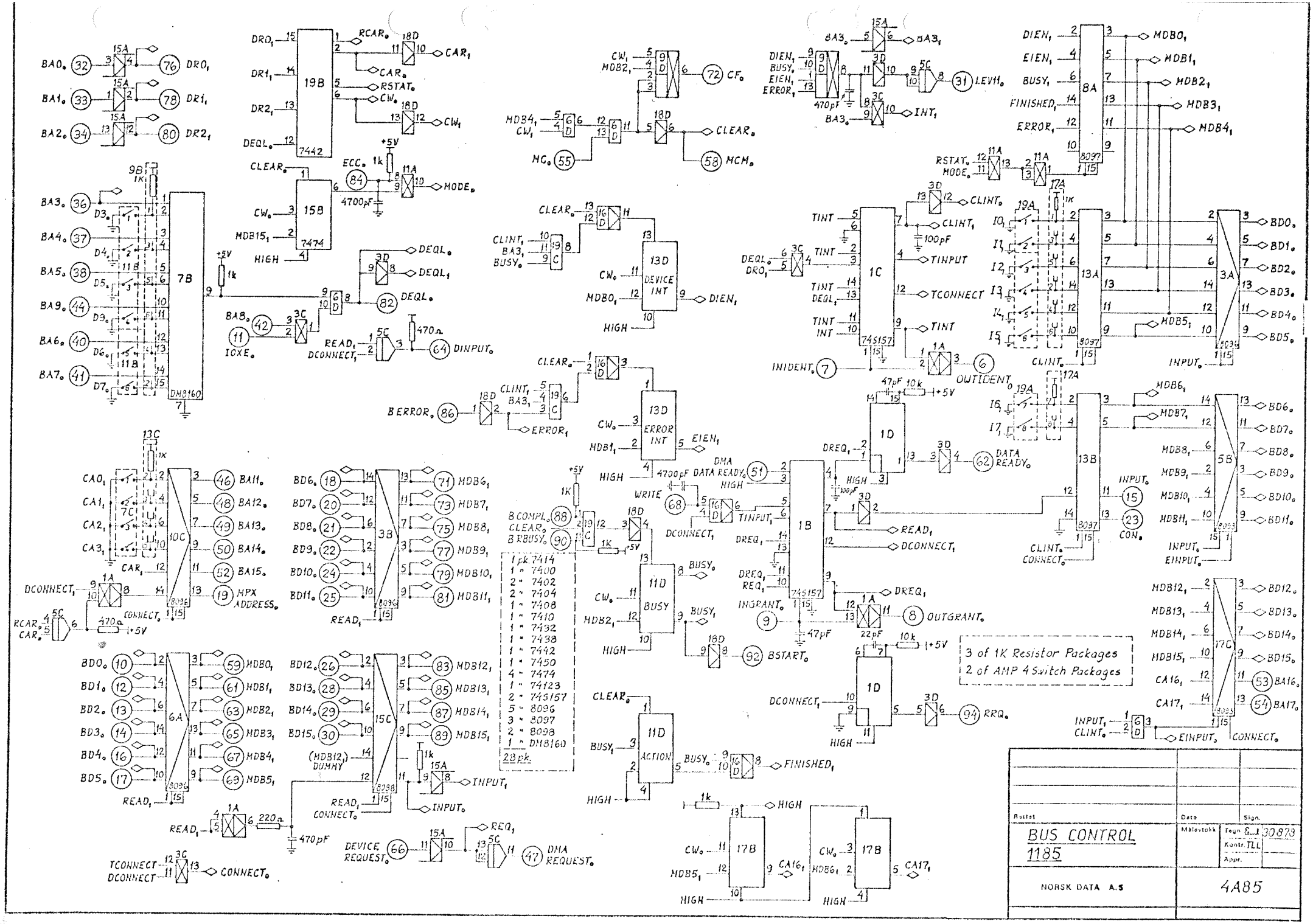
Rettet	Dato	Sign.
HOLC DATA 1181 B Part 1	Målestokk	Tegn. OW 231078 Kontr. TU Appr.
A/S NORSK DATA ELEKTRONIKK	4A81 (1)	

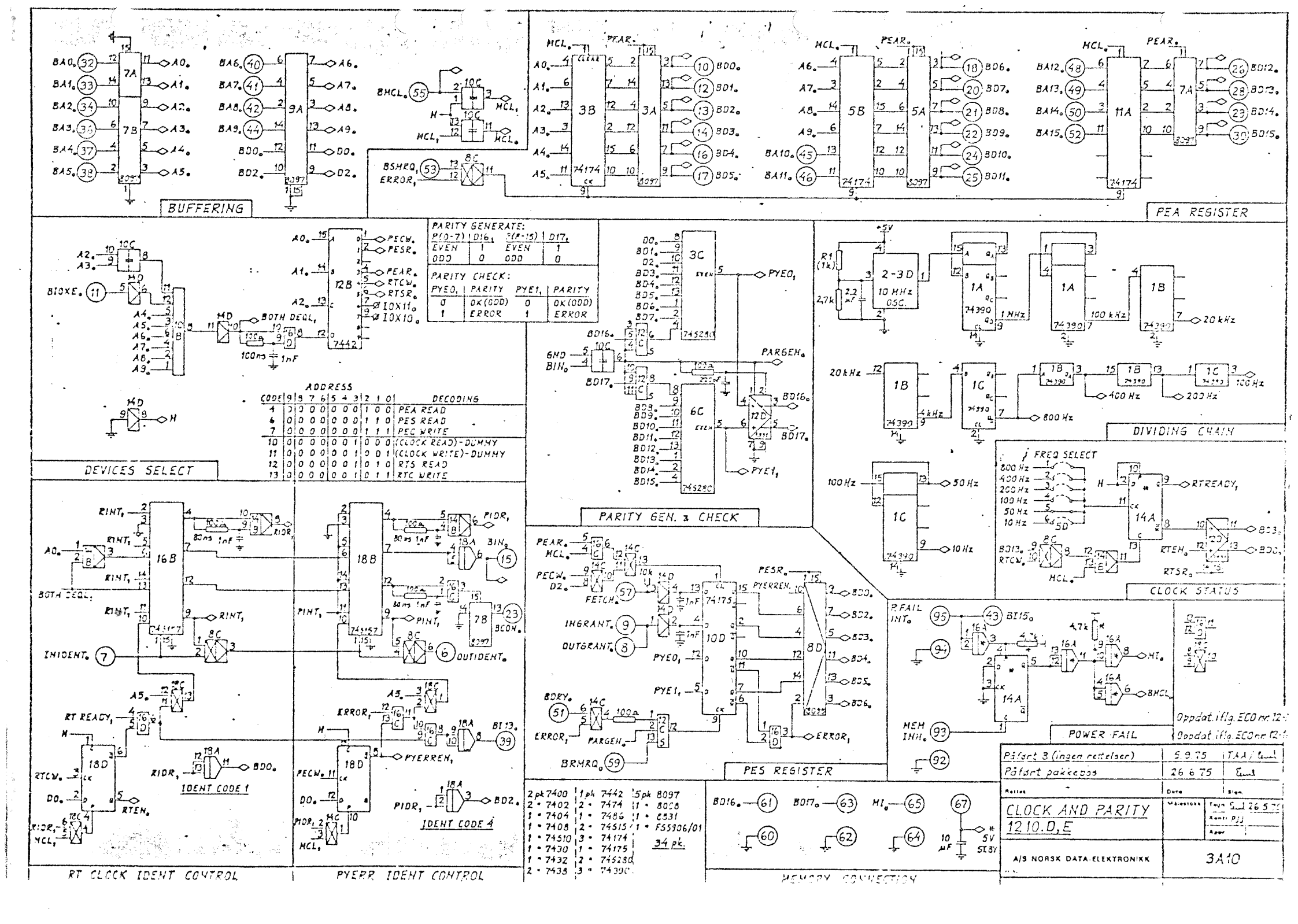

HDLC DATA
1181.B
Part 2
A/S NORSK DATA-ELEKTRONIKK
4A81 (2)
Rettet
Dato
Sign.
Målestokk
Tegn BW 23.10.78
Kontr. TU
Appr.
CCITT V-35
POS EDGE - MIDDLE OF BIT
POS EDGE - START OF BIT
8160
74LS157
74LS155
74LS257
74LS175
74LS74
74LS-393
26LS32
PROM 8578
74LS 257
9A
9B
9C
9E
3B
4A
7C
9638
11F6 11F5
ON ON
19.2/307.2
9.6 / 536
4.8 / 768
2.4 / 384
OSC
307.2 kc s
19.2
307.2
•9.6/153.6
•4.8/76.8
•2.4/38.4
INT
INPUT
OUTPUT
OUTGRANT
OUTIDENT
RSI
RXC
I/TXC
SO
TXC
TXEN
RQTS
RMSC
TMSC
LSFR
MDSR
MTXC
LORQ
IDCEN
IOXCEN
IOXACT
IOXPEN
DPEN
IDCEN2
MIOXADR
IOXADR
CL12
CL13
EN12
EN13
ILEV12
ILEV13
X21
RWB
OBEN
DRQACT
CONNECT
MCL
BMC
MR
RFS
RI
DSR
N.C.
BTSQ
C/TXC
TOA
DTR
CLK
INPEN
IDACT
INIDENT
INGRANT
BOOE
TINT
IDACT
MAINT
PTXEN
BTXA
PRQTS
FOX
H1
H2
+5V
+12V
GND
1N4148
1500pF
100pF
2.2nF

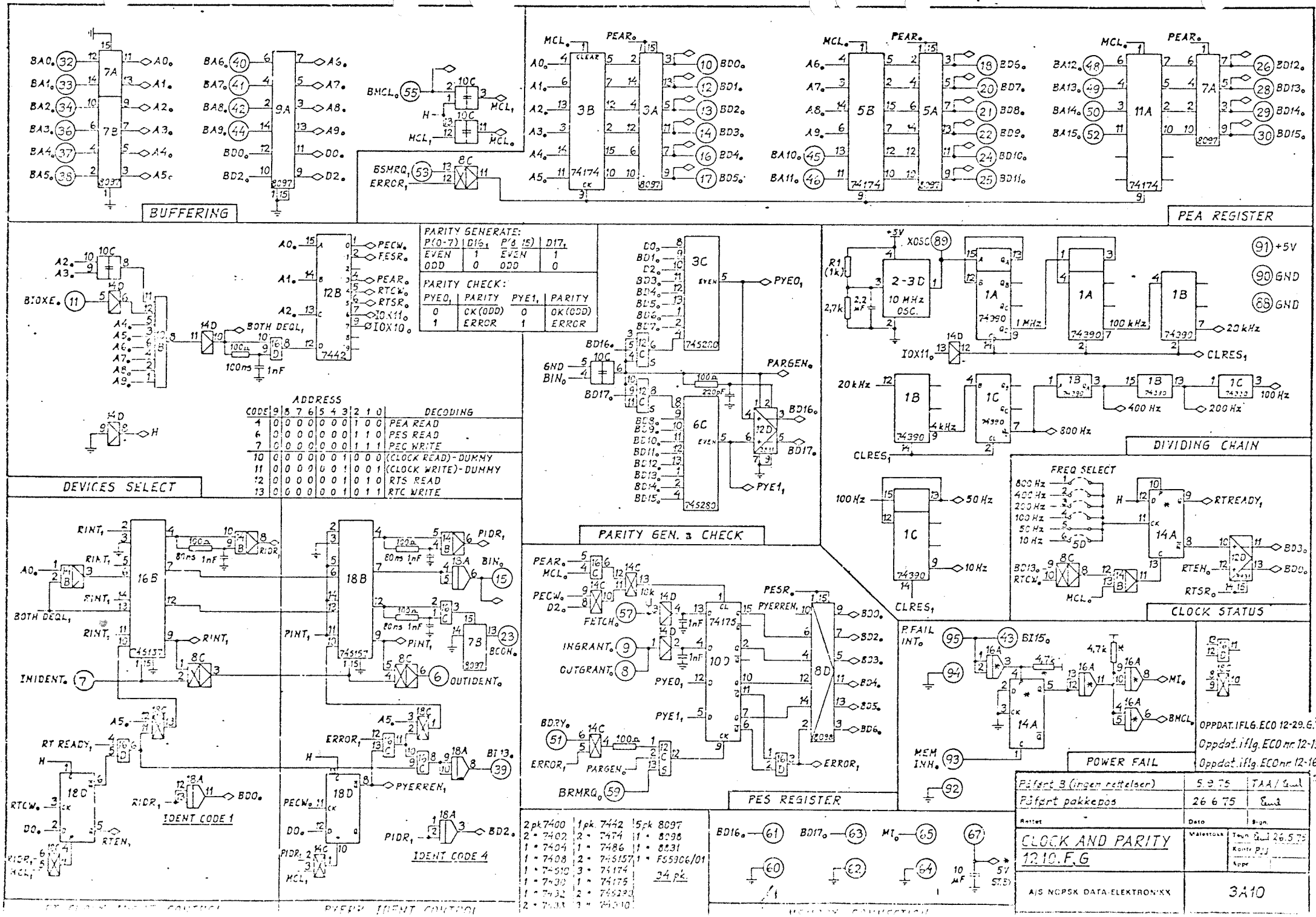

ADDRESS RECEIVE
MEMORY BUS ADDRESS DRIVER
CHANNEL CONTROL
CHANNEL LIMITER
LIMIT CHECK
THUMB WHEELS
1 stk 74S00
1 stk 74LS14
1 stk 74LS08
1 stk 74S74
3 stk 74S240
4 stk 74S283
6 stk 26LS32
4 stk EEC025001A251056G Thumbwheel
MEMORY ADDRESS RECEIVER
1183
A/S NORSK DATA ELEKTRONIKK
4A83

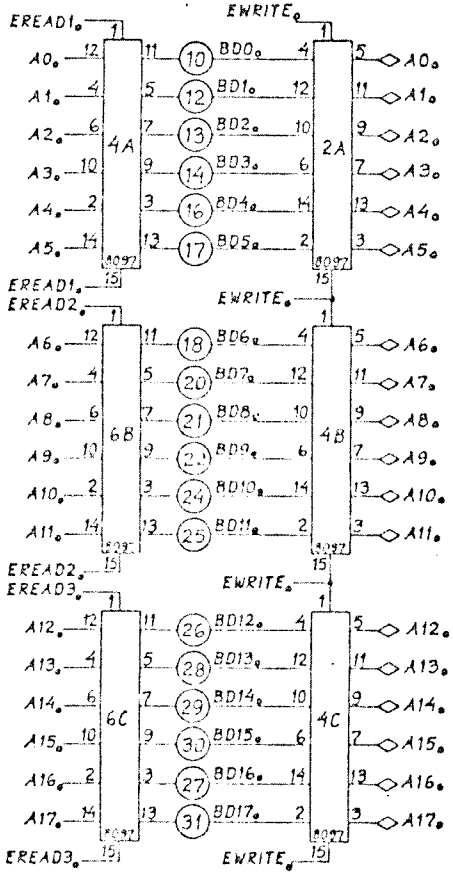


Rettet	Dato	Sign.
MEMORY DATA 1184.	Målestokk	Tegn BW 24 8 78 Kontr SEJ Appr.
A/S NORSK DATA-ELEKTRONIKK	4A84	

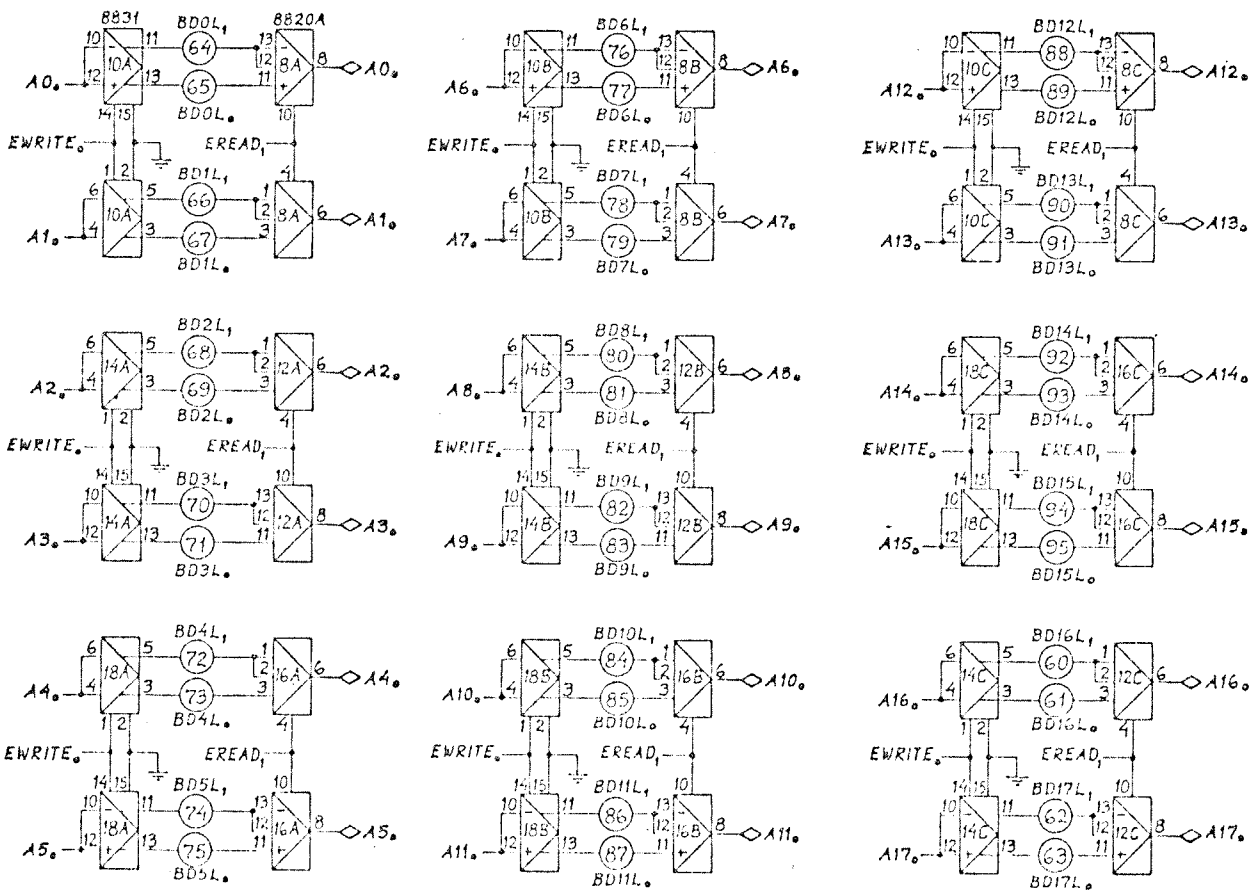

BA0 (32) 15A DR0
BA1 (33) 15A DR1
BA2 (34) 15A DR2
DR0 DR1 DR2 19B 7442 RCAR CAR RSTAT CW 18D CAR CW
DEQL CLEAR CW 15B 7474 MDB15 HIGH +5V 1k ECC (84) 4700pF 11A MODE
9B 1K BA3 (36) BA4 (37) BA5 (38) BA9 (44) BA6 (40) BA7 (41) D3 D4 D5 D9 D6 D7 11B 7B DM8160
+5V 1k 3D DEQL DEQL DEQL (82)
BA8 (42) 3C IOXE (11) READ DCONNECT 5C 470Ω DINPUT (64)
CW MDB2 18D CF (72)
MDB84 CW 6D 18D CLEAR MC (55) MCM (58)
CLEAR 16D 13D DEVICE INT CLINT BA3 BUSY MDB0 HIGH DIEN
DIEN BUSY EIEN ERROR 470pF 3D 5C LEV11 (31) BA3 3C INT
BA3 15A BA3
DIEN EIEN BUSY FINISHED ERROR 8A 8097 MDB0 MDB1 MDB2 MDB3 MDB84
RSTAT MODE 11A 17A 1K
TINT DEQL INT 1C 74S157 3D CLINT 100pF TINPUT TCONNECT TINT 1A INIDENT (7) OUTIDENT (6)
19A IO I1 I2 I3 I4 I5 13A 3A 8034 BD0 BD1 BD2 BD3 BD4 BD5 MDB5 CLINT INPUT
BERROR (86) 18D CLINT BA3 16D 13D ERROR INT CW HDB1 HIGH 4700pF EIEN ERROR
47pF 10k +5V 1D DREQ DMA DATA READY (51) HIGH WRITE (68) DCONNECT TINPUT READ DCONNECT 3D DATA READY (62)
17A 19A I6 I7 13B 8097 INPUT CON (15) (23) CLINT CONNECT MDB6 MDB7 MDB8 MDB9 MDB10 MDB11 5B BD6 BD7 BD8 BD9 BD10 BD11 EINPUT
13C 1K CA0 CA1 CA2 CA3 7C 10C CAR 1A DCONNECT CONNECT 470Ω +5V RCAR CAR 5C BA11 (46) BA12 (48) BA13 (49) BA14 (50) BA15 (52) MPX ADDRESS (19)
BD6 BD7 BD8 BD9 BD10 BD11 3B 8096 READ MDB6 MDB7 MDB8 MDB9 MDB10 MDB11
8 COMPL (88) CLEAR BRBUSY (90) 1K 18D 11D BUSY CW HDB2 HIGH BUSY BSTART (92)
1 pk 7414
1 " 7400
2 " 7402
2 " 7404
1 " 7408
1 " 7410
1 " 7432
1 " 7438
1 " 7442
1 " 7450
4 " 7474
1 " 74123
2 " 74S157
5 " 8096
3 " 8097
2 " 8098
1 " DM8160
28 pk.
DREQ REQ INGRANT (9) 74S157 1A OUTGRANT (8) 47pF 22pF 10k +5V 1D DCONNECT HIGH 3D RRQ (94)
3 of 1K Resistor Packages
2 of AMP 4 Switch Packages
BD0 BD1 BD2 BD3 BD4 BD5 6A 8096 READ MDB0 MDB1 MDB2 MDB3 MDB4 MDB5
BD12 BD13 BD14 BD15 (MDB12) DUMMY 15C 8098 READ CONNECT MDB12 MDB13 MDB14 MDB15 1k 15A INPUT
READ 1A 220Ω 470pF
TCONNECT DCONNECT 3C CONNECT
DEVICE REQUEST (66) 15A REQ 5C (11) (47) DMA REQUEST
HDB12 HDB13 HDB14 HDB15 CA16 CA17 17C 8095 BD12 BD13 BD14 BD15 BA16 (53) BA17 (54)
INPUT CLINT 6D EINPUT CONNECT
CLEAR BUSY ACTION 11D BUSY FINISHED HIGH
1k HIGH CW MDB5 17B CA16 CW MDB6 17B CA17 HIGH HIGH
BUS CONTROL
1185
NORSK DATA A.S
Målestokk
Tegn. nr. 30873
Konstr. TLL
Appr.
4A85


BUFFERING
PEA REGISTER
DEVICES SELECT
PARITY GENERATE:
P(0-7) | D16, | P(8-15) | D17,
EVEN | 1 | EVEN | 1
ODD | 0 | ODD | 0
PARITY CHECK:
PYE0, | PARITY | PYE1, | PARITY
0 | OK(ODD) | 0 | OK(ODD)
1 | ERROR | 1 | ERROR
ADDRESS
CODE | 9 | 8 7 6 | 5 4 3 | 2 1 0 | DECODING
4 | 0 | 0 0 0 | 0 0 0 | 1 0 0 | PEA READ
6 | 0 | 0 0 0 | 0 0 0 | 1 1 0 | PES READ
7 | 0 | 0 0 0 | 0 0 0 | 1 1 1 | PEC WRITE
10 | 0 | 0 0 0 | 0 0 1 | 0 0 0 | (CLOCK READ)-DUMMY
11 | 0 | 0 0 0 | 0 0 1 | 0 0 1 | (CLOCK WRITE)-DUMMY
12 | 0 | 0 0 0 | 0 0 1 | 0 1 0 | RTS READ
13 | 0 | 0 0 0 | 0 0 1 | 0 1 1 | RTC WRITE
PARITY GEN. & CHECK
DIVIDING CHAIN
CLOCK STATUS
RT CLOCK IDENT CONTROL
PYERR IDENT CONTROL
PES REGISTER
POWER FAIL
MEMORY CONNECTION
IDENT CODE 1
IDENT CODE 4
2 pk 7400
2 • 7402
1 • 7404
1 • 7408
1 • 74510
1 • 7430
1 • 7432
2 • 7438
1 pk 7442
2 • 7474
1 • 7486
2 • 74S15
3 • 74174
1 • 74175
2 • 74S280
3 • 74390
5 pk 8097
11 • 8026
1 • 8631
1 • F55906/01
34 pk.
Påført 3 (ingen rettelser) 5.9.75 TAA
Påført pakkepos 26 6 75
CLOCK AND PARITY
1210.D,E
A/S NORSK DATA-ELEKTRONIKK
3A10


BUFFERING
PEA REGISTER
DEVICES SELECT
PARITY GEN. & CHECK
DIVIDING CHAIN
CLOCK STATUS
PES REGISTER
POWER FAIL
PARITY GENERATE:
P(0-7) | BI6₁ | P(8-15) | DI7₁
EVEN | 1 | EVEN | 1
ODD | 0 | ODD | 0
PARITY CHECK:
PYEO₁ | PARITY | PYE1₁ | PARITY
0 | OK(ODD) | 0 | OK(ODD)
1 | ERROR | 1 | ERROR
ADDRESS
CODE | 9 8 7 6 5 4 3 2 1 0 | DECODING
4 | 0 0 0 0 0 0 0 1 0 0 | PEA READ
6 | 0 0 0 0 0 0 0 1 1 0 | PES READ
7 | 0 0 0 0 0 0 0 1 1 1 | PEC WRITE
10 | 0 0 0 0 0 0 1 0 0 0 | (CLOCK READ)-DUMMY
11 | 0 0 0 0 0 0 1 0 0 1 | (CLOCK WRITE)-DUMMY
12 | 0 0 0 0 0 0 1 0 1 0 | RTS READ
13 | 0 0 0 0 0 0 1 0 1 1 | RTC WRITE
IDENT CODE 1
IDENT CODE 4
2pk 7400 | 1pk 7442 | 5pk 8097
2 • 7402 | 2 • 7474 | 1 • 8098
1 • 7404 | 1 • 7486 | 1 • 8831
1 • 7408 | 2 • 745157 | 1 • F55906/01
1 • 74S10 | 3 • 74174 | 34 pk.
1 • 7430 | 1 • 74175
1 • 7432 | 2 • 745280
2 • 7433 | 3 • 74S10
OPPDAT. IFLG. ECO 12-29.6.
Oppdat. iflg. ECO nr 12-13.9.
Oppdat. iflg. ECO nr 12-16.D
Påført 3 (ingen rettelser) | 5.9.75 | TAA / Sund
Påført pakkepos | 26.6.75 | Sund
CLOCK AND PARITY
12.10.F.G
A/S NORSK DATA-ELEKTRONIKK
3A10



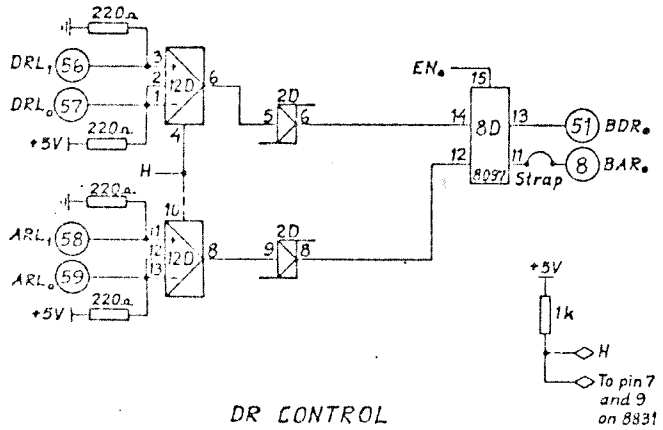
DATA BUFFER



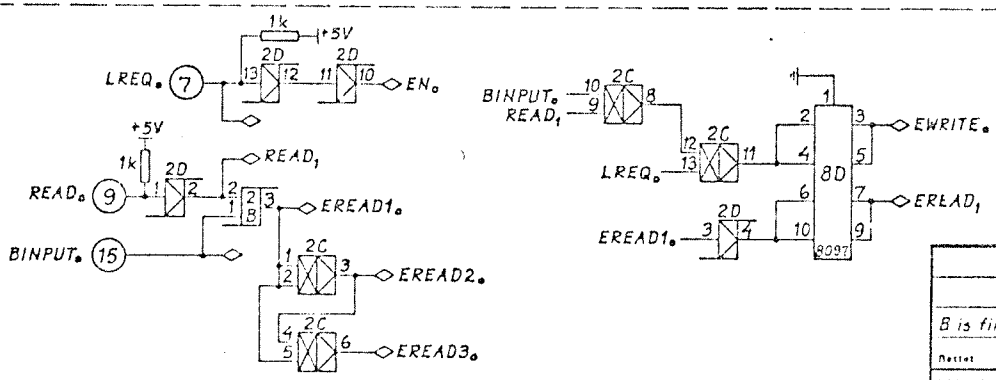
LINE DRIVER-RECEIVER

1 pk. 7414
1 = 7432
1 = 74132
7 = 8097
10 = 8820A
9 = 8831
29 pk.

3 x 1 k.
4 x 220 Ω.



DR CONTROL



CONTROL OF ENABLING

B is first version.

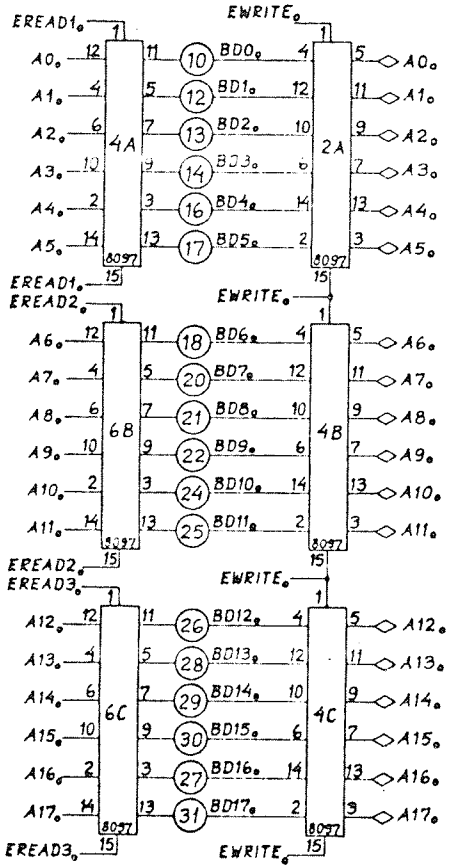
Rettet	Dato	Sign

MPM DATA BUFFER
LINE DRIVER, 1211.B

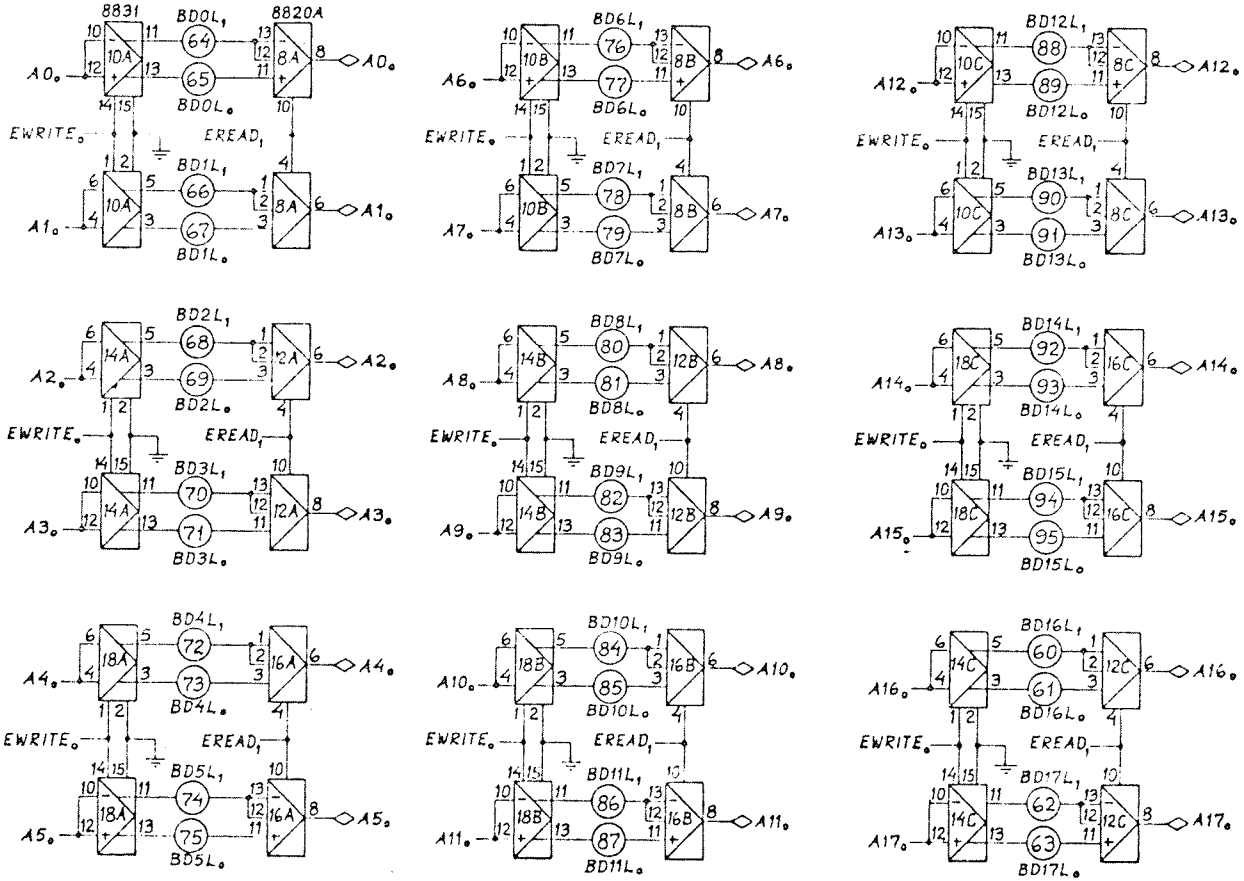
Tegn E.J. 15.9.77
Kontr E.J.
Appr.

A/S NORSK DATA-ELEKTRONIKK

3A11



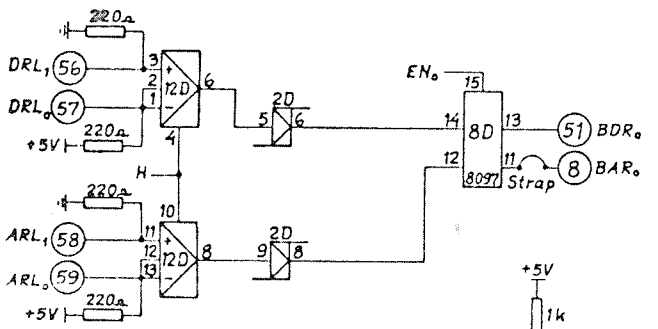
DATA BUFFER



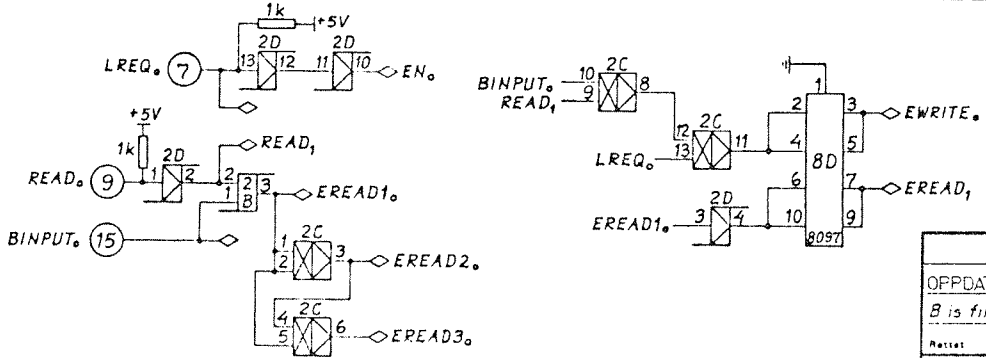
LINE DRIVER-RECEIVER

1 pk. 7414
1 • 7432
1 • 74132
7 • 8097
10 • 8820A
9 • 8831
29 pk

3 x 1kΩ
4 x 220Ω

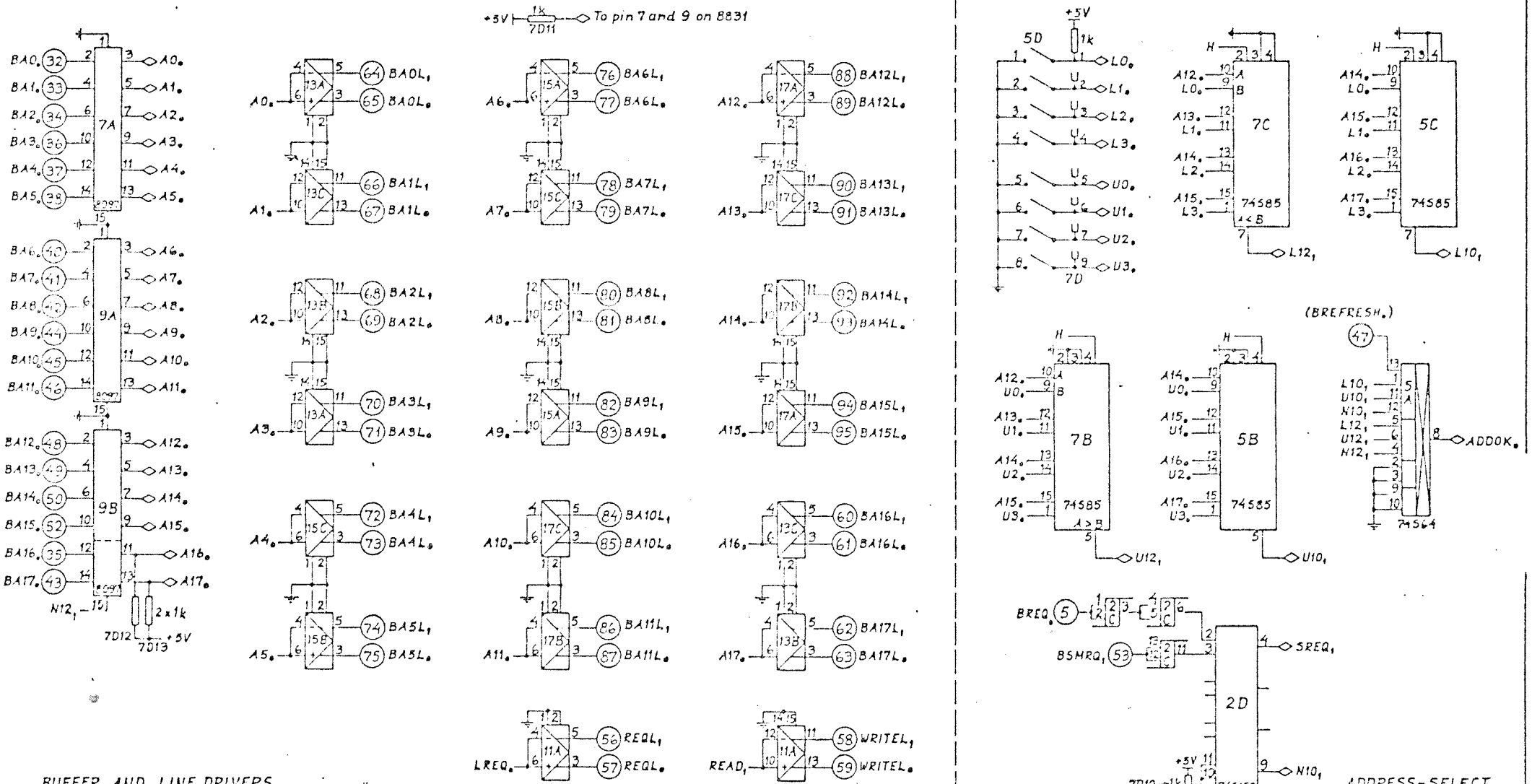


DR CONTROL



CONTROL OF ENABLING

OPPDAT TO M ECO 12-24 PÅFC	23 10 78	BW
B is first version.		
Rettet	Dato	Sign
MPM DATA BUFFER LINE DRIVER, 1211.B,C	Målestokk	Tegn Ø.J 15.9.76 Kontr E.J Appr
A/S NORSK DATA-ELEKTRONIKK		3A11



1pk. 7414	
1 • 74564	
1 • 7474	
4 • 74585	
1 • 74132	
1 • 74S158	
3 • 8097	
10 • 8831	
1 • Switch, 8pos.	
1 • Resistor, 1k	
24pk.	

B is first version.

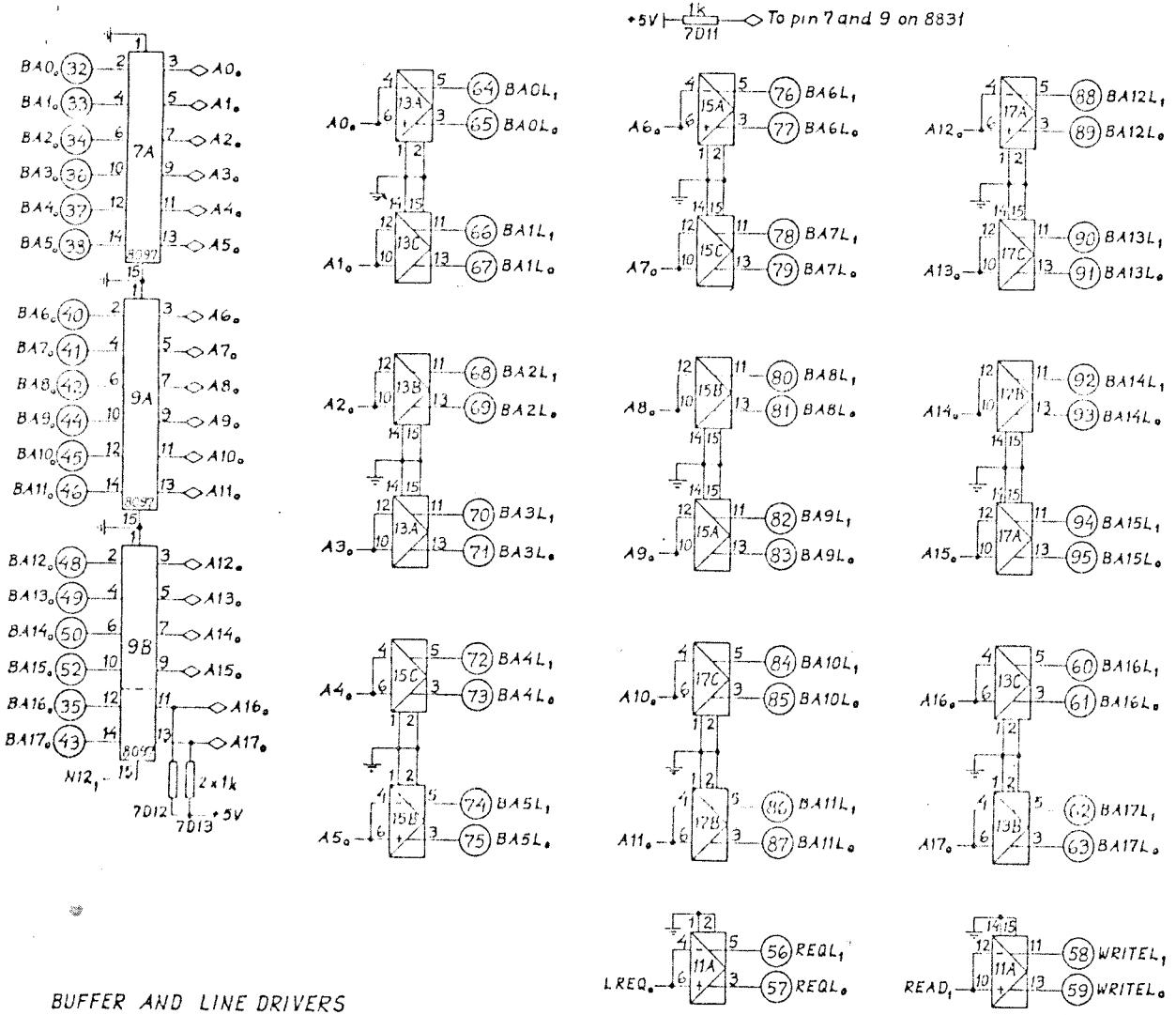
MPM ADDRESS BUFFER LINE DRIVER 1213.B

Tegn C. Gundersen 15.9.76

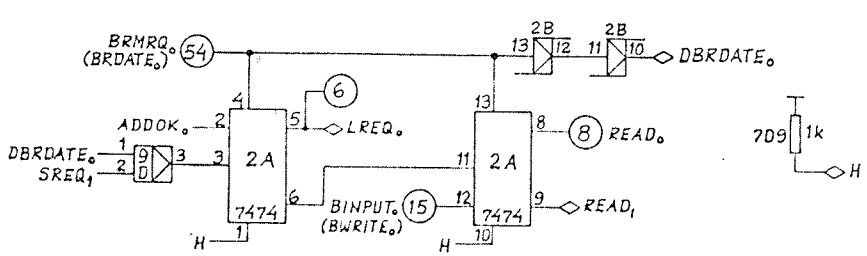
Kontr E.J.

A/S NORSK DATA-ELEKTRONIKK

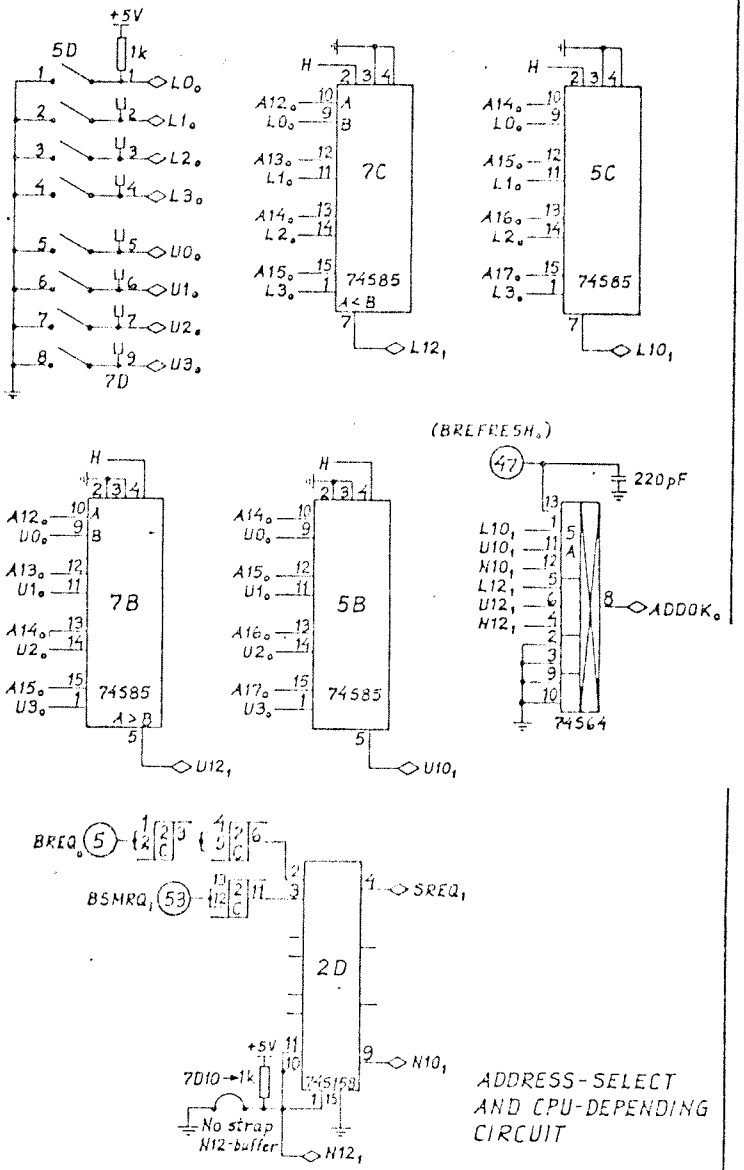
3A13



BUFFER AND LINE DRIVERS

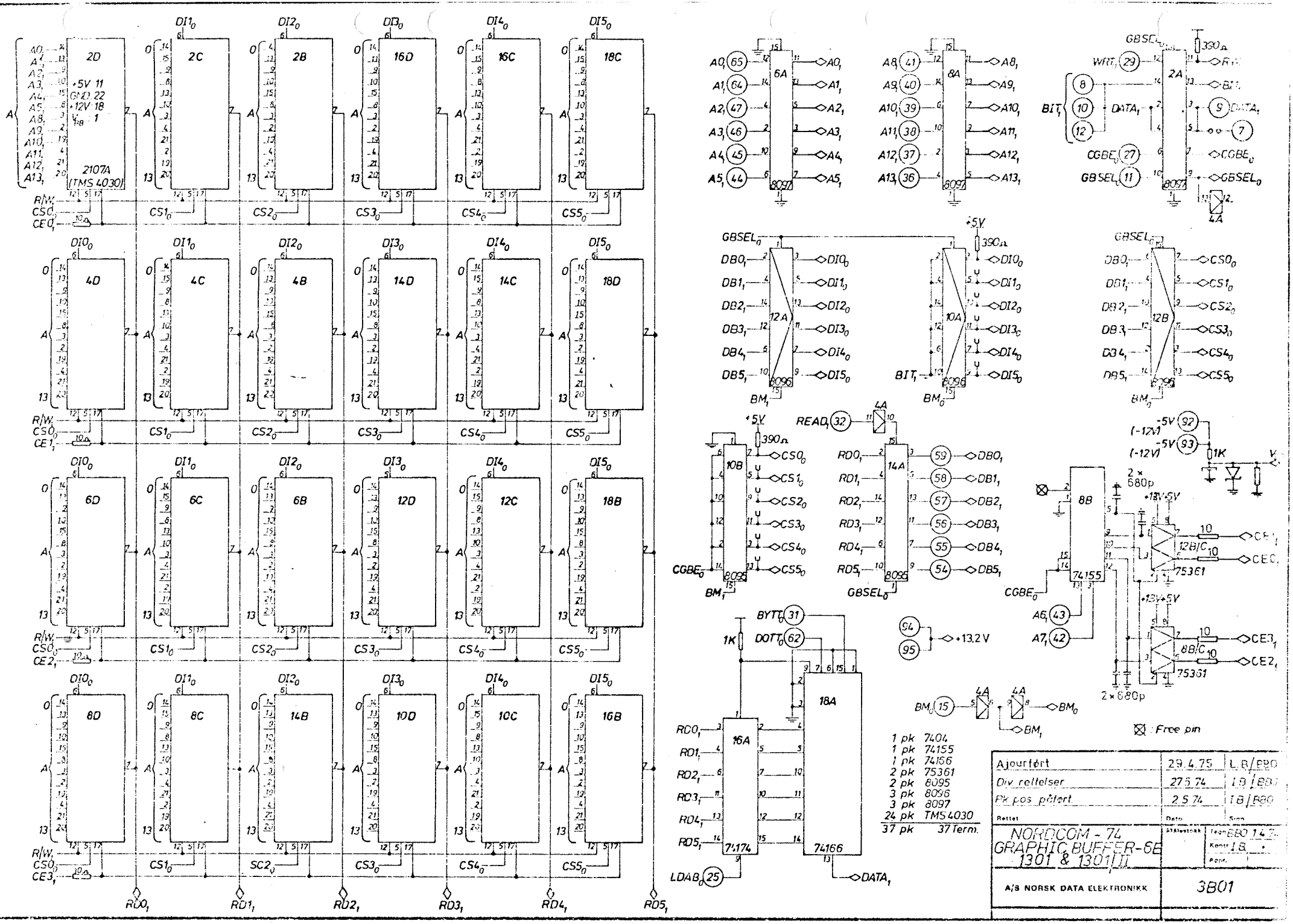


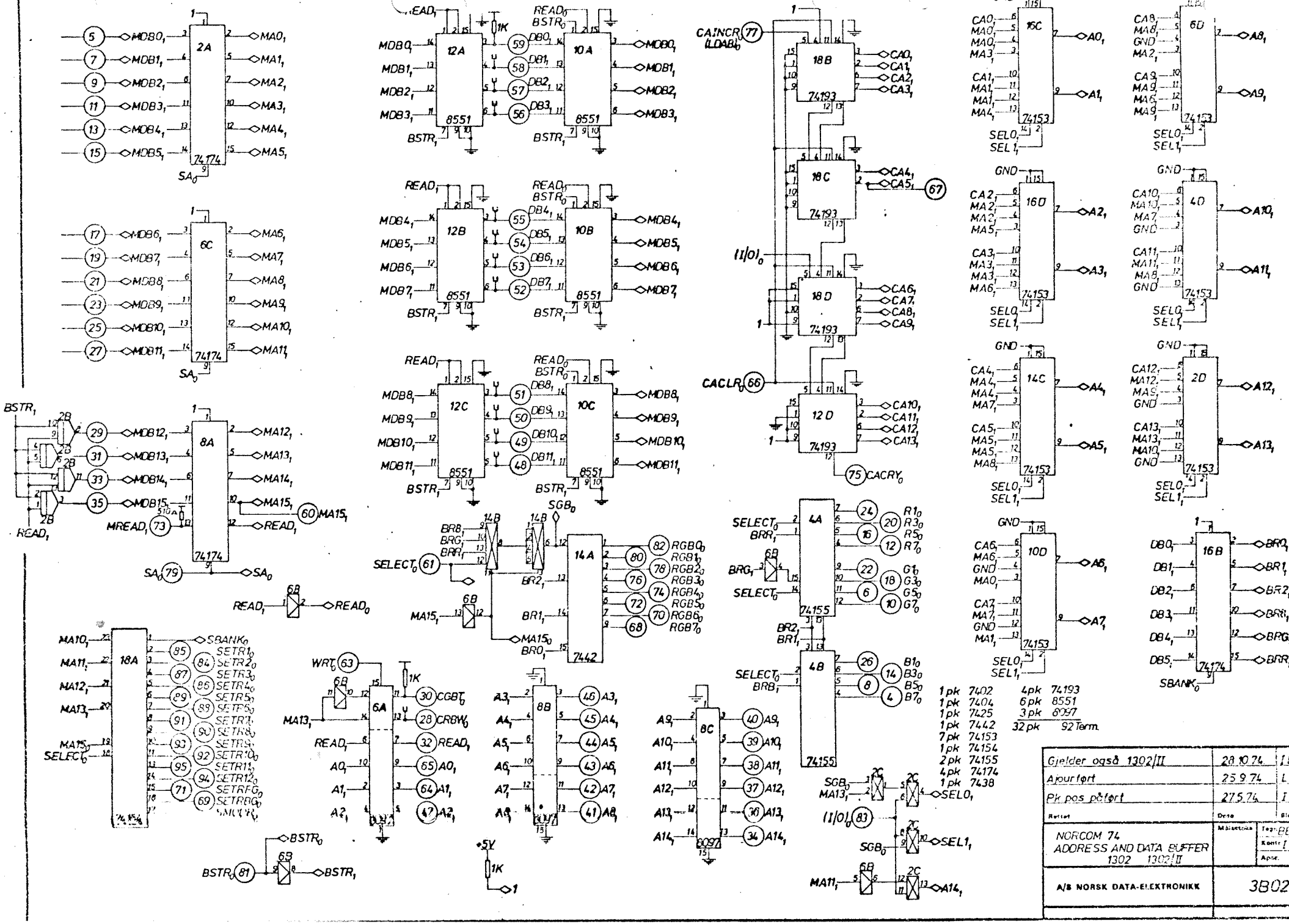
CONTROL

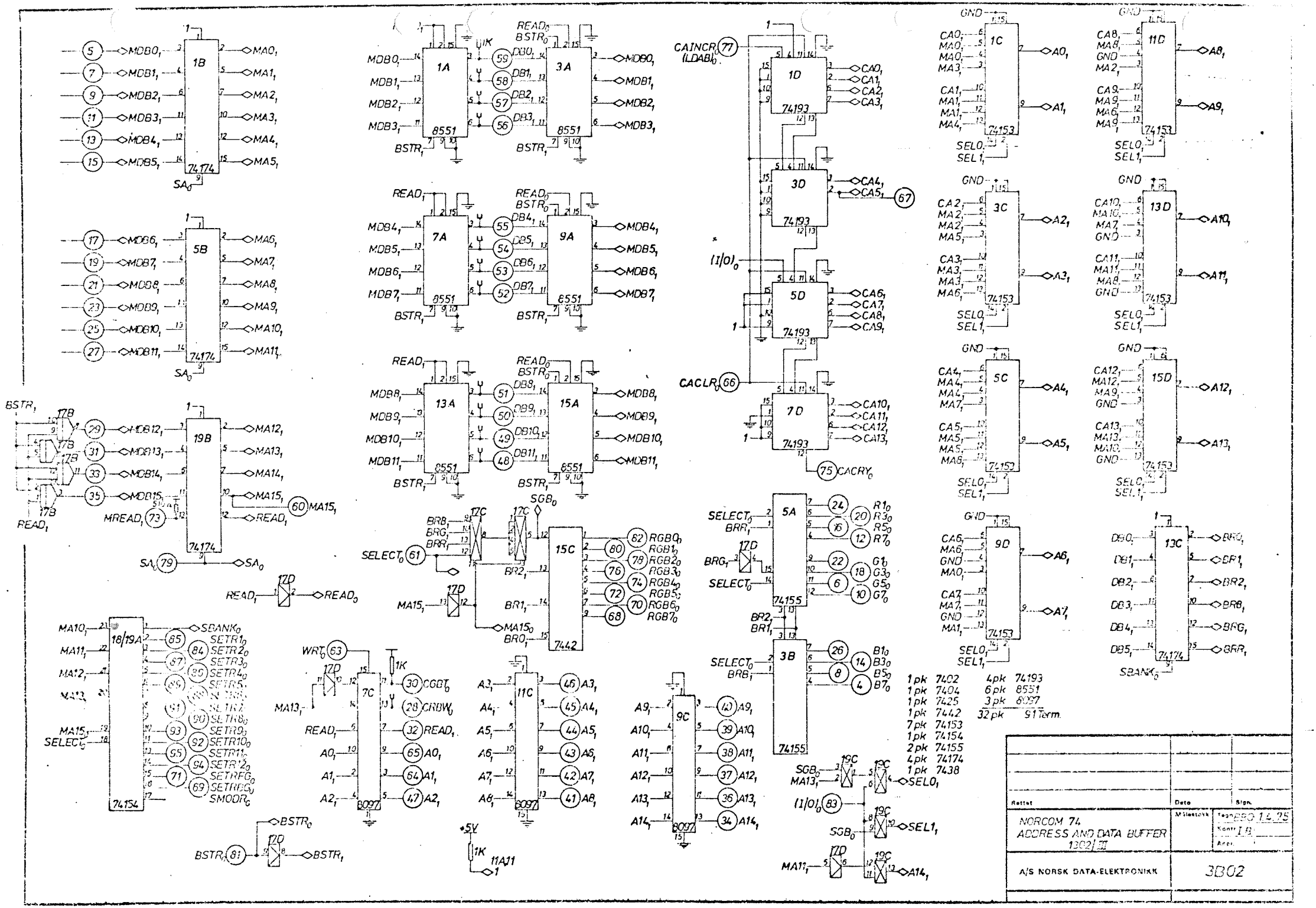


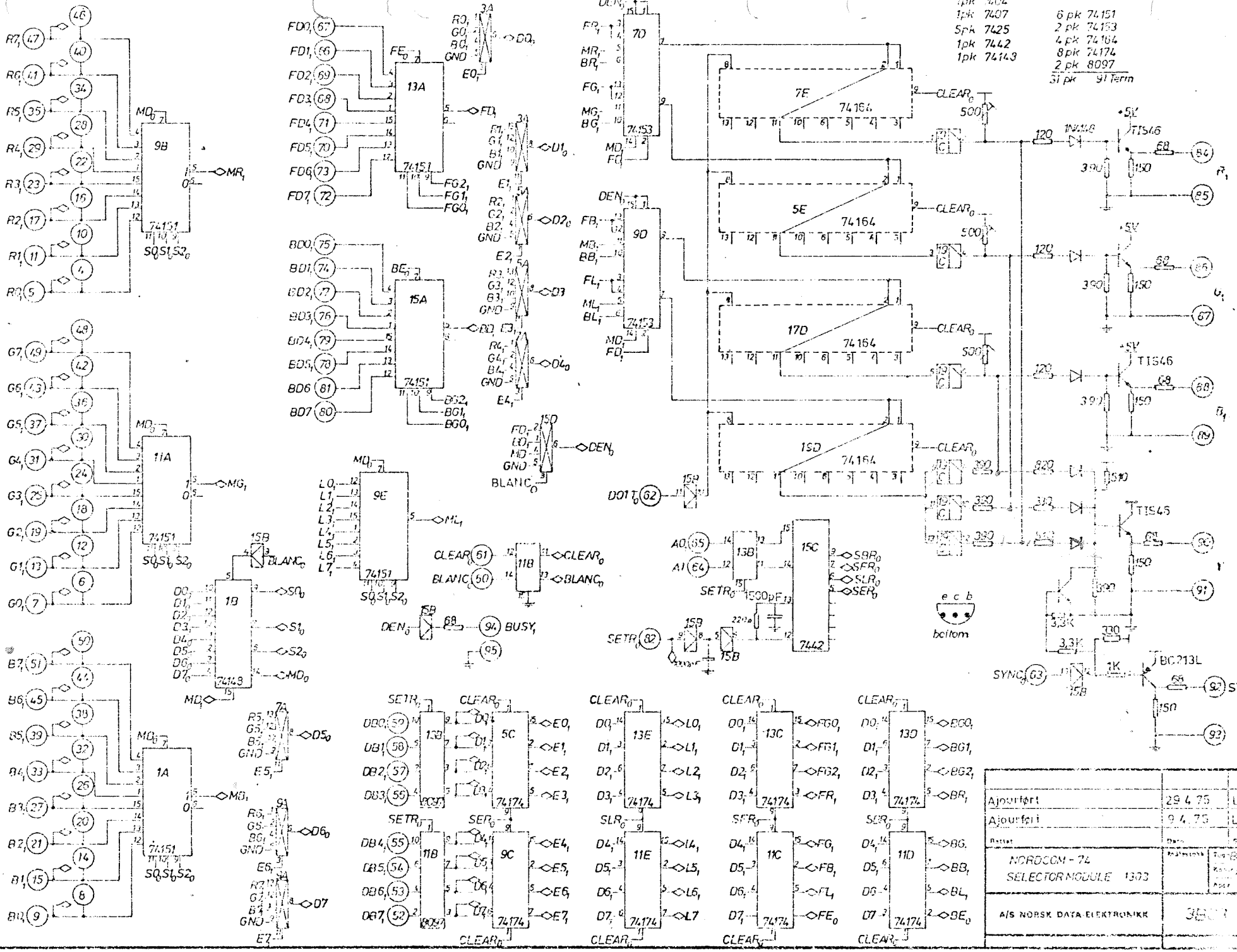
ADDRESS-SELECT AND CPU-DEPENDING CIRCUIT

1pk. 7414	OPPDAT. T.O.M. ECO 12-27 (D)	22.11.77	Emil
1 " 74S64	Pål. E. ECO. 12-30	31.3.78	BW
1 " 7474	B is first version.		
4 " 74S85	Rettet	Dato	Sign
1 " 74132	MPM ADDRESS BUFFER LINE DRIVER 1213.D.E	Målestokk	Tegn. Emil 15.9.76 Kontr. E.J. Appr.
1 " 745158			
3 " 8097			
10 " 8831			
1 " Switch, 8pos.	A/S NORSK DATA ELEKTRONIKK		3A13
1 " Resistor, 1k			
24pk			


2D
2C
2B
16D
16C
18C
4D
4C
4B
14D
14C
18D
6D
6C
6B
12D
12C
18B
8D
8C
14B
10D
10C
16B
2107A
(TMS 4030)
+5V 11
GND 22
+12V 18
V_BB 1
R/W
CS0
CE0
CE1
CE2
CE3
CS1
CS2
CS3
CS4
CS5
SC2
RD0
RD1
RD2
RD3
RD4
RD5
6A
8A
2A
12A
10A
12B
10B
14A
16A
18A
4A
8B
12B/C
8B/C
74155
74166
74174
75361
8095
8096
8097
390Ω
1K
2 × 680p
READ
BYTT
DOTT
LDAB
DATA
BM
BIT
WRT
GBSEL
CGBE
+13.2V
-5V (-12V)
Free pin
1 pk 7404
1 pk 74155
1 pk 74166
2 pk 75361
2 pk 8095
3 pk 8096
3 pk 8097
24 pk TMS4030
37 pk 37 Term.
Ajourført 29.4.75 L.B/BBO
Div rettelser 27.5.74 I.B/BBO
Pk pos påført 2.5.74 I.B/BBO
Rettet Dato Sign
NORDCOM - 74
GRAPHIC BUFFER-6B
1301 & 1301/II
Tegn. BBO 1.4.74
Kontr. I.B.
Godk.
A/S NORSK DATA ELEKTRONIKK
3B01


2A
6C
12A
10A
12B
10B
12C
10C
8A
14A
14B
7442
18A
6A
8B
8C
8097
18B
18C
18D
12D
74193
4A
4B
74155
16C
16D
14C
2D
10D
16B
74153
74174
8551
GND
SELECT0
BRR1
BRG1
MA15
MA13
READ1
BSTR1
WRT0
CAINCR (LDAB)0
CACLR0
CACRY0
SBANK0
SETR1..SETR12
SETRFG0
SETRBG0
SMOVR0
SGB0
1K
5V
1 pk 7402
1 pk 7404
1 pk 7425
1 pk 7442
7 pk 74153
1 pk 74154
2 pk 74155
4 pk 74174
1 pk 7438
4pk 74193
6 pk 8551
3 pk 8097
32 pk 92 Term.
Gjelder også 1302/II 28.10.74 LB/BBO
Ajourført 25.9.74 LB/BBO
Pk pos påført 27.5.74 LB/BBO
NORCOM 74
ADDRESS AND DATA BUFFER
1302 1302/II
Tegn: BBO 3.4.74
Kontr: LB
A/S NORSK DATA-ELEKTRONIKK
3B02


1B
74174
5B
19B
74174
17B
READ1
MREAD1
SA0
1A
3A
8551
BSTR1
READ0
BSTR0
7A
9A
13A
15A
1K
17C
15C
7442
SGB0
SBANK0
SETR1 0
SETR2 0
SETR3 0
SETR4 0
SETR5 0
SETR6 0
SETR7 0
SETR8 0
SETR9 0
SETR10 0
SETR11 0
SETR12 0
SETREG0
SMODR0
18/19A
74154
17D
7C
8097
CGBT0
CRDW0
READ1
11C
11A11
5V
9C
CAINCR0
(LDAB)0
1D
74193
3D
5D
7D
CACLR0
(1/0)0
CACRY0
5A
74155
3B
9D
13C
15D
5C
3C
1C
11D
74153
SEL0
SEL1
1pk 7402
1pk 7404
1pk 7425
1pk 7442
7pk 74153
1pk 74154
2pk 74155
4pk 74174
1pk 7438
4pk 74193
6pk 8551
3pk 8097
32pk 91 Term.
NORCOM 74
ADDRESS AND DATA BUFFER
1302/II
A/S NORSK DATA-ELEKTRONIKK
3B02


1 pk 7404
1 pk 7407
5 pk 7425
1 pk 7442
1 pk 74143
6 pk 74151
2 pk 74153
4 pk 74164
8 pk 74174
2 pk 8097
31 pk 91 Term
9B
13A
15A
9E
11A
1B
1A
7D
9D
7E
5E
17D
15D
74164
74153
74151
74148
74174
74LS46
74LS45
74LS10
74LS00
74LS30
74LS20
74LS04
74LS08
74LS32
74LS86
74LS27
74LS21
74LS14
74LS08
74LS02
74LS03
74LS05
74LS09
74LS10
74LS11
74LS12
74LS13
74LS15
74LS19
74LS22
74LS26
74LS28
74LS33
74LS37
74LS38
74LS39
74LS40
74LS42
74LS48
74LS51
74LS54
74LS55
74LS60
74LS61
74LS62
74LS63
74LS64
74LS65
74LS66
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74LS69
74LS70
74LS71
74LS72
74LS73
74LS74
74LS75
74LS76
74LS77
74LS78
74LS79
74LS80
74LS81
74LS82
74LS83
74LS84
74LS85
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74LS87
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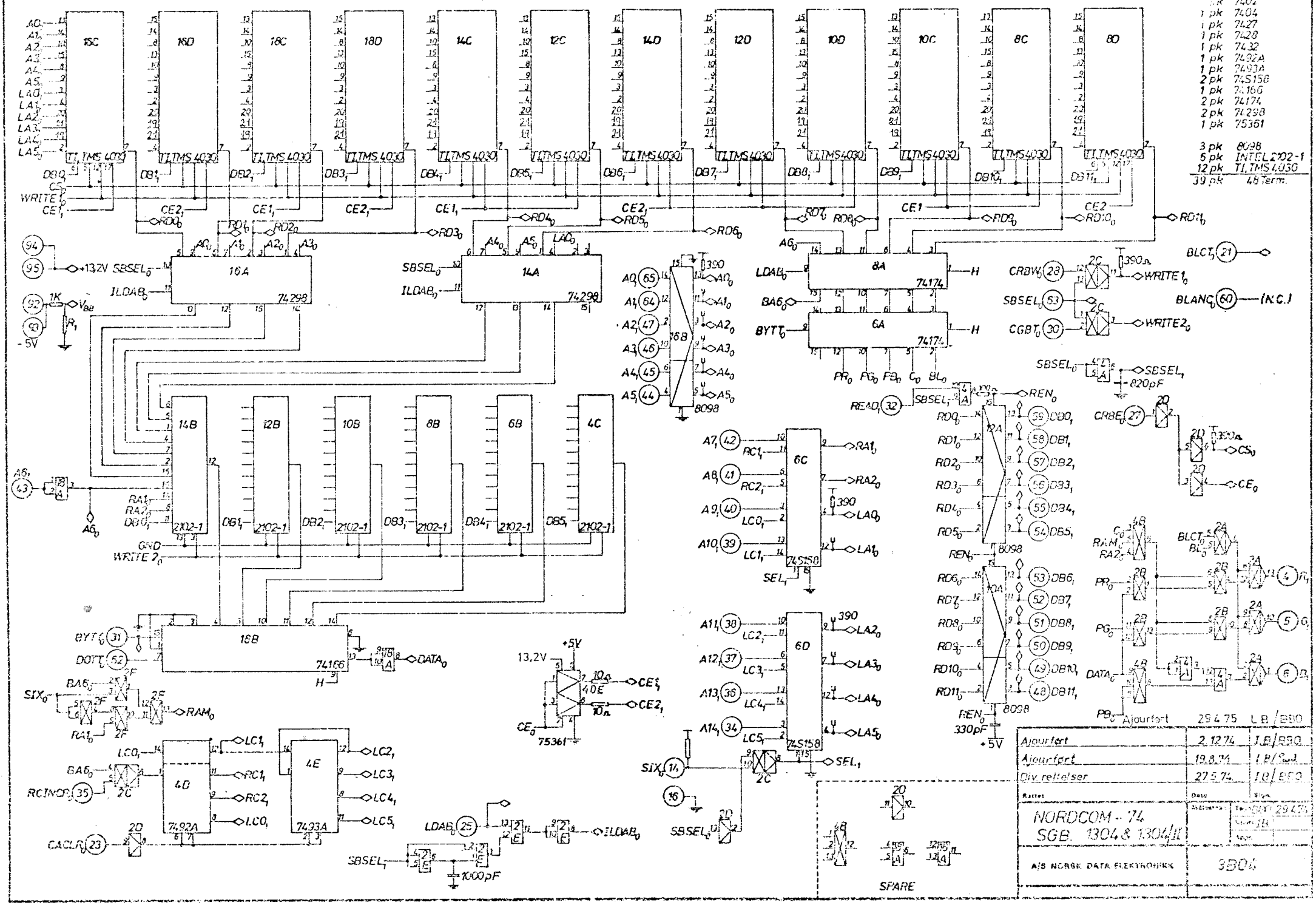
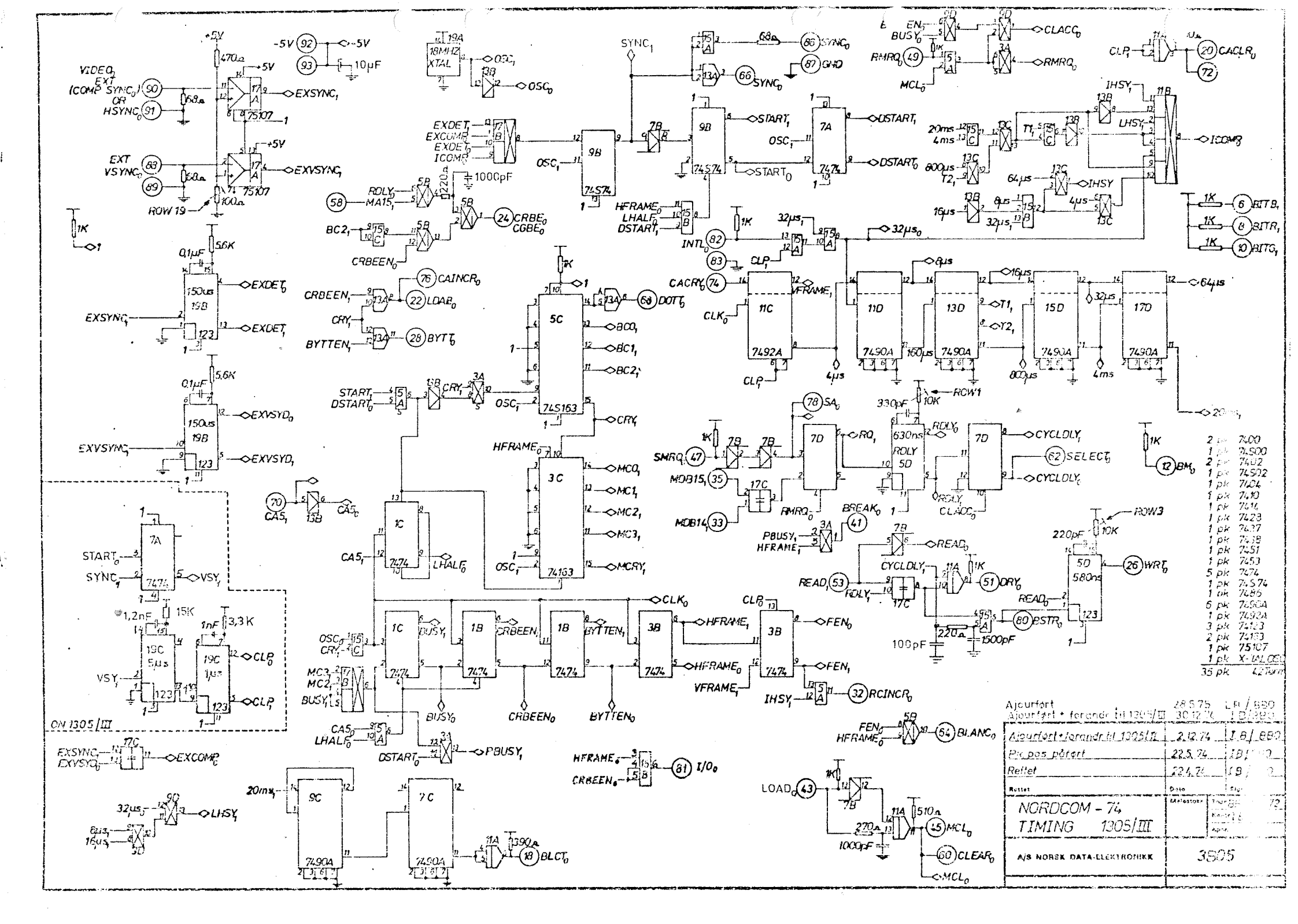
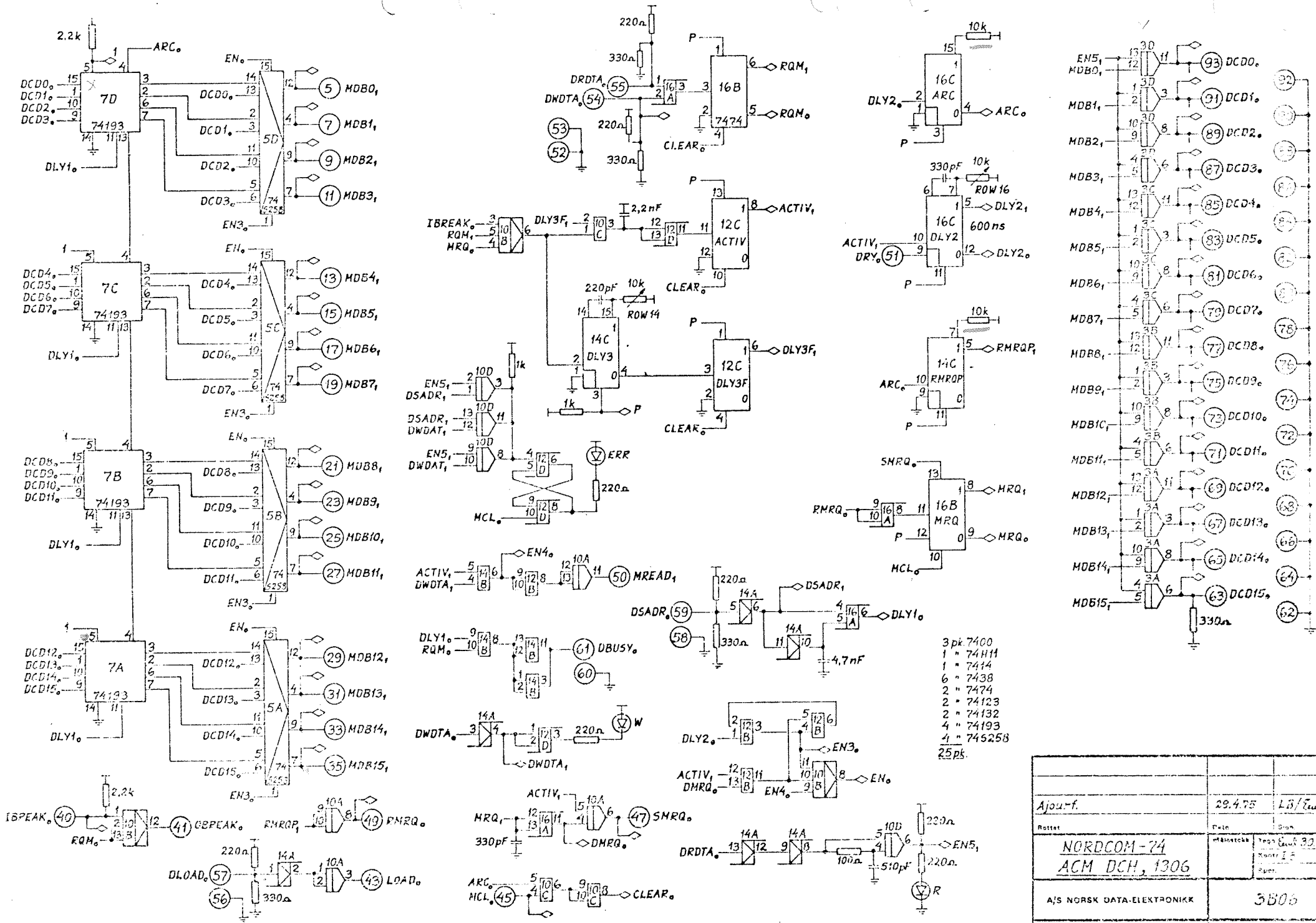
74LS96
74LS97
74LS98
74LS99

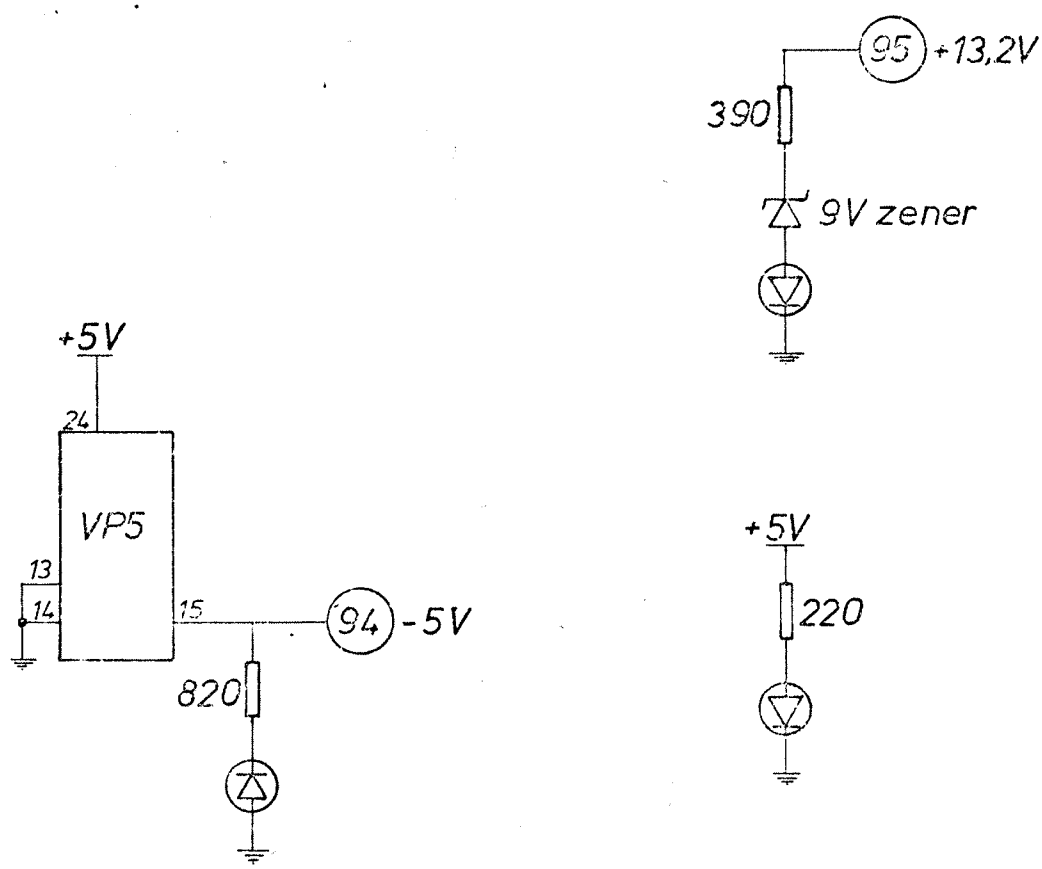


k	7400	
k	7402	
1 pk	7404	
1 pk	7427	
1 pk	7428	
1 pk	7432	
1 pk	7492A	
1 pk	7493A	
2 pk	74S158	
1 pk	74166	
2 pk	74174	
2 pk	74298	
1 pk	75361	
3 pk	8098	
6 pk	INTEL 2102-1	
12 pk	TI TMS 4030	
39 pk	48 Term.	

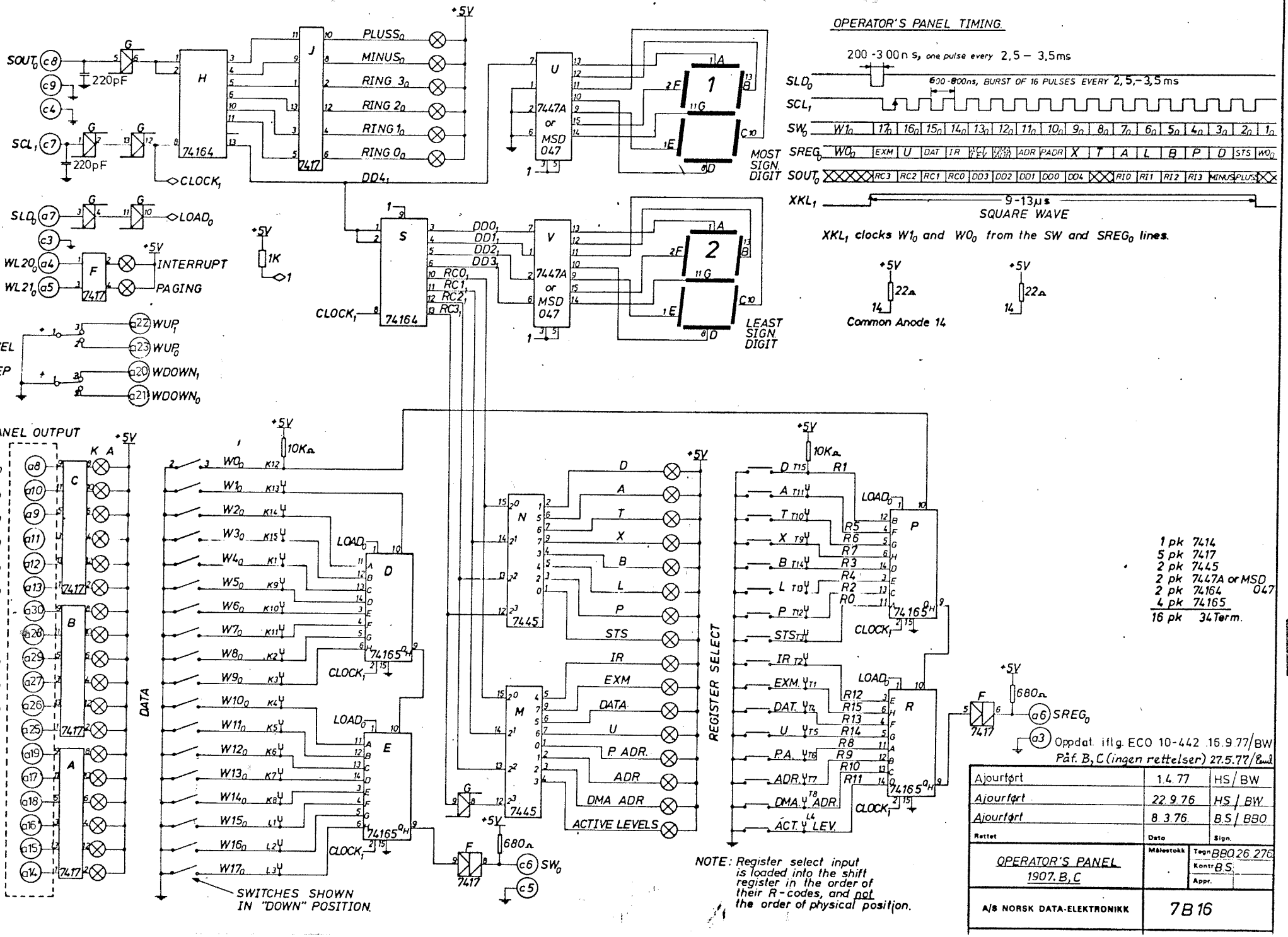

NORDCOM - 74
TIMING 1305/III
A/S NORSK DATA-ELEKTRONIKK
3805
ON 1305/III
2 pk 7400
1 pk 7400
2 pk 7402
1 pk 7402
1 pk 7404
1 pk 7410
1 pk 7414
1 pk 7428
1 pk 7437
1 pk 7438
1 pk 7451
1 pk 7453
5 pk 7474
1 pk 74S74
1 pk 7486
6 pk 7490A
1 pk 7492A
3 pk 74163
2 pk 74193
1 pk 75107
1 pk X-VALGS
35 pk 427mA


7D
7C
7B
7A
74193
5D
5C
5B
5A
74
S258
DCD0₀
DCD1₀
DCD2₀
DCD3₀
DCD4₀
DCD5₀
DCD6₀
DCD7₀
DCD8₀
DCD9₀
DCD10₀
DCD11₀
DCD12₀
DCD13₀
DCD14₀
DCD15₀
DLY1₀
EN₀
EN3₀
ARC₀
2.2k
MDB0₁
MDB1₁
MDB2₁
MDB3₁
MDB4₁
MDB5₁
MDB6₁
MDB7₁
MDB8₁
MDB9₁
MDB10₁
MDB11₁
MDB12₁
MDB13₁
MDB14₁
MDB15₁
16B
7474
CLEAR₀
RQM₁
RQM₀
DRDTA
DWDTA₀
220Ω
330Ω
12C
ACTIV
DLY3F
CLEAR₀
ACTIV₁
DLY3F₁
IBREAK₀
RQM₁
MRQ₀
2,2nF
14C
DLY3
ROW14
220pF
10k
1k
P
10D
EN5₁
DSADR₁
DWDAT₁
ERR
MCL₀
16C
ARC
DLY2
ROW16
330pF
600ns
DLY2₀
DLY2₁
ACTIV₁
DRY₀
14C
RMRQP
RMRQP₁
ARC₀
16B
MRQ
SMRQ₀
MRQ₁
RMRQ₀
MRQ₀
EN4₀
10A
ACTIV₁
DWDTA₁
MREAD₁
14A
DSADR₀
DLY1₀
4,7nF
DBUSY₀
W
DWDTA₀
DWDTA₁
DLY2₀
EN3₀
ACTIV₁
DMRQ₀
EN4₀
EN₀
IBREAK₀
RQM₀
GBREAK₀
RMRQP₁
RMRQ₀
DLOAD₀
LOAD₀
MRQ₁
SMRQ₀
DMRQ₀
ARC₀
MCL₀
CLEAR₀
DRDTA₀
100Ω
510pF
R
93 DCD0₀
91 DCD1₀
89 DCD2₀
87 DCD3₀
85 DCD4₀
83 DCD5₀
81 DCD6₀
79 DCD7₀
77 DCD8₀
75 DCD9₀
73 DCD10₀
71 DCD11₀
69 DCD12₀
67 DCD13₀
65 DCD14₀
63 DCD15₀
3 pk 7400
1 " 74H11
1 " 7414
6 " 7438
2 " 7474
2 " 74123
2 " 74132
4 " 74193
4 " 745258
25 pk
Ajour-f.
29.4.75
LB/Emil
Rettet
Dato
Sign
NORDCOM-74
ACM DCH, 1306
A/S NORSK DATA-ELEKTRONIKK
3806

A/S NORSK DATA-ELEKTRONIKK	Title: -5V GEN 1314	Drawing no.: NORCOM-74



DRAWN BY BBO	Remarks	Replacement for	Date
APPROVED BY L.B.		Replaced by	Date
DATE 30.4.75.			


OPERATOR'S PANEL TIMING
200 -300 ns, one pulse every 2,5 – 3,5ms
600-800ns, BURST OF 16 PULSES EVERY 2,5,–3,5 ms
SLD0
SCL1
SW0
SREG0
SOUT0
XKL1
9-13µs
SQUARE WAVE
XKL1 clocks W10 and WO0 from the SW and SREG0 lines.
MOST SIGN. DIGIT
LEAST SIGN. DIGIT
Common Anode 14
SOUT0
SCL1
SLD0
WL200
WL210
LEVEL
STEP
INTERRUPT
PAGING
CLOCK1
LOAD0
DD41
PLUSS0
MINUS0
RING 30
RING 20
RING 10
RING 00
74164
74174
7447A or MSD 047
220pF
1K
22Ω
680Ω
10KΩ
PANEL OUTPUT
DATA
REGISTER SELECT
SWITCHES SHOWN IN "DOWN" POSITION.
NOTE: Register select input is loaded into the shift register in the order of their R-codes, and not the order of physical position.
ACTIVE LEVELS
DMA ADR
ADR
P ADR.
U
DATA
EXM
IR
STS
P
L
B
X
T
A
D
1 pk 7414
5 pk 7417
2 pk 7445
2 pk 7447A or MSD 047
2 pk 74164
4 pk 74165
16 pk 34 Term.
Oppdat. iflg. ECO 10-442 16.9.77/BW
Påf. B,C (ingen rettelser) 27.5.77/Emil
Ajourført 1.4.77 HS/BW
Ajourført 22.9.76 HS/BW
Ajourført 8.3.76. BS/BBO
Rettet Dato Sign.
OPERATOR'S PANEL
1907.B,C
Målestokk
Tegn BBO 26.276
Kontr BS
Appr.
A/S NORSK DATA-ELEKTRONIKK
7B16